66-week study module

Index — every file, what is in it, and the order to read it

34 files: 671 pages of PDF, a 155-problem tracker, an 11-sheet workbook and seven data files. Keep them all in one folder, including `problem-data`, and every link in the tracker works. Files are numbered in reading order, so a folder listing sorted by name is the reading order.

Start here

Open **study-tracker.html** in Chrome or Firefox. Its Today tab gives the session, the reading desk names the module file and the problems keyed to that week, and the Materials tab lists everything with one-click links. If a link does not open, read `setup-multi-device.pdf` — on an iPad they cannot work until the folder is hosted.

1 · The module, numbered in reading order

File	Weeks	Pages	Contents
module-01-foundations-wk01-10.pdf	1–10	35	Biology foundations for engineers — your original part 1
module-02-core-concepts-wk01-10.pdf	1–10	39	Core concepts for weeks 1–10, compiled: ten chapters, every number computed, 65-term glossary
module-03-framework-wk11-19.pdf	11–19	37	The framework: modelling, control, and the numbers that decide a design
module-04-building-the-strain-wk20-25.pdf	20–25	22	Building the strain: assembly, assays, genome engineering
module-05-designing-the-parts-wk26-31.pdf	26–31	26	Designing the parts: biosensors, regulation, libraries, protein design, natural products
module-06-reading-the-data-wk32-34.pdf	32–34	17	Reading the data: integration, sequencing, RNA-seq, the biofoundry
module-07-constraint-models-wk35-38.pdf	35–38	18	Constraint-based modelling: FBA, COBRApy, reconstruction, MATLAB
module-08-model-to-design-wk39-44.pdf	39–44	24	From model to design: strain design, machine learning, labelling, the bioreactor
module-09-product-recovery-wk45-48.pdf	45–48	20	Getting the product out: toxicity, in situ removal, two-phase modelling, immobilisation
module-10-trusting-numbers-wk49-52.pdf	49–52	19	Trusting the numbers: economics, validation, omics, host defences
module-11-product-classes-wk53-57.pdf	53–57	20	What the field has made: five product classes and their constraints
module-12-hosts-wk58-61.pdf	58–61	18	Hosts: what each organism gives you and what it charges
module-13-the-profession-wk62-66.pdf	62–66	20	The profession: publishing, writing, planning, the proposal, the application

2 · Extension modules, written after the 66 weeks

File	Pages	Contents
module-14-supplementary-chapters.pdf	24	Six supplementary chapters S1–S6: separations, biomass deconstruction, two-substrate kinetics, consortia, quantitative omics, control theory
module-15-protein-products.pdf	15	Protein products — secretion, inclusion bodies, refolding, glycosylation — plus five theory chapters, and problems PR1–PR8
module-16-multiphase-and-doe.pdf	13	Three-phase and solid-state fermentation, gel immobilisation as a process, the pathway-modification playbook, design of experiments as a method, and problems MP1–MP4 and DE1–DE4
module-17-editing-screening-lca.pdf	13	Cas12a, base and prime editing; flavonoids and P450s; FACS and droplets; life cycle analysis; and problems EV1–EV8
module-18-stability-and-screening-funnel.pdf	14	The stability chain vial-to-harvest, ALE campaign design, the screening funnel, contamination and phage, and problems SB1–SB4 and SC1–SC4
module-19-heterogeneity-and-dsp.pdf	9	Single-cell noise, bimodality and persisters wired into sorting; the upstream/downstream capital split for a commodity and a biologic; foam and debris quantified; problems HN1–HN3 and DS1–DS3

Each chapter in these says which weeks it extends, so they can be read when you reach the relevant week rather than at the end.

3 · Problem sets

File	Pages	Contents
problems-01-X1-X24-Y1-Y6.pdf	17	X1–X24 and Y1–Y6 — balances, kinetics, transport, statistics, data, probability, economics; plus six set on real public datasets you download yourself
problems-02-Z1-Z9.pdf	9	Z1–Z9 — downstream separations, quantitative omics, consortia
problems-03-MC1-MC5-R1-R5.pdf	8	MC1–MC5 on model curation, and R1–R5 worked on the supplied instrument data
problems-04-PP1-PP12.pdf	11	PP1–PP12 — the pipeline end to end, host choice through to a costed process
problems-05-corrigendum-and-verification.pdf	2	Corrigendum: the appendix holds 56 problems, not 54 — B4 and G2 were missed by a faulty extraction of mine. B4 verified, the G2 claim retracted, the verification restated for all 56

Modules 15 to 19 carry their own problems as well — PR1–PR8, MP1–MP4 and DE1–DE4, EV1–EV8, SB1–SB4 and SC1–SC4, HN1–HN3 and DS1–DS3. All 155 problems are also inside the tracker with statement, questions and worked solution.

4 · The tracker, your own documents, and the tools

File	Size	Contents
study-tracker.html	—	**THE TRACKER.** 66 weeks, 330 sessions, 155 problems with statement, questions and worked solution, a Materials tab linking every file, a competency tier map, week grid and search
study-tracker-original.html	—	The original tracker, untouched, as a fallback
00-complete-volume.pdf	461	Every PDF merged into one bookmarked volume, for reading on a tablet
curriculum-66-weeks-56-problems.pdf	210	Your curriculum, with the 56-problem appendix from its page 140
lookup-workbook.xlsx	11 sheets	Fillable workbook: the numbers to look up, live formulas, and sheet 10 listing every unverified claim
reading-list-by-tier.pdf	3	Books, papers and tools by competency tier, against module chapters and problems
reading-map-by-week.pdf	4	Textbook chapter per week, with a blank column for your own page numbers
setup-multi-device.pdf	2	How to host the folder so the links work on an iPad and a Mac
problem-data/ (7 files)	200 kB	Data for problems R1–R5, plus the answers file to open only after you have an answer

Cross-references: the old part numbers

The chapters were written as numbered "parts" and cross-reference each other that way. The files are now numbered in reading order instead, so this table maps between them.

File	Called in the text	Weeks	File	Called in the text	Weeks
Module 01	Part 01	1–10	Module 02	Part 14	1–10
Module 03	Part 02	11–19	Module 04	Part 07	20–25
Module 05	Part 08	26–31	Module 06	Part 09	32–34
Module 07	Part 10	35–38	Module 08	Part 11	39–44
Module 09	Part 03	45–48	Module 10	Part 04	49–52
Module 11	Part 05	53–57	Module 12	Part 06	58–61
Module 13	Part 12	62–66	Module 14	Part 13	supplementary
Problems 01	Part 15	—	Problems 02	Part 16	—
Problems 03	Part 17	—	Problems 04	Part 18	—
Module 15	Part 19	—	Module 16	Part 20	—
Module 17	Part 21	—	Module 18	Part 22	—
Module 19	Part 23	—			

Two things to do first

1. Read `problems-05-corrigendum-and-verification.pdf`. It records a counting error of mine and its correction: the appendix

holds 56 problems, not the 54 I reported for several rounds, because a text extraction of mine silently dropped two.

2. Open sheet 10 of the lookup workbook. It lists every claim in the module that came from memory rather than from a calculation, with where to verify each. All arithmetic was computed and re-checked; those inputs were not, because this was written without network access.

Corrigendum — the appendix has 56 problems, not 54, and problem G2 was never missing

This replaces the earlier addendum. A reader checking my work found that the curriculum appendix contains **56** problems rather than the 54 I reported, and that one of them is problem **B4**, which my verification omitted. Checking further, I found a second and worse error: **problem G2 exists in the appendix**, so the G2 I wrote "because it was missing" was never needed. Both errors came from the same cause and it was mine.

What went wrong

I extracted the appendix's problem list with a pattern that looked for an identifier at the start of a line. Two problems — **B4** and **G2** — begin immediately after a page break, so the character before the identifier is a form feed rather than a newline, and the pattern skipped them. Every count downstream inherited the error: the verification table, the two problem-set introductions, the index, the curriculum's filename, and the claim that G2 was absent.

The original tracker was right all along. Its built-in problem list contains all 56, including B4 and G2 — so the discrepancy between "54" in my documents and "56" in the tracker was visible from the start, and I should have treated the disagreement as a signal rather than assuming my own count was correct.

Correction 1 — problem B4, now verified

B4 · A batch fermentation model from scratch: Monod plus Luedeking–Piret (week 16). Monod growth at μ_{max} = 0.5 h^{-1} and K_S = 0.2 g/L, Luedeking–Piret product formation with α = 0.2 g/g and β = 0.05 $g \cdot g^{-1} \cdot h^{-1}$, $Y_{X/S}$ = 0.45 g/g, from 0.1 g/L of biomass and 20 g/L of substrate.

Quantity	Appendix states	I recomputed	
Time to substrate exhaustion	9.0 h	9.01 h	**matches**
Final biomass	8.0 g/L	8.005 g/L	**matches**
Final titer	2.43 g/L	2.433 g/L	**matches**
Observed biomass yield	0.400 g/g	0.400 g/g	**matches**
Product yield	0.122 g/g	0.122 g/g	**matches**
Volumetric productivity	0.270 $g \cdot L^{-1} \cdot h^{-1}$	0.270 $g \cdot L^{-1} \cdot h^{-1}$	**matches**

So B4 is correct as written, and its teaching point is worth restating because the module repeats it elsewhere: **the observed biomass yield (0.400) is not the yield parameter you specified (0.45)**, because product formation draws its own substrate. Any model with more than one product behaves this way, and a yield reported without saying which products it accounts for is ambiguous. That is the same distinction problem X7 makes between yield on substrate consumed and yield on substrate fed.

Correction 2 — problem G2 was never missing

G2 · Sixteen experiments: factorial grid against Bayesian optimisation is present in the appendix, at its week 41. My earlier addendum stated that G2 was "the only referenced problem missing from the appendix" and supplied a replacement. **That claim was false and the replacement was redundant.**

What to do with the version I wrote: treat it as **an additional problem on the same topic**, not as a substitute. It poses the comparison differently — mine sweeps the acquisition function and the noise level, the appendix's compares a fixed factorial grid with a fixed Bayesian run — so working both is not wasted, but the appendix's version is the one the curriculum intends and the one keyed to week 41.

Correction 3 — the count, and where it was wrong

Document	Said	Now says
Index	54-problem appendix; all 54 verified	**56**
Part 15 (problems X1–X24)	"the 54 in your curriculum's appendix"	**56**
Part 16 (problems Z1–Z9)	"together with the 54"	**56**
Curriculum filename	`…-and-54-problems.pdf`	`…-and-56-problems.pdf`

This verification document	"all 54 problems"	**56**, with B4 added

The verification, restated for all 56

Category	Count	Result
Problems whose numerical answer I could recompute independently	**38**	**All 38 matched**, B4 included
Problems verified in structure only (simulation seeds, qualitative answers, code whose output depends on a random draw)	15	Method and reasoning check out; the printed numbers depend on a seed
Problems with a documented convention difference	3	H2 junction counting, W3 lipid accounting, T1/W1 transport assumptions — all three described in the original addendum and unchanged
Total	**56**	

The three convention differences are not errors in the appendix. H2 counts junctions as n − 1 where I counted n; W3 uses 0.30–0.32 g/g for lipid where my accounting gives 0.275; and T1 and W1 assume different transport parameters from the ones I used. In each case the appendix's convention is internally consistent and is the one to follow when working its problems.

What this says about the rest of the module

An extraction error that silently drops two of fifty-six records is exactly the class of mistake the module's own problems warn about — problem MC1's lesson that a carbon balance passes a reaction that breaks the model, and problem Y2's instruction to run a pipeline on shuffled labels to see whether it leaks. I did not apply that discipline to my own extraction, and the check that would have caught it was available and free: **the tracker's own problem count disagreed with mine, and I did not reconcile them**.

The practical consequence for you is small — no arithmetic anywhere in the module depended on the count, and B4 and G2 are both correct as the appendix wrote them. The reason to record it in full is that a committee will ask how you check your own work, and "I reconcile independent counts of the same thing and investigate disagreements" is a better answer than "I was careful".

Reading list, organised by a reviewer's three tiers

The earlier reading map was organised by week. This one is organised by what a PI said they look for, so you can see what to read for each competency, which module part already covers it, and which problems test it. Where a book or paper is named, the title and author are recalled from memory and the edition may have moved — **check each against a library catalogue before buying**. Chapter numbers differ between editions, which is why the topic is named rather than a number given.

How to use it

Read the module chapter first to know what matters, then the book section for the mechanism and figures, then work the problems. The module parts are the files listed on the tracker's Materials tab; problem codes are in its Problems tab.

Tier 1 — the non-negotiable core

Competency	Read	Module	Problems
Central carbon metabolism and redox balancing	Lehninger, *Principles of Biochemistry* — glycolysis, TCA, pentose phosphate, oxidative phosphorylation. Nelson & Cox, any recent edition. Stephanopoulos, Aristidou & Nielsen, *Metabolic Engineering: Principles and Methodologies* — the stoichiometry and yield chapters, which is the book this tier is built on.	Part 14 ch B4–B7 (computed); part 2 ch 18–19; part 15 X1–X4	A2, K1, K3, W2, W3, X1, X2, X3, EV3
Theoretical yields and carbon recovery	Stephanopoulos ch 2–3 again, worked rather than read. For the thermodynamic layer, Noor et al. on max-min driving force (recalled — search "maximum driving force pathway thermodynamics").	Part 2 ch 14, 18; part 14 ch B7	A2, X1, X2, PP2, EV3
CRISPR: Cas9, Cas12a, base editors, prime editing	Doudna & Charpentier's review of CRISPR-Cas9 genome editing (Science, 2014) for the foundation; a recent review of base and prime editing for the mechanisms — this field moves fastest, so prefer a review under two years old. For bacteria specifically: a review of CRISPR-assisted recombineering.	**Part 21 ch W1**; part 7 ch 23–24	**EV1, EV2**, PP5, PP6
Recombineering, markerless editing, integration	Datsenko & Wanner (PNAS, 2000) — the one-step inactivation paper, still the template for how this is done. Court, Sawitzke & Thomason on recombineering (Annu. Rev. Genet.).	Part 7 ch 23; part 18 PP6	PP6, EV1
Enzyme kinetics	Cornish-Bowden, *Fundamentals of Enzyme Kinetics* — steady state, inhibition, two-substrate. Bar-Even et al., "The moderately efficient enzyme" (Biochemistry, 2011) for what real k_{cat} and K_M values look like.	Part 14 ch B3; chapter S3; part 2 ch 12	B1, X5, X6, X8, L2
Directed evolution and rational design	Arnold's Nobel lecture or a recent directed-evolution review for the strategy; Romero & Arnold (Nat. Rev. Mol. Cell Biol., 2009) on exploring protein fitness landscapes.	Part 8 ch 29–30	L1, L2, Y5
Bioprocess: k_La, OUR, fed-batch, burden	Doran, *Bioprocess Engineering Principles* — mass transfer, mixing, heat transfer, worked as problems. Shuler, Kargi & DeLisa for the fermentation chapters. Scott et al. (Science, 2010) and Basan et al. (Nature, 2015) for the growth laws and overflow metabolism.	Part 11 ch 44; part 19 ch T4; part 14 ch B9; part 20 ch V1	D1–D6, X4, X9–X12, PP9, MP1–MP3

Tier 2 — the accelerator skills

Competency	Read	Module	Problems

FBA and genome-scale models	Palsson, *Systems Biology: Constraint-based Reconstruction and Analysis* — the standard text. Orth, Thiele & Palsson, "What is flux balance analysis?" (Nat. Biotechnol., 2010) — read this one first, it is short. Then the COBRApy documentation, worked through on the *E. coli* core model.	Part 10 ch 35–38; part 17 MC1–MC5	E1, E2, MC1–MC5, Y1
Model curation	Thiele & Palsson's protocol for generating a metabolic reconstruction (Nat. Protoc., 2010); the MEMOTE paper for automated testing.	Part 10 ch 37; part 17 ch MC-series	MC1–MC5
Multi-omics literacy	A recent review of multi-omics integration in metabolic engineering. For the statistics that make or break it: Love, Huber & Anders on DESeq2 (Genome Biol., 2014) for count data.	Part 4 ch 51; part 9 ch 33; chapter S5	Y2, Z5, Z6, Z9, R4
^{13}C flux analysis	Wiechert on ^{13}C metabolic flux analysis (Metab. Eng., 2001); Antoniewicz's reviews for the practical side and for natural-abundance correction.	Part 11 ch 43; chapter S5	C1, C2, C3, R4
ML for protein and pathway design	Jumper et al. on AlphaFold (Nature, 2021) — read the results and the confidence metrics, not the architecture. A recent review of protein language models and variant-effect prediction. For the campaign side, a review of ML-guided directed evolution.	Part 8 ch 29–30; part 11 ch 41–42	P1, P2, Y5
Biosensors and high-throughput screening	A review of transcription-factor biosensors for metabolic engineering; Zhang, Jensen & Keasling's work on sensor-driven screening (recalled).	Part 8 ch 26–27	I1, O1, L1
FACS and droplet microfluidics	A methods review on droplet microfluidic screening for enzymes and strains; for FACS, any flow-cytometry handbook's chapters on gating and sort purity.	**Part 21 ch W3**	**EV5, EV6**

Tier 3 — the specialised edge

Competency	Read	Module	Problems
Statistical design of experiments	Box, Hunter & Hunter, *Statistics for Experimenters* — factorials, aliasing, steepest ascent. Myers, Montgomery & Anderson-Cook, *Response Surface Methodology* — canonical analysis and design comparison.	**Part 20 ch V4**; part 19 ch T3; part 2 ch 17	U1, U2, F1, F2, X13–X16, DE1–DE4
Python automation and NGS pipelines	VanderPlas, *Python Data Science Handbook*, free online, for numpy, pandas and plotting. The breseq documentation for variant calling, worked on a real SRA dataset.	Part 9 ch 32, 34; part 10 ch 36	J1, Y2, Y6, R1–R5
Non-conventional chassis	A review of non-model industrial hosts; for autotrophs and C1 substrates, a review of gas fermentation and of methanol bioconversion. Part 6's host table is the summary to argue from.	Part 6 ch 58–61; part 15 X4	K3, K4, N2, W1, X4, PP1
Techno-economic assessment	Humbird et al.'s NREL design report for a cellulosic process (2011) — the worked example everyone cites; a recent TEA review for bio-based chemicals. *See the note below: this belongs higher than Tier 3.*	Part 4 ch 49; part 15 X22	G1, D5, T2, X22, PP12
Life cycle analysis	ISO 14040 and 14044 for the framework; the ecoinvent or GREET documentation for emission factors; one published LCA of a bio-based chemical to see a real system boundary.	**Part 21 ch W4**	**EV7, EV8**

Which file is which "part"

The problem supplements are titled as parts inside the PDF but named by their problem range in the filename. The mapping: **Part 15** = `me-module-supplement-problems-X1-X24.pdf` (X1–X24, Y1–Y6) · **Part 16** = `me-module-supplement-problems-Z1-Z9.pdf` (Z1–Z9, separations, quantitative omics, consortia) · **Part 17** = `me-module-supplement-problems-MC1-R5.pdf` (MC1–MC5, R1–R5) · **Part 18** = `me-module-supplement-problems-PP1-PP12.pdf` (PP1–PP12). Parts 19 to 23 carry their part number in the filename.

Also in Tier 3 and the product classes — Part 16

Competency	Read	Module	Problems
Downstream separations as unit operations	Harrison, Todd, Rudge & Petrides, *Bioseparations Science and Engineering* — chromatography, membranes, disruption; a vendor process-development handbook for breakthrough-curve protocols.	Chapter S1; part 19 ch T5; **part 16**	**Z1 Z2 Z3 Z4**, PP10, PP11, DS3
Quantitative metabolomics and proteomics	A review of absolute protein quantification (AQUA and related); a methods review on metabolite quenching and extraction.	Chapter S5; **part 16**	**Z5 Z6 Z9**, R4
Microbial consortia	A review of synthetic microbial consortia for metabolic engineering, read for the stability arguments rather than the examples.	Chapter S4; **part 16**	**Z7 Z8**

Product-class reading, for the four named classes

Class	Read	Module	Problems
Terpenoids	Paddon et al. on semi-synthetic artemisinin (Nature, 2013) — the canonical end-to-end paper, worth reading twice; a review of isoprenoid pathway engineering.	Part 5 ch 53; part 15 X3	Q1, Q2, S1–S5, X3
Flavonoids and polyphenols	A review of microbial flavonoid production, read specifically for what each group did about malonyl-CoA; a review of P450 functional expression in microbial hosts.	**Part 21 ch W2**	**EV3, EV4**
Tailored biopolymers	A review of PHA production and of the recovery routes; Choi & Lee on PHA economics (recalled).	Part 5 ch 56; part 18 PP10	K1, K2, PP10
Rare amino acids	Becker & Wittmann on systems metabolic engineering of *C. glutamicum* for lysine (recalled); a review of amino-acid fermentation.	Part 6 ch 59; part 18 PP1	W2, PP1
Therapeutic proteins	Baneyx & Mujacic on folding and inclusion bodies (Nat. Biotechnol., 2004); a review of secretory production in *K. phaffii*; a glycoengineering review for humanised yeast.	**Part 19 ch T1–T2**	**PR1–PR8**

The six-month roadmap, mapped onto this module

Phase	The reviewer's focus	What to actually do here
Months 1–2	Advanced systems and flux analysis	Part 14 chapters B4–B7 and part 2 chapters 13–19. Work A1, A2, X1–X3, EV3. Then part 10 and the COBRApy tutorial, finishing with Y1 on the real BiGG model and MC1–MC5 on curation.
Months 3–4	Precision editing and chassis design	Part 7 chapters 20–25 and **part 21 chapter W1**. Work PP5, PP6, EV1, EV2. Then part 8 chapters 26–28 for promoter and RBS tuning, with PP3, PP4 and H1, H2. Add part 6 for the host table and PP1.
Months 5–6	Computational biology and omics	Part 9 and part 11 chapters 41–43, with chapter S5. Work P1, P2, Y2, Y5, Z5, Z6, Z9 and the instrument problems R1–R5 on the supplied data. Finish with part 4 chapter 49, **part 21 chapter W4**, and X22, PP12, EV7, EV8 — so the last thing you do is cost a process in dollars and in carbon.

One deliberate departure. The reviewer places techno-economic assessment in Tier 3 as "easily learned in the first year". Problem G1 and X22 suggest otherwise: the titer–separation relationship decides whether a project is viable at all, and the input that dominates the uncertainty is one a laboratory does not control. A candidate who can say why a strain cannot reach a viable price at any plausible yield is making an argument that first-year catch-up does not replace. This list therefore keeps TEA in Tier 3 where the reviewer put it, and the roadmap schedules it anyway.

On the citations. Author names, journals and years above are recalled from memory and not verified — this environment has no network access. Treat each as a search string rather than a citation: look it up, confirm it exists and is the paper I meant, and replace the entry with the real one. Sheet 9 of the lookup workbook has fields for recording what you actually read.

Reading map — where to find each week in the textbooks

You asked whether the module points you to the right chapter and page in Lehninger, Shuler and the others. The honest answer is: partly, and not to pages. I checked: of the 81 “what to open a book for” boxes in parts 2 to 14, 29 name a chapter number and none names a page. This map fixes what can be fixed.

What this map is built from, and why it has no page numbers

The chapter and section pointers come from your own curriculum, not from my memory. It names the editions — Lehninger 8th and Alberts 7th — and for weeks 1 to 8 it gives a chapter and the section to read in each session, such as “Lehninger Ch. 14, glycolysis through the aldolase step”. Those are more precise than anything I could add, and they are checkable against your copy, so I have used them verbatim in the third column.

Page numbers I cannot give. They change between editions and printings, I have no access to the books, and a wrong page quoted confidently sends you to the wrong place and costs more time than a pointer to a chapter. So the last column is blank: fill in the page in *your* copy the first time you look it up, and the map becomes your own index. For the engineering books — Shuler, Doran, Stephanopoulos, Cornish-Bowden — the curriculum names the topic (“the enzyme-kinetics chapter”) and not a number, and I have kept to that; chapter numbers differ between editions of those books, and I would rather you check the contents page than trust a number I recalled.

The “look up in the index” column is mine. They are the terms to search in a book’s back-of-book index. Suggestions, not citations.

Module location: part 01 and part 14 both cover weeks 1 to 10 (part 14 is the compiled core concepts); S1 to S6 are the supplementary chapters in part 13. File names are in the index.

A — Weeks with a textbook or reading pointer

Wk	In the module	Textbook pointers from your curriculum	Look up in the index	Your page
1	part 01 · ch 1 part 14 · ch B1	· Lehninger Ch. 1, the cellular-foundations section · Lehninger Ch. 1, the rest of the chapter: chemical, physical, genetic and evolutionary foundations · Alberts Ch. 1, skim; then read the central-dogma pages carefully · Alberts Ch. 2, the sections on membranes and organelles	central dogma · prokaryotic cell · macromolecular composition · genetic code	
2	part 01 · ch 2 part 14 · ch B2	· Lehninger Ch. 3, amino acids and peptide bonds · Lehninger Ch. 4, the four levels of structure · Lehninger Ch. 6, the sections before kinetics: what enzymes do and how they lower activation · Lehninger Ch. 6, the cofactor and coenzyme material	amino acids, ionisation · isoelectric point · peptide bond · Ramachandran plot · protein folding · coenzymes, vitamins	
3	part 01 · ch 3 part 14 · ch B3 part 13 · S3	· Lehninger Ch. 6, the enzyme-kinetics section · Lehninger Ch. 6, the regulatory-enzymes section	Michaelis–Menten · Briggs–Haldane · turnover number · inhibition, competitive · Hill equation · allosteric enzymes	
4	part 01 · ch 4 part 14 · ch B4	· Lehninger Ch. 13, the bioenergetics and thermodynamics section · Lehninger Ch. 13, phosphoryl-group transfers and ATP · Lehninger Ch. 13, the biological oxidation–reduction section	free energy, standard · phosphoryl group transfer · ATP hydrolysis · reduction potential · NAD/NADP	
5	part 01 · ch 5 part 14 · ch B5	· Lehninger Ch. 14, glycolysis through the aldolase step · Lehninger Ch. 14, the payoff phase to pyruvate · Lehninger Ch. 14, the pentose phosphate pathway section · Lehninger Ch. 15, principles of metabolic regulation — the sections on glycolytic control	glycolysis, regulation · phosphofructokinase · pentose phosphate pathway · Entner–Doudoroff · phosphotransferase system	
6	part 01 · ch 6 part 14 · ch B6	· Lehninger Ch. 16, the pyruvate dehydrogenase complex · Lehninger Ch. 16, the cycle itself · Lehninger Ch. 16, the anaplerotic reactions section · Lehninger Ch. 19, through ATP synthase	citric acid cycle · pyruvate dehydrogenase complex · anaplerotic reactions · glyoxylate cycle · chemiosmotic coupling	
7	part 01 · ch 7 part 14 · ch B7	· Lehninger Ch. 22, nitrogen assimilation and the glutamate–glutamine node · Lehninger Ch. 21, fatty-acid synthesis · Read the mevalonate pathway in Lehninger Ch. 21 and in KEGG map 00900 · Lehninger Ch. 22, nucleotide biosynthesis, read for the shape rather than every intermediate	fatty acid synthesis · amino acid biosynthesis · glutamine synthetase · mevalonate pathway · nucleotide biosynthesis	
8	part 01 · ch 8 part 14 · ch B8	· Alberts Ch. 5 on replication, then Ch. 6 on transcription · Alberts Ch. 6, translation · Alberts Ch. 7, transcriptional control; then read the lac operon specifically	DNA replication, bacterial · RNA polymerase, sigma factor · Shine–Dalgarno · lac operon · translation initiation	
9	part 01 · ch 9 part 14 · ch B9 part 13 · S4	no chapter assigned; see the tracker	catabolite repression · cAMP receptor protein · stringent response · two-component systems · quorum sensing	
10	part 01 · ch 10 part 14 · ch B10	· Madigan et al. — Brock Biology of Microorganisms, the chapters on microbial growth, diversity and genetics	growth curve · Monod kinetics · chemostat · mutation, spontaneous ·	

		· Snyder & Champness — Molecular Genetics of Bacteria, on selections, transformation and conjugation	Luria–Delbrück · conjugation · plasmid copy number · sterilisation	
11	part 02 · ch 11	· Bailey, “Toward a science of metabolic engineering”, Science 252:1668 (1991) · Stephanopoulos & Vallino, “Network rigidity and metabolic engineering in metabolite overproduction”, Science 252:1675 (… · Nielsen & Keasling, “Engineering cellular metabolism”, Cell 164:1185 (2016)		
13	part 02 · ch 13	· Stephanopoulos Ch. 8, network stoichiometry · Stephanopoulos Ch. 12, determined and underdetermined systems · Stephanopoulos Ch. 13, pathway structure; or the corresponding Palsson chapters	stoichiometric matrix · null space · degrees of freedom · flux balance analysis · elementary flux modes	
14	part 02 · ch 14	· Lehninger Ch. 13, reread the thermodynamics section — this time for ΔG′ rather than ΔG°′ · Lehninger Ch. 11, the passive-transport sections · Lehninger Ch. 11, the active-transport sections · Stephanopoulos Ch. 11, metabolic control analysis, the first half · Stephanopoulos Ch. 11, the rest	free energy of reaction · mass-action ratio · membrane transport · proton antiporter · weak acid, pH partitioning	
15	part 02 · ch 15	no chapter assigned; see the tracker	metabolic control analysis · flux control coefficient · summation theorem · elasticity	
16	part 02 · ch 16 part 13 · S6	· Stephanopoulos, Aristidou & Nielsen, the chapters on cellular-reaction kinetics	kinetic modelling · Haldane relationship · parameter estimation · identifiability	
17	part 02 · ch 17	no chapter assigned; see the tracker	statistical power · multiple comparisons · factorial design · replication	
18	part 02 · ch 18	· Doran — Bioprocess Engineering Principles, the material- and energy-balance chapters	yield coefficient · material balance · theoretical yield · energy balance	
19	part 02 · ch 19	no chapter assigned; see the tracker	NADPH supply · transhydrogenase · oxygen demand · P/O ratio	
23	part 07 · ch 23	· Snyder & Champness — Molecular Genetics of Bacteria, on recombination, selection and counterselection		
30	part 08 · ch 30	· Cornish-Bowden — Fundamentals of Enzyme Kinetics, for what a kinetic constant does and does not tell you	enzyme kinetic constants · k_cat/K_M · control coefficient	
35	part 10 · ch 35	· Install COBRApy and a solver; confirm the install by loading the textbook model · Read Orth, Thiele & Palsson, “What is flux balance analysis?”, Nat. Biotechnol. 28:245		
37	part 10 · ch 37	· Read the Thiele and Palsson protocol for what manual reconstruction actually involves — it is a year		
43	part 11 · ch 43 part 13 · S5	· Doran — Bioprocess Engineering Principles, 2nd ed., the mass-transfer and reactor chapters, and work its problems · Shuler, Kargi & DeLisa — Bioprocess Engineering, the chapters on reactor operation and scale-up	isotopic tracers · mass isotopomer distribution · metabolic flux analysis	
44	part 11 · ch 44 part 13 · S1, S6	no chapter assigned; see the tracker	oxygen transfer · k_L a · power consumption · mixing time · fed-batch · scale-up	
49	part 04 · ch 49 part 13 · S1	no chapter assigned; see the tracker	bioseparations · downstream processing cost · process economics	
59	part 06 · ch 59	· Madigan et al. — Brock Biology of Microorganisms, the cell-envelope chapter	cell envelope, Gram-positive · Corynebacterium	
60	part 06 · ch 60	· Madigan et al. — Brock Biology of Microorganisms, the phototrophy and microbial diversity chapters	phototrophy · cyanobacteria · microbial diversity	

B — Weeks whose sources are papers, reviews and tools

These weeks have no textbook in your curriculum; the first two sources it lists are shown. The full list is on each week’s page of the tracker.

Wk	In the module	First sources named in your curriculum
12	part 02 · ch 12	· NumPy — the absolute basics and linear algebra guides · SciPy — integration and optimization reference
20	part 07 · ch 20	· Addgene, “Plasmids 101: Golden Gate cloning” · Gibson et al., Nat. Methods 6:343 (2009) tooling, Golden Gate design, and yeast in vivo assemb…
21	part 07 · ch 21	see the week’s sources list in the tracker
22	part 07 · ch 22	see the week’s sources list in the tracker
24	part 07 · ch 24	· Cleto et al., “C. glutamicum metabolic engineering with CRISPR interference”, ACS Synth. Biol.… · A metabolite-biosensor review
25	part 07 · ch 25	· Fredens et al., Total synthesis of Escherichia coli with a recoded genome, Nature (2019) · Weinstock et al., Vibrio natriegens as a fast-growing host for molecular biology (preprint, 20…
26	part 08 · ch 26 part 13 · S4	· A review of transcription-factor-based biosensors for metabolic engineering · Kelly et al., Measuring the activity of BioBrick promoters using an in vivo reference standard…
27	part 08 · ch 27 part 13 · S4, S6	· Yoo et al., Metabolic engineering of Escherichia coli using synthetic small regulatory RNAs, N… · CRISPR interference-guided multiplex repression of endogenous competing pathway genes for redi…

28	part 08 · ch 28	· Salis Lab RBS Calculator · MIT OCW 7.344 Directed Evolution: Engineering Biocatalysts
29	part 08 · ch 29	· Dauparas et al., Atomic context-conditioned protein sequence design using LigandMPNN, Nat. Met… · Institute for Protein Design — introducing LigandMPNN
31	part 08 · ch 31	see the week's sources list in the tracker
32	part 09 · ch 32	· breseq — documentation and downloads · breseq on GitHub
33	part 09 · ch 33	· Love, Huber & Anders, Moderated estimation of fold change and dispersion for RNA-seq data with… · Bioconductor — DESeq2 package and vignette
34	part 09 · ch 34	· K-Biofoundry · Global Biofoundry Alliance
36	part 10 · ch 36	· COBRApy documentation — building a model, simulating, deletions, FVA · COBRApy — flux analysis guide
38	part 10 · ch 38	· The COBRA Toolbox — documentation and installation · MATLAB — getting started documentation
39	part 11 · ch 39	· Burgard, Pharkya & Maranas, "OptKnock", Biotechnol. Bioeng. 84:647 (2003) · A ^{13}C-MFA methods review
40	part 11 · ch 40	· Delépine et al., "RetroPath2.0", Metabolic Engineering 45:158 (2018) · A 2025 review on machine learning or AI-powered biofoundries
41	part 11 · ch 41	· Radivojević et al., A machine learning Automated Recommendation Tool for synthetic biology, Na… · Zhang et al., Combining mechanistic and machine learning models for predictive engineering and…
42	part 11 · ch 42	· PubMed search: machine-learning-guided directed evolution of proteins · PubMed search: protein language models and fitness prediction
45	part 03 · ch 45	· Dunlop et al., Engineering microbial biofuel tolerance and export using efflux pumps, Mol. Sys… · Newman et al., High-level production of amorpha-4,11-diene in a two-phase partitioning bioreac…
46	part 03 · ch 46	· Prpich and Daugulis, Polymer development for enhanced delivery of phenol in a solid–liquid two… · Rehmann, Sun and Daugulis, Polymer selection for biphenyl degradation in a solid–liquid two-ph…
47	part 03 · ch 47	· Newman et al., High-level production of amorpha-4,11-diene in a two-phase partitioning bioreac… · Substrate mass transport in two-phase partitioning bioreactors employing liquid and solid non-…
48	part 03 · ch 48	· PubMed search: cell immobilisation in alginate and PVA, reviews of methods and limitations · PubMed search: effectiveness factor and oxygen diffusion limitation in immobilised-cell beads
50	part 04 · ch 50	· A tutorial on analytical method validation for chromatographic assays · A review of quenching and extraction for microbial metabolomics
51	part 04 · ch 51 part 13 · S5	· Ko, Lee et al., "Tools and strategies of systems metabolic engineering", Chem. Soc. Rev. 49:46… · A gas-fermentation or scale-up review
52	part 04 · ch 52	· REBASE — the restriction enzyme database · Wetmore et al., Rapid quantification of mutant fitness in diverse bacteria by sequencing rando…
53	part 05 · ch 53	· Paddon et al., Nature 496:528 (2013) · Paddon & Keasling, Nat. Rev. Microbiol. 12:355 (2014)
54	part 05 · ch 54	· Yim et al., Nat. Chem. Biol. 7:445 (2011) · EPA, 2003 Greener Reaction Conditions Award — DuPont bio-PDO
55	part 05 · ch 55 part 13 · S2	see the week's sources list in the tracker
56	part 05 · ch 56 part 13 · S2	· AIM Lab — KIST Applied and Industrial Microbiology Lab (publications page) · Sun et al., efficient poly(3-hydroxybutyrate) production from lignocellulosic hydrolysates wit…
57	part 05 · ch 57 part 13 · S2	· AIM Lab — KIST Applied and Industrial Microbiology Lab (home) · Firmansyah et al., Upgrading polyethylene plastic waste into a biodegradable polymer: a hybrid…
58	part 06 · ch 58	see the week's sources list in the tracker
61	part 06 · ch 61	· A 2025–2026 review on AI-driven biofoundries or systems metabolic engineering · A cell-free metabolic engineering review
62	part 12 · ch 62	· ICMJE recommendations on authorship and contributorship · COPE, Committee on Publication Ethics: guidance and flowcharts
63	part 12 · ch 63	· Mensh & Kording, Ten simple rules for structuring papers, PLoS Comput. Biol. (2017) · A guide to presenting scientific data and figure design
64	part 12 · ch 64	see the week's sources list in the tracker
65	part 12 · ch 65	· Your own paper log entries · Two recent papers from each of your three target labs
66	part 12 · ch 66	see the week's sources list in the tracker

One habit that makes this work

When a module chapter says "open a book for this", find that week in table A, read the pointer in your curriculum for the section, and write the page in your copy into the last column. After ten weeks you have a page-level index of your own editions, which is the thing I cannot write for you.

Metabolic engineering: the study module

Part 1 — Biology foundations for engineers (weeks 1 to 10)

This is the reading, written out, so that you can work through weeks 1 to 10 without opening four books in parallel. It is written for someone who already has thermodynamics, kinetics, mass balances and transport, and who needs the biology those tools will be applied to. Every chapter maps onto one week of the tracker.

What this is not: it is not a substitute for Lehninger or Alberts, and it does not contain their text, their figures or their problem sets, which are copyrighted. I wrote every word here myself. Where their treatment is genuinely better than anything I can compress into a few pages — the three-dimensional structure figures, the full mechanisms, the pathway maps you will want to annotate — the chapter says so and names the chapter to open. Use this module as the spine and the books as the reference.

How to use it: read the chapter, then do that week's sessions in the tracker. The self-check questions at the end of each chapter are there to be answered on paper before you look at the answer. If you cannot answer them, reread rather than continuing: phase 1 compounds, and weeks 13 onward are unusable without it.

Ch.	Week	Chapter
1	1	The cell, the central dogma, and a map of the territory
2	2	Protein structure and enzyme function
3	3	Enzyme kinetics, in your own language
4	4	Bioenergetics and cofactors: the currency of the cell
5	5	Central carbon metabolism I: glycolysis and the pentose phosphate pathway
6	6	Central carbon metabolism II: the TCA cycle, respiration and overflow
7	7	Biosynthetic metabolism: where products come from
8	8	Molecular biology and gene regulation
9	9	Regulation at the system level: what will fight your design
10	10	Microbial genetics and microbiology for engineers

Chapter 1 · tracker week 1

The cell, the central dogma, and a map of the territory

> *A cell is a chemical reactor that builds its own catalysts, controls its own feed rates and makes copies of itself. Everything in this field follows from taking that description literally.*

What kind of reactor you are dealing with

A bacterial cell is about 1 µm across and holds roughly 1 femtolitre of cytoplasm — measured values for E. coli range from about 0.4 to 1.8 µm³, so take 1 fL as the round number and the range as the honest one. At a dense culture of 10^9 cells per millilitre, all that cytoplasm together is about 0.1% of the broth volume. That single number explains a great deal: a product kept inside the cell is capped at a fraction of a gram per litre however hard you push it, while a secreted product is limited only by what the cells can make and what the medium tolerates.

Inside, the cytoplasm is not dilute. Measured cytoplasmic protein in E. coli is 200 to 320 g/L, rising with external osmolarity, with total macromolecules near 300 g/L, and the whole interior is crowded enough that diffusion is slower than in water and weak associations between molecules are favoured. When you later read that an enzyme has a Km of 0.1 mM, remember that the cell is operating in a medium nothing like the assay buffer it was measured in.

Two architectures matter for production. Prokaryotes (bacteria, and the archaea you will mostly not use) have no internal membrane-bound compartments: one cytoplasm, one membrane in Gram-positives, two in Gram-negatives with a periplasm between them. Eukaryotes (yeasts, algae, mammalian cells) partition their chemistry into a nucleus, mitochondria, an endoplasmic reticulum and Golgi for the secretory route, peroxisomes and a vacuole. Compartments are both an opportunity and a tax, and week 58 of the tracker returns to this with numbers.

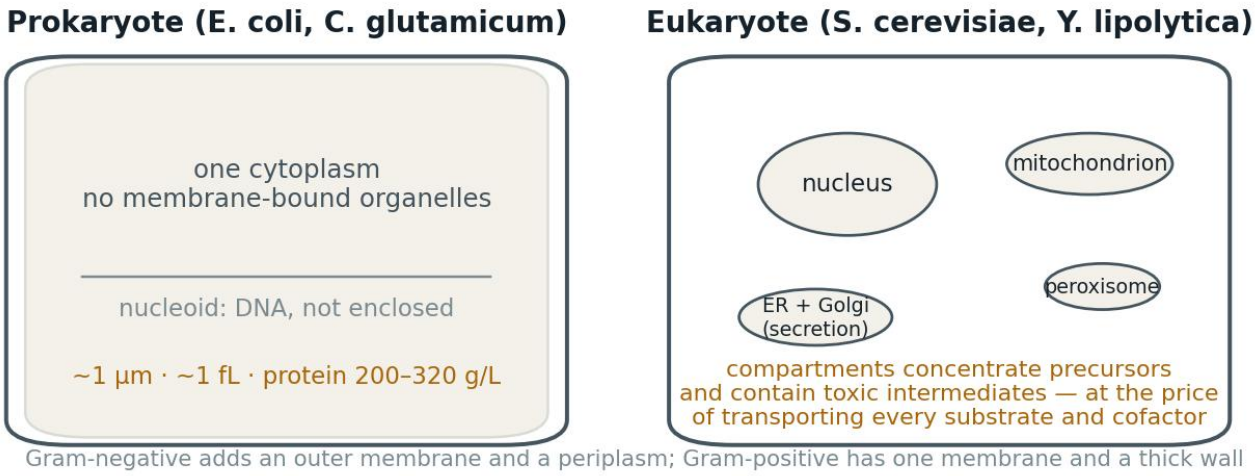


Figure 1.1 Two architectures. Everything in the module that depends on compartments, secretion or transport traces back to this picture. Drawn for this module.

The central dogma, stated as an engineer needs it

DNA is copied to DNA (replication), transcribed to RNA, and RNA is translated to protein. The direction matters because it is where your control points are. You can change the gene sequence, the amount of mRNA made from it, how efficiently that mRNA is translated, and how long the protein survives. Each of these is a separate engineering handle with a different cost and a different response time.

Three numbers worth carrying. Transcription in E. coli runs at roughly 40 to 80 nucleotides per second and translation at 10 to 20 amino acids per second, so a 1,000-codon gene takes on the order of a minute to express. Most bacterial mRNAs have half-lives of a few minutes, which is why transcriptional control acts quickly. Most proteins are stable over a cell generation, so once a protein is made, turning off its gene removes it mainly by dilution as the cells grow. That last point catches people out: a knockdown at the RNA level does not clear the protein, and a fast-growing culture dilutes it faster than a slow one.

The vocabulary you actually need

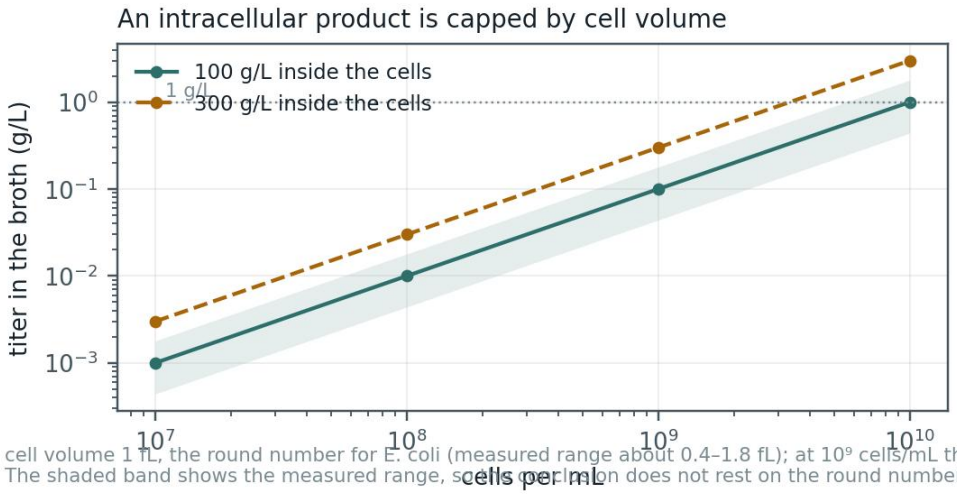
A gene is a stretch of DNA encoding a product. An open reading frame is the part translated into protein, read in codons of three bases. A promoter is the DNA where RNA polymerase binds and starts transcription; its strength sets how much mRNA is made. A ribosome binding site, in bacteria, is the sequence where the ribosome engages, and it sets translation initiation rate. A terminator ends transcription. An operon is several genes transcribed from one promoter as a single mRNA.

A plasmid is a small circular DNA molecule that replicates independently of the chromosome; its copy number, from a handful to hundreds per cell, multiplies the dose of whatever it carries. An integrated construct sits in the chromosome at one copy, which is more stable and usually lower expression. Choosing between them is a real design decision and reappears in week 20.

Where the carbon goes

A heterotroph takes in a reduced carbon source, oxidises part of it to make energy and reducing power, and uses the rest as building blocks. Three fates compete for every carbon atom you feed: biomass, product, and CO_2. Nothing you do in metabolic engineering escapes that three-way split, and your yield is a statement about it.

Biomass is not free. Roughly half of dry cell weight is protein, with nucleic acids, lipid and cell wall making up most of the rest, and building all of it consumes carbon, ATP and reducing power. This is why a strain that grows beautifully often produces poorly: the two demands draw on the same pool. The tension is quantified in weeks 18 and 19.



cell volume 1 fL, the round number for E. coli (measured range about 0.4–1.8 fL); at 10^9 cells/mL the cytoplasm is 0.1% of the broth. The shaded band shows the measured range, so the conclusion does not rest on the round number.

Figure 1.2 Why an intracellular product is capped. Computed from a 1 fL cell volume with the measured E. coli range shaded; the dotted line is 1 g/L, which most processes must clear to be worth running.

The equations from this chapter

Cell volume fraction	$\varphi_{cell} = N \cdot V_{cell}$, so 10^9 cells/mL × 1 fL = 0.001 L per L of broth
Carbon split	$C_{substrate} = C_{biomass} + C_{product} + C_{CO_2}$, always

Worked example

A culture reaches 10^9 cells/mL. Your product is retained in the cytoplasm at 150 g/L inside the cells. What titer do you measure in the broth?

The intracellular volume is 10^9 cells/mL × 10^3 mL/L × 1×10^{-15} L = 0.001 L per litre of broth, so 0.1%. The broth titer is 150 g/L × 0.001 = 0.15 g/L, and even at the top of the measured cell-volume range, 1.8 fL, it is only 0.27 g/L. Even an implausibly high intracellular concentration gives a titer that would be dismissed as a failure. This is the single strongest argument for secretion, and it is why hosts are chosen partly on whether they export.

Three things people get wrong

- ‘The cell is a bag of dilute enzymes.’ It is not: 200 to 320 g/L of protein, crowded, with slower diffusion than water.
- ‘Turning off the gene removes the protein.’ Mostly it does not; growth dilutes it.
- ‘More plasmid copies is always more product.’ More copies also means more burden and more instability.

Self-check — answer on paper first

1. Why does a secreted product have no cell-volume ceiling, and an intracellular one does?

Because the intracellular product occupies only the cytoplasmic volume, which is about 0.1% of the broth at 10^9 cells/mL, so the broth concentration is the intracellular concentration times that fraction. A secreted product occupies the whole liquid volume, so its ceiling is set by solubility, toxicity and what the cells can make, not by cell volume.

2. Name the four levels at which you can set how much of a protein a cell contains, and the timescale of each.

Gene sequence and copy number (permanent, set at construction); transcription rate via the promoter (minutes, because mRNA turns over in minutes); translation initiation via the ribosome binding site (minutes); protein degradation via a degradation tag (otherwise slow, since most proteins are cleared mainly by dilution through growth).

3. Why is a strain that grows well often a poor producer?

Because biomass and product draw on the same carbon, ATP and reducing power. Carbon has three fates: biomass, product and CO_2. Fast growth claims the first, so unless the pathway is decoupled from growth or the process is staged, the two compete directly.

What to open a book for

Alberts Ch. 1–2 for the cell and its compartments, and Lehninger Ch. 1 for the chemical logic, are worth reading for their figures rather than their text. If you buy only one thing for this week, buy nothing: MIT OCW 7.05 lectures 1 and 2 cover it free.

Chapter 2 · tracker week 2

Protein structure and enzyme function

An enzyme is a catalyst whose selectivity comes from a folded shape. Everything you will later do to enzymes — engineering them, expressing them, failing to express them — depends on that shape being fragile and context-dependent.

Four levels, and why each one can ruin your day

Primary structure is the amino acid sequence, written from the gene. Secondary structure is the local pattern the backbone takes, chiefly α-helices and β-sheets, held by hydrogen bonds along the backbone. Tertiary structure is the three-dimensional fold of one chain, held by the burial of hydrophobic side chains, hydrogen bonds, salt bridges and sometimes disulfide bonds. Quaternary structure is the assembly of several chains into one functional unit.

For your purposes: primary structure is what you design and order. Tertiary structure is what fails when you express a plant enzyme in E. coli at 37 °C and get an inactive pellet. Quaternary structure is why some enzymes lose activity when you tag them at the wrong terminus, and why an enzyme can be present on a gel and dead in an assay.

The twenty side chains, grouped usefully

You do not need to memorise structures, but you do need the groups, because they predict where a residue will sit and what a mutation will do. Non-polar side chains (glycine, alanine, valine, leucine, isoleucine, methionine, proline, phenylalanine, tryptophan) are mostly buried in the core, and substituting a buried large residue for a small one leaves a cavity and destabilises the fold. Polar uncharged side chains (serine, threonine, cysteine, tyrosine, asparagine, glutamine) sit at the surface or in active sites; serine, threonine, tyrosine and cysteine are the usual nucleophiles and the usual sites of modification. Acidic (aspartate, glutamate) and basic (lysine, arginine, histidine) side chains sit at the surface and form salt bridges; histidine is special because its pKa near 6 means it can be protonated or not at physiological pH, which is why it appears so often in catalysis and in metal binding.

Two practical consequences. First, cysteine is reactive: in the reducing cytoplasm it stays free, so a protein that needs disulfide bonds will not fold properly there, which is one reason secreted and extracytoplasmic proteins often need a different compartment or host. Second, proline breaks helices and constrains the backbone, so a proline substitution in a helix is rarely neutral.

What catalysis actually is

An enzyme does not change the equilibrium of a reaction. It lowers the activation barrier for both directions equally, so it changes the rate, not the position. If you remember one thing from this week, remember that: no amount of enzyme engineering makes an unfavourable reaction favourable, and a pathway step that is thermodynamically stuck stays stuck. Week 14 gives you the arithmetic for deciding whether a step is stuck.

How the barrier is lowered: by binding the transition state more tightly than the substrate, by positioning reacting groups correctly so that the entropic cost of getting them together is paid once at binding, by providing acid and base groups to move protons, by covalent catalysis through a nucleophilic residue, and by metal ions that stabilise charge. The common thread is that the active site is a pocket complementary to the transition state, not to the substrate.

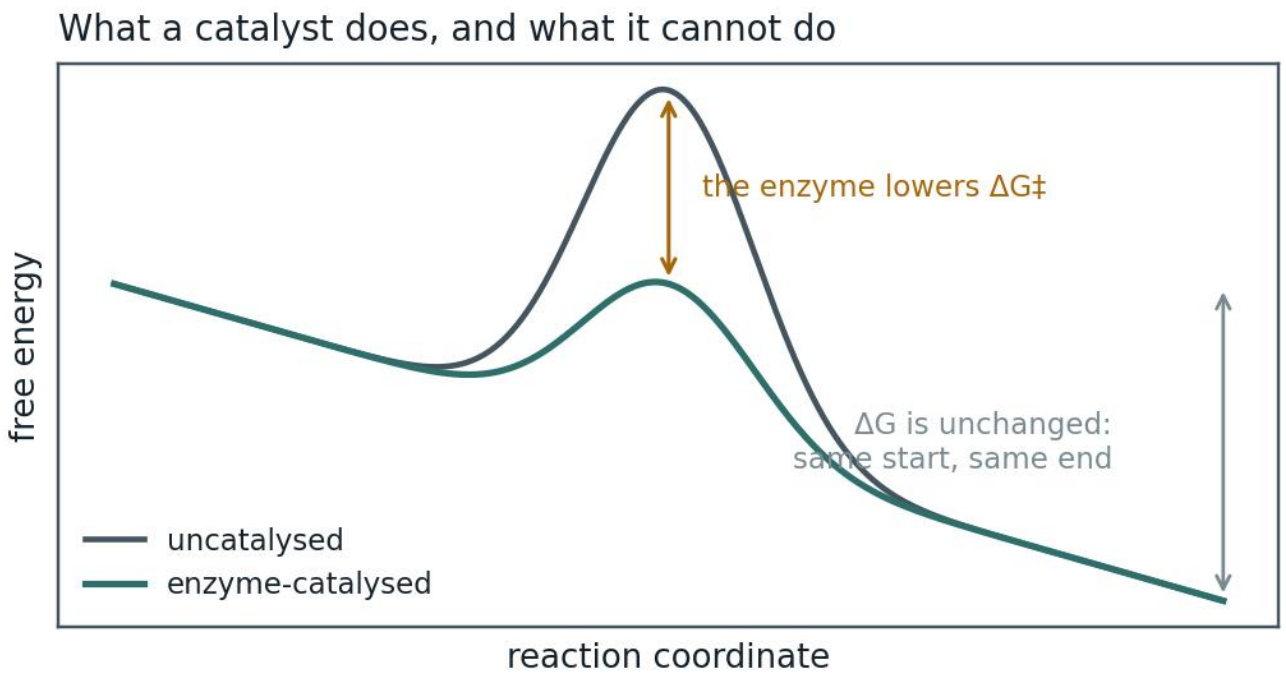


Figure 2.1 A catalyst lowers the activation barrier in both directions and leaves ΔG untouched. If you remember one diagram from week 2, this is the one.

Cofactors, because your pathway will need them

Many enzymes cannot work alone. Cofactors divide into metal ions (magnesium with nearly every phosphoryl transfer, iron in haem and iron-sulfur clusters, zinc, manganese, copper) and organic cofactors, which are either tightly bound prosthetic groups or freely diffusing cosubstrates. The ones you will meet constantly are NAD^+ and $NADP^+$ as hydride carriers, ATP as the phosphoryl donor, coenzyme A as the acyl carrier, and thiamine pyrophosphate, pyridoxal phosphate, biotin and tetrahydrofolate for specific chemistry.

The engineering point is that a cosubstrate must be regenerated. When you install a pathway that consumes NADPH, you have not just added a reaction, you have added a demand on the cell's reductant supply, and that demand competes with everything else. Week 4 makes this quantitative and week 19 makes it a design constraint.

The equations from this chapter

Catalysis	`k_cat/K_M is the specificity constant; an enzyme lowers ΔG‡, never ΔG`
Why histidine	`pKa ≈ 6 means it is a usable acid or base at pH 7`

Worked example

You express a plant enzyme in E. coli. It appears as a strong band at the right size, but there is no activity in the lysate. Give the three hypotheses you would test first, with the test for each.

First, misfolding into inclusion bodies: run soluble and insoluble fractions side by side on a gel, since a strong total band says nothing about which fraction it is in. Second, a missing cofactor or metal, or a missing redox partner, which is common for plant cytochrome P450 enzymes: add the cofactor to the assay, or co-express the partner. Third, incorrect disulfide bonding or a missing post-translational modification, since the E. coli cytoplasm is reducing: test expression in a compartment or host that permits it. A fourth, often overlooked, is that the assay conditions are wrong for this enzyme: vary pH and temperature before concluding the protein is dead.

Three things people get wrong

- ‘A band on a gel means the enzyme works.’ It means something of that size is present.
- ‘Enzymes make reactions go.’ They make reactions go faster, in both directions.
- ‘Codon-optimise everything.’ Translation that is too fast can fold worse; this is a hypothesis to test, not a rule.

Self-check — answer on paper first

1. Why can an enzyme not make a thermodynamically unfavourable reaction proceed?

Because a catalyst lowers the activation barrier for the forward and reverse directions equally, leaving the equilibrium constant unchanged. What determines direction is $\Delta G'$ at the actual concentrations, so the only way to drive an unfavourable step is to change the concentrations, by removing the product or coupling the step to a favourable reaction.

2. Why does the reducing cytoplasm of E. coli matter when you express a eukaryotic secreted protein?

Because such proteins often depend on disulfide bonds for their fold, and the cytoplasm keeps cysteines reduced, so the bonds do not form and the protein misfolds. The usual answers are to target the protein to the periplasm, to use a strain engineered with an oxidising cytoplasm, or to use a eukaryotic host with an endoplasmic reticulum.

3. Name the cofactor role you would check first for a pathway consuming four NADPH per product, and why.

Whether the host can supply NADPH at the required rate, because NADPH is a cosubstrate that must be regenerated, usually through the oxidative pentose phosphate pathway, which costs carbon as CO_2. A pathway with a large NADPH demand is a demand on central metabolism, not just an addition to it.

What to open a book for

This is the week where the books earn their price: Lehninger Ch. 3 to 6 for amino acids, protein structure and enzymes, and the structure figures in Alberts. No prose substitute shows you a fold. Cornish-Bowden is the reference if you want the kinetics done properly.

Chapter 3 · tracker week 3

Enzyme kinetics, in your own language

You already know rate laws, steady-state approximations and reactor design. This week is mostly a translation exercise, plus one piece of biology that has no chemical-engineering analogue: allosteric regulation.

Michaelis–Menten, derived the way you would derive it

Take E + S ⇌ ES → E + P. Apply the steady-state approximation to ES, exactly as you would to a reactive intermediate, and impose the enzyme balance $E_{total} = E + ES$. The result is $v = V_{max}\cdot[S]/(K_M + [S])$, with $V_{max} = k_{cat}\cdot E_{total}$ and $K_M = (k_{-1} + k_{cat})/k_1$. That is the whole derivation, and it is the same algebra as a Langmuir–Hinshelwood rate law on a catalyst surface; the enzyme is simply a one-site catalyst.

Read the two constants physically. V_{max} is capacity: it scales with how much enzyme you have, so doubling expression doubles it. K_M is the substrate concentration giving half of V_{max}, and it tells you where on the curve the cell operates. If the intracellular substrate concentration is well above K_M the enzyme is saturated and adding substrate does nothing; if it is well below, the rate is roughly proportional to substrate and the enzyme behaves first order with apparent rate constant $k_{cat}\cdot E/K_M$.

What the constants do and do not tell you

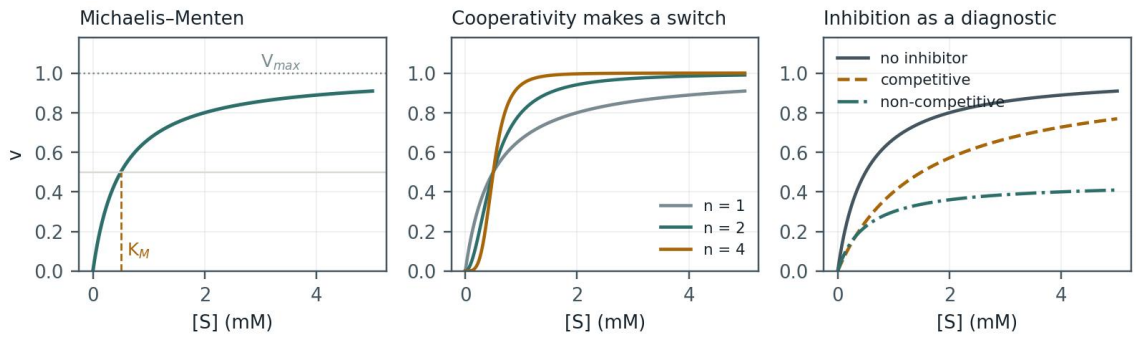
k_{cat}/K_M, the specificity constant, is the figure of merit for an enzyme working at low substrate, and it is bounded above by the rate at which substrate and enzyme can meet by diffusion, around 10^8 to 10^9 $M^{-1}\,s^{-1}$. An enzyme at that bound cannot be improved by protein engineering; it can only be given more substrate or more copies.

A warning you will need in week 31. A higher k_{cat} measured in a cuvette often does not raise titer in a cell, because the enzyme may not be the step that controls flux, because expression or solubility may fall as activity rises, or because the substrate concentration in the cell sits in a different regime from the assay. Kinetic constants are necessary, not sufficient.

Inhibition, read as a diagnostic

Competitive inhibition raises the apparent K_M and leaves V_{max} unchanged: the inhibitor competes for the site, and enough substrate outruns it. Uncompetitive inhibition lowers both, binding only the enzyme-substrate complex. Non-competitive or mixed inhibition lowers V_{max}, because the inhibitor binds elsewhere and reduces catalysis whatever the substrate does. The practical value of these patterns is diagnostic: measure the rate at several substrate and inhibitor concentrations, see which constant moves, and you know where the inhibitor acts.

Product inhibition is the case you will meet most often in a fermentation, and it is usually competitive or mixed. It is also the reason that the two-phase and in situ removal strategies of weeks 44 to 47 exist.



competitive inhibition raises apparent K_M and leaves V_{max} alone; non-competitive lowers V_{max}. Measure which constant moves and you know where the inhibitor binds.

Figure 3.1 The three curves you must be able to read: the hyperbola, the sigmoid that makes a switch, and the two inhibition patterns that tell you where an inhibitor binds.

Allosteric regulation, the part that is not chemical engineering

Many regulatory enzymes do not follow Michaelis–Menten at all. They have several subunits, and substrate binding at one site changes the affinity at the others, giving a sigmoidal rather than hyperbolic curve. The standard empirical description is the Hill equation, v = V_max·[S]ⁿ/(Kⁿ + [S]ⁿ), where n is a measure of cooperativity rather than a number of sites.

Why it matters: a sigmoidal response is a switch. A small change in substrate or effector concentration produces a large change in rate, which lets a cell respond sharply to a threshold. Feedback inhibition, where the end product of a pathway inhibits the first committed enzyme allosterically, is the most common regulatory motif you will fight. The classic engineering answer is to use a feedback-resistant variant of that enzyme, and you will see this in nearly every amino-acid production paper you read.

The equations from this chapter

Michaelis–Menten	`v = V_max·[S] / (K_M + [S]), V_max = k_cat·[E]_total`
Hill	`v = V_max·[S]ⁿ / (Kⁿ + [S]ⁿ), n > 1 means cooperative`
Diffusion bound	`k_cat/K_M ≤ about 10⁸ to 10⁹ M⁻¹ s⁻¹`

Worked example

An enzyme has k_cat = 50 s⁻¹ and K_M = 0.5 mM. The intracellular substrate concentration is 50 µM. What fraction of V_max is the enzyme achieving, and what would you do to raise flux through this step?

v/V_max = [S]/(K_M + [S]) = 0.05/(0.5 + 0.05) = 0.091, so about 9% of capacity. The enzyme is far from saturated and is operating in its first-order regime, so flux here is set by substrate supply as much as by enzyme amount. Options, in order of likely payoff: raise the intracellular substrate concentration by improving the upstream step or reducing a competing drain; express more enzyme, which raises V_max proportionally and therefore raises the first-order rate too; or engineer a lower K_M, which is the hardest route. Before any of them, check in week 15 whether this step actually controls the flux.

Three things people get wrong

- ‘K_M is the enzyme’s affinity.’ It is a composite of three rate constants and only approximates affinity when k_cat is small.
- ‘A better k_cat means more product.’ Only if that enzyme controls the flux and expression holds up.

- ‘Sigmoidal curves are just noisy hyperbolas.’ They are the signature of cooperativity, and they are how cells build switches.

Self-check — answer on paper first

1. Derive, in one line each, why V_max scales with enzyme concentration but K_M does not.

V_max = k_cat·E_total comes from the enzyme balance, so it is proportional to how much enzyme is present. K_M = (k_{-1} + k_cat)/k_1 is a ratio of rate constants belonging to a single enzyme molecule, so it is a property of the enzyme and conditions, not of its amount.

2. You measure that an inhibitor raises the apparent K_M but leaves V_max unchanged. What does this tell you, and what follows for a process?

It is competitive inhibition, so the inhibitor binds the free enzyme at the substrate site. It follows that raising the substrate concentration can overcome it, which in a process means the inhibition matters most when substrate is limiting, as in a fed-batch running at low residual sugar.

3. Why is a feedback-resistant enzyme variant such a common feature of amino-acid production strains?

Because the end product allosterically inhibits the first committed enzyme of its own pathway, so a strain that accumulates the product switches off its own synthesis. A variant that no longer binds the product at its allosteric site keeps the pathway on, which is usually the single most effective change in such a strain, at the cost of losing the regulation that protected the cell.

What to open a book for

Lehninger Ch. 6 covers this, but you will find Cornish-Bowden or the enzyme-kinetics chapter of Shuler, Kargi & DeLisa closer to how you think. Shuler in particular derives the same equations in engineering notation and connects them straight to reactor design.

Chapter 4 · tracker week 4

Bioenergetics and cofactors: the currency of the cell

This is the most important week in phase 1 for what comes later. Almost every design failure in metabolic engineering is, underneath, an energy or redox balance that does not close.

ΔG, and why the textbook number misleads

Tabulated standard free energies, ΔG°′, assume 1 M concentrations of everything, pH 7 and 25 °C. A cell operates at micromolar to millimolar concentrations. What decides direction is ΔG′ = ΔG°′ + RT·ln Q, where Q is the actual mass-action ratio of products to substrates. The consequence is that a reaction with an unfavourable standard free energy runs perfectly well if its product is kept low, and a favourable one stalls if its product accumulates.

This single idea does more work than any other in the field. It is why removing a product in situ can pull a pathway forward; it is why an intermediate that accumulates tells you where a pathway is stuck; and it is why you should never declare a designed pathway infeasible from standard free energies alone. Week 14 gives you the full calculation with real concentrations.

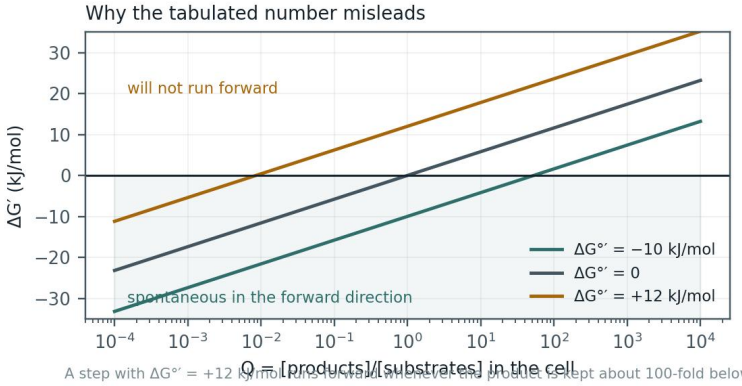


Figure 4.1 ΔG′ against the mass-action ratio, computed at 30 °C. The shaded region is where a reaction runs forward. This is the argument for in situ product removal, drawn.

ATP, properly understood

ATP is not a store of energy in any useful sense; the cell holds only seconds to a minute of demand in its ATP pool. It is a carrier, constantly regenerated. What matters is the rate of regeneration and the ratio ATP to ADP and phosphate, which sets the free energy actually available per mole transferred. Compilations give roughly 40 to 60 kJ/mol in a living cell against the 30.5 kJ/mol standard value, and a measurement in growing E. coli put it near 47 kJ/mol. Quote a range, not a number.

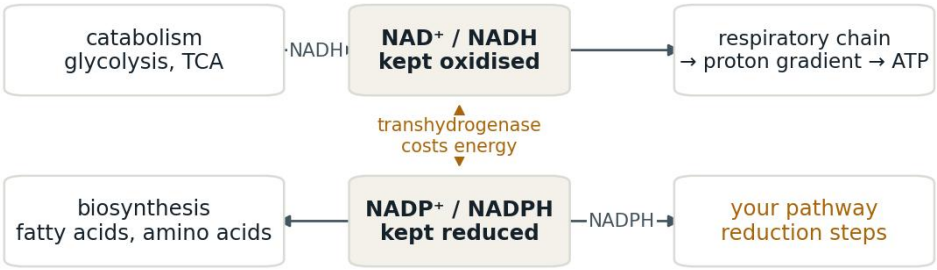
Count ATP in equivalents, not molecules. Hydrolysis to ADP is one equivalent; hydrolysis to AMP, as in acyl-CoA activation, is two, because regenerating ATP from AMP costs two phosphoryl transfers. Getting this wrong by a factor of two in a pathway balance is a common and consequential mistake.

NADH and NADPH are not interchangeable

Both carry a hydride. The difference is a single phosphate group, and the cell uses it as a label: NADH is kept relatively oxidised and is the carrier for catabolism, feeding the respiratory

chain to make ATP, while NADPH is kept relatively reduced and is the carrier for biosynthesis. The ratios differ by orders of magnitude, which is the point: it lets the cell run oxidation and reduction at the same time without futile cycling.

For you this means that a pathway requiring NADPH cannot simply draw on the NADH your host makes abundantly. The options are a transhydrogenase, which converts one to the other at an energetic cost; routing flux through the oxidative pentose phosphate pathway, which generates NADPH but loses carbon as CO_2; or engineering the enzyme's cofactor specificity. Each appears in real strain papers, and each has a cost you can quantify.



Same hydride, different label. The cell holds the two pools at deliberately different ratios so that it can oxidise and reduce at once.

Figure 4.2 Two hydride carriers, deliberately held at different ratios. A pathway needing NADPH cannot simply draw on the NADH the host makes.

How respiration actually makes ATP

Oxidation of NADH by the respiratory chain pumps protons across the membrane, creating an electrochemical gradient — the proton motive force — and ATP synthase lets protons back in, turning that gradient into ATP. The yield is not a fixed integer: it depends on which dehydrogenases and oxidases the organism uses, and the P/O ratio differs between organisms and conditions. Treat any number you are given for ATP per NADH as a modelling assumption to be stated, not a constant.

The proton motive force also does other work: it drives many transporters, including the efflux pumps of week 44, and it powers flagella. When you export a product using a proton antiporter, you are spending the same gradient that would otherwise have made ATP.

The equations from this chapter

Real free energy	`ΔG′ = ΔG°′ + RT·ln Q; R = 8.314 J/mol/K, RT ≈ 2.5 kJ/mol at 30 °C`
ATP equivalents	`ATP → ADP + Pi = 1 equivalent; ATP → AMP + PPi = 2 equivalents`
Redox balance	`sum of NAD(P)H produced = sum consumed, per unit of product, or the strain cannot run`

Worked example

A pathway makes one mole of product from one mole of glucose, consuming 4 NADPH and 2 ATP, and releasing 2 CO_2. Glycolysis to two pyruvate yields 2 NADH and 2 ATP net. Is this balanced, and what follows?

Carbon: glucose has 6, two CO_2 leave, so 4 carbons reach the product; check that against the product's formula. ATP: glycolysis supplies 2, the pathway spends 2, so ATP is break-even before any maintenance, which means there is nothing left for growth — a red flag. Redox: glycolysis supplies 2 NADH, the pathway needs 4 NADPH, so the strain is short by 4 NADPH and has 2 NADH it does not need. Nothing about this balances by itself. The realistic resolutions are to route some glucose through the oxidative pentose phosphate pathway to make NADPH at a cost in carbon, to install a transhydrogenase at a cost in energy, or to change the pathway enzymes' cofactor preference to NADH. Which you choose is a quantitative question, and week 19 answers it.

Three things people get wrong

- ‘ATP stores energy for the cell.’ It carries it; the pool lasts seconds.
- ‘NADH and NADPH are the same thing.’ They are kept at deliberately different ratios and serve opposite purposes.
- ‘Standard free energy tells me whether my pathway works.’ Only $\Delta G'$ at real concentrations does.

Self-check — answer on paper first

1. Explain, using the equation, how in situ product removal can make a pathway step feasible.

$\Delta G' = \Delta G^{\circ\prime} + RT \ln Q$, and Q contains the product concentration in the numerator. Removing the product lowers Q, which lowers $\Delta G'$ and can turn a positive value negative. The step itself is unchanged; only the concentrations are. This is the thermodynamic half of the argument for two-phase fermentation, the other half being relief of toxicity.

2. Why does activating a fatty acid as an acyl-CoA cost two ATP equivalents rather than one?

Because the reaction cleaves ATP to AMP and pyrophosphate rather than to ADP and phosphate. Regenerating ATP from AMP requires two phosphoryl transfers, so the true cost is two equivalents. Counting it as one understates the energy demand of every acyl-CoA-dependent pathway, which includes polyketides and fatty acid products.

3. A colleague gives you a model that assumes 3 ATP per NADH oxidised. What is your response?

That it is an assumption rather than a constant: the ATP yield depends on which respiratory complexes the organism uses, the proton stoichiometry of its ATP synthase, and the conditions, and the proton motive force is also spent on transport. The right response is to state it as an assumption, check how sensitive the conclusion is to it, and look for a measured P/O value for that organism.

What to open a book for

Lehninger Ch. 13 is the chapter to read, and it is worth reading twice. Doran's material- and energy-balance chapters cover the accounting side in engineering notation, with problems you can actually work.

Central carbon metabolism I: glycolysis and the pentose phosphate pathway

Central metabolism is the feedstock supply system for every pathway you will install. You need to know it well enough to say, from memory, where a precursor comes from and what making it costs.

Glycolysis as a two-stage process

The first stage invests: glucose is phosphorylated and split, consuming 2 ATP, to give two molecules of glyceraldehyde-3-phosphate. The second stage pays back: each three-carbon unit is oxidised and dephosphorylated to pyruvate, yielding 2 ATP and 1 NADH per unit. Net per glucose: 2 ATP, 2 NADH, 2 pyruvate. Those three numbers are worth knowing cold, because almost every balance you write starts from them.

The intermediates matter more than the steps. Glucose-6-phosphate is the branch point to the pentose phosphate pathway. Fructose-6-phosphate and glyceraldehyde-3-phosphate are where the non-oxidative pentose phosphate reactions rejoin. Dihydroxyacetone phosphate is the precursor of glycerol and therefore of lipid backbones. Phosphoenolpyruvate and pyruvate are the precursors of aromatic amino acids, of the anaplerotic reactions, and of a large fraction of engineered products. When you read a pathway paper, it is these names, not the enzyme names, that tell you what is being competed for.

Alternative routes, which exist for a reason

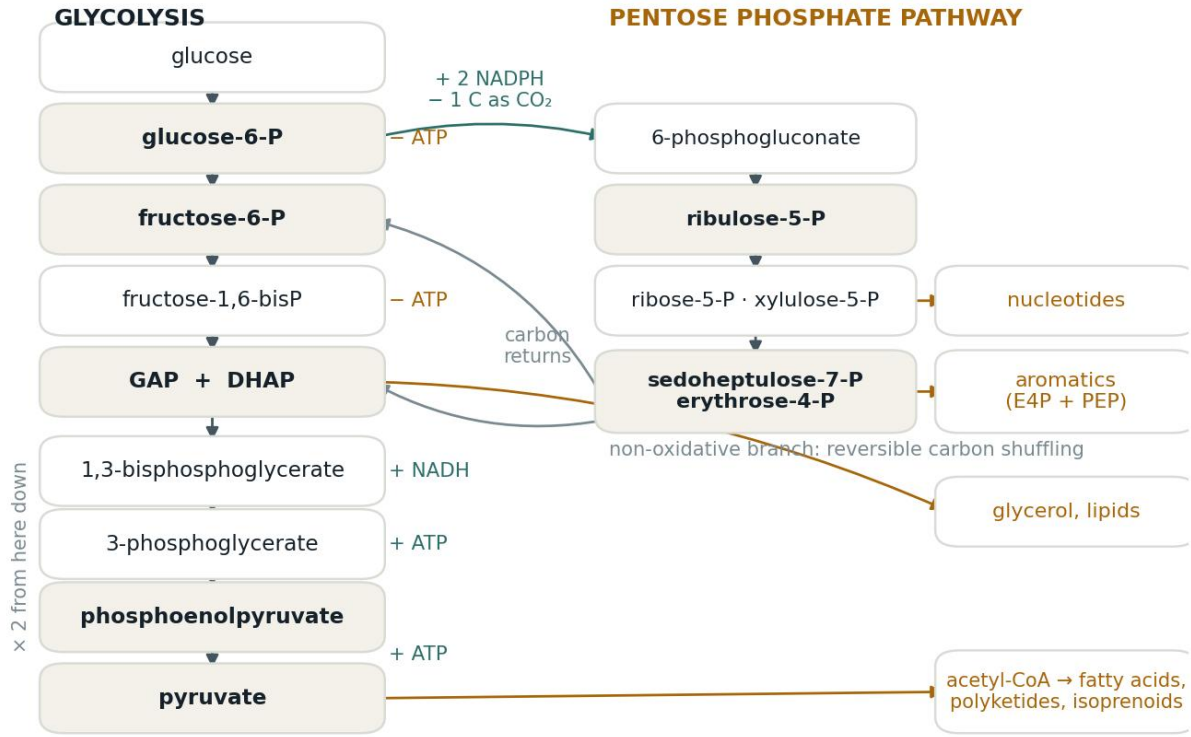
Not every organism uses the glycolysis you learned. The Entner–Doudoroff pathway, used by Pseudomonas and others, yields only 1 ATP per glucose but is calculated to need several-fold less enzymatic protein for the same glucose throughput, which is a sensible trade when carbon is plentiful and protein is expensive (Flamholz et al., PNAS 110:10039, 2013). The phosphoketolase pathway, found in some bacteria, splits sugars to give acetyl phosphate and avoids the two-carbon loss of pyruvate decarboxylation, which is why it appears in engineered strains aiming at high acetyl-CoA yield.

The lesson is not to memorise variants but to ask, for any host, which route it uses, because the ATP and redox yields differ and your balance depends on them.

The pentose phosphate pathway: the NADPH tap

The oxidative branch takes glucose-6-phosphate, oxidises it twice and decarboxylates it, producing 2 NADPH and one CO_2 per glucose entering, and leaving ribulose-5-phosphate. The non-oxidative branch is a set of reversible carbon-shuffling reactions, transketolase and transaldolase, which interconvert sugar phosphates of three to seven carbons and let the cell return carbon to glycolysis or draw ribose-5-phosphate for nucleotides and erythrose-4-phosphate for aromatic amino acids.

This is the pathway you will manipulate most often, because it is where NADPH comes from. The cost is explicit: every glucose sent through the oxidative branch for reductant loses a carbon as CO_2. That is the trade-off behind the lysine calculation in week 59 and behind the lipid ceiling in week 58. Flux through this branch is not fixed; it is regulated by NADPH demand, which is why strains with a high NADPH requirement naturally pull carbon this way, and why deleting phosphoglucose isomerase forces them to.



Net per glucose: 2 ATP, 2 NADH, 2 pyruvate.

Shaded boxes are the branch points every engineered pathway competes for. Drawn for this module, not traced from any source.

Figure 5.1 Glycolysis and the pentose phosphate pathway, with the branch points marked. The arithmetic of the NADPH-versus-carbon trade-off starts here.

The equations from this chapter

Glycolysis, net	`glucose + 2 ADP + 2 Pi + 2 NAD⁺ → 2 pyruvate + 2 ATP + 2 NADH + 2 H⁺ + 2 H₂O`
Oxidative PPP	`glucose-6-P + 2 NADP⁺ → ribulose-5-P + CO₂ + 2 NADPH`
The trade	`2 NADPH per glucose through the oxidative branch, at the cost of 1 carbon`

Worked example

A product needs 4 NADPH per molecule and the only NADPH source is the oxidative pentose phosphate pathway. How much glucose does the reductant alone require per product molecule, and what does that do to the carbon yield?

Each glucose through the oxidative branch gives 2 NADPH, so 4 NADPH needs 2 glucose. Those 2 glucose each lose one carbon as CO_2 in the process, and their remaining five-carbon skeletons return to central metabolism rather than disappearing, so the carbon is not all lost — but the decarboxylation is. The practical consequence is that the carbon yield of such a product is always below the naive one-to-one bound, which is exactly what the verified lysine figure shows: a theoretical maximum of 0.81 mol per mol rather than 1.0, because reductant has to be paid for in carbon.

Three things people get wrong

- ‘Glycolysis makes a lot of ATP.’ It makes 2 net per glucose; respiration makes most of it.
- ‘The pentose phosphate pathway is for nucleotides.’ It is also the main NADPH source, which is usually why you care.

- ‘Every organism runs the same glycolysis.’ Entner–Doudoroff and phosphoketolase routes have different yields.

Self-check — answer on paper first

1. Name the five intermediates of central metabolism that most engineered pathways branch from, and one product from each.

Glucose-6-phosphate (sugar alcohols, and entry to the pentose phosphate pathway); fructose-6-phosphate and glyceraldehyde-3-phosphate (entry points for returning carbon, and precursors for amino sugars); dihydroxyacetone phosphate (glycerol and lipid backbones); phosphoenolpyruvate and erythrose-4-phosphate together (aromatic amino acids and aromatics); pyruvate and acetyl-CoA (fatty acids, polyketides, isoprenoids, organic acids). Credit any defensible list that names the branch point rather than the enzyme.

2. Why does a strain lacking phosphoglucose isomerase still grow on glucose, and what is its advantage?

Because glucose-6-phosphate cannot be isomerised into glycolysis, so carbon is forced through the oxidative pentose phosphate pathway and returns to lower glycolysis through the non-oxidative branch. It grows more slowly, because of the extra steps and the carbon lost as CO_2, but it generates far more NADPH, which is an advantage for an NADPH-demanding product.

3. Why is ATP per glucose a poor way to compare glycolytic routes for a production strain?

Because what you usually need is precursor supply and reductant, not ATP. A route with a lower ATP yield may be preferable if it avoids a decarboxylation, conserves carbon, or requires less enzyme protein. The right comparison is against the specific demands of your pathway, written as a balance.

What to open a book for

Lehninger Ch. 14 and 15. Read them with a blank sheet and redraw the pathway from memory afterwards; this is the one piece of memorisation in the whole curriculum that genuinely pays.

Chapter 6 · tracker week 6

Central carbon metabolism II: the TCA cycle, respiration and overflow

Where the rest of the carbon goes, where most of the ATP comes from, and why a well-fed culture excretes acetate.

The cycle, and its two jobs

Pyruvate is oxidised to acetyl-CoA, losing a carbon as CO_2 and reducing NAD^+. Acetyl-CoA then condenses with oxaloacetate and the resulting six-carbon acid is oxidised round to oxaloacetate again, releasing two CO_2 and generating three NADH, one reduced flavin and one nucleoside triphosphate per turn. Fully oxidising one glucose this way yields a large amount of reducing power, which the respiratory chain converts to ATP.

But the cycle has a second job that matters more for engineering: it is a supplier of precursors. α-Ketoglutarate is the nitrogen entry point and the precursor of glutamate and therefore of most amino nitrogen. Oxaloacetate is the precursor of aspartate and of the lysine and threonine families. Succinyl-CoA feeds haem synthesis. Drawing these off means the cycle loses carbon it needs to keep turning, which brings us to anaplerosis.

Anaplerosis, the word that will keep appearing

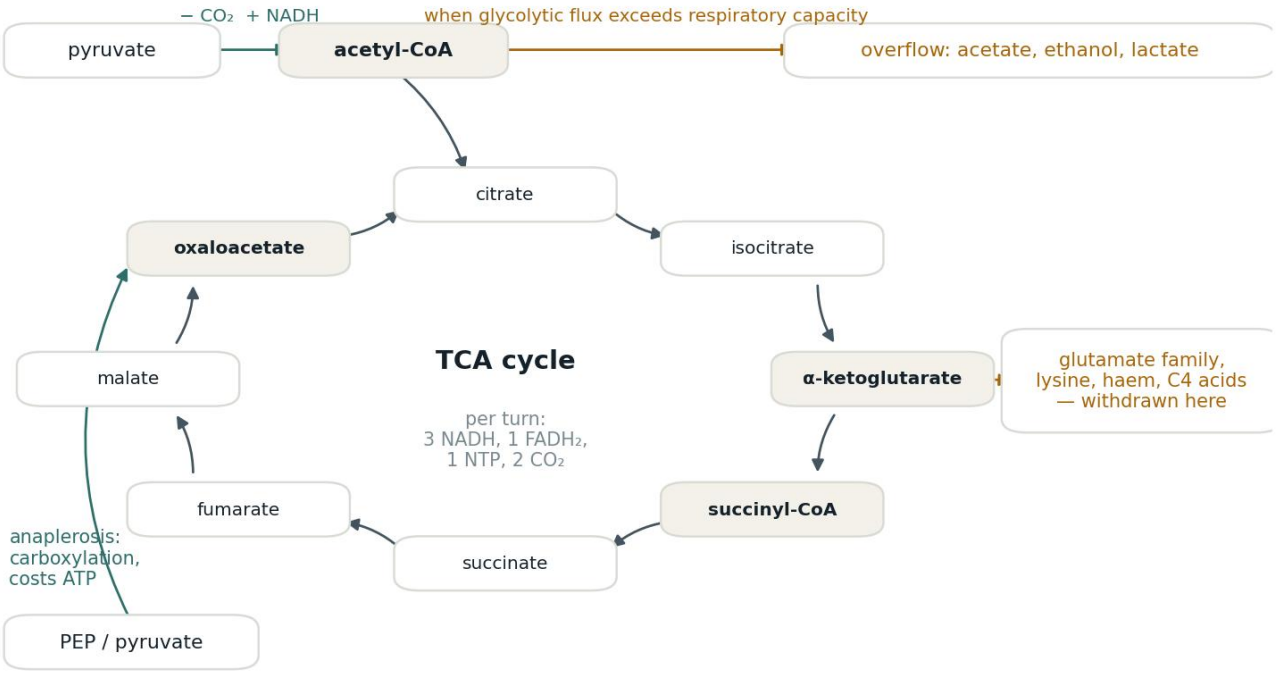
If you withdraw an intermediate, the cycle runs down, so cells have reactions that replenish them: carboxylation of pyruvate or phosphoenolpyruvate to oxaloacetate, and the glyoxylate shunt when growing on two-carbon substrates. These anaplerotic reactions are a frequent target in strain engineering, because a product drawn from oxaloacetate or α-ketoglutarate is limited by how fast the cell can refill the cycle, not by the pathway you installed.

This is one of the places where an engineer's instinct is right: it is a recycle loop with a purge, and the purge rate sets the make-up rate. If you are drawing product from the loop, you must feed it.

Overflow metabolism, and why your culture smells of vinegar

Feed E. coli plenty of glucose with plenty of oxygen and it still excretes acetate. This is not a defect, and the modern explanation is sharper than the older ‘limited respiratory capacity’ one. Making ATP by respiration costs more proteome per unit of energy than making it by fermentation, so above a threshold growth rate the cell gets more ATP per gram of protein invested by fermenting part of the carbon and excreting the product. Measurements in E. coli show acetate excretion rising linearly with growth rate above a threshold near 0.76 h^{-1}, and zero below it (Basan et al., Nature 528:99, 2015). The same logic produces ethanol in Crabtree-positive yeasts and lactate in mammalian cells. Either way the operational conclusion is the same: it is set by the specific uptake rate, so the feed rate controls it.

Three consequences for you. First, carbon you paid for leaves as a by-product, so yield falls. Second, acetate is itself inhibitory, so growth and production suffer at high concentrations. Third, the standard answer in practice is a fed-batch: feed slowly enough that glycolytic flux stays below the respiratory capacity. That is why nearly every industrial fermentation is fed-batch, and it is a process answer to a metabolic problem.



The cycle is a recycle loop with a purge: every intermediate you withdraw must be replaced, or the loop runs down.

Figure 6.1 The TCA cycle as a recycle loop with withdrawals, anaplerotic make-up, and the overflow outlet that appears when the feed outruns respiration.

The equations from this chapter

Pyruvate oxidation	`pyruvate + CoA + NAD⁺ → acetyl-CoA + CO₂ + NADH`
Per turn of the cycle	`3 NADH + 1 reduced flavin + 1 NTP + 2 CO₂`
Overflow condition	`excretion begins when glycolytic flux exceeds respiratory capacity`

Worked example

Your fed-batch accumulates 5 g/L acetate and growth has slowed. Your colleague proposes deleting the acetate pathway. What do you say?

That deleting the outlet does not remove the cause. Acetate appears because glycolytic flux exceeds respiratory capacity, so blocking the excretion route backs the pressure up into acetyl-CoA and the cycle, which commonly slows growth further or shifts the excretion to another by-product such as pyruvate or lactate. The first intervention is the feed: lower the feed rate so flux stays within respiratory capacity, which is the standard reason fed-batch exists. If a genetic change is wanted, raising respiratory or TCA capacity, or reducing glucose uptake rate, addresses the cause rather than the symptom. Deletion can work, but as part of a rebalanced strain, not on its own.

Three things people get wrong

- ‘Acetate means something went wrong.’ It means the feed exceeded respiratory capacity, which is normal and predictable.
- ‘The TCA cycle is for making ATP.’ It is also the precursor supply for nitrogen metabolism and several amino-acid families.
- ‘Deleting a by-product pathway raises yield.’ Often it moves the problem; fix the flux imbalance.

Self-check — answer on paper first

1. Why does withdrawing α-ketoglutarate for a product require an anaplerotic reaction?

Because the TCA cycle is a closed loop: every intermediate withdrawn is one fewer oxaloacetate returning to condense with acetyl-CoA, so the cycle runs down and both ATP generation and precursor supply fall. Anaplerotic carboxylation of pyruvate or phosphoenolpyruvate refills oxaloacetate, and the required anaplerotic flux is set by the withdrawal rate.

2. Explain why fed-batch operation is a metabolic answer and not merely a process convenience.

Because overflow metabolism is triggered by glycolytic flux exceeding respiratory capacity. By feeding substrate at a controlled rate, the fed-batch holds the specific uptake rate below the threshold, so carbon is fully oxidised or directed to product instead of being excreted as acetate or ethanol. It is a direct manipulation of the cell's flux distribution through the feed rate.

3. Name two products drawn from oxaloacetate and state what their yield depends on besides the installed pathway.

Aspartate-family products such as lysine, threonine and methionine, and C4 acids such as succinate and malate. Their yield depends on anaplerotic capacity, since oxaloacetate must be replenished by carboxylation, and that carboxylation costs ATP and requires a CO_2 supply; it also depends on competition from the cycle itself and from amino-acid biosynthesis.

What to open a book for

Lehninger Ch. 16 to 19. Alberts is weaker here. For overflow metabolism and proteome allocation, read a recent review rather than a textbook, since the quantitative account is recent.

Biosynthetic metabolism: where products come from

Twelve precursors from central metabolism build essentially everything. Knowing which one your target comes from tells you immediately where in metabolism you will be fighting.

The twelve precursors

Biosynthesis in a bacterium starts from about twelve compounds, all of them intermediates you met in weeks 5 and 6: glucose-6-phosphate, fructose-6-phosphate, ribose-5-phosphate, erythrose-4-phosphate, glyceraldehyde-3-phosphate, 3-phosphoglycerate, phosphoenolpyruvate, pyruvate, acetyl-CoA, α-ketoglutarate, succinyl-CoA and oxaloacetate. Every amino acid, nucleotide, lipid and cofactor is assembled from these, with ATP and reducing power supplied separately.

Treat this as a lookup table rather than a memory exercise. When you choose a target molecule, the first question is which precursor it derives from, because that determines which part of central metabolism must be amplified, what competes with you, and what the theoretical yield can be.

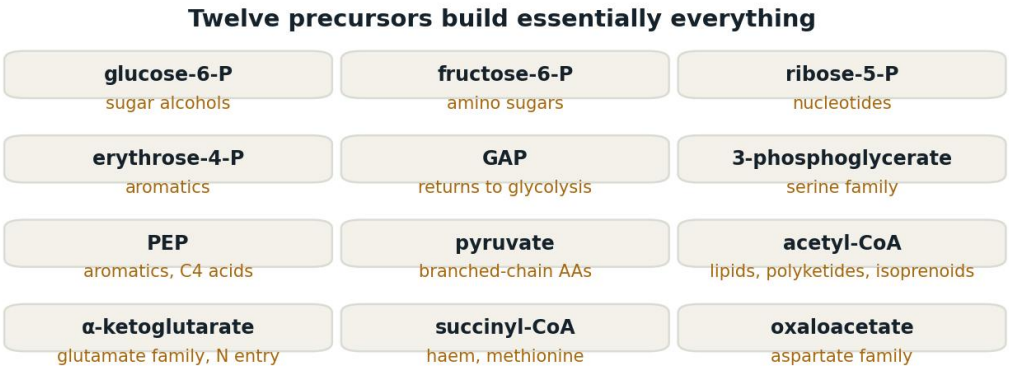
The four families you will actually meet

Fatty acids and polyketides are built by iterative condensation of two-carbon units from acetyl-CoA, or methylmalonyl-CoA for polyketides, with reduction at each cycle. They are therefore acetyl-CoA sinks and NADPH sinks, and the arithmetic in week 58 shows how severe the reductant demand is.

Isoprenoids are built from five-carbon units, either by the mevalonate pathway from acetyl-CoA, which costs ATP and NADPH, or by the methylerythritol phosphate pathway from pyruvate and glyceraldehyde-3-phosphate. Which route your host has, and which you install, changes the balance substantially. Carotenoids, including the astaxanthin you worked on, are in this family.

Amino acids come in families named after their precursor: the aspartate family from oxaloacetate, the glutamate family from α-ketoglutarate, the aromatic family from phosphoenolpyruvate plus erythrose-4-phosphate, and others from pyruvate or 3-phosphoglycerate. Aromatic compounds, including flavonoids and many pharmaceutical intermediates, come through the shikimate pathway from that phosphoenolpyruvate and erythrose-4-phosphate junction, which is why the supply of those two precursors dominates aromatic product engineering.

Nucleotides and their derivatives come from ribose-5-phosphate with amino acid and one-carbon donors. You will meet them less often as products, but their demand is a fixed cost of growth.



The first question about any target molecule is which of these it comes from: that decides what you amplify and what you compete with.

Figure 7.1 The twelve precursors and what each one builds. Use it as a lookup: find your target's precursor and you know which part of metabolism you are about to fight.

Nitrogen, sulfur and one-carbon units

Nitrogen enters metabolism almost entirely through glutamate and glutamine, made from α-ketoglutarate and ammonium. Every amino group in the cell is transferred from there by transamination, which is why α-ketoglutarate availability couples nitrogen assimilation to the TCA cycle. If your product carries nitrogen, your balance needs this.

Sulfur enters through cysteine and methionine, and reducing sulfate to sulfide is expensive in reducing power, a cost often omitted from back-of-envelope balances for sulfur-containing products such as glutathione. One-carbon units move on tetrahydrofolate and on S-adenosylmethionine, which is the methyl donor for nearly every methylation, including the ones that decorate natural products.

The equations from this chapter

Fatty acid cycle	each round: 1 acetyl-CoA (or malonyl-CoA) + 2 NADPH → chain extended by 2 carbons
Nitrogen entry	α-ketoglutarate + NH_4^+ + NAD(P)H → glutamate, then transamination everywhere
Methylation	S-adenosylmethionine is the methyl donor, and it is made from methionine and ATP

Worked example

You want to produce a methylated aromatic compound. Trace the precursor demands and name three things that will limit you.

The aromatic ring comes through the shikimate pathway, which condenses phosphoenolpyruvate and erythrose-4-phosphate, so you need both: phosphoenolpyruvate from glycolysis and erythrose-4-phosphate from the non-oxidative pentose phosphate pathway. The methyl group comes from S-adenosylmethionine, made from methionine and ATP. The three limits, in the order I would check them: erythrose-4-phosphate supply, which is usually the scarcer of the two ring precursors; competition for phosphoenolpyruvate, which is also consumed by the sugar uptake system in many bacteria, so taking it for your product directly impairs glucose uptake; and S-adenosylmethionine regeneration, since methylation leaves a product that must be recycled and the pool is small. A fourth, often decisive, is feedback inhibition on the first shikimate enzyme by the aromatic amino acids.

Three things people get wrong

- ‘Precursor supply is a detail.’ It is usually the binding constraint, and pathway enzymes are rarely the problem.
- ‘Methylation is cheap.’ Each one costs an ATP-derived cofactor that must be regenerated.
- ‘Sulfur is just another element.’ Reducing sulfate costs substantial reducing power.

Self-check — answer on paper first

1. Why does engineering an aromatic product in E. coli often interfere with glucose uptake?

Because the phosphotransferase system that imports glucose consumes phosphoenolpyruvate, which is also one of the two precursors of the shikimate pathway. Diverting phosphoenolpyruvate to the product therefore competes directly with sugar uptake, which is why strains for aromatics often replace the uptake system or re-route it.

2. A target molecule derives from acetyl-CoA and needs many reduction steps. Which two supplies do you check first, and why?

Cytosolic or cellular acetyl-CoA supply, because it is a small, fast-turning pool competed for by the TCA cycle, fatty acids and amino acids; and NADPH, because iterative reduction is the dominant reductant demand and the pentose phosphate pathway pays for it in carbon. Both are supply problems, not pathway-enzyme problems.

3. Where does nitrogen enter metabolism, and what does that couple it to?

Through glutamate and glutamine, formed from α-ketoglutarate and ammonium. That couples nitrogen assimilation to the TCA cycle, since α-ketoglutarate is a cycle intermediate, so nitrogen-rich products draw on the same pool as energy metabolism.

What to open a book for

Lehninger Ch. 20 to 22 covers the chemistry. For the twelve-precursor framing and the yield arithmetic that follows from it, Stephanopoulos Ch. 2 and 3 are better, and that is the book you will keep using.

Molecular biology and gene regulation

This is the week that turns the central dogma from a slogan into a set of parts you can specify, which is what weeks 20 onward will have you assembling.

Transcription, and what a promoter really is

RNA polymerase binds a promoter, melts the duplex, and synthesises RNA from the template strand. In bacteria, promoter recognition is conferred by a sigma factor, an exchangeable subunit that lets one core polymerase read different promoter classes: a housekeeping sigma for most genes, and alternative sigmas for heat shock, stationary phase, nitrogen limitation and so on. That architecture is why a strong promoter in exponential phase can be silent in stationary phase.

A promoter is therefore not simply ‘on’ or ‘off’ but a quantitative part, characterised by its strength, its dependence on sigma factors and on activators and repressors, and its dynamic range from the uninduced to the induced state. When you later choose a promoter from a library, these are the numbers you are choosing among, and week 27 has you measuring them.

Translation and the ribosome binding site

In bacteria the ribosome engages an mRNA at a short sequence upstream of the start codon, and the strength of that interaction, together with the local folding of the mRNA, sets the initiation rate. Because initiation is usually rate-limiting, the ribosome binding site is a second, independent dial on expression. This is practically important: two genes on the same transcript can be expressed at very different levels by changing only their binding sites, which is how you set the stoichiometry of a multi-enzyme pathway without changing its promoter.

A complication to remember: the mRNA sequence immediately after the start codon also folds, so the same binding site can perform differently in front of different genes. This is why predictive calculators for translation initiation are useful but imperfect, and why libraries exist.

Regulation as a circuit

A repressor binds an operator and blocks transcription; an inducer binds the repressor and relieves it. An activator binds and recruits polymerase. Combine them and you have logic: the classic lactose operon is repressed in the absence of lactose and poorly activated in the presence of glucose, so it is expressed well only when lactose is present and glucose is not. Read that as an AND gate implemented in chemistry.

For engineering, three things follow. Inducible systems let you separate growth from production, which matters whenever the product is toxic or the burden is high. Leakiness — expression in the uninduced state — is a real and quantifiable property, and week 28 shows that leak sets a hard ceiling on how far a repressor can turn a gene down. And the inducer itself has a cost and a diffusion problem at scale, which is why density-triggered and self-inducing systems are attractive industrially.

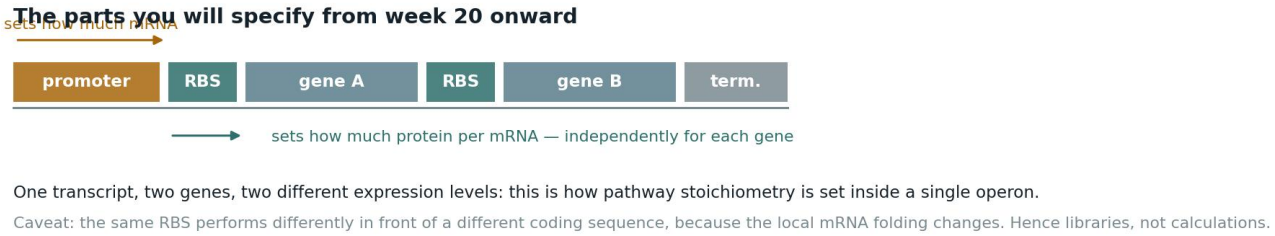


Figure 8.1 The genetic parts you will specify from week 20 onward, and the two independent dials on expression.

Beyond transcription

Expression is also controlled by mRNA stability, by small regulatory RNAs that pair with a transcript and change its translation or degradation, by riboswitches in the untranslated region that change conformation on binding a metabolite, and by protein degradation tags. Each of these is a tool you will use later: small RNAs for tunable knockdown without editing the genome, riboswitches as metabolite sensors, degradation tags for fast clearance. Week 28 treats them as an engineering toolkit.

The equations from this chapter

Expression, loosely	`protein ≈ (transcription rate) × (translation initiation rate) / (degradation + dilution)`
Dilution	`in a growing culture, protein is cleared mainly at the growth rate μ, not by proteolysis`

Worked example

You need three enzymes of a pathway at a ratio of roughly 1 : 5 : 1. Give two ways to achieve this and the trade-off of each.

One: put all three in a single operon under one promoter and set the ratio with ribosome binding sites of different strengths. This keeps the stoichiometry fixed relative to each other whatever the promoter does, which is robust, but it is constrained by context effects — the same binding site performs differently in front of a different coding sequence — so it usually needs a small library and a measurement rather than a calculation. Two: use three separate promoters of different strengths. This is easier to reason about and to change one gene at a time, but the ratio now depends on three promoters that may respond differently to growth phase and conditions, so it can drift. In practice people combine both and screen, which is why week 33 is about plate-based screening rather than calculation.

Three things people get wrong

- ‘The promoter sets expression.’ The promoter and the ribosome binding site each set part of it, independently.
- ‘Uninduced means off.’ Leak is real, measurable, and often the limiting factor.
- ‘A part behaves the same everywhere.’ Context, especially local mRNA folding, changes it.

Self-check — answer on paper first

1. Why can two genes in the same operon be expressed at very different levels?

Because transcription gives them the same mRNA, but translation initiation is set separately for each by its own ribosome binding site and the local folding around its start codon. Differential mRNA stability along the transcript and premature termination also contribute. This is what lets a designer set pathway stoichiometry within one operon.

2. What does a sigma factor do, and why does it matter for a production process?

It confers promoter specificity on RNA polymerase, so which promoters are read depends on which sigma factors are active. It matters because the sigma factor complement changes with growth phase and stress, so a promoter that is strong in exponential growth may be weak in stationary phase or under the stress your product causes, which is precisely when you want production.

3. Why is an inducible system attractive for a toxic product, and what are its two practical drawbacks?

Because it separates a growth phase, in which the pathway is off and the cells accumulate biomass unburdened, from a production phase. The drawbacks are leak, which means the pathway is never fully off and sets a limit on how toxic a product can be tolerated during growth, and the inducer itself, which costs money and must be distributed uniformly in a large vessel, which is why density- or nutrient-triggered systems are preferred at scale.

What to open a book for

Alberts Ch. 4 to 8 selectively, and Lehninger Ch. 24 to 28 for the chemistry of the information pathways. Snyder & Champness is the better book if you want bacterial regulation and genetics specifically, and it is written around experiments rather than mechanisms.

Chapter 9 · tracker week 9

Regulation at the system level: what will fight your design

Your pathway is a tenant in a cell that is optimising for something else. This week is about the landlord.

Carbon catabolite repression

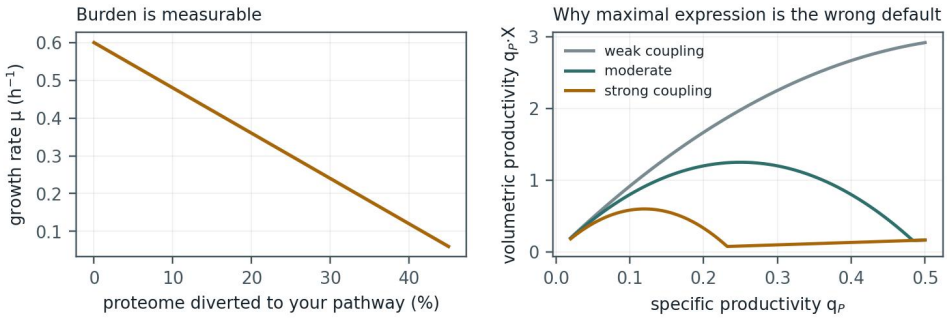
Given two carbon sources, a cell usually consumes the better one first and represses the machinery for the other, producing the familiar two-phase growth curve with a lag between them. In E. coli the mechanism runs through the phosphotransferase system and cyclic AMP; in Bacillus it runs through a different regulator entirely. The details differ by organism, but the principle does not: the cell has a global preference and will enforce it.

Why you care. If your process uses a mixed feedstock — a lignocellulosic hydrolysate with glucose and xylose, for example — catabolite repression means the sugars are consumed sequentially rather than together, which lengthens the fermentation and complicates the feed. Engineering simultaneous utilisation is a recognised and non-trivial objective, and it is a genuine research topic in the Korean groups on your shortlist.

Proteome allocation, and why burden is not a vague complaint

A cell has a finite ribosome capacity and a finite proteome budget. Expressing your pathway means a fraction of that budget is spent on proteins that do nothing for growth, and the cell responds by growing more slowly. This is measurable: the relationship between growth rate and the fraction of proteome allocated to ribosomes, to catabolism and to everything else follows reproducible patterns, and heterologous expression displaces the other sectors.

The engineering consequence is a trade-off you cannot escape by cleverness: more pathway enzyme means less growth, and volumetric productivity depends on both. It is why the right answer is often a two-stage process rather than maximal expression, and why measuring growth rate with and without induction should be among your first experiments.



Illustrative shapes, not measured data: the point is that volumetric productivity is $q_P \cdot X$, so a gain in q_P paid for with biomass can lower output. Two-stage operation is the usual escape.

Figure 9.1 Burden, and why maximal expression is the wrong default. Illustrative shapes rather than measured data: the point is the shape of the trade-off.

Stress responses, and the stringent response

Cells have dedicated programmes for heat, oxidative stress, envelope damage, and nutrient starvation. The one to know by name is the stringent response: on amino acid starvation, an alarmone signal reprogrammes transcription away from ribosomes and growth and towards

survival. Several things you will do — overexpressing a protein that drains an amino acid pool, accumulating a toxic intermediate, running a nutrient-limited fed-batch — can trigger it, and when it fires, production usually falls with growth.

The practical reading of all this: when a strain underperforms, the cause is as likely to be a global regulatory response as a pathway defect. Measuring growth rate, by-product formation and a stress reporter will often tell you more than measuring another pathway intermediate.

The equations from this chapter

Burden	`μ falls roughly in proportion to the proteome fraction diverted to a non-growth protein`
Volumetric productivity	`q_P × X; raising q_P at the expense of X can lower the product`

Worked example

Induction raises your specific productivity by 60% but lowers growth rate from 0.45 to 0.20 h^{-1}. Is induction worth it?

It depends on the process, and the calculation is the answer rather than the biology. Volumetric productivity is specific productivity times biomass, so the question is how much biomass you reach and when. In a batch where induction happens at the start, a growth rate of 0.20 instead of 0.45 means roughly half the doublings in the same time, so you lose far more biomass than the 60% gain in specific rate recovers. In a two-stage process — grow uninduced to the target biomass, then induce — you keep both, which is exactly why decoupled production exists. The general rule: compare on volumetric productivity and on titer at harvest, not on specific rate, and design the staging rather than accepting the trade.

Three things people get wrong

- ‘Burden is hand-waving.’ It is a measurable reduction in growth rate from proteome allocation.
- ‘Mixed sugars are consumed together.’ Usually they are not, and making them so is a project.
- ‘A poor strain means a poor pathway.’ Often it means a global regulatory response you have triggered.

Self-check — answer on paper first

1. Why does catabolite repression matter for a lignocellulosic process?

Because the hydrolysate contains glucose and pentoses, and the cell will consume glucose first and repress pentose utilisation, giving sequential rather than simultaneous consumption. That lengthens the fermentation, leaves residual pentose if the process is stopped early, and complicates feeding, so simultaneous utilisation is an explicit engineering objective.

2. State the trade-off that makes maximal pathway expression the wrong default.

Expressing more pathway protein raises specific productivity but consumes proteome and ribosome capacity, lowering growth rate and therefore biomass. Volumetric productivity is the product of specific rate and biomass, so beyond a point more expression lowers output. The usual resolution is to decouple growth from production in two stages rather than to maximise either alone.

3. A strain underperforms. Name three system-level measurements you would make before touching the pathway.

> Growth rate with and without induction, to quantify burden; by-product formation, especially acetate or ethanol, to see whether flux exceeds respiratory capacity; and substrate consumption with a carbon balance, to find where the carbon actually goes. A stress or stringent-response reporter, and intracellular amino acid or intermediate pools, are reasonable additions.

What to open a book for

No single chapter covers this well. Alberts Ch. 7 for the mechanism of bacterial regulation, then a recent review on proteome allocation and burden, which is where the quantitative account lives. Brock is useful for the physiology.

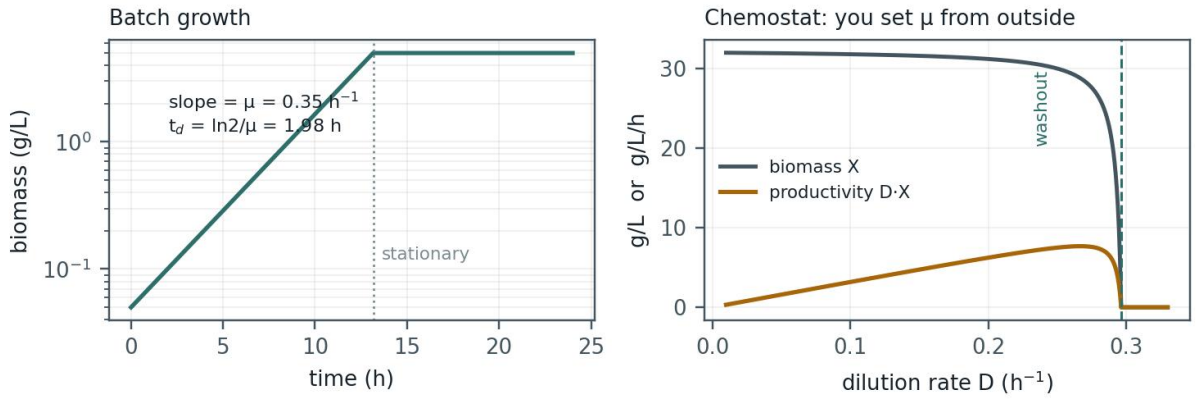
Microbial genetics and microbiology for engineers

The practical biology: how you move DNA, how you select what you want, and how you grow and count cells. This is also the week that ends with the phase 1 checkpoint.

Growth, measured honestly

Batch growth has a lag, an exponential phase in which biomass follows X = X₀·e^(μt), a stationary phase and death. The specific growth rate μ and the doubling time t_d = ln2/μ are the two numbers you quote. In continuous culture at steady state the specific growth rate equals the dilution rate, which is why a chemostat is such a clean instrument: you set the growth rate from outside. Above a critical dilution rate the cells wash out, and week 60 uses that fact.

Three ways of counting disagree, and you should know why. Optical density measures light scattering, which tracks biomass and cell size rather than number, so a culture of filamentous or enlarged cells reads high. Colony counts measure cells able to grow on that medium, so cells that are alive but not culturable are missed. Dry cell weight is the measurement your balances actually need, and it is the one most often skipped. For a yield calculation, use dry weight or a calibration to it, not raw optical density.



Assumed parameters. Note that the most productive dilution rate sits just below the washout cliff, which is an uncomfortable place to run a plant.

Figure 10.1 Batch growth and the chemostat. Note where the most productive dilution rate sits relative to washout.

Getting DNA into cells, and the selection that follows

Three routes matter. Transformation is uptake of free DNA, natural in some species and forced by chemical treatment or electroporation in others. Conjugation is transfer from a donor cell through direct contact, and it is the standard route into organisms such as Streptomyces that resist transformation. Transduction uses a phage to carry DNA. Each has a characteristic efficiency, measured in transformants per microgram, and that efficiency decides whether a library approach is even possible, as the calculation in week 26 shows.

What follows delivery is the part engineers under-appreciate: selection. A selectable marker lets only the cells that took up your DNA grow, usually antibiotic resistance or complementation of an auxotrophy. A counterselectable marker does the opposite: it kills cells that still carry it, which is how you remove a marker after use and how two-step genome editing works. Designing a clean selection is often the difference between a one-week

experiment and a three-month one, and it is why Snyder & Champness is a more useful book here than a molecular biology text.

Mutation, repair and why strains drift

Mutations arise from replication error, chemical damage and mobile elements, and are corrected by several repair systems. Two practical consequences. First, a population is not homogeneous: in a large culture, variants exist, and if a variant grows faster than your producer — typically one that has lost the burden of your pathway — it will take over. That is genetic instability, and it is a routine cause of a strain that performs well at bench scale and fails at production scale, where many more generations elapse. Second, insertion sequences can inactivate your construct, which is why strains with reduced mobile-element content exist.

The defence is design: integrate rather than carry a plasmid where stability matters, avoid unnecessary burden, use a selection that remains in place, and sequence the strain after the fermentation, not only before it.

Sterility, contamination and the things that actually stop work

Aseptic technique is the skill the tracker cannot teach you. The relevant knowledge is: what a sterile medium is and how it is made, why autoclaving and filtration suit different components, what contamination looks like in a plate and in a culture, and that phage contamination in a production plant is a serious event rather than a nuisance. The Bench log tab exists because this is learned at a bench with someone watching, not from a page.

The equations from this chapter

Exponential growth	$X = X_0 \cdot e^{(\mu t)}$; $t_d = \ln 2/\mu$
Chemostat	at steady state $\mu = D$; washout above $\mu_{max} \cdot S_f/(K_s + S_f)$
Library coverage	clones for probability P of seeing a given variant: $n = \ln(1-P)/\ln(1-1/N)$

Worked example

A strain gives 8 g/L at 1 L but 3 g/L in a 1,000 L run from the same inoculum protocol. Give the explanation you would test first, and how.

Genetic instability from the extra generations. Scaling from 1 L to 1,000 L adds roughly ten doublings of inoculum build-up, and if a variant that has lost or silenced the pathway grows even a few per cent faster, it can become a large fraction of the population by the time the production vessel is full. The tests are to plate the final broth and screen colonies for production, to check plasmid retention without selection, and to sequence the construct from the end of the run rather than from the seed. If instability is confirmed, the answers are integration instead of a plasmid, lower burden, or a selection that persists. Alternative explanations worth excluding are oxygen transfer, which falls on scale-up, and gradients in substrate and pH in a large vessel — but instability is the one that matches ‘same protocol, fewer generations, better result’.

Three things people get wrong

- ‘Optical density is biomass.’ It is a correlate; your balances need dry weight.
- ‘A plasmid stays put.’ Without selection and with burden, it does not.
- ‘Scale-up is a process problem.’ Part of it is a genetics problem: more generations, more drift.

Self-check — answer on paper first

1. Why does a chemostat let you set the growth rate, and what is the limit?

Because at steady state the specific growth rate equals the dilution rate: cells must divide exactly as fast as they are removed. The limit is washout, which occurs when the dilution rate exceeds the maximum growth rate achievable on the feed concentration, and the most productive operating point sits just below it, which is an unstable place to run a process.

2. Explain why a clean selection scheme can decide whether an experiment takes a week or three months.

Because the number of cells you must examine depends on whether the desired event is selected or merely screened. A selection lets only the correct cells grow, so a rare event is recovered from a large population in one step. A screen requires you to examine every candidate individually, so the same rare event needs thousands of assays. Designing a selection, including a counterselection for removing markers, is therefore a design task of the same importance as the construct itself.

3. Name the three cell-counting methods and the question each one answers best.

Optical density, for rapidly following the course of a culture; colony counts, for viability in the sense of being able to grow; and dry cell weight, for mass balances and yields. They can disagree legitimately, and quoting a yield from optical density without a calibration to dry weight is a common error.

What to open a book for

Brock is the reference for this entire week, and it is the book most worth having on the shelf for phase 1. MIT OCW 7.13 Experimental Microbial Genetics covers the genetics practically, and Snyder & Champness covers selection design better than anything else.

Before you go on to phase 2

Phase 2 starts from the assumption that you can write a balanced reaction with cofactors, name where a precursor comes from, and say what a promoter and a ribosome binding site each do. If any of those is shaky, the modelling weeks will feel like algebra rather than biology, and the modelling is the part that distinguishes you as a chemical engineer. Go back rather than on.

The phase 1 checkpoint in the tracker's week 10 is the test. Answer it from memory, on paper, before opening the model answers. The three things it asks you to reconstruct — the central carbon map, the global regulators, and a balanced reaction with its cofactors — are exactly the three things weeks 11 to 19 will use on every page.

On the numbers in this module. I checked the quantitative claims against primary compilations rather than from memory, and two were wrong in the first draft: I had given a bacterial cell volume of 2 fL, where measured E. coli values are about 0.4 to 1.8 μm³ and 1 fL is the right round number, and I had narrowed the in-vivo free energy of ATP hydrolysis to 45–55 kJ/mol where the sourced range is about 40 to 60. Both are corrected here, and the cell-volume correction strengthens rather than weakens the conclusion it supports. Where a figure rests on an assumption rather than a measurement, the caption says so. If a number here matters to a calculation you depend on, check it against BioNumbers or the primary source; the mistakes in this document are mine, not a publisher’s.

What this module deliberately leaves out: structural figures, full enzyme mechanisms, and the pathway maps you should annotate yourself. Those are what the books are for, and the chapter notes above say which to open. The module is the spine; it is not a replacement for a library.

Appendix — figures you can legally reuse, and the ones I drew

Every figure in this module I drew myself from the underlying facts. That is legitimate: stoichiometry, pathway topology and rate laws are facts, and facts are not copyrightable — only a particular drawing of them is. What I cannot do is reproduce a figure from Lehninger or Alberts, and I have not.

Three kinds of image are beyond what a drawing can give you: real protein structures, electron and light micrographs of cells, and photographs of equipment and colonies. For those, the sources below are either public domain or openly licensed, so you can download, print and put them in your own notes and even in a thesis, provided you follow the licence. Always check the licence on the individual file rather than trusting the site as a whole, and keep the attribution with the image.

Protein Data Bank (RCSB PDB) rcsb.org	Experimental structure coordinates for essentially every enzyme you will study. The coordinate files are released without copyright restriction, and the site's own images are free to use with attribution. You can also open a structure in free software (Mol*, which runs in the browser, or PyMOL) and make your own figure, which is what most students do.
PDB-101 pdb101.rcsb.org	The teaching arm of the PDB, including the Molecule of the Month series, which explains one enzyme at a time with original illustrations. Stated to be usable for teaching with attribution. It is the closest free equivalent to a structure chapter in a textbook.
Wikimedia Commons commons.wikimedia.org	Pathway maps, micrographs and apparatus photographs under Creative Commons or public domain. Quality is uneven and some diagrams contain errors, so check a map against a primary source before you trust it. Each file states its own licence; respect the attribution and share-alike terms.
OpenStax Biology and Microbiology openstax.org	Full textbooks under a Creative Commons Attribution licence, with figures you may reuse with credit. The level is introductory, which is exactly right for the cell biology and microbiology in weeks 1 and 10.
LibreTexts libretexts.org	Openly licensed chemistry and biology texts, including Fundamentals of Biochemistry. Licences vary by page, commonly Creative Commons with a non-commercial clause, so check before reusing in anything commercial.
MIT OpenCourseWare ocw.mit.edu	Lecture notes, slides and video for 7.05, 7.03, 7.13 and 7.344, under a Creative Commons non-commercial share-alike licence. This is the single best free substitute for lectures in phase 1.
KEGG and MetaCyc pathway databases genome.jp/kegg · metacyc.org	Authoritative pathway and reaction data. Use them to check a map and to get reaction stoichiometry. Note that both have licensing restrictions on bulk reuse and on redistribution of their images: read the maps, then draw your own.

BioRender and similar tools free tiers exist	Figure-drawing tools with icon libraries. Convenient, but the free tiers usually restrict publication use, so check the terms before putting an output in a thesis or a paper.

A practical recommendation. For the three gaps in this module, do this: for protein structure, open one enzyme of your own pathway in Mol* on the PDB site and rotate it until the active site makes sense, which teaches more than any printed figure; for cell micrographs, take them from OpenStax or Commons; and for pathway maps, redraw the central carbon map yourself from my Figure 5.1 and check it against KEGG. The redrawing is the part that makes it stick, and it is also the part that makes the figure yours to use.

Part 14 — Core concepts for weeks 1 to 10: the biology an engineer needs, compiled

You asked why I could not simply write the roughly 900 pages of textbook reading that weeks 1 to 10 assign. There are three reasons, and only one of them is a limit. Copyright is not the limit: facts are not copyrightable, and what follows is original prose that states the same facts. The real limits are that I cannot check descriptive biology from memory — the arithmetic in this module was computed and re-verified, but "which enzyme catalyses which step" cannot be computed, and the more of it I write, the more unverifiable claims you inherit with no way to tell them from the verified ones; that I have no access to protein structures or micrographs, so the figures are schematics; and that volume is not the same as understanding, so 900 pages from me would be a worse copy of a book that has been proofread for decades.

What I can do well, and what this part does, is compile **the core ideas** of those ten weeks, in the form an engineer can use: each concept stated plainly, each derivable number computed, and the connection to the rest of the module made explicit. It is about 40 pages and about 1.7 times the density of part 1 (2,100 words a week against 1,270), and it is meant to be read *beside* Lehninger, Alberts and Brock, not instead of them: use it to know what matters, and the books for the mechanisms, structures and figures it deliberately leaves out.

How to read it

The dagger. Every constant marked † is a standard textbook value that I recalled rather than computed — a pK_a, a redox potential, a molecular count per cell, a reaction stoichiometry in a specific organism. They are listed with where to verify them in the table at the end. Everything *without* a dagger was either derived from the code or computed from the daggered inputs, and the computation is shown. So when a result looks surprising, check the daggered inputs first.

The eight ideas. The previous reply named the core concept — product flux is set by whichever constraint binds first — and eight ideas under it. This part supplies the biology each of them rests on:

Core idea	Biology it rests on	Chapters here
1 Mass balance and the yield ceiling	What a cell is made of and what it costs to make	B1, B5, B7
2 Redox and ATP balance	Cofactors, the proton motive force, respiration	B2, B4, B6
3 Thermodynamics decides direction	Free energy, coupling, the ATP hierarchy	B4, B5
4 Distributed control; regulation fights you	Enzyme kinetics and allostery; regulatory networks	B3, B8, B9
5 The cell is resource-limited	Ribosomes, proteome allocation, growth laws	B1, B8, B9
6 Transport and scale	Membranes, oxygen demand, respiratory quotient	B4, B6, B10
7 Measurement and the DBTL loop	Counting, selection, mutation and its statistics	B10
8 Economics	The carbon cost of biomass; why yield has a ceiling	B7

Each chapter ends with a short **what to carry forward** table and three self-check questions with answers. A glossary of 65 terms follows chapter B10, and it is the fastest way to find out whether you know a word the tracker assumes.

What is not here. Mechanisms drawn atom by atom, three-dimensional structures, every step of every pathway with its enzyme and cofactor, and the experimental history. Those are what the books are for. Where a chapter points you to one, it says which.

Ch.	Week	Chapter
B1	1	The cell as a reactor: contents, the central dogma and the genetic code, in numbers
B2	2	Proteins and enzymes: structure, charge, stability and catalysis
B3	3	Enzyme kinetics: the derivations and the regimes
B4	4	Bioenergetics: free energy, ATP, the proton motive force and the ATP yield
B5	5	Glycolysis and the pentose phosphate pathway
B6	6	The TCA cycle, respiration and overflow
B7	7	Biosynthesis: what a gram of cells costs
B8	8	Molecular biology: replication, transcription, translation and their regulation
B9	9	System-level regulation and the growth laws

B10	10	Microbial genetics and microbiology: growth, mutation, gene transfer, sterility
		Glossary · Recalled constants to verify

Chapter B1 · tracker week 1

The cell as a reactor: contents, the central dogma and the genetic code, in numbers

Part 1 described the cell as a reactor that builds its own catalysts. This chapter puts numbers on that sentence — how many molecules, how fast, at what cost — because every later calculation in the module uses one of them.

What is in a cell

An *E. coli* cell at moderate growth weighs about 0.3 pg dry†, of which roughly 55% is protein, 20% RNA, 9% lipid and 3% DNA, with the remainder lipopolysaccharide, peptidoglycan, glycogen and small molecules†. Its genome is 4.64 Mb† encoding about 4,300 proteins†, and about 88% of the sequence is coding † — bacterial genomes are compact, with short intergenic regions that carry the promoters, ribosome binding sites and terminators of part 1 chapter 8.

From those inputs, what follows is computed. Protein per cell is 0.3 × 0.55 = 0.165 pg. At an average 36 kDa† that is **2.8 million protein molecules**, or **9.0×10^8 amino-acid residues** at 110 Da each. Those two numbers are the denominators of most of what follows.

Copy number and concentration are the same quantity in different units. One molecule in a 1 fL cell is $1/(6.022 \times 10^{23} \times 10^{-15}$ L) = **1.66 nM**. So a regulatory protein present at ten copies is 17 nM, a metabolite at 1 µM is 600 molecules, and a strongly expressed enzyme at 30,000 copies is 50 µM. This conversion is why the repressor of part 8 chapter 27, with K = 10 nM and a handful of copies per cell, is in a sensible regime — and why a cell has no concentration to speak of for anything it holds fewer than a few tens of copies of; stochastic effects dominate there.

The central dogma, as an information-flow budget

DNA is replicated to DNA, transcribed to RNA, and translated to protein; information does not flow back from protein to nucleic acid. The exceptions are real and bounded: reverse transcription, RNA-dependent RNA replication in viruses, and catalytic RNA. For the organisms in this module the dogma holds as stated, and what matters for an engineer is that each arrow has a rate and a cost.

Process	Rate	Consequence
DNA replication	~1,000 nt/s per fork†, two forks	4.64 Mb ÷ (2 × 1,000) = 2,320 s = **39 minutes**
Transcription	40–80 nt/s†	A 1 kb gene is transcribed in 12–25 s
Translation	10–20 aa/s†	A 330-aa protein takes 17–33 s to synthesise
mRNA lifetime	half-life of a few minutes†	Transcriptional control acts within minutes
Protein lifetime	mostly longer than a generation	Loss of a protein is mainly dilution by growth

The replication arithmetic is worth dwelling on, because it is a computed result that matches a measured one: 39 minutes is almost exactly the 40-minute replication period (the C period) that part 9 chapter 32 used to compute how gene dose varies with position on the chromosome. And it explains something that looks paradoxical, a doubling time of 20 minutes in a cell whose chromosome takes 40 minutes to copy: the cell starts new rounds of replication before the previous round has finished, so a fast-growing cell carries overlapping replication forks and more copies of origin-proximal genes than terminus-proximal ones.

The genetic code, and what it implies for engineering

Sixty-four codons encode 20 amino acids and a stop: 61 sense codons and 3 stops. The mapping is degenerate — nine amino acids have two codons, one (isoleucine) three, five have four, three (leucine, serine, arginine) six, and two (methionine, tryptophan) only one. I built the standard table in code and asked what a random single-base substitution does to a sense codon. Of the 549 possible substitutions:

Outcome	Fraction	Reading
Synonymous — same amino acid	**24.4%**	Roughly one in four point mutations is silent at the protein level
Missense — different amino acid	**71.4%**	The large majority change the protein
Nonsense — premature stop	**4.2%**	A few per cent truncate it

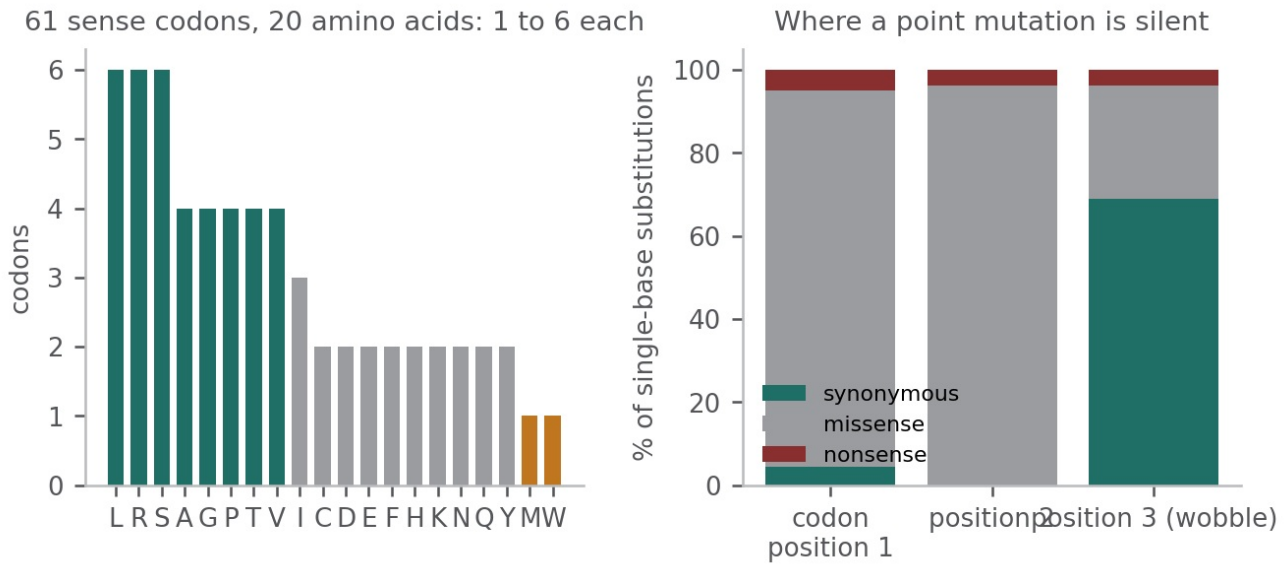


Figure B1.1 Left: codons per amino acid in the standard code, from 1 (Met, Trp) to 6 (Leu, Ser, Arg). Right: the fate of every possible single-base substitution by codon position — silent only at the third position, never at the second. Computed from the standard code for this module.

The position dependence is the useful part. **At the third position 68.9% of substitutions are synonymous; at the first, 4.4%; at the second, none.** That is the wobble: the third base is the least constrained, which is why codon usage and GC content are free to drift between organisms — a high-GC genome such as *Streptomyces* at 72% GC (part 8 chapter 31) is rich in G or C at the third position — and why a gene moved to a new host must be re-coded rather than just transferred. It also means a synonymous change, which is what domestication (part 7 chapter 21) and codon harmonisation (part 8 chapter 30) rely on, is almost entirely a third-position change.

Two caveats the textbook version skips. Synonymous does not mean neutral: codon choice affects translation speed, mRNA folding near the start codon, and in some genes co-translational folding, which is why part 8 chapter 30 treated "optimise the codons" as a hypothesis. And the code is not universal in every organism — mitochondria and some bacteria reassign codons — which is part of what makes genome recoding (part 7 chapter 25) possible at all.

What it costs to run: the ribosome budget

Here is the calculation that links this chapter to the growth laws of chapter B9. A cell must make all 9.0×10^8 residues of its protein once per doubling. At exponential growth rate μ, the required synthesis rate is μ times the protein content, so the number of ribosomes needed is

```
N_ribosomes = μ · N_aa / (f_active · k_elongation)
```

With f_{active} = 0.8 and k = 18 aa/s†, a cell growing at 0.6 h^{-1} needs about **10,500 ribosomes**. A cell growing at 2.0 h^{-1} is also bigger — taking twice the protein — and needs about **70,000**: a 3.3-fold faster growth rate costs a 6.7-fold larger ribosome population. Both agree with measured ribosome counts of order 10^4 and 7×10^4 at those rates †, which is a consistency check on the model rather than a measurement. It says the ribosome pool is superlinear in growth rate, so the cell commits a rapidly increasing share of its proteome to translation as it speeds up — and any pathway protein you add competes with it.

The energy bill is as follows. Each peptide bond costs about four high-energy phosphate bonds†: two to charge the tRNA (ATP to AMP is two equivalents, part 1 chapter 4), one GTP for EF-Tu delivery and one for EF-G translocation. A 330-aa protein therefore costs about **1,300 ATP-equivalents**, and the cell's protein polymerisation comes to 5 mmol of residues per gram dry weight × 4 = 20 mmol ATP-equivalents, against the 59.8 mmol the core model of part 10 charges as growth-associated maintenance. So **protein synthesis is about a third of the growth-associated energy budget** by this count — before the cost of making the amino acids, which chapter B7 adds.

Prokaryote and eukaryote, as far as an engineer needs

	Bacteria	Yeast and other eukaryotes
Genome organisation	One circular chromosome; genes grouped into operons, transcribed together as polycistronic mRNA	Several linear chromosomes; one gene per mRNA
Introns, splicing	Essentially none	Present; splicing before export

Transcription–translation	Coupled — ribosomes load while the mRNA is still being made	Separated by the nuclear envelope
mRNA processing	None; the 5′ end is the start	5′ cap, poly(A) tail
Replication origins	One	Many per chromosome
Compartments	Cytoplasm only (periplasm in Gram-negatives)	Nucleus, mitochondria, ER, Golgi, peroxisomes, vacuole

The coupling of transcription and translation is why a bacterial promoter and ribosome binding site are two independent dials on expression (part 1 chapter 8) and why a polar mutation in an operon can silence genes downstream. The compartmentalisation is why a pathway that works in *E. coli* may need re-engineering in yeast — acetyl-CoA is mitochondrial, as part 6 chapter 58 showed.

Cell contents	`0.3 pg dry weight; 55% protein → 2.8 × 10⁶ proteins, 9.0 × 10⁸ residues`
Copy number ↔ concentration	`1 molecule in 1 fL = 1.66 nM; 600 molecules = 1 µM`
Replication time	`genome ÷ (2 forks × 1,000 nt/s) = 39 min`
Point mutations, standard code	`24.4% synonymous, 71.4% missense, 4.2% nonsense; third position 68.9% silent`
Ribosomes required	`µ·N_aa/(f·k): ~10,500 at 0.6 h⁻¹, ~70,000 at 2.0 h⁻¹`
Cost of a protein	`~4 ATP-equivalents per residue: ~1,300 for 330 aa`

Common confusions

- **"Silent mutations do nothing."** They are silent in the amino-acid sequence only; they change translation speed and mRNA structure.
- **"Concentration is the natural unit."** At a few tens of copies per cell it is not; count molecules.
- **"A 20-minute doubling time means a 20-minute replication."** Replication takes 40 minutes; overlapping rounds are what let the cell double faster.

Self-check — answer on paper first

1. A repressor is present at 30 copies per cell. What is its concentration, and what does that imply about its operator binding?

30 × 1.66 nM = 50 nM. If the dissociation constant for its operator is at or below that, the repressor is largely bound; if it is far above, the operator is rarely occupied. That ratio, not the absolute count, sets the fold repression of part 8 chapter 27, and at 30 copies the cell-to-cell variation is substantial because a Poisson count of 30 has a standard deviation of 5.5 — about 18%.

2. Why do point mutations in a protein-coding gene so rarely give a synonymous change at the second codon position?

Because every change at the second base alters the amino acid in the standard code — the computed fraction is exactly zero. The second position largely determines the chemical class of the residue (U at that position gives the hydrophobic amino acids), so a change there almost always moves the residue into a different class, which is why the position is conserved more strongly than the third.

3. A colleague says a cell making a 36 kDa protein at 1% of its proteome needs only a trivial amount of energy. Check it.

1% of 2.8 × 10^6 proteins is 28,000 copies, each costing about 1,000 to 1,300 ATP-equivalents to synthesise, so about 3 × 10^7 ATP-equivalents per generation — about 1% of the protein-synthesis budget, and therefore about 0.3% of the growth-associated total. It is small in energy and not small in ribosome time, because the ribosomes that make it are not making growth proteins, which is why burden shows up as a growth penalty rather than an energy penalty.

What to open a book for. Alberts, *Molecular Biology of the Cell*, chapters 1 to 7 for the cell and the dogma, read for the figures; BioNumbers for every daggered constant in this chapter; Neidhardt's *Escherichia coli and Salmonella* for the definitive composition tables. MIT OCW 7.05 lectures 1 and 2 cover this chapter free.

Proteins and enzymes: structure, charge, stability and catalysis

A protein is a chain of twenty kinds of building block that folds because it is cheaper in free energy to fold than not to — by a margin so small that a single mutation can cost a tenfold change in how much is unfolded.

The twenty amino acids, grouped by what they do

All twenty share a backbone — an amino group, a carboxyl group and an α-carbon — and differ in the side chain. The grouping that predicts behaviour is chemical, and part 1 chapter 2 gave it; here is the quantitative layer that decides what you can do with it.

Class	Residues	Side-chain pK_a†	Where it sits and what it does
Non-polar	G A V L I M P F W	—	Buried in the core; the hydrophobic effect drives folding; proline breaks helices, glycine allows tight turns
Polar, uncharged	S T N Q C Y	Cys 8.3; Tyr 10.1	Surface and active sites; Ser, Thr, Tyr, Cys are the usual nucleophiles and modification sites
Acidic	D E	3.9; 4.2	Negative at physiological pH; salt bridges, metal binding
Basic	K R H	Lys 10.5; Arg 12.5; His 6.0	Positive at physiological pH; His is the one residue near neutral and is therefore a catalytic acid–base

The backbone terminals ionise too, near pK_a 2 for the carboxyl and 9 for the amino group†, though inside a chain those groups are consumed in peptide bonds and only the two ends remain.

Charge is a titration curve, and the arithmetic is useful

A group with pK_a p is protonated to the fraction $1/(1 + 10^{pH - p})$, so the net charge of a protein at a given pH is a sum of such terms. Writing that sum for a few sequences and finding where it crosses zero gives the isoelectric point, pI:

Sequence	pI	Net charge at pH 5	at 7.5	at 8
20 residues, one of each	7.11	+1.12	−0.14	−0.43
DDEEDEGSDE (acidic)	2.51	−7.16	−8.03	−8.09
KRKRHKRGSK (basic)	12.81	+7.91	+7.00	+6.91
HHHHHH (a His_6 tag)	7.90	+5.46	+0.15	−0.03

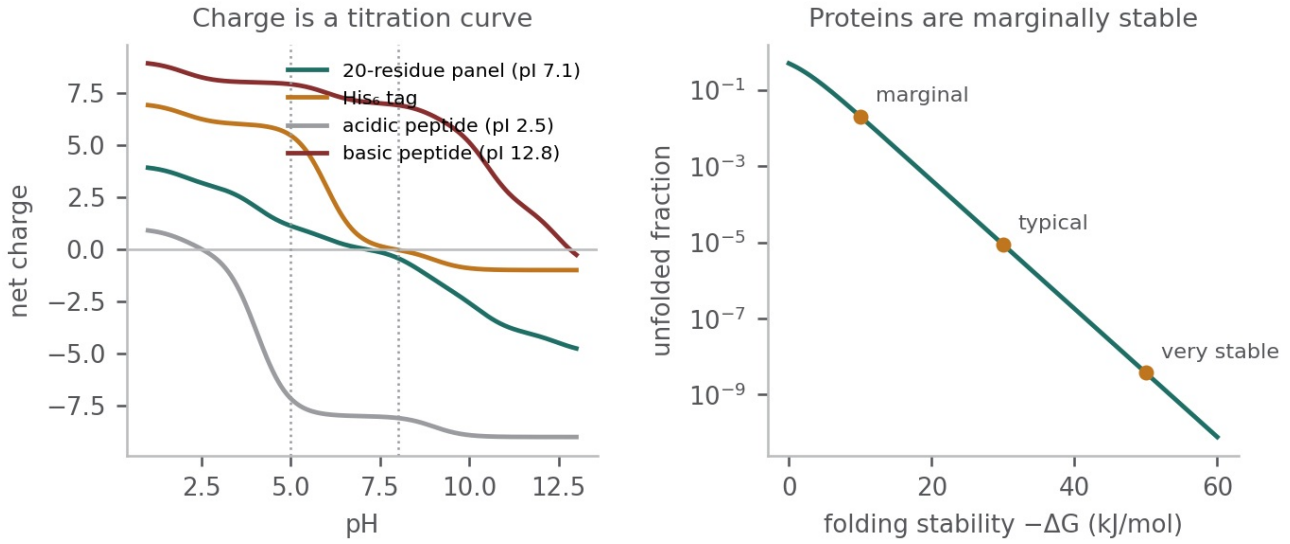


Figure B2.1 Left: net charge against pH for four peptides, with pH 5 and pH 8 marked. Right: fraction of a protein unfolded against its folding stability, with three typical stabilities marked. Computed from the pK_a values in the table and from $K = \exp(-\Delta G/RT)$ at 310 K.

The His_6 row is the one that earns its keep. A histidine is neutral — and therefore able to coordinate a nickel ion on a

resin — when it is deprotonated: at pH 8 that is 99% of residues, at pH 5 only 9%. So a His-tagged protein binds a nickel resin at pH 8 and is eluted by dropping the pH toward 5 (or by competing imidazole), which is the arithmetic behind a purification step that every protein lab runs without computing it. The same table says why a protein at its pI aggregates — net charge zero removes electrostatic repulsion — and why a buffer pH within a unit of the pI is a poor choice for storage.

Structure, in the four levels that matter

Primary structure is the sequence. **Secondary** structure is the local backbone pattern, stabilised by hydrogen bonds between backbone groups: the α-helix has 3.6 residues per turn, a rise of 1.5 Å per residue and a pitch of 5.4 Å, with each carbonyl bonded to the amide four residues further on †; the β-sheet has strands about 3.3 Å per residue joined side by side. A 20-residue helix is therefore 30 Å long, which is the thickness of a membrane's hydrophobic core — the reason transmembrane segments are about twenty residues of hydrophobic helix. **Tertiary** structure is the three-dimensional fold of one chain, driven by burial of hydrophobic side chains and held by hydrogen bonds, salt bridges and sometimes disulfides. **Quaternary** structure is the assembly of several chains, which is what makes cooperativity possible (chapter S3).

The peptide bond is planar, because the C–N bond has partial double-bond character, so the backbone's only freedom is rotation about two angles per residue, φ and ψ. The allowed combinations — the Ramachandran plot — are restricted by steric clashes, which is why helices and sheets exist at all and why proline, whose ring fixes φ, and glycine, which has no side chain to clash, behave differently.

Why proteins fold, and how marginal the result is

The dominant driving force is the **hydrophobic effect**: burying non-polar groups releases ordered water, and the entropy gain of that water pays for the loss of the chain's own conformational entropy. Against it, a folded protein's net stability is small — typically 20 to 60 kJ/mol† — the difference between two enormous numbers. The equilibrium constant is K = exp(−ΔG/RT), with RT = 2.58 kJ/mol at 310 K:

ΔG_{fold}	K = [folded]/[unfolded]	Unfolded fraction
−10 kJ/mol	48	2.0%
−20 kJ/mol	2,340	4.3×10^{-4}
−30 kJ/mol	114,000	8.8×10^{-6}
−50 kJ/mol	2.7×10^{8}	3.8×10^{-9}

Two consequences. A 5 kJ/mol destabilising mutation — one lost hydrogen bond, one hydrophobic residue exposed — **multiplies the unfolded fraction by exp(5/2.58) = 7.0**. A protein at −10 kJ/mol that loses 5 is at 13% unfolded, and an unfolded protein is a substrate for proteases and a seed for aggregation. Hence the engineering fact that most random mutations are destabilising, that a protein carries only a limited stability budget to spend on new function, and that stabilising mutations are valuable because they buy room for activity-improving ones.

The second is Levinthal's argument. A 100-residue chain with three conformations per residue has $3^{100} = 5 \times 10^{47}$ states, and sampling them at 10^{13} per second would take 5×10^{34} seconds, 10^{27} years. Real proteins fold in milliseconds to seconds, so folding cannot be a random search: the energy landscape is a funnel, and each local contact formed lowers the energy and narrows the search. Chaperones (GroEL/GroES, DnaK) do not direct folding; they prevent aggregation of intermediates and give a misfolded chain another attempt, which is why co-expressing them rescues some heterologous proteins (part 8 chapter 30).

Catalysis: how an enzyme lowers a barrier

Transition-state theory gives the rate of a reaction as $k = (k_B T/h)\cdot\exp(-\Delta G^{\ddagger}/RT)$. The prefactor $k_B T/h$ is 6.5×10^{12} s^{-1} at 310 K and is the maximum frequency at which any barrier can be crossed. Inverting it turns a rate into a barrier:

Rate constant	Barrier $\Delta G^{\ddagger}$
10^{-9} s^{-1} (an uncatalysed reaction taking decades)	129 kJ/mol
10^{-3} s^{-1}	94 kJ/mol
10^{3} s^{-1} (a good enzyme)	58 kJ/mol
10^{5} s^{-1}	46 kJ/mol

Each tenfold change in rate is RT ln 10 = **5.93 kJ/mol** of barrier, so a 10^{12} rate enhancement — a typical enzyme — is 71

kJ/mol, less than the energy of a few hydrogen bonds. That is the scale to hold in mind: catalysis is not a large energy effect, it is a small energy effect that is exponentiated. It is also why part 2 chapter 15's point that an enzyme lowers the barrier in both directions equally follows directly — the equilibrium constant depends on the difference between the reactant and product states, which the enzyme does not change.

The mechanisms, which the books treat atom by atom, fall into five groups: **proximity and orientation** — holding the reactants at the right geometry, so the entropic cost is paid at binding; **transition-state stabilisation** — the active site is complementary to the transition state, not the substrate; **general acid–base catalysis** — residues such as His, Asp and Glu move protons; **covalent catalysis** — a nucleophile such as Ser or Cys forms a transient bond; and **metal-ion catalysis** — stabilising charge, as magnesium does for every phosphoryl transfer.

Cofactors, and the vitamins behind them

Cofactor	Vitamin precursor†	What it carries
NAD^+/NADH, $NADP^+$/NADPH	Niacin	Hydride (two electrons); catabolic and anabolic respectively
FAD, FMN	Riboflavin	One or two electrons
Coenzyme A	Pantothenate	Acyl groups
Thiamine pyrophosphate	Thiamine	Aldehyde groups; the two-carbon unit in decarboxylations and transketolase
Pyridoxal phosphate	Vitamin B_6	Amino groups in transamination
Biotin	Biotin	CO_2 in carboxylations
Tetrahydrofolate	Folate	One-carbon units
Cobalamin	Vitamin B_{12}	Radical rearrangements and methyl groups
S-adenosylmethionine	(made from methionine and ATP)	Methyl groups
Haem; iron–sulfur clusters	—	Electron transfer; O_2 chemistry

Enzymes are named by the Enzyme Commission into seven classes†: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases, and, since 2018, translocases. An EC number such as 1.1.1.42 reads as class 1 (oxidoreductase), subclass 1 (acting on a CH–OH group), and so on down to the specific enzyme; it is the key to look an enzyme up in BRENDA.

Charge	`fraction protonated = 1/(1 + 10^(pH − pK_a)); sum over groups; pI where net = 0`
Helix geometry	`3.6 residues/turn, 1.5 Å/residue; 20 residues = 30 Å`
Folding	`K = exp(−ΔG/RT); −30 kJ/mol → 9 × 10⁻⁶ unfolded; a 5 kJ/mol loss → ×7.0`
Transition-state theory	`k = (k_BT/h)exp(−ΔG‡/RT); 5.93 kJ/mol per decade at 310 K`
A typical enzyme	`10¹² enhancement ≈ 71 kJ/mol of barrier`

Common confusions

- **"A stable protein is far from unfolding."** It is typically 20 to 60 kJ/mol from it, which is a few hydrogen bonds.
- **"The pI is where the protein is most stable."** It is where it is least soluble.
- **"Enzymes make reactions thermodynamically favourable."** They change the barrier and not the equilibrium.

Self-check — answer on paper first

1. A His_6-tagged protein will not bind the resin at pH 6.0. Why, and what do you change?

At pH 6 half of each histidine is protonated (the pK_a is 6.0), so only half of the imidazole groups are available to coordinate nickel and the tag's effective binding is far weaker than at pH 8, where 99% are neutral. Raise the pH to 7.5–8, and keep imidazole out of the binding buffer beyond a few millimolar. If the protein itself has a surface cluster of histidines or a low pI that makes it bind nonspecifically, the fix is the other way, in the wash.

2. A variant is 8 kJ/mol less stable than wild type and the wild type is −25 kJ/mol. What happens to the unfolded fraction?

Wild type: $K = \exp(25/2.58) = 1.6 \times 10^4$, unfolded fraction 6×10^{-5}. Variant at −17 kJ/mol: $K = \exp(17/2.58) = 720$, unfolded fraction 1.4×10^{-3}. The unfolded fraction rises 22-fold, from 0.006% to 0.14%. That is still small, which is why the variant may express and assay fine at 30 °C, and why it can fail in a fermenter at 37 °C with proteases present or on a modest rise in temperature, since the stability falls as the temperature approaches the melting point.

3. An enzyme speeds a reaction by 10^9. Convert to a barrier change and say what it implies about the binding energy.

9 decades × 5.93 kJ/mol = 53 kJ/mol. The enzyme must therefore bind the transition state more tightly than the substrate by about that much, which is the quantitative content of "the active site is complementary to the transition state". It also says a substrate-analogue inhibitor designed to resemble the transition state could in principle bind up to 10^9 times more tightly than the substrate does, which is how transition-state analogues achieve their very low dissociation constants.

What to open a book for. Lehninger chapters 3 to 6 for amino acids, structure and enzymes, and Alberts chapter 3 — both are worth reading for their structure figures, which nothing in this module replaces. Open one enzyme of your own pathway in Mol* on the PDB site and rotate it until the active site makes sense, as part 1's appendix suggested; it is the single best use of this week's reading time.

Enzyme kinetics: the derivations and the regimes

Michaelis–Menten is three lines of algebra and four pages of consequences. The algebra is in part 1 chapter 3; this chapter supplies the consequences — what the constants mean, when an assay is measuring them, and what changes with pH and temperature.

The derivation, and the assumption that matters

For $E + S \rightleftharpoons ES \rightarrow E + P$ with rate constants k_1, k_{-1} and k_{cat}, the Briggs–Haldane treatment sets $d[ES]/dt = 0$ and imposes the enzyme balance $E_{total} = [E] + [ES]$. The result is

$$v = V_{max}[S]/(K_M + [S]), \quad V_{max} = k_{cat} \cdot E_{total}, \quad K_M = (k_{-1} + k_{cat})/k_1$$

Michaelis and Menten's original derivation assumed instead that the binding step is at equilibrium, which gives $K_M = k_{-1}/k_1 = K_d$. The two agree only when $k_{cat} \ll k_{-1}$ — when catalysis is slow compared with dissociation. In general **K_M is not a dissociation constant**, and reading it as an affinity is wrong whenever k_{cat} is comparable with k_{-1}, which for fast enzymes it is. This is the first of part 1 chapter 3's "things people get wrong", and the derivation is why.

The three parameters answer three different questions. **k_{cat}**, the turnover number, is the maximum rate per active site — the capacity. **K_M** is the substrate concentration at half-maximal rate — where on the curve the cell is operating. And **k_{cat}/K_M** is the rate constant at low substrate, where $v = (k_{cat}/K_M)[E][S]$, so it is the figure of merit for an enzyme working below saturation, and it has an upper bound near 10^8 to 10^9 $M^{-1}s^{-1}$ set by how fast substrate and enzyme can meet by diffusion†. For scale, typical enzymes have a median k_{cat} around 10 s^{-1} and a median K_M around 100 µM, and so a median k_{cat}/K_M of order 10^5 $M^{-1}s^{-1}$† — three to four orders below the diffusion limit, which is the room protein engineering has to work in.

The regimes, and why the form of the rate law changes

Three regimes follow from the same equation. For $[S] \ll K_M$, $v \approx (V_{max}/K_M)[S]$ — first order in substrate, and linear in enzyme, so doubling the enzyme or the substrate doubles the rate. For $[S] \gg K_M$, $v \approx V_{max}$ — zero order in substrate, so the rate is set by enzyme amount alone and adding substrate does nothing. Near K_M it is mixed. The cell's intracellular concentrations usually sit around or below K_M, which is why part 2 chapter 15 found that substrate supply matters as much as enzyme amount.

The regime also decides how a reaction proceeds in time. Integrating the Michaelis–Menten equation has a closed form involving the Lambert W function, and evaluating it for V_{max} = 1 mM/min and K_M = 1 mM gives the time to convert 10% and 90% of the substrate:

Starting substrate	10% converted	90% converted	First-order estimate	Zero-order estimate
0.1 K_M	0.12 min	2.39 min	2.30 min	0.09 min
1 K_M	0.21 min	3.20 min	2.30 min	0.90 min
10 K_M	1.11 min	11.30 min	2.30 min	9.00 min

Read the columns together. At low substrate the reaction is first order and the 90% time is the first-order estimate, $\ln 10 \cdot K_M/V_{max}$; at high substrate it is close to the zero-order estimate, $0.9[S]_0/V_{max}$; in between neither is right. And the time to reach 10% conversion — the window within which an *initial rate* is valid — varies from 0.12 to 1.1 minutes across the three cases. That is why the standard instruction is to measure initial rates at less than 5–10% conversion and to check linearity of product against time, not to assume it: the window depends on $[S]_0/K_M$, and an assay designed at one substrate concentration is not valid at another.

Inhibition, and what the patterns mean

Type	Binds	Apparent K_M	Apparent V_{max}	Double-reciprocal lines
Competitive	Free enzyme, at the substrate site	Rises: $K_M(1 + [I]/K_i)$	Unchanged	Intersect on the 1/v axis
Uncompetitive	The ES complex only	Falls	Falls	Parallel
Non-competitive	E and ES equally, elsewhere	Unchanged	Falls	Intersect on the 1/[S] axis

Mixed	E and ES with different affinities	Changes	Falls	Intersect off both axes

The practical content is the one part 1 chapter 3 stated: the pattern tells you where the inhibitor acts, and competitive inhibition is overcome by substrate. A further kind needs one line because it appears in production hosts — **substrate inhibition**, $v = V_{max}[S]/(K_M + [S] + [S]^2/K_{si})$, where a second substrate molecule binds the ES complex unproductively. The rate peaks at $[S]^* = \sqrt{(K_M K_{si})}$, and at that point the enzyme reaches only $V_{max}/(1 + 2\sqrt{(K_M/K_{si})})$: for $K_M = 0.1$ and $K_{si} = 10$, the optimum is $[S] = 1$ and the maximum rate is 83% of V_{max}. A fed-batch that overshoots the substrate concentration lowers the rate, which is one more reason feed control matters.

Reversible reactions and the Haldane relation

An enzyme catalyses both directions, and its kinetic parameters in the two directions are not independent. The Haldane relation fixes the equilibrium constant in terms of them: $K_{eq} = (V_f K_{M,P})/(V_r K_{M,S})$. Its use is exactly the one part 2 chapter 16 gave: **take K_{eq} from thermodynamics and let the kinetics respect it**, rather than fitting a K_{eq} that the enzyme cannot violate. The net rate of a reversible enzyme is its forward rate times $(1 - Q/K_{eq})$, which goes to zero as the mass-action ratio reaches equilibrium — the kinetic expression of the point that a near-equilibrium step carries little net flux and cannot be a control point (part 2 chapter 14).

Temperature and pH

A rate constant follows the Arrhenius form $k = A \cdot \exp(-E_a/RT)$. The practical measure is Q_{10}, the factor per ten degrees, which for activation energy E_a between 30 °C and 40 °C is $\exp[E_a/R \cdot 10/(T_1 T_2)]$:

E_a	Q_{10} (30 to 40 °C)
30 kJ/mol	1.46
50 kJ/mol	1.88
80 kJ/mol	2.76

The commonly quoted "doubling per ten degrees" corresponds to about 50 kJ/mol and is a convenient average rather than a rule. The activity of a real enzyme then falls at higher temperature because the protein unfolds — chapter B2's marginal stability — so an activity–temperature curve is the product of a rising Arrhenius term and a falling folded fraction, with an optimum below the melting point. Activity against pH is a bell curve for a similar reason: the catalytic residues have pK_a values, the rate needs one protonated and one deprotonated, and the product of the two ionisation fractions peaks between the two pK_a values. Both curves are measured, not assumed, and the pH and temperature of an assay should be stated beside every K_M and k_{cat} you report.

What an assay actually measures

A kinetic constant is only meaningful with its conditions, and the list of what to fix is short: temperature, pH and buffer, ionic strength, the concentration of every cofactor and metal, enzyme concentration and purity, the substrate range — which must bracket K_M on both sides — and the conversion at which the rate was read. For a two-substrate enzyme the second substrate's concentration is part of the K_M (chapter S3). And in vitro constants differ from in vivo: the cytoplasm is crowded at 200 to 320 g/L of protein (part 1 chapter 1), so a K_M measured in dilute buffer is the starting point for a model, not an input to trust.

Michaelis–Menten	$v = V[S]/(K_M + [S])$; $K_M = (k_{-1} + k_{cat})/k_1 \neq K_d$ in general
Regimes	$[S] \ll K_M$: first order · $[S] \gg K_M$: zero order
Initial-rate window	10% conversion at 0.12 / 0.21 / 1.1 min for $S_0 = 0.1 / 1 / 10\ K_M$
Substrate inhibition	optimum $[S]^* = \sqrt{(K_M K_{si})}$; rate $V/(1 + 2\sqrt{(K_M/K_{si})})$
Haldane	$K_{eq} = V_f K_{M,P}/(V_r K_{M,S})$
Q_{10}	$\exp[E_a/R \times 10/(T_1 T_2)]$; 1.88 at 50 kJ/mol

Common confusions

- **"K_M is affinity."** It is $(k_{-1} + k_{cat})/k_1$; it equals K_d only when catalysis is slow.
- **"I measured the rate over five minutes."** If conversion exceeded 10%, you measured an average over a falling rate.
- **"The enzyme's optimum temperature is a constant."** It depends on the assay time, because unfolding is time-

dependent.

Self-check — answer on paper first

1. An enzyme has k_{cat} = 400 s^{-1} and K_M = 20 µM. Compute k_{cat}/K_M and compare it with the diffusion limit.

400 s^{-1} ÷ 2 × 10^{-5} M = 2 × 10^{7} $M^{-1}s^{-1}$. That is about 2% to 20% of the 10^{8} to 10^{9} diffusion limit, so it is already a very efficient enzyme: there is between about five- and fifty-fold of headroom left before the physical limit, and it is two orders of magnitude above the median 10^{5} — so a campaign to improve this enzyme further would be working near the physical limit, and the control-coefficient test of part 8 chapter 30 should come first.

2. You measure initial rates at [S] = 5 K_M over 3 minutes and see 40% conversion. What is wrong, and what do you change?

Initial-rate validity requires under about 10% conversion, and 40% conversion means most of the three minutes lay outside that window, so the reading is an average over a rate that has fallen substantially — the apparent rate is too low and the K_M derived from several such points will be biased. Shorten the assay or lower the enzyme concentration until conversion is under 10%, confirm product is linear in time, and only then fit the rates. Where the signal is too small at 10% conversion, use a more sensitive detection rather than a longer reaction.

3. A dehydrogenase shows a clear maximum in rate as the substrate is raised from 0.1 to 10 mM. Give two explanations and a test for each.

Substrate inhibition — a second substrate molecule binding the ES complex — which predicts an optimum at $\sqrt{(K_M K_{si})}$ and a smooth decline fitted by the substrate-inhibition equation; and a contaminant or ionic-strength effect from the high substrate concentration, which predicts the same decline when an unrelated salt or the substrate's counter-ion is added at the same level. Test the first by fitting both models and checking whether the substrate-inhibition term is significant; test the second by holding the substrate at its optimum and varying ionic strength with an inert salt.

What to open a book for. Cornish-Bowden, *Fundamentals of Enzyme Kinetics*, the early chapters on steady-state kinetics and inhibition — and, for the engineering view, Shuler, Kargi & DeLisa on enzyme kinetics, which connects directly to reactor design. Part 2 chapter 12 of this module gives the fitting code, and chapter S3 extends the story to two substrates.

Bioenergetics: free energy, ATP, the proton motive force and the ATP yield

The "ATP per glucose" in every textbook is an assumption about a membrane gradient, and the assumption can be computed. This chapter computes it, and in doing so shows that the 32 you were taught is the high end of a range that starts near 20.

Free energy, equilibrium, and the thing that decides direction

The standard free-energy change is related to the equilibrium constant by $\Delta G'^{\circ} = -RT \ln K_{eq}$, and the free-energy change at actual concentrations is $\Delta G' = \Delta G'^{\circ} + RT \ln Q$ (part 1 chapter 4). A reaction runs forward when $\Delta G'$ is negative, regardless of the sign of $\Delta G'^{\circ}$. Two implications to keep from the textbook: free energies are additive for coupled reactions, so an unfavourable step can be driven by a favourable one that shares an intermediate or a carrier; and a pathway's driving force is set by its worst step, which is the max–min idea of part 2 chapter 14.

The energy currency, ATP, is worth computing rather than quoting. With $\Delta G'^{\circ} = -30.5$ kJ/mol† for ATP hydrolysis, and illustrative cytoplasmic concentrations of 3 mM ATP, 0.5 mM ADP and 10 mM phosphate†:

```
ΔG′ = −30.5 + 2.58 · ln[(0.0005)(0.010)/(0.003)] = −30.5 − 16.5 = −47.0 kJ/mol
```

That is the −47 kJ/mol that part 1 chapter 4 quoted from a measurement in growing *E. coli*, and the computation shows why the real value exceeds the standard one: the cell keeps ATP far above equilibrium with ADP and phosphate, so the ratio term contributes another −16.5 kJ/mol. **The cell is not storing energy in ATP; it is maintaining a disequilibrium**, and the disequilibrium is what every ATP-driven reaction taps.

Reduction potentials, and what an electron is worth

For redox reactions the free energy is $\Delta G'^{\circ} = -nF\Delta E'^{\circ}$, with $F = 96.485$ kJ·V^{-1}·mol^{-1} and n the number of electrons. Standard potentials† arranged from most reducing to most oxidising set the direction of electron flow:

Couple	E'° (V)†	Role
NAD^+/NADH	−0.320	Catabolic electron carrier
$NADP^+$/NADPH	−0.324	Anabolic electron carrier — nearly the same potential, held at a very different ratio
Fumarate/succinate	+0.031	Anaerobic terminal acceptor; part of the TCA cycle
Ubiquinone/ubiquinol	+0.045	Membrane electron carrier
Cytochrome c (Fe^{3+}/Fe^{2+})	+0.254	Respiratory chain (in organisms that have it)
O_2/H_2O	+0.816	Terminal acceptor of aerobic respiration

NAD and NADP have almost identical standard potentials, which is the point of part 1 chapter 4's statement that they are "held at deliberately different ratios": since the chemistry cannot distinguish them, the cell does it with kinetics and with enzymes that discriminate the 2′-phosphate. For NADH oxidised by oxygen, $\Delta E'^{\circ} = 0.816 - (-0.320) = 1.136$ V and $\Delta G'^{\circ} = -2 \times 96.485 \times 1.136 =$ **−219 kJ/mol**: roughly four times the energy needed to make one ATP, so the respiratory chain breaks the drop into steps, each coupled to pumping protons.

The proton motive force

Respiration pumps protons out of the cell, building a gradient with two parts: an electrical potential across the membrane, Δψ, interior negative, and a chemical pH difference. The proton motive force is

```
Δp = Δψ − (2.303RT/F)·ΔpH      with ΔpH = pH_in − pH_out;  2.303RT/F = 61.5 mV per pH unit at 310 K
```

For a typical growing *E. coli* — Δψ = −140 mV, interior at pH 7.6 and exterior at 7.0†:

```
Δp = −140 − 61.5 × 0.6 = −177 mV      → energy per mole of protons returning = 96.485 × 0.177 = 17.1 kJ/mol
```

The two parts are in different proportions in different organisms, and acid-tolerant ones hold a large ΔpH with a small Δψ. The total, 177 mV, is what drives ATP synthesis, transport and the flagellar motor. That sharing is what part 1 chapter 4

meant by "when you export a product using a proton antiporter, you are spending the same gradient that would otherwise have made ATP".

The ATP synthase, and a limit you can compute

ATP synthase lets protons flow back through a rotating ring of c-subunits, and the number of protons per ATP is the ring's c-subunit count divided by three (three ATP per full rotation of the catalytic head): c_{10} → 3.33, c_8 → 2.67, c_{12} → 4.0†. Whether a ring is thermodynamically possible is a one-line check — the protons must deliver at least as much energy as the ATP needs:

Ring	H⁺/ATP	Energy delivered at Δp = 177 mV	ATP needs	Efficiency
c_8	2.67	46 kJ	47 kJ	**103% — impossible here**
c_{10} (*E. coli*)	3.33	57 kJ	47 kJ	83%
c_{12}	4.00	68 kJ	47 kJ	69%

The minimum is 47/17.1 = **2.75 protons per ATP** at this driving force, so a c_8 ring cannot make ATP at 177 mV. It works in mitochondria because their proton motive force and phosphorylation potential are both larger† — and the table is a clean demonstration that thermodynamics sets a floor on the stoichiometry, not just a direction on the reaction. It also shows the cost of a large ring: a c_{12} synthase is 69% efficient, spending more gradient per ATP, and trades efficiency for faster rotation or different regulation.

The ATP yield per glucose: an assumption with a range

Glucose is oxidised to six CO_2 with 4 ATP from substrate-level phosphorylation (2 in glycolysis, 2 in the TCA cycle), 10 NADH and 2 $FADH_2$ (chapters B5 and B6). The oxidative part depends on the P/O ratio — ATP made per pair of electrons — which is the protons pumped per electron pair divided by the protons per ATP. Taking $FADH_2$ at two thirds of NADH's yield because it enters the chain after the first proton pump:

```
ATP per glucose = 4 + 10·(P/O)_NADH + 2·(2/3)(P/O)_NADH
```

H⁺ pumped per NADH	H⁺ per ATP	P/O (NADH)	ATP per glucose
6	4.00	1.50	21.0
6	3.33	1.80	24.4
8	4.00	2.00	26.7
8	3.33	2.40	31.2
10	4.00	2.50	32.3
10	3.33	3.00	38.0

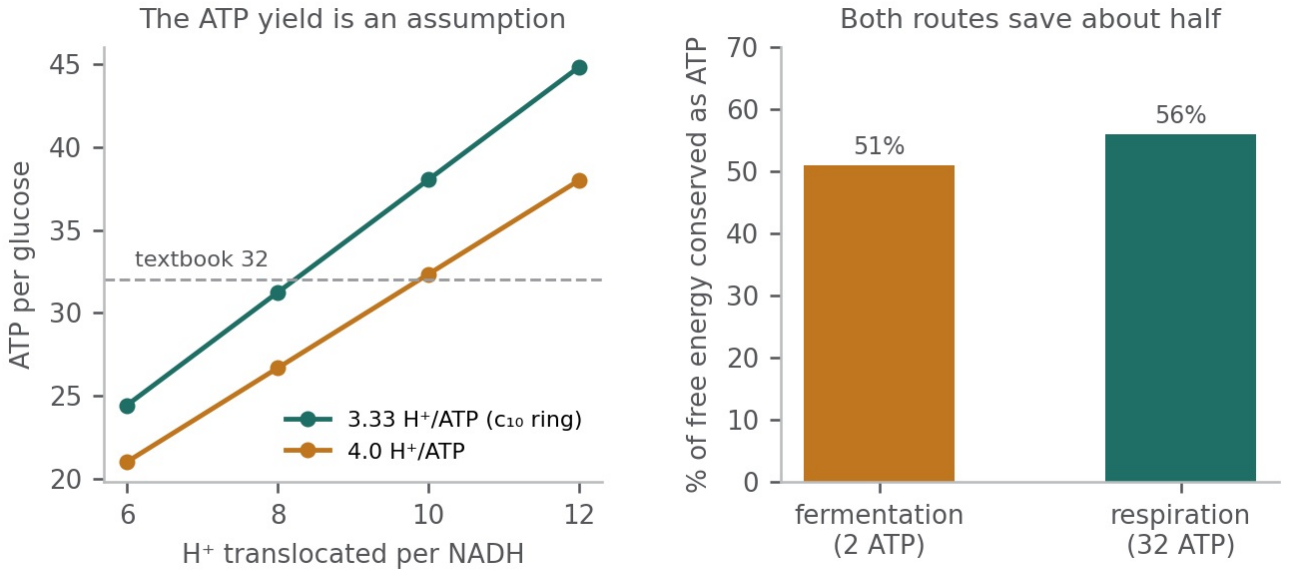


Figure B4.1 Left: ATP per glucose against the number of protons pumped per NADH, for two ATP synthase stoichiometries. Right: the share of the

glucose's free energy conserved as ATP by fermentation (2 ATP, ΔG ≈ −196 kJ/mol to lactate) and by respiration (32 ATP, ΔG ≈ −2,870 kJ/mol), at 50 kJ per ATP. Computed for this module.

The "32" is the row with P/O = 2.5 and 1.5, and it needs 10 protons per NADH and 4 per ATP. *E. coli* has a branched chain with variants that pump different numbers of protons — a proton-pumping NADH dehydrogenase and a non-pumping one, and terminal oxidases that differ† — so its effective yield depends on which are expressed, and that is a regulated choice. The consequence for the module is the one stated in part 1 chapter 4 and used in part 10: **treat the ATP yield as a stated modelling assumption and check how much your conclusion depends on it**. Between 21 and 32 ATP the oxidative branch's contribution changes by 50%, and a yield that scales with it moves with it.

One comparison puts the numbers in perspective. Fermentation to lactate conserves 2 × 50 = 100 kJ of the 196 kJ released, 51%; full respiration conserves 32 × 50 = 1,600 kJ of 2,870, 56%. **Both conserve about half the free energy**, which is a typical efficiency for a biological energy converter; respiration is not more efficient per joule, it is just able to extract far more of the substrate's energy because it removes the electrons rather than discarding them in a reduced end product.

Where the electrons go: NADH against NADPH, again

Because the carriers have the same potential but different cellular roles, converting between them costs something. Transhydrogenases move hydride between the pools; the membrane-bound one, PntAB, is driven by the proton motive force and pumps protons while making NADPH from NADH — which is the "energy cost" of part 2 chapter 19 stated in its mechanism — while the soluble one, UdhA, is the reverse direction and spends nothing. Expressing both strongly builds a futile cycle that dissipates the gradient, which is the trap that chapter named. Part 10's core model used an assumed 0.33 ATP per NADPH for PntAB†, which follows from roughly one proton per hydride at a few protons per ATP.

ATP in the cell	$\Delta G' = -30.5 + RT \ln([ADP][P_i]/[ATP]) = -47.0$ kJ/mol at 3, 0.5, 10 mM
Redox	$\Delta G^{\circ\prime} = -nF\Delta E^{\circ\prime}$; NADH + ½O₂: −219 kJ/mol
Proton motive force	$\Delta p = \Delta\psi - 61.5 \cdot \Delta pH$ (mV at 310 K) = −177 mV → 17.1 kJ/mol H^+
Minimum H^+/ATP	$\lvert\Delta G_{ATP}\rvert/(F\Delta p) = 2.75$; c_{10} ring = 3.33 (83% efficient)
ATP per glucose	4 + 10(P/O) + 1.33(P/O); 21 to 38 over the plausible range, 32 as the textbook case
Energy conserved	fermentation 51%, respiration 56% of the free energy released

Common confusions

- **"ATP stores energy."** It carries it, and the disequilibrium it maintains is the energy.
- **"A glucose yields 38 ATP."** That is the high edge of a range; 30 to 32 is the usual figure, and 21 to 26 is quite possible in a bacterium.
- **"NADH and NADPH are interchangeable because their potentials match."** The cell holds them at different ratios precisely because their chemistry cannot distinguish them.

Self-check — answer on paper first

1. A cell has Δψ = −110 mV and a ΔpH of 0 (neutral). What is the proton motive force, and can a c_{10} ATP synthase make ATP at ΔG = −47 kJ/mol?

Δp = −110 mV, so each mole of protons delivers 96.485 × 0.110 = 10.6 kJ. A c_{10} ring needs 3.33 protons per ATP, delivering 35.4 kJ against the 47 kJ the ATP requires — efficiency 133%, which is impossible, so the synthase would run backwards, hydrolysing ATP to pump protons. The minimum Δp that could drive it is 47/(3.33 × 96.485) = 146 mV, which is a useful number: it is why uncouplers, which dissipate the gradient, stop ATP synthesis even though the chain keeps running.

2. Your model assumes 32 ATP per glucose. Your organism pumps 6 H^+ per NADH and its synthase has a c_{12} ring. What do you correct, and by how much?

H^+ per ATP is 4.0, so $(P/O)_{NADH}$ = 6/4 = 1.5 and ATP per glucose = 4 + 10 × 1.5 + 1.33 × 1.5 = 21.0, a third lower than 32. Every quantity downstream that depends on ATP — the maintenance requirement in units of glucose, the growth yield, the oxygen demand — changes accordingly; a yield calculation that was ATP-limited falls by up to a third, and one that was carbon-limited barely moves, which is how to tell which regime you are in.

3. Why does fermentation conserve about the same fraction of free energy as respiration despite making 16 times less ATP?

Because it releases far less energy to begin with. Glucose to two lactate releases about 196 kJ/mol and conserves 100 kJ

of it, 51%; complete oxidation releases 2,870 kJ and conserves 1,600, 56%. The ATP number differs by a factor of 16 because the fraction of the substrate's energy that is released at all is 7%, the rest being left in lactate. The efficiency per joule is similar; the difference is how much of the substrate you are willing to leave unoxidised, which is the carbon-and-electron bookkeeping of part 2 chapter 18.

What to open a book for. Lehninger chapter 13 and the chapters on oxidative phosphorylation, read twice; Nicholls & Ferguson, *Bioenergetics*, for the proton motive force treated properly. Doran's energy-balance chapter does the accounting in engineering notation, and it is where problem-solving practice is.

Chapter B5 · tracker week 5

Glycolysis and the pentose phosphate pathway

Know the three irreversible steps, the net stoichiometry, and where the branch points sit. The rest of the pathway you can look up; those are what you will be asked to engineer around.

The pathway, by what each step does

Glycolysis converts one glucose (C_6) into two pyruvate (C_3) in ten steps, in two phases. The **preparatory phase** invests two ATP: glucose is phosphorylated (in *E. coli*, by the phosphotransferase system, which spends a phosphoenolpyruvate instead of an ATP), isomerised to fructose-6-phosphate, phosphorylated again by phosphofructokinase, and cleaved by aldolase into two triose phosphates that interconvert. The **payoff phase** oxidises glyceraldehyde-3-phosphate with NAD^+ and phosphate to 1,3-bisphosphoglycerate, then to 3-phosphoglycerate (making ATP), to 2-phosphoglycerate, to phosphoenolpyruvate, and finally to pyruvate (making ATP). The net equation is the one in part 1 chapter 5:

$$\text{glucose} + 2\ NAD^+ + 2\ ADP + 2\ P_i \rightarrow 2\ \text{pyruvate} + 2\ NADH + 2\ ATP + 2\ H^+ + 2\ H_2O$$

Step	Enzyme	Character
1	Glucose phosphorylation (PTS in *E. coli*; hexokinase in yeast)	**Irreversible** — commits glucose; regulated
2	Phosphoglucose isomerase	Near equilibrium
3	Phosphofructokinase	**Irreversible** — the main control point; allosterically regulated
4	Aldolase	Near equilibrium
5	Triosephosphate isomerase	Near equilibrium
6–7	GAPDH; phosphoglycerate kinase	Near equilibrium; make NADH and the first ATP
8–9	Phosphoglycerate mutase; enolase	Near equilibrium
10	Pyruvate kinase	**Irreversible** — regulated; makes the second ATP

The three irreversible steps are the ones that sit far from equilibrium (part 2 chapter 14), they are the points at which the pathway is regulated, and in gluconeogenesis they are bypassed by different enzymes — fructose-1,6-bisphosphatase, and phosphoenolpyruvate synthetase or carboxykinase — which is why a cell can run glycolysis forwards and backwards without a futile cycle, provided both sets are not on at once. The seven near-equilibrium steps are the ones that cannot be control points.

Alternatives that exist for a reason

The Entner–Doudoroff pathway converts glucose-6-phosphate to 6-phosphogluconate, dehydrates it to 2-keto-3-deoxy-6-phosphogluconate and cleaves that into pyruvate and glyceraldehyde-3-phosphate. Its net yield is **1 ATP, 1 NADH and 1 NADPH** per glucose† — half the ATP, but with NADPH generated for free — and, as part 1 chapter 5 reported, the pathway needs several-fold less enzyme protein for the same flux. That is the trade: ED is the better choice when protein is expensive and ATP is not, which is why it is common in organisms growing aerobically on rich carbon. Phosphoketolase pathways cleave a pentose or hexose directly to acetyl phosphate and a triose, avoiding the decarboxylation at pyruvate dehydrogenase — which is why they are engineered in when acetyl-CoA yield is the target.

The pentose phosphate pathway, as a bookkeeping exercise

The oxidative branch converts glucose-6-phosphate to ribulose-5-phosphate in two oxidations and one decarboxylation, **making 2 NADPH and releasing 1 CO_2**. The non-oxidative branch — transketolase and transaldolase — shuffles carbon between sugar phosphates of three to seven carbons and is reversible, so it can run in either direction depending on demand. Its net stoichiometry, when the cell needs only NADPH, is the cycle

$$6\ \text{glucose-6-P} + 12\ NADP^+ \rightarrow 5\ \text{glucose-6-P} + 6\ CO_2 + 12\ NADPH + P_i$$

which is the clean form of "every glucose that goes round loses all six carbons as CO_2 and yields twelve NADPH" — in practice the cycle recycles carbon back into glycolysis, so the carbon is not lost on the first pass, but the decarboxylation is. When the cell needs ribose-5-phosphate for nucleotides, the non-oxidative branch runs from fructose-6-phosphate and glyceraldehyde-3-phosphate backwards and the oxidative branch is bypassed; when it needs erythrose-4-phosphate for

aromatics, a transketolase makes it (part 1 chapter 7).

How much NADPH, and so how much carbon

The calculation worth carrying out here is the one that explains why the oxidative branch matters, using numbers computed in part 10. The core model's biomass reaction demands 13.03 mmol of NADPH per gram dry weight†. At the growth rate of 0.754 h^{-1} that model reaches, the demand is 13.03 × 0.754 = **9.8 mmol NADPH per gram per hour**. If the oxidative pentose phosphate pathway were the *only* source, at 2 NADPH per glucose it would have to carry 4.9 mmol of glucose per gram per hour — **49% of the 10 mmol uptake**. That is an upper bound, because real cells make NADPH elsewhere too: isocitrate dehydrogenase in the TCA cycle, NADP-dependent malic enzyme, and the transhydrogenase of chapter B4. The measured oxidative-branch share in *E. coli* is a fraction of the bound†, and the bound is what makes the cofactor accounting of part 2 chapter 19 a real constraint rather than a bookkeeping detail: a product needing four NADPH per molecule on top of the biomass demand competes for a supply that is already half-committed.

The carbon cost follows directly. Each NADPH from the oxidative branch costs half a carbon, so the biomass demand alone, 13.03 mmol NADPH per gram, would cost 6.5 mmol of carbon per gram if met entirely there. Against the roughly 40 mmol of carbon in a gram of biomass† (chapter B7), that is 16% — the size of the penalty that makes the pentose phosphate route "the NADPH tap" and why deleting phosphoglucose isomerase, which forces all glucose through it, cost a third of the growth rate in part 10's deletion scan.

Regulation, and what to engineer around

Phosphofructokinase in *E. coli* is activated by ADP and inhibited by phosphoenolpyruvate† — the second is feedback from the pathway's own product, which keeps the pool of PEP from collapsing. The phosphotransferase system couples glucose uptake to the PEP/pyruvate ratio and is the entry point of catabolite repression (chapter B9). Glucose-6-phosphate dehydrogenase, the first oxidative step, is inhibited by NADPH, so the cell's NADPH demand regulates its own supply: when NADPH is consumed the enzyme speeds up. That coupling is why a strain with a high NADPH demand pulls carbon through the oxidative branch without any genetic change, as part 1 chapter 5 noted.

Glycolysis, net	glucose + 2 NAD^+ + 2 ADP + 2 P_i → 2 pyruvate + 2 NADH + 2 ATP
Irreversible steps	glucose phosphorylation, phosphofructokinase, pyruvate kinase
Entner–Doudoroff	1 ATP, 1 NADH, 1 NADPH per glucose; several-fold less enzyme protein
Oxidative PPP	glucose-6-P → ribulose-5-P: 2 NADPH + 1 CO_2
NADPH demand	13.03 mmol/gDW × 0.754 h^{-1} = 9.8 mmol/gDW/h; oxidative PPP alone would need 49% of uptake
Carbon price of NADPH	0.5 C per NADPH; 6.5 mmol C/gDW if all from the oxidative branch

Common confusions

- **"Glycolysis makes a lot of ATP."** Two net per glucose; respiration makes most.
- **"The PPP is a nucleotide pathway."** Its oxidative branch is the main NADPH source, and the non-oxidative branch supplies pentoses and erythrose-4-phosphate.
- **"Every enzyme of glycolysis is a candidate for overexpression."** Seven of the ten are near equilibrium and carry little control.

Self-check — answer on paper first

1. Why does the PTS-based uptake in *E. coli* give the same net ATP as hexokinase, and why does it matter anyway?

Hexokinase spends one ATP phosphorylating glucose, giving net 2 ATP overall from the 4 made minus the 2 spent in the preparatory phase. The PTS spends one PEP instead; a PEP would otherwise have made an ATP at pyruvate kinase, so the opportunity cost is the same one ATP and the net is again 2. What differs is *which pool the cost comes out of*: PEP is also the precursor for aromatic amino acids and for the anaplerotic carboxylation, so PTS-based uptake competes directly with those, which is part 1 chapter 7's point about aromatic products impairing glucose uptake.

2. A product needs 4 NADPH per molecule. Estimate how much glucose passes through the oxidative branch per molecule if no other NADPH source were available, and say what that does to the carbon yield.

At 2 NADPH per glucose through the oxidative branch, four NADPH need two glucose. Each loses a carbon to CO_2 on the first pass, so 2 of the 12 carbons are decarboxylated before the remainder returns to glycolysis and is used. The yield ceiling falls from the naive carbon limit — part 1 chapter 5 gave 0.81 mol/mol for lysine instead of 1.0 — and the correct answer depends on how much NADPH can be sourced elsewhere, which is why a transhydrogenase or a cofactor swap is

attractive.

3. Why is an irreversible step a good place to look for control, and a near-equilibrium step not?

At a near-equilibrium step, forward and reverse fluxes are both large and nearly equal, and the net flux is their small difference, set by how far the mass-action ratio sits from K_{eq}; raising the enzyme raises both fluxes and leaves the net almost unchanged. An irreversible step has a negligible reverse flux, so the net flux is set by the enzyme's own activity and its regulation — which is also why that is where the cell puts its regulators, and where an intervention can change flux.

What to open a book for. Lehninger chapters 14 and 15: redraw the pathway from memory on a blank sheet afterwards, which is the one piece of memorisation in the curriculum that pays. For ED and the alternative pathways, Flamholz et al., PNAS 110:10039 (2013). Stephanopoulos chapters 2 and 3 for the stoichiometric accounting that part 2 chapter 18 applies.

The TCA cycle, respiration and overflow

The cycle has two jobs — oxidise acetyl-CoA to feed the chain, and supply precursors for biosynthesis — and they pull against each other. Overflow metabolism is what the cell does when the supply of carbon outruns the cell's capacity to respire it.

The cycle, as carbon bookkeeping

Pyruvate is oxidised to acetyl-CoA by pyruvate dehydrogenase, releasing one CO_2 and one NADH. Acetyl-CoA (C_2) condenses with oxaloacetate (C_4) to form citrate (C_6), which is isomerised to isocitrate and then oxidatively decarboxylated to α-ketoglutarate (C_5), losing a CO_2 and making an NADH. α-Ketoglutarate is oxidatively decarboxylated again to succinyl-CoA (C_4), losing a second CO_2 and making another NADH. The remaining steps convert succinyl-CoA to succinate (making one GTP or ATP by substrate-level phosphorylation), then to fumarate (making $FADH_2$, or reducing quinone), to malate, and back to oxaloacetate (making a third NADH).

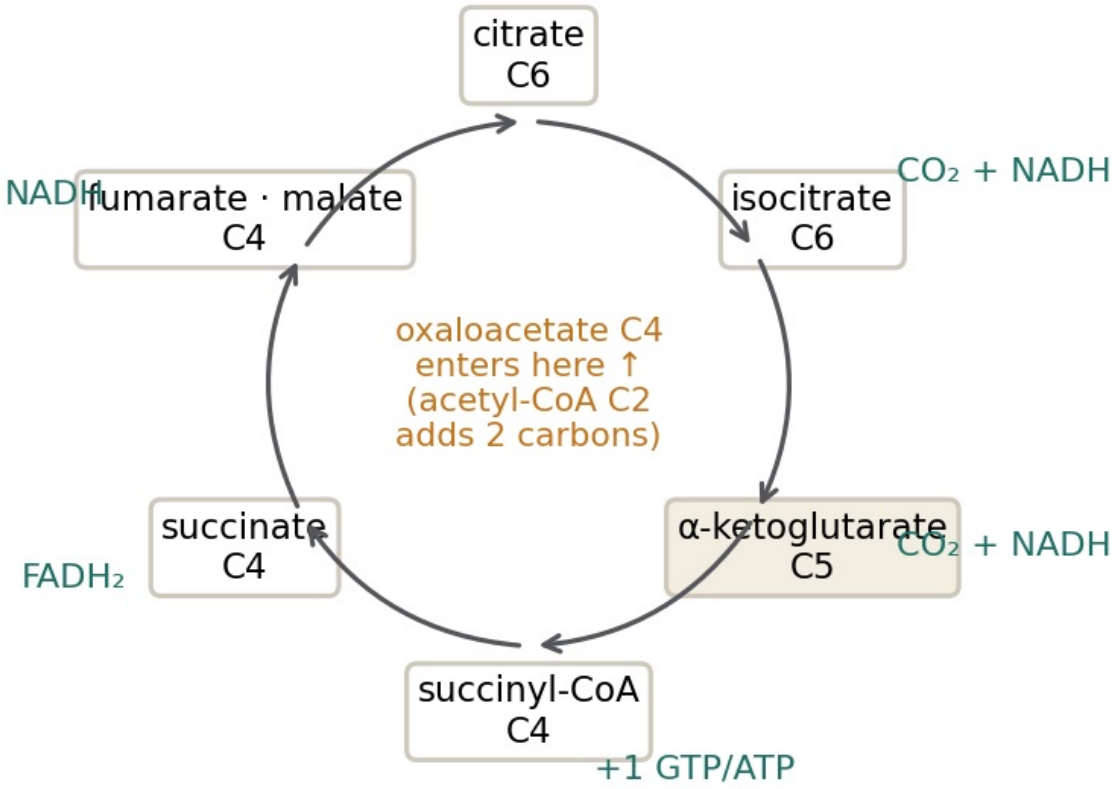


Figure B6.1 The TCA cycle as a carbon ledger: carbon counts at each intermediate, the two CO_2 released, and where reducing equivalents form. α-Ketoglutarate, shaded, is the nitrogen entry point. Drawn for this module.

Per turn of the cycle, the net is: acetyl-CoA + 3 NAD^+ + FAD + GDP + P_i + 2 H_2O → 2 CO_2 + 3 NADH + $FADH_2$ + GTP + CoA. Adding pyruvate dehydrogenase and glycolysis gives the totals that chapter B4 used: glucose → 6 CO_2 + **10 NADH + 2 $FADH_2$ + 4 ATP** (2 NADH and 2 ATP from glycolysis, 2 NADH from pyruvate dehydrogenase, 6 NADH, 2 $FADH_2$ and 2 ATP from two turns of the cycle).

A fact worth knowing because it surprises: the two carbons released as CO_2 in a turn are **not** the two that entered as acetyl-CoA in that turn. Citrate is a symmetric molecule, but aconitase treats its two arms differently, so it is the carbons that came from oxaloacetate that are lost; the acetyl-derived carbons are carried into succinate, which is symmetric and scrambles them, and are released only in later turns. It matters in practice for labelling experiments (part 11 chapter 43): a label on acetyl-CoA reaches CO_2 with a delay, not immediately.

The cycle as a supplier of precursors

Withdrawing an intermediate for biosynthesis drains the cycle: α-ketoglutarate to glutamate and the glutamate family; oxaloacetate to aspartate and the aspartate family; succinyl-CoA to lysine and methionine precursors and to tetrapyrrole synthesis; citrate to fatty acids in organisms that export it. To keep the cycle turning, **anaplerotic reactions** replenish oxaloacetate: PEP carboxylase (PEP + CO_2 → oxaloacetate, costing no ATP but consuming the PEP), pyruvate carboxylase in organisms that have it — *Corynebacterium* and *Bacillus*, though not *E. coli*† — which costs an ATP, and the glyoxylate shunt when growing on two-carbon substrates, which bypasses both decarboxylations (isocitrate lyase and malate synthase) and converts two acetyl-CoA into one four-carbon unit.

The engineering fact is part 1 chapter 6's: the cycle is a recycle loop with a purge, and the purge rate sets the required make-up. A product drawn from α-ketoglutarate or oxaloacetate competes with the cycle for carbon and with PEP for its anaplerotic route, and the yield is limited by the anaplerotic capacity and its ATP cost as much as by the pathway installed.

One note on a correction. Part 1 chapter 6 said succinyl-CoA feeds haem synthesis. That is true for yeast, animals and some bacteria, which make δ-aminolevulinate from succinyl-CoA and glycine (the Shemin route); *E. coli* and most bacteria and plants instead make it from glutamate through the C5 pathway, via glutamyl-tRNA†. Which one your host uses decides which precursor a haem-protein product competes for, which is a real consideration for the leghemoglobin of part 6 chapter 55.

Anaerobic operation: the split cycle

Without oxygen, *E. coli* does not run the cycle as a loop; it runs it as two branches. An oxidative branch carries oxaloacetate through citrate to α-ketoglutarate to supply glutamate-family precursors, and a reductive branch carries oxaloacetate through malate and fumarate to succinate, regenerating NAD^+ by using fumarate as an electron acceptor. That is the route the growth-coupled succinate design of part 11 chapter 39 exploited, and it is why succinate is an anaerobic product: the reductive branch consumes the NADH that anaerobic glycolysis cannot otherwise dispose of.

Respiration, oxygen, and the respiratory quotient

Complete oxidation of one glucose consumes six O_2 and releases six CO_2, so the **respiratory quotient**, RQ = CO_2 produced ÷ O_2 consumed, is 1.0 for a pure glucose oxidation. It is below 1 when the substrate is more reduced than glucose (fatty acids, 0.7), or when the product is reduced; and above 1 when part of the substrate is fermented, since fermentation releases CO_2 without consuming oxygen. RQ is a cheap, online physiological diagnostic from off-gas analysis, and a drift above 1 in a fed-batch is the first sign of overflow. In part 10's core model, 9.5 mmol of O_2 is consumed per 10 mmol of glucose while making 0.754 gDW per hour, which is **12.6 mmol of O_2 per gram of biomass formed, or 0.40 g O_2 per g** — the figure that sized the k_La requirement of part 11 chapter 44.

Overflow metabolism

When glucose uptake exceeds what respiration can handle, *E. coli* excretes acetate even in the presence of oxygen; *S. cerevisiae* makes ethanol (the Crabtree effect); mammalian cells make lactate (the Warburg effect). The older explanation was limited respiratory capacity. The modern one, from part 1 chapter 6, is allocation: respiration costs more protein per ATP than fermentation does, so above a threshold growth rate near 0.76 h^{-1}† the cell gains more ATP per gram of protein by fermenting part of the carbon. Part 11 chapter 39 reproduced the behaviour on the core model by adding a proteome budget with illustrative costs, and the onset of by-products as uptake rose was the result.

The three consequences stand: carbon leaves as by-product and yield falls; acetate inhibits growth, more at low pH where the weak acid enters the cell (part 2 chapter 14); and the operational answer is a fed-batch that holds the specific uptake rate below the threshold. Deleting the by-product pathway moves the problem, as part 1 chapter 6's worked example showed, rather than removing it.

Per turn	acetyl-CoA → 2 CO_2 + 3 NADH + $FADH_2$ + GTP
Per glucose	6 CO_2 + 10 NADH + 2 $FADH_2$ + 4 ATP (substrate-level)
Anaplerosis	PEP + CO_2 → oxaloacetate (no ATP); pyruvate + CO_2 + ATP → oxaloacetate
RQ	CO_2/O_2 = 1.0 for glucose oxidation; above 1 signals fermentation
Oxygen per biomass	12.6 mmol O_2 per gram formed in the part 10 model (0.40 g/g)
Overflow threshold	~0.76 h^{-1} in *E. coli*†; set by specific uptake rate

Common confusions

- **"The TCA cycle makes ATP."** Directly it makes one per turn; its main output is reducing equivalents and precursors.

- **"The carbons that enter as acetyl-CoA leave as CO_2 in the same turn."** They are released in later turns.
- **"Acetate means something went wrong."** It is the predicted response to uptake outrunning respiratory capacity.

Self-check — answer on paper first

1. A fed-batch shows an RQ of 1.15 and 2 g/L of acetate. What do you conclude and what do you change?

An RQ above 1 means CO_2 is being produced without a matching oxygen consumption, which is what fermentation does; together with acetate, it says the cells are running overflow. The change is the feed: lower the substrate feed rate so the specific uptake rate falls below the threshold, and the RQ should return to about 1.0 and the acetate be re-consumed. It is a quantitative diagnosis you can make online, without a sample, which is why the RQ is worth logging.

2. Why does withdrawing α-ketoglutarate for glutamate production require an anaplerotic reaction?

Because the cycle is closed: every α-ketoglutarate drawn off is a four-carbon unit that does not return as oxaloacetate to condense with the next acetyl-CoA, so the pool of cycle intermediates falls and the cycle slows. Replenishing oxaloacetate — by PEP carboxylase or pyruvate carboxylase — restores it at the cost of the carboxylation (and an ATP for the carboxylase), and that carboxylation, not the glutamate pathway itself, often sets the product yield.

3. A yeast and an *E. coli* both overflow. What differs in the mechanism and in the by-product?

The yeast makes ethanol, because it has pyruvate decarboxylase and alcohol dehydrogenase and is Crabtree-positive, so overflow begins at much lower glucose concentration; the bacterium makes acetate through phosphotransacetylase and acetate kinase, gaining an extra ATP as it does so. In both the driver is that fermentation yields more ATP per unit of protein invested than respiration when growth is fast; the by-product differs, and so does its toxicity and its value, which is why part 6 chapter 58 treated the Crabtree status of a yeast as a deciding host property.

What to open a book for. Lehninger chapters 16 to 19; Alberts is weaker here. For overflow metabolism and proteome allocation, Basan et al., Nature 528:99 (2015), plus a recent review since the quantitative account is recent. Part 11 chapter 39 of this module reproduces the effect on a model.

Chapter B7 · tracker week 7

Biosynthesis: what a gram of cells costs

Twelve precursors build everything. The useful number is not how many reactions that takes but how much glucose a gram of cells consumes — about 2.4 g — and where half of the carbon goes.

The cost of a gram, computed

Part 10's core model grows at 0.754 h^{-1} on 10 mmol of glucose per gram per hour, which means it uses 10/0.754 = **13.3 mmol of glucose to make one gram of biomass — 2.39 g of glucose**, a yield of 0.42 g/g. That figure comes out of the stoichiometry rather than being measured, and it is close to measured aerobic yields. The decomposition is the instructive part. A gram of cells contains about 48% carbon†, which is 40 mmol; as glucose that is 6.7 mmol. So **50% of the glucose carbon ends up in biomass and the other 50% leaves as CO_2**, for two reasons: carbon is lost during biosynthesis wherever a decarboxylation occurs (pyruvate dehydrogenase, the TCA cycle, the oxidative pentose phosphate pathway), and carbon is burned to make the ATP and NADPH that biosynthesis consumes.

That 50% is the baseline against which every product's yield is judged. A strain that makes a product from the same glucose competes with biomass for the same carbon, ATP and reducing power, which is the three-way split of part 1 chapter 1 — and it is why a growing cell can never convert more than about half its substrate carbon to cells, and why a product pathway must be cheaper than the biomass it displaces.

The twelve precursors, and the families they define

All biosynthesis begins from twelve compounds of central metabolism (part 1 chapter 7, figure 7.1): glucose-6-P, fructose-6-P, ribose-5-P, erythrose-4-P, glyceraldehyde-3-P, 3-phosphoglycerate, PEP, pyruvate, acetyl-CoA, α-ketoglutarate, succinyl-CoA and oxaloacetate. The core model's biomass reaction lists the demand in mmol per gram dry weight: acetyl-CoA is the largest at 3.7, pyruvate 2.8, oxaloacetate 1.8, α-ketoglutarate (with glutamate) the nitrogen entry, 3-phosphoglycerate 1.5, and so on†. In part 10's rounded version of that reaction the demands sum to 70.9 mmol of carbon per gram, more than the 40 in the biomass, because the extra carbon is lost on the way to the monomers — and that difference is the biosynthetic decarboxylation loss.

Family	Precursor	Amino acids	Notes
Glutamate	α-ketoglutarate	Glu, Gln, Pro, Arg	The nitrogen entry point; ammonium is assimilated here
Aspartate	Oxaloacetate	Asp, Asn, Lys, Thr, Met, Ile	Feedback-inhibited aspartokinases — the lysine and threonine engineering target
Aromatic	PEP + erythrose-4-P	Phe, Tyr, Trp	The shikimate pathway; competes with the PTS for PEP
Pyruvate	Pyruvate	Ala, Val, Leu	Valine and leucine feed the 2-ketoacid alcohol route of part 5 chapter 55
Serine	3-phosphoglycerate	Ser, Gly, Cys	Gly and Ser are the main one-carbon donors
Histidine	Ribose-5-P (via PRPP)	His	Shares PRPP with nucleotide synthesis

The costs differ per amino acid, and the published estimates of the ATP-equivalents spent range from roughly a dozen for the cheapest to several tens for the most expensive†, depending on whether the accounting includes the opportunity cost of the carbon and reducing power used. The engineering use is relative: the aromatic and branched-chain amino acids are expensive, glycine and alanine are cheap, and a protein product enriched in the expensive ones is correspondingly costlier to make.

Nitrogen and sulfur

Nitrogen enters through glutamate and glutamine. Glutamate dehydrogenase makes glutamate from α-ketoglutarate, ammonium and NADPH at a cost of no ATP but with low affinity for ammonium, so it works when ammonium is plentiful. When it is scarce the cell uses glutamine synthetase, which spends one ATP to attach ammonium to glutamate, then glutamate synthase (GOGAT) to transfer the amide nitrogen back to α-ketoglutarate; the pair has a much higher affinity and costs an ATP plus an NADPH per nitrogen. Every other amino group in the cell is then moved by transamination, using pyridoxal phosphate. Nitrogen assimilation is therefore tied to the TCA cycle through α-ketoglutarate, and a nitrogen-rich product draws on that pool (part 1 chapter 7).

Sulfur enters as sulfate and must be reduced to sulfide — an eight-electron reduction — before it can be put into cysteine. The cost is two ATP to activate sulfate to PAPS and four NADPH for the reductions†. That is an NADPH price of the same order as a fatty-acid elongation and is routinely omitted from back-of-envelope balances for sulfur-containing products such as glutathione or methionine. Methyl groups come from S-adenosylmethionine, made from methionine

and ATP, and each methylation leaves S-adenosylhomocysteine that must be recycled.

Lipids, nucleotides and the cell envelope

Fatty acids are built from acetyl-CoA. Acetyl-CoA carboxylase, a biotin enzyme, makes malonyl-CoA at a cost of one ATP; in bacteria a dissociated synthase (type II) then adds two carbons per cycle using two reductions — a ketoacyl reduction that uses NADPH and an enoyl reduction that *E. coli*'s FabI does with NADH †. A palmitate (C_{16}) therefore costs 8 acetyl-CoA, 7 ATP and 14 reducing equivalents, split about 7 NADPH and 7 NADH in *E. coli*; yeast's fatty acid synthase takes both from NADPH, giving the 14 NADPH of part 6 chapter 58, and adds the ATP-citrate lyase step so pays 15 ATP. **Phospholipids** add glycerol-3-phosphate and a headgroup, with CDP-activated intermediates.

Nucleotides start from phosphoribosyl pyrophosphate, made from ribose-5-phosphate and ATP. Purines are assembled on a ribose scaffold through inosine monophosphate, a ten-step pathway using glycine, glutamine, aspartate, one-carbon units from folate and a good deal of ATP; pyrimidines build the ring first and attach the ribose after, through orotate. Ribonucleotide reductase makes deoxyribonucleotides from ribonucleotides, and thymidylate synthase, which needs methylene-tetrahydrofolate, makes the thymidine — which is why folate antagonists are antimicrobials.

The **envelope** is the other large sink: peptidoglycan, whose synthesis is the target of the β-lactams and which is a substantial part of a Gram-positive cell; and lipopolysaccharide in Gram-negatives, which is the endotoxin that part 6 chapter 59 listed as an advantage of Gram-positive hosts for injectables. For product engineering the point is that the envelope is a fixed cost of growth, and a host with a smaller or simpler one has less to build.

Cost of a gram	13.3 mmol glucose = 2.39 g per gram of cells; yield 0.42 g/g
Where the carbon goes	48% C → 40 mmol → 6.7 mmol glucose-equivalent: 50% to biomass, 50% to CO_2
Precursor demand	70.9 mmol C/gDW in the core biomass reaction; the surplus over 40 is decarboxylation
Nitrogen	GDH: no ATP, low affinity · GS-GOGAT: 1 ATP + 1 NADPH per N, high affinity
Sulfate to sulfide	2 ATP + 4 NADPH (8 electrons)
Palmitate	8 acetyl-CoA + 7 ATP + 14 reducing equivalents (7 NADPH + 7 NADH in *E. coli*†; 14 NADPH in yeast)

Common confusions

- **"Biomass is cheap."** A gram costs 2.4 g of glucose and half the carbon is respired.
- **"Succinyl-CoA feeds haem."** In yeast and animals; *E. coli* uses glutamate through the C5 route.
- **"Ammonium assimilation is free."** It depends on the route; the high-affinity one costs an ATP and an NADPH per nitrogen.

Self-check — answer on paper first

1. Your strain makes a product at a carbon yield of 60%. Is that better or worse than making biomass, and why does the comparison matter?

Better: biomass converts only about 50% of the glucose carbon, so a product at 60% uses carbon more efficiently than growth does. The comparison matters because it is the three-way split of part 1 chapter 1 — carbon goes to cells, product or CO_2 — and a growth-coupled design (part 11 chapter 39) works precisely because product formation can be arranged to be the better use of carbon than growth. A product whose carbon yield is below 50% is competing at a disadvantage and needs its value to compensate.

2. A palmitate-derived product needs 14 NADPH per molecule. How many glucose does the reductant cost if it all comes from the oxidative pentose phosphate pathway, and what is the carbon penalty?

14/2 = 7 glucose pass through the oxidative branch (each gives 2 NADPH). In part 6 chapter 58 the corresponding figure is 1.17 glucose when the pentose phosphate pathway is fully cycled, because there the six carbons recycle through the non-oxidative branch giving 12 NADPH per glucose — the answer depends on which accounting you use, 2 per pass or 12 per cycle, and the accounting is the assumption. At 2 per pass the carbon penalty is 0.5 C per NADPH, 7 carbons for 14 NADPH.

3. Why is sulfate reduction usually omitted from yield estimates, and when does that matter?

Because the demand for sulfur in cell mass is small — a few per cent of the amino acids carry sulfur — so the NADPH cost is a minor part of the biomass budget. It matters when the product itself is sulfur-rich: glutathione, methionine, biotin, coenzyme A derivatives or iron–sulfur-dependent products. For glutathione, which has one cysteine per molecule, each molecule carries a sulfate-reduction cost of four NADPH and two ATP that a balance restricted to the glutamate and glycine steps will miss.

What to open a book for. Lehninger chapters 20 to 22 for the pathways; Stephanopoulos chapters 2 and 3 for the twelve-precursor yield arithmetic, which is the book you will keep using. The KEGG and MetaCyc pathway maps are the right reference for every enzyme step this chapter deliberately skips — read the map, then draw your own.

Molecular biology: replication, transcription, translation and their regulation

A gene's output is a product of a promoter, a ribosome binding site and two half-lives. Put numbers on each and you find that "strongly expressed" means about one per cent of the proteome — and that a consensus sequence alone predicts nothing.

Replication, in four facts that matter

Bacterial replication starts at one origin, *oriC*, where the initiator protein DnaA binds and opens the duplex; the helicase DnaB loads, and DNA polymerase III extends from RNA primers, continuously on the leading strand and in Okazaki fragments of one to two kilobases† on the lagging strand. Two forks proceed in opposite directions and meet at the terminus, where Tus protein blocks them. Four facts carry into engineering. The two forks together copy the genome in about 39 minutes (chapter B1). Initiation is regulated by cell mass, so a faster-growing cell initiates more often per generation and carries more than one set of origin-proximal genes (part 9 chapter 32). A plasmid has its own origin and its own copy-number control, independent of the chromosome (chapter B10). And replication errors are the main source of mutation, repaired by proofreading and mismatch repair at a residual rate near 10^{-10} per base per generation†.

Transcription: the promoter as a quantitative part

RNA polymerase is a core of five subunits ($\alpha_2\beta\beta'\omega$) that cannot start transcription alone. A sigma factor binds the core and recognises the promoter. The housekeeping factor σ^{70} recognises two hexamers, a −35 element near TTGACA and a −10 element near TATAAT, separated by a spacer of about 17 base pairs †; initiation melts the duplex around the −10 element, and the polymerase leaves the promoter after synthesising a short RNA. Termination is either intrinsic — a hairpin followed by a run of uridines destabilises the complex — or Rho-dependent, where a helicase chases the polymerase off. Alternative sigma factors redirect the same core to different promoter classes: σ^{38} in stationary phase and stress, σ^{32} for heat shock, σ^{54} for nitrogen-regulated genes, σ^{28} for flagella, σ^{24} for envelope stress†. The consequence for a production strain is the one part 1 chapter 8 gave: a promoter that is strong in exponential phase may be silent when the culture is stressed, which is when you want production.

A consensus match is not a promoter. How often would a six-base motif occur by chance in a 4.64 Mb genome? Counting both strands, an exact match to a given hexamer is expected 2,266 times; allowing one mismatch, which real promoters tolerate freely, the expected count is **43,047**. For a 12-base motif — a −35 and a −10 element together, ignoring the spacer — exact matches are expected 0.55 times. So a consensus-matching hexamer is almost always a coincidence, and only the combination of two elements at the right spacing, in the right context, with activators or repressors present, makes a promoter. It is the reason computational promoter prediction is unreliable, and the reason a measured, characterised part (part 8 chapter 26) is worth so much more than a sequence.

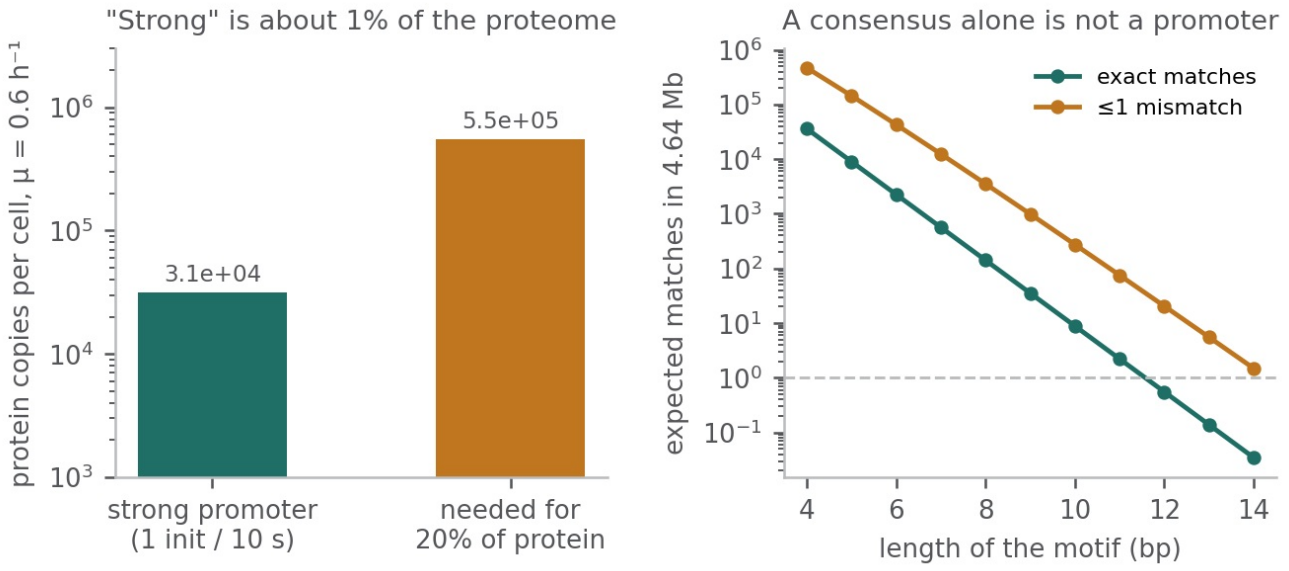


Figure B8.1 Left: protein copies per cell from a strong promoter at μ = 0.6 h⁻¹ (31,000, about 1.1% of the proteome) against the copies needed to make a protein 20% of the proteome (550,000). Right: expected random matches to a motif of the stated length in 4.64 Mb, exact and with one mismatch allowed. Computed for this module; the initiation rates and half-life are assumptions.

Translation: the second dial

The small ribosomal subunit binds the mRNA at the **Shine–Dalgarno sequence**, about AGGAGG, which pairs with the 3′ end of 16S rRNA five to nine bases upstream of the start codon†; initiator tRNA carrying formylmethionine is placed on AUG (the dominant start codon; GUG and UUG are used less often†) with the help of initiation factors IF1 to IF3, and the large subunit joins. Elongation cycles through aminoacyl-tRNA delivery by EF-Tu (spending GTP), peptide-bond formation, and translocation by EF-G (spending another). Termination occurs at a stop codon recognised by release factor RF1 (UAA, UAG) or RF2 (UAA, UGA), with RF3 assisting. Because initiation is usually rate-limiting, the ribosome binding site and the mRNA structure around the start codon set the protein level per mRNA — the independent dial of part 1 chapter 8, and the quantity the RBS calculator of part 8 chapter 28 predicts.

Codon choice matters at the margin. A handful of codons are rare in *E. coli* — AGA, AGG, AUA, CUA, CGA, CCC† — and a gene rich in them stalls ribosomes, depletes the matching tRNAs and can misincorporate or truncate. Whether to replace them is the "optimise or harmonise" question of part 8 chapter 30, and the rule remains to compare constructs and measure soluble active protein.

What a strong gene costs in numbers

Combine the two dials with the two half-lives, as part 1 chapter 8's expression equation did, and compute a concrete case. A strong promoter initiating once every 10 seconds makes 6 transcripts a minute; with an mRNA half-life of 3 minutes (decay constant 0.231 min^{-1}) the steady state is 6/0.231 = **26 mRNA copies**. A strong ribosome binding site loads a ribosome every 5 seconds, so each message yields 12 proteins a minute, and the cell makes 26 × 12 = 312 proteins a minute. A protein that is stable is lost only by dilution, at the growth rate: at $\mu = 0.6\ h^{-1}$ (0.01 min^{-1}) the steady state is 312/0.01 = **31,200 copies, 1.1% of the 2.8 million protein molecules**; at $\mu = 1.2\ h^{-1}$ it halves to 15,600, 0.56%.

Three conclusions. That is within a factor of a few of the most abundant proteins in the cell — the translation factor EF-Tu is about 5% of the protein† — so "strong promoter plus strong RBS" is already at the top of the natural range. Reaching 20% of the proteome needs 5.5×10^5 copies, **eighteen times more**, which a single chromosomal copy cannot supply and a high-copy plasmid with a very strong promoter can only approach by crowding out growth (chapter B9). And dilution works in the engineer's disfavour: the faster the cells grow, the lower the steady-state level of every stable protein, which is why a pathway expressed in exponential phase falls as a fraction of the proteome when growth accelerates.

Regulation by repressors and activators: the *lac* operon as the template

The operon *lacZYA* encodes β-galactosidase, a permease and an acetyltransferase from one promoter. The repressor LacI, a tetramer, binds an operator overlapping the promoter and blocks transcription; the inducer allolactose — or the non-metabolisable IPTG used in the lab — binds LacI and releases it. A second input comes from the catabolite activator protein CRP, which in complex with cyclic AMP binds upstream and recruits polymerase, and cyclic AMP is high only when glucose is absent. So the operon is expressed strongly only when lactose is present **and** glucose is not — an AND gate, in chemistry — as part 1 chapter 8 put it. Every inducible expression system in the module is a variation on this architecture, with the same properties: leak when the repressor is saturated, a Hill response set by cooperativity, and an inducer that must enter the cell.

Other layers act after transcription: RNase E degrades mRNA, with the half-life set partly by structure at the 5′ end; small RNAs pair with a transcript, usually with the chaperone Hfq, to change translation or decay; riboswitches in the leader fold differently on binding a metabolite and so regulate their own gene; and proteins are removed by proteases such as Lon and ClpXP, recruited by degradation tags like the ssrA sequence†. Each is a tool in part 8 chapter 27 and a source of context effects you must control.

σ^{70} promoter	−35 ≈ TTGACA, 17 bp, −10 ≈ TATAAT†; recognition by σ factor
Random matches, 4.64 Mb	hexamer exact 2,266; ≤1 mismatch 43,047; 12-mer exact 0.55
Shine–Dalgarno	≈ AGGAGG, 5–9 nt before AUG; initiation is rate-limiting
Expression budget	26 mRNA × 12 proteins/min ÷ 0.01 min^{-1} = 31,200 copies = 1.1% at μ 0.6 h^{-1}
To reach 20%	5.5×10^5 copies, 18× the strong-promoter case

Common confusions

- **"A sequence that matches the consensus is a promoter."** It is one of tens of thousands of matches.
- **"The promoter sets the expression level."** It sets transcription; the ribosome binding site and the two half-lives set the rest.

- **"A stronger promoter always gives more protein."** Not when growth slows — the level is production divided by (decay + dilution).

Self-check — answer on paper first

1. Your protein of interest is stable, expressed from a strong promoter, and cells grow at 0.3 h^{-1} in a nutrient-limited fed-batch. Estimate the level relative to the 0.6 h^{-1} case.

With production unchanged the steady-state level is production/dilution, so halving the growth rate doubles the level: 62,400 copies, 2.3% of the protein. The assumption that production stays the same is the weak one, since a slower-growing cell has fewer ribosomes and a different promoter activity (chapter B9) — so this is an upper bound, and the real level lies between the two cases. It does show why a protein that looks well expressed in a rich-medium shake flask can behave very differently in a slow fed-batch.

2. Two genes in one operon give protein ratios of 1:8 although they share a promoter. Why?

They share one mRNA, so transcription cannot differ; what differs is translation initiation, set by each gene's own ribosome binding site and local mRNA structure, and possibly differential mRNA decay along the transcript. A 1:8 ratio is a ribosome binding site difference of about eightfold, well within what the RBS calculator spans, which is how pathway stoichiometry is tuned inside one operon (part 1 chapter 8's worked example).

3. Why does the AND-gate logic of the lac operon matter for a production process that uses IPTG?

IPTG bypasses the lactose arm — it releases the repressor regardless of the sugar supply — but the CRP–cAMP arm still operates, so expression from a lac-type promoter can be lower in glucose-grown cultures than in glycerol- or lactose-grown ones even at saturating IPTG. A fed-batch on glucose can therefore show weaker induction than the same strain on another carbon source, and the cure is either a promoter without CRP dependence or a carbon source that does not repress.

What to open a book for. Alberts chapters 4 to 8 selectively; Lehninger chapters 24 to 28 for the chemistry of the information pathways. Snyder & Champness for bacterial gene regulation written around experiments, which is the better route into the lac operon than any mechanism-first chapter.

System-level regulation and the growth laws

A cell with a fixed protein budget grows at a rate set by how much of it is ribosomes. Every protein you add that does not help growth is paid for out of that budget — and the price, computed here, is about 1.8% of growth rate for each 1% of proteome.

Global preferences: catabolite repression

Given two carbon sources, *E. coli* uses the better one first and represses the machinery for the other, giving the diauxic growth curve with a lag between phases. The mechanism runs through the phosphotransferase system. While glucose is being imported, one component of the PTS, $EIIA^{Glc}$, is mostly dephosphorylated; in that state it fails to activate adenylate cyclase, so cyclic AMP is low and CRP–cAMP cannot activate the operons for other sugars, and it also blocks the lactose permease directly (inducer exclusion)†. When glucose is exhausted, $EIIA^{Glc}$ becomes phosphorylated, cyclic AMP rises, and the alternative operons switch on after a lag while the enzymes are made. The engineering consequence from part 1 chapter 9 stands: a mixed feedstock is consumed sequentially, and simultaneous utilisation is a project, not a default.

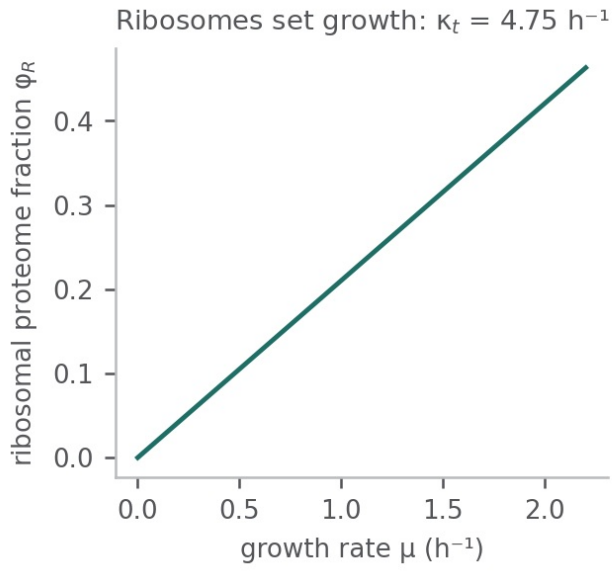
The proteome is a budget, and ribosomes set the growth rate

Chapter B1 computed the number of ribosomes a given growth rate requires. Now invert it into the relation that organises the field. If a fraction φ_R of the protein is ribosomal protein, and each ribosome (7,500 residues of protein per ribosome†) elongates at k = 18 aa/s with a fraction f = 0.55 of them active, the protein synthesis rate per unit protein is $(k \cdot f/L_R) \cdot \varphi_R$, which is the growth rate:

```
μ = κ_t · φ_R,      κ_t = k·f/L_R = 18 × 0.55 / 7,500 × 3,600 = 4.75 h⁻¹
```

Growth rate μ (h^{-1})	Growth-dependent ribosomal proteome fraction $\varphi_R = \mu/\kappa_t$
0.3	0.06
0.6	0.13
1.0	0.21
1.5	0.32
2.0	0.42

A linear relation of ribosome content to growth rate is exactly what is measured, with a slope in the region of 4.5 h^{-1} for the translational capacity†; my computed 4.75 h^{-1} is an order-of-magnitude check on the model, built from three recalled constants, not a measurement. What it says is the point: **the doubling time is set by how much of the proteome is devoted to making proteins**. At 2 h^{-1} over 40% of the protein is ribosomal; there is no room for a large uncommitted expression load.



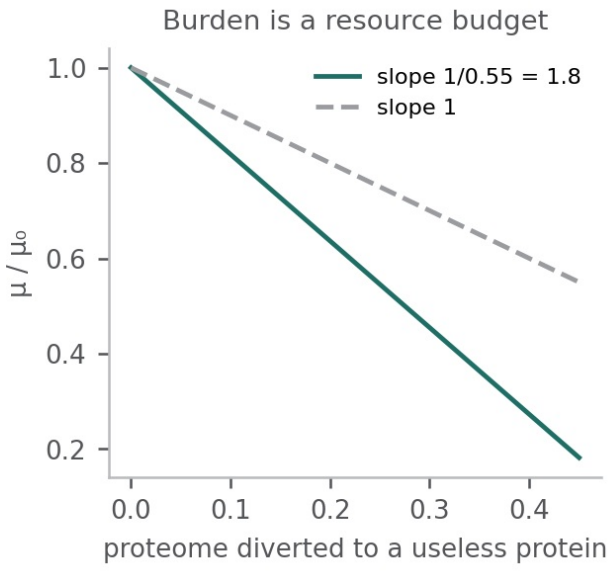


Figure B9.1 Left: the growth-dependent ribosomal proteome fraction against growth rate, from $\mu = \kappa_t \varphi_R$. Right: the fall in growth rate when a fraction of the proteome is diverted to a useless protein, for a maximum diversion $\varphi_{max} = 0.55$ (slope 1.8) and for slope 1. Computed for this module from the stated constants.

The price of burden, in numbers

Because the proteome is a fixed budget, expressing a protein that does not contribute to growth displaces the growth-supporting sectors and slows the cell. The simplest form, which fits measurements of cells expressing a useless protein†, is $\mu/\mu_0 = 1 - \varphi_X/\varphi_{max}$, with φ_{max} about 0.55†, so the slope is 1/0.55 = 1.8. Computed:

Proteome diverted, φ_X	μ/μ_0	Growth lost
1.1% (the strong gene of chapter B8)	0.98	2%
5%	0.91	9%
10%	0.82	18%
20%	0.64	36%
30%	0.45	55%

So **a single very strong gene costs about 2% of the growth rate, a pathway taking a fifth of the proteome costs a third, and one taking 30% halves it**. This is the quantitative form of part 1 chapter 9's "burden is measurable". It also connects to part 8 chapter 28, where the library model swept a burden slope and found the best combination changing at 2.4 and at 3.6: a slope near 1.8 puts a heavy pathway just below the first crossover, so the expression-everywhere design is on the edge of being the wrong one — the library is how you find out on which side your pathway sits. The burden depends on the medium, as chapter B8's discussion of slow growth implies; the slope is steeper in poorer media†.

Stress responses and the stringent response

The stringent response is triggered when amino acids run short. An uncharged tRNA entering the ribosome activates RelA, which synthesises the alarmone ppGpp (and SpoT makes and degrades it); ppGpp, with its partner DksA, binds RNA polymerase and shuts down transcription of ribosomal RNA and tRNA while raising that of amino acid biosynthesis genes. The cell stops investing in growth and shifts to maintenance. Anything that drains an amino-acid pool — overexpressing a protein rich in one residue, a toxic intermediate, a starved fed-batch — can fire it, and production falls with growth (part 1 chapter 9).

The other global responses to know by name: the heat-shock response through σ^{32}, which induces chaperones and proteases and is triggered by misfolded protein — so heavy overexpression induces it; the general stress response through σ^{38} in stationary phase; the envelope stress response through σ^{24} and the Cpx system; the SOS response to DNA damage, where RecA stimulates cleavage of the LexA repressor and raises mutation rates; and oxidative stress through OxyR and SoxRS. A strain that underperforms for no pathway reason is often showing one of them.

Sensing: two-component systems and the global regulators

A two-component system is a sensor histidine kinase in the membrane and a response regulator in the cytoplasm. The sensor autophosphorylates on a histidine when it detects its signal and transfers the phosphoryl group to an aspartate on the regulator, which then binds DNA — EnvZ and OmpR for osmolarity, PhoR and PhoB for phosphate, NtrB and NtrC for nitrogen. The others to know: Fnr, which senses oxygen through an iron–sulfur cluster and switches on anaerobic genes; ArcAB, which senses the quinone redox state; Fur, for iron; and the quorum-sensing systems, where cells secrete a small diffusible signal — an acyl-homoserine lactone in many Gram-negatives, a peptide in Gram-positives — and respond above a density threshold, which is what part 8 chapter 27 used as a clock.

When a strain underperforms: the diagnostic

The measurements to take before touching the pathway, as in part 1 chapter 9, now with numbers to interpret them: growth rate with and without induction, and compare it with $\mu/\mu_0 = 1 - \varphi_X/0.55$ to infer how much proteome you are using; by-product formation and the respiratory quotient (chapter B6), to see whether the specific uptake rate has crossed the overflow threshold; a carbon balance, to find where the carbon went; and a stress reporter, to see whether a global response has fired. If the growth rate fell by more than the expression level predicts, a global response or a toxic intermediate is the likelier cause than a pathway defect.

Growth law	$\mu = \kappa_t \varphi_R$, $\kappa_t = kf/L_R = 4.75\ h^{-1}$ (computed); ~4.5 h^{-1} measured†
Ribosome fraction	0.13 at 0.6 h^{-1}; 0.21 at 1.0; 0.42 at 2.0

Burden	$\mu/\mu_0 = 1 - \varphi_X/\varphi_{max}$, $\varphi_{max} \approx 0.55$ → slope 1.8
Price of expression	1.1% of proteome → 2% growth; 20% → 36%; 30% → 55%
Stringent response	uncharged tRNA → RelA → ppGpp → rRNA transcription down, amino-acid synthesis up

Common confusions

- **"Burden is vague."** It is a budget with a measured slope near 1.8 per unit of proteome.
- **"A mixed sugar medium is consumed together."** Usually sequentially, because CRP–cAMP and inducer exclusion enforce the order.
- **"The stringent response is a starvation curiosity."** A heavy overexpression can trigger it and production falls with growth.

Self-check — answer on paper first

1. A pathway takes 12% of the proteome. How much slower do the cells grow, and what would you do if that is too much?

$\mu/\mu_0 = 1 - 0.12/0.55 = 0.78$, so 22% slower. If that is unacceptable, the options of part 1 chapter 9 apply: stage the process so growth finishes before expression begins, which keeps both; lower the expression of the enzymes with low control coefficients (part 2 chapter 15), since those cost proteome and buy little flux; or move the least-needed enzymes to a second strain (chapter S4), which halves each one's burden.

2. Why does the ribosome fraction have to rise faster than linearly in the number of ribosomes when growth speeds up?

Two effects multiply. The protein synthesis rate required is μ times the protein content, so it rises with μ; and at higher μ the cell is larger and holds more protein, so the content rises too. Chapter B1's calculation showed a 3.3-fold growth rate increase needing a 6.7-fold ribosome increase; the proteome fraction — the ratio of ribosomal to total protein — rises linearly with μ, but the absolute count rises faster, and both are what the growth law and the ribosome count say.

3. Your IPTG-induced strain stops growing 30 minutes after induction. Name the first three things you check.

The proteome fraction the protein takes, by gel or by a ratio of its band to total protein — if it exceeds 25 to 30%, the burden alone explains it; the ppGpp level or a stringent-response reporter, since a protein rich in a scarce amino acid can drain it; and whether the protein is toxic in its own right, by testing an inactive point mutant at the same expression level, which separates burden from activity. The heat-shock response is a fourth, with misfolded protein as its trigger.

What to open a book for. Scott et al., Science 330:1099 (2010), which is the paper behind the growth laws and is short and readable; Basan et al., Nature 528:99 (2015) for the overflow link; Alberts chapter 7 for the regulatory mechanisms, and a recent review of proteome allocation for the quantitative account.

Chapter B10 · tracker week 10

Microbial genetics and microbiology: growth, mutation, gene transfer, sterility

Mutations arise before you select, not because you select — and that one result, derived below from a simulation with a variance-to-mean ratio of 649, is why every large culture already contains the variant that will take it over.

Growth, yield and maintenance

Batch growth has a lag, an exponential phase in which biomass follows $X = X_0 e^{\mu t}$, a stationary phase and decline; the doubling time is $t_d = \ln 2/\mu$. The dependence of μ on substrate is Monod's: $\mu = \mu_{max} S/(K_s + S)$, a Michaelis–Menten form with $\mu = \mu_{max}/2$ at $S = K_s$. In a chemostat at steady state $\mu = D$, so the residual substrate is $S = K_s D/(\mu_{max} - D)$; with $\mu_{max} = 0.9$ h⁻¹ and $K_s = 0.5$ g/L, a dilution rate of 0.3 h⁻¹ leaves 0.25 g/L and 0.8 h⁻¹ leaves 4.0 g/L — the sharp rise that precedes washout.

The observed biomass yield falls at low growth rates because the cell spends part of its substrate on maintenance. Pirt's relation is $1/Y_{obs} = 1/Y_{true} + m/\mu$. With an illustrative true yield of 0.50 g/g and a maintenance of 0.058 g glucose per g biomass per hour — which is the core model's 8.4 mmol ATP per g per h at about 26 ATP per glucose — the observed yield is:

Growth rate μ	Observed yield
0.60 h⁻¹	0.48 g/g
0.10 h⁻¹	0.39 g/g
0.05 h⁻¹	0.32 g/g
0.02 h⁻¹	0.20 g/g

That table is the quantitative reason a very slow fed-batch is expensive, which part 11 chapter 44's feeding discussion balanced against overflow: below a growth rate of about 0.1 h⁻¹ the maintenance term takes a growing share of the carbon, and a product that is not growth-coupled competes with it.

Counting, and the 30 to 300 rule

Three measurements of "how many cells" disagree for reasons part 1 chapter 10 gave: optical density tracks biomass and cell size, colony counts measure cells able to grow on that medium, and dry weight is what a balance needs. A colony count is a Poisson variable, so its relative uncertainty is $1/\sqrt{N}$:

Colonies counted	Coefficient of variation
10	32%
30	18%
100	10%
300	5.8%
1,000	3.2%

The conventional 30 to 300 counting window is the compromise between that statistical floor at the bottom and the crowding that makes colonies merge at the top. The arithmetic of a plate count is: 150 colonies from 0.1 mL of a 10^{-6} dilution is $150/(0.1 \times 10^{-6}) =$ **1.5×10^9 CFU/mL**, with an uncertainty of 8% from the count alone — and larger once pipetting and dilution error are added, which is why replicate plates at two adjacent dilutions are standard.

Mutation, and why it is always ahead of you

Do mutations arise because a selective condition calls them forth, or do they arise at random before it, and the selection merely finds them? Luria and Delbrück answered by noting that the two models predict different *distributions* of mutant counts across parallel cultures. If mutations were induced at the moment of selection, each culture would give a Poisson number of mutants, with variance equal to the mean. If they arise at random during growth, a mutation that happens early produces a large clone — a jackpot — and the variance greatly exceeds the mean.

I simulated it: 4,000 cultures each grown from 10^3 cells to 1.3×10^8 over 17 generations, with a mutation rate of 2×10^{-9} per cell per division.

	Mean mutants	Variance / mean	Largest culture
Random mutation during growth (simulated)	1.70	**649**	**2,048**
Induced at selection (Poisson, same mean)	1.70	1.0	8

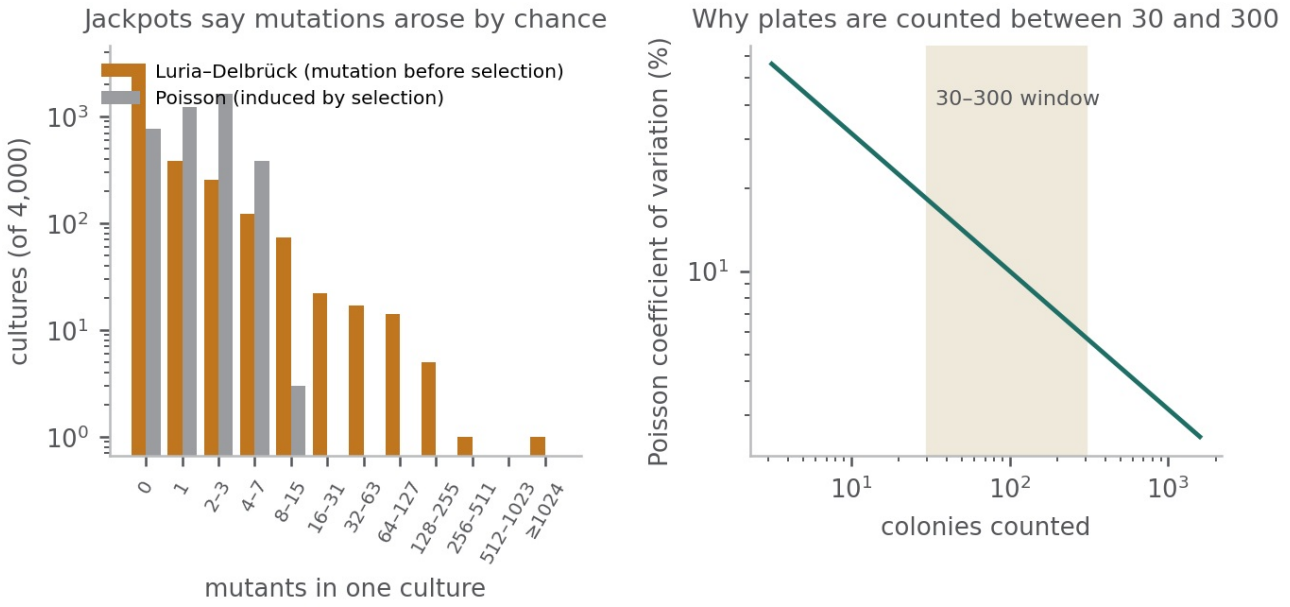


Figure B10.1 Left: the distribution of mutant counts across 4,000 simulated cultures under the Luria–Delbrück model against a Poisson distribution with the same mean. Right: the Poisson coefficient of variation of a colony count, with the 30–300 counting window. Computed for this module.

The difference is enormous, and it is the result: **a variance-to-mean ratio of 649 against 1.0, and a jackpot of 2,048 mutants where a Poisson distribution gives at most 8**. The experiment's data match the first. The simulation also gives the practical estimator: the fraction of cultures with no mutants is $p_0 = e^{-m}$, where $m = \mu N_f$ is the expected number of mutation events, so $\mu = -\ln p_0 / N_f$. Here $p_0 = 0.777$ and $N_f = 1.31 \times 10^8$ give $\mu = 1.9 \times 10^{-9}$, against the true 2.0×10^{-9} — the "p_0 method" recovers the rate from the number of empty cultures alone.

The engineering consequence is the one part 9 chapter 32 stated as instability. A loss-of-function mutation in a given gene occurs at roughly 10^{-10} per base per generation × 1,000 bases × about half of the mutations being inactivating† ≈ 5×10^{-8} per generation. A 1,000 L fermenter at 10^{10} cells/mL holds 10^{16} cells, so about **5×10^8 new loss-of-function mutants of any one gene appear every generation**. Whether they take over depends on their growth advantage — a pathway-inactivating mutation that removes burden has one — and on the number of generations. That is the reason scale-up adds a genetics problem, and the reason designs that couple production to growth (part 11 chapter 39) are valuable: they make the inactivating mutant lose the race.

Moving DNA between cells

Route	What happens	Engineering relevance
Transformation	Uptake of free DNA — natural competence in some species; $CaCl_2$/heat shock or electroporation in *E. coli*	Efficiency in transformants per µg decides whether a library is possible (part 7 chapter 25)
Conjugation	DNA transferred from donor to recipient through a pilus; needs an origin of transfer and a helper that supplies the transfer genes	The standard route into *Streptomyces* and other resistant hosts (part 8 chapter 31)
Transduction	A phage carries host DNA — any fragment in generalised transduction (P1), neighbouring genes in specialised (λ)	Moving a marked allele between strains

Plasmids, markers and compatibility

A plasmid replicates from its own origin, and the origin sets its copy number†: pUC-type high-copy ColE1 derivatives at several hundred per cell, pBR322 at 15 to 20, p15A at 10 to 12, pSC101 at about five, and F-factor bacterial artificial chromosomes at one or two. The ColE1 family controls copy number with an antisense RNA, RNA I, that pairs with the RNA II primer and prevents it from priming replication — a natural regulatory circuit of exactly the kind chapter B8 described. Two plasmids with the same type of origin are **incompatible** and one is lost, so co-expression plasmids use different families — a ColE1, a p15A and a pSC101 can coexist.

Marker	Gene and mechanism	Note
Ampicillin	*bla*, a β-lactamase that degrades the drug	Secreted, so it clears the drug around a colony and lets satellite colonies grow
Kanamycin	*aph*, an aminoglycoside phosphotransferase	Stable and widely used
Chloramphenicol	*cat*, an acetyltransferase	Bacteriostatic; works at lower plasmid copy numbers
Tetracycline	*tetA*, an efflux pump	The pump runs on the proton motive force (chapter B4), so it costs energy and burden
Spectinomycin	*aadA*, an adenylyltransferase	Common for integrative constructs

None of these belongs in a production process, so the standard replacements are auxotrophic complementation — a plasmid that restores a gene deleted from the chromosome — and integration. Counterselection markers, which kill cells that still carry them, such as *sacB* (lethal on sucrose) or *ccdB*, are what make marker removal and two-step genome editing work.

Sterility, computed

Killing by heat is first order, so the survivors fall as $N = N_0 \cdot 10^{-t/D}$, with D, the decimal reduction time, the minutes to kill 90%. The most heat-resistant contaminants are bacterial spores, with D_{121} of the order of 1.5 minutes for *Geobacillus stearothermophilus*†. A 100 L medium containing 10^6 spores per mL has 10^{11} spores. To make the probability of a surviving spore only 10^{-3} per batch needs a reduction of 10^{14}, **14 D**, which at 121 °C is 14 × 1.5 = **21 minutes**. The dependence on temperature is exponential: with a z-value — the temperature rise that cuts D tenfold — of 10 °C†, 14 D takes 21 minutes at 121 °C, 2.1 minutes at 131 °C and 0.21 minutes at 141 °C.

The reason plants use short high-temperature sterilisation is also computable. The nutrient damage — the browning and degradation of sugars and vitamins — has a much larger z-value, of the order of 30 °C †, so it rises with temperature far more slowly than spore killing does. Raising the temperature from 121 to 131 °C cuts the exposure time to a tenth while the damage rate rises only by $10^{10/30} = 2.15$, so the total damage falls to 0.1 × 2.15 = 0.215 — **4.6 times less** — for the same sterility. That asymmetry is the whole design logic of a continuous high-temperature, short-time steriliser.

Heat-labile components are filter-sterilised through a 0.2 µm membrane, which removes bacteria but not viruses; phage can pass, which is why phage contamination in a plant is managed by prevention and strain resistance and not by filtration alone (part 1 chapter 10). Aseptic technique itself, and the biosafety framework that surrounds it, are in part 9 chapter 32.

Monod	$\mu = \mu_{max} S/(K_s + S)$; chemostat $S = K_s D/(\mu_{max} - D)$
Maintenance	$1/Y_{obs} = 1/Y_{true} + m/\mu$; 0.48 → 0.20 g/g from µ 0.6 to 0.02
Counting	$CV = 1/\sqrt{N}$; 18% at 30 colonies, 5.8% at 300
Luria–Delbrück	variance/mean 649, jackpot 2,048 against Poisson 1.0 and 8; $\mu = -\ln p_0/N_f$
Mutants at scale	5×10^8 new inactivating mutants per generation in 10^{16} cells
Thermal death	$N = N_0 10^{-t/D}$; 14 D = 21 min at 121 °C; HTST gives 4.6× less nutrient damage

Common confusions

- **"Resistance appears because we applied the antibiotic."** The variants were already there; selection found them.
- **"A plate with 12 colonies is a good count."** Its uncertainty is 29%; count between 30 and 300.
- **"Longer sterilisation is always safer."** It is safer for spores and worse for the medium; the high-temperature short-time optimum exists because the two have different z-values.

Self-check — answer on paper first

1. Of 30 parallel cultures, 24 contain no resistant mutants after 10^8 cells each. Estimate the mutation rate.

$p_0 = 24/30 = 0.8$, so $m = -\ln 0.8 = 0.223$ and $\mu = 0.223/10^8 = 2.2 \times 10^{-9}$ per cell per division. With only 30 cultures the uncertainty is large — the standard error of p_0 is $\sqrt{(0.8 \times 0.2/30)} = 0.073$, which moves µ by roughly ±30% — so report the interval, and use 50 to 100 cultures if the rate matters. If the other six cultures show a few mutants and one shows hundreds, the jackpot confirms the mutations were not induced.

2. A 1,000 L fermentation needs a 10^{-4} chance of a surviving spore from 10^8 spores per batch. How many D-values, and what time at 121 °C with D = 1.5 min?

$\log_{10}(10^8/10^{-4}) = 12$ D; 12 × 1.5 = 18 minutes. Because the medium takes time to heat and cool, the lethality

accumulated during the ramps counts too, and the real hold time is usually set by integrating the lethal rate over the whole temperature profile rather than by the hold alone.

3. Why can two plasmids with ColE1-type origins not be maintained together, and what do you do?

They share the same regulatory RNA I and priming mechanism, so each controls the other's copy number as well as its own, and one is gradually diluted out as the cell enforces a single plasmid copy budget — incompatibility. Choose plasmids from different families, such as a ColE1 and a p15A origin, with different antibiotic markers; or put one of the genes on the chromosome by integration, which removes the compatibility question and the need for the second antibiotic.

What to open a book for. Brock *Biology of Microorganisms* is the reference for this week and the book most worth having for phase 1. MIT OCW 7.13 Experimental Microbial Genetics for the genetics in practice, and Snyder & Champness for selection design. Doran for sterilisation design and the thermal death arithmetic done as engineering problems.

Glossary

Each definition is original wording of a standard concept. Terms are those the tracker assumes.

Term	Meaning
Allosteric	Regulated by a ligand binding at a site other than the active site, changing activity
Anaplerosis	Reactions that replenish cycle intermediates drawn off for biosynthesis
Antiporter / symporter	Transporters that move two substances in opposite / the same direction
Auxotroph	A strain unable to make a compound and therefore needing it supplied
Catabolite repression	Repression of the use of secondary carbon sources while a preferred one is present
Chaperone	A protein that assists folding and prevents aggregation without becoming part of the product
Chemostat	Continuous culture at constant feed and removal; $\mu = D$ at steady state
Codon	Three nucleotides specifying an amino acid or a stop
Cofactor / coenzyme	A non-protein helper of an enzyme; a coenzyme is an organic one, often a vitamin derivative
Conjugation	Transfer of DNA between cells in contact
Cooperativity	Binding at one site changing affinity at others, giving a sigmoidal response
CRP (CAP)	cAMP receptor protein, a global activator of genes for alternative carbon sources
Crabtree effect	Fermentation to ethanol by yeast despite oxygen, at high sugar
D-value	Time at a temperature to kill 90% of a population
Dilution rate	Feed flow divided by culture volume; equals μ in a chemostat
Doubling time	$\ln 2/\mu$
EC number	Four-part code classifying an enzyme by the reaction it catalyses
Electron transport chain	Membrane complexes passing electrons to a terminal acceptor while pumping protons
Entner–Doudoroff pathway	An alternative glycolysis giving 1 ATP, 1 NADH and 1 NADPH per glucose
Fermentation	Energy metabolism without an external terminal electron acceptor
Flux	Rate of conversion through a reaction or pathway, per unit biomass or volume
Gram-negative / positive	Two envelope types: outer membrane plus thin wall, or one membrane plus thick wall
Hill coefficient	An empirical steepness of a sigmoidal response; not a count of sites
Hydrophobic effect	The tendency of non-polar groups to cluster to release ordered water; the main folding force
Inducer	A molecule that turns a gene on, usually by releasing a repressor
Isoelectric point (pI)	The pH at which a molecule has zero net charge
Isozyme	Different proteins catalysing the same reaction
k_{cat}	Turnover number: maximum rate per active site
K_M	Substrate concentration at half-maximal rate; $(k_{-1} + k_{cat})/k_1$
Lag phase	The adjustment period before exponential growth
Lysogeny	A phage integrating into the host genome instead of lysing it
Maintenance	Energy spent to stay alive, independent of growth
Mass-action ratio (Q)	Product of product concentrations over substrates; compared with K_{eq}
Missense / nonsense / synonymous	Mutations changing an amino acid / creating a stop / leaving the amino acid unchanged
Monod equation	$\mu = \mu_{max} S/(K_s + S)$
Operator	The DNA site where a repressor binds
Operon	Genes transcribed together from one promoter
Overflow metabolism	Excretion of fermentation products despite sufficient oxygen, when uptake outruns respiration

Oxidative phosphorylation	ATP synthesis driven by the proton motive force from respiration
P/O ratio	ATP made per pair of electrons passed to oxygen
Peptide bond	The amide link between amino acids; planar
Plasmid	An extrachromosomal replicon with its own origin
Polycistronic mRNA	One transcript encoding several proteins
Promoter	The DNA site where RNA polymerase starts transcription
Proton motive force (Δp)	Electrochemical proton gradient across a membrane, $\Delta\psi - 61.5 \cdot \Delta pH$ mV
PTS	Phosphotransferase system: imports and phosphorylates sugars using PEP
Quorum sensing	Density-dependent gene regulation through a secreted signal
Redox potential ($E^{\circ\prime}$)	A measure of the tendency of a couple to accept electrons
Regulon	Genes at different locations controlled by one regulator
Repressor	A protein that blocks transcription by binding an operator
Respiratory quotient	CO_2 produced over O_2 consumed
Ribosome binding site	The mRNA sequence, including Shine–Dalgarno, where translation starts
Riboswitch	An mRNA element that changes fold on binding a metabolite
Sigma factor	A subunit directing RNA polymerase to a class of promoters
Stringent response	Amino-acid starvation response via ppGpp that shuts down ribosome synthesis
Substrate-level phosphorylation	ATP made directly by a reaction, without a gradient
Terminator	A sequence ending transcription
Transduction	DNA transfer by a phage
Transformation	Uptake of free DNA by a cell
Transhydrogenase	An enzyme converting NADH and NADPH to each other
Two-component system	A sensor kinase and response regulator pair that transmits a signal by phosphorylation
Vitamin	An organic nutrient that an organism cannot make and needs for a cofactor
Wobble	Flexibility in pairing at the third codon position
Yield coefficient	Product or biomass formed per substrate consumed
z-value	The temperature rise that reduces D tenfold

Recalled constants to verify

Every value marked † in this part was recalled from standard references rather than computed. They are listed here with where to check each. A computed result that depends on one of them inherits its uncertainty, and chapters B1, B4, B8 and B9 are the ones most exposed.

Constant	Value used	Where to verify
Dry weight per cell, protein fraction, composition	0.3 pg; 55% protein; 20% RNA; 9% lipid; 3% DNA	BioNumbers; Neidhardt, *Escherichia coli and Salmonella*
Genome size, gene count, coding fraction	4.64 Mb; ~4,300; ~88%	EcoCyc; NCBI genome record
Fork speed, transcription and translation rates	1,000 nt/s; 40–80 nt/s; 10–20 aa/s	BioNumbers; Alberts chapters 5–6
mRNA half-life; average protein mass	3 min; 36 kDa	BioNumbers; proteome-wide measurements
ATP cost per peptide bond; ribosome counts	~4; ~10^4 at 0.6 h^{-1}, ~7 × 10^4 at 2 h^{-1}	Lehninger chapter 27; Bremer & Dennis, EcoSal
Amino-acid pK_a values	D 3.9, E 4.2, H 6.0, C 8.3, Y 10.1, K 10.5, R 12.5; termini 2.0, 9.0	Lehninger chapter 3
Helix geometry; folding stability range	3.6/turn, 1.5 Å; 20–60 kJ/mol	Lehninger chapter 4; Alberts chapter 3
Median k_{cat}, K_M; diffusion limit	~10 s^{-1}; ~100 µM; 10^8–10^9 $M^{-1}s^{-1}$	Bar-Even et al., Biochemistry 50:4402 (2011)
Vitamins and groups carried; EC classes	as tabulated	Lehninger chapter 6; the IUBMB enzyme list
Standard potentials	NAD −0.320, NADP −0.324, O_2 +0.816, etc.	Lehninger chapter 19
ATP $\Delta G^{\circ\prime}$ and cell concentrations	−30.5 kJ/mol; 3, 0.5, 10 mM	BioNumbers; Park et al., Nat. Chem. Biol. 12:482 (2016)
Δψ and pH values; c-ring sizes	−140 mV; 7.6/7.0; c_8, c_{10}, c_{12}	Nicholls & Ferguson; Lehninger chapter 19
E. coli respiratory chain variants; H^+/e^-	NDH-1, NDH-2, cytochromes bo_3 and bd	Unden & Bongaerts; EcoCyc
ED yield; PFK regulation; PTS role	1 ATP, 1 NADH, 1 NADPH; ADP/PEP	Lehninger chapter 14; EcoCyc
Biomass NADPH; carbon content	13.03 mmol/gDW; 48%	BiGG core model biomass reaction; Neidhardt
Anaplerotic enzymes by host; haem C5 route	PYC absent in *E. coli*; C5 in most bacteria	EcoCyc; MetaCyc
Amino-acid cost range; sulfate reduction; FabI cofactor	~12–75 ATP-eq.; 2 ATP + 4 NADPH; NADH	Akashi & Gojobori, PNAS (2002); MetaCyc
EF-Tu abundance; rare codons	~5% of protein; AGA AGG AUA CUA CGA CCC	Ecomics/BioNumbers; codon-usage tables
Promoter and SD consensus; start-codon usage	TTGACA/TATAAT/17 bp; AGGAGG; AUG ≫ GUG > UUG	Alberts chapter 6; Snyder & Champness
Growth-law constants	7,500 aa/ribosome; φ_{max} 0.55; κ_t ~4.5 h^{-1}	Scott et al., Science 330:1099 (2010)
Overflow threshold	0.76 h^{-1}	Basan et al., Nature 528:99 (2015)
Mutation rate; inactivation fraction	~10^{-10}/bp/gen; ~50% of mutations	Lee et al., PNAS 109:E2774 (2012)
Plasmid copy numbers	pUC ~500; pBR322 15–20; p15A 10–12; pSC101 ~5	Addgene plasmid reference
D and z values	D_{121} 1.5 min; z 10 °C (spores); z 30 °C (nutrients)	Doran; a food-microbiology reference

Closing notes to part 14

What this part adds, and what it still cannot

Ten chapters, 39 pages, about 1.7 times part 1's density per week. It supplies the quantitative core that the textbooks leave you to assemble: the codon-position structure of point mutations, the ribosome budget and what a protein costs, the ionisation arithmetic behind a His-tag purification, the barrier arithmetic of catalysis, the regime-dependence of an assay, the proton motive force and the minimum H^+/ATP, the decomposition of an ATP yield into its assumptions, the NADPH bound on the pentose phosphate pathway, the carbon cost of a gram of cells, the 1% of proteome that a strong gene occupies, the growth law and the price of burden, the Luria–Delbrück statistics, and thermal death. All of it is computed, with the recalled inputs marked.

What it still does not replace is the textbook's work: the mechanisms drawn atom by atom, the structures, the full pathway maps with every enzyme and cofactor, the experimental history that makes the facts stick. For weeks 1 to 10 you still need Lehninger for chapters 3 to 6, 13 to 22 and 24 to 28, Alberts for the cell and the information pathways, and Brock for week 10. This part is meant to be read first, to know what matters in each, and then reread afterwards as a check that you can reproduce the numbers.

The six numbers worth being able to reproduce cold

Number	Chapter	Why that one
2.8×10^6 proteins, 9×10^8 residues, 1 molecule = 1.66 nM	B1	The denominators of every per-cell calculation
71 kJ/mol for a 10^{12} rate enhancement; 5.93 kJ/mol per decade	B2	The scale of catalysis, and why barriers are small numbers exponentiated
$\Delta p = -177$ mV → 17.1 kJ per H^+; minimum 2.75 H^+/ATP	B4	Thermodynamics setting a floor on stoichiometry
2.39 g glucose per g of cells; 50% of carbon to CO_2	B7	The baseline against which every product yield is judged
1.1% of proteome from a strong gene; 2% growth cost; 20% → 36%	B8, B9	The price of expression
Variance/mean 649 against 1; 5×10^8 mutants per generation	B10	Why every large culture already contains the variant that will take it over

On the numbers in this part

Everything without a † was computed from the standard genetic code, from the recalled constants listed in the table before this note, or from numbers already established in parts 2 to 13. The recalled constants are the weak point, and the ones that carry the most weight are the ribosome parameters (7,500 residues per ribosome, 18 aa/s, 55% active), which set κ_t; $\varphi_{max} = 0.55$, which sets the burden slope; the P/O assumptions, which set the ATP yield; and the 48% carbon content, which sets the 50% carbon-to-biomass figure. If one of those is wrong by 20%, the chapter's conclusion survives and its third digit does not.

I also corrected two statements from earlier parts as I went. Part 1 chapter 6 said succinyl-CoA feeds haem synthesis, which is the route in yeast and animals but not in *E. coli*, which uses glutamate through the C5 pathway (chapter B6). And part 14 uses 7 NADPH + 7 NADH for an *E. coli* palmitate where a shorthand elsewhere in the module says 14 NADPH — the latter is right for yeast (chapter B7).

Part 2 — The framework: modelling, control and the numbers that decide a design (weeks 11 to 19)

This part is written at a little over twice the density of part 1 — about 2,700 words a week against 1,265 — and deliberately. Phase 1 competes with Lehninger and Alberts, which are better than anything I can compress into five pages a week. Phase 2 competes with nothing you own: Stephanopoulos is excellent and thirty years old, Palsson assumes you want the full constraint-based apparatus, and the parts you most need — the algebra done in your notation, the code that runs, the arithmetic of what a design costs — are scattered across chapters, documentation and papers. So here they are in one place, worked rather than summarised.

What you should expect. Every chapter derives its results rather than quoting them, every number in the figures was computed by a script rather than drawn by hand, and every worked example is one I actually ran. Where a figure rests on assumed parameters rather than measured ones, the caption says so and names the assumption. The code in chapters 12, 13 and 16 is written to be typed in and run, not read; it is short on purpose.

How to use it. Read the chapter before that week's sessions, then do the sessions. The self-checks at the end are the week's exam, and the phase 2 checkpoint in week 19 assumes all of them. One instruction matters more here than in part 1: when a chapter gives a number, reproduce it yourself before you believe me. This is the phase where your chemical-engineering training makes you faster than a biologist, and the way to cash that in is to check the arithmetic rather than to memorise the result.

Ch.	Week	Chapter
11	11	What the field claims to be: Bailey, network rigidity, and the design loop
12	12	Python for modelling I: the linear algebra, the solvers and the fits you will actually use
13	13	Stoichiometric modelling: S·v = 0 and what its null space is telling you
14	14	Driving force and transport: whether a step can run, and whether the product can leave
15	15	Metabolic control analysis: why the rate-limiting step is a myth
16	16	Kinetic and dynamic modelling: dc/dt = S·v(c), and what your data can and cannot identify
17	17	Experimental design and statistics: when a difference is real
18	18	Titer, rate, yield: the three numbers and the arithmetic behind them
19	19	Cofactor balancing, redox and toxicity — and the phase 2 checkpoint

What the field claims to be: Bailey, network rigidity, and the design loop

Metabolic engineering is not genetic engineering applied to metabolism. The difference is that it measures the network before it changes it, and judges the change by a flux rather than by a band on a gel.

The definition, and the word that carries it

Bailey's 1991 paper defined the field as the improvement of cellular activities by manipulating enzymatic, transport and regulatory functions with recombinant DNA. Read it carefully and the operative word is *improvement* — a quantity, measured before and after — rather than *modification*. Classical strain improvement also improves cells, by mutagenesis and screening, and it worked: most of the antibiotic and amino-acid titres of the twentieth century came from it. What it could not do was say why. A production strain from a twenty-round mutagenesis programme carries hundreds of mutations, of which a handful matter, and nobody knows which. You cannot transfer that knowledge to the next molecule.

So the claim of the field is epistemic rather than technical: that a strain should be the result of a stated hypothesis about where flux is limited, tested by a measurement that could have falsified it. That is why the discipline's name is engineering and not breeding, and it is also why the field's literature is so preoccupied with flux measurement. If you cannot measure the distribution of flux, you cannot tell a successful design from a lucky mutation.

Two things follow for how you read papers from week 51 onward. First, a paper that reports a titer and a list of genetic changes, with no flux or balance analysis, is a classical strain improvement paper wearing modern clothes — which is not a criticism, but it tells you how much of its reasoning transfers. Second, the useful papers tend to report a failed intervention alongside the successful one, because the failure is where the network told you something.

Network rigidity: the single most useful idea from 1991

Stephanopoulos and Vallino, in the companion paper to Bailey's, asked why overproduction attempts so often fail despite correct biochemistry. Their answer was that metabolic networks resist redistribution of flux at their branch points, and that this resistance is structural rather than accidental. They called it network rigidity, and they classified branch points — nodes — by how much the split ratio moves when you push on it.

A flexible node redistributes freely: push flux toward one branch and it goes. A rigid node holds its split ratio almost regardless of what you do to the enzymes, so the carbon you liberated upstream arrives and then leaves by the route it always used. Three mechanisms produce rigidity, and you can diagnose which one you have.

- **Allosteric feedback.** The branch you want is regulated by its own end product, so raising its enzyme raises flux briefly, product accumulates, and the enzyme switches off. You met this in week 3. The fix is a feedback-resistant variant, and it is the single most common intervention in the amino-acid literature.
- **Kinetic competition at the node.** Two enzymes draw on the same metabolite with different K_M values. If the competing enzyme is saturated and yours is not, raising the pool concentration helps yours disproportionately; if yours is the saturated one, raising the pool does nothing at all and you must raise V_{max}. This is decidable from the two K_M values and the measured pool concentration, which is why week 3's arithmetic matters.
- **Thermodynamic constraint.** The branch you want is close to equilibrium, so flux through it is set by the concentration ratio rather than by the enzyme. No amount of expression moves it. Week 14 tells you how to spot this before you build the strain.

The three have different costs to fix, and the diagnosis is cheap compared with the construction. Writing down which of the three you believe you are facing, before you build, is the habit that separates a designed experiment from an expensive guess.

What changed by 2016, and what did not

Nielsen and Keasling's 2016 review is the natural checkpoint against Bailey. What changed is the cost of the build and test steps: DNA synthesis went from a bottleneck to a line item, multiplexed genome editing replaced one-gene-at-a-time construction, genome-scale models exist for every industrial host, and analytics went from one metabolite at a time to a few hundred. What did not change is that a design still fails for reasons the host does not tell you about, that tolerance to the product is still mostly evolved rather than designed, and that the gap between a titer in a flask and a titer in a plant is still measured in years.

The honest summary is that the field got much better at build and test and only modestly better at design and learn. That is worth knowing because it tells you where to point your own effort: the bottleneck is not your ability to make strains.

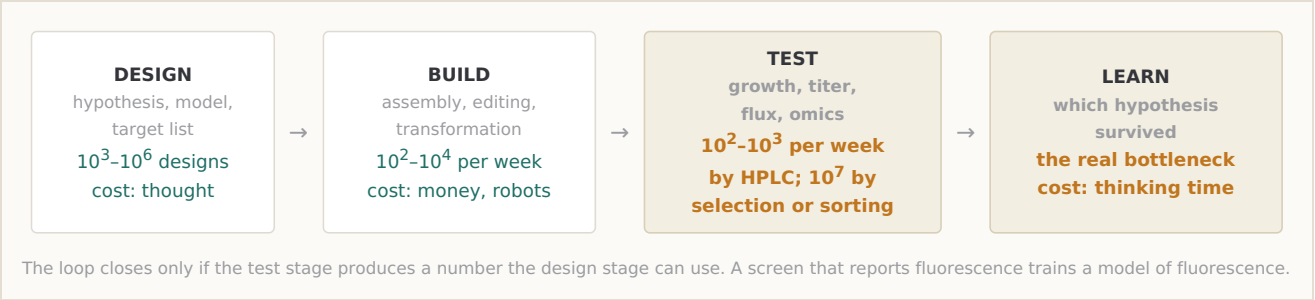


Figure 11.1 The design–build–test–learn cycle with throughputs attached. The orders of magnitude are indicative of a well-equipped academic lab rather than measured; the point is that they differ by four orders between build and analytical test, which is what biofoundries and selection-based screens exist to reconcile. Drawn for this module.

DBTL as an engineering loop, with arithmetic

Written as a loop, DBTL is familiar: design, build, test, learn, iterate. Treated as an engineering problem it has three parameters — cycle time, parallelism, and information gained per cycle — and the reason to be quantitative about them is that they trade against each other.

A campaign that builds 10^5 variants and screens them by a selection gains enormous parallelism and almost no information: you learn that one variant won, and typically not why. A campaign that builds twelve rationally chosen strains and characterises each with a flux measurement gains little parallelism and a great deal of information, because each strain tests a stated hypothesis. Neither is right in general. What decides is how good your prior is: if your model of the pathway is poor, parallelism buys more than thought, and if it is good, the reverse. Most industrial campaigns run both, with large libraries early and characterised strains late.

Cycle time is the parameter engineers under-rate. A loop that takes six weeks and a loop that takes six days differ by a factor of seven in the number of hypotheses you can test before your funding ends, which compounds more powerfully than any single improvement in hit rate. That is the real argument for automation, and it is the argument you should be able to make in an interview: not that a robot is more accurate than you, but that it shortens the loop.

What counts as evidence in this field

You will read perhaps four hundred papers over this curriculum, and the single skill that compounds is knowing what a claim needs before you believe it. Four claims recur, and each has a standard of evidence you can hold it to.

- **"Enzyme X was limiting."** Needs more than a titer increase on overexpression. The minimum is that the intermediate upstream of X fell and the one downstream rose; better is a titration — several expression levels, showing that the response saturates — because that is the shape control analysis predicts and a single point cannot distinguish from an artefact of the construct.
- **"Flux was redirected."** Needs a flux measurement, not a product measurement. Product up with substrate uptake also up is not redirection, it is more throughput; the carbon balance distinguishes them, and isotope labelling settles it.
- **"The pathway is now balanced."** Needs the cofactor arithmetic shown, per mole of product. A paper that reports a redox fix without stating the demand it was fixing has reported an intervention, not a result.
- **"The strain is stable."** Needs a measurement at the end of the fermentation, not the beginning: plasmid retention without selection, or sequencing of the construct from the final broth, over the number of generations the process actually runs.

Applying these to a paper takes a few minutes and is the main thing the paper log is for. The useful by-product is that it tells you where the field's soft spots are, which is where a new entrant can contribute: the claims that are routinely made without the evidence above are, almost by definition, the ones where a careful measurement is publishable.

A related habit: when a paper reports a fold-improvement, convert it into the two numbers that matter — the absolute titer and the yield as a fraction of theoretical. A tenfold improvement from 0.08 to 0.8 g/L is a result about a research strain; a 20% improvement from 85 to 102 g/L is a result about a process. The literature mixes the two freely and the abstracts rarely distinguish them.

You can build 200 strains a month by hand, or 2,000 a month on the foundry at ten times the cost per strain. Your screen is HPLC at 400 samples a month. Where is the bottleneck, and what should you change?

The bottleneck is the screen, and the foundry would make it worse. Building 2,000 strains a month into a 400-sample screen means five months of backlog accumulating per month of building, so the extra build capacity converts directly into unmeasured strains. Before buying build throughput you buy test throughput, and the cheapest route is almost never a second HPLC: it is a proxy that correlates with the product — a biosensor coupled to fluorescence, a colorimetric assay, a growth coupling — validated against HPLC on a few dozen strains and then used to triage thousands.

The cost of that route is the correlation. A proxy screens what it measures, not what you want, so a library optimised against a biosensor returns variants that optimise the biosensor, including ones that have found a way to light it up without making product. The standard discipline is to confirm every hit by HPLC and to keep the confirmation rate as a reported number, because the day it falls is the day your screen stopped measuring your product.

Three things people get wrong

- ‘DBTL is a project-management diagram.’ It is a loop with a measurable cycle time, and shortening it is the highest-leverage intervention in most campaigns.
- ‘Rigidity means the enzyme is too slow.’ Usually it means feedback, saturation at the node, or a near-equilibrium step — three different fixes.
- ‘Modern tools solved design.’ They solved build. Design and learn are where the field still loses.

Self-check — answer on paper first

1. Give the distinction between metabolic engineering and classical strain improvement in one sentence, then give the consequence for the next molecule you work on.

Metabolic engineering changes a defined target chosen from a measured picture of the network, so the result is attributable; classical improvement selects improved phenotypes without knowing the cause. The consequence is transferability: a mechanism you demonstrated in one strain is a hypothesis you can carry to the next product, whereas a mutagenised strain's advantage is locked inside its genome.

2. You raise the enzyme at a branch point fivefold and the split ratio does not move. Name the three candidate explanations and the measurement that distinguishes them.

Allosteric feedback from the end product, kinetic competition in which the competing enzyme is unsaturated and yours is saturated, and a near-equilibrium step whose flux is set by the concentration ratio. Measure the intracellular concentration of the branch-point metabolite and of the end product: feedback shows as inhibition relieved in a feedback-resistant variant; saturation is decided by comparing the pool concentration with both K_M values; a near-equilibrium step is identified from $\Delta G'$ computed at the measured concentrations, which is week 14's calculation.

3. Your group can either halve the cycle time or double the hit rate of its screen. Which do you argue for?

Halving the cycle time, in most cases, because it doubles the number of cycles and every cycle also raises the hit rate by what you learned in the last one, so the two effects compound. Doubling hit rate is a one-off gain on a fixed number of attempts. The exception is when cycle time is already dominated by the fermentation or the analytics of a necessarily slow test — a sixty-hour fed-batch cannot be shortened by a robot — in which case hit rate is the only handle left.

What to open a book for. Nothing: this week is three papers, and they are short. Read Bailey (Science 252:1668) and Stephanopoulos & Vallino (Science 252:1675) in the same sitting, then Nielsen & Keasling (Cell 164:1185) as the thirty-year report card. If you want the history properly, the introduction chapters of Stephanopoulos, Aristidou & Nielsen cover it from the inside.

Python for modelling I: the linear algebra, the solvers and the fits you will actually use

You do not need to learn Python. You need to learn the four or five calls that metabolic modelling actually makes, and to know what each one does when the problem is ill-conditioned, stiff, or under-determined — which is most of the time.

The environment, and why the versions are part of the result

Create a virtual environment per project, install numpy, scipy, pandas and matplotlib into it, and freeze the versions. This is not hygiene for its own sake: solver defaults change between releases, and a notebook that produced one answer in 2024 can produce a different one in 2026 without any edit. A result you cannot reproduce is not a result, and the cheapest insurance is a `requirements.txt` and a note of the Python version in the README.

One habit worth adopting now, because it will save you in week 51: restart the kernel and run the notebook top to bottom before you believe any figure in it. Notebooks accumulate state, and most wrong figures in science are not wrong calculations but stale variables.

Arrays, and the two mistakes everyone makes

A numpy array is a typed, fixed-shape block of memory; a Python list is not. The two mistakes are shape and dtype. Shape: a vector of length 3 has shape `(3,)`, not `(3,1)`, and broadcasting will silently turn a sloppy combination of the two into a 3×3 matrix rather than raising an error. If a result is suddenly a matrix where you expected a vector, print the shapes, and do it before you print the values. Dtype: an array created from integers stays integer, so dividing it can truncate on some operations and silently promote on others. Write `np.array([...], dtype=float)` for anything numerical.

```
import numpy as np
S = np.array([[ 1, -1,  0,  0],
              [ 0,  1, -1,  0],
              [ 0,  0,  1, -1]], dtype=float)   # 3 metabolites x 4 reactions
v = np.array([1.0, 1.0, 1.0, 1.0])
print(S.shape, v.shape, (S @ v))                 # (3, 4) (4,) [0. 0. 0.]
```

The linear algebra metabolic modelling actually uses

Four calls cover almost everything in weeks 13 to 16.

- `np.linalg.solve(A, b)` for a square, determined system. Use this, not `inv(A) @ b`.
- `np.linalg.lstsq(A, b, rcond=None)` for an over-determined system — more measurements than unknown fluxes, which is the good case in flux analysis, because the residual tells you whether your measurements are consistent with your network.
- `np.linalg.matrix_rank(S)` for the number of independent balances, which is what you actually need for a degree-of-freedom count rather than the number of rows.
- `scipy.linalg.null_space(S)` for the space of steady-state flux vectors, which is chapter 13's entire subject.

Why not invert. Inversion computes more than you asked for, costs more, and is less accurate, but the real reason is diagnostic: `solve` will tell you when the matrix is singular, whereas a nearly singular matrix inverts quietly and returns enormous numbers that look like fluxes. Metabolic matrices are frequently nearly singular, because two reactions in your network are nearly the same reaction, or because a metabolite appears in only one balance. Check `np.linalg.cond(A)`: a condition number near 10^k means you can lose about k digits of the 16 a float carries, so 10^{10} is a warning and 10^{14} means your answer is noise.

```
A = np.array([[1.0, 1.0], [1.0, 1.0 + 1e-10]])
print(np.linalg.cond(A))            # ~4e10: ten digits gone
b = np.array([2.0, 2.0])
print(np.linalg.solve(A, b))        # returns something; believe none of it
```

Integrating differential equations

One call: `scipy.integrate.solve_ivp(fun, t_span, y0, ...)`, where `fun(t, y)` returns dy/dt. Note the argument order — t first, then y — which is the opposite of the old `odeint` convention and the single most common source of a model that runs and is wrong. Pass parameters with `args=(p,)` rather than by closing over globals, so the same function can be

called by a fitting routine later.

Two features earn their keep. Events stop integration on a condition, which is how you find the time at which a substrate is exhausted rather than inspecting the output afterwards; set `event.terminal = True` and, if you need the crossing direction, `event.direction`. And `dense_output=True` gives you an interpolant you can evaluate anywhere, which is what you want when fitting to data taken at times that are not your solver's steps.

Stiffness is the thing you must recognise. A system is stiff when it contains processes on widely separated timescales, which metabolic models always do: enzyme steps equilibrate in milliseconds and pools fill over minutes. An explicit solver such as RK45 is then forced to take steps set by the fastest process even when nothing interesting is happening, and the cost is brutal. The figure measures it on the standard stiff test problem: RK45 needed 243,050 evaluations of the right-hand side, LSODA needed 305 and BDF 329, for the same tolerances and the same answer. That is a factor of roughly 800.

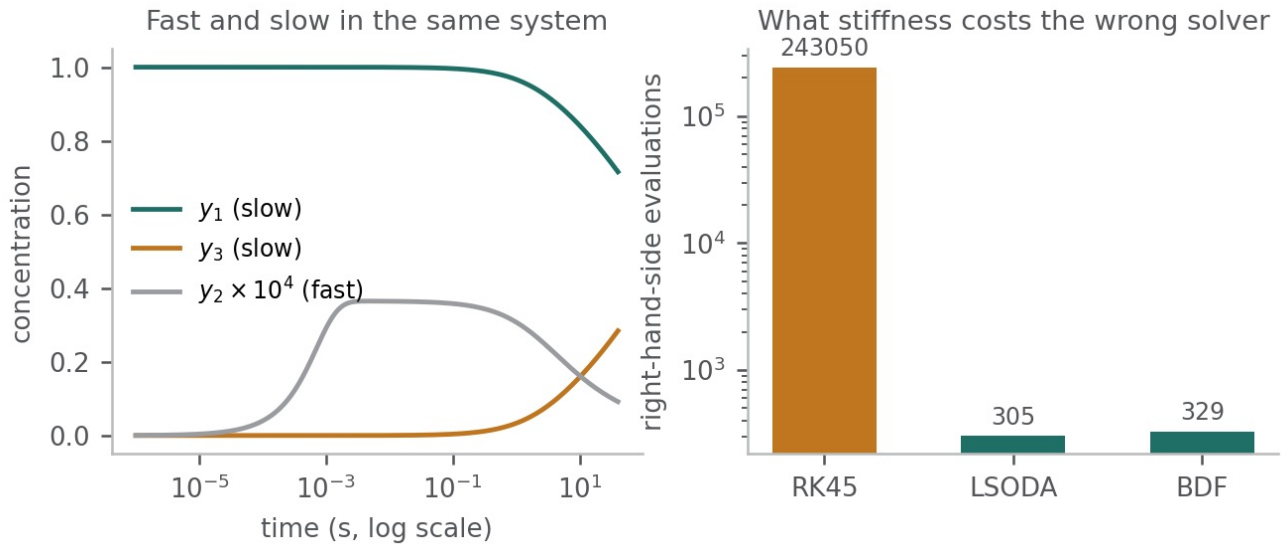


Figure 12.1 Left: the Robertson problem, the standard stiff benchmark, whose middle species moves four orders of magnitude faster than the other two. Right: cost of solving it to the same tolerance (rtol 10^{-6}, atol 10^{-9}) with an explicit and two implicit solvers, counted as right-hand-side evaluations. Computed for this module with SciPy 1.x.

The practical rule: start with `method='LSODA'`, which detects stiffness and switches, and reach for `'BDF'` or `'Radau'` when you know the system is stiff and want the Jacobian used properly. If a model takes minutes to integrate, you have a solver problem and not a biology problem.

Fitting, and reading the uncertainty you get back

`scipy.optimize.curve_fit` returns the parameters and their covariance matrix. The standard errors are the square roots of its diagonal, and the off-diagonal entries matter more than most people notice: normalise the covariance to a correlation matrix and you will frequently find two parameters correlated at 0.99, which means your data constrain their ratio but not either one. This is the first appearance of identifiability, and week 16 treats it properly.

```
from scipy.optimize import curve_fit
def mm(S, Vmax, Km):
    return Vmax * S / (Km + S)

popt, pcov = curve_fit(mm, S_data, v_data, p0=[1.0, 0.5],
                       bounds=([0, 0], [np.inf, np.inf]))
se   = np.sqrt(np.diag(pcov))
corr = pcov / np.outer(se, se)
print(popt, se, corr[0, 1])
```

Three cautions. First, bounds change the problem: with bounds, `curve_fit` switches algorithm, and a parameter pinned at a bound has a meaningless standard error — if V_{max} comes back at exactly its upper bound, the fit failed, it did not succeed. Second, `curve_fit` assumes constant Gaussian error unless you pass `sigma`; enzyme assays usually have error proportional to the rate, so pass `sigma=0.05*v_data` and `absolute_sigma=False` rather than letting the high-rate points dominate. Third, fitting Michaelis–Menten to a transformed linear form — Lineweaver–Burk and its relatives — is a nineteenth-century convenience that distorts the error structure; fit the hyperbola directly.

Five checks that catch most modelling errors

Models fail silently, and the failures are nearly always one of five things. Build the habit of running these before interpreting any output; together they take a minute and they have saved me more time than any debugging skill.

- **Units and scale.** Print the steady-state concentrations and fluxes and ask whether they are physically possible: a metabolite at 10 M or a flux implying a doubling time of 20 seconds means a parameter is in the wrong units. Most model errors are unit errors wearing a biological costume.
- **Conservation.** Sum what should be conserved — total cofactor, total carbon entering against leaving — and confirm it is constant to numerical precision over the integration. A drift means a sign error in S.
- **The zero test.** Set a V_{max} to zero and confirm the flux through that branch goes to zero and that the model does something sensible rather than returning negative concentrations. Negative concentrations are always a modelling error, never a result.
- **The steady-state cross-check.** Integrate to steady state, then solve for the steady state directly with a root finder and confirm they agree. Disagreement means the integration did not converge or there are multiple steady states — both worth knowing.
- **Independence from the initial condition.** Rerun from different starting concentrations. If the steady state moves, you have multistability, which is a real phenomenon but one you should discover deliberately rather than by accident in a figure.

```
from scipy.optimize import fsolve

def steady_state(p, guess):
    f = lambda c: rhs(0.0, c, p)          # dc/dt = 0
    c, info, ok, msg = fsolve(f, guess, full_output=True)
    assert ok == 1, msg
    assert np.all(c > 0), "negative concentration: check the model, not the solver"
    return c, rates(c, p)
```

Note the two assertions. A model that silently returns a failed solve or a negative pool will produce a figure, and a figure is persuasive; an assertion that fires costs you a minute and saves you a conclusion.

A repository someone else can run

The week's deliverable is not code, it is a repository: a README saying what the thing is, a pinned requirements file, one notebook that runs top to bottom from a clean kernel, figures written to files rather than only displayed, and results exported as CSV rather than read off a plot. The test is social, not technical: a colleague clones it and reproduces your figure without asking you a question. That standard is also what week 51 and your capstone will be judged by, and it is much cheaper to adopt now than to retrofit.

Determined system	`np.linalg.solve(A, b)` — never `inv(A) @ b`
Over-determined	`np.linalg.lstsq(A, b, rcond=None)`; the residual is a consistency check
Conditioning	`cond(A) ≈ 10^k` means you lose about k of your 16 digits
Steady-state fluxes	`scipy.linalg.null_space(S)`; dim = `n_reactions − rank(S)`
Integration	`solve_ivp(f, [t0, t1], y0, args=(p,), method='LSODA')` # `f(t, y)`
Standard errors	`se = sqrt(diag(pcov))`; `corr = pcov / outer(se, se)`

You fit V_{max} and K_M to eight points, all at substrate concentrations below $0.2 \cdot K_M$. The fit converges and reports $V_{max} = 4.1 \pm 2.8$ and $K_M = 1.9 \pm 1.4$, correlated at 0.997. What do you report?

That the data determine k_{cat}/K_M and nothing else. At substrate far below K_M the rate law collapses to $v \approx (V_{max}/K_M)\cdot[S]$, so the data constrain only the ratio; the fitter can raise both parameters together along that ratio and the residual barely changes, which is exactly what a correlation of 0.997 and relative errors near 70% are telling you. Report V_{max}/K_M with its own confidence interval, state that the individual parameters are not identifiable from this design, and — the important part — say what would fix it: points at and above K_M, which means an experiment, not a better optimiser.

The general lesson, which returns in week 16: a converged fit is not evidence that the parameters are determined. Always look at the correlation matrix, and prefer to report the combination the data actually constrain.

Three things people get wrong

- ‘The fit converged, so the parameters are right.’ Convergence says the optimiser stopped; the covariance says whether the data had anything to say.

- ‘The model is slow because it is big.’ Three ODEs can be slow if stiff and you chose an explicit solver. Try LSODA before you blame the model.
- ‘`inv` and `solve` are the same thing.’ They differ exactly when the matrix is nearly singular, which is when you most need to be told.

Self-check — answer on paper first

1. Your stoichiometric matrix has 12 rows and 15 columns, and `matrix_rank` returns 10. How many degrees of freedom does the system have, and what does the gap between 12 and 10 mean?

Fifteen reactions minus a rank of 10 leaves five degrees of freedom, so five fluxes must be measured or assumed. The gap between 12 rows and rank 10 means two of the metabolite balances are not independent — typically a conserved moiety such as a cofactor pair whose total is fixed, or a metabolite that appears in only one reaction so its balance duplicates another. Counting rows rather than rank would have made you look for two measurements you do not need.

2. Why does passing `args=(p,)` matter rather than reading parameters from the enclosing scope?

Because the same right-hand-side function then has an explicit parameter interface, which is what a fitting routine, a sensitivity analysis and a parameter sweep all need: each must call it with different values without editing the function or mutating a global. The practical symptom of getting this wrong is a sensitivity analysis that reports zero sensitivity, because every call silently used the same global parameter set.

3. Give two signs, visible without any biological knowledge, that a model you have been handed is stiff.

The solver takes a very large number of small steps relative to the simulated interval, visible as a huge `nfev` or a dense `sol.t`; and the solution contains variables whose characteristic times differ by several orders of magnitude, visible as a species that reaches a plateau immediately on a plot where the others are still moving. A third sign is that switching to LSODA or BDF changes the run time by orders of magnitude while leaving the answer unchanged.

What to open a book for. No book. The NumPy absolute-basics and linear-algebra guides, the SciPy integrate and optimize references, and the ten-minute pandas tour are the sources, and they are better than any textbook treatment because they are versioned with the software. Read the `solve_ivp` page in full once; you will use it for the next fifty weeks.

Stoichiometric modelling: S·v = 0 and what its null space is telling you

S·v = 0 is a mass balance around every metabolite in the cell, written once. You have been writing these since your second year; the only new part is that there are two thousand of them and you solve them with a matrix.

Why the steady state is legitimate

The claim is that intracellular metabolite concentrations are constant, so for every internal metabolite, production equals consumption. That is obviously false in detail — pools move constantly — and it is nonetheless a good approximation for a reason you can quantify. Metabolite pools are small and turnover is fast: a pool at 0.1 mM carrying a flux of 1 mM/s turns over in 0.1 s, while the biomass around it doubles in an hour. Any transient in the pools is over before the culture has changed measurably, so on the timescale on which you make and measure a fermentation, the network is at a pseudo-steady state. The honest statement is that the concentrations are quasi-stationary relative to the growth and feeding timescale, and that is why the approximation is the *pseudo*-steady state rather than the steady state.

It fails where you would expect it to fail: immediately after an induction or a substrate pulse, during a shift between substrates, and for any pool that is not small compared with its flux — a storage polymer, a product accumulating in the cytoplasm. In those cases you need chapter 16.

Building S, and reading it as what it is

Rows are metabolites, columns are reactions, and the entry S_{ij} is the stoichiometric coefficient of metabolite i in reaction j, negative for consumption and positive for production. Then $(S \cdot v)_i$ is the net rate of accumulation of metabolite i, and setting it to zero is the mass balance. Nothing in that sentence is biology.

Take a five-reaction toy network: substrate enters as A, A branches to B and to C, and both branches leave as product.

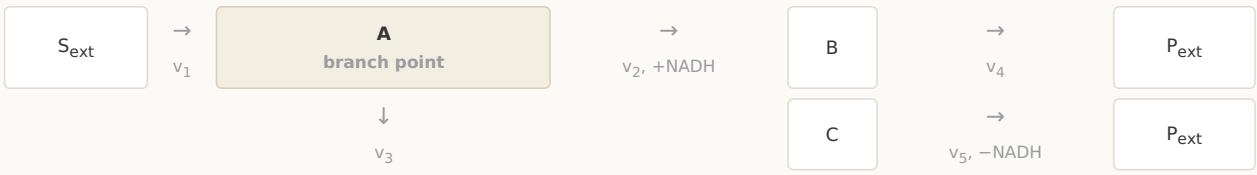
```
v1: S_ext -> A          A: [ 1 -1 -1  0  0 ]
v2: A -> B              B: [ 0  1  0 -1  0 ]
v3: A -> C              C: [ 0  0  1  0 -1 ]
v4: B -> P_ext
v5: C -> P_ext          rank(S) = 3, so 5 - 3 = 2 degrees of freedom
```

Solve it by hand: the A balance gives v1 = v2 + v3, the B balance gives v2 = v4, the C balance gives v3 = v5. Two quantities remain free — the overall rate, and how A splits between the two branches. That is exactly what the null space dimension of 2 says, and it is worth pausing on, because it is the general lesson: **the null space is the set of flux distributions consistent with your network, and its dimension is the number of things the network does not determine**. Stoichiometry constrains; it does not decide.

Now add a cofactor and watch the geometry change. Suppose v2 produces one NADH and v5 consumes one, and that NADH must balance like any other metabolite:

```
NADH: [ 0  1  0  0 -1 ]     rank(S) = 4, null space dimension = 1
null space basis, scaled:  v = (2, 1, 1, 1, 1)
```

The split ratio is no longer free. The redox balance has fixed it at one-to-one, because the only NADH sink is the lower branch and the only source is the upper one. This is the thing to take from the week: in real networks the cofactor rows do most of the constraining, which is why a pathway design that ignores them has far more apparent freedom than the cell does. When someone tells you a flux distribution is "possible", ask whether they balanced NAD(P)H and ATP or only carbon.



Without the cofactor: two degrees of freedom — rate and split. With it: one — rate only. Carbon says the split is free; redox says it is not.

Figure 13.1 The toy network of this chapter. Ranks and the null-space basis (2, 1, 1, 1, 1) were computed with `scipy.linalg.null_space` rather than

asserted; reproduce them in six lines before going on. Drawn for this module.

Degrees of freedom, and the three honest ways to close them

Count the degrees of freedom as (number of reactions) − rank(S), not minus the number of rows: rows are frequently dependent, typically through a conserved moiety — the total of NAD^+ and NADH is fixed if no reaction makes or destroys the ring — or because a metabolite appears in only one reaction. Using rank saves you looking for measurements you do not need.

With f degrees of freedom you need f pieces of outside information, and there are exactly three kinds.

- **Measure fluxes.** Exchange rates are the cheap ones: substrate uptake, oxygen uptake, CO_2 evolution, product secretion, growth rate. Each is a boundary flux you can fix from a fermentation balance, and together they frequently close most of the gap. This is the classical metabolic flux analysis route, and it requires nothing but a well-run fermentation.
- **Assume.** Fix a split ratio from literature, set a reaction's flux to zero because the gene is deleted, or constrain a reversible reaction's direction. Assumptions are legitimate and they must be stated, because an assumed flux returns in your answer looking exactly like a measured one.
- **Optimise.** Declare an objective — most often growth — and ask which point in the null space maximises it subject to capacity bounds. This is flux balance analysis, and it is a hypothesis about what the cell is doing, not a deduction from the network. It is remarkably good for wild-type growth on a single substrate and frequently poor for an engineered strain carrying a burden, which is a distinction week 51 will make you defend.

If you have more measurements than degrees of freedom the system is over-determined, which is the good case: solve it with `lstsq` and the residual becomes a consistency check on your measurements. A large residual means your carbon balance does not close, and the usual causes are an unmeasured by-product, a wrong biomass composition, or an uncalibrated dry-weight conversion — all of which you would rather learn now than after you have published the flux map.

Elementary flux modes, and when they are worth the trouble

A null-space basis is mathematically arbitrary: the vectors returned above are orthonormal and contain negative entries, which makes them useless as pathways. Elementary flux modes fix this. An elementary flux mode is a flux vector that satisfies the steady state, respects the irreversibility of every reaction in it, and is minimal, in the sense that removing any of its reactions destroys the steady state. Those three conditions make each mode a genuine route through the network, and every feasible steady-state flux distribution is a non-negative combination of modes.

For the toy network without the cofactor, the modes are the two obvious ones: (1, 1, 0, 1, 0) and (1, 0, 1, 0, 1), the two routes from substrate to product. Add the NADH balance and there is a single mode, (2, 1, 1, 1, 1), which says the only way the network can run at steady state is both branches together in a fixed ratio. Extreme pathways are a closely related construction with extra conventions on how exchange reactions are split; the practical difference rarely matters, and the distinction is worth knowing mainly because both terms appear in the literature.

The catch is combinatorial. The number of elementary modes grows faster than exponentially with network size, and genome-scale networks have more modes than can be enumerated. So modes are a tool for a pathway or a subnetwork of a few dozen reactions — where they are genuinely illuminating, because they tell you every route and therefore every theoretical yield — and not for a whole organism, where constraint-based optimisation is the only tractable approach.

Biomass as a reaction, and the linear programme it makes possible

The step that converts a stoichiometric network into a usable model of a growing cell is the biomass reaction: one artificial reaction that consumes the twelve precursors of week 7, plus amino acids, nucleotides, lipid and cofactors, in the proportions measured for that organism, together with the ATP needed to polymerise them, and produces one gram of dry cell mass. Its flux is the growth rate, in h^{-1}.

Two things about it are worth knowing, because both are routinely misused. First, the coefficients are measurements — of a particular strain, in a particular medium, at a particular growth rate — and they shift with conditions: protein fraction rises with growth rate, which changes the amino acid demand. A model using a 1990s composition for a modern engineered strain in a defined medium is making an assumption you should state. Second, the ATP term splits into growth-associated maintenance, which scales with biomass produced, and non-growth-associated maintenance, a constant drain that the cell pays whether it grows or not. The second is what makes a slow fed-batch expensive: at low growth rate, maintenance consumes a growing fraction of the substrate, and it is why yield on substrate falls as you slow the feed.

With a biomass reaction and capacity bounds, the under-determined network becomes a linear programme:

```
maximise    c·v           (usually the biomass flux)
subject to  S · v = 0     (the balances of this chapter)
            lb ≤ v ≤ ub   (irreversibility, measured uptake, capacity limits)
```

That is flux balance analysis, and recognising it as an LP is most of what you need: the solution sits at a vertex of the feasible region, which is why FBA solutions tend to be sparse and all-or-nothing, and why alternative optima are common — several vertices can give the same objective value with different flux distributions. If you report an FBA flux map, check for alternative optima before interpreting any individual flux, because the one the solver returned is an arbitrary choice among them.

The dual of the LP is where the engineering insight is. Each constraint has a shadow price: the improvement in the objective per unit of relaxation. The shadow price on oxygen uptake tells you what another unit of oxygen transfer is worth in growth or product — directly comparable with the cost of supplying it — and a shadow price of zero tells you a constraint is not binding, so effort spent relaxing it is wasted. This is the same reduced-cost reasoning you would apply to any process optimisation, and it transfers without translation.

Two standard variants are worth naming now because you will meet them constantly. Flux variability analysis fixes the objective at its optimum and then minimises and maximises each flux in turn, returning the range each flux can take without changing the objective — which is the honest way to report an FBA result. And minimisation of metabolic adjustment, and its relatives, replace the growth objective for engineered strains with the assumption that a mutant stays as close as it can to the wild-type flux distribution, which predicts knockout phenotypes better than growth maximisation does, precisely because an engineered cell has not had time to evolve toward optimality.

The balance	`S·v = 0, one row per internal metabolite, one column per reaction`
Degrees of freedom	`f = n_reactions − rank(S); rank, not row count`
Solution space	`v ∈ null(S); dimension f is what the network does not determine`
Over-determined	`more measurements than f → lstsq, and the residual tests your data`
Yield from a mode	`Y_max = max over modes of (product flux / substrate flux)`

Your 42-reaction model of central metabolism has rank 38 and you have measured glucose uptake, oxygen uptake, CO_2 evolution, acetate secretion and growth rate. Is the system determined, and what do you do with the answer?

Forty-two reactions minus rank 38 leaves four degrees of freedom and you have five measurements, so the system is over-determined by one. Solve it as a least-squares problem rather than discarding a measurement: the single redundancy gives you one consistency check, and the residual on it is the only internal evidence you will get that the flux map means anything. If the residual is small, you report fluxes with some confidence; if it is large, one of the five numbers is wrong, and the candidates in order of likelihood are the CO_2 measurement (gas-phase calibration drifts), the biomass composition used to convert growth rate into precursor drains (usually taken from a paper on a different strain), and an unmeasured by-product.

What you must not do is quietly drop the redundancy to make the system exactly determined. That converts your one piece of validation into a number that cannot be wrong, which is the same as having no validation at all.

Three things people get wrong

- ‘The null space gives the flux distribution.’ It gives the set of candidates; measurements or an objective choose among them.
- ‘Degrees of freedom equal reactions minus metabolites.’ Minus the *rank*. Conserved moieties make rows dependent.
- ‘Elementary modes scale to genome models.’ They do not; their number explodes. Use them on pathways.

Self-check — answer on paper first

1. Explain S·v = 0 to a chemical engineer who has never seen it, in three sentences.

Each row is a steady-state material balance around one chemical species inside the cell: what is made equals what is consumed, with the stoichiometric coefficients supplying the conversion factors. The vector v is the set of reaction rates, so the matrix product is the accumulation term, set to zero exactly as in any continuous process at steady state. The only unfamiliar part is scale — a few thousand simultaneous balances — and the consequence that the system is usually under-determined, so the balances bound the fluxes rather than fixing them.

2. Your network has one free degree of freedom after all measurements, and the free parameter is the split at a branch point. Give two different ways to resolve it, and state what each one costs in credibility.

Measure it, by feeding a labelled substrate and inferring the split from labelling patterns — expensive in time and instrumentation but it is a measurement. Or assume it, from a published value or by assuming the cell maximises growth, which costs nothing and makes every downstream number conditional on that assumption; the cost is that a reader cannot distinguish your assumed split from your measured fluxes unless you label it clearly, and conclusions that depend on it must be reported with that dependence.

3. Why does adding a cofactor balance usually reduce the apparent freedom of a designed pathway more than adding another carbon reaction?

Because cofactors appear in many reactions across the whole network, so their balance couples parts of the network that share no carbon, whereas a carbon reaction adds one column and one or two entries. A single NADPH balance can tie a biosynthetic branch to the pentose phosphate pathway and to respiration simultaneously, which is why designs that balance only carbon look feasible and designs that balance carbon, redox and ATP usually do not.

What to open a book for. Stephanopoulos Ch. 8, 12 and 13 are the core, and Ch. 12 on determined and under-determined systems is the one to work with a pen. Palsson Ch. 1 to 4 give the same material in the constraint-based language you will meet in the software, and his treatment of the null space is the clearer of the two. Do the algebra by hand first and in numpy second; the agreement is the lesson.

Driving force and transport: whether a step can run, and whether the product can leave

Two questions decide more designed pathways than any kinetic argument: can this step carry flux in the direction I need at the concentrations the cell can hold, and can the product get out of the cell at the rate I am making it.

ΔG′, computed rather than quoted

Week 4 gave you ΔG′ = ΔG°′ + RT·ln Q. This week you use it on real numbers, and the exercise changes how you read pathways. With classic erythrocyte metabolite concentrations, hexokinase comes out near −27 kJ/mol and phosphofructokinase near −26: both far from equilibrium, effectively irreversible, both regulated. Triosephosphate isomerase comes out near zero, as do phosphoglucose isomerase, aldolase, phosphoglycerate mutase and enolase. Pyruvate kinase is far from equilibrium again, near −14 kJ/mol.

The pattern is the point: the three far-from-equilibrium steps of glycolysis are precisely the three regulated steps. That is not a coincidence, it is a design principle you can use. A reaction near equilibrium cannot be a control point, because its net flux is a small difference between two large opposing fluxes and is set by the concentration ratio rather than by the enzyme; doubling such an enzyme doubles both directions and changes the net almost not at all. A far-from-equilibrium step is where control can sit, and therefore where regulation evolved and where your intervention has a chance.

Two cautions. The values are organism- and condition-specific, so quote your source — and eQuilibrator, which applies a consistent component-contribution estimate and lets you set pH and ionic strength, is the right source rather than a table copied from a textbook. And the classification is about the magnitude of ΔG′, not its sign: a step at −2 kJ/mol runs forward, but only just, and a modest accumulation of its product will stop it.

The driving force of a pathway, and the max-min idea

A designed pathway can have a comfortably negative overall ΔG′ and still be unrunnable, because what stops it is its worst step. The max-min driving force, from Noor and colleagues, formalises this: choose the intermediate concentrations, within physiologically allowed bounds — conventionally 1 µM to 10 mM, with cofactor ratios fixed at measured values — so as to maximise the smallest driving force across all steps. The result is a single number per pathway, and it ranks competing designs.

Read the output rather than the number alone. If the optimum has the smallest −ΔG′ still negative in the wrong direction, the pathway cannot run at any feasible concentration and no enzyme will rescue it. If the optimised minimum is a few kJ/mol, the pathway runs but only with intermediates pinned at the top or bottom of their allowed range, which tells you two useful things: which intermediate must be kept low, and that the design is fragile to anything that lets that intermediate accumulate. And the solution vector tells you which concentrations the design depends on — the diagnostic, not the score, is what you want.

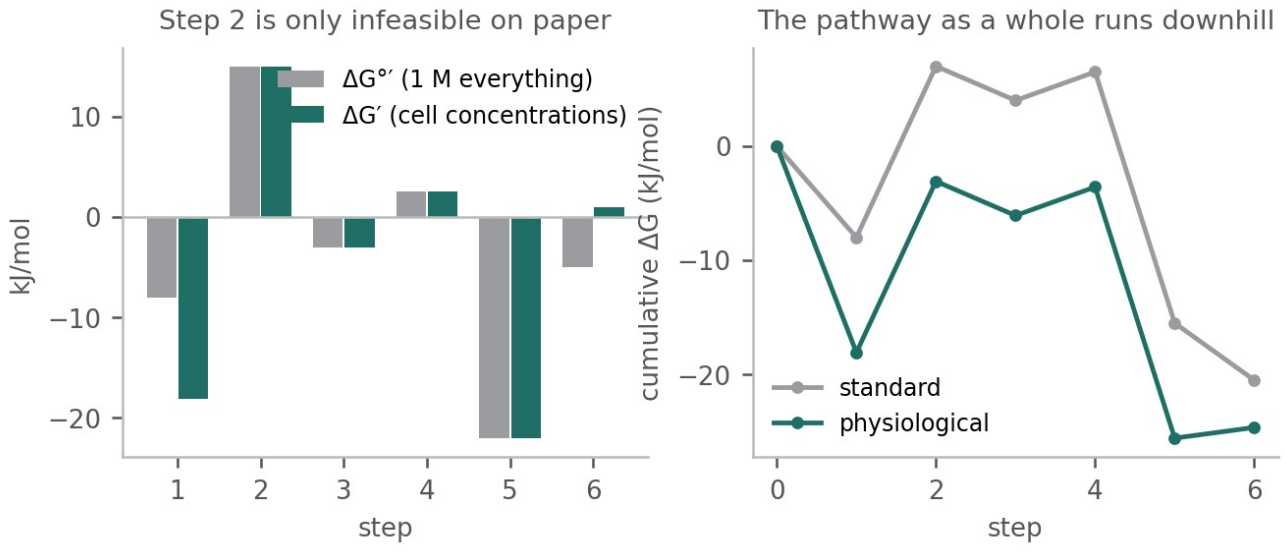


Figure 14.1 A six-step pathway with assumed standard free energies, computed at 37 °C with substrate at 5 mM, intermediates at 0.1 mM and product at 1 mM. Step 2 is positive under standard conditions and negative at cell concentrations — the usual situation and the reason standard values alone never condemn a design. Parameters are illustrative, the arithmetic is not.

Getting out: diffusion, and why it is usually not enough

Fick's law across a membrane is J = P·ΔC, where the permeability coefficient P collects the partition coefficient between water and lipid, the diffusivity in the membrane, and the membrane thickness. The numbers are unforgiving. For a small neutral solute, P is of order 10^{-6} cm/s, which is 10^{-8} m/s; with a 10 mM inside-to-outside difference, that is 10 mol/m³, so J = 10^{-7} mol·m^{-2}·s^{-1}, about 0.36 mmol per square metre per hour.

Whether that is enough depends on how much membrane a cell has per unit of production. A 1 µm cell has roughly 3 µm² of surface, and a cell producing at 2 fmol per hour — about 0.2 g/L/h of a 100 Da product at 10^9 cells per mL — needs a flux of about 1.6 × 10^{-7} mol·m^{-2}·s^{-1}. So for a small neutral molecule, passive diffusion is of the same order as the requirement: marginal, and it only works because the concentration gradient can be made large. For a polar molecule P falls by two orders of magnitude and for a charged one by four to six, and diffusion stops being an option at all.

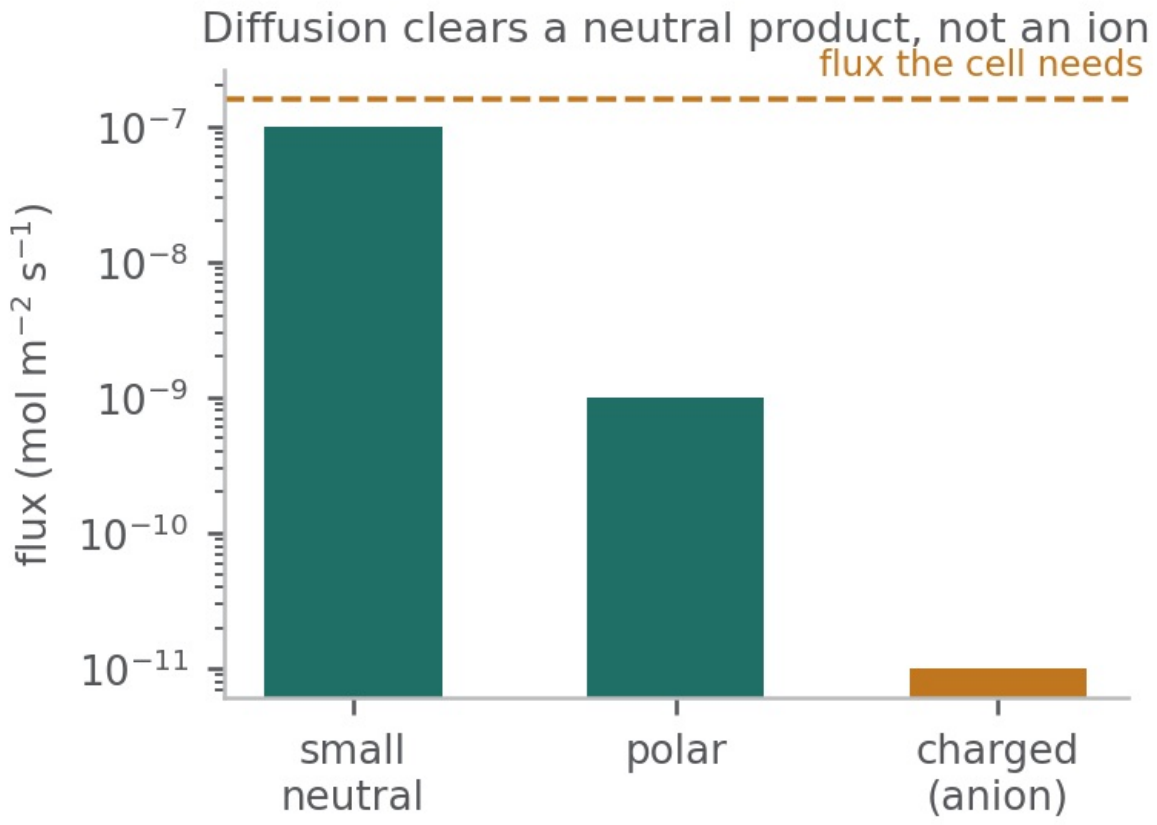


Figure 14.2 Diffusive flux at a 10 mM gradient for three permeability classes, against the flux a producing cell requires. Assumptions, all stated in the text: P = 10^{-8}, 10^{-10} and 10^{-12} m/s; 3 µm² of membrane per cell; 2 fmol per cell per hour. Computed for this module.

This is why the question "is my product charged at cytoplasmic pH?" belongs on the first page of a design. Almost every organic acid, amino acid and phosphorylated compound is, and each therefore needs a transporter, which means either finding one, engineering specificity into one, or accepting intracellular accumulation with the ceiling that part 1 computed.

Weak acids, pH partitioning and the trap

A weak acid crosses as its neutral form and dissociates on the other side, so its equilibrium distribution follows the pH on each side. For an acid of pK_a 4.8 with a cytoplasm at pH 7.6, the ratio of total acid inside to outside is $(1 + 10^{pH_{in} - pK_a}) / (1 + 10^{pH_{out} - pK_a})$: about 4 at external pH 7, about 38 at pH 6, and about 245 at pH 5.

Two consequences, which are the same physics read in two directions. For an organic acid product, acidifying the broth — which you may want to do for downstream processing, or which may happen because the product is an acid — traps the product inside the cell as the anion and dissipates the pH gradient, because each neutral molecule that diffuses in releases a proton into the cytoplasm. That is the mechanism of weak-acid toxicity, and it is why acetate at 5 g/L hurts far more at pH 5.5 than at pH 7. For an acid you are feeding or an inhibitor you are trying to exclude, the same arithmetic tells you what the external pH buys you.

Carriers, pumps and what export costs

A transporter is an enzyme with a V_{max}, which is the structural difference from diffusion: diffusive flux rises indefinitely with the gradient, carrier flux saturates. So a strain that works at low titer can hit a transport ceiling at high titer, and the symptom — intracellular accumulation with flux falling as titer rises — looks exactly like product inhibition of the pathway. Distinguishing them requires measuring the intracellular concentration.

The thermodynamic cost of moving product against a gradient is $RT \cdot \ln(c_{out}/c_{in})$: 5.8 kJ/mol per tenfold at 30 °C, so concentrating from 10 mM inside to 1 M outside costs 11.6 kJ/mol. Against ATP hydrolysis at roughly 40 to 60 kJ/mol in vivo, that looks almost free, and in the reversible limit one ATP would move a molecule against many orders of magnitude. Real transporters run far from reversibility, so budget one ATP, or one to two protons of the motive force, per molecule exported, and put it in the yield calculation. For a product made at 1 mol per mol of glucose, one ATP per molecule exported is a few per cent of the cell's energy budget — not decisive, but not negligible either, and it belongs in the balance rather than in a footnote.

One more number: why the cell is well mixed and the vessel is not

The characteristic diffusion time is $t \approx L^2/D$. With $D \approx 5 \times 10^{-10}$ m²/s and L = 1 µm, t ≈ 2 ms, which is faster than most enzyme turnovers: the cytoplasm is well mixed, and treating it as a single compartment in your model is justified. Put L = 10 m into the same expression and you get about 6,300 years. Mixing in a vessel is therefore entirely mechanical, and the gradients in substrate, oxygen and pH that a cell experiences as it circulates are real and are the subject of weeks 40 onward. The same two-line calculation justifies your model's central assumption and condemns the scale-up.

Cofactor ratios are part of Q, and they are not free parameters

Most interesting reactions involve a cofactor pair, and the pair's ratio enters the mass-action ratio directly: for a reduction, Q contains $[NAD(P)^+]/[NAD(P)H]$, so a reaction's driving force depends on the redox state of the pool it draws on, not merely on the metabolites you were thinking about. The two pools are held at deliberately different ratios — the catabolic pool oxidised, the anabolic pool reduced, as part 1 chapter 4 explained — and the difference between their ratios is orders of magnitude.

Three consequences follow, and they are the reason this section exists rather than being a footnote to week 4.

- A reduction driven by NADPH has a substantially larger driving force than the same reduction driven by NADH, simply because the pool it draws on is more reduced. So a cofactor swap made for supply reasons can weaken the thermodynamics of the step you swapped. Check it; it takes two minutes.
- The ratios are set by the cell's own balance of production and consumption, so they are not parameters you can choose in a model. Fix them from measurements for the organism and condition, and when you optimise concentrations for a driving-force calculation, hold the ratios fixed — which is exactly what the standard max-min driving force formulation does, and why.
- Your pathway changes them. A large NADPH demand pulls the anabolic pool toward oxidised, which reduces the driving force of every NADPH-dependent step in the cell, including the ones you did not install. That coupling is invisible in a pathway-level calculation and is one of the standard reasons a design behaves worse in the cell than on paper.

Turning thermodynamics into constraints a model can use

The practical payoff of this week is that ΔG′ becomes a bound rather than an observation. In a constraint-based model, a reaction whose driving force is positive in the forward direction at every feasible concentration can have its lower bound set to zero: you have eliminated a direction, and therefore a whole family of spurious solutions, without any experimental data. Applied systematically this is thermodynamic flux balance analysis, and its main practical effect is to remove the internal cycles that plain FBA permits — loops that carry flux round a set of reactions with no net conversion, which are thermodynamically impossible and which plain FBA will happily return because $S \cdot v = 0$ does not forbid them.

The reverse use is a design filter. Before committing to a heterologous route, compute the max-min driving force for every candidate route to the same product and discard those whose optimised worst step is positive. This costs an afternoon with eQuilibrator and routinely eliminates designs that would otherwise consume a year, which makes it, hour for hour, the highest-return calculation in the curriculum.

Real driving force	`ΔG′ = ΔG°′ + RT·ln Q; RT = 2.58 kJ/mol at 37 °C, 2.52 at 30 °C`
Near equilibrium	`|ΔG′| small → net flux set by concentrations, not by enzyme amount`
Max-min driving force	`maximise min(−ΔG′) over concentrations in 1 µM–10 mM`

Diffusion	`J = P·ΔC; P ≈ 10⁻⁸ m/s neutral, 10⁻¹² m/s charged`
pH partitioning	`c_in/c_out = (1 + 10^(pH_in−pKa)) / (1 + 10^(pH_out−pKa))`
Export cost	`RT·ln(c_out/c_in) = 5.8 kJ/mol per tenfold at 30 °C`
Mixing time	`t ≈ L²/D: 2 ms across a cell, ~6,300 years across 10 m`

Your product is a C6 dicarboxylic acid, pK_a values near 4.4 and 5.6, and you are targeting 80 g/L. Walk through the transport questions in the order you would ask them.

First: at cytoplasmic pH 7.6 both carboxyls are deprotonated, so the species inside the cell is a dianion and passive diffusion is excluded — P for a doubly charged species is some six orders below a neutral solute, and figure 14.2's arithmetic says that is five orders short of the required flux. So an exporter is mandatory, not optional, and finding or engineering one is a project in its own right rather than a detail of the pathway.

Second: the cost. Eighty grams per litre of a 146 Da acid is about 0.55 M outside; against an intracellular concentration of, say, 10 mM that is a 55-fold gradient, so RT·ln 55 ≈ 10 kJ/mol at the thermodynamic minimum, and realistically one ATP or one to two protons per molecule. Put that in the ATP balance before you decide the pathway is energetically viable.

Third: pH. The fermentation will be titrated to keep the broth near neutral, which is fortunate, because at pH 5 the mono-protonated form is abundant outside and will diffuse back in, releasing protons and collapsing the pH gradient — the weak-acid toxicity mechanism. This is why organic-acid processes are run either at neutral pH with a base feed and a salt to acidify downstream, or in a host engineered or selected for low-pH tolerance. Both choices are made here, on thermodynamic grounds, not in the plant.

Three things people get wrong

- ‘The pathway is thermodynamically fine, the overall ΔG is very negative.’ What stops a pathway is its worst step, which is what the max-min driving force reports.
- ‘Small molecules diffuse out.’ Neutral ones, marginally. Charged ones, never — and most products are charged.
- ‘Transport is a detail for the process engineers.’ It has a V_{max} and an ATP cost, both of which belong in your yield calculation.

Self-check — answer on paper first

1. Why can a near-equilibrium reaction not be a control point, and what does that imply for target selection?

Because its net flux is the small difference between large forward and reverse fluxes, and that difference is set by how far the mass-action ratio sits from the equilibrium constant. Raising the enzyme scales both directions equally and leaves the net almost unchanged. The implication is that overexpression targets should be chosen from the far-from-equilibrium steps, which is a thermodynamic screen you can run from eQuilibrator before any cloning — and which, combined with the control analysis of week 15, eliminates most candidate targets on paper.

2. A colleague proposes relying on passive diffusion to export a zwitterionic amino acid at 20 g/L. Give the calculation that settles it.

A zwitterion has no appreciable neutral fraction at cytoplasmic pH, so its membrane permeability is in the 10^{-11} to 10^{-12} m/s range. At the most favourable gradient the cell could hold, say 100 mM = 100 mol/m³, $J = P \cdot \Delta C \approx 10^{-10}$ mol·m^{-2}·s^{-1}, against a requirement of order 10^{-7} for a producing cell: three orders of magnitude short, and that is before noting that a 100 mM intracellular concentration is itself near the limit of what the cell tolerates. The conclusion is an exporter, and in fact the amino-acid industry's exporters were identified precisely because of this arithmetic.

3. Your intracellular product concentration rises while the pathway flux falls, as titer increases. Name the two explanations and the experiment that separates them.

Either the exporter has saturated — it has a V_{max}, unlike diffusion — or the product is inhibiting an enzyme of the pathway. Both give falling flux with rising internal concentration. Separate them by overexpressing the transporter alone: if flux recovers and the internal pool drops, transport was the ceiling. The complementary test is in vitro, measuring pathway enzyme activity against product concentration to see whether the inhibition exists at the concentrations you measured inside the cell.

What to open a book for. Lehninger Ch. 11 for membranes and transport, read for the mechanisms rather than the numbers, and Ch. 13's thermodynamics section reread with concentrations in hand. The indispensable tool is not a book but eQuilibrator, and the paper to read is Noor et al., PLoS Comput. Biol. 10:e1003483 (2014), which is short and will change how you look at a proposed pathway.

Metabolic control analysis: why the rate-limiting step is a myth

Control over a flux is a property of the whole pathway, shared out among its steps, and the shares sum to one. That single sentence, once you believe it with numbers in front of you, rewrites your list of engineering targets.

Two coefficients, and the difference between them

A **flux control coefficient** measures how much the pathway's steady-state flux responds to a small change in one enzyme:

```
C^J_i = (∂J / ∂E_i) · (E_i / J)     — a fractional response to a fractional change
```

It is dimensionless, it is a property of the system at a particular state, and it is what you want to know before you clone anything: C = 1 means a 10% increase in that enzyme gives 10% more flux, C = 0.1 means it gives 1%, and C = 0 means it gives nothing.

An **elasticity coefficient** measures how much one enzyme's rate responds to a metabolite concentration, with everything else held fixed:

```
ε^i_S = (∂v_i / ∂S) · (S / v_i)     — a local, enzyme-level property
```

The distinction is the whole conceptual content of the subject. Elasticities are local and measurable in a cuvette: they follow from the rate law of the isolated enzyme. Control coefficients are systemic and cannot be measured on an isolated enzyme at all, because they depend on how the rest of the network responds to the pool changes your perturbation causes. Metabolic control analysis is the machinery that computes the systemic quantities from the local ones, and the reason that matters is that an enzymologist's data can be turned into an engineer's answer.

The summation theorem, and the easiest way to believe it

The flux control coefficients of all steps in a pathway sum to one:

```
Σ_i C^J_i = 1
```

The proof is a scaling argument, and you can verify it computationally in a minute. If you multiply every enzyme in the pathway by the same factor λ, every rate scales by λ at fixed concentrations, so the steady state is unchanged and the flux scales by λ exactly. A fractional change of λ in all enzymes therefore produces a fractional change of λ in flux, and since the total response is the sum of the individual responses, the coefficients must sum to one.

I ran exactly that check on the three-enzyme model used for the figure. Doubling all four steps together — three enzymes and the export reaction — changed the flux by 100.0%, as the theorem requires. Perturbing them one at a time by 1% gave control coefficients of 0.327 for enzyme 1, 0.102 for enzyme 2, 0.108 for enzyme 3 and 0.455 for export, summing to 0.993. The 0.7% shortfall is the finite-difference error in a 1% perturbation, not a failure of the theorem, and the fact that it is small is itself a check that the perturbation was small enough to be local.

What the theorem forbids is the thing everybody says. If one enzyme had a coefficient of 1 — a true rate-limiting step — every other enzyme in the pathway would have to have exactly zero, which would mean that halving any of them changed nothing. That is not what happens in real pathways. Control is distributed, usually with a few steps carrying most of it, and the distribution shifts as soon as you perturb it.

The connectivity theorem, and what it is for

The second theorem links the two kinds of coefficient. For a metabolite S, summing over the enzymes that act on it:

```
Σ_i C^J_i · ε^i_S = 0
```

Its practical use is computational: with the elasticities, which you can get from rate laws, the summation and connectivity theorems give you enough simultaneous equations to solve for all the control coefficients without ever perturbing the real cell. That is the classical route, and it is worth working through once by hand on a two- or three-step pathway so that the algebra is not mysterious. In practice, when you have a kinetic model, you will get the same

numbers faster by numerical perturbation — chapter 16 does it in five lines — and the theorems then serve as the check that your numerics are right.

There is a qualitative reading of connectivity worth carrying. An enzyme with a large elasticity toward its substrate — one far from saturation, responding steeply to its pool — tends to carry *little* control, because it absorbs pool changes rather than transmitting them. Saturated, insensitive enzymes carry control. This is the opposite of the intuition that the "fastest" enzyme matters least, and it is the reason that measuring a pool concentration against the K_M of the enzyme that consumes it is such a cheap diagnostic.

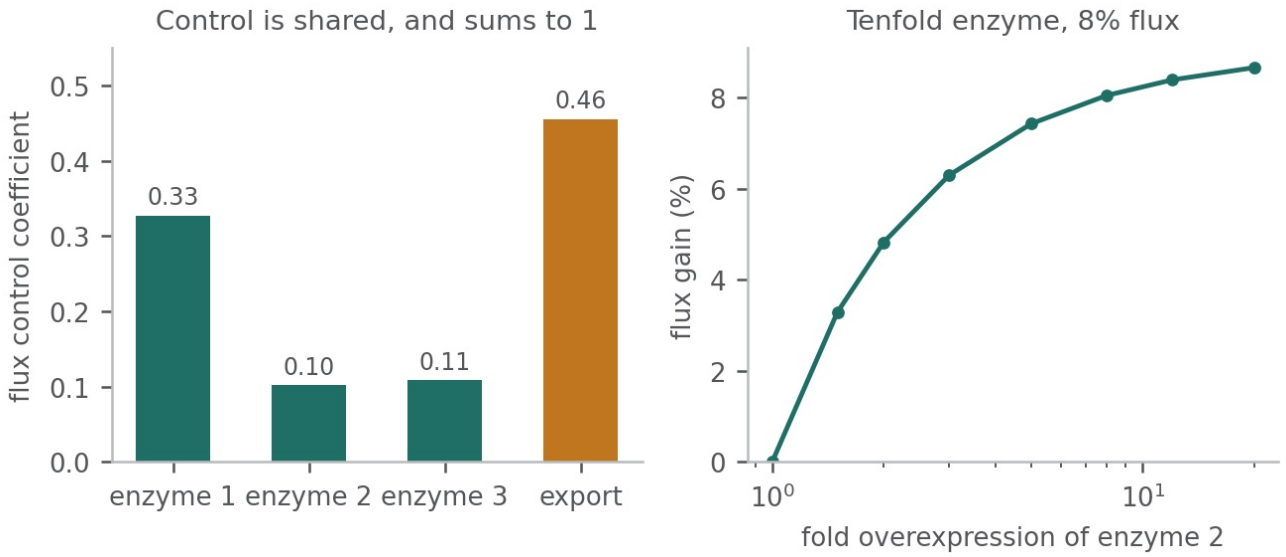


Figure 15.1 Left: flux control coefficients of a three-enzyme pathway with product export and feedback inhibition on step 1, computed by 1% perturbation of each step; they sum to 0.993, with the shortfall being finite-difference error. Right: flux gain against fold overexpression of enzyme 2, which saturates near 8%. Parameters are assumed, the model is in the text of chapter 16, and both panels were computed rather than sketched.

Why overexpressing one enzyme usually fails, in numbers

Enzyme 2 in the model carries a control coefficient of 0.102, so the first 1% of overexpression buys 0.1% more flux. Overexpress it tenfold and you do not get tenfold or even 100% more: you get 8.3%. The curve saturates, and the reason is visible in the model's own output. As $V_{max,2}$ rises, its substrate pool falls from 1.55 to 0.43 in model units while the pool downstream rises, and its own control coefficient collapses from 0.102 to 0.004 — by the time you have overexpressed it twentyfold, that enzyme has no control at all and the limitation has moved somewhere else.

This is the general shape of the result, and it is why the single most common outcome in the literature is a modest gain from a large overexpression. It is also why the answer is not to find the right single enzyme but to change several steps at once — and that is the real argument for multiplexed genome editing in week 23 and for library-based tuning in week 33. Not that editing many genes is impressive, but that distributed control makes it necessary.

One more result from the same model, and it is the most useful one. Removing the feedback inhibition on step 1 — a one-residue change in a real strain — raised flux by 43%, four times what tenfold overexpression of enzyme 2 achieved. And it redistributed control: export control fell from 0.455 to 0.098 while enzyme 3's rose from 0.108 to 0.398. That is what regulation does. It holds control at the demand side, so that the pathway tracks demand rather than supply, and releasing it hands control back to the enzymes — which is exactly why the feedback-resistant variant is so often the first and best intervention, and why the second intervention should be chosen after re-measuring, not from the original analysis.

Where the formalism breaks, and what to do about it

Four limits are worth knowing before you quote a control coefficient at an interview.

- **It is local.** Coefficients are derivatives at one state. A tenfold overexpression is not a small perturbation, and the coefficient that predicted your first 1% does not predict your tenth fold, as the figure shows.
- **Enzyme amount is not V_{max}.** The formalism assumes rate is proportional to enzyme, which fails when the enzyme aggregates, when expression saturates the folding machinery, or when the burden of expression slows growth. In a real strain, overexpression has a cost that the model does not contain.
- **It assumes a steady state exists and is reached.** Where a pathway oscillates or the culture is changing fast, the coefficients are not defined.

- **Boundaries absorb control.** If your model fixes a boundary concentration — substrate held constant, cofactor pool clamped — the control that would have sat on that boundary step disappears, and your coefficients will not sum to one. This is the most common reason a numerical summation check fails, and it is diagnostic rather than fatal.

The two-step case, done by hand

Do this derivation once and the formalism stops being abstract. Take a linear two-step pathway with a fixed external substrate S, one intermediate X, and a fixed external product, with both steps first order in their substrate: $v_1 = k_1(S - X/K_1)$ and $v_2 = k_2X$, where k_i is proportional to enzyme i.

At steady state $v_1 = v_2 = J$, so $k_1S - k_1X/K_1 = k_2X$, giving

```
X = k1·S / (k2 + k1/K1)        and        J = k1·k2·S / (k2 + k1/K1)
```

Differentiate J with respect to k_1 and scale, and after one line of algebra:

```
C^J_1 = k2 / (k2 + k1/K1)          C^J_2 = (k1/K1) / (k2 + k1/K1)
C^J_1 + C^J_2 = 1                  exactly, for any parameter values
```

Three readings, all of which generalise. The summation theorem falls out of the algebra rather than being imposed. The control sits with whichever step is *slow relative to the other* — raise k_2 and control moves to step 1 — which is the migration you saw numerically in the figure. And the equilibrium constant K_1 appears in the answer: the closer step 1 is to equilibrium, the smaller k_1/K_1 has to be before it holds control, which is the thermodynamic argument of week 14 arriving inside the control analysis.

Response coefficients: the version you actually need for a design

Enzymes are not the only thing you change. The response coefficient R^J_P measures how flux responds to an external effector P — an inhibitor, an inducer concentration, a feed rate — and the partitioned response theorem factorises it:

```
R^J_P = C^J_i · ε^i_P     summed over the enzymes that P acts on
```

That is the equation that connects the biochemistry you can measure to the process variable you can set. The effect of an inhibitor on pathway flux is its effect on the enzyme it binds — an elasticity, measurable in vitro — times the control that enzyme holds in the pathway. So a potent inhibitor of an enzyme with little control does almost nothing to flux, which explains a great many disappointing results with pathway inhibitors, and conversely a weak effect on a high-control step can be decisive.

How control coefficients are actually measured

You will see the numbers quoted and should know where they come from, because the method determines how much to trust them.

- **Titratable expression.** Put the enzyme under an inducible promoter, measure flux across a series of induction levels, and take the slope of ln J against ln E at the wild-type point. This is the cleanest method and the most work, and it gives you the whole response curve rather than one derivative — which is better, since the curve is what you need.
- **Specific inhibitor titration.** Titrate an inhibitor of known mechanism and infer the coefficient from the initial slope. Fast, and limited to enzymes with a usable specific inhibitor.
- **Genetic series.** Compare strains with different copy numbers or promoter strengths. The pitfall is that each strain differs in burden as well as in the target enzyme, so some of the response you measure is proteome allocation (week 9) rather than pathway control.
- **From a kinetic model.** Numerical perturbation, as in chapter 16. Free, instant, and only as good as the model's parameters — which is why the answer is a prediction rather than a measurement, and should be described as one.

Flux control coefficient	`C^J_i = (∂J/∂E_i)(E_i/J); dimensionless, systemic`
Elasticity	`ε^i_S = (∂v_i/∂S)(S/v_i); local, from the rate law`
Summation	`Σ_i C^J_i = 1 over all steps, including transport`
Connectivity	`Σ_i C^J_i · ε^i_S = 0 for each metabolite S`

Concentration control	`Σ_i C^S_i = 0`: pools are buffered, flux is not

A reviewer asks why you propose editing six genes when your own model says enzyme 1 carries a control coefficient of 0.33. Answer in a paragraph.

Because 0.33 is the response to an infinitesimal change at the current state, and it decays as soon as the change is made. Raising enzyme 1 alone redistributes control to the steps downstream and to export, so the second 10% buys less than the first; the summation theorem guarantees that no single-enzyme route can reach the flux we need, because the most it can deliver is the fraction of control it holds, and that fraction shrinks as we use it. Six coordinated changes keep each individual perturbation small, which keeps each coefficient near its measured value and keeps the response close to additive. The design is therefore not six guesses but one hypothesis — that control is distributed as measured — tested six ways at once.

The honest addition, which a good reviewer will want: a multiplexed edit also multiplies the ways the strain can go wrong, and the burden of six overexpressions is not in the control analysis at all. That is why the design includes a staged construction with the flux measured after each round, rather than all six at once with a single titer at the end.

Three things people get wrong

- ‘Find the rate-limiting step.’ The summation theorem forbids a single step holding all control in a real pathway.
- ‘Control coefficients are enzyme properties.’ They are system properties, and they change when you change anything, including the thing you measured.
- ‘A coefficient of 0.1 means the enzyme is unimportant.’ It means a small change buys 10% of itself; halving that enzyme may still be catastrophic, because the coefficient is a local derivative, not a global statement.

Self-check — answer on paper first

1. State the summation theorem and its engineering implication, from memory, in the form you would give in an interview.

The flux control coefficients of all the steps in a pathway sum to one. Since control is therefore shared, no single enzyme can deliver more than its share, and using that share redistributes control to other steps, so single-gene overexpression gives sub-linear gains that decay. Useful improvements require several coordinated changes, which is the argument for multiplexed editing and library-based tuning rather than for finding the one correct target.

2. Your numerical control coefficients sum to 1.4. Give three candidate causes and how to test each.

The perturbation was too large, so you measured a secant rather than a derivative: halve it and see whether the sum moves toward one. A conserved moiety or clamped boundary in the model is absorbing or injecting flux: check whether any concentration is held fixed, and whether a cofactor total is conserved, since a conserved pool changes the theorem's form. Or the steady state was not actually reached before you read off the flux: integrate longer, or solve for the steady state directly and compare.

3. Why does an enzyme with a high elasticity toward its substrate tend to carry little control?

Because a steep response to its substrate means the enzyme absorbs changes in that pool: if flux through it rises, its substrate falls slightly and the rate self-corrects, so the perturbation is damped rather than transmitted to the rest of the pathway. Saturated enzymes, whose elasticity is near zero, cannot self-correct, so a change in their amount propagates and they carry the control. This is why comparing a measured pool concentration with the consuming enzyme's K_M is such a cheap way to guess where control sits.

What to open a book for. Stephanopoulos Ch. 11 covers the formalism and has problems worth working all the way through. Fell's *Understanding the Control of Metabolism* is where the theorems are derived properly and is the better book if you want to be able to defend them. Work two or three problems by hand this week; you will do the rest numerically for the next fifty, and the hand calculation is what makes the numerics trustworthy.

Kinetic and dynamic modelling: dc/dt = S·v(c), and what your data can and cannot identify

A kinetic model is the only tool that predicts a transient, and the most expensive one to earn. The judgement this week teaches is knowing when you have earned it and when constraint-based modelling is the honest answer.

From the steady state to the dynamic form

Chapter 13 solved S·v = 0. The dynamic form removes the zero:

```
dc/dt = S · v(c)
```

Everything new is in the parenthesis. In the stoichiometric case v was an unknown vector; here it is a *function* of the concentrations, which means you now need a rate law for every reaction and parameters for every rate law. A 20-reaction model needs perhaps 60 parameters, and that is the whole difficulty of the subject: the structure is free, the parameters are not.

Here is the complete model used for the figures in chapters 15 and 16 — three enzymes, reversible kinetics, feedback inhibition from the product onto step 1, and a first-order export. Type it in and you can reproduce every number I have quoted.

```
import numpy as np
from scipy.integrate import solve_ivp

S0 = 5.0
p = dict(V1=2.0, K1=1.0, K1p=1.0, Keq1=10.0, Ki=0.6, n=2.0,
         V2=2.5, K2=0.5, K2p=0.5, Keq2=5.0,
         V3=2.0, K3=0.5, K3p=0.5, Keq3=8.0, kP=0.6)

def rates(c, p):
    X1, X2, P = c
    v1 = p['V1']*(S0 - X1/p['Keq1'])/(1 + S0/p['K1'] + X1/p['K1p']) \
         / (1 + (P/p['Ki'])**p['n'])            # feedback inhibition
    v2 = p['V2']*(X1 - X2/p['Keq2'])/(1 + X1/p['K2'] + X2/p['K2p'])
    v3 = p['V3']*(X2 - P /p['Keq3'])/(1 + X2/p['K3'] + P /p['K3p'])
    return np.array([v1, v2, v3, p['kP']*P])    # export is the fourth step

def rhs(t, c, p):
    v = rates(c, p)
    return [v[0]-v[1], v[1]-v[2], v[2]-v[3]]

sol = solve_ivp(rhs, [0, 20000], [0.1, 0.1, 0.1], args=(p,),
                method='LSODA', rtol=1e-11, atol=1e-13)
c_ss = sol.y[:, -1]
print(c_ss, rates(c_ss, p))   # X1 1.545  X2 1.338  P 0.787 ; all four rates 0.4722
```

Rate laws, and how much to spend on them

Irreversible Michaelis–Menten is the default and is wrong for most steps in a pathway, because most steps are near enough to equilibrium that the reverse rate matters. The reversible form, $v = V_f(S - P/K_{eq}) / (1 + S/K_S + P/K_P)$, is barely more work and has the property that it cannot violate thermodynamics: flux goes to zero at equilibrium and reverses beyond it. The Haldane relation, $K_{eq} = (V_f K_P) / (V_r K_S)$, ties the kinetic parameters to the thermodynamics you computed in week 14, so fitting them independently is over-fitting: fix K_{eq} from eQuilibrator and let the kinetics respect it.

Beyond that, add only what you can justify: a competitive term for an inhibitor you measured, a Hill exponent for a step you know is cooperative, and nothing else. Convenience kinetics — a generic reversible rate law with one parameter per substrate and product — exists precisely to avoid inventing mechanisms you have no evidence for, and for a model whose purpose is to explore behaviour rather than to predict a number, it is the right default.

Numerics: stiffness, and verifying against the steady state

Metabolic models are stiff, for the structural reason given in chapter 12: enzymatic steps equilibrate orders of magnitude faster than pools fill. Use LSODA or BDF, and treat a slow integration as a solver diagnosis rather than a biological one.

There is one check you should run on every model you build, and almost nobody does: integrate to steady state, then confirm that the resulting flux vector satisfies S·v = 0 to numerical precision and matches what the stoichiometric

analysis of chapter 13 predicted. In the model above, all four rates come out equal to 0.4722, as a linear pathway at steady state requires. If they do not, you have a sign error in S or a rate law that does not do what you think, and finding it now is far cheaper than explaining a wrong result later.

Parameter estimation, and the identifiability that bites

The exercise that teaches this is to generate noisy data from your own model and fit the parameters back. You know the truth, so you can see exactly which parameters return and which do not, and the answer is always the same: some come back accurately, some come back with enormous confidence intervals, and some come back wrong with narrow confidence intervals, which is the dangerous case.

Two kinds of non-identifiability, and the distinction matters because the remedies differ. **Structural** non-identifiability is a property of the model: two parameters appear only as a product or a ratio, so no amount of data of that type can separate them. V_{max} and K_M at substrate far below K_M, from chapter 12, is the textbook case; no better experiment of the same design fixes it, only a different measurement. **Practical** non-identifiability is a property of your data: the parameter is identifiable in principle but your measurements are too few, too noisy, or taken at the wrong times. More or better data fixes it.

The diagnostics, in increasing order of effort: look at the correlation matrix from the covariance; re-fit from several starting points and see whether you land in the same place; and compute a profile likelihood for the parameter you care about, by fixing it at a grid of values, re-optimising everything else, and plotting the best achievable residual. A profile that is flat over orders of magnitude is the honest picture of a parameter your data do not determine, and it is the figure to show rather than a confidence interval that implies precision you do not have.

Sensitivity analysis, and getting control coefficients for free

A local sensitivity analysis perturbs each parameter by 1% and records the effect on the output you care about. Do it on the steady-state flux with respect to each V_{max} and you have computed the flux control coefficients numerically:

```
def fcc(p, keys=('V1','V2','V3','kP'), eps=0.01):
    J0 = steady(p)[1][-1]
    out = {}
    for k in keys:
        q = dict(p); q[k] = p[k]*(1+eps)
        out[k] = ((steady(q)[1][-1] - J0)/J0)/eps
    return out            # 0.327, 0.102, 0.108, 0.455 — sum 0.993
```

That is the whole of chapter 15, executed. Check the sum against one, and if it misses, work through the four causes in chapter 15's list before you blame the biology. Global sensitivity analysis — sampling the whole parameter space rather than perturbing around one point — is the honest version when your parameters are uncertain, and ensemble modelling takes it further still: instead of one parameter set, keep a population of sets, all consistent with the available data, and report the spread of predictions. A prediction that is robust across the ensemble is worth acting on; one that depends on which member you picked is not a prediction.

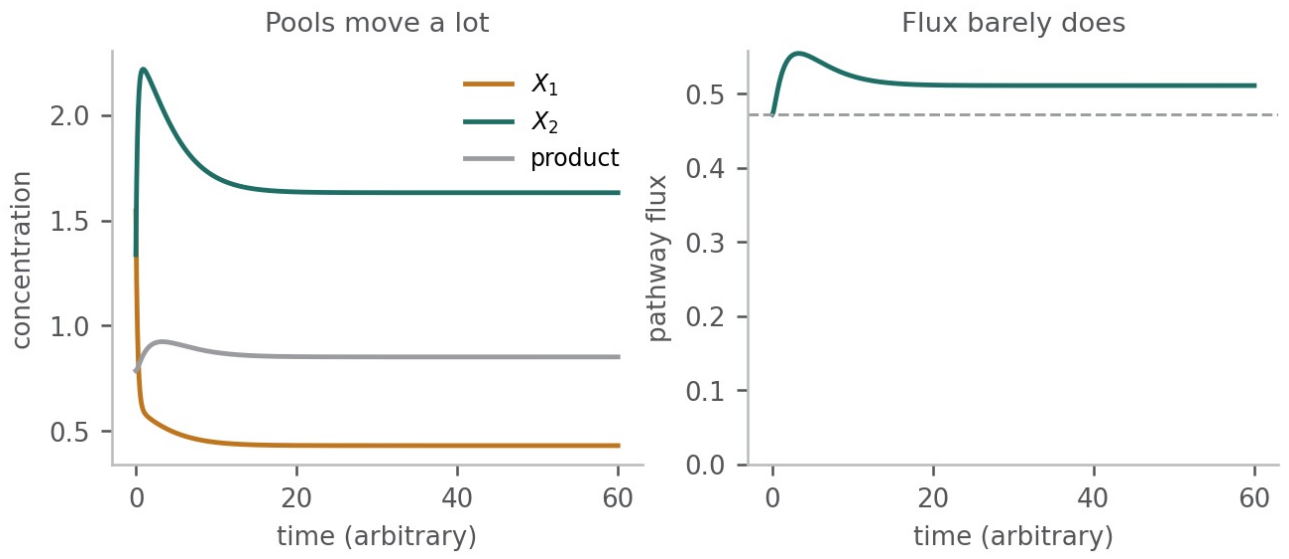


Figure 16.1 The model's response to a tenfold step increase in enzyme 2 at t = 0, starting from the steady state. The intermediate pools move substantially — X_1 falls by a factor of three — while the pathway flux rises by 8%. This is metabolic control analysis seen rather than proved, and it is the

result to expect from a single overexpression. Computed with the model listed above.

Dynamic flux balance analysis, and when to stop modelling kinetically

Dynamic FBA is the pragmatic middle ground: solve a constraint-based model for the current external conditions, use its predicted uptake and secretion rates to advance the extracellular concentrations and biomass by a time step, and repeat. You get substrate depletion, growth, product accumulation and diauxic shifts over a batch, with no intracellular kinetic parameters at all. For simulating a fermentation it is usually the right tool, and COBRApy implements it in a few lines.

So when does a kinetic model earn its parameters? When the question is about a transient, a pool concentration, or a regulatory mechanism — how fast the pathway responds to induction, whether a toxic intermediate accumulates, whether a dynamic controller will be stable. Those questions are invisible to constraint-based methods, which have no concentrations and no time. When the question is about yields, flux distributions at steady state, or the consequences of a deletion on growth, a constraint-based model answers it with assumptions you can state, and a kinetic model answers it with sixty parameters you cannot defend. Choosing the weaker tool is the expert move, and being able to say why in one sentence is a question you will be asked.

Model reduction: making the stiffness go away instead of solving through it

Stiffness comes from timescale separation, and timescale separation is also an opportunity. If a step equilibrates far faster than the rest of the model, you can replace its differential equation with an algebraic one — the quasi-steady-state assumption you already use for reactive intermediates — and the stiffness disappears with it. Michaelis–Menten itself is this reduction applied to the enzyme–substrate complex, which is worth noticing: you have been doing model reduction since week 3 without calling it that.

Two rules make it safe. Reduce only what is genuinely fast at the concentrations your model actually visits, which you check by comparing the reduced model's trajectory with the full one over the perturbation you care about rather than at steady state, where they agree by construction. And remember that the reduction discards the fast dynamics, so any question about the first milliseconds — a rapid-mixing experiment, a burst phase — must be asked of the full model.

The complementary move is non-dimensionalisation: scale concentrations by their K_M values and time by a characteristic turnover time. It costs twenty minutes and pays twice. Numerically, the solver sees variables of order one instead of quantities spanning six decades, which improves conditioning. Conceptually, the parameters collapse into a smaller number of dimensionless groups — ratios of timescales, saturation numbers — and those groups, not the raw parameters, are what the behaviour depends on. A model with fifteen parameters often has four groups that matter, and finding them tells you which measurements are worth making.

Designing the experiment the fit needs

The usual order is backwards: run the experiment, then try to fit it. Reverse it. Given a candidate model, simulate the measurement you are considering and compute the sensitivity of each proposed data point to each parameter — the same 1% perturbation as before, but recorded at each time point rather than at steady state. The result is a map of where in time each parameter is visible, and three things fall out of it.

Points where every sensitivity is near zero carry no information about parameters, however beautiful the data look: typically, long plateaus after the system has settled. Points where two parameters have proportional sensitivity curves cannot separate those two parameters, which is structural non-identifiability diagnosed before the experiment rather than after. And the highest-value sampling times cluster where the sensitivities are largest and least parallel — usually during the transient, which is exactly the part of a time course that is hardest to sample and therefore usually under-sampled.

For a fermentation this analysis typically says: take many samples in the first few hours after the perturbation and fewer later, which is the opposite of the convenient schedule. Knowing that in advance is worth more than any improvement in the fitting algorithm, and it is the sort of argument that gets a chemical engineer taken seriously in a biology group.

Dynamic balance	`dc/dt = S·v(c); same S as week 13, rate laws added`
Reversible MM	`v = V_f(S − P/Keq)/(1 + S/K_S + P/K_P)`
Haldane	`Keq = (V_f·K_P)/(V_r·K_S) — fix Keq, do not fit it`
Numerical FCC	`C^J_i ≈ (ΔJ/J)/(ΔE/E) at 1%; check Σ ≈ 1`
Identifiability	`structural: no data of this kind can help. practical: more data can`

Your model predicts an 8% flux gain from tenfold overexpression of enzyme 2. The experiment gives a 2.3-fold gain. Which is wrong, and how do you find out?

Probably neither the arithmetic nor the experiment: probably the model's structure. A gain that large from one enzyme means its control coefficient in the real cell is far above the 0.1 the model assumed, and the usual structural reasons are that the real pathway has a step the model does not — an intermediate that is unstable or leaks to a side reaction, so that accelerating its consumption recovers carbon the model never lost — or that the model's export term is wrong, since in the model export carries 0.46 of the control and in the cell it may carry much less.

How to find out, in order of cost: check whether the intermediate pool fell as the model predicts, because the pools are where the mechanism shows; measure the product and the by-products, since a gain in product with no change in substrate uptake means carbon was recovered from somewhere and that somewhere is the missing reaction; and re-fit the model with the new data point included, which will tell you whether any parameter set within the existing structure can reproduce both observations. If none can, the structure is wrong, which is the most useful possible outcome — a model that cannot be reconciled with data has told you something, whereas a model you re-fit until it agrees has told you nothing.

Three things people get wrong

- ‘More mechanism is a better model.’ Every term you cannot constrain adds parameters that your data will fit to noise.
- ‘The confidence interval is narrow, so the parameter is right.’ Narrow intervals on correlated parameters are an artefact; look at the correlation matrix or the profile likelihood.
- ‘Kinetic models are the serious ones and FBA is a shortcut.’ They answer different questions; using FBA where it suffices is the stronger answer.

Self-check — answer on paper first

1. What exactly changes between $S \cdot v = 0$ and $dc/dt = S \cdot v(c)$, and what does it cost you?

Structurally, nothing but the zero: the same stoichiometric matrix, now with accumulation on the left. Conceptually, v becomes a function of the concentrations, so every reaction needs a rate law and every rate law needs parameters — typically three to five each. The cost is that the model's conclusions now depend on numbers you must measure, fit or assume, so a dynamic model's predictions are only as good as its least identifiable parameter, whereas a stoichiometric model's conclusions depend only on the network.

2. Distinguish structural from practical identifiability and give the remedy for each.

Structural non-identifiability means the model's equations determine only a combination of the parameters, such as a ratio, so no quantity of data from that experiment can separate them; the remedy is a different measurement — a different substrate range, a direct measurement of one parameter, or a reparameterisation that fits the combination the data constrain. Practical non-identifiability means the parameter is determinable in principle but your data are too sparse or too noisy; the remedy is more replicates, more time points, or points placed where the model is sensitive to that parameter.

3. A collaborator wants a kinetic model of central metabolism to predict the yield of a deletion strain. What do you advise?

That a constraint-based model answers this question better. Yield at steady state is a stoichiometric quantity, decided by the network and the cofactor balances, and a genome-scale model with the deletion imposed as a bound will give it with assumptions that can be written down. A kinetic model of central metabolism would need hundreds of parameters, most unmeasured for this organism, and its yield prediction would be dominated by those choices. The kinetic model becomes the right tool when the question changes to whether an intermediate accumulates to toxic concentrations, or how fast the strain reaches its new steady state after the deletion.

What to open a book for. Stephanopoulos's chapters on cellular reaction kinetics for the formalism; the COBRApy documentation for dynamic FBA, which is better than any textbook because it is runnable. For identifiability, read a methods paper on profile likelihood rather than a textbook — the subject is younger than the textbooks that mention it. Keep the model listed in this chapter: weeks 31 and 51 both reuse it.

Experimental design and statistics: when a difference is real

Most strain comparisons in the literature are under-powered, and most error bars are drawn on the wrong quantity. This is the cheapest week in the curriculum to get an advantage from, because the arithmetic is easy and almost nobody does it before the experiment.

What an error bar means, and what counts as n

A biological replicate is an independent realisation of the thing you are claiming about: an independent colony, transformed independently, grown in its own flask. A technical replicate is a repeated measurement of the same realisation: the same culture injected three times on the HPLC. They estimate different variances — biological variation plus measurement error, against measurement error alone — and they are not interchangeable.

The consequence is the one people resist. Three flasks inoculated from a single colony are, for the question "is strain B better than strain A", a single biological replicate: whatever that colony happened to be — a plasmid rearrangement, a suppressor mutation, an unusual expression level — is shared by all three flasks and will be reported as if it were the strain. This is pseudoreplication, and it inflates your apparent precision while leaving your actual uncertainty untouched. The practical rule: decide what the independent unit of your claim is, and count that.

Then choose which bar to draw. A standard deviation describes the spread of the population you sampled and answers "how variable are clones of this strain". A standard error, SD/√n, describes the precision of your estimate of the mean and answers "how well do I know this strain's mean". On a figure comparing two strains, the standard error is what the comparison rests on; on a figure characterising variability between clones, the standard deviation is. Say which one the bar is, in the caption, always — a figure whose caption does not say is uninterpretable, and reviewers are entitled to assume the more flattering one was chosen.

Comparing two strains, worked all the way through

Take the tracker's example: strain A at 0.60, 0.64, 0.62 g/g and strain B at 0.65, 0.69, 0.64 g/g, three biological replicates each. The means are 0.620 and 0.660, so the difference is 0.040 g/g — about 6.5% better. The standard deviations are 0.020 and 0.026, the pooled standard deviation is 0.0235, and the standard error of the difference is 0.0191.

The t statistic is 0.040/0.0191 = 2.09 on 4 degrees of freedom, giving p = 0.105; Welch's correction, which you should prefer by default since it costs nothing and does not assume equal variances, gives p = 0.110. The 95% confidence interval on the difference runs from −0.013 to +0.093 g/g. Cohen's d — the difference in pooled standard deviations — is 1.71, which is a large effect by any convention.

Now read those four numbers together, because the reading is the lesson. The effect is large, the p-value is above 0.05, and the confidence interval contains zero but is wildly asymmetric about it: the data are consistent with B being no better, and also with B being 15% better. The honest conclusion is not "no significant difference" — which implies the strains are the same — but "this experiment could not resolve a difference of this size". Those are different claims, and only the second tells a reader what to do next, which is to run more replicates.

What to put in a paper: the effect size and its confidence interval, with the p-value third if at all. A p-value compresses the two things a reader needs — how big the effect is, and how precisely you measured it — into one number from which neither can be recovered.

Power, before the experiment rather than after

Power is the probability that your experiment finds a real effect of a given size. For the comparison above, with d = 1.71, the power at n = 3 is 0.36. That is the sentence worth carrying out of this week: with the standard three replicates, a large, real effect is missed about two times in three. Reaching the conventional 80% requires n = 7, and n = 10 gives 0.95.

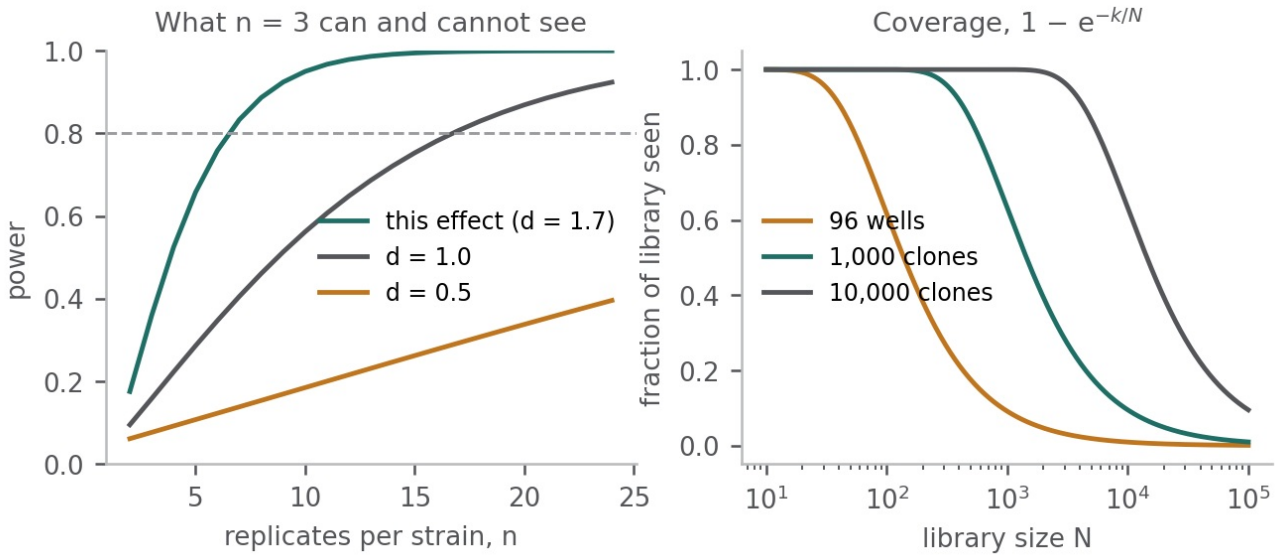


Figure 17.1 Left: power of a two-sample t-test against replicate number for three effect sizes, with the tracker's strain comparison marked (d = 1.71); the dashed line is the conventional 80%. Right: library coverage, $1 - e^{-k/N}$, for three screening efforts. Computed for this module with SciPy; the power curve uses the non-central t distribution rather than a normal approximation.

The constructive use of this is to invert it. Before running the experiment, decide the smallest difference that would change your decision — not the smallest you would like to detect, the smallest that would alter what you do next — and compute the n that detects it with 80% power. If that n is unaffordable, you have learned something important before spending anything: this experiment cannot answer this question, and either the question or the measurement precision has to change.

The same inversion rescues negative results. "We saw no difference" is an absence of information; "we would have detected a 10% difference with 80% power and saw none" is a measurement. Report the second and your negative results become publishable.

Screening many strains: multiplicity and coverage

Two different arithmetics govern screens, and they are often confused.

Multiplicity. Compare 96 strains against a control at $\alpha = 0.05$ and you expect about five false positives even if every strain is identical to the control. If you then pick the best three for follow-up, you have selected the three luckiest wells. The defences are to control the false discovery rate rather than the per-test error, and — more powerful in practice — to re-test hits in an independent experiment, which is the only thing that reliably separates a real effect from a lucky well. Budget the retest from the start.

Coverage. For a library of N variants and k clones screened at random, the expected fraction of the library you have seen at least once is $1 - e^{-k/N}$. A three-gene ribosome binding site library with 20 variants each has N = 8,000, so a 96-well plate covers 1.2% of it, and 1,000 clones cover 12%. Covering 95% needs about 3N — 24,000 clones. This is the calculation that decides whether your screen is a search or a sample, and it is the sentence to have ready in an interview: screening 96 wells of a combinatorial library is not searching the library, it is drawing 96 samples from it, and the right design is therefore either to shrink the library to what you can cover or to couple the phenotype to selection so that the cells do the screening.

Error propagation into a reported yield

A yield is a quotient of two measured quantities, each built from several others, and for independent relative errors the relative errors add in quadrature. List them honestly for a typical yield: HPLC peak area 3%, calibration curve 2%, dry cell weight 5%, culture volume 1%, dilution factor 2%. The total is $\sqrt{(9 + 4 + 25 + 1 + 4)} = 6.6\%$.

Two conclusions. First, a yield of 0.40 g/g carries ±0.026, so it should be reported as 0.40 ± 0.03 and not as 0.4023 — which is the commonest form of overclaiming in the field, and it is visible to any reader who counts significant figures. Second, dry cell weight dominates, contributing 25 of the 43 units of variance, so the way to improve this number is a better biomass measurement, not a better chromatogram. That is a result you can act on, and the calculation that produced it took two minutes.

Design of experiments, in the time it takes to do one factor at a time

Three medium components at two levels each is a 2^3 factorial: eight runs, from which you get all three main effects and all four interactions, each estimated from all eight runs rather than from a pair. The half-fraction, 2^{3-1}, needs four runs and gives you the three main effects — but each is now aliased with a two-factor interaction, so an apparent effect of A is really A + BC and you cannot tell which. That is the trade, stated exactly: fractional designs buy runs with the assumption that high-order interactions are negligible.

One factor at a time is worse than merely inefficient. With interactions present, its answer depends on the order in which you tested, because each factor is optimised at whatever level the others happened to be fixed at; test in a different order and you get a different optimum. And it cannot report an interaction at all, which is a problem when the thing you are optimising — a medium, a feed, an induction profile — is interaction-heavy by nature. The defensible uses are narrow: screening a factor for an effect of any kind at all, or a situation where changing two things at once is physically impossible.

Finally, randomise and block. Plate position, day, batch, operator and instrument are real factors with real effects, and if your strains are laid out so that all of strain A is in the left half of the plate, the edge effect becomes your result. Randomise the assignment, block on the nuisance factors you cannot randomise, and write the plan down before the experiment — a randomisation decided after seeing the data is not randomisation.

The calibration curve, which decides more numbers than the statistics do

Every titer and every yield in your thesis will rest on a calibration curve, and the curve is usually treated as a formality. Four points of practice, in order of how often they are violated.

- **Calibrate over the range your samples occupy, and dilute into it.** Extrapolating above the top standard is the commonest source of a wrong titer, because detector response saturates and the extrapolation is invisible in the fitted line. If a sample reads above the highest standard, dilute and re-inject; do not trust the number.
- **Replicate standards and look at residuals, not R^2.** An R^2 of 0.999 is compatible with obvious curvature, which shows immediately in a residual plot and not at all in the coefficient.
- **Report limits honestly.** With σ the standard deviation of the blank or of the intercept, and m the slope, the conventional detection limit is 3.3σ/m and the quantification limit 10σ/m. A value between them is evidence that something is there and not a measurement of how much, and should be reported as such rather than as a small number with error bars.
- **Use an internal standard where injection volume or extraction efficiency varies.** It converts a systematic error you cannot see into a ratio you can.

The reason this belongs in a statistics chapter: calibration uncertainty is systematic across every sample in a run, so it does not shrink with replication, and no amount of n fixes it. In the error budget above, the 2% calibration term is the one that stays when everything else has been averaged away.

Multiple comparisons, as a procedure rather than a warning

When you test m strains against a control, the two usable corrections are easy to apply by hand. Bonferroni compares each p-value against α/m: conservative, appropriate when a false positive is expensive — when a hit triggers a fermenter campaign — and nothing else. Benjamini–Hochberg controls the false discovery rate instead: sort the p-values ascending, find the largest i for which $p_{(i)} \le (i/m)\cdot q$, and declare the first i tests significant. With m = 96 and q = 0.10, the smallest p-value is compared against 0.001, the tenth against 0.0104, and the fiftieth against 0.052.

Which to use is a decision about what the screen is for. In a discovery screen, where every hit will be re-tested anyway, false discovery rate control at q = 0.1 to 0.2 is right: you are willing to carry a few false hits into a cheap confirmation step in exchange for not discarding real ones. In a confirmatory experiment, where a positive result is the conclusion, Bonferroni or a pre-registered single comparison is right. Deciding this before you look at the data is what makes either defensible.

Standard error	`SE = SD/√n; SE on comparisons, SD on variability`
Difference	`t = Δ/SE_Δ; report Δ and its 95% CI, not p alone`
Effect size	`d = Δ / pooled SD; d = 1.71 here, and still p > 0.05 at n = 3`
Power	`n = 3 gives 0.36 at d = 1.71; 80% needs n = 7`
Coverage	`fraction seen = 1 − e^(−k/N); 95% needs k ≈ 3N`
Propagation	`(σ_Y/Y)² = Σ (σ_i/x_i)² for independent relative errors`
Aliasing	`2^(3−1): A is confounded with BC, B with AC, C with AB`

A colleague reports that their new strain is significantly better, p = 0.03, from three flasks per strain, and wants to take it to the 5 L fermenter. What do you ask?

First, what the three flasks were: if they came from one colony each, the comparison is sound as far as it goes; if all three came from one colony per strain, n is 1 and the p-value is meaningless regardless of its size. Second, the effect size and its confidence interval, because at n = 3 a p-value of 0.03 implies a very large observed difference and a correspondingly wide interval — the lower bound is what the fermenter campaign will actually deliver. Third, whether this was the only comparison made: if it was the best of twelve strains tested, the effective α is twelve times larger and p = 0.03 is unremarkable.

None of this is an argument against the fermenter run, which may be worth doing on a weak signal if the run is cheap relative to the information. It is an argument for stating the expectation honestly before the run, so that a result at the low end of the interval is recognised as consistent with the flask data rather than as a scale-up failure.

Three things people get wrong

- ‘n = 3 is standard, so n = 3 is enough.’ At a large effect size it has 36% power. Standard is not the same as adequate.
- ‘Not significant means no difference.’ It means this experiment could not resolve one; the confidence interval says what sizes remain possible.
- ‘Error bars are error bars.’ SD and SE differ by √n, and a caption that omits which is being shown cannot be read.

Self-check — answer on paper first

1. Why are three flasks from one colony n = 1, and what is the practical remedy when colonies are expensive?

Because the claim is about the strain, and all three flasks share whatever is peculiar to that one colony — a plasmid rearrangement, a second-site mutation, an unusual expression state — so they measure flask-to-flask variation, not strain variation. The remedy when colonies are expensive is a nested design: a few independent colonies, each with a few flasks, analysed so that the colony-level variance is the one used for the comparison. That recovers the correct error term without demanding many independent colonies, and it also tells you how much of your variance is clonal, which is useful in itself.

2. You screen a 10,000-member library in two 96-well plates and find a variant three times better than the parent. What can and cannot you conclude?

That you have found a better variant, which is a real and useful result. You cannot conclude anything about the library: 192 clones cover under 2% of 10,000, so the best variant in the library is almost certainly not in your sample, and the distribution you observed is a sample of 192 from a distribution you have barely touched. You also cannot conclude the variant is three times better until it is re-tested independently, since a single well at the tail of a 192-well distribution is exactly where measurement noise concentrates.

3. Your yield has 6.6% relative uncertainty, dominated by dry cell weight. Give two ways to reduce it and say which is cheaper.

Replicate the dry-weight measurement, which reduces its contribution by √n and needs nothing but more filters and time; or replace it with a better-calibrated proxy, such as an optical-density-to-dry-weight calibration built from many paired measurements, which shifts the uncertainty from a single noisy measurement into a calibration curve that can be made very precise once and reused. The second is cheaper over a campaign and is the standard practice; the first is the right answer for the one definitive measurement you report at the end.

What to open a book for. No textbook; the *Points of Significance* columns in Nature Methods are the best short treatment and each takes ten minutes. Use scipy.stats for the tests and statsmodels when you need power calculations or a designed experiment analysed properly. If you want one book, a design-of-experiments text aimed at process engineers will serve you better than a biostatistics one.

Titer, rate, yield: the three numbers and the arithmetic behind them

Every strain is eventually reduced to three numbers, and which of them binds is decided by the product's price, not by the biology. Knowing which one you are optimising is the difference between a project and a hobby.

The definitions, with units, and what a flask actually measures

Titer is the product concentration in the broth at harvest, in g/L. **Rate** is volumetric productivity, g/L/h, usually quoted as titer divided by total fermentation time — including the lag and any growth phase, which is the honest version. **Yield** is product per substrate consumed, in g/g, mol/mol, or as a fraction of the theoretical maximum, which is the most informative form because it says how much room is left.

Specific productivity, q_P, in g per g of biomass per hour, is the fourth number and belongs with them: volumetric productivity is $q_P \cdot X$, so a strain can improve on one and worsen on the other, which is the trade you met in week 9.

A shake flask measures titer and yield reasonably and rate badly. It has no pH control, no dissolved-oxygen control and no feed, so the culture spends part of its time in conditions a fermenter would never allow, and the time axis is therefore not comparable to a controlled process. Treat flask rates as a screening ranking rather than as a number, and never extrapolate them to a plant.

Theoretical maximum yield, computed twice

The theoretical maximum is a stoichiometric statement: the best any strain could do with this pathway, with all carbon not required for the product's own chemistry conserved, and with carbon, redox and ATP balances closed. Compute it before you build, because it tells you what fraction of the achievable you have reached and therefore whether to keep working on the strain or on the process.

Ethanol. One glucose gives two pyruvate, each decarboxylated to acetaldehyde and reduced to ethanol, so 2 mol per mol: $2 \times 46.07 / 180.16 = 0.511$ g/g, with two CO_2 carrying the lost carbon. The redox balance closes exactly — glycolysis makes 2 NADH, the two reductions consume 2 NADH — which is precisely why this is a fermentation and can run anaerobically with no electron acceptor. Real processes reach roughly 90 to 93% of the maximum, the gap going to biomass and glycerol, which is itself the cell's redox valve.

Isobutanol, as a case where the balance does not close so comfortably. Two pyruvate condense to acetolactate, losing one CO_2; reduction and dehydration give 2-ketoisovalerate; decarboxylation loses a second CO_2 and a final reduction gives isobutanol, $C_4H_{10}O$. So one mole per mole of glucose: $74.12 / 180.16 = 0.411$ g/g. Carbon closes — six carbons in, four out plus two CO_2. Redox closes in total: glycolysis supplies 2 NADH and the pathway consumes two hydrides. But one of those reductions is NADPH-dependent in the natural enzyme, so the cell must convert one NADH to NADPH per product molecule, and the real cost of that conversion is chapter 19's subject. The arithmetic says the yield is reachable; the cofactor identity says what it will cost to reach it.

The general method, in order, and it is the one the phase 2 checkpoint asks for: write the balanced synthesis from glucose, close carbon, then close NADH and NADPH *separately*, then close ATP. A surplus of NADH means the pathway is more oxidised than the substrate and needs an electron sink or respiration; a deficit of NADPH means pentose phosphate flux or a transhydrogenase, at a cost you must state. Doing redox as one lumped "reducing equivalents" line is the single most common error in a back-of-envelope balance, and it hides exactly the problem that will dominate the project.

Where the money is: two plots you can do on an envelope

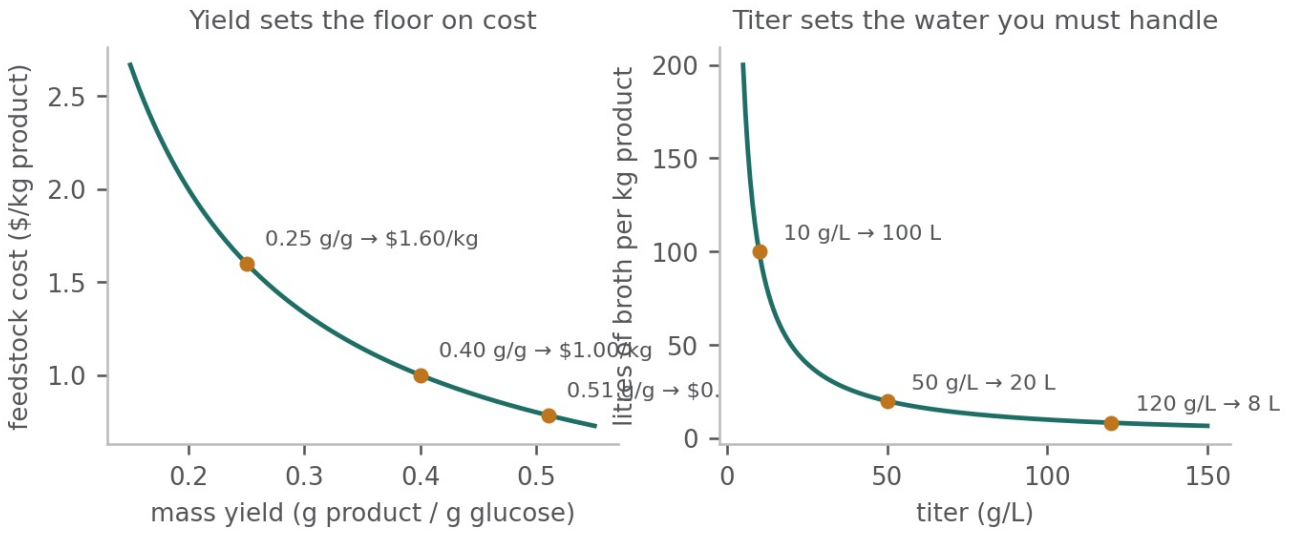


Figure 18.1 Left: feedstock cost per kilogram of product against mass yield, at an assumed glucose price of $0.40/kg; the three marked points are a mediocre strain, a good one and ethanol's theoretical maximum. Right: litres of broth that must be processed per kilogram of product against titer. Both are arithmetic, not data; the glucose price is the only assumption and it moves the left panel vertically without changing its shape.

Feedstock cost per kilogram of product is simply the substrate price divided by the mass yield. At $0.40/kg glucose, a yield of 0.25 g/g means $1.60 of sugar per kilogram of product, and 0.40 g/g means $1.00. For a commodity selling at $1.50 to $2.00/kg, the difference between those two strains is the difference between a business and a demonstration — which is why yield is the binding metric for bulk chemicals, and why the first question about any commodity paper is the yield as a fraction of theoretical, not the titer.

Titer binds differently: it sets the volume of water you must handle per unit of product. At 10 g/L you process 100 litres of broth per kilogram; at 120 g/L you process 8.3. Every one of those litres must be heated, separated, and disposed of, and downstream processing is typically the larger half of the capital cost for a dilute product. This is the quantitative version of a sentence you will read everywhere: a low titer does not merely cost more, it changes which separation technology is feasible at all.

For a high-value pharmaceutical, neither dominates. A few per cent of sugar is irrelevant against a product worth thousands per kilogram, and what matters is total output, purity, and the regulatory reproducibility of the process. The engineering conclusions invert accordingly: for a commodity you optimise yield and accept complexity in the strain; for a pharmaceutical you keep the strain and the process simple and conservative, because a change costs a regulatory filing.

The gap between a paper and a plant

The 1,4-butanediol work of Yim and colleagues is the standard teaching case because the route was designed computationally, built, and then taken to commercial scale, so the whole gap is documented. The paper's titer is tens of grams per litre; commercial operation required several times that, plus years of strain and process development that the paper does not contain. Nothing went wrong in that story — that is what the gap always looks like.

Where does the gap come from? Oxygen transfer falls as vessels get larger; mixing times go from seconds to tens of seconds, so cells experience gradients in substrate, oxygen and pH; the number of generations between the seed vial and the production vessel roughly doubles, which is week 10's genetic instability problem; and the feed profile that was hand-tuned in a 1 L fermenter has to be re-tuned for a vessel whose heat and oxygen limits are different. None of these is a biology problem you can fix in a flask, which is why the industrial literature reads as though it were about a different subject from the academic one.

By-products, and how they destroy a yield while the titer looks fine

The usual suspects in an *E. coli* fermentation, and where each comes from: acetate, from overflow when glycolytic flux exceeds respiratory capacity (week 6); lactate and formate, from fermentative routes under oxygen limitation, including the micro-anaerobic pockets of a large vessel; succinate, from the reductive branch of the TCA cycle under the same conditions; pyruvate, when its consumption is slower than its supply, which is common after a pathway deletion; and glycerol or other reduced compounds, as redox valves when NADH accumulates.

The reason to know the list is diagnostic. Each by-product is an arrow on the metabolic map, so measuring which one appears localises the imbalance: acetate says the feed outran respiration, lactate says oxygen, pyruvate says the step you deleted had a job, glycerol says redox. A fermentation in which you measure only glucose and product is a fermentation in which you cannot diagnose anything, and the marginal cost of adding organic acids to the HPLC method

is an hour.

What rate buys: the arithmetic nobody writes down

Yield sets the cost of sugar and titer sets the cost of water; rate sets the cost of steel. Work it through once and you will never again treat volumetric productivity as the third number.

Take a 100 m³ fermenter — 100,000 L — running at 1 g/L/h. That is 100 kg of product per hour, so with 8,000 operating hours a year and 80% of them productive after cleaning, sterilisation, filling and turnaround, the vessel makes about 640 tonnes a year. Halve the productivity to 0.5 g/L/h and the same vessel makes 320 tonnes, so meeting the original target needs a second fermenter, a second seed train and the air, cooling and downstream capacity to match. Rate does not appear in the cash cost per kilogram directly; it appears in how much plant you had to build to produce the kilograms at all, which lands in the capital charge and is usually the largest single line after feedstock.

This is also where the three metrics stop being independent. Pushing titer by extending the fermentation lowers the rate, because the denominator is total elapsed time: a run reaching 100 g/L in 60 h and one reaching 130 g/L in 100 h have productivities of 1.7 and 1.3 g/L/h, and which is better depends on whether the plant is capital-limited or separation-limited. Pushing yield by slowing the feed to suppress overflow lowers the rate too, for the reasons in week 6. There is no strain that is best on all three, and the useful question in any project is which one your particular process is currently losing money on.

A serviceable cash-cost model you can build in a spreadsheet in an hour, and which is enough to keep a conversation with a process economist honest:

```
cost per kg = substrate price / yield            (feedstock)
            + other variable cost per kg          (nutrients, base, enzymes, utilities)
            + annual fixed cost / annual output   (labour, maintenance, capital charge)

annual output = volume x titer x utilisation / batch time, summed over vessels
```

Put your own numbers in and note which term dominates. For a commodity it is nearly always feedstock, which is why yield binds; for a mid-value specialty it is often the fixed term divided by a small output, which is why rate binds; for a biologic it is almost always downstream and quality, which is why neither of the first two appears in the discussion.

Titer	g/L at harvest
Rate	volumetric productivity g/L/h = q_P · X; quote total elapsed time
Yield	g/g, mol/mol, and as a fraction of theoretical — the informative form
Ethanol maximum	2 mol/mol = 0.511 g/g, redox closed exactly
Isobutanol maximum	1 mol/mol = 0.411 g/g, redox closed but cofactor-mismatched
Feedstock cost	substrate price ÷ mass yield: $0.40/kg ÷ 0.40 g/g = $1.00/kg product
Broth volume	1,000/titer litres per kg: 100 L at 10 g/L, 8.3 L at 120 g/L

Your strain reaches 12 g/L at 0.18 g/g in 72 h. A colleague proposes a campaign to double the titer. The product is a commodity selling near $1.80/kg. What do you propose instead, and with what numbers?

Yield, and the arithmetic settles it. At 0.18 g/g the feedstock alone costs $0.40/0.18 = $2.22 per kilogram of product against a selling price of $1.80: the process loses money on sugar before any capital, labour, utilities or separation. Doubling the titer to 24 g/L at the same yield halves the water to process, which is worth real money downstream, and leaves the feedstock cost unchanged, so the process still loses money. Raising the yield to 0.40 g/g brings the sugar cost to $1.00/kg and makes everything else worth arguing about.

The constructive version of the proposal: find out where the carbon currently goes, with a closed carbon balance including by-products and biomass, because at 0.18 g/g — probably a quarter of theoretical — the large majority of the carbon is going somewhere identifiable. Then yield work has a target rather than a slogan. Titer becomes the priority once yield is within reach of theoretical, and at that point the usual levers are process rather than genetic: fed-batch profile, in situ removal, tolerance.

Three things people get wrong

- ‘Titer is the headline number.’ For a commodity, yield decides viability and titer decides separation cost; titer alone tells you almost nothing.
- ‘Volumetric productivity is titer over fermentation time.’ Only if that time includes the lag and growth phase. Quoting

the production phase alone flatters the number by a factor of two.

- ‘A clean broth means good yield.’ Carbon can disappear into biomass and CO_2 without a single by-product peak. Close the carbon balance.

Self-check — answer on paper first

1. Compute the theoretical maximum mass yield of ethanol on glucose and state the balance that makes it a fermentation.

Two moles of ethanol per mole of glucose, so 2 × 46.07/180.16 = 0.511 g/g, with the remaining two carbons leaving as CO_2. It is a fermentation because the redox balance closes internally: glycolysis generates 2 NADH, and the two reductions of acetaldehyde consume 2 NADH, so no external electron acceptor is needed and the route can run anaerobically. Any product whose balance does not close this way requires respiration or an electron sink, which is why the list of true fermentation products is short.

2. Explain, with a number, why a by-product can destroy a yield while the titer looks acceptable.

Because titer measures what you made and yield measures what it cost. A strain making 30 g/L of product while excreting 20 g/L of acetate has consumed sugar for both; if the acetate represents a third of the carbon taken up, the yield is a third below what the same titer would imply in a clean fermentation. The titer is unchanged by the leak, and for a commodity the leak is the whole economics. It is also recoverable: the carbon going to acetate is carbon you could redirect, which is why the by-product measurement is the first place to look for yield.

3. Which metric would you optimise for a monoclonal antibody, and why is the answer different from a bulk chemical?

Total output and product quality — the correct glycosylation, aggregate content, reproducibility — with titer mattering mainly because it sets the scale of the purification train. Yield on glucose is nearly irrelevant: sugar is a negligible fraction of the cost of a product selling for orders of magnitude more per kilogram. The inversion is driven entirely by price, and it is why strategies that are obviously right for ethanol are obviously wrong for a biologic.

What to open a book for. Doran's material- and energy-balance chapters, for the accounting done in engineering notation with problems you can work. Read Yim et al., Nat. Chem. Biol. 7:445 (2011) for the numbers rather than the pathway, and log it in the paper log — it is the case study you will be asked about.

Cofactor balancing, redox and toxicity

This is the week where phase 1's biology and phase 2's arithmetic meet. Almost every design that fails for a reason nobody predicted fails here: the carbon balanced, the redox did not, and the intermediate was toxic.

Diagnosing the imbalance, in a fixed order

Take your target's synthesis from glucose and work through four balances in this order, because each one uses the result of the last: carbon, NADH, NADPH, ATP. Keep the two hydride carriers separate — this is the whole point — and express the result per mole of product, so it composes with the yield calculation of week 18.

Three outcomes, each with a standard reading. A **surplus of NADH** means your pathway is more oxidised than the substrate, so the cell must dispose of electrons: aerobically through respiration, which is usually fine and costs oxygen transfer; anaerobically it needs an electron sink, which means a by-product and a yield loss. A **deficit of NADPH** is the common case for reduced products, and it is paid for in carbon or in energy, as below. A **deficit of ATP** means there is nothing left for growth and maintenance, which is a red flag rather than an impossibility — it usually means the process must be two-stage, with growth paid for separately from production.

The three fixes, and what each actually costs

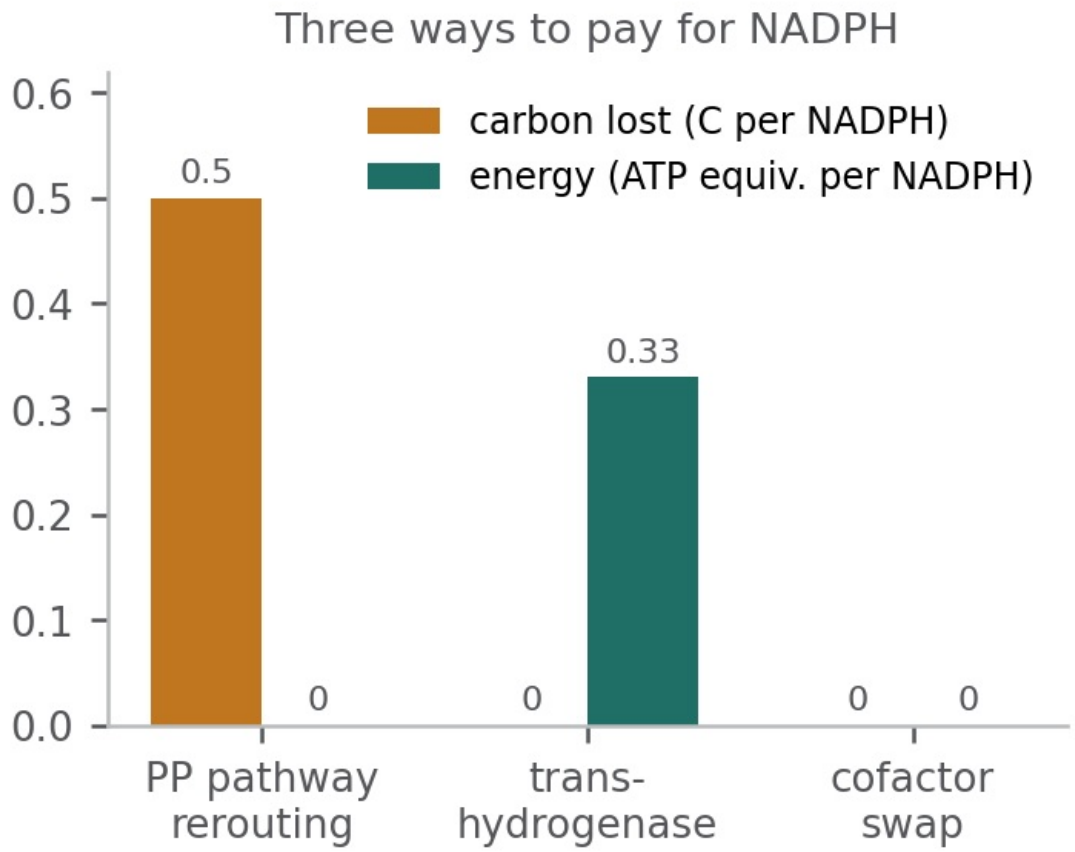


Figure 19.1 The cost of supplying one NADPH by each route. Pentose phosphate rerouting loses one carbon as CO_2 per two NADPH, so 0.5 C per NADPH; a proton-translocating transhydrogenase spends roughly one proton of the motive force per hydride, which at three to four protons per ATP is about 0.3 ATP equivalents; a cofactor swap has no metabolic cost at all, which is why it is attractive and why it is hard. Stoichiometric arithmetic, not measured data.

- **Reroute through the oxidative pentose phosphate pathway.** Each glucose-6-phosphate through the oxidative branch gives 2 NADPH and loses one carbon as CO_2, so half a carbon per NADPH. For a product needing 4 NADPH, that is two carbons lost before you have made anything, and it is why part 1's lysine figure tops out at 0.81 mol/mol rather than 1.0. Easy to implement — deleting phosphoglucose isomerase forces it — and expensive in exactly the currency a commodity process cannot spare.
- **Install or amplify a transhydrogenase.** The membrane-bound, proton-translocating enzyme converts NADH to

NADPH at the cost of the proton motive force, roughly one proton per hydride, so about a third of an ATP equivalent. Cheap in carbon, which is why it is the right first move for a product whose yield is carbon-limited rather than energy-limited. The trap is the soluble transhydrogenase, which runs the other way: express both strongly and you have built a futile cycle that burns ATP equivalents and produces nothing, a mistake that appears in the literature more than once.

- **Swap the enzyme's cofactor specificity.** Change the pathway enzyme so it accepts NADH rather than NADPH. There is no metabolic cost whatsoever, which makes it the best answer in principle; the cost is protein engineering, usually a handful of residues in the binding pocket, and the risk is that the variant has a lower k_{cat} than the parent, so you trade a cofactor problem for a kinetic one. It has been done many times and is now a reasonable first attempt rather than a research project.

Which to choose is a quantitative question and the answer is set by what limits you. Carbon-limited commodity: transhydrogenase or swap. Energy-limited or oxygen-limited process: pentose phosphate, since carbon is cheaper than oxygen transfer. High-value product where neither binds: whichever is fastest to build.

Driving force again, in a design context

Week 14's max-min driving force belongs in this diagnosis, because a cofactor fix can change the thermodynamics. Reductions driven by NADPH have a different driving force from the same reactions driven by NADH, since the two pools are held at very different ratios — NADPH kept reduced, NADH kept oxidised, as part 1 chapter 4 explained. A cofactor swap that solves your supply problem can therefore weaken the driving force of the very step you swapped, and the check is two minutes in eQuilibrator with the appropriate ratio. The same applies in reverse: a step that is marginal with NADH may become comfortable with NADPH, which is occasionally a reason to *increase* your NADPH demand deliberately.

Toxicity, and why dynamic control beats static expression

Two different problems travel under the word toxicity. **Product toxicity** sets the titer ceiling and is mostly a tolerance problem, addressed by in situ removal (weeks 44 to 47), by two-phase fermentation, or by evolving the host — and tolerance is still, honestly, evolved rather than designed, because it is usually polygenic and involves membrane composition and efflux in ways that resist rational design. **Intermediate toxicity** is a pathway-balance problem and is much more tractable, because it arises from a mismatch between the rate of the step that makes the intermediate and the step that consumes it.

That is why dynamic control helps where static overexpression does not. Statically, you choose one expression level for the whole fermentation, and you must choose it low enough that the toxic intermediate never accumulates dangerously — which means it is too low during the phase when the downstream enzyme is abundant. Dynamically, a sensor of the intermediate, or of growth phase, or of cell density, raises the downstream enzyme when the intermediate rises and throttles the upstream one when it rises too far. The pathway then holds the intermediate near a set point instead of at a compromise, which is a control problem of a type you already know how to analyse: a feedback loop with a sensor, an actuator and a set point, and with the same stability questions you would ask of any controller.

Compartmentalisation addresses the same problem differently and should be named alongside it: confining the pathway to an organelle or a protein scaffold raises the local intermediate concentration where the enzymes are while keeping the cytosolic concentration — the one that poisons the cell — low. The costs are transport of every substrate and cofactor across the compartment boundary, and they are quantified in week 58.

Where the surplus electrons actually go, and what that costs in oxygen

A surplus of NADH is usually waved away with the word respiration, but respiration is a transfer to oxygen and oxygen has to be delivered. The stoichiometry is simple: each NADH oxidised by the chain consumes half a mole of O_2, so a pathway with a surplus of two NADH per product imposes one mole of oxygen demand per mole of product on top of whatever growth already needs.

Make it concrete. Producing 2 g/L/h of a 100 Da product is 20 mmol/L/h of product, so a surplus of two NADH apiece is 40 mmol/L/h of NADH and therefore 20 mmol/L/h of extra oxygen demand. Industrial fermenters achieve oxygen transfer rates of the order of 100 to 300 mmol/L/h, depending on vessel, agitation and pressure, so this particular demand is affordable — but it is a tenth to a fifth of the budget before the cells have grown, and a design with a surplus of eight NADH per product at the same rate would not be. The calculation takes a minute and belongs in the same page as the carbon balance, because oxygen transfer is the hard physical ceiling of aerobic fermentation and the thing that degrades most on scale-up.

Two refinements worth carrying. Oxygen transfer costs power, roughly as a strong function of agitation rate, so the last increments are disproportionately expensive, and pushing oxygen demand is a decision about operating cost and not

only about feasibility. And the heat follows the oxygen: oxidising substrate releases heat in near-fixed proportion to oxygen consumed, so an oxygen-demanding process is also a cooling-demanding one, which is the real constraint in a large vessel in a warm climate. Both are quantified in phase 4, and both are decided here, in the redox balance.

Tolerance: why it is still evolved rather than designed

The honest state of the art deserves stating plainly, because it is an exception to everything else in this part. Product tolerance is usually polygenic, involving membrane lipid composition, efflux capacity, proton homeostasis, chaperones and general stress responses, and the causal mutations found by evolution are frequently in regulators whose connection to the phenotype is not obvious even in hindsight. Rational design has a poor record here, and adaptive laboratory evolution — serial passage under increasing product concentration, followed by sequencing the survivors — has a good one.

Treat that as a method rather than a defeat, and run it properly. Evolve in the condition you actually care about, including the product at the concentration you need and the medium you will use, because evolution optimises what you select for and nothing else. Keep the populations replicate and independent, so that convergent mutations across lineages identify real targets rather than one lineage's history. Sequence and then reconstruct the candidate mutations individually in the clean parent, because an evolved strain carries passengers as well as drivers, and only the reconstruction tells you which is which. Done that way, evolution becomes a hypothesis generator feeding the design stage of chapter 11's loop, and the mutations it finds are reusable knowledge.

Order of balances	`carbon → NADH → NADPH → ATP, separately, per mole of product`
PP pathway cost	`2 NADPH per glucose-6-P, 1 CO₂ lost → 0.5 C per NADPH`
Transhydrogenase cost	`≈1 H⁺ per hydride ≈ 0.3 ATP equivalents per NADPH`
Cofactor swap cost	`zero metabolic cost; risk is k_cat of the engineered variant`
Dynamic control	`hold the intermediate at a set point instead of at a compromise`

Your pathway consumes 4 NADPH and 2 ATP per product and yields 4 of glucose's 6 carbons. Diagnose it and propose a route to the highest yield you can defend.

Carbon first: 4 of 6 carbons reach the product, so the naive ceiling is 0.67 mol/mol on carbon alone. Redox: 4 NADPH are needed and glycolysis supplies NADH, not NADPH. If you pay for all of it through the oxidative pentose phosphate pathway, 4 NADPH needs 2 glucose-6-phosphate through the branch and costs 2 carbons as CO_2 — which, with the carbon skeletons returning to glycolysis, drops the achievable yield well below the carbon ceiling, which is exactly the lysine result from part 1. ATP: glycolysis nets 2 and the pathway spends 2, so there is nothing for growth before maintenance, which forces a two-stage process or a partial respiration of some carbon.

The defensible route: swap as much of the NADPH demand to NADH as the enzymes allow, since glycolysis already makes NADH and the swap costs no carbon; cover the residual demand with a proton-translocating transhydrogenase, paying ~0.3 ATP per NADPH rather than 0.5 carbon; and accept that the ATP balance now requires a growth phase that is decoupled from production. Then state the assumption that will decide whether this works — that the engineered enzymes retain enough activity — and design the experiment that tests it before building the rest.

Three things people get wrong

- ‘Reducing equivalents balance.’ NADH and NADPH balance separately, and lumping them hides the problem that will dominate the project.
- ‘Add a transhydrogenase and the redox problem goes away.’ It moves it into the energy budget, and expressing both directions builds a futile cycle.
- ‘Toxicity is a tolerance problem.’ Product toxicity is; intermediate toxicity is a pathway balance problem and is fixable by control.

Self-check — answer on paper first

1. Give the cost, in its own units, of each of the three standard NADPH fixes, and name the process condition that decides between them.

Pentose phosphate rerouting costs half a carbon per NADPH, lost as CO_2; a proton-translocating transhydrogenase costs roughly a third of an ATP equivalent per NADPH, taken from the proton motive force; a cofactor swap costs nothing metabolically and risks the enzyme's turnover number. What decides is which resource binds: carbon for a commodity whose economics are feedstock-dominated, energy or oxygen transfer for a process already at its respiratory limit.

2. Why does a toxic intermediate argue for dynamic rather than static control, and what does the controller need?

Because static expression forces one compromise level for the whole run: low enough to keep the intermediate safe at the worst moment, and therefore too low whenever conditions are better, so the pathway is throttled most of the time. A dynamic controller holds the intermediate near a set point by raising consumption or lowering production as it rises. It needs three things: a sensor that responds to the intermediate or to a correlate of it, an actuator — a promoter, small RNA or degradation tag that changes an enzyme's level fast enough — and a loop whose gain and delay do not make it oscillate, which is the stability question you would ask of any controller.

3. A design balances carbon and total reducing equivalents but needs 3 NADPH and produces 3 NADH per product. What is wrong and what are the options?

Nothing is wrong on paper and everything is wrong in the cell: the two pools are held at deliberately different ratios and are not interchangeable, so the design has a 3 NADPH deficit and a 3 NADH surplus simultaneously. Options: a transhydrogenase, which solves both at once and is the natural answer here, at roughly an ATP equivalent in total; swapping pathway enzymes to NADH, which solves both at no metabolic cost if the enzymes allow it; or oxidising the NADH surplus through respiration while paying for NADPH through the pentose phosphate pathway, which solves both while losing 1.5 carbons and is the worst of the three on yield.

What to open a book for. Hartline et al., Metabolic Engineering 63:126 (2021), for dynamic control — it is the review to know by name. Re-read Nielsen & Keasling (2016) with cofactors in mind; the second reading is a different paper. For the balances themselves, part 1 chapter 4 and Stephanopoulos Ch. 2 and 3 are all you need.

The phase 2 checkpoint

Answer all four from memory, on paper, before reading the model answers. If the first one is shaky, the problem is the balances in part 1 chapter 4 and you should go back to them rather than forward.

1. For a target of your choice, write the maximum-yield stoichiometry from glucose and identify the cofactor surplus or deficit.

Method, not answer. Write the balanced synthesis from glucose; close the carbon balance; close NADH and NADPH separately; close ATP. A surplus of NADH means the pathway is more oxidised than the substrate and needs an electron sink or respiration. A deficit of NADPH means pentose phosphate flux or a transhydrogenase must supply it, at the carbon or energy cost you should state in the answer. An ATP deficit means nothing is left for growth, which forces a staged process. A good answer states the number and its cost; an excellent one states which of the four balances binds.

2. State the MCA summation theorem and its engineering implication.

The flux control coefficients of all enzymes in a pathway sum to 1. Control is therefore shared: no single enzyme has a coefficient of 1, so raising one enzyme gives a sublinear flux gain and the limitation simply moves to another step — in the model of chapter 15, tenfold overexpression of a step with a coefficient of 0.10 bought 8% more flux and left that step with no control at all. Useful improvements require several coordinated changes, which is the argument for multiplexed editing and for library-based tuning.

3. Define titer, rate and yield, and say which dominates for a bulk chemical and which for a high-value pharmaceutical.

Titer is concentration at harvest (g/L); rate is volumetric productivity (g/L/h), quoted over the whole fermentation; yield is product per substrate (g/g or mol/mol), best reported as a fraction of theoretical. For a bulk chemical, yield binds because feedstock dominates cost — $0.40/kg glucose at 0.25 g/g is $1.60 of sugar per kilogram of product — and titer matters next because a dilute broth makes separation expensive. For a high-value pharmaceutical, total output and purity dominate and a few per cent of sugar is irrelevant.

4. Read one Metabolic Engineering abstract, predict the design logic, then check yourself against the paper.

No fixed answer. You are checking whether, from the abstract alone, you could name the host, the likely bottleneck and the type of intervention before reading the paper. If you could not, reread weeks 11 to 19 rather than pressing on.

Before you go on to phase 3

Phase 3 is the technique phase: assembly, editing, screening, analytics. It is much more concrete than what you have just done, and the risk is the opposite one — that you will enjoy the protocols and stop asking what they are for. The discipline to carry forward is that every technique in phase 3 exists to widen one stage of the loop in chapter 11, and the question to ask of each is which stage, and by how much.

Three things from this phase you will use on nearly every page of what follows: that stoichiometry constrains and measurements decide, so every flux number you read was partly assumed; that control is distributed, so every single-gene result you read should be checked against the fold-change it required; and that the honest reporting of an experiment includes what it could have detected. If you keep those three, you will read the literature more usefully than most of the people writing it.

On the numbers in this part

Everything quantitative here was computed by a script rather than remembered, and the scripts are short enough to retype. The three-enzyme model in chapter 16 generated figures 15.1 and 16.1 and every control coefficient quoted in chapter 15; its parameters are assumed, and the captions say so. The stiffness comparison in chapter 12 was measured on the Robertson problem at stated tolerances — 243,050 right-hand-side evaluations against 305 — and will vary a little with SciPy version. The statistics in chapter 17 come from the tracker's own example data, computed with scipy.stats, with power from the non-central t distribution rather than a normal approximation. The thermodynamic and transport numbers in chapter 14 use the assumptions stated beside them, and the permeability coefficients are order-of-magnitude class values rather than measurements for any particular molecule.

Where I have given a figure that matters to a decision, check it: eQuilibrator for free energies, BioNumbers for physiological concentrations and cell parameters, and the primary paper for anything attributed to one. The errors in this document are mine. What it leaves out deliberately is the full derivation of the control-analysis theorems, the detailed treatment of labelling-based flux analysis, which belongs with the analytics in phase 3, and genome-scale modelling practice, which needs the software in front of you rather than a page.

Part 7 — Building the strain: assembly, everyday assays and genome engineering (weeks 20 to 25)

I said when I handed over part 6 that a written module adds least to the technique weeks, because a protocol you can run beats a chapter about it. That is true of the wet-lab steps and false of everything around them, which is what this part covers: the arithmetic that tells you whether an assembly will work before you order it, the controls that make a daily assay interpretable, and the probability calculations that decide how many MAGE cycles or how much DNA a project needs. None of that is in a protocol, and all of it is the difference between running a technique and understanding one.

So these chapters deliberately do not teach you to pipette. They teach the five things a protocol leaves out: why a method exists, what its failure mode is, what number tells you in advance whether it will work, which control makes its output mean something, and when to choose a different method. Where there is code — the sequence work in chapter 21 — it is written to be typed in, because a design check should be a script that runs in a second rather than an afternoon with a sequence viewer.

One structural note. Weeks 20 to 25 are where the module's earlier parts stop being theory. The control analysis of part 2 chapter 15 says you must change several steps at once, and MAGE in chapter 23 is how; the leak argument of part 1 chapter 8 says a repressor sets a floor, and CRISPRi in chapter 24 is how far down that floor actually goes; the restriction barriers of part 4 chapter 52 decide whether chapter 25's non-model host is reachable at all. The cross-references are not decoration; they are the same problems arriving with tools attached.

Ch.	Week	Chapter
20	20	DNA assembly: four methods, and how to choose between them
21	21	Multigene pathways in code: cutting geometry, overhang design and in silico verification
22	22	Everyday assays: what the daily measurements actually mean
23	23	Genome engineering I: recombineering, MAGE, and Cas9 as a counterselection
24	24	Genome engineering II: CRISPRi, base editing, biosensors and dynamic control
25	25	Genome design and non-model hosts: edit counts, error budgets and toolkit arithmetic

DNA assembly: four methods, and how to choose between them

The methods differ in one thing that matters for pathway work: what they leave behind at the junction, and whether you can take the construct apart again afterwards.

Scars, and why they are the organising idea

A scar is sequence left at a junction by the assembly chemistry itself — a restriction site, a linker, anything that is there because of how the parts were joined rather than because the design wanted it. In a single-gene construct a scar is cosmetic. In a multi-part construct it is three problems at once.

- **It occupies a position where sequence matters.** Between a ribosome binding site and a start codon, or inside the 5′ untranslated region, the local sequence sets translation initiation — part 1 chapter 8 — so a scar there is not neutral, it is an uncontrolled part.
- **It repeats.** The same scar at every junction means repeated sequence in one molecule, which is a substrate for recombination and therefore for the instability of part 1 chapter 10.
- **It consumes a site.** If the scar is a restriction site, that enzyme can no longer be used elsewhere in the construct, which compounds as the construct grows.

The four methods line up on exactly that axis, and the comparison below is the week's deliverable.

Method	How it works	Junction	Best for	Fails when
Restriction–ligation	Cut both pieces with the same enzyme, ligate the compatible ends	Leaves the recognition site — a scar by construction	A single subclone into an existing vector; moving a cassette between standard backbones	The insert contains the site; more than two or three parts; sequence at the junction matters
Gibson assembly	One isothermal reaction: a 5′ exonuclease chews back to expose single strands, the overlaps anneal, a polymerase fills, a ligase seals	Scarless — the junction is whatever you designed	Joining a few large fragments, any sequence, PCR-generated ends; circularising a construct	Overlaps are short, repetitive or structured; many fragments at once; parts must be swapped repeatedly
Golden Gate	A Type IIS enzyme cuts *outside* its recognition site, so the site is removed with the cut and the four-base overhang is user-chosen; cutting and ligation run in one pot because correct joins destroy the site	Scarless, with a defined four-base junction	Many parts in a defined order, one reaction, high efficiency	Parts contain internal sites; overhangs are badly chosen (chapter 21)
MoClo and similar hierarchies	Golden Gate with a standard: fixed overhang syntax for part types, two enzymes alternating by level, parts to transcription units to multigene constructs	Defined and standard	Combinatorial libraries; any construct whose parts will be swapped repeatedly; foundry work	The project is one-off, or the parts cannot be domesticated cheaply

Gibson, read as chemistry

Three enzymes in one tube at 50 °C, each doing a job you should be able to state. **T5 exonuclease** chews the 5′ ends back, exposing 3′ single-stranded overhangs on both fragments — it is also heat-labile, which is why the reaction self-terminates and why the order of events works out. **The annealing** itself is not enzymatic: the exposed complementary overlaps find each other. **A high-fidelity polymerase** fills the gaps left behind, and **Taq ligase** seals the nicks, chosen because it is thermostable and discriminates against mismatched nicks.

Practical envelope: overlaps of 15 to 40 bp, with 20 to 25 the usual choice, and reliability falling off at both ends. Too short and the annealing is unstable at reaction temperature; too long and you are paying for primer length and increasing the chance that an overlap is similar to another one in the mixture. The failure modes are specific: GC-rich or structured overlaps that form hairpins instead of annealing, repeated sequences that let fragments join in the wrong order, and more than four or five fragments at once, where the number of possible wrong pairings outgrows the right one.

Golden Gate, and the trick that makes one pot work

A Type IIS enzyme such as BsaI recognises an asymmetric site and cuts at a fixed distance *outside* it. Two consequences follow and together they are the whole method. First, **the overhang sequence is yours** rather than the enzyme's, so you can choose a different four-base overhang at every junction and thereby fix the order of parts. Second, **a correct**

ligation destroys the recognition site, because the site sat outside the cut and ends up in the discarded fragment; so a correctly assembled junction cannot be cut again while an incorrectly assembled one — or an unreacted part — can. Running cutting and ligation together in one tube with cycling temperatures therefore pushes the reaction toward the correct product, which is why efficiencies are high enough to join six or ten parts in one reaction.

A four-overhang scheme joining five parts in a defined order, as the week asks you to draw: part A carries overhang 1 on its left (matching the vector) and 2 on its right; part B carries 2 and 3; C carries 3 and 4; D carries 4 and 5; E carries 5 and 6, with 6 matching the vector's other end. Each junction is unique, so A cannot ligate to C and no part can insert backwards. Chapter 21 is about choosing those six four-base sequences so that they behave, which turns out to be less obvious than it looks.

MoClo, and why foundries standardised

MoClo adds a syntax to Golden Gate: each part type — promoter, 5′ untranslated region, coding sequence, terminator — has fixed, agreed overhangs, so any promoter from the collection drops into any construct in the promoter position. Levels alternate enzymes, typically BsaI to assemble parts into transcription units and BsmBI to assemble units into multigene constructs, so a level-2 reaction cannot cut its own products.

The reason biofoundries standardised on this is not elegance, it is logistics. A standard syntax makes parts interchangeable so that a library of promoters can be combined with a library of coding sequences without redesigning anything; every reaction runs at the same temperature with the same enzyme mix, so a robot can do them all; and the design step becomes a computation over part identifiers rather than a bespoke primer design. The cost is paid up front, in domesticating every part to remove internal sites, which chapter 21 quantifies.

Choosing, with the week's two questions answered

Six parts, each to be swapped independently later. MoClo-style hierarchical Golden Gate, and the deciding word is *independently*. Scarless matters because some junctions are in translation-initiation territory; defined overhangs matter because six parts in a defined order in one reaction is exactly what the method is for; and the standard syntax matters most of all, because "swapped later" means you are building a family of constructs, so the up-front domestication cost is amortised over every future variant. Gibson would assemble the first construct about as well and would require new primers for every swap.

The one case for restriction–ligation. When you are moving a single cassette between two backbones that already carry compatible sites, no junction sequence matters, and the construct will not be rebuilt. It needs no primers, no sequence design and no domestication; it is twenty minutes of work with reagents already in the freezer. The mistake is not using it, it is using it for a multi-part pathway and then discovering that the scars are in the untranslated regions and that two of your parts contain the site.

One habit that applies whichever method you choose: design the verification before you order anything. For each junction, name the primer pair that would amplify across it and the band size you expect, and know which junctions a gel can confirm and which need sequencing — chapter 22 draws that line. A construct that assembles and cannot be verified cheaply is a construct you will not trust.

Scar	`sequence left by the chemistry, not the design; repeats, occupies regulatory space, consumes a site`
Gibson	`T5 exonuclease + polymerase + Taq ligase, 15–40 bp overlap, scarless`
Type IIS	`cuts outside its recognition site → user-chosen 4 bp overhang, and the site is destroyed by a correct join`
One-pot logic	`correct products cannot be re-cut, so cycling drives the reaction toward them`
Choose by	`how many parts, whether junction sequence matters, and whether parts will be swapped again`

Three things people get wrong

- ‘Scarless is a nicety.’ A scar in a 5′ untranslated region is an uncontrolled part, and the same scar at ten junctions is ten repeats inviting recombination.
- ‘Golden Gate works because Type IIS enzymes are efficient.’ It works because a correct junction destroys the site, so the reaction is driven rather than merely fast.
- ‘Gibson handles any number of fragments.’ Reliability falls off past four or five, because wrong pairings grow faster than right ones.

Self-check — answer on paper first

1. Explain in two sentences why a Type IIS enzyme permits one-pot assembly while a Type II enzyme does not.

A Type IIS enzyme cuts outside its recognition sequence, so the site is carried away on the discarded fragment and a correctly ligated junction no longer contains it — the product is therefore immune to further cutting while unreacted parts are not, which lets cutting and ligation proceed together and drives the equilibrium toward the assembly. A Type II enzyme cuts within its site, so every correct ligation recreates a substrate for the enzyme and the reaction cannot be run in one pot without the product being destroyed as fast as it forms.

2. You must join three 4 kb fragments amplified from genomic DNA, once, with no plan to rebuild. Which method and why?

Gibson. The parts are PCR products, so overlaps can be added in the primers at no extra step; three fragments is well within its reliable range; it needs no domestication, which matters because genomic fragments of 4 kb will almost certainly contain internal Type IIS sites — an expected 1.5 to 2 per 4 kb for a 6-base recogniser, counting both strands; and since nothing will be rebuilt, the standardisation that would justify MoClo buys nothing.

3. Why is a repeated junction sequence a stability problem rather than only an aesthetic one?

Because identical sequences in one molecule are substrates for homologous recombination, which can delete everything between two copies in the same orientation or invert it if they are opposed. In a multigene construct with the same scar at every junction, the most likely recombination product is a shorter construct missing one or more genes, and because that product has lost the expression burden it frequently outgrows the intact one — part 1 chapter 10's instability argument, arriving through the assembly design rather than through the host.

What to open a book for. Addgene's "Plasmids 101: Golden Gate cloning", which is short and better than any textbook treatment, and Gibson et al., Nat. Methods 6:343 (2009), read for the enzymology rather than the protocol. A MoClo or Yeast Toolkit methods paper for the syntax. No book is needed this week.

Multigene pathways in code: cutting geometry, overhang design and in silico verification

A design check should be a script that runs in a second, not an afternoon in a sequence viewer. This week is where sequences stop being pictures and become data you can test.

Sequences as data

Biopython gives you three objects and that is most of what pathway design needs: `Seq` for a sequence with biological methods on it, `SeqRecord` for a sequence with features and annotations, and `SeqIO` to read and write FASTA and GenBank. The useful habit is to put features into a pandas DataFrame — one row per feature, columns for type, start, end and strand — at which point every check you want to run is a query rather than a visual scan.

```
from Bio import SeqIO
import pandas as pd

rec = SeqIO.read("plasmid.gb", "genbank")
rows = [{"type": f.type, "start": int(f.location.start), "end": int(f.location.end),
         "strand": f.location.strand,
         "name": f.qualifiers.get("label", [""])[0]} for f in rec.features]
df = pd.DataFrame(rows).sort_values("start")

for f in rec.features:                       # check every CDS translates to its annotation
    if f.type == "CDS":
        prot = f.extract(rec.seq).translate(to_stop=True)
        print(f.qualifiers.get("label", ["?"])[0], len(prot))
```

That last loop is worth running on any plasmid you are handed. An annotation that disagrees with its own sequence is common, usually because someone edited the sequence in a viewer without updating the feature, and finding it before you build is free.

Cutting geometry, drawn once and then coded

Draw BsaI on paper until it is automatic: it recognises GGTCTC, cuts one base after the site on the top strand and five after it on the bottom, leaving a four-base 5′ overhang whose sequence is whatever you put in those four positions. BsmBI recognises CGTCTC with the same one-and-five spacing; BbsI recognises GAAGAC with two-and-six. The spacing is the only thing that differs, and it is why the enzymes are interchangeable in a design as long as the spacer bases are right.

Then the thing people forget: **a recognition site exists on both strands**. The reverse complement of GGTCTC is GAGACC, and a site in that orientation cuts just as well, on the other side. Any site-finding function that searches only the given strand will miss half of them, and the symptom is an assembly that fails for no visible reason.

```
from Bio.Seq import Seq

def find_sites(seq, site="GGTCTC", spacer=1, overhang=4):
    """Return (position, strand, overhang) for every occurrence on both strands."""
    s, hits = str(seq).upper(), []
    rc = str(Seq(site).reverse_complement())
    for i in range(len(s) - len(site) + 1):
        if s[i:i+len(site)] == site:
            a = i + len(site) + spacer
            hits.append((i, "+", s[a:a+overhang]))
        if s[i:i+len(site)] == rc:
            b = i - spacer - overhang
            hits.append((i, "-", str(Seq(s[b:b+overhang]).reverse_complement())))
    return hits
```

Cross-check it against `Bio.Restriction`, and expect a discrepancy that is not an error: Biopython reports *cutting* positions while the function above reports *recognition* positions. Reconciling the two by hand once is the exercise, because confusing them is how a domestication edit gets made in the wrong place.

Domestication, and how often you will need it

Removing an internal site means changing a base without changing the protein — a synonymous codon substitution — then confirming by translation that the protein is identical. How often does this come up? For a six-base recogniser, the

expected number of sites on both strands of a sequence of length L is $2L/4^6$:

Part length	Expected BsaI sites (both strands)	Probability of none
1 kb	0.49	61%
3 kb	1.46	23%
30 kb pathway	14.6	effectively zero

So a typical 3 kb coding sequence has a three-in-four chance of needing domestication for one enzyme, and a ten-gene pathway will contain around fifteen sites between the two enzymes a MoClo scheme uses. This is the up-front cost that chapter 20 said standardisation charges, and it is why parts collections exist: the domestication is done once, by someone, and then never again.

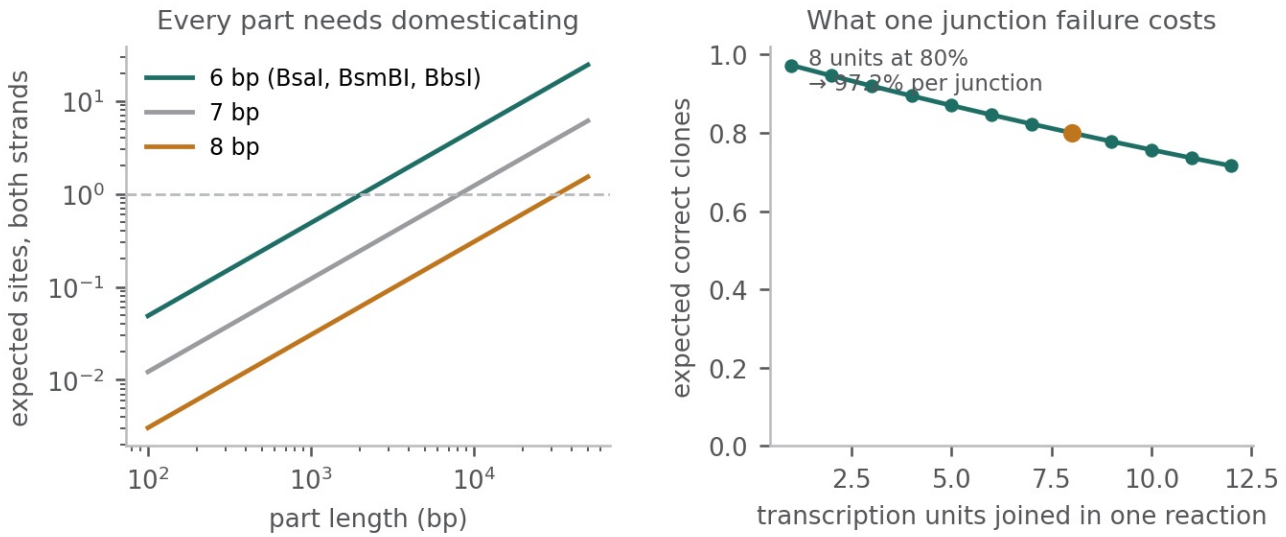


Figure 21.1 Left: expected recognition sites on both strands against part length for three motif lengths — a 3 kb part has about a three-in-four chance of containing a six-base site. Right: expected fraction of correct clones against the number of transcription units joined in one reaction, at the per-junction fidelity implied by 80% success for eight units. Computed for this module; the left panel assumes random equal base composition, so count your actual sequence rather than trusting it.

Overhang design, which is where assemblies silently fail

Four-base overhangs are not interchangeable. T4 DNA ligase joins some pairs faithfully and others promiscuously, and the comprehensive profiling of all four-base overhang ligations established the two practical facts: fidelity varies substantially between overhangs, and **mismatched pairs do ligate**, particularly when they differ at only one position. So a set of overhangs chosen because they look different can give an assembly that produces mostly wrong products, with no diagnostic other than disappointing colonies.

The rules a checker should enforce, and the arithmetic behind how many overhangs you can have:

- No palindromes. A palindromic overhang is its own reverse complement, so a part can join to itself or insert backwards. There are 16 palindromic four-mers out of 256.
- No duplicates and no reverse-complement pairs in the set, which leaves 120 mutually non-complementary overhangs as the absolute maximum.
- At least two base differences from every other overhang *and* from every other overhang's reverse complement, because one-base mismatches are the ones that ligate. Imposing that by greedy search collapses the maximum from 120 to **24** — which is why published high-fidelity sets contain roughly that many, and why a set of thirty overhangs chosen by eye cannot be faithful.
- A complexity filter: no homopolymers, no extreme GC content, since both correlate with poor ligation.

Write the checker, then hand-pick seven overhangs and run it; the usual outcome is that two of your seven are one base apart from each other's complement, which is exactly the failure the fidelity data describe. Generate the set by search instead, and assign the result to the junctions of your transcription unit as a table — part, left overhang, right overhang — which becomes the specification you order against.

Yeast in vivo assembly, and what a connector does

Yeast assembles DNA by homologous recombination: give it fragments whose neighbours share 40 to 60 bp of overlap

and it will join them in order inside the cell, including the vector, in one transformation. The method's strength is that it does not care about internal restriction sites and can join many fragments; its weakness is that each junction is a recombination event with its own probability, and that similar overlaps mispair.

The hybrid approach — build transcription units in vitro by Golden Gate, then join the units in yeast through standard connector sequences — exists because each method fails where the other works. Golden Gate struggles when the construct grows to many units because the number of unique faithful overhangs is limited, as the arithmetic above shows; yeast recombination struggles when the fragments are many and similar, which transcription units sharing promoters and terminators inevitably are. A connector is a short, designed, non-repetitive sequence placed at the ends of each unit, whose only job is to be the unique homology that pairs that unit with its neighbour — which removes the mispairing between similar units and makes the junction count the only remaining limit. The 2026 paper from KRIBB that the tracker names reports up to eight transcription units assembled at 80% efficiency in one transformation this way.

Now the arithmetic the week asks for, which is the useful part. If eight units at 80% overall implies independent junctions, the per-junction success is $0.8^{1/8}$ = **97.2%**. Under that figure, three units give 92%, five give 87%, eight give 80% and twelve would give 72%. Two readings follow. The method is remarkably good — 97% per junction is excellent — and the decline with size is gentle enough that the limit is not efficiency but screening: at 80% correct you pick four colonies and expect three to be right.

What the independence model leaves out, and you should say so whenever you quote it: mispairing between overlaps that are similar rather than identical, which is not independent across junctions because one mispairing event consumes two junctions at once; recombination between repeated sequences *within* the assembled construct, such as the same terminator used in four units, which produces clones that assembled correctly and then deleted part of themselves; and point mutations in correct-looking clones, which no junction PCR will detect. The model gives you the floor, not the answer.

Checking the build before you order it

The week's last session is the habit worth keeping for the rest of your career: assemble the construct in code, from the part sequences, in the order and with the overlaps you designed, and then test the assembled sequence. Translate every open reading frame and confirm each junction reads in frame. Scan the final molecule for stray BsaI and BsmBI sites, since domestication is only complete once the assembled sequence — including the junctions you created — is clean. List the colony-PCR primers for each junction with expected band sizes. Then order.

This costs an hour and catches the errors that cost a month: a part cloned in the wrong orientation because its overhangs were palindromic, a frameshift at a junction where the overhang length was miscounted, a site reintroduced by the junction itself, and a verification strategy that turns out to give indistinguishable band sizes for the right and wrong products.

Site frequency	`expected sites ≈ 2L·(1/4)^k; 1.46 for a 6-mer in 3 kb`
BsaI geometry	`GGTCTC, cut +1 top / +5 bottom → 4 bp 5′ overhang; reverse strand is GAGACC`
Overhang set limit	`256 four-mers → 16 palindromes → 120 non-RC → 24 with ≥2 differences from all others and their complements`
Junction success	`overall = p^n; 80% at 8 units → p = 0.972; 5 units → 87%`
Before ordering	`assemble in silico, translate every ORF, rescan for sites, list verification primers`

Three things people get wrong

- ‘I searched the sequence for the site.’ On one strand. The reverse complement cuts too.
- ‘The overhangs are all different.’ Different is not enough; one-base mismatches ligate, so the requirement is at least two differences from every other overhang and its complement.
- ‘The assembly worked, so the construct is right.’ Junction PCR confirms topology, not sequence. Point mutations and internal recombination survive it.

Self-check — answer on paper first

1. A colleague designs a 12-part Golden Gate assembly with 13 overhangs chosen by eye. What do you predict and how would you check?

That some fraction of the products will be mis-assembled, because thirteen overhangs chosen without a distance constraint will very likely contain at least one pair differing by a single base from each other or from a reverse complement, and such pairs ligate. The check is the set checker: no palindromes, no duplicates or reverse-complement pairs, and a minimum Hamming distance of two against every other overhang and every other overhang's complement.

If it fails, replace the offending overhangs from a published high-fidelity set — there are about 24 mutually compatible ones, which is enough for twelve parts but not by a wide margin.

2. Derive the per-junction success rate implied by 60% correct clones for five transcription units, and say what it predicts for eight.

With five junctions treated as independent, $p = 0.6^{1/5} = 0.902$, so eight units would give $0.902^8 = 43\%$. The practical reading is that at 43% you still only need to screen a handful of colonies, so efficiency is not the barrier; what the lower per-junction figure suggests is a systematic problem — overlaps too short, similar overlaps mispairing, or repeated internal sequences — which is worth fixing because it will get worse with every unit added.

3. Why should the final assembled sequence be rescanned for Type IIS sites even if every part was domesticated?

Because a junction creates new sequence that did not exist in either part: the four-base overhang plus the flanking bases of both neighbours can spell a recognition site that neither part contained. The same applies to any sequence feature you care about — a cryptic promoter, a terminator, an internal ribosome binding site or a start codon out of frame — which is why the in silico assembly is scanned as a whole rather than part by part.

What to open a book for. The Biopython Tutorial's `Bio.Seq`, `Bio.SeqIO` and restriction chapters, read with an editor open. Potapov et al., ACS Synth. Biol. (2018) for the ligation-fidelity result that justifies the overhang rules; Lee et al., ACS Synth. Biol. (2015) for the Yeast Toolkit syntax; and Seong et al., Trends Biotechnol. (2026) for the connector approach and the eight-unit figure.

Everyday assays: what the daily measurements actually mean

These are the measurements a lab makes every day, and not recognising them marks someone who has not been at a bench. The thread through all of them: every assay has a control that makes it interpretable, and the first question about any result is which controls were run.

qPCR, and what a fold change rests on

The measurement is a cycle threshold, C_t: the cycle at which fluorescence crosses a threshold, which is later for a less abundant template. Everything else is arithmetic resting on one assumption — how efficiently each target amplifies.

Efficiency comes from a standard curve of C_t against the log of template concentration. Its slope gives $E = 10^{-1/\text{slope}} - 1$, so a slope of −3.32 is 100% efficiency (a perfect doubling per cycle), −3.60 is 90%, and −3.10 is 110% — the last usually meaning inhibition at high template rather than better-than-perfect chemistry.

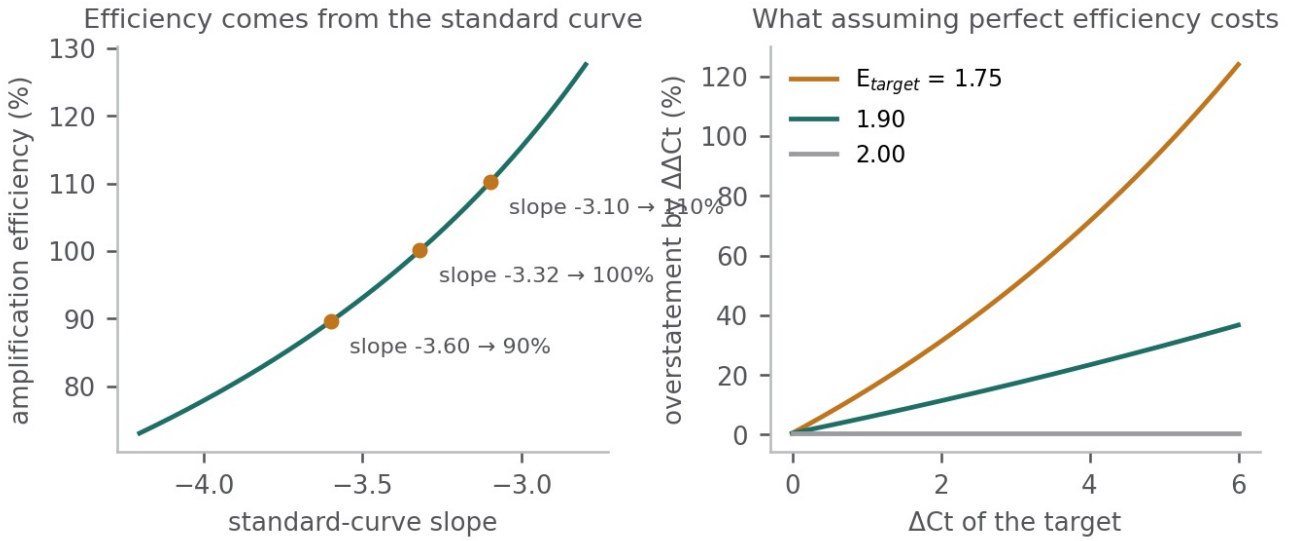


Figure 22.1 Left: amplification efficiency from the standard-curve slope, with three slopes marked. Right: how much the $\Delta\Delta C_t$ method overstates a fold change when the target does not amplify at 100% efficiency, against the size of the change, computed against the Pfaffl correction with a reference efficiency of 2.05. Computed for this module.

The $\Delta\Delta C_t$ method assumes both target and reference amplify at exactly 100%, so a fold change is $2^{-\Delta\Delta Ct}$. The Pfaffl method uses the measured efficiencies instead. The difference is not academic: for a ΔC_t of 3 with a target efficiency of 1.90 against a reference of 2.05, $\Delta\Delta C_t$ reports 6.96-fold where Pfaffl reports 5.94 — **17% overstated**; at a target efficiency of 1.75 it reports 6.96 against 4.64, **50% overstated**. And because the error compounds with the exponent, it grows with the size of the change you are reporting, which is the opposite of convenient. The rule: measure both efficiencies once per assay pair and use $\Delta\Delta C_t$ only when they agree with each other and with 100%.

The controls, and what each rules out. **No-template control**: contamination and primer dimers. **No-reverse-transcriptase control**: genomic DNA contamination, which otherwise appears as expression and is the commonest reason an "expressed" gene is not. **Melt curve**: specificity — one product gives one peak, and a second peak or a low-temperature shoulder means dimers or a second amplicon. **Standard curve**: efficiency and linear range. And the reference gene must be **validated in your own conditions**, because a gene that is stable across growth phases in one organism on one medium need not be stable across induction, nutrient limitation or your own pathway's burden — which is exactly what you are perturbing. Normalising to a reference that itself responds to induction converts a real change into no change, or manufactures one.

Gels, read as evidence

An agarose gel is a cheap, fast answer to one question: is there DNA of about the right size? Read the common outcomes:

- **A band at the wrong size** — mispriming, a wrong template, or a construct that is not what you think. If it is

consistently wrong by the same amount, suspect the construct rather than the PCR.

- **A smear** — degraded template, too much template, or over-cycling; also what sheared genomic DNA looks like.
- **Two bands where one was expected** — mispriming, a heteroduplex, a partially digested plasmid, or a genuine mixture, which in colony PCR often means a mixed colony.
- **A bright front** — primer dimers, which also consume reagents and depress the real product; the no-template control tells you immediately.

Failed PCR against failed gel is distinguished by the ladder and a positive control: if the ladder resolves, the gel ran; if a known-good template amplifies in the same run, the reaction chemistry works. And the line this draws for chapter 21's verification: a gel confirms the **presence and approximate size** of a junction, so it can tell you a part is in and roughly where. It cannot tell you the sequence, so a junction whose correctness depends on four bases of overhang, a reading frame, or a point mutation needs sequencing. The sensible division is colony PCR to triage, sequencing to confirm the clone you will keep.

Protein: present, soluble, active — three different claims

Running soluble and insoluble fractions side by side on SDS-PAGE answers the question a total lysate cannot: part 1 chapter 2's first hypothesis, that the protein is there and in inclusion bodies. A strong band in the insoluble lane and nothing in the soluble one is the classic plant-enzyme-in-*E. coli* result, and it points at expression temperature, induction strength, chaperone co-expression or a different host — the triage of week 30.

A Western blot establishes that something of approximately that size binds your antibody. It does not establish quantity without a loading control and a standard curve, and it does not establish activity at all. Three reasons a band can be present while the enzyme is dead, all from part 1 chapter 2: it is misfolded, aggregated or proteolysed into a fragment the antibody still binds; it is missing a cofactor, a metal or a redox partner; or it requires a post-translational modification or disulfide the host cannot make. The assay to add is an activity assay — initial rate, in the linear range, with a blank — because activity is the only measurement that answers the question you actually have.

Cells: distributions, not averages

Flow cytometry plots one event per cell, so a dot plot of two parameters and a histogram of one show you the *distribution*. Gating — drawing the boundary that defines a population — is subjective, and two competent people gate the same data differently, which is why gates must be shown and justified rather than described. Autofluorescence sets the floor, so an unlabelled control is not optional.

What a population average hides is the shape: a culture whose mean fluorescence doubled may have doubled uniformly, or may have split into an induced and an uninduced subpopulation — a bimodal response, which is common with positive feedback and with leaky inducible systems, and which means something entirely different for a biosensor-based screen. This is the connection to the biosensor work of week 26: a sensor is useful for sorting only if its single-cell distribution separates producers from non-producers, and a mean shift with overlapping distributions cannot sort.

Microscopy gives an image, and an image is not a measurement until it is quantified with stated rules. Brightfield and phase contrast show morphology and visible inclusion bodies — which is a real and useful observation, since inclusion bodies are large enough to see as refractile poles; fluorescence shows localisation. What magnification buys is resolution of structures, not certainty, and the honest limit on a fluorescence image of cells with inclusion bodies is that you can conclude the protein is aggregated and localised, not how much of it there is or whether the soluble remainder is active.

Finally the three growth measurements, which part 1 chapter 10 introduced and which disagree legitimately. Optical density is light scattering, so it tracks biomass and cell size rather than number. Colony counts measure cells able to grow on that medium, missing viable-but-non-culturable cells. Live/dead staining measures membrane integrity, which is neither growth nor number. For a yield calculation, use dry weight or a calibration to it; for the course of a culture, optical density; for viability after a stress, colony counts, with staining as the fast proxy you validated against them.

Reading a results figure like a reviewer

The week's last session is a habit: for each figure, write what was measured, which controls appear, and what claim it actually supports. Then the question you would ask in a lab meeting. A short list of the questions that are nearly always worth asking: what is the error bar and is it SD or SE (part 2 chapter 17); how many biological replicates, and were they independent colonies; what is the loading control or reference gene and was it validated here; where is the no-template or unlabelled control; is this an endpoint or an initial rate; and is the axis a log scale that flatters a small difference.

And the one you should ask of your own planned work: which artefact would most easily be mistaken for my result? For a strain-engineering project the usual answers are a growth-rate difference being read as a productivity difference, a measurement drift aligned with sample order, and a construct that recombined — and each has a control you can run in

advance rather than a defence you mount afterwards.

qPCR efficiency	`E = 10^(−1/slope) − 1; slope −3.32 → 100%`
$\Delta\Delta C_t$ error	`17% overstated at E = 1.90, 50% at E = 1.75, for a` ΔC_t `of 3`
Required controls	`no template, no RT, melt curve, standard curve, validated reference gene`
Gel can confirm	`presence and size; sequence needs sequencing`
Band ≠ activity	`soluble/insoluble fractions, then an initial-rate assay`
Average ≠ distribution	`a bimodal population and a uniform shift give the same mean`

Three things people get wrong

- '$\Delta\Delta C_t$ is the standard method.' It is, and it assumes both assays amplify perfectly; at 1.75 efficiency it overstates a threefold ΔC_t change by half.
- 'The reference gene is a housekeeping gene.' It is a hypothesis about stability that must be validated in the conditions you are perturbing.
- 'The mean fluorescence went up.' Which is consistent with every cell responding a little and with a third of them responding a lot — and only one of those can be sorted.

Self-check — answer on paper first

1. Your target amplifies at 85% efficiency and your reference at 100%. What does $\Delta\Delta C_t$ do to a reported sixfold induction, and what do you do instead?

It overstates it, because the true per-cycle multiplication of the target is 1.85 rather than 2, so each cycle of ΔC_t represents less change than assumed, and the error compounds with the exponent — for a ΔC_t near 2.6 the overstatement is of order 30 to 40%. Use the Pfaffl formulation, which raises each measured efficiency to its own ΔC_t, and report the efficiencies alongside the fold change. Better still, redesign the target assay: an efficiency of 85% usually means primer or amplicon problems that are worth fixing rather than correcting for.

2. A Western blot shows a strong band for your enzyme and the lysate has no activity. Give three hypotheses and the experiment for each.

Misfolding into inclusion bodies — run soluble and insoluble fractions side by side, since the blot of a total lysate cannot distinguish them. A missing cofactor, metal or redox partner — add the candidate cofactor to the assay or co-express the partner. Degradation to a fragment that retains the epitope, or the wrong protein cross-reacting — check the band size precisely against a marker and confirm identity by mass spectrometry or by a tagged construct. A fourth, often the real one, is that the assay conditions are wrong for this enzyme, which is cheapest to test by varying pH and temperature before concluding anything.

3. Why can optical density, colony counts and live/dead staining disagree without any of them being wrong?

Because they measure three different things. Optical density measures light scattering, which depends on biomass and cell size, so filamentous or enlarged cells read high while dividing into more, smaller cells can read lower. Colony counts measure the ability to form a colony on that medium, so stressed but living cells that cannot grow under those conditions are missed. Live/dead staining measures membrane integrity, which a cell can retain while being metabolically dead or lose transiently. The question decides which to trust: mass balances need dry weight, viability after stress needs counts, and the course of a culture needs optical density.

What to open a book for. No book: the MIQE guidelines for reporting qPCR, the Pfaffl paper for the efficiency-corrected formulation, and the literatures on reference-gene validation and on Western blot loading controls. Read them alongside four figures from your own paper log, one of each type, which is the exercise that makes them stick.

Genome engineering I: recombineering, MAGE, and Cas9 as a counterselection

Three things get confused in every discussion of CRISPR in bacteria: cutting, editing and regulating. Cas9 cuts. Recombineering edits. In a bacterium the cut is not the edit — it is the selection that finds it.

Recombineering: short homology, because a phage does the work

Lambda Red is three phage proteins that let a linear DNA fragment with short homology arms replace a chromosomal sequence. Each has one job and you should be able to state all three. **Exo** is a 5′→3′ exonuclease that chews back one strand of the incoming double-stranded fragment, exposing 3′ single-stranded ends. **Beta** is a single-strand annealing protein that binds those ends and pairs them with complementary sequence at the replication fork. **Gam** inhibits the host's RecBCD nuclease, which would otherwise degrade the linear fragment before anything else could happen.

That mechanism is why 40 to 50 bp of homology suffices where classical homologous recombination needs a kilobase: the pairing is done by a dedicated protein at a single-stranded region of a replication fork, not by the host's homology search. Practically, it means the homology arms fit on the PCR primers, so a knockout cassette is one PCR from a template.

The one-step inactivation workflow, drawn end to end: design primers with 40 to 50 bp of homology to the regions flanking your target gene, followed by sequence annealing to a resistance cassette flanked by FRT sites; amplify the cassette; transform the PCR product into cells expressing Red from an inducible plasmid; select on the antibiotic; verify by colony PCR with primers outside the homology arms, which gives a different band size for the wild type and the replacement; then, if you want the marker out, express FLP recombinase to excise it, leaving a scar, and cure both plasmids by growth at the non-permissive temperature. Each step has a failure mode worth knowing: no colonies usually means the Red system was not induced or the PCR product was contaminated with template plasmid, which is why the template carries a different marker and why DpnI digestion of the PCR is standard.

MAGE: many loci, by cycling oligonucleotides

MAGE drops the double-stranded cassette and the selection. Synthetic single-stranded oligos, each about 90 bases with the desired change in the middle, are introduced by electroporation into cells expressing Beta alone; Beta anneals them at the replication fork as Okazaki-fragment mimics, so the change is incorporated into one daughter strand. No selection is applied. Then the cycle repeats — grow, electroporate, grow — and because many different oligos can be introduced in the same electroporation, many loci are edited in parallel across a population.

What limits it is arithmetic, and the arithmetic is the point of this chapter. If the per-cycle allele-replacement frequency at a locus is p, then after N cycles the probability that a given locus is edited in a given cell is $1 - (1 - p)^N$, and for k independent loci the probability that all of them are edited is that quantity to the power k.

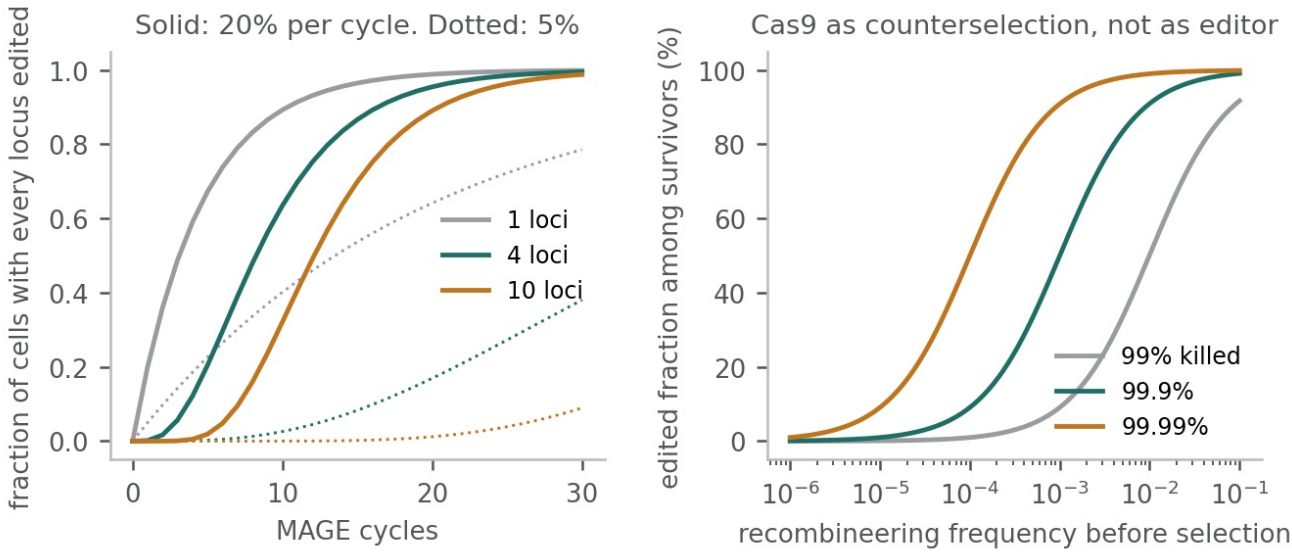


Figure 23.1 Left: fraction of the population with every target locus edited against MAGE cycles, at 20% per-cycle replacement (solid) and 5% (dotted), for one, four and ten loci. Right: the fraction of survivors that carry the intended edit when a Cas9 cut is used to kill unedited cells, against the

recombineering frequency, for three killing efficiencies. Computed for this module; the per-cycle frequency and the killing efficiency are the assumptions, and both vary with oligo design and strain.

At a realistic 20% per cycle: four loci need ten cycles to reach 64% of the population and five cycles give only 20%; ten loci give 32% at ten cycles and need twenty to reach 89%. At 5% per cycle — a poorly designed oligo, a secondary-structured target, or a strain with active mismatch repair — the same targets need three to four times as many cycles. Two engineering conclusions follow. **Mismatch repair is the first thing to disable**, which is why MAGE strains are mismatch-repair deficient and why that in turn raises the background mutation rate, so you sequence the final strain rather than trusting it. And **the oligo design matters more than the cycle count**, since p enters the exponent: targeting the lagging strand, avoiding secondary structure, and using mismatches that evade repair all raise p, and raising p from 5% to 20% saves more cycles than any amount of patience.

The connection to part 2 chapter 15 is the reason MAGE exists at all. Control is distributed, so a useful flux improvement needs several changes at once, and the combinatorial space of "which ribosome binding sites, at which of twenty genes" is far too large to build one construct at a time. MAGE generates that diversity in a population and lets a screen or selection find the combination — which is the parallelism argument of part 2 chapter 11 implemented at the chromosome.

Cas9, and the thing to get right about bacteria

SpCas9 cuts double-stranded DNA 3 bp upstream of a 5′-NGG-3′ protospacer-adjacent motif, guided by a 20-nucleotide spacer, leaving blunt ends. Cas12a recognises a T-rich PAM, 5′-TTTV-3′, cuts further from it and leaves staggered ends, and processes its own guide array from a single transcript — which makes it the better choice when you want many guides from one promoter, or when your target region is AT-rich and lacks NGG sites. SpCas9 remains the default for a single well-placed cut because its targeting range is denser in most genomes and its variants are better characterised.

Now the part that is genuinely different from mammalian work. **Bacteria have no efficient non-homologous end joining**, so a chromosomal double-strand break is not repaired and is lethal. There is no "cut and let the cell make an indel": there is cut and die, unless the cell happens to carry a repair template that restores a sequence the guide can no longer recognise. Which makes Cas9's role in bacteria a **counterselection** — it kills everything that has not been edited by recombineering — rather than an editing enzyme. CRMAGE and similar hybrids are exactly this: Red recombineering supplies the edit, Cas9 removes the unedited majority.

Size the benefit. If recombineering edits a fraction f of the population and the Cas9 cut kills a fraction of the unedited cells, the edited fraction among survivors is f/(f + (1 − f)(1 − kill)):

Recombineering frequency f	Cas9 kills 99.9%	Cas9 kills 99.99%
10^{-4}	9% edited among survivors	50%
10^{-3}	50%	91%
10^{-2}	91%	99%

So a counterselection converts an unscreenable 0.1% into a 50 to 91% hit rate, which is the difference between sequencing thousands of colonies and sequencing four — part 1 chapter 10's selection-versus-screening argument, with numbers. And it explains the standard failure: if the "edited" colonies are mostly unedited, the guide is not cutting efficiently, or the cells have escaped by mutating the target, the PAM or the Cas9 plasmid — which is why escapers should be sequenced rather than discarded, since what they tell you is which part of the system failed.

Designing guides, and what to record

For a knockout, place guides in the first third of the coding sequence, so that any survivor carrying a small repair-template-derived change is still disrupted; design three, because guide activity varies several-fold for reasons that are still not fully predictable; and check each against the genome for near-matches, counting mismatches and their positions, since mismatches in the PAM-proximal seed region abolish cutting while distal ones may not. Record every PAM site in the target region and the predicted off-targets for each guide, because that record is what a reviewer or a supervisor will ask for.

Then the repair template, which in a bacterium is not optional: a double-stranded cassette or a single-stranded oligo carrying the edit, with homology arms, and — this is the detail that is frequently missed — **designed so that the edited sequence can no longer be cut**. Either the edit itself destroys the protospacer or the PAM, or you must introduce a silent change that does. Otherwise Cas9 cuts the repaired allele and the edited cells die along with the rest.

Lambda Red	`Exo chews 5′, Beta anneals, Gam blocks RecBCD; 40–50 bp arms suffice`
MAGE accumulation	`P(all k loci after N cycles) = [1 − (1 − p)^N]^k`

MAGE numbers	`p = 0.2: 4 loci need 10 cycles for 64%, 10 loci need 20 for 89%`
PAMs	`SpCas9 NGG, blunt cut 3 bp upstream; Cas12a TTTV, staggered, self-processing array`
Cut ≠ edit	`no NHEJ in bacteria, so an unrepaired cut is lethal: Cas9 is the counterselection`
Enrichment	`edited among survivors = f/(f + (1−f)(1−kill))`

Three things people get wrong

- ‘CRISPR edits the genome.’ In bacteria it cuts, and the cut kills; recombineering does the editing and Cas9 selects for it.
- ‘More MAGE cycles will get there.’ p is in the exponent — improving the oligo from 5% to 20% per cycle saves more than tripling the cycles.
- ‘The repair template restores the gene, so we are done.’ If the repaired allele still matches the guide and PAM, it gets cut too. Destroy the site with the edit.

Self-check — answer on paper first

1. How many MAGE cycles are needed for 80% of the population to carry edits at six loci, at 20% per cycle?

Require $[1 - 0.8^N]^6 \geq 0.8$, so $1 - 0.8^N \geq 0.8^{1/6} = 0.9637$, giving $0.8^N \leq 0.0363$ and $N \geq \ln(0.0363)/\ln(0.8) \approx 14.8$, so fifteen cycles. The useful corollary is the sensitivity: at 10% per cycle the same target needs about 32 cycles, so the oligo design decides whether this is two weeks of work or five.

2. Explain why Cas9 is lethal in *E. coli* but produces indels in a mammalian cell line.

Because the repair capability differs. Mammalian cells have efficient non-homologous end joining, which rejoins a broken chromosome imprecisely and produces the insertions and deletions that are the usual editing outcome. Most bacteria lack an efficient NHEJ pathway, so a chromosomal double-strand break cannot be rejoined and the cell dies — which makes the cut a selection rather than an edit, and means a repair template must be supplied for anything to survive.

3. Your CRISPR–recombineering experiment gives plenty of colonies and almost none are edited. Name the three explanations and how to distinguish them.

The guide is not cutting, so there is no counterselection — test by transforming the guide plasmid without a repair template, which should kill nearly everything if cutting works. The cells have escaped by losing or mutating the Cas9 or guide plasmid — test by re-isolating plasmid from survivors and sequencing the guide cassette. Or the recombineering frequency is far below what you assumed, so even good killing leaves few edited cells among survivors — test by applying a conventional selectable marker edit in parallel to measure f directly. The enrichment table above tells you which is plausible: with good killing, f must be very low indeed to give this result.

What to open a book for. Snyder & Champness on recombination, selection and counterselection, which remains the best treatment of why selection design matters. Then Datsenko & Wanner, PNAS 97:6640 (2000) for the one-step workflow and Wang et al., Nature 460:894 (2009) for MAGE, plus a 2023-or-later CRISPR-in-metabolic-engineering review.

Genome engineering II: CRISPRi, base editing, biosensors and dynamic control

Regulating a gene is not editing it, and for an essential gene it is the only option you have. This chapter is about the tools that change expression without changing sequence — and about closing a loop with them.

CRISPR interference, and why it beats a knockout

Mutate the two nuclease domains of Cas9 and you get dCas9: a programmable DNA-binding protein that still finds its target and no longer cuts. Bound at a promoter or in the early coding sequence, it blocks RNA polymerase — initiation if it sits on the promoter, elongation if it sits in the gene — and the gene is repressed. The repression is **tunable**, through the guide's affinity and position and through dCas9 expression level; **reversible**, by removing the inducer or curing the plasmid; **multiplexable**, since many guides can share one dCas9; and it requires no change to the chromosome, which is why it works in hosts whose editing tools are poor — the Cleto study in *C. glutamicum* being the standard demonstration.

Three reasons it beats a knockout, and the first is decisive. **Essential genes cannot be knocked out**, so for the large fraction of central-metabolism genes that are essential — and those are precisely the ones whose flux you want to redirect — a knockdown is the only experiment available. **A dose-response is more informative than an on-off answer**: a graded series of guides tells you the shape of the flux response, which is part 2 chapter 15's control coefficient measured directly. And **it is fast**: a new target is a new 20-base spacer, ordered as an oligo pair, rather than a cassette, a selection and a curing step.

What it cannot do: reach zero. Repression of 90 to 99% is typical and complete silencing is not, so a residual flux remains — which is part 1 chapter 8's leak argument from the other direction, and which matters when the residual activity of an enzyme is enough to spoil a selection. It also acts on transcription, so the protein already present is cleared only by dilution, as part 1 chapter 1 insisted: a knockdown takes several generations to show its phenotype, and in a slow-growing or stationary culture it may never fully show it. And guides can repress neighbouring genes in an operon, so a polar effect must be excluded before a phenotype is attributed.

CRISPR activation is the same machinery with an activator domain fused to dCas9, recruiting polymerase instead of blocking it. It works well in yeast, where eukaryotic activation domains are potent and the available upstream space is large; in bacteria it is more constrained, because activation depends on positioning the fused domain within a narrow window upstream of the promoter and the achievable fold-activation is modest. Treat it as a real tool in yeast and a developing one in bacteria.

Base editing: a change without a break

Fuse a deaminase to a catalytically impaired Cas9 — a nickase, or dead Cas9 — and it chemically converts a base within a small window of the protospacer without ever cutting both strands: cytosine deaminases give C→T, adenine deaminases give A→G, with the edit typically falling in a window of a few bases positioned at a fixed distance from the PAM.

When it beats a double-strand break, which is the question the week asks. When the organism cannot survive the break, which in bacteria is always — a base editor needs no repair template and kills nothing, so the editing efficiency is not multiplied by a survival probability. When you want a specific point change: a feedback-resistant allele of an allosteric enzyme (part 1 chapter 3), a promoter mutation, a premature stop codon to inactivate a gene without deleting it. And when you want to make many such changes at once, since base editors multiplex well and each edit is independent of a template.

What it cannot do: insert or delete anything; reach a target with no suitably positioned PAM, since the editing window's position relative to the PAM is fixed; or avoid bystander edits, because any editable base in the window will be edited — so a target with two cytosines in the window gives a mixture of products. Checking the window for bystanders is a design step, exactly like checking for internal restriction sites in chapter 21.

Biosensors: turning a metabolite into a signal

A transcription-factor biosensor is a natural regulatory protein repurposed. The transcription factor binds your metabolite, which changes its affinity for its operator, which changes transcription of a promoter you have placed in front of a fluorescent protein. The output is fluorescence per cell that varies with intracellular metabolite concentration — and, with that, the test stage of part 2 chapter 11 jumps from hundreds of samples a week on HPLC to 10^7 an hour by sorting.

Metabolite classes with well-characterised sensors include aromatic amino acids and their intermediates, several organic acids including muconate and malonyl-CoA, and sugars and sugar phosphates; plus riboswitch-based sensors for a handful of metabolites that bind RNA directly. The parameters to characterise before trusting one, which are the biosensor equivalent of a calibration curve: the **dynamic range** between uninduced and fully induced, the **operating range** of concentrations over which the output actually varies, the **leak**, and the **single-cell distribution** at each concentration — because, as chapter 22 said, a sensor whose distributions overlap cannot sort however well its means separate.

The failure mode to design against is the one part 2 chapter 11 named: cells optimise what is selected. A sorting campaign against a biosensor returns variants that light up the sensor, including ones that have mutated the sensor, lost the degradation of the fluorescent protein, or found an unrelated metabolite that cross-activates the transcription factor. Confirm every hit by the real assay and report the confirmation rate.

Dynamic control, with the loop drawn

Part 2 chapter 19 argued that a toxic intermediate calls for dynamic rather than static control, because static expression forces one compromise level for the whole run. Now the components exist, and the loop is drawable: a biosensor for the intermediate drives a promoter; that promoter expresses either a small RNA or a dCas9 guide against the upstream enzyme's transcript, or a protease-recognition tag system against the enzyme itself; so as the intermediate rises, the upstream enzyme is throttled, and as it falls, the throttle releases. The downstream half of the loop is the mirror image: the same sensor drives expression of the consuming enzyme, so supply and demand are matched at a set point rather than at a guess.

What you would measure to prove the loop works — and this is the question that separates a control system from a decorated construct. **The intermediate pool itself**, which should hold near a set point in the closed-loop strain and rise or oscillate in the open-loop one; without that measurement you have not demonstrated control, only an outcome. **The actuator's output**, meaning the upstream enzyme's level over time, which should visibly track the sensor. **The response to a disturbance**: pulse the substrate or step the inducer and show that the loop returns the intermediate to its set point, with a settling time you can quote — this is the test any control engineer would demand and almost no paper reports. And **the comparison that matters**: titer and yield against the best static expression level, found by titration, because a loop that beats a badly chosen constant is not evidence of anything.

The stability question is also real and is worth raising before you build: a feedback loop with high gain and a long delay oscillates, and the delays here are substantial — transcription and translation of the actuator, then dilution of the existing enzyme, which part 1 chapter 1 put at a generation time. A loop whose actuator can only act by dilution is a loop with a very slow response, which is one reason degradation tags appear in these designs.

The toolkit, consolidated

Tool	What it does	Choose it when
Restriction–ligation	Joins parts with a scar at a shared site	One subclone into an existing backbone, junction sequence irrelevant
Gibson	Scarless join of a few fragments by overlap	Two to four fragments, any sequence, no rebuild planned
Golden Gate / MoClo	One-pot ordered assembly of many parts with defined 4 bp junctions	Many parts, combinatorial variants, repeated swaps, foundry work
Yeast in vivo assembly	Joins fragments by homologous overlap inside the cell	Many or large fragments; joining transcription units via connectors
Lambda Red recombineering	Replaces a chromosomal sequence using short homology arms	A defined deletion or insertion with a selectable marker
MAGE	Cycled oligos edit many loci across a population without selection	Combinatorial tuning of many loci where a screen or selection exists
Cas9 / Cas12a	Programmable double-strand break — in bacteria, a counterselection	Enriching rare recombineered edits; curing plasmids; killing unedited cells
CRISPRi (dCas9)	Tunable, reversible, multiplexable knockdown	Essential genes; dose–response; hosts with poor editing tools
Base editing	Single-base change with no double-strand break	Point mutations such as feedback-resistant alleles; multiplexed point edits; hosts that cannot survive cuts
Biosensor + sorting	Converts an intracellular metabolite into a sortable signal	Libraries too large to screen by HPLC; and as the sensor in a dynamic loop
Dynamic control	Holds an intermediate at a set point by feedback	Toxic intermediates; growth–production decoupling; mismatched pathway segments

Three things people get wrong

- ‘CRISPRi knocks the gene out.’ It knocks it down, typically by 90 to 99%, and the residual flux is often what decides the experiment.
- ‘Base editing is CRISPR without the mess.’ It cannot insert or delete, needs a PAM at the right distance, and edits every editable base in its window.
- ‘The dynamic circuit raised the titer, so the loop works.’ Without the intermediate measured and a disturbance test, you have an outcome, not a demonstrated controller.

Self-check — answer on paper first

1. Why is CRISPRi the only available experiment for an essential gene, and what is the cost of using it?

Because a deletion of an essential gene cannot be recovered — the cells die — so there is no strain to measure, whereas a knockdown leaves the gene present and reduces its expression to a chosen level, giving a viable strain and a dose–response. The costs are that repression is incomplete, so a residual flux persists and a phenotype may be attenuated; that the effect appears only after the existing protein is diluted out over several generations; and that a guide can exert polar effects on downstream genes in the same operon, which must be excluded before the phenotype is attributed to the intended target.

2. You want a feedback-resistant allele of an allosteric enzyme, changing one codon. Compare base editing with a CRISPR–recombineering edit.

Base editing is likely the better route if the required change is a C→T or A→G transition and a PAM sits at the right distance to put the target base in the editing window: no repair template, no double-strand break, no killing, and it multiplexes if you want the same change in several genes. CRISPR–recombineering handles any substitution, insertion or deletion and does not depend on window positioning, but needs a repair template designed to destroy the protospacer, and its yield is the recombineering frequency enriched by the counterselection. Check the window for bystander editable bases first — that check decides it as often as anything else.

3. Design the minimum experiment that would convince you a dynamic control loop is working.

Three measurements against an open-loop control. The intermediate pool over time, which should sit near a set point in the closed loop and drift or accumulate in the open one. The actuator's level over the same time course, showing it tracks the sensor rather than merely being expressed. And a disturbance: a substrate pulse or an induction step, with the intermediate returning to its set point and a settling time reported. Then the comparison that makes it worth having — titer and yield against the best static expression level found by titration, not against an arbitrary constant.

What to open a book for. Cleto et al., ACS Synth. Biol. 5:375 (2016) for CRISPRi in a non-model industrial host, a metabolite-biosensor review for the sensor parameters, and Hartline et al., Metab. Eng. 63:126 (2021) reread with the tools in hand — it reads differently once you know what the actuators are.

Genome design and non-model hosts: edit counts, error budgets and toolkit arithmetic

Two apparently different questions with the same arithmetic underneath: how many edits can you make before verification becomes the bottleneck, and how many cycles does a new host's toolkit have to repay?

Why rewrite a genome

Fredens and colleagues synthesised a 4 Mb *E. coli* genome in which two serine codons, TCG and TCA, were replaced by AGC and AGT, and the TAG stop codon by TAA — **18,214 edits** in total, compressing the genetic code from 64 to 61 codons. The purposes are three and they are worth separating.

- **Free codons for new chemistry.** A codon with no assigned tRNA or release factor can be reassigned to a non-canonical amino acid, which is how you put chemistry into a protein that the twenty natural residues cannot do.
- **A barrier against viruses and foreign DNA.** A phage genome arriving in a recoded host carries codons the host can no longer translate — which is part 4 chapter 52's restriction-barrier logic implemented in the code itself rather than in a nuclease.
- **A test of what is essential.** Building a genome from a design is the only way to find out whether the design was right, which is a different kind of knowledge from deleting genes one at a time.

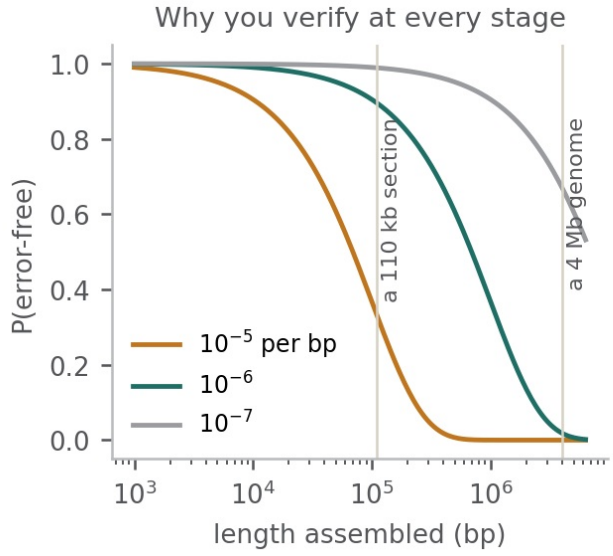
The costs are visible in the same paper and should be quoted alongside: 18,214 edits is an enormous design and verification burden; overlapping genes, where one gene's coding sequence runs into another's on the other strand, had to be corrected by hand because a synonymous change in one frame is not synonymous in the other; and the resulting strain grew more slowly than its parent, which is the honest summary of what recoding costs a cell.

Counting the edits, and the error budget

Do the arithmetic the week asks for. 18,214 edits across about 4 Mb is **4.6 edits per kilobase** — roughly one per 220 bases, which means almost every synthesised fragment carries several. Against roughly 1.3 million codons in the genome, the edits touch about **1.4% of all codons**. And built in sections of 91 to 136 kb, a 4 Mb genome is about **37 sections**, each of which must be assembled, verified and installed.

Now the part that explains the whole build strategy. If synthesis and assembly leave a residual error rate r per base pair, the probability that a sequence of length L is error-free is approximately e^{-rL}:

Residual error rate	P(a 110 kb section is clean)	P(a 4 Mb genome is clean)
10^{-5} per bp	33%	4×10^{-18}
10^{-6} per bp	90%	1.8%
10^{-7} per bp	99%	67%



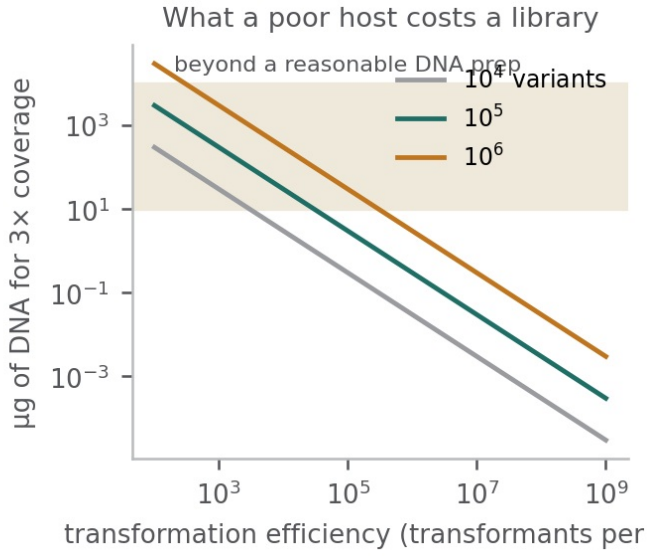


Figure 25.1 Left: probability that an assembly is error-free against its length, for three residual error rates, with a 110 kb section and a 4 Mb genome marked. Right: micrograms of DNA needed for 3× coverage of a library against the host's transformation efficiency, with the region beyond a reasonable preparation shaded. Computed for this module.

The conclusion is forced rather than chosen: **at any achievable error rate, a whole genome assembled without intermediate verification is certain to carry errors, while a 110 kb section is usually clean.** So the build hierarchy — oligonucleotides to fragments to sections of about 100 kb, with sequence verification at every stage, then replacement of the natural genome piece by piece with selection and counterselection — is not caution, it is the only design that can work. The same logic scales down to your own projects: verify at the level where the probability of being clean is high, not at the end where it is not.

Minimal genomes get one paragraph and it is the important one. **"Essential" depends on the conditions.** A gene is dispensable in rich medium at 37 °C and essential in minimal medium, or under osmotic stress, or at 42 °C, or in a strain carrying your pathway. So a minimal genome is minimal for something, and the strains produced this way are typically fragile outside the conditions they were minimised in — which is a feature if the condition is your process and a liability if it is not.

Choosing a non-model host, with the two calculations

The reasons to leave *E. coli* are the host properties of part 6: growth rate, tolerance, substrate range, native pathways, secretion and regulatory status. The reason not to is that everything has to be rebuilt. Two calculations decide it.

The toolkit repayment. Suppose a faster host shortens your design–build–test cycle from five days to four — a 20% gain, which for *V. natriegens* doubling in roughly half *E. coli*'s time is plausible. Building the toolkit first — transformation method, replicon, markers, characterised promoters, an editing tool, curing, a reporter — costs perhaps six months. At one day saved per cycle, that is **180 cycles to break even**. Unless your project will run many hundreds of cycles, or unless someone else has already built the toolkit, the arithmetic says stay where the tools are. This is the calculation that most enthusiasm for new hosts does not survive, and it is worth doing explicitly because the conclusion flips entirely when a toolkit already exists: then the cost is weeks, not months, and the same 20% gain is worth taking.

The library coverage. Transformation efficiency decides whether a library approach is possible at all, as part 1 chapter 10 said. For 3× coverage of a 10^5-member library you need 3×10^5 transformants: at 10^8 per microgram that is 3 ng of DNA, at 10^6 per microgram it is 0.3 µg, and at 10^4 per microgram it is 30 µg — a preparation that is possible but painful, and at 10^3 it becomes 300 µg, which is not a library experiment. So a host's transformation efficiency does not merely slow you down; below a threshold it removes an entire class of experiment, and that is the number to measure in week one rather than week ten.

Host	Attractive because	What it lacks
E. coli	Fastest cycle, every tool, deepest literature	No secretion to the medium, endotoxin, no GRAS status, acetate overflow
Vibrio natriegens	Divides about twice as fast as *E. coli*, naturally competent, grows on formate and acetate	A young toolkit, salt requirement, less literature, unfamiliar physiology
C. glutamicum	Native amino-acid export, high tolerance, GRAS, little overflow	Slower growth, fewer parts, conjugation or electroporation with modest efficiency
Y. lipolytica	Oleaginous, Crabtree-negative, hydrophobic substrates, secretory	NHEJ-dominated targeting, dimorphism, fewer characterised parts
C. necator	PHA physiology, autotrophic growth on H_2 and CO_2	Slow growth, conjugation-based delivery, restriction barriers in wild isolates
Streptomyces	Native secondary metabolism, secretion, large biosynthetic gene clusters	Mycelial growth, slow cycles, conjugation only, GC-rich sequence design

And the answer to the week's last framing question — when to stay in *E. coli* and move only the final strain: when the project is an iteration-heavy design problem rather than a physiology problem. Build, tune and screen in the host with the shortest cycle, then transfer the finished pathway into the production host once and accept that it will need re-tuning there. The cases where that fails are the ones where the host's physiology is the point: you cannot prototype a PHA accumulation strategy in a host that does not accumulate, or a secretion strategy in a host that does not secrete.

A toolkit checklist, and three first experiments

For any host new to you, the checklist is: a transformation method with a measured efficiency; a replicon that is maintained, or an integration site; a selectable marker that works in that organism's antibiotic-resistance background; a set of characterised promoters and ribosome binding sites with measured strengths in your conditions; an editing tool; a way to cure the plasmids you used; a reporter that can be read in that host's background fluorescence; and an

annotated genome, ideally including its restriction–modification systems from part 4 chapter 52.

The first three experiments, each with a control and a pass mark, because a plan without pass marks is a wish. **One**: transformation efficiency with a known replicating plasmid, controlled against a no-DNA mock, with a pass mark of 10^5 transformants per microgram — above that libraries are possible, below it they are not. **Two**: expression of a fluorescent reporter from two or three promoters, controlled against an empty vector for background, with a pass mark of a clearly resolved signal and a measured fold-range between the promoters. **Three**: a single targeted edit — a deletion of a non-essential, easily scored gene — controlled against an untargeted template, with a pass mark of verified edits among a handful of screened colonies.

And what would make you abandon the host: transformation efficiency stuck below about 10^3 per microgram after the obvious fixes, which eliminates library work and makes every construct a project; no working marker or no plasmid maintained; a doubling time so long that the cycle exceeds the project's patience; or restriction barriers that defeat every bypass in part 4 chapter 52. Writing those criteria down in advance is what lets you abandon a host after three months instead of eighteen.

Recoding scale	`18,214 edits in 4 Mb = 4.6 per kb ≈ 1.4% of codons; ~37 sections of 91–136 kb`
Error budget	`P(clean) ≈ e^(−rL): 90% for 110 kb at r = 10⁻⁶, 1.8% for 4 Mb`
Build rule	`verify at the length where P(clean) is high, not at the end`
Toolkit repayment	`(months to build × days) ÷ (days saved per cycle) = cycles to break even; ~180 for a 20% gain`
Library feasibility	`3N/efficiency = µg of DNA; 10⁵ variants needs 0.3 µg at 10⁶/µg, 30 µg at 10⁴/µg`
"Essential"	`conditional on medium, temperature and what the strain is carrying`

Three things people get wrong

- ‘We'll sequence the final construct.’ At 10^{-6} per base a 4 Mb assembly is clean 1.8% of the time. Verify in 100 kb pieces.
- ‘A faster host speeds up the project.’ Only after the toolkit repays itself, which at a 20% cycle gain takes of order 180 cycles.
- ‘A minimal genome is a clean chassis.’ It is minimal for the conditions it was minimised in, and frequently fragile outside them.

Self-check — answer on paper first

1. At a residual error rate of 10^{-6} per base, what fraction of 50 kb assemblies are error-free, and how does that justify the build hierarchy?

$e^{-0.05}$ = 95%, so nineteen of twenty are clean and a single verified clone is usually obtainable from a handful. At 500 kb it falls to 61% and at 4 Mb to 1.8%, so the probability of success collapses with length while the cost of a failed assembly rises. Building in pieces of tens to low hundreds of kilobases and verifying each one means every stage has a high success probability and every error is caught where it is cheap to fix — which is the same logic as verifying a Golden Gate transcription unit before assembling it into a pathway.

2. Your group proposes moving to a host with a 15% faster cycle and no existing toolkit. Make the argument for or against with numbers.

Against, unless the project is unusually long. A 15% gain on a five-day cycle saves 0.75 days per cycle; a toolkit of transformation, markers, characterised parts, editing and curing is realistically four to six months, say 150 days of work, so break-even is around 200 cycles — far more than most PhD-scale projects run. The arguments that would flip it: a property other than speed that the new host has and *E. coli* cannot be given, such as native secretion or a substrate range; an existing toolkit from another group, which cuts the investment from months to weeks; or a plan to use the host for many projects, which amortises the toolkit across all of them.

3. Why does transformation efficiency remove a class of experiment rather than merely slow it down?

Because library-based approaches require covering the library, and coverage is transformants divided by library size: at 3× coverage of 10^5 variants you need 3×10^5 transformants, which at 10^4 per microgram means 30 µg of DNA per experiment and at 10^3 means 300 µg — quantities that are not preparable for routine work. Below that threshold, every variant must be built and tested individually, so the whole parallelism argument of part 2 chapter 11 disappears and the project becomes rational design only, which also means it needs a far better prior.

What to open a book for. Fredens et al., Nature (2019), read for the design rules and the edit count rather than the

methods. The *V. natriegens* preprint and the 2024 PNAS Nexus paper on natural plasmid transformation for the fast-host case, and the SEVA broad-host-range vector literature for what a transferable toolkit looks like. Part 4 chapter 52 of this module is the chapter to read beside this one.

Closing notes to part 7

The six weeks in one page

Week 20. Methods differ in what they leave at the junction and whether the construct can be taken apart again. Gibson is scarless and good to four or five fragments; Golden Gate works because a correct join destroys the recognition site; MoClo adds the syntax that makes parts interchangeable and robots possible. Restriction–ligation survives for the single subclone where no junction matters.

Week 21. A 3 kb part has a 77% chance of containing a six-base site and needs domesticating. Of 256 four-base overhangs, 16 are palindromic, 120 are mutually non-complementary, and imposing a two-base minimum distance against every other overhang and its complement leaves **24** — which is why published high-fidelity sets are that size. Eight transcription units at 80% implies 97.2% per junction; the independence model gives the floor, not the answer. Assemble in silico and translate every frame before ordering.

Week 22. Efficiency comes from the standard-curve slope, and assuming it is perfect overstates a threefold ΔC_t change by 17% at E = 1.90 and 50% at E = 1.75. A band is evidence of size and antibody binding, never of activity. A mean fluorescence shift and a bimodal split look identical in a population average and completely different in a sort.

Week 23. Exo, Beta, Gam, and 40–50 bp arms. MAGE accumulates as $[1 - (1 - p)^N]^K$, so four loci need ten cycles at 20% and the oligo design beats the cycle count because p is in the exponent. In bacteria an unrepaired cut is lethal, so Cas9 is a counterselection: it turns a 0.1% edit frequency into a 50 to 91% hit rate among survivors.

Week 24. CRISPRi is the only experiment available for an essential gene and reaches 90–99% rather than zero. Base editing needs no break and no template but cannot insert, delete or avoid bystanders in its window. A dynamic loop is demonstrated by the intermediate's set point and a disturbance test, not by a titer.

Week 25. 18,214 edits, 4.6 per kb, 1.4% of codons, 37 sections. At 10^{-6} per base a 110 kb section is clean 90% of the time and a 4 Mb genome 1.8%, which forces verification at every stage. A 20% faster host with no toolkit takes about 180 cycles to repay, and a transformation efficiency below 10^4 per microgram removes library work altogether.

On the numbers in this part

The site-frequency, overhang-set, junction-success, qPCR-efficiency, MAGE-accumulation, Cas9-enrichment, error-budget and library-coverage figures were all computed for this module from the assumptions stated beside them. Four carry most of the weight and should be checked against your own system: the per-cycle MAGE replacement frequency, which varies several-fold with oligo design and strain; the Cas9 killing efficiency, which varies with guide and promoter; the residual error rate of your synthesis and assembly supplier; and your host's actual transformation efficiency, which is the first thing to measure and the number most often quoted from a catalogue rather than a bench. The overhang-set result of 24 is a greedy construction under a stated distance rule, so a better search may find a slightly larger set; the order of magnitude is the point.

What remains

Written so far: weeks 1 to 25 and 45 to 61 — 42 of 66. What is left is weeks 26 to 44, which is the parts-and-circuits block (26 to 31), the data and foundry block (32 to 34), the modelling-software block (35 to 38), and the design-algorithms, machine-learning and fluxomics block (39 to 44); plus the professional weeks 62 to 66. I will keep going in tracker order and take them in those blocks, which are the natural groupings — parts and circuits next.

Part 8 — Designing the parts: biosensors, regulation, libraries, protein design and natural-product hosts (weeks 26 to 31)

Part 7 was about building a construct. This part is about deciding what to build: which part, with what strength; which control layer; which library, at what size; which enzyme is worth improving and by how much; and what a natural-product project costs before the first clone is made. Every one of these weeks has an arithmetic spine that a protocol leaves out, and as in part 7 the chapters are organised around it: what the method is, what number tells you in advance whether it will work, and which control makes its output mean something.

A note on the problems. The tracker refers to numbered problems for these weeks — I1, O1, L1 to L4. I have not worked from their datasets, so wherever a calculation here would have used one I have used a constructed example with its parameters printed beside it, and said so. The method transfers; the numbers will differ from the tracker's model answers, and where they do, trust the tracker's data and check my method against it.

Two results in this part were not what I expected when I computed them, and both are reported as they came out rather than tidied. In chapter 26, the baseline of a biosensor fit is determined to about 60% if you fit it the default way and to about 6% if you weight it, which is the calibration lesson of part 4 chapter 50 arriving in a new place. In chapter 28, a burden-aware model of a three-enzyme library has "everything high" as its winner at modest burden, and the winner only moves when burden is steeper — so the honest claim is not that maximal expression is wrong, but that whether it is wrong depends on a number you do not know in advance, which is the whole case for screening a library rather than reasoning to an answer.

Ch.	Week	Chapter
26	26	Genetic parts, transcription-factor biosensors and circuits
27	27	Regulatory control: how far a gene can be turned down, and which layer to use
28	28	Libraries, expression tuning and evolution
29	29	Structure prediction and computational enzyme design: literacy
30	30	Enzyme and protein engineering for pathways
31	31	*Streptomyces* and natural-product pathway engineering

Genetic parts, transcription-factor biosensors and circuits

A part's number is only worth something if it would come out the same on someone else's instrument. A sensor's dynamic range is only worth quoting with its uncertainty. A circuit's speed is set by dilution, whatever the diagram implies.

Characterising a part so the number travels

Fluorescence from a plate reader is in arbitrary units: it depends on the gain setting, the excitation and emission filters, the path length of the well, the volume, the medium's autofluorescence and the age of the lamp. Two labs measuring the same strain report numbers that differ by a factor of ten and mean the same thing. The fix, from the in vivo reference standard approach for standard biological parts, is to measure every part against a reference promoter *on the same plate* and report the ratio — relative promoter units. The instrument's gain divides out because it multiplies both numerator and denominator.

A measurement plan for three promoters that would survive a reviewer: the reference construct in at least three wells of every plate, so plate-to-plate and well-position effects show up in its spread; at least three independent colonies per promoter, not three wells of one culture, for the reason in part 2 chapter 17; all strains in the same medium, grown to the same optical density and read in exponential phase, because a promoter's output changes with growth rate and sigma-factor availability (part 1 chapter 8); a blank and an empty-vector control for background; and normalisation to cell density, preferably checked by flow cytometry so that you are reporting fluorescence *per cell* and not per millilitre.

Three ways the same promoter behaves differently somewhere else, and what to hold fixed. **Copy number**: the same promoter on a high-copy and a low-copy plasmid differs by the copy ratio, so characterise on the vector you will use or integrate. **Upstream and downstream sequence**: the sequence before the promoter can act as an enhancer or a competing promoter, and the sequence after it sets the mRNA's 5′ structure, which sets translation initiation — part 1 chapter 8's caveat that the same ribosome binding site performs differently in front of a different gene. **Host and conditions**: a different strain, medium or growth phase changes the sigma-factor pool. The practical rule is to quote strength together with the context it was measured in, and to re-measure when any of the three changes.

Detecting that a part slows growth is the burden question of part 1 chapter 9, and the cheapest test is a growth-rate comparison of the strain with and without the part, in the same plate and medium, with enough replicates to see a few per cent. A subtler and more sensitive version is a competition: mix the two strains and follow their ratio over several generations, since a 2% growth disadvantage that is invisible in a growth curve is a factor of two in the ratio after about thirty-five generations.

The architecture of a transcription-factor biosensor

Four components: a regulator that binds the metabolite, an operator it binds on DNA, a promoter whose activity the binding changes, and a reporter. In a repressor-type sensor the regulator sits on the operator and blocks the promoter until the ligand arrives and releases it; in an activator-type sensor the ligand converts the regulator into a form that recruits polymerase. Either way a metabolite concentration becomes a transcription rate becomes, after the reporter's own maturation and dilution, a fluorescence per cell.

Regulator	Ligand	Native or first built in	Worth knowing
FapR	Malonyl-CoA	*Bacillus subtilis*, then used in *E. coli* and yeast	The sensor behind dynamic control of fatty-acid and polyketide pathways; reports a pool that is tiny and tightly regulated, so it is a sensitive reporter of acetyl-CoA carboxylation
BenM	*cis,cis*-Muconate	*Acinetobacter baylyi*	Also responds to benzoate, a close analogue — an example of why a sensor's specificity table must be measured
LysG	L-Lysine	*C. glutamicum*	Responds to lysine, arginine and histidine; useful for screening lysine producers in the industrial host, with the cross-reaction as the limit
TtgR	Naringenin and other flavonoids	*Pseudomonas putida*	A multidrug-efflux repressor whose promiscuity is the reason it senses several compounds

These are recalled from the biosensor literature rather than recomputed, and the ligand ranges are exactly the kind of cell to verify against the primary paper before building on it.

The failure modes are four, and each has a test. **Leak**: reporter output with no ligand, which sets the floor and makes the dynamic range smaller than it looks — test with the sensor in a strain that cannot make the ligand. **Saturation at low**

concentration: a sensor whose half-maximal concentration is far below your product's range gives full output for every producer and sorts nothing — test across a titration. **Host-specific behaviour**: a regulator evolved in one organism may respond differently, or to different analogues, in another, so characterise in the host you will screen in. **Ligand uptake**: if the compound is not taken up, the sensor sees only what the cell itself makes, and if it is exported, the sensor sees the intracellular pool and not the titer — which is both its value and its limitation.

Fitting a dose–response, and the uncertainty that comes with it

The four-parameter Hill function, $b + (m - b)\cdot L^n/(K^n + L^n)$, has a baseline b, a maximum m, a half-maximal concentration K and a cooperativity n. I generated nineteen points over six decades of ligand with three replicates at zero, from a known function (b = 120, m = 5,200, K = 8 µM, n = 1.6) with 8% proportional noise plus a small additive term — the same noise structure as part 4 chapter 50's calibration — and fitted it two ways.

	Baseline	Maximum	K (µM)	n	Dynamic range, 5–95% interval
True value	120	5,200	8.0	1.60	43
Default `curve_fit`	160 ± 95	4,950 ± 130	7.9 ± 0.7	1.83 ± 0.27	31 (15 to 147)
Weighted by the noise model	116 ± 7.5	4,979 ± 191	7.6 ± 0.65	1.60 ± 0.09	43 (38 to 49)

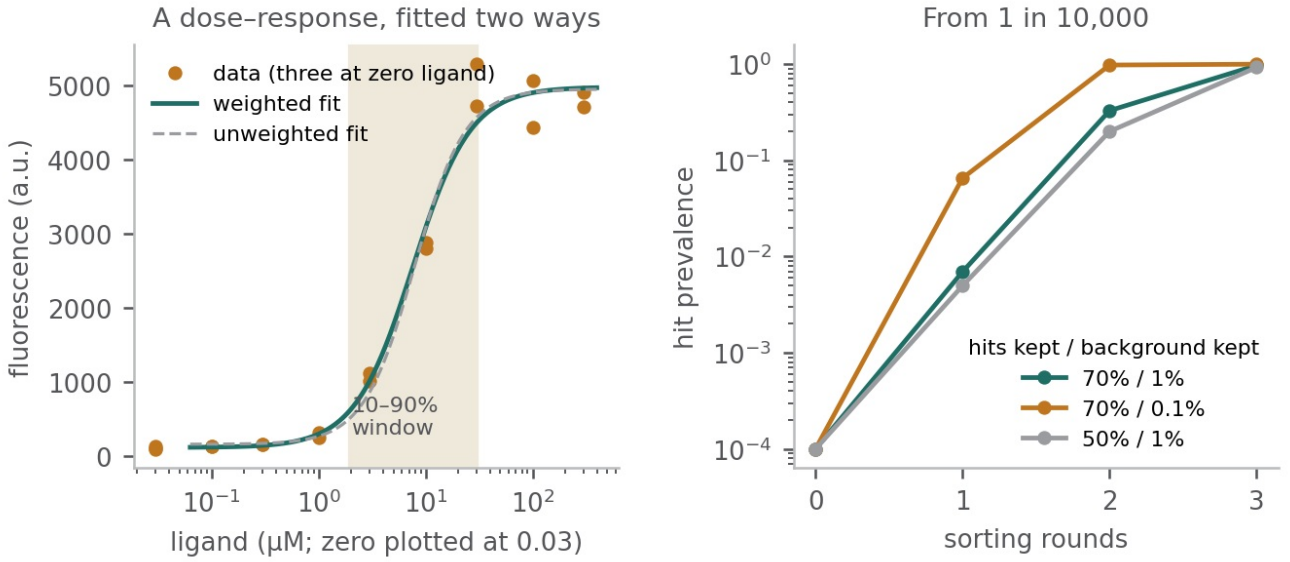


Figure 26.1 Left: the dose–response data with the weighted and unweighted fits and the 10–90% operating window shaded. The two curves are nearly indistinguishable by eye. Right: hit prevalence after one to three rounds of sorting from one in ten thousand, for three combinations of the fraction of hits kept and the fraction of background kept. Synthetic data, computed for this module; the fit parameters in the table are what the fitting routine returned, not the values I started from.

The two fits agree on K and the curve's shape and disagree by a factor of thirteen in how well the baseline is known. The reason is the lesson of chapter 50 in a new place: the default routine assumes constant noise, the high-concentration points have 8% error on a value of 5,000, and those large residuals set the scale of the uncertainty that the baseline inherits. Weighting by the noise model recovers a baseline known to about 6%. And since dynamic range is the ratio of maximum to baseline, its uncertainty is essentially the baseline's — which is why the week's phrase is correct, that dynamic range is the worst-determined of the derived quantities: it is a ratio, its denominator is the parameter pinned by the fewest points, and in the default fit it was uncertain by an order of magnitude.

Dropping the zero-ligand replicates raised the baseline's standard error by about 1.5× in the weighted fit (7.5 to 11) and left K and n essentially unchanged. The single extra measurement that would tighten the answer most is therefore **more replicates at zero and at the lowest concentrations**, since the baseline dominates the dynamic range's variance by a factor of about three over the maximum.

The operating window has a clean formula worth memorising. The ligand concentrations giving 10% and 90% of full response are $K\cdot(1/9)^{1/n}$ and $K\cdot 9^{1/n}$, so the window spans $\mathbf{81^{1/n}}$**-fold**: 81× for n = 1, 9× for n = 2, 3× for n = 4. A cooperative sensor is a switch — excellent for a sorting threshold, because a small change in metabolite produces a large change in signal, and poor as a graded readout, because it saturates within a factor of three. Choose by what the sensor is for. With the weighted fit's n = 1.6 and K = 7.6 µM the window spans 15.6-fold, from 1.9 to 30 µM.

From sensor to screen: the arithmetic of sorting

A gate sorts cells by fluorescence. Two numbers define it: the fraction of true hits that pass, r, and the fraction of background that passes, b. The enrichment per round is r/b, and the prevalence p after a round follows from Bayes: $p' = r{\cdot}p \,/\, (r{\cdot}p + b{\cdot}(1 - p))$. Starting from one hit in 10,000 with a gate keeping 70% of hits and 1% of background — 70-fold per round — prevalence is 0.7% after one round, 33% after two and 97% after three. Tighten the gate to 0.1% background and two rounds suffice.

The cost is in the other direction and is usually left out: the fraction of true hits *retained through all the rounds* is r^k, so $0.7^3 = 0.34$ — two thirds of the genuine hits are lost by the third round. The trade between purity and recovery is the same one a chromatographer makes, and the way to manage it is to sort a generous first gate, then tighten, and to keep every round's sorted population frozen so that a hit lost to a tight gate can be recovered from the one before.

Coverage and time: ten-fold coverage of a million-member library is 10^7 events, which at an assumed 20,000 events per second is 500 seconds — eight minutes. Sorting is so fast that the bottleneck is never the instrument; it is the library, the gate and the sensor, and — as part 7 chapter 24 insisted — the fact that a sorted hit is a hypothesis. The sensor reports the intracellular signal, not the titer, so every hit is confirmed by the real assay, and the confirmation rate is reported.

Circuits: a first model, and why speed costs level

The simplest circuit is a Hill-activated gene with production and loss: $dx/dt = a{\cdot}f(L) - \gamma{\cdot}x$, where f(L) is the sensor's dose–response scaled to its maximum, a is the maximal expression rate and γ is the total loss rate. The steady state is $x^* = (a/\gamma){\cdot}f(L)$: the output follows the sensor's input with a gain of a/γ. And γ is dilution plus decay. For a stable protein in a growing culture, $\gamma \approx \mu$, so the approach to steady state has a half-time of $\ln 2/\mu$ — the doubling time. At $\mu = 0.6\ h^{-1}$ that is 69 minutes, at $0.1\ h^{-1}$ it is nearly seven hours.

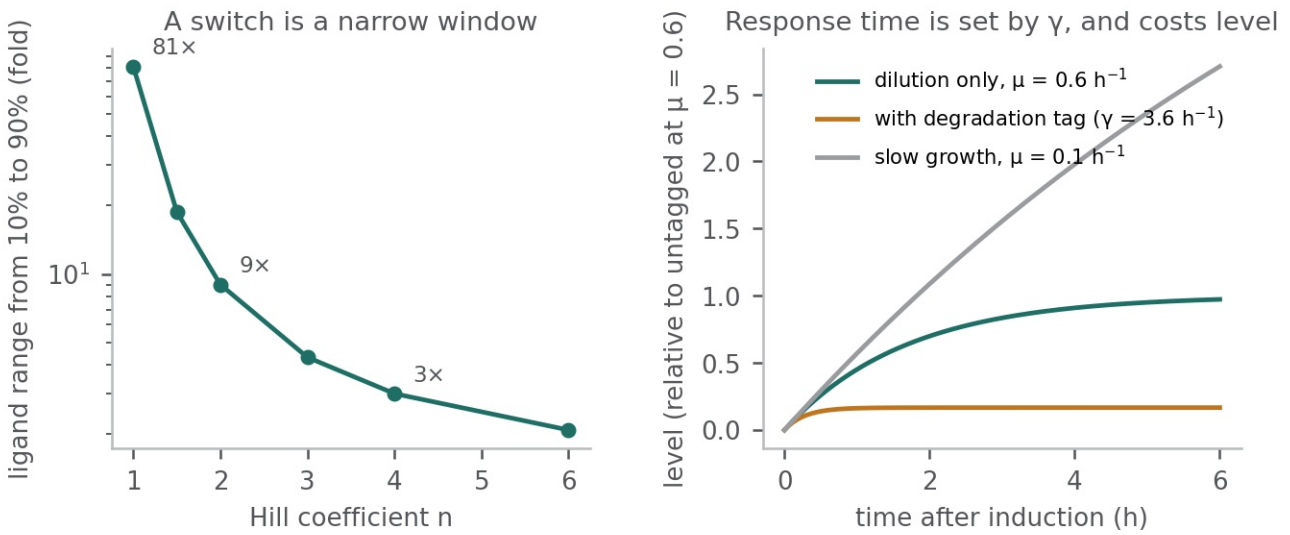


Figure 26.2 Left: the ligand range from 10% to 90% of full response against the Hill coefficient, equal to $81^{1/n}$. Right: response of a constitutively-made protein after induction, for three loss rates. A degradation tag adding $3\ h^{-1}$ cuts the half-time from 69 to 12 minutes and the steady-state level to one sixth. Computed from the first-order model in the text.

That is the engineering content of a feedback circuit and the thing a diagram hides: **a loop is as fast as its actuator can clear the protein it made**. If the control acts by dilution, the loop has a doubling-time response and a controller built on it will oscillate or lag badly. A degradation tag speeds it six-fold and costs a six-fold lower steady-state level at the same production rate — so the faster the loop, the more production capacity you must give the promoter, and the more burden you buy. There is no free speed.

Burden is the other half. A strong constitutive promoter diverts a fraction of the proteome to a protein that does not grow the cell, and part 1 chapter 9's relation says growth falls roughly in proportion, with a slope that depends on the medium. A sensor-controlled promoter wins when the protein is needed only part of the time, or only when the metabolite is present: it pays the burden in proportion to the time and level that the sensor asks for it. It loses when the sensor's own components cost more than they save, or when the loop's delay lets the cell over- and under-shoot.

Relative promoter units	`RPU = (part / reference), same plate, same conditions`
Hill response	`b + (m − b)·Lⁿ/(Kⁿ + Lⁿ); weight by the noise model`
Operating window	`L₉₀/L₁₀ = 81^(1/n); 81× at n = 1, 9× at n = 2, 3× at n = 4`

Sorting	`enrichment r/b per round; p′ = rp/(rp + b(1−p)); hits kept = r^k`
Circuit steady state	`x* = (a/γ)·f(L); half-time ln2/γ; γ = μ + k_deg`

A group reports a sensor with a 40-fold dynamic range and sorts a library with it. After two rounds the top clones show three times the fluorescence of the parent, and HPLC shows no change in titer. Give the explanations and how to separate them.

The most likely is that the sort selected the sensor rather than the product. Mutations in the reporter, in the regulator's affinity, in the plasmid's copy number or in the reporter's degradation raise fluorescence without raising the metabolite. The test is cheap: isolate the sensor plasmid from a sorted clone and put it into a fresh, unsorted parent. If fluorescence stays high, the hit was on the plasmid; if it falls back, the hit was in the chromosome and may be real. Second, a close analogue of the target may have been what the sensor saw, which is the cross-reaction column of the table above: measure the response to the analogues directly. Third, the sensor reads the intracellular pool, so a clone that has lost an exporter shows a higher signal and a lower titer — which HPLC of the broth would show as no gain, and HPLC of the pellet would show as a gain. Fourth, the gate may have enriched noise: if the single-cell distributions of parent and producer overlap heavily, sorting the top few per cent draws mostly from the parent's tail. Report the confirmation rate, because the day it falls is the day the screen stopped measuring your product.

Three things people get wrong

- ‘The sensor has a 40-fold dynamic range.’ Say with what interval: the same data gave 31 with a range of 15 to 147 and 43 with a range of 38 to 49, depending only on whether the fit used the noise model.
- ‘A steeper sensor is a better sensor.’ Steeper means a narrower window — 3× at n = 4 — which is right for a threshold and wrong for a graded readout.
- ‘The feedback loop will respond in minutes.’ Only if something removes the protein in minutes. Dilution alone gives a response time equal to the doubling time.

Self-check — answer on paper first

1. A sensor has K = 20 µM and n = 3. What ligand range does it cover between 10% and 90%, and would you use it for a graded readout?

The window spans $81^{1/3}$ = 4.3-fold, from 20 × $(1/9)^{1/3}$ = 9.6 µM to 20 × $9^{1/3}$ = 41.6 µM. That is a narrow range, so it would be a poor graded readout — a product varying over one order of magnitude would be saturated at one end and unresolved at the other — and a good threshold sensor, for example to trigger a switch or to set a sorting gate at a defined level.

2. You sort a library with a gate that keeps 60% of hits and 0.5% of background, from a hit prevalence of 10^{-5}. How many rounds, and what do you lose?

Enrichment is 60/0.5 = 120-fold per round. Prevalence goes 10^{-5} → 0.12% → 12.6% → 94%, so three rounds. Hits retained are 0.6^3 = 22%, so more than three quarters of the true hits are discarded along the way. To limit the loss, use a looser first gate and tighten in later rounds, and freeze the sorted pool after every round.

3. Why does adding a degradation tag to a feedback circuit's actuator make it faster and also weaker, and what do you do about the second?

Because the protein's loss rate γ sets both the response time, ln 2/γ, and the steady-state level, a/γ. Raising γ by tagging shortens the time and lowers the level by the same factor. To keep the same level you must raise the production rate a — a stronger promoter or ribosome binding site — which raises the burden and the leak, so the loop's speed is bought with expression capacity. The alternative is to accept a slower loop where the disturbance it must reject is itself slow.

What to open a book for. A review of transcription-factor-based biosensors for the architecture and the sensor catalogue, and Kelly et al., J. Biol. Eng. (2009) for the reference-standard method. SciPy's `curve_fit` documentation for `sigma` and `absolute_sigma` — the two arguments that turned a 95-unit baseline uncertainty into 7.5. The iGEM Registry for parts, read with the caveat that registry entries are characterised unevenly.

Regulatory control: how far a gene can be turned down, and which layer to use

Every regulator has a ceiling, and the ceiling is nearly always lower than the diagram suggests. Computing it before you build tells you whether one layer is enough or whether you must stack two.

Choosing a layer

Gene expression has levels, and part 1 chapter 8's expression equation — protein ≈ (transcription rate × translation initiation rate) / (degradation + dilution) — names every handle. Each is a control layer with its own speed, cost and ceiling.

Layer	Acts on	Speed	Typical limit	Cost and portability
Promoter, repressor	Transcription	Minutes (mRNA turns over)	Leak sets a ceiling on repression	Genome edit or plasmid; the part must work in the host
Riboswitch, sRNA	mRNA stability and translation	Minutes	Needs a degradation rate far above the native one for large knockdown	No genome edit for sRNA; works in many hosts; requires the RNA chaperone Hfq in *E. coli*
Ribosome binding site	Translation initiation	Set at construction	Context-dependent; a library is needed	A sequence change; very portable within a host
Degradation tag	Protein half-life	Sets response time	Lowers level in proportion to the added loss	Needs the host's protease; the tag must be tuned
Quorum sensing	Population density	Hours — it follows biomass	Signal is diluted by feeding; delay	Two genes plus a promoter; portable between related hosts
CRISPRi	Transcription, by blocking	Minutes to hours; protein clears by dilution	90–99% repression, not complete	One guide per target; works where editing is poor

The week's four goals, each answered with the layer and the reason. **Turn a competing pathway down, not off**: an sRNA or CRISPRi, because both give a graded reduction and neither needs a genome edit; a deletion would remove the flux entirely, which for many competing branches kills growth. **Switch production on at high density**: quorum sensing or a stationary-phase promoter, because the trigger you want is the population state. **Tune an enzyme across a ten-fold range**: a promoter or ribosome binding site library, because the target is a quantity you set once and screen, not a dynamic response. **Knock down an essential gene**: CRISPRi or an sRNA, because a knockout is not recoverable and a knockdown is.

How far can a regulator turn a gene down?

An sRNA acts by adding decay, and a knockdown of F-fold in steady-state mRNA needs the total loss rate to rise by that factor. With the native mRNA decay constant k and growth dilution μ, the extra decay needed is $(F - 1)\cdot(k + \mu)$. For a native half-life of three minutes — $k = 13.9\ h^{-1}$ — and $\mu = 0.6\ h^{-1}$:

Knockdown	Extra decay rate	Multiple of the native rate	mRNA half-life becomes
2-fold	14 h^{-1}	1×	86 s
10-fold	130 h^{-1}	9×	17 s
100-fold	1,430 h^{-1}	103×	1.7 s

A two-fold knockdown needs the sRNA to double the mRNA's decay; ten-fold needs an increase of nearly an order of magnitude and is achievable; a hundred-fold needs an mRNA half-life of under two seconds, which no degradation mechanism delivers on its own. The implication for design is the one the week asks: **do not ask a single layer for a hundred-fold knockdown**. Layers multiply, because protein level is a product of independent factors. An sRNA that cuts mRNA ten-fold and blocks translation initiation ten-fold gives a hundred-fold at the protein, from two modest mechanisms — which is in fact how most sRNAs work, since they both recruit degradation and occlude the ribosome binding site.

A repressor is limited by its leak. Even with the operator saturated, a small fraction l of promoter activity gets through, so expression relative to the unrepressed state is $l + (1 - l)/(1 + (R/K)^n)$ and the best possible repression is 1/l. A 5% leak caps repression at 20-fold, 1% at 100-fold, 0.1% at 1,000-fold. And the approach to the ceiling is slow,

because the repression curve is a Hill function. Take l = 2%, K = 10 nM and n = 2, so the ceiling is 50-fold:

Target repression	Repressor needed	As a multiple of K
2-fold	10 nM	1.0×
10-fold	34 nM	3.4×
30-fold	85 nM	8.5×
50-fold	unreachable — it is the ceiling	

Going from ten- to thirty-fold costs two and a half times the repressor, and thirty- to fifty-fold costs infinitely more. That is diminishing returns written as arithmetic, and the engineering consequence is to stack: a leaky promoter repressed to its ceiling and then weakened with a weak ribosome binding site gets a product of ceilings that no single part reaches.

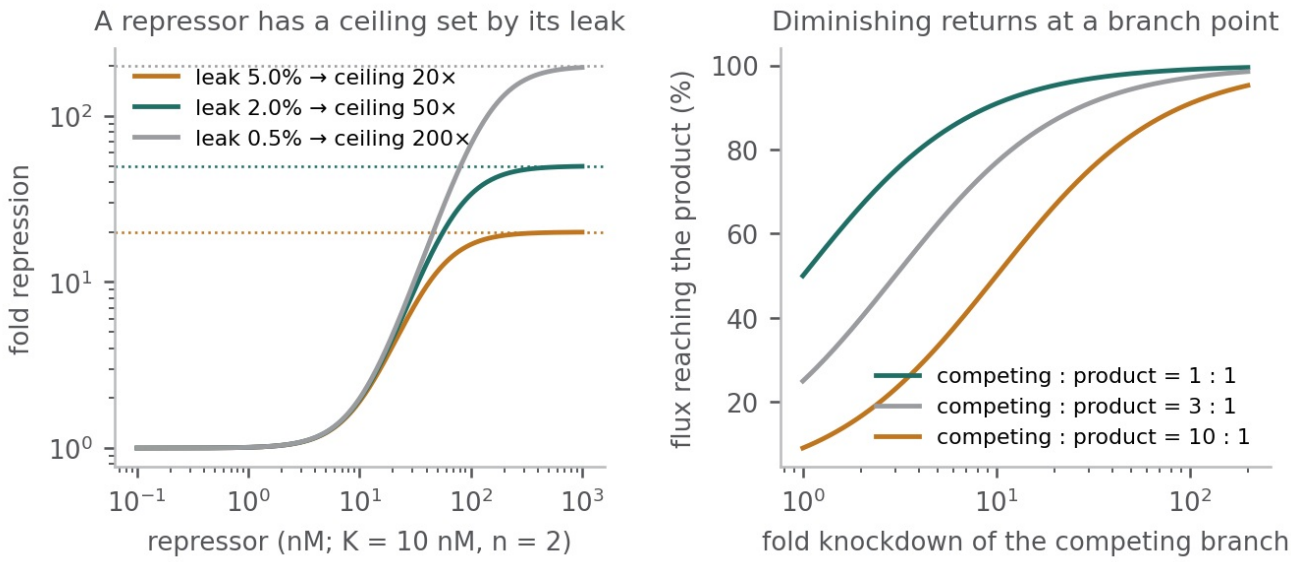


Figure 27.1 Left: fold repression against repressor amount for three leaks, with each ceiling dotted; K = 10 nM and n = 2 are assumed. Right: the fraction of flux reaching the product as a competing branch is knocked down, for three initial ratios of competing to product flux, using the first-order partition $v_p/(v_p + v_c/F)$. Computed for this module.

How much knockdown does a branch actually need?

Whether ten-fold is enough depends on how much flux the competitor takes. At a node where a metabolite is split between a product branch and a competing branch, and both enzymes are well below saturation so flux divides in proportion to activity, the fraction reaching the product after knocking the competitor down F-fold is 1/(1 + r/F), with r the initial ratio of competing to product flux.

Competing : product flux	F = 1	F = 2	F = 10	F = 100
1 : 1	50%	67%	91%	99%
3 : 1	25%	40%	77%	97%
10 : 1	9%	17%	50%	91%

Read it as a design table. When the competitor takes half, a ten-fold knockdown captures 82% of the available gain and a hundred-fold captures 98%, so the extra order of magnitude is almost worthless — ten-fold is enough, and an sRNA or a CRISPRi guide delivers it. When the competitor takes ten times the product's flux, a ten-fold knockdown only reaches half the flux and the answer is to stack layers or, if the branch is not essential, delete it. The assumptions are the ones to state alongside it: both enzymes unsaturated, no feedback, and no change in the shared pool. Part 2 chapter 15 is the reminder that the pool will move when you perturb it, and this table is an estimate to be checked against flux, not a prediction to be believed.

CRISPRi, multiplexed, and the controls it needs

Chapter 24 gave the mechanism. For several competing genes at once, a guide array expresses one guide per target and one dCas9 serves all of them, which is the multiplex repression the tracker names. The controls an experiment needs, because every one of them has caught a published result: a **non-targeting guide** in the same vector, to show the phenotype is the targeting and not the presence of dCas9 and a guide RNA; a **dCas9-only** strain, since dCas9

expression is itself a burden and some guide sequences are toxic even without a perfect genomic match; **two or more guides per target**, to exclude an off-target effect of one; an **RT-qPCR on the neighbouring genes** in the operon, to exclude polar effects; and **growth rate** of every strain, because a flux change without a growth-rate measurement may be a growth-rate change.

Dynamic control and quorum sensing: a sensor or a clock

The growth–production trade-off of part 1 chapter 9 says that expressing the pathway from the start costs growth, and a two-stage process keeps both. Quorum sensing is the autonomous version: cells secrete a small signal, it accumulates with density, and at a threshold it activates the regulator that turns the pathway on. The tracker's example controls a natural product in *S. coelicolor* this way.

The question the week asks is when to use a sensor and when a clock. **Use a sensor when the thing that should trigger the response is a metabolite** — a toxic intermediate that should throttle its own production, or a product whose accumulation should switch a pathway off. Chapter 24's loop is that. **Use a clock — a density or time trigger — when what matters is the growth phase** and the metabolic state is predictable enough that the trigger does not need to read it. The clock is simpler, cheaper and has no ligand to measure, and its weaknesses are the ones to name: the signal is diluted by a fed-batch, so the threshold density depends on the feed; at scale, mixing and gradients change how the signal accumulates; and a population under selection to grow faster will evolve strains that ignore the signal, which is the instability of part 1 chapter 10 reaching a control circuit.

A knockdown plan, with the arithmetic and the decision tree

Take a competing gene in a capstone pathway. State the layer, the fold, and why. For a non-essential competing branch that takes about as much flux as the product, ten-fold from an sRNA or a CRISPRi guide is the target, because the table above says ten-fold captures 82% of the available gain, and ten-fold needs about a nine-fold increase in mRNA decay, which is within what synthetic sRNAs are built to deliver — measure it rather than assume it.

Measure at three levels: **mRNA** by RT-qPCR, with the validated reference gene of week 22, to show the knockdown happened; **protein** by Western blot or mass spectrometry, to show it propagated, since the existing protein is cleared only by dilution; and **flux** by a carbon balance on the by-product and the product, to show it mattered. Add a pass mark in advance: for instance, at least five-fold at the mRNA, a measurable fall in the competing by-product, and a rise in product yield of at least the amount the partition table predicts, less a stated tolerance.

If flux did not move, the possibilities in order of likelihood. The enzyme you knocked down carried little control, which part 2 chapter 15 makes the commonest outcome: check its control coefficient or simply the by-product titer. The protein was still there because dilution had not cleared it: sample at a later time or after more generations. The flux re-routed through another competing branch, which the carbon balance shows as a by-product that rose while the targeted one fell. Or the knockdown reduced the enzyme but the flux was set by supply, not by the enzyme's amount, which is the saturated-enzyme case. Each has a distinct signature in the three measurements, which is the reason for taking all three.

Expression	`protein ≈ (transcription × initiation)/(degradation + dilution)`
sRNA knockdown	`extra decay = (F − 1)(k + μ); 100-fold needs ~103× the native rate`
Layers multiply	`10× at the mRNA × 10× at translation = 100× at the protein`
Repressor ceiling	`best fold = 1/leak; 5% → 20×, 1% → 100×`
Branch partition	`fraction to product = 1/(1 + r/F), r = competing : product flux`
Sensor or clock	`metabolite state → sensor; growth phase → density or time`

A colleague wants a hundred-fold knockdown of a competing gene using a single sRNA. What do you tell them, and what would you build?

That the arithmetic does not support it. A hundred-fold at steady state needs the total mRNA loss rate to rise a hundredfold, which with a native three-minute half-life means an mRNA half-life of under two seconds — beyond what the sRNA's recruitment of the degradation machinery can deliver on its own. The sRNA's real mechanism is two things at once, though: it recruits degradation and it blocks the ribosome binding site, and those multiply at the protein. So a hundred-fold at the protein is plausible from a ten-fold mRNA effect times a ten-fold translation block — which means the design question is whether this sRNA binds the ribosome binding site region, and the experiment is to measure protein and mRNA separately, since protein knockdown much larger than mRNA knockdown is the signature of the translational component.

If a hundred-fold is really needed, stack a second layer. A weaker ribosome binding site, a degradation tag on the enzyme, or a CRISPRi guide at the promoter each contribute an independent factor, and two ten-fold layers do the job with neither near its ceiling. And first check whether a hundred-fold is needed at all: if the competing branch takes half the flux, the partition table says ten-fold already captures 82% of the available gain.

Three things people get wrong

- ‘A hundred-fold knockdown needs a hundred-fold effect.’ From one mechanism, yes; layers multiply, so two ten-fold effects do it.
- ‘A stronger repressor gives more repression.’ Up to 1/leak. Past about three times K there is little left to gain, and a leaky promoter has a hard ceiling.
- ‘The knockdown worked because mRNA fell.’ The protein falls only as fast as dilution clears it, and the flux moves only if that enzyme held control.

Self-check — answer on paper first

1. A promoter has a 3% leak. What is the best repression a repressor can give, and what combination would reach 200-fold?

1/0.03 = 33-fold, whatever the repressor level. To reach 200-fold, stack a second independent layer worth at least 6-fold — a weak ribosome binding site that lowers initiation six-fold, or a degradation tag — since 33 × 6 = 200. The two ceilings multiply because transcription and translation are separate factors in the expression equation.

2. A competing branch carries four times the product's flux. You can deliver a ten-fold knockdown. What fraction of flux reaches the product, and is it enough?

1/(1 + 4/10) = 71%, up from 20% before. Whether that is enough depends on the economics, but by the diminishing-returns shape, going to a hundred-fold would add only about 25 further points — 96% — at the cost of a second layer. A sensible plan is the ten-fold knockdown, measured, with a second layer held in reserve if the measured yield falls short of the table.

3. When would you choose quorum sensing over a biosensor, and what is the first thing that can go wrong at scale?

Quorum sensing when the right moment to switch is a growth state — enough biomass has accumulated — and the metabolic state is predictable, so no metabolite needs to be read. A biosensor when the trigger is the metabolite itself. At scale the first thing to go wrong is the threshold: the signal is produced per cell and diluted by the feed, so the density at which a fed-batch crosses it differs from the one measured in a batch flask, and the switching time moves with it. Verify it by measuring signal concentration, not only reporter output, in the production vessel.

What to open a book for. Yoo et al., Nat. Biotechnol. (2013) for synthetic sRNAs in metabolic engineering, and the 2017 Microb. Cell Fact. paper on CRISPRi-guided multiplex repression, both named by the tracker — read the abstracts for which goal each was built for, then come back to the arithmetic here. The tracker's literature searches cover quorum sensing in *S. coelicolor* and CRISPRi in *V. natriegens*.

Libraries, expression tuning and evolution

When you cannot predict the optimum you search for it, and the quality of a search is decided before the first plate: by how the library is built, what the screen measures, and how many clones the arithmetic says you need.

Why a library, rather than a calculation

Part 2 chapter 15 established that control is distributed, so the best pathway is a combination of changes and not a single gene's overexpression; part 1 chapter 8 established that the same ribosome binding site performs differently in front of a different gene, so the effect of a part cannot be looked up. Together these mean the optimal expression level of each enzyme is unknown in advance and depends on the others — which is the definition of a problem that is solved by building many variants and measuring them.

The two handles are the promoter and the ribosome binding site, and they are independent, so a library can vary either or both. Promoter libraries are built by randomising bases around the −35 and −10 elements of a constitutive promoter, giving a graded range of strengths from one construct. Ribosome binding site libraries are built by randomising the Shine–Dalgarno region and its spacing to the start codon. Their advantage over a promoter library is that they change the translation initiation rate without changing the mRNA level, so within an operon each gene can be set independently — the point of the worked example in part 1 chapter 8.

The RBS Calculator, and what to expect of it

The Salis Lab calculator models translation initiation as a free-energy problem: it sums the hybridisation of the mRNA to the 16S rRNA, the cost of the spacing between them, the energy needed to unfold the local mRNA structure around the start codon, and a standby-site contribution, and converts the total into a predicted relative initiation rate. It works in two directions. Given a sequence it *predicts* a rate; given a target rate and a coding sequence it *designs* a ribosome binding site, which is the "design three spanning a hundred-fold range" exercise of the week.

What to expect, honestly: it is useful as a ranking and as a way to generate a spread, and it is less reliable as an absolute predictor, for the reason part 1 chapter 8 gave — the downstream coding sequence changes the local folding, and the model is an approximation of that folding. The reported accuracy is on a logarithmic scale with scatter of a few-fold, and I am not going to quote a figure from memory; read the validation in the primary paper and note which benchmark it was measured on and whether your host and gene resemble it. The correct use is therefore to *design a library with predicted rates spanning the range you want, then measure*, rather than to trust a prediction.

Three sites spanning a predicted hundred-fold range for one gene: aim for predicted relative rates of roughly 1, 10 and 100, evenly spaced on a log scale, rather than 1, 50 and 100, because expression effects are multiplicative and a log-spaced set samples each order of magnitude once. Record the predicted rates and the sequences, then include a reference construct with a known rate on the same plate so the measurement can be normalised as in chapter 26.

Combining them, and what the model says about it

Three enzymes at three expression levels each is $3^3 = 27$ combinations, and 95% coverage of 27 variants needs about $3 \times 27 = 81$ clones. That is a few plates, which is why a three-enzyme pathway is an ideal first combinatorial library and a ten-enzyme pathway at the same three levels (59,049) is not. To see what such a library finds, I used the three-enzyme model of part 2 chapter 16 — with the export step held fixed — set each enzyme to 0.3×, 1× or 3× its base activity, and added a burden term: a pathway proteome of 4% at baseline, shared equally, and growth falling in proportion to the proteome fraction diverted, with a slope s. Output is pathway flux multiplied by the growth factor, relative to the baseline strain.

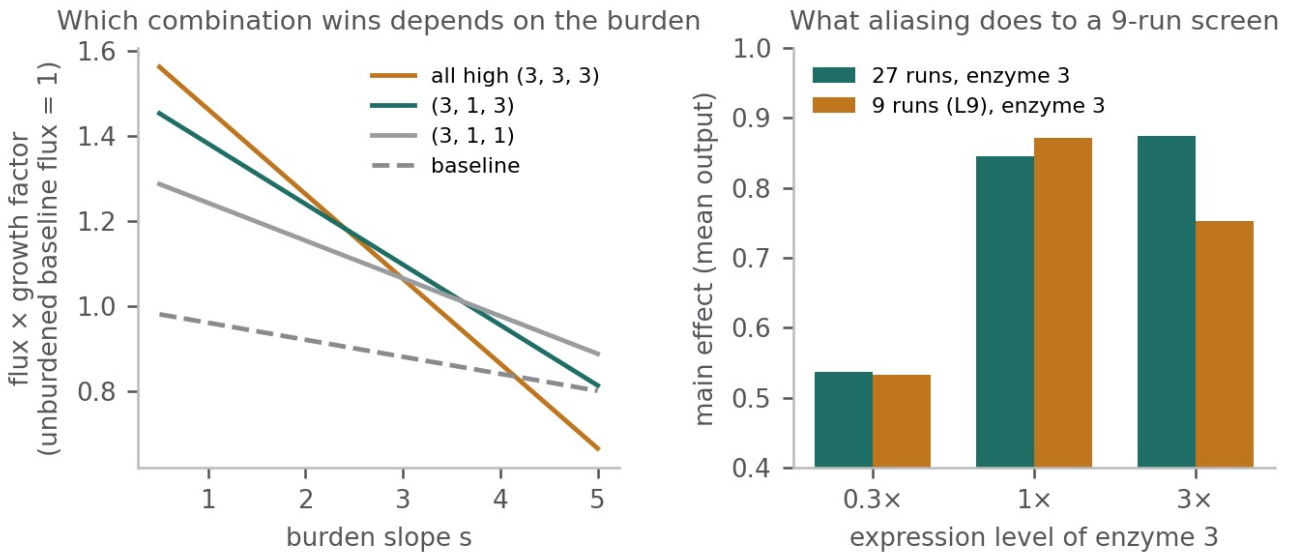


Figure 28.1 Left: output of four of the 27 combinations against the burden slope, the fraction of growth lost per unit of proteome fraction diverted; the best combination changes at about s = 2.4 and again at 3.6. Right: the effect of enzyme 3's expression level estimated from all 27 combinations and from a nine-run orthogonal array. Computed from the model of part 2 chapter 16 with the export step fixed; the enzyme levels, the 4% baseline proteome and the burden slope are assumptions.

The results, as they came out. At a burden slope of 1 — growth lost in proportion to the proteome diverted — **the best of the 27 combinations is all three enzymes at 3×**, with a net output of 1.46 against 0.96 for the baseline (which pays its burden and gains nothing). The next three are all combinations with enzyme 1 at 3×. So at modest burden, "express everything high" is correct, and my earlier warning that maximal expression is the wrong default is not supported by this model at that slope.

It becomes true as the burden steepens. Above a slope of about 2.4 the winner is (3, 1, 3): enzyme 2 held at baseline. Above about 3.6 it is (3, 1, 1), with only enzyme 1 pushed. The quantity that changes the answer is the burden slope — which part 1 chapter 9 said depends on the medium and is steeper in poor media — and nothing about the biochemistry changes. **Enzyme 1 is at its highest level in every regime**, because it carries the most control and sits before the feedback inhibition; the question is only how much of the rest of the proteome to spend on enzymes 2 and 3, which have low control and cost the same per gram.

That is the argument for a library, stated honestly: not that the optimum is surprising, but that **you cannot know the burden slope in advance**, so you cannot know which regime you are in, and building 27 variants and measuring them finds the answer without needing to. A reasoning-to-the-answer approach would have required a number that only the experiment supplies.

The second lesson is about how many runs you need. A nine-run orthogonal array — the standard L9 for three factors at three levels, a third of the full factorial — estimated enzyme 3's effect as 0.53, 0.87 and 0.75 at its three levels, a maximum at the middle level; the full 27-run design gives 0.54, 0.85 and 0.88, a maximum at the top. The array recommended (3, 3, 1) when the truth was (3, 3, 3). This is the aliasing of part 2 chapter 17 made visible: the nine-run design estimates each main effect confounded with interactions between the other factors, and in a pathway the interactions are not small because the enzymes share pools. Fractional designs are the right tool for screening many factors cheaply; for three enzymes at three levels, the full factorial is only 81 clones and there is no reason to accept the confounding.

Directed evolution

The loop is mutate, express, screen, select the best, repeat. The mutagenesis can be random — error-prone PCR, tuned to a mean number of amino-acid substitutions per gene — or focused, at positions chosen from structure or sequence conservation (chapter 30). The mutations per clone follow a Poisson distribution, which is worth tabulating because it sets the quality of a random library:

Mean mutations per gene	No change	Exactly 1	Exactly 2	3 or more
1	37%	37%	18%	8%
2	14%	27%	27%	32%
4	2%	7%	15%	76%

At two mutations per gene, 14% of the library is wild type and a third carries three or more changes, most of which will

be deleterious or neutral hitchhikers. The usual advice — aim for one to three amino-acid changes per gene — follows from this: more mutations per clone raises the rate of a beneficial mutation arriving and raises the load of harmful ones faster. And because random mutations are mostly neutral or harmful, the proportion of improved clones is small, which is why the screen must be able to find a rare positive among a large majority of negatives.

You get what you screen for. The central rule of the method is that selection acts on the measured property, exactly and only. An enzyme evolved for activity on a surrogate chromogenic substrate improves on the surrogate, and on the real substrate by whatever fraction of the improvement transfers, which may be none. An enzyme screened as a purified protein can improve in vitro and fail in the cell because its expression or solubility fell. A growth-coupled selection can be won by a strain that raised the uptake of a nutrient rather than the pathway's activity. A biosensor-based sort can be won by mutating the sensor — chapter 26's box. The defence is the same in every case: screen under conditions as close as possible to the use, on the real substrate, in the real host, and confirm every hit on the property you actually care about.

Adaptive laboratory evolution

ALE applies a selective pressure to a growing population over many generations and sequences the survivors. A daily 1:100 dilution gives $\log_2(100) = 6.6$ generations a day, so a hundred days is about 660 generations; a 1:1,000 dilution gives 10 a day and a thousand generations. It is the method of choice for phenotypes that are polygenic and hard to design — product tolerance above all, as part 2 chapter 19 argued — because evolution searches the whole genome without needing a model.

The supply arithmetic explains both its power and its limit. With a mutation rate of about 10^{-10} per base per generation — the order of magnitude for wild-type *E. coli*, to be checked against the literature for your strain — a culture of 10^{10} cells produces each of the possible single-base changes about once per generation: $N \cdot \mu = 1$ per site. So **any single mutation that helps is present almost immediately and its spread is limited by selection, not by supply**. A specific *pair* of mutations, though, arises at $N \cdot \mu^2 = 10^{-10}$ per generation, which over a thousand generations is 10^{-7} — essentially never, unless one of the pair is already established. So adaptation proceeds as a sequence of single steps, each fixed before the next arrives, which is why an evolved strain carries a handful of mutations after hundreds of generations and why a mutator, with a hundred-fold higher rate, speeds the search at the price of more passengers.

Finding the causal mutations is the real work after the evolution. Sequence the evolved population or clones from several independent lineages and look for **convergence**: the same gene or pathway mutated in different lineages is very unlikely to be coincidence. Then reconstruct each candidate in the unevolved parent, individually and in combination, because an evolved clone carries hitchhikers and the only test of causality is to put the mutation into a clean background and measure. Tolerance is "evolved rather than designed" because the causal mutations are often in global regulators and membrane functions whose connection to the phenotype is not obvious in advance — which makes ALE a hypothesis generator for the design stage rather than a replacement for it.

Selection against screening

A screen examines every candidate; a selection lets only the right ones survive. Part 1 chapter 10 made the point and the arithmetic is the whole case: a screen is limited by how many clones you can assay — 10^3 to 10^4 a day on plates, 10^7 an hour by sorting — and a selection is limited by how many cells you can plate, 10^9 or more. Coupling the product to growth is the way to make the cells do the work.

A selection that would enrich a capstone product has the same structure whatever the product: the product, or an intermediate downstream of it, becomes necessary for growth or for survival under a stress. Concretely: a transcription-factor sensor for the product drives an antibiotic resistance gene, so that producers survive a drug; or the pathway is placed so that its by-product complements an auxotrophy; or a product that detoxifies a medium component is the cell's only way to grow in it. The failure modes are the ones above, and they are specific: the cells will mutate the sensor, the resistance cassette, the transporter that admits the drug, or the copy number of the plasmid carrying it — and the defence is a second, independent confirmation and a selection stringency that is raised gradually rather than set at maximum.

Library size	`levels^enzymes; 3³ = 27 variants, ~81 clones for 95% coverage`
Spacing	`equal on a log scale: 1, 10, 100`
Poisson mutations	`P(k) = e^(−μ)μᵏ/k!; μ = 2 gives 14% wild type, 32% with three or more`
ALE generations	`log₂(dilution) per transfer; 1:100 daily = 6.6 per day`
Mutation supply	`N·μ per site; N·μ² for a specific pair; 1 and 10⁻¹⁰ at N = 10¹⁰`
Screen against selection	`10³–10⁷ clones per day against 10⁹ and above per plating`

You have a three-enzyme pathway. A colleague proposes a ten-level RBS library for each gene — 1,000 combinations — screened by HPLC at 80 samples a day. Is that sensible?

Not as stated. Ninety-five per cent coverage of 1,000 combinations is about 3,000 clones, which at 80 a day is 37 days, and that is before replicates. Two cheaper routes exist. First, three or four log-spaced levels per gene give 27 to 64 combinations and 81 to 192 clones — a week or less — and the model above says the response is smooth enough that the best of a coarse grid is near the best of a fine one. Then refine around the winner with a finer library over only the genes that mattered. Second, a proxy screen: a colorimetric or biosensor-based assay on the plate to triage the 3,000 down to the top fifty, and HPLC only on those, with the confirmation rate reported.

The deeper point is that ten levels is false precision. The measurement noise on a titer is a few per cent and the response between adjacent levels at ten levels is smaller than that, so the library is mostly samples that cannot be told apart. The sensible design uses as many levels as the assay can distinguish.

Three things people get wrong

- ‘Maximal expression is wrong.’ Not at modest burden — in the model, everything-high won up to a burden slope near 2.4. It depends on a number you do not know, which is why you screen.
- ‘A nine-run design finds the optimum.’ It found enzyme 3's best level wrongly, because main effects are confounded with interactions that pathways have.
- ‘ALE needs a high mutation rate.’ At $N = 10^{10}$ every single change is present each generation; it is the double mutants that are limiting, which is why adaptation goes step by step.

Self-check — answer on paper first

1. How many clones give 95% coverage of a library with four enzymes at four levels, and how long does it take at 200 clones a day?

$4^4 = 256$ combinations, so roughly $3 \times 256 = 770$ clones — more exactly $\ln(0.05)/\ln(1 - 1/256) \approx 765$ — which at 200 per day is about four days. That is a perfectly practical campaign. Increasing to six levels per enzyme gives 1,296 combinations and about 3,900 clones, twenty days, which is the point where a proxy screen begins to earn its keep.

2. Why can an ALE experiment find a single beneficial mutation quickly but need hundreds of generations to find a beneficial pair?

Because supply scales as $N \cdot \mu$ for a single change and $N \cdot \mu^2$ for a specific pair. At 10^{10} cells and 10^{-10} per base per generation, each single substitution appears about once per generation, but a specific double appears at 10^{-10} per generation, so it is effectively never generated from the wild-type background. A pair is found only when one member arises, spreads under selection until it is common, and then the second arises on that background — a sequence of single steps whose duration is set by how fast selection fixes each one.

3. A tolerance-evolved strain carries nine mutations. How do you decide which matter?

First sequence independent lineages and look for genes or pathways hit in more than one: convergence is strong evidence of causality. Then reconstruct candidates individually in the unevolved parent, measuring tolerance against both parent and evolved strain, and in combinations to test for interactions. A mutation that reproduces most of the phenotype alone is a driver; one that matters only in combination is a partner; one with no effect in a clean background is a passenger. Mutations in a mismatch-repair gene deserve a check of their own — a mutator phenotype explains a long list of passengers.

What to open a book for. The Salis Lab RBS Calculator documentation and its primary paper for the model and its reported accuracy; MIT OCW 7.344 on directed evolution; an ALE review, preferably one on solvent tolerance. Part 4 chapter 17 of this module is the chapter to have open for the coverage and aliasing arithmetic.

Structure prediction and computational enzyme design: literacy

A confident fold is not a confident active site, and a score with no experiment behind it is a hypothesis. This week is about knowing which tool answers which question, and what each is silent about.

What a structure prediction is, and how to read one

AlphaFold2 predicts a protein's three-dimensional structure from its sequence, using evolutionary information from a multiple sequence alignment. You rarely run it yourself: the AlphaFold database holds precomputed models for a very large number of proteins, and ColabFold gives a faster route for a sequence of your own by replacing the slow alignment step. AlphaFold3 extends the approach to complexes including small molecules, ions and nucleic acids; the terms of access for the three differ, so check the current ones before depending on any of them for a project.

Two confidence measures matter, and reading them is the first skill. **pLDDT** is a per-residue score from 0 to 100. Broadly, above 90 is very high confidence, 70 to 90 is confident for the backbone, 50 to 70 is low, and below 50 is very low and usually means the region is disordered or the model is guessing. **PAE**, the predicted aligned error, is a residue-by-residue matrix in ångströms giving the expected error in the relative position of two residues; it is what tells you whether two domains are positioned correctly relative to each other even when each is confidently folded. For a complex, **ipTM** scores the confidence in the interface.

The warning of the week, and the one most worth carrying out of it: **a high pLDDT tells you very little about the geometry that matters for catalysis.** It reports confidence in the local backbone conformation, usually to within an ångström or two. Catalysis depends on the position of side chains in the pocket to a fraction of an ångström, on the presence and placement of a cofactor or metal that the model does not contain, and on whether the enzyme is in its open or closed state. A prediction can have a pLDDT of 95 across the active site and still place a catalytic residue in a conformation the real enzyme never adopts. Treat the model as a good map of the fold and a poor measurement of the pocket.

Which tool answers which question

Question	Tool	Output	What it cannot establish
What fold does this sequence have?	AlphaFold database, ColabFold	A model with pLDDT and PAE	Ligand geometry, cofactor position, oligomeric state, conformational change
Do these two proteins associate?	AlphaFold multimer or AlphaFold3	A complex model with ipTM	Affinity; whether they associate in the cell
Design a sequence for this backbone	ProteinMPNN	Sequences predicted to fold to it	Anything involving a ligand — it sees only backbone atoms
Redesign a pocket around a cofactor or metal	LigandMPNN	Sequences conditioned on ligand atoms	Kinetics, expression, whether the ligand pose was right
Which substitutions are tolerated?	Protein language-model zero-shot scores	A score per variant	A reliable ranking — the correlation with measured activity is partial
How might a ligand sit in the pocket?	Docking	Poses and scores	A measured affinity or the true pose; scoring functions track affinity poorly
Is a structure already known?	The Protein Data Bank	An experimental structure	Look here first — an experimental structure beats any prediction for the same protein

The single distinction most student proposals get wrong. ProteinMPNN is an inverse-folding model: given a backbone, it proposes sequences likely to fold to it, and it works from backbone atoms only. It is ligand-blind. A sequence it designs for a cofactor-binding pocket has been optimised to stabilise the fold, not to hold the cofactor, and may place residues where the cofactor must sit. LigandMPNN models the atoms of cofactors, metals and small molecules explicitly, so its designs are conditioned on the ligand's presence. If your proposal redesigns a pocket and names ProteinMPNN, it has the wrong tool; if it names LigandMPNN, it has the right one, and it also has to say what the ligand coordinates came from.

Why a cofactor swap needs a ligand-aware tool

An NADPH-to-NADH swap, the engineering step part 2 chapter 19 flagged and part 5 chapter 55 applied to ketol-acid

reductoisomerase in isobutanol strains, is a pocket redesign. In Rossmann-fold enzymes the discrimination between the two cofactors is made by a small number of residues around the adenosine ribose: an NAD-specific enzyme typically has an acidic residue at the end of the second β-strand that hydrogen-bonds the 2′- and 3′-hydroxyls of the ribose, and an NADP-specific enzyme lacks it and carries basic residues that bind the 2′-phosphate. Swapping specificity means removing the basic residues, introducing the acidic one, and keeping the rest of the pocket intact — a design problem whose entire content is the cofactor's atoms, which is why a ligand-blind model cannot do it.

The redesign you would request for one step of a pathway has a fixed form. **The step**: the reduction that consumes NADPH and is the first to fail the redox balance. **The residues**: those within about 5 Å of the cofactor's adenosine ribose and 2′-phosphate, from an experimental structure or a model with the cofactor docked, since the apo prediction does not show them. **The cofactor**: modelled explicitly, NADH for the target and ideally NADPH for the counter-design. **What you measure**: the k_{cat}/K_M for each cofactor of each variant, because the aim is a change in the *ratio* without a collapse in turnover, which is the cost part 2 chapter 19 named.

Variant-effect scores: what a partial correlation is worth

Protein language models trained on sequence databases assign each substitution a score — roughly the log-likelihood of the mutant against the wild type — that correlates with measured fitness across deep-mutational-scanning data sets without any training on the enzyme in question. That is the zero-shot property. The correlation is partial: benchmark values are of the order of 0.4 to 0.5 as rank correlations, and I have not checked the current leaderboard, so look up ProteinGym rather than quoting my figure. What a partial correlation is *worth* can be computed.

Model the score and the true activity as correlated normal variables with correlation r, and ask: if you take the top 1% of variants by score, how many are in the true top 1%? Random selection gives 1%. Simulating 400,000 variants:

Correlation r	True top-1% variants in the top 1% by score	Enrichment over random
0.2	3.2%	3×
0.3	5.2%	5×
0.5	12.7%	13×
0.7	27.1%	27×

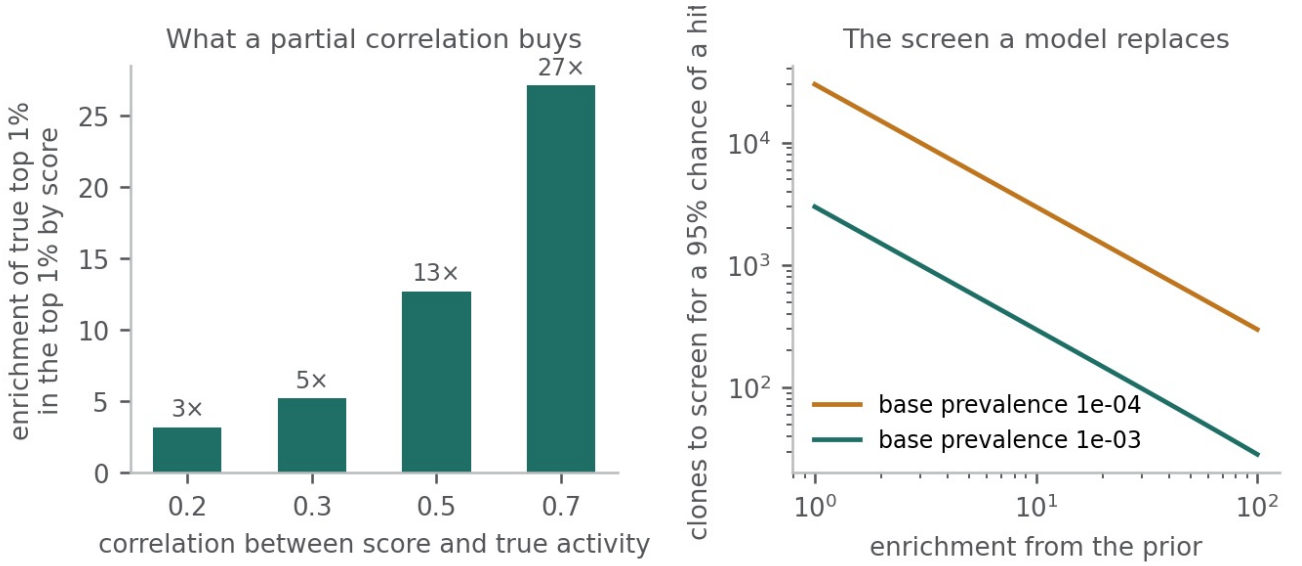


Figure 29.1 Left: enrichment of true top-1% variants in the top 1% ranked by a score with the stated correlation, from a bivariate-normal simulation. Right: the number of clones to screen for a 95% chance of at least one hit, against the enrichment a prior provides, at two base prevalences. Computed for this module; the normal model is an idealisation of real score distributions.

Now the second half. With a hit prevalence of 10^{-4}, a 95% chance of finding at least one hit takes about 30,000 clones. A prior that enriches by 13× — a correlation near 0.5 — brings that to about 2,300, and one that enriches by 100× to 300. So a score with only a partial correlation is worth an order of magnitude in screening effort (I applied an enrichment measured for the top 1% to a rarer hit class; for rarer classes the enrichment from a correlated score is generally larger, so this is on the conservative side, but simulate your own case), which is why the tracker's description of the real workflow is exactly right: **narrow a library, then still screen**. A model is a way to choose which clones to build, not a replacement for measuring them. And a high score on one variant is worth almost nothing on its own, because at r = 0.5 the variant with the top score is still more likely than not to be an unremarkable one.

Docking, and why it generates hypotheses

Docking places a ligand in a binding site and scores the pose. Poses are useful for asking which residues could contact the substrate and therefore which positions to randomise in chapter 30, and scores are useful for excluding clearly impossible placements. They are poor predictors of affinity or of catalytic competence, because scoring functions approximate solvation and entropy crudely and treat the protein as nearly rigid. The right use is to generate a short list of residues for a library, and the wrong use is to rank variants by docking score.

Writing the paragraph

The week ends with a paragraph for a proposal, and the test of it is that it names a step, a problem, a tool and a measurement. A model version: *The ketol-acid reductoisomerase step in the isobutanol pathway consumes NADPH while glycolysis supplies NADH, imposing a cofactor mismatch that costs yield (chapter 19). To shift cofactor preference we will model NADH in the experimentally determined pocket with LigandMPNN, restricting redesign to residues within 5 Å of the adenosine ribose and 2′-phosphate, and generate a library of about 200 sequences from the lowest-scoring designs. Each will be expressed in E. coli and screened in a plate assay for NADH-dependent activity against NADPH-dependent activity, selecting variants with a raised ratio and preserved k_{cat}, and the best three characterised by Michaelis–Menten kinetics with both cofactors before being returned to the production strain.*

Check it against the rule: it has the right tool, because the problem is a pocket and a cofactor; it restricts the design to a defined region, which keeps the library small; and it has an assay and a pass criterion, so the design is a plan and not a hope. A paragraph that stops at "we will use AlphaFold to redesign the enzyme" fails all three.

pLDDT bands	`>90 very high · 70–90 confident backbone · 50–70 low · <50 very low, often disordered`
PAE	`expected error in ångströms between residue pairs; domain placement`
ProteinMPNN / LigandMPNN	`backbone only (ligand-blind) / conditioned on ligand, metal and cofactor atoms`
Score enrichment	`r = 0.3 → 5×, 0.5 → 13×, 0.7 → 27× in the top 1%`
Screen size	`n = ln(0.05)/ln(1 − p·E): 30,000 at p = 10⁻⁴; 2,300 at 13×`
Workflow	`model narrows the library; the assay decides`

A student's proposal says: "We will use ProteinMPNN to redesign the cofactor-binding pocket of the reductase to accept NADH." What is wrong and what do you suggest?

The tool is blind to the thing being redesigned. ProteinMPNN sees backbone atoms only, so it has no representation of the cofactor and will propose pocket sequences optimised for stability of the fold — which can place a bulky residue where the adenosine ribose needs to sit. The right tool is LigandMPNN, or a structure-based design workflow in which the cofactor is explicit. Then three requirements the proposal does not meet: a source for the ligand coordinates, since the predicted apo model does not contain the cofactor, so an experimental structure with a cofactor or a documented docking step is needed; a limited residue set, because redesigning the whole protein is neither plausible nor testable; and an assay with a pass mark — the ratio of NADH to NADPH activity, with a floor on turnover.

I would also ask for the counter-design: a model that merely makes NADH binding better does not guarantee NADPH binding is worse, and a cofactor *swap* needs both. And I would ask what the fallback is if the designed library gives nothing, which is saturation mutagenesis of the three or four residues the literature on Rossmann-fold specificity already names — the oldest method, and the one with a track record.

Three things people get wrong

- ‘pLDDT is 95, so the active site is right.’ pLDDT is confidence in local backbone placement; the pocket geometry that sets catalysis is finer than that and the model has no cofactor.
- ‘ProteinMPNN designs enzyme pockets.’ It is ligand-blind. LigandMPNN is the tool that sees the cofactor.
- ‘The model's top-ranked variant will be the best.’ At a correlation of 0.5 it is more likely unremarkable than outstanding; the value is in enriching a library.

Self-check — answer on paper first

1. A model with a correlation of 0.3 to measured activity is used to pick the top 1% of a 10^5-variant library. Hit prevalence is 10^{-4}. How many clones must you screen with and without it?

Without the prior, $\ln(0.05)/\ln(1 - 10^{-4}) \approx 30{,}000$ clones for a 95% chance of at least one hit. With 5× enrichment the prevalence in the selected subset is 5×10^{-4}, so about 6,000. The 1,000 variants selected from 10^5 can be screened

outright, but the probability of any hit among them is then $1 - (1 - 5 \times 10^{-4})^{1000} \approx 39\%$ — so a correlation of 0.3 narrows the search but does not remove the need to screen deep.

2. Why does an AlphaFold model usually contain no information about whether your enzyme will accept an alternative substrate?

Because substrate specificity is decided by the shape and chemistry of the pocket at sub-ångström resolution and by the dynamics of the open and closed states, and the model predicts the apo backbone with a confidence that does not extend to side-chain positions in a binding site. It also has no ligand in it, so nothing says how a substrate would sit. A prediction can propose which residues line the pocket; whether a different substrate fits is a docking-and-experiment question.

3. Name the three things a computational enzyme-engineering paragraph must contain to count as a plan.

A named step with a stated problem — which reaction, and what limits it. A tool that is appropriate to the problem, correctly named, with its inputs specified: a cofactor-binding pocket needs a ligand-aware method and a source of ligand coordinates. And the experiment that tests the prediction: the assay, the variants measured, and the criterion for success. A design with no assay is a hypothesis, not a plan.

What to open a book for. Dauparas et al., Nat. Methods (2025) for LigandMPNN and the Institute for Protein Design's announcement of it; the ColabFold paper for what structure prediction does and does not give you; and the AlphaFold database entry for one enzyme in your own pathway — fetch it, read its pLDDT and PAE, and write what it cannot tell you. For variant-effect benchmarks, the ProteinGym leaderboard is the place to check the current correlations.

Enzyme and protein engineering for pathways

Before spending a campaign on an enzyme, compute what the improvement is worth. For a chain of first-order steps that calculation is exact, takes one line, and usually says the answer is smaller than the enthusiasm.

Is this enzyme worth engineering?

Part 2 chapter 15 derived the two-step case by hand: with fixed substrate and first-order steps, the flux is $J = k_1k_2S/(k_2 + k_1/K_1)$, which is a flux driven through two *resistances in series*. Write each step's resistance as $R_i \propto 1/(\text{enzyme activity})$. Then $J \propto 1/\Sigma R_i$, the control coefficient of step i is $C_i = R_i/\Sigma R$, and multiplying enzyme i's activity by m shrinks its resistance by m. The new flux relative to the old is

```
J_new / J = 1 / (1 − C_i·(1 − 1/m))        and as m → ∞:  J_max / J = 1 / (1 − C_i)
```

That is exact for a chain of first-order steps and it is the engineer's answer to "what is a faster enzyme worth". A perfect enzyme — infinite activity, zero resistance — can raise flux by at most $1/(1 - C)$, no matter how much protein engineering is spent on it.

Control coefficient	2× activity	10× activity	Ceiling (infinite)
0.05	+2.6%	+4.7%	+5.3%
0.10	+5.3%	+9.9%	+11.1%
0.30	+17.6%	+37%	+43%
0.60	+43%	+117%	+150%

Check it against a model that is not first order. Part 2 chapter 16's three-enzyme pathway has saturable steps, a reversible kinetic form and feedback, and enzyme 2 there carried a control coefficient of 0.102. The formula predicts +5.4% for a doubling, +8.9% for fivefold, +10.7% for twentyfold and a ceiling of +11.4%; the full model gave +4.8%, +7.4%, +8.7% and a plateau near 9%. The formula overstates by one to two percentage points because real control coefficients fall as you raise the enzyme and feedback absorbs part of the gain — so it is an *upper* estimate, which is the right direction to err when deciding whether a campaign is worth starting.

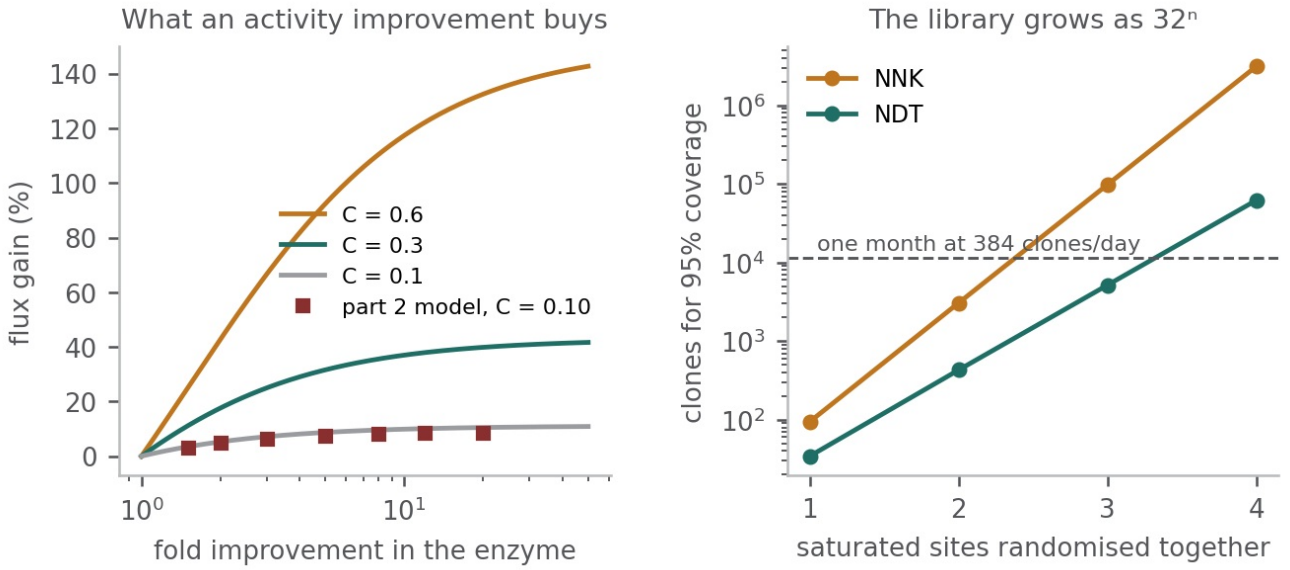


Figure 30.1 Left: flux gain against fold improvement of one enzyme for three control coefficients, from the series-resistance formula, with the full nonlinear model of part 2 chapter 16 overlaid for its C = 0.10 step. Right: clones for 95% coverage of a saturated library against the number of sites randomised together, for NNK and NDT codons, with a month at an assumed 384 clones a day marked. Computed for this module.

The sentence the week asks for, with numbers: *I would engineer enzyme X because it holds 35% of the control, and a ten-fold improvement in its activity would raise flux by 46%, with a ceiling of 54% whatever I do.* If the enzyme held 5% instead, the sentence reads: *a ten-fold improvement buys 4.7% and the ceiling is 5.3%, so I would not* — and the next question is what to check instead, in the order the tracker lists: expression and solubility of the enzyme, cofactor supply, a competing pathway draining its substrate. Those are usually where the real control sits.

An assay you can trust, and a number that says so

Design the assay for one enzyme before anything else, because a library is only as good as its assay: a named substrate, a detection method with a linear range wide enough to cover wild type and the improved variants, initial rates rather than endpoints (part 1 chapter 3, and chapter 22's warning that an endpoint can mislead), a blank, a positive control and a negative control.

The quality of a screening assay has a standard single number, the Z′ factor, from the means and standard deviations of positive and negative controls:

```
Z′ = 1 − 3·(σ_pos + σ_neg) / |μ_pos − μ_neg|
```

Above 0.5 is an excellent assay, 0 to 0.5 is marginal, and below 0 means the two control distributions overlap and the assay cannot separate them. Two constructed examples, since the tracker's data are its own. Assay A: positives 1,000 ± 60, negatives 100 ± 40, so Z′ = 1 − 3(100)/900 = 0.67 — usable for screening. Assay B: positives 1,000 ± 180, negatives 400 ± 150, so Z′ = 1 − 3(330)/600 = −0.65 — the distributions overlap, and no number of replicates fixes an assay whose controls cannot be told apart without improving its precision or its signal. What it implies for a hit, connecting to part 4 chapter 50: you need the calibration, the linear range and the recovery in the real matrix before a number from the assay is a measurement.

Library size, and what a screen can actually find

Saturating a position means randomising its codon. NNK (N = any base, K = G or T) encodes all twenty amino acids with 32 codons and one stop; NDT (D = A, G or T) encodes twelve amino acids with 12 codons and no stop, giving a smaller, chemically balanced alphabet. Clones for 95% coverage of V codon variants at s sites is $\ln(0.05)/\ln(1 - 1/V^s)$, about $3 \cdot V^s$:

Sites	NNK variants	NNK clones	NDT variants	NDT clones
1	32	94	12	34
2	1,024	3,066	144	430
3	32,768	98,163	1,728	5,175

NNK counts are codon-level; because several amino acids have two or three NNK codons, coverage at the amino-acid level needs fewer clones, so these are conservative.

At an assumed 384 clones a day, two NNK sites take eight days, three NNK sites take 256 days, and three NDT sites take 13.5 days. That is the practical reason not to randomise three sites at once with NNK: the library is twenty times the size of the two-site one and well beyond a month's screening. The standard strategy is **iterative saturation mutagenesis**: randomise one or two sites, take the best, make it the template for the next sites, and repeat — which turns a multiplicative search space into an additive one, at the cost of missing combinations whose individual members are not improvements. Choose the sites first from the structure or model of chapter 29: positions within a few ångströms of the substrate, cofactor or the entrance of the channel.

Then the false-hit arithmetic that part 2 chapter 17 introduced, applied to a 10,000-clone screen. If activity is normally distributed among clones that are not truly improved, the one-sided number of false hits above a threshold is $10{,}000 \times (1 - \Phi(z))$: **228 at two standard deviations, 13.5 at three, and 0.3 at four.** A threshold of three standard deviations therefore returns about fourteen impostors, and the first thing to do with any hit list is re-test it in triplicate; the second is to ask how many of the survivors would be expected by chance. That is why a screen's hit rate must be reported with its confirmation rate.

Cytochrome P450 enzymes, and the three limits that are not the enzyme

The tracker's two papers are a good pair. Tyrian purple, a dibrominated indigoid, was produced from tryptophan in *E. coli* with a pathway involving an oxidising enzyme step; the CYP105D18 work engineers a peroxygenase to produce indigo. What they illustrate is what a P450 needs: electrons, delivered either as NADPH through a reductase partner — a two-component system with its own stoichiometry — or, in a peroxygenase mode, directly from hydrogen peroxide, which removes the redox partner and adds a peroxide problem. Both need haem.

Three things that limit the titer, each separable from the enzyme's intrinsic activity. **Electron supply**: the NADPH or the reductase is short, or the reductase-to-P450 ratio is wrong. **Haem supply**: apo-P450 is inactive, and haem is expensive and tightly regulated, so adding the haem precursor δ-aminolevulinic acid to the medium is both a diagnostic and a fix. And **the oxidant** in the peroxygenase case: peroxide inactivates the enzyme and damages the cell, so there is an optimum concentration and a delivery problem — a slow, steady supply rather than a bolus. A fourth, product and intermediate toxicity or insolubility, is the case for the removal strategies of part 3.

What can go wrong between a plant P450 gene and a working enzyme, in the order you would check it:

- **The N-terminal membrane anchor.** Plant P450s are anchored in the endoplasmic reticulum by a hydrophobic N-terminus that misbehaves in bacterial membranes; replacing it with a short, well-behaved anchor, or truncating it, often makes the difference between an aggregated protein and a functional one.
- **The reductase partner**, expressed at a matched ratio and in the right membrane context.
- **Haem supply**, as above.
- **Codon use** — especially of the first codons, where AT-rich, low-structure sequences raise initiation, as part 1 chapter 8 explained.
- **The host.** Yeast has an endoplasmic reticulum and native reductase machinery, which is why it is often the easier host for a plant P450, as part 6 chapter 58 said.

To tell an enzyme limit from a supply limit, **measure active P450 directly**: the reduced carbon-monoxide difference spectrum has a peak at 450 nm for properly folded, haem-bound P450 and at 420 nm for the inactive form, and its amplitude quantifies the functional enzyme per milligram of protein. Then run a lysate assay with saturating NADPH, reductase and substrate. If the lysate activity under those conditions is high and the in vivo flux is low, the limit is supply — electrons, haem or substrate — and engineering the enzyme would be wasted. If the lysate is low even with everything in excess, the enzyme itself, or its folding, is the problem.

Getting a heterologous enzyme to fold

An enzyme that is made and inactive — a strong band, no activity — has a triage, in order of cheapness. Run **soluble against insoluble fractions** to see which; then **lower the temperature and the inducer**, which slows synthesis and gives folding time; then try a **solubility tag** — maltose-binding protein, NusA or SUMO, which are large, soluble fusion partners that raise the yield of soluble protein, with the cost that the tag can interfere with activity or must be cleaved; then **co-express a chaperone**, GroEL–GroES or DnaK–DnaJ–GrpE. The tracker's instruction is to change one thing at a time, and for a triage — where the aim is to find which factor matters — that is right. Once you know which matter, the combination should be optimised by a small factorial, because temperature and inducer interact: a lower temperature tolerates a higher inducer, and testing one at a time hides that. One-at-a-time to diagnose, factorial to optimise.

On codon optimisation: replacing rare codons with the host's most frequent ones raises translation speed, and speed is not always an advantage. Rare codon clusters appear to slow the ribosome at points that help a nascent chain fold co-translationally, so uniformly fast translation can worsen folding. **Codon harmonisation** — matching the pattern of fast and slow codons to the source organism's — is a hypothesis about that, and the way to treat both it and "optimise the codons" is as constructs to be compared: make the native, the optimised and the harmonised versions, express them under identical conditions, and measure soluble active protein, not total protein.

From a hit to a result

A hit from a screen is a hypothesis about one genotype and has five steps to becoming a result. **Purify** the variant and the parent and measure k_{cat} and K_M in vitro — the screen measured something correlated with activity, and this measures activity. **Check expression and solubility**, because an apparent improvement in a lysate screen is often more enzyme, not a better enzyme. **Return it to the host** in the pathway context. **Measure titer against the parent**, with the parent expressed at the same level — otherwise a variant that is merely more highly expressed passes as improved. And include the controls that stop a screen artefact passing as an improvement: an empty-vector strain, the parent assayed alongside the variant on every plate, and a second assay that uses a different detection principle.

Series-resistance gain	`J_new/J = 1/(1 − C(1 − 1/m)); ceiling 1/(1 − C)`
Z′ factor	`1 − 3(σ_p + σ_n)/|μ_p − μ_n|; >0.5 excellent, <0 unusable`
Coverage	`clones = ln(0.05)/ln(1 − 1/V^s) ≈ 3V^s; NNK V = 32, NDT V = 12`
False hits	`N·(1 − Φ(z)): 228, 13.5, 0.3 at z = 2, 3, 4 for N = 10⁴`
Active P450	`CO difference spectrum: 450 nm active, 420 nm inactive`
Folding triage	`fractions → temperature and inducer → tag → chaperone; one at a time, then factorial`

Your pathway's rate-limiting-looking enzyme has a K_M ten-fold above the substrate concentration and a control coefficient you estimate at 0.15. A collaborator proposes a three-site NNK saturation library. What do you say?

The first number settles it. At C = 0.15 even an infinitely improved enzyme raises flux by at most 1/(1 − 0.15) = 18%, and a realistic ten-fold gain gives about 16%. A three-site NNK library needs about 98,000 clones — 256 days at 384 a day — to buy a gain that is smaller than the run-to-run variation of many fermentations. I would not run it.

What I would do instead: with the K_M ten-fold above the substrate concentration the enzyme is operating in its first-order regime, where $v \approx (V_{max}/K_M)\cdot[S]$ and the rate is proportional to both enzyme amount and substrate. So *express more of it*, which raises V_{max} in proportion and costs a promoter change rather than a campaign, and *raise the substrate pool* by improving the upstream step. If the enzyme is then still limiting — the control coefficient will tell — a one-site NNK library at the single residue lining the substrate entrance is a week's work and a reasonable test. The general rule is that a campaign should be proportionate to the ceiling it can reach, and the ceiling is a number you can compute before you start.

Three things people get wrong

- ‘A ten-fold better enzyme gives a ten-fold better pathway.’ It gives 1/(1 − C(1 − 0.1)) — 4.7% at C = 0.05 and a ceiling of 5.3%.
- ‘A band means the enzyme is there, so the problem is activity.’ A P450 band can be apo, misfolded or the 420 nm form; measure the CO spectrum before engineering anything.
- ‘Optimise the codons.’ Fast translation can worsen folding. Compare native, optimised and harmonised versions and measure soluble active protein.

Self-check — answer on paper first

1. An enzyme holds C = 0.4 of the control. What flux gain does a five-fold activity increase give, and what is the ceiling?

1/(1 − 0.4 × (1 − 1/5)) = 1/(1 − 0.32) = 1.47, so +47%. The ceiling is 1/(1 − 0.4) = 1.67, +67%. So five-fold captures 70% of what any improvement could ever deliver, and a further ten-fold beyond that would add little — which is the diminishing returns that part 2 chapter 15 showed from the other side.

2. Your assay has positives at 800 ± 50 and negatives at 300 ± 60. Compute Z′ and say whether you would screen 5,000 clones with it.

Z′ = 1 − 3(50 + 60)/500 = 1 − 0.66 = 0.34, marginal. A hit would have to sit well outside the negative distribution to be called with confidence, so I would expect a hit list rich in false positives and would build in a confirmatory re-screen in triplicate; or improve the assay first — reducing σ by more replicates per well, or widening the separation of the means — because moving from 0.34 to above 0.5 is worth more than screening more clones with a noisy readout.

3. A plant P450 gives a strong band in *E. coli* and a CO difference spectrum with a peak at 420 nm only. What does that tell you and what next?

That the protein is present but not as functional haem-bound P450 — it is the inactive form, or apoprotein with damaged or absent haem. So engineering the enzyme's catalytic residues is premature. Supplement δ-aminolevulinic acid or overexpress the haem pathway, lower the expression temperature and inducer, replace the N-terminal anchor, and re-measure the spectrum: the target is a peak at 450 nm with an amplitude that scales with expression. Only then does an activity assay mean anything.

What to open a book for. Cornish-Bowden's *Fundamentals of Enzyme Kinetics* for what a kinetic constant does and does not tell you; Zhang, Chung & Oldenburg, J. Biomol. Screen. (1999) for the Z′ factor, which is short; the two P450 papers the tracker names for what each enzyme needs; and the literature searches on directed evolution library design, solubility tags and chaperone co-expression. Part 2 chapter 15 is the chapter to read beside the first section.

Streptomyces and natural-product pathway engineering

A natural-product cluster is a pathway you do not have to design and still have to clone, express, feed and sell. The arithmetic of each of those four steps is available before the first experiment, and one of them — cloning a large cluster whole — is worse than intuition suggests.

From cluster to compound, balanced

The tracker starts at the cleanest case: the 6-deoxyerythronolide B synthase, the polyketide synthase that builds the macrolactone core of erythromycin. One propionyl-CoA starter unit and six methylmalonyl-CoA extender units are joined by the assembly line, with reduction steps consuming NADPH. Write it out and check the elements, which is the problem the tracker sets:

```
propionyl-CoA + 6 methylmalonyl-CoA + 6 NADPH + 6 H⁺
       →  6-deoxyerythronolide B (C₂₁H₃₈O₆) + 6 CO₂ + 7 CoA + 6 NADP⁺ + H₂O

carbon   3 + 6×4 = 27   →   21 + 6 = 27
hydrogen 5 + 30 + 12 = 47  →   38 + 7 + 2 = 47        (12 H from six NADPH + six H⁺)
oxygen   1 + 18 = 19   →   6 + 12 + 1 = 19            (acyl parts, CoA-thioesters counted separately)
```

It balances by element count. Three things are worth reading from it. Each methylmalonyl-CoA loses a CO_2 in the condensation, which is why carbon in the product is 3 + 6 × 3 = 21. The single water is released by the one dehydratase in the assembly line. And note the formula: 6-dEB is $C_{21}H_{38}O_6$, molar mass 386.5 g/mol; it is easy to confuse with erythronolide B, which carries one more oxygen and is $C_{21}H_{38}O_7$.

Now the supply bill, which is the reason a precursor or cofactor limit shows up as low titer and not as a failed enzyme. Seven propionyl-derived units go into each molecule — the starter and, via carboxylation, the six extenders — so per gram of 6-dEB:

Titer	Product	Propionyl-CoA units	NADPH	ATP for carboxylation
0.1 g/L	0.26 mmol/L	1.8 mmol/L	1.6 mmol/L	1.6 mmol/L
1 g/L	2.6 mmol/L	18 mmol/L	15.5 mmol/L	15.5 mmol/L
5 g/L	12.9 mmol/L	90 mmol/L	78 mmol/L	78 mmol/L

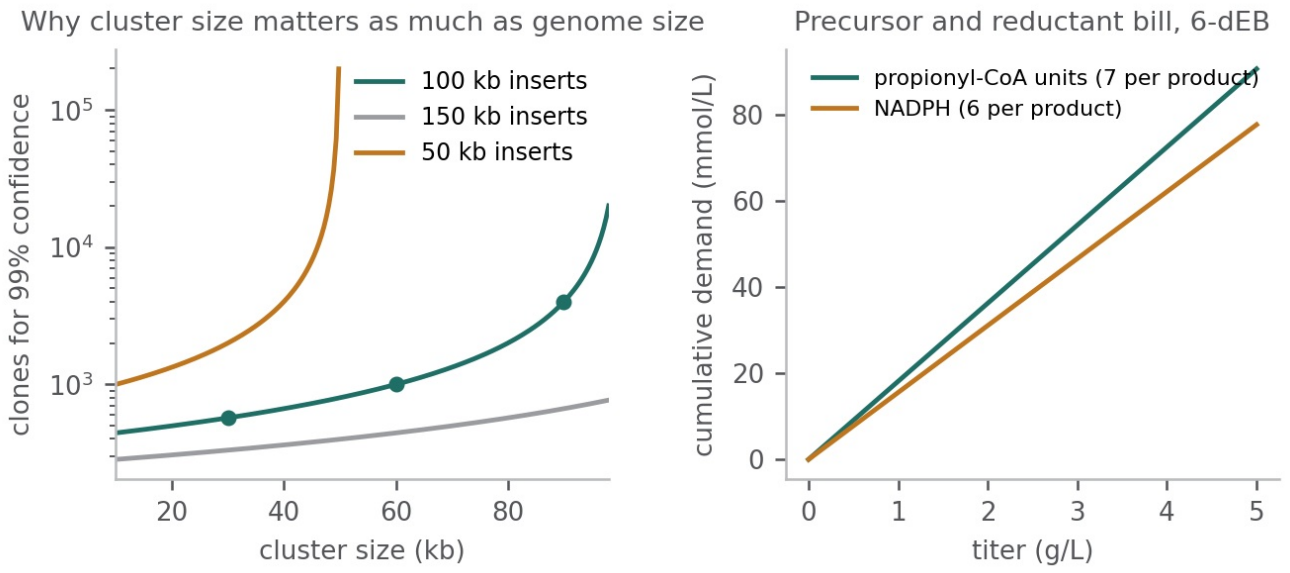


Figure 31.1 Left: clones needed to capture a cluster whole at 99% confidence against cluster size, for three insert sizes, with 30, 60 and 90 kb marked for 100 kb inserts from an 8.7 Mb genome. Right: cumulative propionyl-CoA and NADPH demand against 6-dEB titer. Computed for this module; the cloning model assumes random, unbiased fragment positions.

Converted to mass: 7 × 74 g/mol of propionate per 386.5 g of product, so each gram of 6-dEB consumes 1.34 g of propionate-equivalent, and the ceiling is a mass yield of 0.745 g/g on propionate. At 5 g/L the product alone represents

6.7 g/L of propionate — a weak acid that is toxic to growth at gram-per-litre concentrations at low pH (part 3 chapter 45), so the supply cannot simply be dumped in. It comes from the host's own metabolism — methylmalonyl-CoA from succinyl-CoA by a mutase, propionyl-CoA from propionate or amino-acid catabolism — or is fed slowly. And NADPH at 78 mmol/L, set against part 1 chapter 4's conclusion that NADPH is regenerated at a cost in carbon, explains why polyketide titers are limited by supply long before the synthase is saturated.

Genetics: getting into *Streptomyces*

Streptomyces is a filamentous, spore-forming soil bacterium with a large, linear, high-GC chromosome — about 8.7 Mb and 72% GC for the model *S. coelicolor*. Transformation by electroporation is unreliable, so DNA enters by **conjugation from *E. coli***. The standard donor is a methylation-deficient strain (*dam*⁻ *dcm*⁻ *hsdS*⁻) carrying a non-transmissible helper plasmid that supplies the transfer machinery, because *Streptomyces* carries restriction systems that cleave methylated DNA — the Type IV logic of part 4 chapter 52, and the reason the first thing to try against a methylation-sensitive barrier is a methylation-free donor.

Vectors fall into two kinds. **Integrative vectors** carry a phage attachment site and an integrase — the φC31 system is the standard — so they insert at a defined chromosomal site and are stable without selection; they suit a large cluster. **Replicative vectors** are maintained as plasmids and are copy-number-variable, which is a stability problem for a burdensome cluster. The Streptomyces artificial chromosome (SAC) combines a bacterial-artificial-chromosome replicon, which carries inserts of 100 kb and more stably in *E. coli*, with the integrase and transfer functions for *Streptomyces*, which is why the tracker's pikromycin and neotetrafibricin papers use that kind of system for clusters too large for a conventional plasmid.

Cloning a giant cluster: the arithmetic that surprises

A cluster of size c is captured whole in a clone of insert size i only if the insert's start falls in a window of length (i − c) along the genome of length G. So for random, unbiased fragment positions, the probability that one clone contains the whole cluster is (i − c)/G, and the clones needed for confidence P are ln(1 − P)/ln(1 − (i − c)/G). For 100 kb inserts from an 8.7 Mb genome, at 99% confidence:

Cluster size	P(one clone holds it whole)	Clones for 99%	Genome equivalents
30 kb	0.80%	570	6.6
60 kb	0.46%	999	11.5
90 kb	0.11%	4,004	46

The pattern is the result: **cluster size matters as much as genome size.** Capturing a cluster that almost fills the insert is four times harder than capturing a 60 kb one, because the window of acceptable start positions shrinks to 10 kb. A 90 kb cluster needs 46 genome equivalents of library, not the 3 to 5 that a rule of thumb for random clones suggests, and a cluster larger than the insert cannot be captured whole at all and must be reassembled from overlapping clones.

The model flatters reality in three ways: real libraries have cloning bias, since some regions are poorly cloned or toxic in *E. coli*; a GC-rich genome with uneven restriction-site distribution gives uneven fragment positions; and a clone is only useful if it is verified. Verification is by restriction pattern, junction PCR across both cluster boundaries, and sequencing, and each can miss something — a restriction pattern can match a deleted clone, junction PCR confirms ends and not the interior, and sequencing a 60 kb insert is the only complete check. Compare the alternative from part 7 chapter 21: direct cloning by yeast recombination with overlapping fragments amplified from genomic DNA avoids the library and its bias but needs cluster-specific primers and verification of every junction. Try the library first if a tool exists for your cluster's size and host, and the yeast route when the cluster is near the insert limit.

Silent clusters, and expressing them

Genome mining — antiSMASH and similar tools — finds clusters, and most are silent under laboratory conditions: they are not transcribed, because their pathway-specific regulators are off, or transcribed and not made. Four ways to raise expression, in the order of increasing effort: **replace the promoters** in front of the cluster's operons with constitutive or inducible ones, which refactors the cluster away from its native regulation; **activate a regulator**, often a pathway-specific activator such as a member of the SARP family, by overexpressing it; **move the cluster to a heterologous host** engineered for the job — chassis strains in which the native clusters have been deleted to lower background and free precursors; and **add quorum-sensing control**, as the tracker's *S. coelicolor* example does.

To tell a cluster that is not transcribed from one that is transcribed but not made: RT-qPCR on a biosynthetic gene from each operon, with the validated reference of chapter 22, shows whether transcription happens; LC-MS of culture extracts, with the calibration discipline of part 4 chapter 50, shows whether product exists. Not transcribed points to the

regulators and promoters. Transcribed and not made points to precursor supply, an inactive enzyme, post-translational maturation — a carrier-protein modification, for instance — or product degradation or sequestration; the precursor bill above is the first place to look.

Editing, and why a break is harsh

Cas9 works in *Streptomyces*, with the caveats of part 7 chapter 23 amplified. The nuclease is toxic when expressed strongly, so it must be under tight control and the plasmid must be cured afterwards — typically by a temperature-sensitive replicon. A double-strand break is poorly repaired in the absence of a template, so cutting kills and the repair template carries the edit, with homology arms of one to two kilobases rather than the 40 to 50 base pairs of *E. coli* recombineering. The tracker names a 2025 Cas9 paper and a 2023 review for what had to be engineered to make editing work; I have not summarised their specific modifications, so read the abstracts and log them.

The design the week asks for — a knockout with a homology template, controls and verification: arms of 1 to 2 kb flanking the target on either side, cloned into the editing plasmid with a guide placed within the region to be deleted; **controls** a non-targeting guide, no template, and the empty vector; **verification** by PCR with primers outside the arms, which gives a smaller band for the deletion, then sequencing of the junction, and finally phenotype — loss of the product on LC-MS. Because of the harsh break and the multinucleate mycelium, expect mixed colonies, and plan a round of single-spore purification before any measurement.

Process, and what the product must earn

Filamentous growth changes the bioreactor problem of part 5 chapter 57 in a third way. Mycelium grows as dispersed filaments or as pellets, and both are bad for transport. Dispersed mycelium gives a viscous, shear-thinning broth in which oxygen transfer falls steeply and mixing is poor. Pellets of a millimetre or more have an anaerobic core: with oxygen penetrating a hundred micrometres or so at realistic cell loadings — the dead-core arithmetic of part 3 chapter 48 — a pellet much larger than a few hundred micrometres has idle or dying centres. Morphology control is therefore part of the process, by inoculum density, medium additives and agitation, not an afterthought.

Finally the price. Take part 4 chapter 49's cost model with its parameters unchanged — $3M installed per vessel, $120 per m³ of annual downstream throughput, $3M a year of labour, a 10% discount rate over twenty years — and change only what *Streptomyces* forces: a 100 m³ vessel, a seven-day batch with a day of turnaround, an output of 100 tonnes a year, and propionate at an assumed $1.50 per kg at the 0.745 g/g yield, which is $2 of precursor per kilogram of product:

Titer	Vessels	Capital	Feedstock	Fixed	Capital charge	MPSP
0.5 g/L	48	$168M	$2	$97	$197	**$297/kg**
1 g/L	24	$84M	$2	$64	$99	**$164/kg**
2 g/L	12	$42M	$2	$47	$49	**$98/kg**
5 g/L	5	$17M	$2	$37	$20	**$59/kg**

Utilities add a further $0.15 per kilogram in every row and are not shown.

The lesson is the inverse of chapter 54's. For a commodity, feedstock dominates and yield decides; here feedstock is $2 of a $164 price, and **capital and fixed cost, which scale as 1/titer, are 99% of it.** A 1 g/L process must sell at about $160 per kilogram to break even, which is plausible for an antibiotic intermediate or an agrochemical and impossible for anything priced as a chemical. So the right question about a natural-product project is not "what is the yield" but "what titer do we need for the price this molecule can command" — and the answer is the one that sets the strain-engineering target. These are order-of-magnitude figures from a deliberately simple model, with an assumed $1.50 propionate price and a vessel cost taken from a different product class; the shape of the result, not the third digit, is what transfers.

6-dEB balance	propionyl-CoA + 6 mm-CoA + 6 NADPH + 6 H^+ → $C_{21}H_{38}O_6$ + 6 CO_2 + 7 CoA + 6 $NADP^+$ + H_2O
Mass yield on propionate	386.5/(7 × 74.08) = 0.745 g/g; 1.34 g propionate per g product
Demand per g/L	18 mmol/L propionyl units, 15.5 mmol/L NADPH, 15.5 mmol/L ATP
Cluster capture	clones = ln(1 − P)/ln(1 − (i − c)/G); 570, 999, 4,004 for 30, 60, 90 kb
Pellet limit	oxygen penetrates ~100–200 µm; pellets beyond a few hundred µm have a dead core
Price at low titer	capital + fixed ∝ 1/titer; $164/kg at 1 g/L in this model

You have identified a 75 kb cluster in an *S. albus*-like strain with an 8 Mb genome. A collaborator proposes a cosmid library with 40 kb inserts. What do you say?

That the cluster cannot be captured whole in a clone of 40 kb at all: an insert smaller than the cluster contains at most part of it, so the library would deliver the cluster in pieces that have to be reassembled from overlapping clones, with the junction verification that implies. The arithmetic for a sufficiently large insert is the point: with 100 kb inserts, (100 − 75)/8,000 = 0.31% of clones hold the whole cluster, so about ln(0.01)/ln(1 − 0.0031) ≈ 1,470 clones for 99% confidence — roughly 18 genome equivalents — which is a library of a size that is routine in a BAC system and impractical as cosmids.

So the options are a BAC or SAC-type library with inserts comfortably larger than the cluster — which also gives you the integrase and conjugation functions for the next step — or direct cloning by yeast recombination, which avoids the library entirely, at the price of designing and verifying the overlaps. I would check the cluster's boundaries by genome annotation first, since a 75 kb estimate with uncertain ends changes the insert size you need, and then choose the BAC route if a suitable vector is available for the host, since the host-ready vector saves the transfer step. In either case the verification plan is restriction pattern, junction PCR at both boundaries, and full sequencing of the clone that goes forward.

Three things people get wrong

- ‘A 90 kb cluster in 100 kb inserts is just a bit harder than a 60 kb one.’ It is four times as many clones, 46 genome equivalents against 11.5, because the permissible start window shrinks to 10 kb.
- ‘Low polyketide titer means the synthase is the problem.’ Seven propionyl units and six NADPH per molecule make supply the usual limit, long before the enzymes saturate.
- ‘The product is cheap to make, so the price can be low.’ At 1 g/L the cost is dominated by capital and fixed cost, which scale as 1/titer; it needs a high-value molecule to clear.

Self-check — answer on paper first

1. Balance the production of one molecule of 6-dEB from one propionyl-CoA and six methylmalonyl-CoA, and state the NADPH, CO_2 and water.

propionyl-CoA + 6 methylmalonyl-CoA + 6 NADPH + 6 H^+ → $C_{21}H_{38}O_6$ + 6 CO_2 + 7 CoA + 6 $NADP^+$ + H_2O. Carbon is 27 on each side, hydrogen 47, oxygen 19. Six NADPH, six CO_2 lost at the condensations, and one water from the dehydratase. If your hydrogen count disagrees, check you are counting the reductant's two hydrogens per NADPH and that the product is $C_{21}H_{38}O_6$ and not the more oxidised erythronolide B.

2. How many clones for 99% confidence of capturing a 45 kb cluster in 120 kb inserts from a 10 Mb genome?

P(one clone) = (120 − 45)/10,000 = 0.0075; clones = ln(0.01)/ln(1 − 0.0075) = 4.605/0.007528 ≈ 612, which is about 7.3 genome equivalents. Verification still has to follow: both boundaries by PCR and the interior by restriction and sequencing.

3. A cluster gives no product. RT-qPCR shows the biosynthetic genes are expressed at a good level. What are your hypotheses, in order?

Precursor supply first: seven propionyl-derived units and six NADPH per molecule for a polyketide, which a heterologous host may not provide at the rate needed — test by feeding propionate or overexpressing the carboxylase. Then post-translational maturation: carrier proteins need a phosphopantetheinyl transferase to be activated, and a missing one gives a fully expressed, inactive assembly line. Then product degradation or an undetected product — a different compound from the one the LC-MS method looks for, or one below the quantification limit of part 4 chapter 50. Finally an inactive or mis-folded enzyme, tested by lysate activity with excess precursor.

What to open a book for. The IUBMB enzyme entry for 6-deoxyerythronolide-B synthase and Lowry et al.'s in vitro reconstitution for the chemistry; Pyeon et al., Microb. Cell Fact. (2017) for the pikromycin cluster in the artificial-chromosome system and Kim et al., Front. Bioeng. Biotechnol. (2022) for a giant cluster by BAC; the tracker's 2025 and 2023 *Streptomyces* editing sources for what had to be engineered. Part 3 chapter 48 and part 4 chapter 49 are the two chapters to reread beside the process section.

Closing notes to part 8

The six weeks in one page

Week 26. A part's number travels only as a ratio to a reference on the same plate. A Hill fit gave a baseline uncertain by 95 units with the default routine and 7.5 when weighted by the noise model, so dynamic range spanned 15 to 147 in one and 38 to 49 in the other. The operating window is $81^{1/n}$-fold. A sorting gate keeping 70% of hits and 1% of background takes one in 10,000 to 97% in three rounds, retaining 34% of the true hits. A circuit's speed is its loss rate: dilution alone gives the doubling time.

Week 27. A hundred-fold mRNA knockdown needs the decay rate to rise about a hundredfold — a 1.7-second half-life — so layers must stack. A repressor's ceiling is 1/leak. At a branch point a ten-fold knockdown captures 82% of the available gain when the competitor takes half the flux, and half of it when the competitor takes ten times as much.

Week 28. A three-enzyme library of three levels is 27 variants and about 81 clones. In the burden-aware model, everything-high wins up to a burden slope of about 2.4 and enzyme 1 stays high in every regime. A nine-run orthogonal array picked the wrong level for enzyme 3 because of aliasing. In ALE, single mutations are supplied every generation at 10^{10} cells and specific doubles essentially never.

Week 29. A high pLDDT is confidence in the backbone, not the pocket. ProteinMPNN is ligand-blind; LigandMPNN is not. A score correlated at 0.5 with activity enriches the top 1% thirteen-fold and turns a 30,000-clone screen into roughly 2,300.

Week 30. A faster enzyme raises flux by $1/(1 - C(1 - 1/m))$, ceiling $1/(1 - C)$: 4.7% at 10× for C = 0.05. Two NNK sites are eight days at 384 clones a day; three are 256. A 10,000-clone screen at three standard deviations returns 13.5 impostors. Measure active P450 by the CO difference spectrum before engineering it.

Week 31. 6-dEB balances with six NADPH and one water. Seven propionyl units per molecule make supply the limit. Capturing a 90 kb cluster takes 46 genome equivalents against 11.5 for 60 kb. At 1 g/L the price must be about $160 per kilogram, almost all of it capital and fixed cost.

On the numbers in this part

Every figure and table was computed for this module from the assumptions printed beside it, except where a number is explicitly recalled from the literature and marked so — the biosensor ligand assignments, the pLDDT bands, the benchmark correlation range for variant-effect scores and the wild-type mutation rate of about 10^{-10} per base per generation. Those should be verified against primary sources before they carry a decision. The problem data the tracker refers to (I1, O1, L1 to L4) were not available to me, so the biosensor, assay and library examples use constructed data and stated parameters; the methods transfer, the numbers will differ. Four assumptions carry the most weight: the 4% baseline pathway proteome and the burden slope in chapter 28, which move the winning library member; the 20,000 events a second and 384 clones a day throughputs, which set every time quoted; the random unbiased fragment positions in the cluster-capture model, which real libraries violate; and the $3M vessel and $1.50 propionate in chapter 31's price, which set the dollar figures but not the conclusion that capital and fixed cost dominate.

What remains

Written so far: weeks 1 to 31 and 45 to 61 — 48 of the tracker's 66. What is left is weeks 32 to 44 and 62 to 66. The first group is the data and foundry block (genomic integration, RNA-seq, biofoundry workflows: 32 to 34), the modelling-software block (COBRApy, genome-scale reconstruction, the MATLAB toolbox: 35 to 38), and the design-algorithm, machine-learning and fluxomics block with the process weeks (39 to 44). The second is the publishing, proposal and application weeks. I will take them in tracker order, starting with weeks 32 to 34.

Part 9 — Reading the data and running the loop: integration, sequencing, RNA-seq and the biofoundry (weeks 32 to 34)

Weeks 20 to 31 were about making things. These three are about finding out what you made and whether you can believe the answer: sequencing a strain, counting transcripts, and reading a plate of fluorescence from a robot. The common thread is that each measurement arrives with a bias that is easy to miss and easy to compute — a read-depth threshold, a normalisation assumption, an edge effect — and each week's useful content is the arithmetic that tells you how large the bias is before you act on the result.

A limit on what is in here. This environment has no network access, so I could not run breseq, DESeq2, salmon, the Opentrons simulator or a download from the Sequence Read Archive. Where those tools would produce output, the chapters give you the command, explain what its output means and tell you what to look for, and are marked as not run. Everything quantitative is a first-principles calculation or a simulation I ran, with the parameters printed beside it — the sequencing statistics are analytic, the RNA-seq and plate-effect results come from simulated data with known truth, and the plate plan is code that was executed. That has an advantage for teaching: with simulated data you know the right answer, which is the only way to see how much a threshold or a normalisation costs.

The tracker's problems for these weeks (J1, J2 and the troubleshooting questions) have their own data; I have used constructed examples with the same structure and said so. The methods transfer and the numbers will differ from the tracker's model answers.

Ch.	Week	Chapter
32	32	Genomic integration, sequencing data and wet-lab reality
33	33	RNA-seq and differential expression: what a count matrix assumes
34	34	The biofoundry as a data pipeline: plate plans, protocols and position effects

Genomic integration, sequencing data and wet-lab reality

Industrial strains do not carry pathways on plasmids, and the arithmetic of why is a two-line recursion. The same discipline — compute the failure rate before you trust the result — is what makes a sequencing table or a failed assembly interpretable.

Why integrate: the plasmid-loss recursion

Part 1 chapter 10 said a plasmid does not stay put without selection and burden makes it worse. Put numbers on it. Let p be the fraction of cells carrying the plasmid, l the probability that a division produces a plasmid-free daughter, and s the growth advantage of a plasmid-free cell over a plasmid-bearing one — the burden. After one generation the carrying fraction is

```
p′ = p(1 − l) / [ p(1 − l) + (1 − p + p·l)(1 + s) ]
```

Iterating it, starting from a fully carrying population:

Loss per division	Advantage of plasmid-free cells	Carrying at 40 generations	at 70	Below 50% at
10^{-4}	5%	98.7%	94.2%	—
10^{-3}	2%	94.1%	86.2%	—
10^{-3}	5%	88.5%	60.6%	generation 79
10^{-2}	2%	56.1%	29.3%	generation 46
10^{-2}	5%	37.5%	8.6%	generation 33

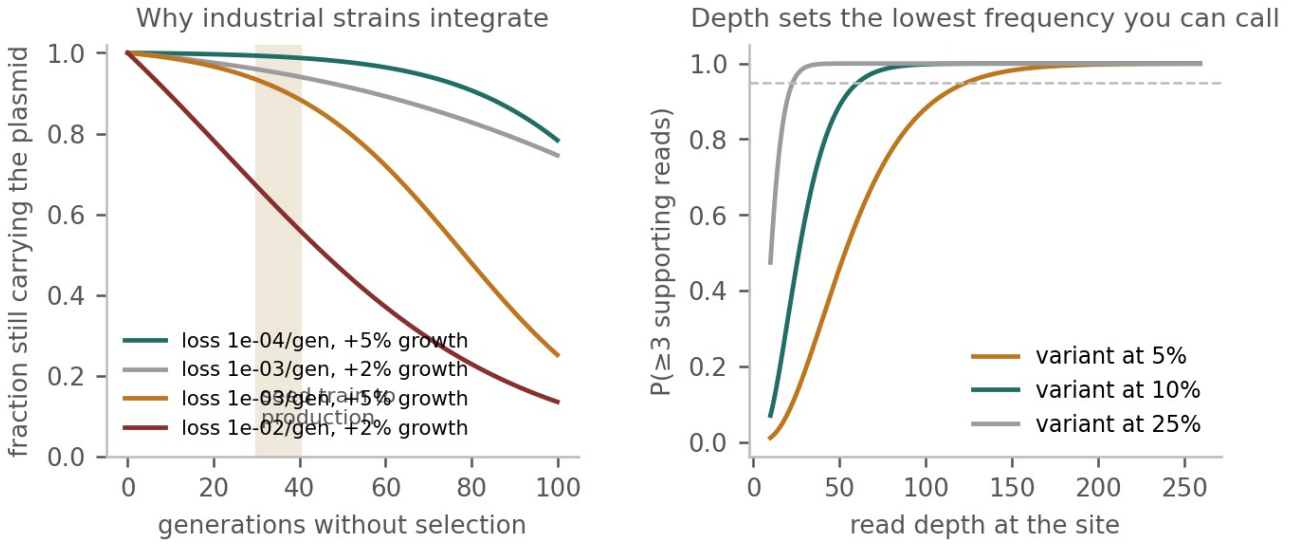


Figure 32.1 Left: the fraction of cells still carrying a plasmid against generations without selection, for four combinations of loss rate and burden; the shaded band is a vial-to-production seed train of thirty to forty generations. Right: probability of at least three reads supporting a variant against depth, at three variant frequencies. Computed for this module; the loss rates and burdens are illustrative, and measuring your own is a plate-count experiment.

Read the third row against the shaded band in the figure. A plasmid that is stable in a bench flask over a few generations can be a majority-free culture by the time it fills a production vessel, because the two rates compound: a loss that is modest per generation is amplified by the growth advantage of the cells that lost it. A seed train from a vial to a 1,000 L vessel is a few tens of generations, which is squarely in the region where a leaky or burdensome plasmid starts to fail. That is the quantitative content of "industrial strains integrate". The other reasons are dosage and antibiotics — a single chromosomal copy is a defined dose, whereas plasmid copy number varies from cell to cell, and no one wants to add an antibiotic to a hundred-cubic-metre fermentation.

Serine integrases and landing pads

Serine integrases such as Bxb1 and φC31 recombine two short, different sites: **attB** on the bacterial side and **attP** on the phage side. The integrase binds both, cleaves each strand, rotates one half of the complex by 180° and religates, producing two hybrid sites, **attL** and **attR**, that flank the inserted DNA. The thing to understand is why this matters for engineering: attL and attR are *not substrates for the integrase alone*. Reversal needs a second protein, a recombination directionality factor, which you do not supply. So integration is effectively irreversible, which buys three things: the integrant does not excise itself during a fermentation, you can use the integrase transiently and lose it, and the integrase will not insert a second copy into the product's own sites. Compare that with a resolvase or a tyrosine recombinase, where the reaction is readily reversible and an insertion can excise.

A **landing pad** is an attB (or attP) site installed in advance at a chosen chromosomal locus, so that every later construct goes to the same place by the same reaction. A safe harbour is a locus where inserting DNA does not disrupt a gene, an essential regulatory element or a transcription–replication conflict — and, an engineering point that is often missed, where the *dosage* suits you. In a fast-growing bacterium replication starts at the origin before the previous round has finished, so genes near the origin are present in more copies than genes near the terminus. For a gene at a fractional position x along the replichore from origin (0) to terminus (1), the copy number is $2^{(C/\tau)(1\,-\,x)}$, with C the replication time and τ the doubling time. With C ≈ 40 minutes (roughly constant at fast growth — an assumption) the origin-to-terminus ratio is 4 at a 20-minute doubling time, 2.5 at 30 minutes and 1.6 at 60. So the same integrated cassette gives two-to-four-fold different dose depending on where it sits and how fast the cells grow, which is worth knowing before you compare two integrants at different loci and attribute the difference to the promoter.

A one-step integration of a four-gene cassette, drawn on paper: a **donor plasmid** carrying attP, the four-gene cassette, and a selectable marker flanked by FRT sites for later removal, on a replicon that cannot replicate in the recipient (a conditional origin that needs a host-supplied initiator protein, so the donor is a dead end unless it integrates); a **helper plasmid**, temperature-sensitive, expressing the integrase; a recipient with attB at the safe-harbour locus. Transform the helper, then the donor; select on the marker; the survivors have integrated, since the donor cannot otherwise be maintained. Verify with two PCRs, one across attL and one across attR, each with one primer in the chromosome and one in the cassette — which also confirms orientation. Express the flippase to excise the marker, leaving one FRT scar; cure the helper by growth at the non-permissive temperature; and sequence the whole cassette, because a long integration is exactly where a mutation or a deletion would hide, as chapter 25's error budget said. Cassettes of five to ten genes have been integrated this way in one step; efficiency falls with size, so the first run for a large cassette is a test of what you actually get.

Editing without a double-strand break

	Recombineering (Red + oligo or cassette)	Cas9 nuclease with a template	Base editing
What it changes	Any defined substitution, deletion or insertion	Any defined edit, enriched by killing the unedited	A single base in a window — C→T or A→G
What it needs	Red proteins; homology arms; an oligo or cassette	Cas9, a guide, a template that destroys the target site	A deaminase–Cas fusion and a guide with a suitable PAM
Failure mode	Low frequency without selection; mismatch repair removes oligo changes	Escapers; a lethal cut if repair fails; plasmid toxicity	Bystander edits in the window; no insertions or deletions; PAM position
Bacterial context	Native to bacteria; works without a break	No efficient end-joining, so an unrepaired cut kills	No break needed, so it avoids the lethality

Prime editing, which dominates attention in the mammalian literature, is a tool that transfers poorly to bacteria, as the tracker says. Bacterial labs instead use base editors for point changes and CRISPR–recombineering for anything else — the combination of chapters 23 and 24. The practical advice is to choose by the edit you need: a point mutation within reach of a PAM, base editing; a deletion or insertion, recombineering with Cas9 counterselection; many loci at once, MAGE.

Sequencing data, from a file to a table

A FASTQ file holds four lines per read: an identifier, the sequence, a plus sign, and a quality string. Each character of the quality string encodes a **Phred score** $Q = -10\,\log_{10}(P_{error})$, stored as the character with ASCII code Q + 33. So `?` (code 63) is Q30, an error probability of 0.1%, and `I` (73) is Q40. A 150-base read with an average Q30 has 0.15 expected errors; at Q20 it has 1.5, which means most reads carry one.

The workflow, which I have not run here: `fastqc reads_1.fastq.gz reads_2.fastq.gz` for the quality report, whose per-base quality plot and adapter-content plot tell you what to trim; align with `bwa mem ref.fasta reads_1.fastq.gz reads_2.fastq.gz > aln.sam`; convert and sort with `samtools sort -o aln.bam aln.sam` and index with `samtools index aln.bam`; view coverage with `samtools depth` or a genome browser. The file formats are worth being able to say: FASTQ holds reads with qualities; SAM is the text alignment, one line per read; BAM is its compressed binary form and the index

(.bai) lets you jump to a region; a reference is FASTA or GenBank, and variants are reported as VCF or, in breseq's case, as an HTML table.

How much depth. For N reads of length L on a genome of size G, the mean depth is c = N·L/G and the fraction of the genome with no reads is e^{-c} (the Lander–Waterman estimate). For a 4.6 Mb genome and 150-base reads, 10^5 reads give 3× depth and leave 3.8% uncovered — 176 kb with no data; 5×10^5 give 16× and leave under one base uncovered; 10^6 give 33× and 2×10^6 give 65×. Coverage is therefore cheap; what depth buys is the ability to call *variants*, and that needs more than coverage.

What depth buys, and what it costs. A variant present in a fraction f of reads is called if at least k reads support it. With k = 3 and binomial sampling, the probability of calling a variant at 10% frequency is 59% at 30× depth, 95% at 60× and essentially certain at 100×; at 5% frequency it is 19% at 30×, 58% at 60×, 88% at 100× and 100% at 200×. Those are the numbers behind figure 32.1's right panel, and they say that for a clonal isolate, where every real variant is at or near 100% frequency, 30× is plenty, whereas for a *population* sample where you want to see subpopulations at a few per cent, you need 100 to 200×.

The cost is false calls. If sequencing errors were independent at 10^{-3} per base, then requiring three reads to agree on the same wrong base would still produce, over 4.6 million sites and three possible wrong bases each, an expected **2.3 false calls at 30× depth, 83 at 100× and 2,100 at 300×**. At 10^{-4} the numbers are 0.002, 0.09 and 2.3. This is an idealised independent-error model, and real errors are not independent — they cluster in particular sequence contexts, which makes some sites worse than the average — but it shows the structure: deeper sequencing finds rarer real variants and also finds more noise, so a caller must also use base quality, strand balance and a minimum frequency. It is why polymorphism-calling modes demand more depth *and* produce more false positives than consensus modes.

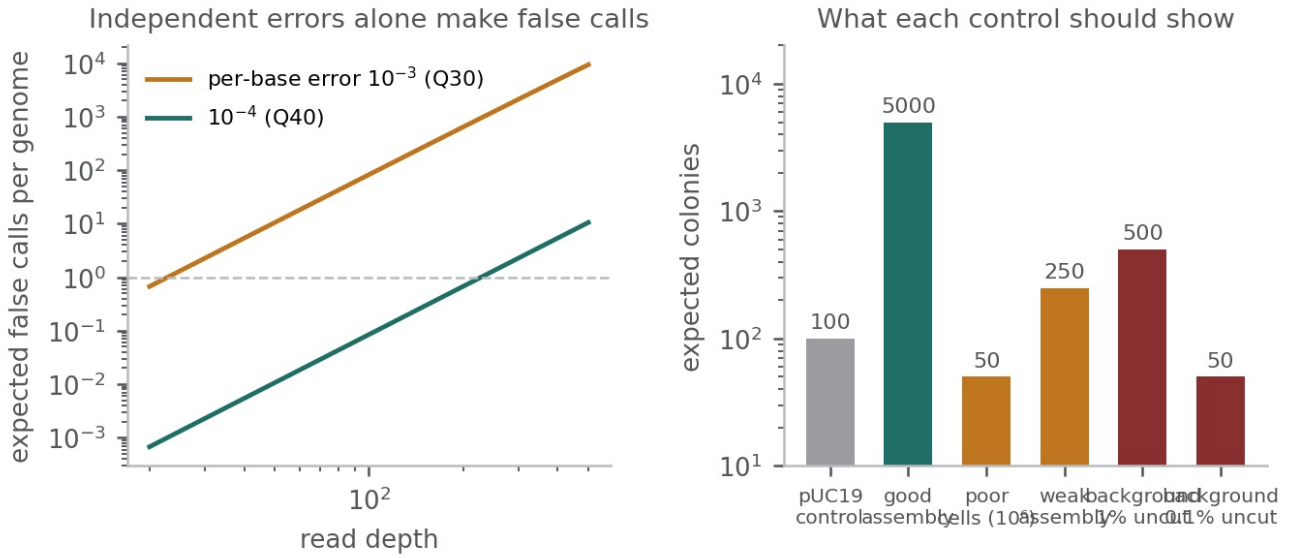


Figure 32.2 Left: expected false variant calls per genome against depth for two per-base error rates, in an independent-error model requiring three supporting reads. Right: the number of colonies each control should give in a transformation, from the assumptions in the text. Computed for this module.

breseq maps reads to a reference and predicts mutations in haploid microbial genomes, including the *new junctions* that betray a mobile-element insertion — reads that span from the reference into an insertion sequence. It has two modes. Consensus mode reports mutations present in essentially all reads, which is right for a clonal isolate. Polymorphism mode (the `-p` option) reports mixed sites down to a few per cent frequency, which is what you want for a population sample from an evolution experiment, and needs the depth above. The tracker's example command is `breseq -j 8 -r REL606.gbk SRR2584524.fastq.gz`, with a note to confirm the accession. The HTML output has three evidence tables: mutation predictions (point mutations and small indels, with the gene, the amino-acid consequence and the frequency), missing coverage (deletions) and unassigned new junction evidence (insertions and rearrangements that breseq saw reads for but could not place) — and the habit to build is to read the third table as well as the first.

To argue which one or two mutations of an evolved strain are causal, use chapter 28's tests: is the same gene or pathway hit in independent lineages; does the mutation fall in a regulator, a transporter or the selected pathway; does a loss-of-function mutation make sense for the selection applied; and then reconstruct each candidate in the parent. A mobile-element insertion into a gene is a gene knockout, and in a long evolution experiment insertion sequences are a common route to loss of function — the point of reading the junction table.

Wet-lab controls, and troubleshooting by arithmetic

Molar ratio. Convert mass to moles with pmol = ng/(0.66 × bp), since a base pair weighs about 660 g/mol: 50 ng of a 5.0 kb vector is 0.0152 pmol, and a 3:1 molar excess of a 1.0 kb insert is 0.0455 pmol, which is 30 ng. For a 300 bp insert at 5:1 it is 15 ng. Many failed assemblies are a mass ratio used where a molar ratio was intended, and the small fragment is then in vast molar excess or deficit.

The control set for a Gibson or Golden Gate reaction. A positive transformation control — 10 pg of pUC19 — which reports the competence of the cells independently of your assembly; a negative control of linearised vector with no insert, which counts the colonies that arise from uncut vector, self-ligation and carry-over; and the assembly itself. With these three, the colony counts decide what went wrong. The expected number of colonies is the correctly assembled DNA transformed, times the efficiency of the cells, times the fraction plated. With assumed values — 50 ng vector, 10% conversion to correct circles, 10% of the reaction transformed and 10% of the recovery plated, into cells at 10^8 colonies per microgram — a good assembly gives about 5,000 colonies, the same reaction in cells at 10^6 gives 50, and a weak assembly at 0.5% conversion gives 250. The pUC19 control under the same fractions gives about 100. And background: if 1% of the vector is uncut, the vector-only control gives about 500 colonies; at 0.1% uncut, 50.

Zero colonies, in order. First look at the pUC19 plate. If it also has none, the problem is the cells, the antibiotic or the heat-shock and recovery — not your DNA. If the control grew, the assembly failed: the likely causes in order are the molar ratio and the DNA amount, an inhibitor in the DNA (column carry-over of ethanol or salt, or too much PCR product in the reaction), fragments with non-compatible or damaged ends (check the overhangs of chapter 21 and whether a 5′ end was phosphorylated where it needed to be), a toxic insert, and finally assembly enzymes that have lost activity. The control that distinguishes the last from the others is assembling a known-good pair of fragments in parallel.

Only background colonies, all empty vector. Compare the assembly plate with the vector-only control. If they are about equal, the background is the whole signal, and the vector was not fully cut or not dephosphorylated, or the digest was incomplete: gel-purify the cut vector, add DpnI to remove template plasmid after PCR, and use a counterselectable marker in the vector's stuffer such as *ccdB* so that uncut vector kills the cell. If the assembly plate has many more colonies but all test as empty vector by colony PCR, the insert is joining but not in the intended way — an overhang that allows self-ligation, a vector that religates through a compatible end, or a bad insert that is not in the reaction. Colony PCR with one primer in the backbone and one in the insert gives a band only when the insert is present and in the expected orientation, with size equal to the distance from each primer to the junction; a missing band is therefore ambiguous, between empty vector and wrong orientation, which is a reason to pair it with a second primer set.

What a Sanger trace adds. A `.ab1` file holds the chromatogram — the four fluorescence traces behind the base calls. Read it by eye for what the calls hide: overlapping double peaks from the point of a mixed colony or a heterozygous population, a trace that falls off in quality after a repeat, poor quality in the first thirty or so bases and past about 800 to 1,000, and the quality values themselves. A gel confirms size; the trace confirms sequence and shows you whether the colony was pure.

Biosafety, and the conventions you will be asked for

Read your institution's framework before you order anything, because you will meet an institutional biosafety committee in your first month. In outline: biosafety level 1 covers well-characterised agents not known to cause disease in healthy adults, with standard microbiological practice and an open bench; level 2 covers agents of moderate hazard, with restricted access, biohazard signage, protective equipment, decontamination by autoclaving and safety cabinets for procedures that make aerosols. Most laboratory *E. coli* and *S. cerevisiae* work is level 1. The committee reviews recombinant and genetically modified organism work before it starts — the host, the inserted genes, whether the organism can survive outside the laboratory, and the containment and waste route. The national framework differs by country (in Korea, the Living Modified Organisms Act is the relevant statute — check the current text and your institution's implementation), and the hazardous cases are the ones you should raise before you plan them: a gene encoding a toxin, a pathway that makes a controlled compound, and any work that could change an organism's host range.

The data conventions that save you a year: a **naming scheme** that encodes type and a serial number — `S0001` for strains, `p0001` for plasmids, `P0001` for plates — with the serial never reused; an **inventory table** with one row per physical item and columns for ID, parent ID, genotype, antibiotic, date, person, freezer, rack, box and position; an **electronic notebook** entry for every experiment, with the IDs it used; and a rule for **where raw data live**, immutable and backed up, with analysis reading from it and never writing to it. Then apply it retrospectively to everything you have built, which is the exercise that shows how much of what you "know" about your strains is in your memory rather than your records.

Plasmid loss	$p' = p(1-l)/[p(1-l) + (1-p+pl)(1+s)]$
Dosage by position	$\text{copies} = 2^{(C/\tau)(1-x)}$; 4 → 1 at τ = 20 min, C = 40 min
Phred	$Q = -10 \log_{10} P$; ASCII = Q + 33; Q30 = 0.1%

Coverage	`c = N·L/G; uncovered fraction e^(−c)`
Call probability	`P(≥3 reads) binomial(depth, f); 59% at 30× for f = 10%`
Moles	`pmol = ng/(0.66 × bp)`
Colonies	`correct DNA (µg) × efficiency × fraction transformed × fraction plated`

Your Golden Gate gave 12 colonies. The vector-only control gave 9, and the pUC19 control gave about 100. Colony PCR on the 12 shows one with an insert. What do you conclude, and what do you do?

The cells are fine — the pUC19 control gave its expected 100 — and the assembly is mostly background: nine of the twelve colonies are what uncut or religated vector gives with no insert at all, so the assembly contributed three colonies and one confirmed correct clone. The signal-to-background is poor and it is the vector that is at fault. First step is to cut the vector more completely: increase digestion time and enzyme, gel-purify the cut backbone, and check that the stuffer fragment is removed by running the digest on a gel. Second, change the vector or the design so that uncut vector cannot give colonies — a *ccdB* stuffer in a *ccdB*-sensitive host. Third, check the molar ratio: if the insert is in low molar excess, empty vector religation wins.

The one correct clone is not wasted: sequence it, and if it is right you have your construct and the question of efficiency becomes a question of cost rather than success. But I would not scale this up before the background is under control, because the next construct, in a bad week, will give twelve colonies with none correct.

Three things people get wrong

- ‘The plasmid is stable, we grew it for ten generations.’ Production is thirty to forty, and a burden of a few per cent compounds a loss rate that looked harmless at ten.
- ‘More depth always finds more.’ It finds rarer real variants and more false ones; the caller's filters, not the depth alone, decide the table.
- ‘Zero colonies means the assembly failed.’ Look at pUC19 first; if the control also failed, the DNA is not the problem.

Self-check — answer on paper first

1. A plasmid is lost at 5×10^{-4} per division and plasmid-free cells grow 4% faster. Roughly what fraction carries it after 50 generations, and what does that imply for a production run?

Iterating the recursion, about 92%: the effective net loss rate per generation is a few times 10^{-3} once the growth advantage acts on the lost fraction, so over fifty generations the carrying fraction falls by about eight per cent, and by a hundred it is near 60%. That is tolerable for a short run and not for a long continuous one, where the same effect continues to compound; and a plasmid carrying a burdensome pathway has a larger s than this, so the number would be worse. The check is to plate the culture with and without selection at the end of the run.

2. You sequence a clonal isolate at 40× and a mixed population at 40×. For each, what can you call, and what can you not?

At 40× the clonal isolate's real variants are at about 100% frequency and are called with certainty, so you can build a reliable mutation table; the only concern is false calls from mapping artefacts in repeats. In the population, a variant at 10% frequency has about a 78% chance of reaching three supporting reads and one at 5% about 32%, so you cannot see minor subpopulations reliably and a negative is uninformative below about 10 to 15%. For the population you want 100× or more and polymorphism mode — and then a stricter filter, since the noise grows with depth.

3. Explain why the serine integrase's irreversibility is useful in a production strain and what you would have to supply to reverse it.

The product sites attL and attR are not recognised by the integrase alone, so once the cassette is in it will not excise during the fermentation, whatever the cell does with the integrase, and the integrase can be lost with its helper plasmid without risk. To reverse it you need the phage's recombination directionality factor, which changes the integrase's preference toward attL × attR and allows excision — useful if you want to remove a cassette deliberately, and a reason not to express it by accident.

What to open a book for. A review of serine integrases and landing pads for the mechanism and the toolkit; the breseq documentation and Deatherage & Barrick, Methods Mol. Biol. 1151:165 (2014); the NCBI Sequence Read Archive for the dataset; and Addgene's protocols on colony PCR and cloning troubleshooting, which are short and good. Part 7 chapters 21 to 25 are the chapters to have open beside the wet-lab section.

RNA-seq and differential expression: what a count matrix assumes

A count matrix looks like data. It is data that has already passed through five assumptions, one of which — that most genes do not change — a production strain can violate by design. This chapter shows each assumption failing, on data where the right answer is known.

From reads to counts, and what is baked in

A bacterial RNA-seq experiment turns RNA into cDNA, sequences it, and asks for each gene how many reads came from it. Three choices made before any statistics determine what the numbers can mean. **rRNA depletion**: ribosomal RNA is most of the total RNA in a bacterial cell, so without removal most reads are wasted on it and the mRNA that is left is sampled shallowly; with removal, the residual rRNA fraction varies between libraries, which becomes a source of technical variation. **Strandedness**: a stranded protocol records which strand a read came from, which separates a gene from an antisense transcript or from a gene on the opposite strand overlapping it — important in the compact, overlapping gene arrangements of bacteria and for detecting the small regulatory RNAs of chapter 27. **Depth**: with about 4,500 genes, 5 million reads of which perhaps 60% are mRNA gives a mean of roughly 650 reads per gene, but the distribution is wide, and the median gene has far fewer, so the genes you care about at low expression may have counts in the tens.

Quantification turns reads into counts. **Alignment-based** methods map reads to the genome and count those overlapping each gene's annotation. **Selective-alignment** methods such as salmon assign reads to transcripts directly, handle ambiguous reads probabilistically, and correct for fragment length and sequence-composition bias; in a bacterium without splicing the advantage is smaller than in a eukaryote, but it helps with overlapping genes and operons. Either way the count matrix hides assumptions you should be able to write down: that the annotation is complete and correct; that reads that map to more than one place have been handled by a stated rule; that a read is counted for the gene it overlaps and not for a longer transcript it belongs to; and that the library is a random sample of the RNA, which is not true of any protocol with a size-selection or amplification step.

Why raw counts lie: composition

The central fact of RNA-seq is that you sequence a fixed number of reads in total, so the counts are *relative*: a gene's count is proportional to its share of the library, not to its absolute amount in the cell. If one gene takes a larger share, every other gene's share must fall, even if their absolute levels did not change.

Demonstrate it, with a simulated experiment where the answer is known. Take 5,000 genes with broad expression, and two conditions with three replicates each at 5 million reads. In the second condition a single gene — your pathway — is 20% of all mRNA; nothing else changes. Compare each gene's share of the library between the two conditions with simple library-size scaling:

```
apparent log₂ fold change of every unchanged gene = log₂(0.80 / 0.9999) = −0.32
```

That is what the simulation gives: a median apparent change of −0.323, and **86% of the genes that did not change have an apparent |log₂FC| above 0.25**. A DESeq2-style *median-of-ratios* normalisation estimates a size factor for each library as the median across genes of the ratio of the gene's count to its geometric mean over all libraries. Because the median gene is unchanged, the size factor ignores the one outlier gene and scales the library by the right amount: the unchanged genes now have a median log₂FC of 0.000 and only 3% exceed 0.25, which is the noise.

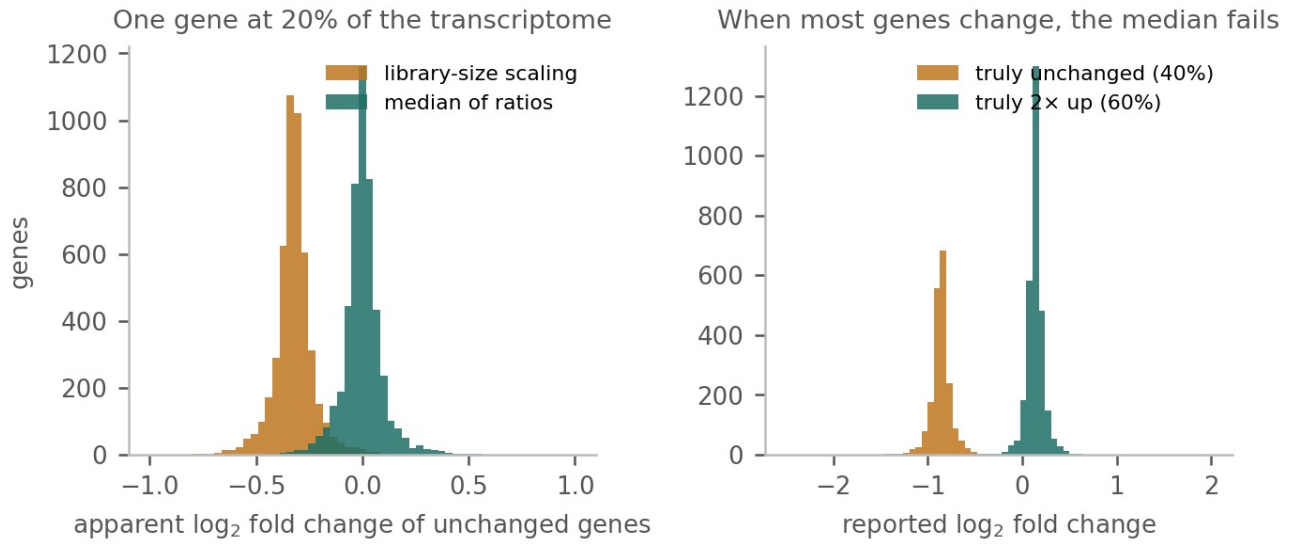


Figure 33.1 Left: apparent log₂ fold changes of unchanged genes with library-size scaling and with median-of-ratios, when one gene takes 20% of the mRNA. Right: reported log₂ fold changes after median-of-ratios normalisation when 60% of genes are truly doubled. Simulated with 5,000 genes, 5 million reads per library and three replicates; all parameters assumed.

When the assumption fails. Median-of-ratios assumes that most genes do not change — the median gene is a reliable internal standard. A production strain can violate that. Suppose 60% of all genes truly go up two-fold, as in a global shift between growth phases or a stress response. The median gene is now one of the changed ones, so the size factor is set by them and rescales the library so that *they* look unchanged. The result in the simulation is that the truly-doubled genes are reported at a median log₂FC of +0.14 — essentially unchanged — and the genes that truly did not change are reported at −0.86, nearly two-fold *down*, with 98% of them past a 1.5-fold cut. The experiment has inverted the biology.

The tracker's specific warning for production strains follows from the first case. A pathway expressed to 20% of the transcriptome is a strongly asymmetric composition shift, and while median-of-ratios handles one dominant gene, it does not handle a pathway-driven global change, and the standard defence is a spike-in: a known quantity of an external RNA added in proportion to cell number, which gives a normaliser independent of the cell's own composition. If your question is about a global shift, say so in the design and add the spike-in; if it is not, check that the median-ratio assumption holds by looking at the distribution of fold changes, which should be centred on zero.

Differential expression: what the model does

DESeq2 and edgeR model each gene's counts as negative binomial, in which the variance exceeds the mean by a term that grows with the square of the mean:

```
Var(count) = μ + α·μ²        α = dispersion, the squared coefficient of variation of the biological variability
```

At low counts the Poisson term μ dominates; at high counts the biological variability α μ² dominates, which is why a highly expressed gene is not measured arbitrarily precisely however deep you sequence. The practical consequence is a floor on the precision of a log fold change. For n replicates per condition, the standard error of the log fold change is approximately $\sqrt{[(2/n)(1/\mu + \alpha)]}$, and as μ grows it tends to $\sqrt{(2\alpha/n)}$, set by biology alone. With α = 0.1 and n = 3 that floor is 0.26 natural-log units, or 0.37 in log₂, however many reads you buy.

The second thing DESeq2 does is the reason n = 3 is workable at all. With three replicates the per-gene dispersion is badly estimated, so the method fits a trend of dispersion against mean across all genes and *shrinks* each gene's estimate toward the trend — borrowing information across 4,500 genes to stabilise each one. Fold changes can then also be shrunk, which reduces the inflation of large estimates in low-count genes. This works, but only so far: it is a statistical rescue of an experiment that is under-replicated, and the arithmetic of what three replicates can see is sobering.

What n = 3 can and cannot find

With the delta-method standard error above, the power to call a given fold change at a critical value corresponding to α = 10^{-3} (a stand-in for an FDR-adjusted threshold on thousands of genes; the critical z is 3.29) is:

Mean count	2-fold: n = 3	n = 6	n = 12	1.5-fold: n = 3	n = 6	n = 12
10	0.08	0.27	0.69	0.01	0.04	0.14
100	0.23	0.63	0.97	0.04	0.12	0.38

1,000	0.27	0.69	0.98	0.04	0.14	0.43

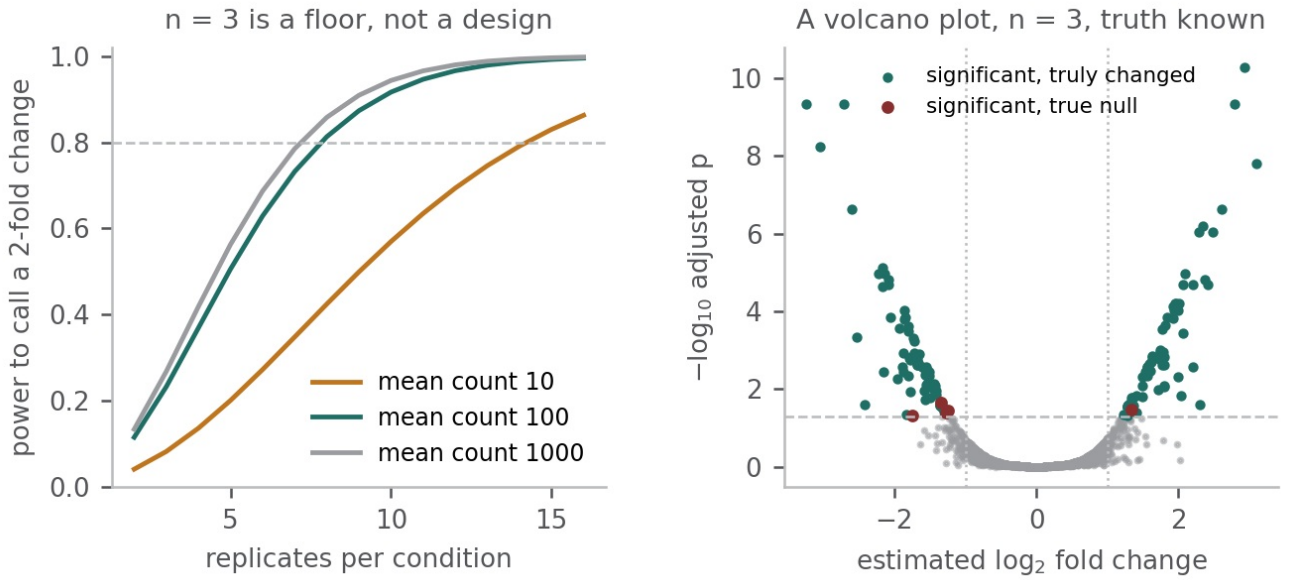


Figure 33.2 Left: power to call a 2-fold change against the number of replicates at three expression levels, from the delta-method standard error with a dispersion of 0.1 and a critical z of 3.29. Right: a volcano plot of a simulated n = 3 experiment in which 10% of genes truly change, coloured by truth. Computed for this module; the dispersion and the effect-size distribution are assumptions.

So at three replicates and a typical mean of a hundred, a real two-fold change is detected 23% of the time and a real 1.5-fold change 4%; it takes about six replicates to reach 63% and twelve to reach 97% for the two-fold case. "n = 3 is the practical floor" is true in that three is the minimum for a variance estimate, and false in that it is an adequate design: a three-replicate experiment is a survey for large effects, and a failure to see a modest one is no evidence that it is absent — the part 2 chapter 17 argument, now with numbers for RNA-seq.

Multiple testing, and what a threshold hides

A genome-wide test is thousands of tests at once. In the simulation, 4,524 genes had no true change, and with an *unadjusted* p < 0.05 there were 484 calls, of which 234 — **48%** — were true nulls. That is about 5% of the 4,524 nulls. A list of 6,000 genes tested at the same cutoff with no true changes at all would return 300 "hits". The Benjamini–Hochberg procedure controls the expected fraction of false discoveries among the calls instead: sort the p-values, and call every gene whose p-value lies below (rank/G) × q.

Now vary the cutoffs on the same simulated data, at two replicate numbers, and read off what changes. Of 5,000 genes, 10% truly changed, with true $\log_2$ changes drawn from a normal distribution of standard deviation 1 — many of them small.

Replicates	Adjusted p below	Fold-change cut (|log₂|)	Genes called	True nulls among them	True changes found
3	0.10	none, 0.585 or 1	165	7.9%	32%
3	0.05	none, 0.585 or 1	133	3.8%	27%
3	0.01	none, 0.585 or 1	93	0%	20%
8	0.05	none or 0.585	247	5.3%	49%
8	0.05	1	170	0.6%	36%
8	0.01	1	163	0.6%	34%

Two results are worth the table. **At n = 3 the fold-change cutoff does nothing**: every gene that reaches adjusted p < 0.05 already has an estimated |log₂FC| of at least 1.23, because with a standard error near 0.4 only a large estimated change can be significant. So "p < 0.05 and 2-fold" is the same list as "p < 0.05", and quoting the two-fold cutoff gives the impression of a stringency you did not apply. At n = 8 the cutoff matters — it removes 77 genes and drops the false fraction from 5.3% to 0.6% — and it also discards true changes that were small, so it is a real trade, which is what the tracker means by "a two-fold cutoff is a convention rather than a biological threshold": a gene 1.5-fold up that is the entry step of your pathway matters more than a ten-fold change in a stress gene, and no cutoff knows that.

The second result is the winner's curse. Among the true changes called at n = 3, the mean estimated |log₂FC| was

1.21 times the true one; at n = 8, 1.05. Genes are selected for being significant, and at low precision the genes that cross the line are disproportionately those whose noise pushed them across, so their estimates are inflated. That is exactly the bias that shrinkage of fold changes corrects, and it is why a paper's list of "top genes by fold change" from a three-replicate experiment overstates its effects.

Defend a pair of thresholds in a sentence, as the week asks: *adjusted $p < 0.05$ to hold the expected false discoveries to about one in twenty of the calls, and no fold-change cutoff at $n = 3$ because significance already implies a large change, with a note that the experiment is powered for changes above about two-fold.*

Running it, and reading it as a biologist

The analysis in R, not run here: build a `DESeqDataSet` from the count matrix and a sample table with `DESeqDataSetFromMatrix(countData, colData, design = ~ condition)`, run `DESeq(dds)`, extract `results(dds, alpha = 0.05)`, and apply `lfcShrink` before ranking by fold change. In Python, `pydeseq2` implements the same model. Public count matrices for a strain comparison are on the Gene Expression Omnibus; reanalyse one and record how many genes pass at your thresholds and what the volcano plot's shape says about the replication.

Transcript is not flux

Three reasons a transcript level can rise while flux does not, each quantifiable from earlier parts. **Translation and turnover**: part 1 chapter 8's expression equation puts protein at transcription × initiation ÷ (degradation + dilution), so a doubled transcript can give an unchanged or only modestly changed protein if initiation or degradation changed. **Enzyme activity**: the enzyme can be present and inactive — misfolded, missing a cofactor, post-translationally modified or inhibited — the point of chapter 22's band-is-not-activity warning. **Metabolite control**: even a fully active, doubled enzyme changes flux only by its control coefficient — for an enzyme with C = 0.10, doubling the activity buys at most 5% (chapter 30), and for one saturated by its substrate the gain is smaller still. A transcriptome therefore tells you what the cell regulated, which is useful for diagnosing a stress response or a mis-regulated promoter, and it is a weak proxy for what the pathway did.

Take your top differentially expressed genes and say what you would measure next: for each, the protein (by Western blot or proteomics), then the activity in a lysate under saturating substrate, then — if it matters — the intracellular substrate concentration against K_M to decide whether the enzyme is in its first-order regime, and finally the flux, by a carbon balance or ^{13}C labelling (week 43). That sequence is the week 51 argument in the order you would run it.

Composition shift	apparent $\log_2 FC = \log_2[(1 - x)/(1 - x_0)]$ for a gene taking share x
Median of ratios	size factor = median over genes of count/geometric mean; fails if most genes change
Negative binomial	$\mathrm{Var} = \mu + \alpha\mu^2$; $\mathrm{SE}(\ln FC) \approx \sqrt{[(2/n)(1/\mu + \alpha)]}$
Power, n = 3	0.23 for a 2-fold change at mean 100, $\alpha = 0.1$, $z_{crit} = 3.29$
Unadjusted false calls	5% of the true nulls — 234 of 4,524 here
Winner's curse	estimated/true = 1.21 at n = 3, 1.05 at n = 8, among called genes

A colleague compares an engineered strain expressing a pathway at roughly 25% of its mRNA against the parent, n = 3, and reports 600 genes "significantly down at least two-fold". How do you read that?

With a pathway taking about a quarter of the transcriptome, naive library-size scaling would make every other gene look about 25% lower — a $\log_2$ shift of about -0.4 — so a large share of "down" calls could be composition, not biology. First check which normalisation was used and look at the distribution of $\log_2$ fold changes for the bulk of genes: if it is centred on -0.4, normalisation did not correct it; if it is centred on zero, median-of-ratios did, and the 600 are more credible. Second, the genes truly downregulated by a heavy burden are exactly what you expect — ribosomal proteins and growth-related genes fall under a stringent-response-like state — so the biology is plausible, but plausible is not demonstrated.

Third, at n = 3 the two-fold cutoff is redundant with significance, as the table shows, so the claim of "at least two-fold" is carried by the significance threshold alone, and the estimated fold changes of the significant genes are inflated by about 20%. What I would ask for is the $\log_2 FC$ histogram, the normalisation, the adjusted-p cutoff, and whether a spike-in or a second normalisation method was used as a check — and, if the question is global, a spike-in in the next run.

Three things people get wrong

- ‘Counts are abundance.’ They are shares of a fixed library; one dominant gene lowers every other gene's apparent level until normalisation is checked.
- ‘$p < 0.05$ and two-fold is a strict filter.’ At three replicates significance already implies a large change, so the second

condition adds nothing.

- ‘The gene is upregulated, so the pathway flux rose.’ Two lossy steps and a control coefficient sit between them.

Self-check — answer on paper first

1. One gene takes 40% of the mRNA in condition B and 1% in A. With library-size scaling, what apparent $\log_2$ change does an unchanged gene show?

Its share in B is multiplied by $(1 - 0.40)/(1 - 0.01) = 0.606$, so $\log_2(0.606) = -0.72$, an apparent 1.65-fold decrease in every unchanged gene. Median-of-ratios would remove this, provided most genes are in fact unchanged; with a spike-in the correction does not depend on that assumption.

2. You have a budget for either 3 replicates of each of 4 strains or 6 of each of 2. Which and why?

Six of two, if the question is a pairwise comparison, because power for a two-fold change at a mean of 100 rises from 23% to 63% on doubling the replicates, whereas the four-strain design buys breadth at a power below a quarter. If the question needs all four strains, a reduced design — a reference and three test strains, with more replicates of the reference — is better than equal n = 3 throughout, since every comparison involves the reference and its precision is shared.

3. Your 3-replicate experiment returns 20 genes at adjusted $p < 0.05$, and you expected hundreds. What do you do before concluding the strain is unchanged?

Compute what the design could have seen: at n = 3 and a dispersion of 0.1, a real two-fold change at a mean of 100 is found about a quarter of the time and a 1.5-fold change about 4%, so a short list is what a weak experiment returns whether or not there is biology. Look at the dispersion estimates — a high one means the replicates disagreed, which is an experimental problem to fix — and at the raw fold-change distribution, where a shift of many genes by small amounts is invisible to gene-by-gene testing and visible to a pathway-level analysis. Then replicate to at least six.

What to open a book for. Love, Huber & Anders, Genome Biol. (2014) for DESeq2, read for dispersion estimation and shrinkage rather than the software; Patro et al., Nat. Methods (2017) for salmon; the Bioconductor DESeq2 vignette, which is the best tutorial there is; and the Gene Expression Omnibus for a count matrix to reanalyse. Part 4 chapters 17 and 51 are the two chapters of this module that this chapter is built on.

The biofoundry as a data pipeline: plate plans, protocols and position effects

A biofoundry is a data pipeline with robots at some of the nodes. Most of what goes wrong is not the robot: it is an identifier that was lost, a plan whose volumes were not counted, or a plate in which position was mistaken for biology.

The cycle as a pipeline, and why identifiers come first

The design–build–test–learn cycle of part 2 chapter 11 becomes, in a foundry, a sequence of files. A **design file** lists each variant and its parts. A **build plan** turns it into a table of reactions with source wells, volumes and destination wells. An **automation protocol** executes the plan. A **test** produces plate-reader or chromatography data keyed by well. A **results table** joins the data back to the design, and the next round's design is computed from it. K-Biofoundry, which the tracker names as KRIBB's facility, and the Global Biofoundry Alliance it belongs to, exist so that groups can share the equipment, the standards and the people for this loop; what an academic group gets from membership that a single laboratory does not is throughput, standardisation and the discipline of a pipeline that someone else has already debugged.

The load-bearing component is the **identifier**. Every strain, plasmid, plate and well needs one that is created once, never reused, and carried through every file. A workable scheme: a design ID per variant (`V00417`), a plate barcode per physical plate (`PL0231`), a well address (`B07`), and a sample ID that is their concatenation (`PL0231-B07`). The results table is then a join of the design table and the data table on sample ID, and every result traces back to its design by a key rather than by someone's memory.

What goes wrong if two plates share a barcode: the join is many-to-many, so a well on one plate silently takes data from the other. Nothing errors, every row has a value, and the results are plausible and wrong — which is the worst failure a pipeline has. The defences are that barcodes are generated and checked for uniqueness, that a join asserts a one-to-one match and fails loudly if it does not, and that the plate map is stored in the same file as the data it describes.

From a variant list to a plate plan

The week's concrete exercise: a 24-variant combinatorial assembly — four promoters, three coding sequences, two terminators. The code below builds the plan and counts what the robot will need; I ran it, and the numbers in the table that follow are its output.

```
import itertools, pandas as pd
prom, cds, term = ['P1','P2','P3','P4'], ['C1','C2','C3'], ['T1','T2']
plan = pd.DataFrame(list(itertools.product(prom, cds, term)), columns=['prom','cds','term'])
plan['dest'] = [f"{'ABCDEFGH'[i//12]}{i%12+1}" for i in range(len(plan))]     # 24 of 96 wells

rxn_uL, overage, dead_uL = 10, 1.10, 10
# per 10 uL reaction: 1 backbone + 3 parts (1 uL each) + 1 buffer + 1 enzyme mix + 4 water
common = 1 + 1 + 1 + 4                                  # in the master mix: backbone, buffer, enzyme, water
mm_total = common * len(plan) * overage                 # 184.8 uL
for name, lst in (('prom', prom), ('cds', cds), ('term', term)):
    uses = len(plan) // len(lst)                        # reactions each part appears in
    print(name, uses, 'uL +', dead_uL, 'dead =', uses + dead_uL, 'uL per source well')
print('transfers', len(plan) * 4, '; fresh-tip transfers for parts', 3 * len(plan))
```

Item	Count or volume	Why
Reactions	24	4 × 3 × 2
Master mix	7 µL × 24 × 1.10 = 184.8 µL, plus dead volume	Backbone, buffer, enzyme mix and water are common to every reaction
Each promoter well	6 µL + 10 dead = 16 µL	Each appears with 3 coding sequences × 2 terminators = 6 reactions
Each coding-sequence well	8 µL + 10 dead = 18 µL	4 promoters × 2 terminators = 8 reactions
Each terminator well	12 µL + 10 dead = 22 µL	4 × 3 = 12 reactions
Source wells / destination wells	9 / 24 of 96	4 + 3 + 2 distinct parts
Transfers	96	3 parts + 1 master mix per reaction

Tips	72 for parts; 73 if one tip serves the master mix, 96 if each reaction gets its own	A fresh tip for every part avoids cross-contamination of the stock

Three things in the table repay attention. **Dead volume** dominates the small stocks: the 10 µL left at the bottom of a well that the pipette cannot reach is more than the volume you actually use of each promoter, so the stock you must prepare is 16 µL for 6 µL consumed. That ratio worsens as the experiment gets smaller and is the reason plate plans are written as a table before a protocol is. **The master mix** is where most of the volume is, and its overage is an insurance against the pipette's imprecision at the end of a long dispense; 10% is conventional and should be tested against what is actually left. And **the number of tips** is a real consumable budget — a thousand-sample screen at 96 tips per 24 reactions is four thousand tips.

What changes in a 384-well plate: the reaction volume falls, often to 2 to 5 µL, so the dead volume is a larger fraction of everything and several parts will fall below the reliable volume of a standard pipette, which typically has a minimum of about 1 µL; the source plate, if also 384-well, has smaller wells with different dead volumes and a different liquid-height-to-volume relationship; the destination layout needs a plan for the edge effect below; and evaporation, discussed next, becomes a first-order concern.

Writing the protocol, and what the simulator cannot check

The Opentrons Python API declares labware, loads pipettes, and expresses a plan as transfers. A skeleton for the plan above, which I have not run — check the labware definitions and API level against the current documentation and run it through the simulator before it goes near a robot:

```
from opentrons import protocol_api
metadata = {'protocolName': 'GG 24-variant assembly', 'apiLevel': '2.15'}

def run(protocol: protocol_api.ProtocolContext):
    tips  = protocol.load_labware('opentrons_96_tiprack_20ul', 1)
    src   = protocol.load_labware('nest_96_wellplate_100ul_pcr_full_skirt', 2)    # parts
    dest  = protocol.load_labware('nest_96_wellplate_100ul_pcr_full_skirt', 3)    # reactions
    mm    = protocol.load_labware('opentrons_24_tuberack_nest_1.5ml_snapcap', 4)  # master mix
    p20   = protocol.load_instrument('p20_single_gen2', 'right', tip_racks=[tips])
    # plan: a table of (source well, destination well, volume) generated from the pandas plan above
    for part_well, dest_well, vol in transfers:
        p20.transfer(vol, src[part_well], dest[dest_well], new_tip='always')
```

The simulator checks that the protocol parses, that labware exists, that every transfer is geometrically possible and that you have enough tips. It cannot verify what only the machine does: **the calibration** of the deck positions, which is the responsibility of whoever calibrated the robot; **liquid class**, the way a viscous master mix with glycerol-containing enzyme aspirates differently from water and needs a slower speed and a delay; **tip behaviour**, whether tips seat and eject cleanly, and whether a droplet clings; **evaporation** during a long run; and **real liquid levels**, so that a source well that has run low is detected rather than aspirated dry. Each needs a named owner, and each is checked by a test run with a dye or a gravimetric check rather than by trusting the plan.

The PyLabRobot paper's argument is the software-engineering one: a protocol written against one vendor's API is locked to that vendor's machine, and a hardware-agnostic layer lets the same script drive different robots. The problem it solves is the portability of a protocol across a foundry's mixed fleet, and it reflects a view that protocols should be treated as versioned code — reviewed, tested, tagged — not as a file that lives on one laptop.

Miniaturisation, and what it saves and costs

For a 384-variant library with a reaction in which one tenth of the volume is enzyme mix and one tenth is buffer — so 2 µL of the 10 µL is expensive reagent — the totals are:

Reaction volume	Total reaction volume	Enzyme + buffer for 384	Saving versus 10 µL
10 µL	3,840 µL	768 µL	1×
2 µL	768 µL	154 µL	5×
500 nL	192 µL	38 µL	20×

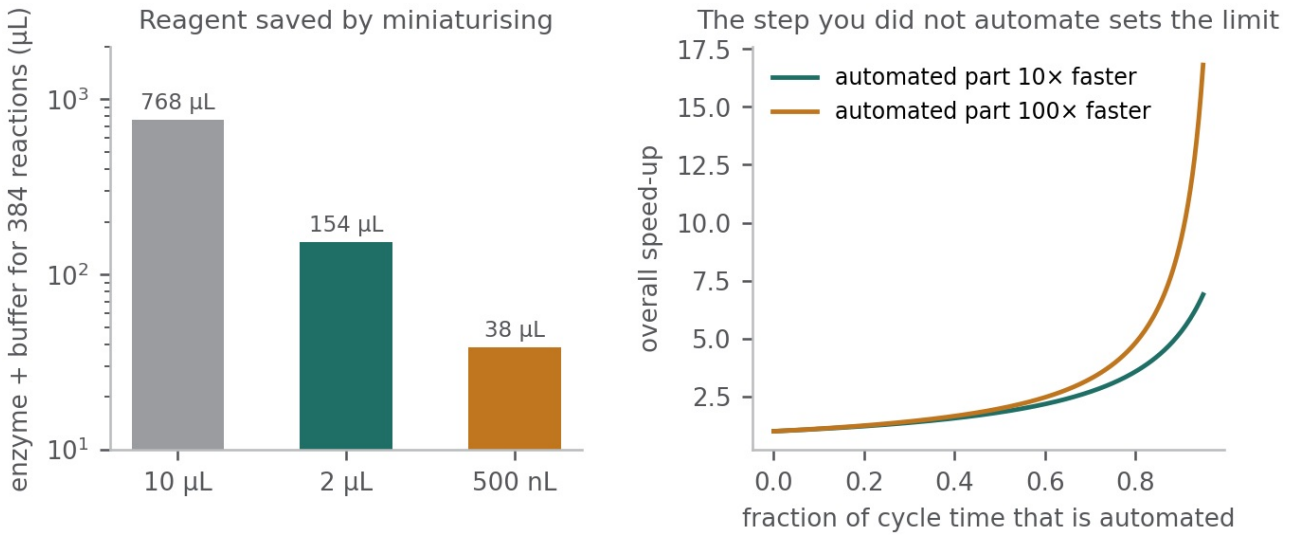


Figure 34.1 Left: enzyme and buffer needed for 384 reactions at three scales, on a log scale. Right: overall speed-up of a workflow against the fraction of its time that is automated, for an automated step that is ten and a hundred times faster than manual — Amdahl's law. Computed for this module.

The reagent saving is real and proportional to volume. What it costs is **evaporation**. A well with a fixed opening loses liquid at a roughly constant rate in volume per hour, so the fractional loss goes as one over the volume. If an open well loses 0.1 µL an hour — an assumed figure that depends on humidity, temperature and the seal — a 10 µL reaction loses 1% an hour, a 2 µL reaction 5% and a 500 nL reaction 20%. At that point concentration changes over a run and varies by position, which is the next section's subject. The mitigations are the ones chemical engineers know: seal the plate, work at a low temperature or high humidity, keep the plate on the deck as short a time as possible, and put an inert oil overlay on very small volumes.

The steps that cannot be miniaturised are the ones that involve living cells or space: transformation into cells, plating and colony picking, and growth in deep-well blocks. They set the throughput, and the arithmetic is Amdahl's law: if a fraction f of the cycle time is made s times faster, the overall speed-up is $1/[(1 - f) + f/s]$. If assembly set-up is 30% of the cycle time and automation makes it ten times faster, the whole cycle is only 1.37 times faster, and a hundredfold gain takes it to 1.42; the limit, with f = 0.3, is 1.43 whatever the robot does. That is the same lesson as part 2 chapter 11's foundry that builds 2,000 strains into a 400-sample screen: speed up the bottleneck, not the step that was already fast.

Plate data, honestly: the edge effect

Edge wells of a plate evaporate faster and sit at a different temperature, so in a growth or fluorescence assay they read high. The question is how badly that corrupts a ranking, and the answer is simulable. Take a 96-well plate with 36 edge and 60 inner wells, 96 variants whose true values vary by about 15% (lognormal), placed in random wells, edge wells reading 20% high, and 5% measurement noise.

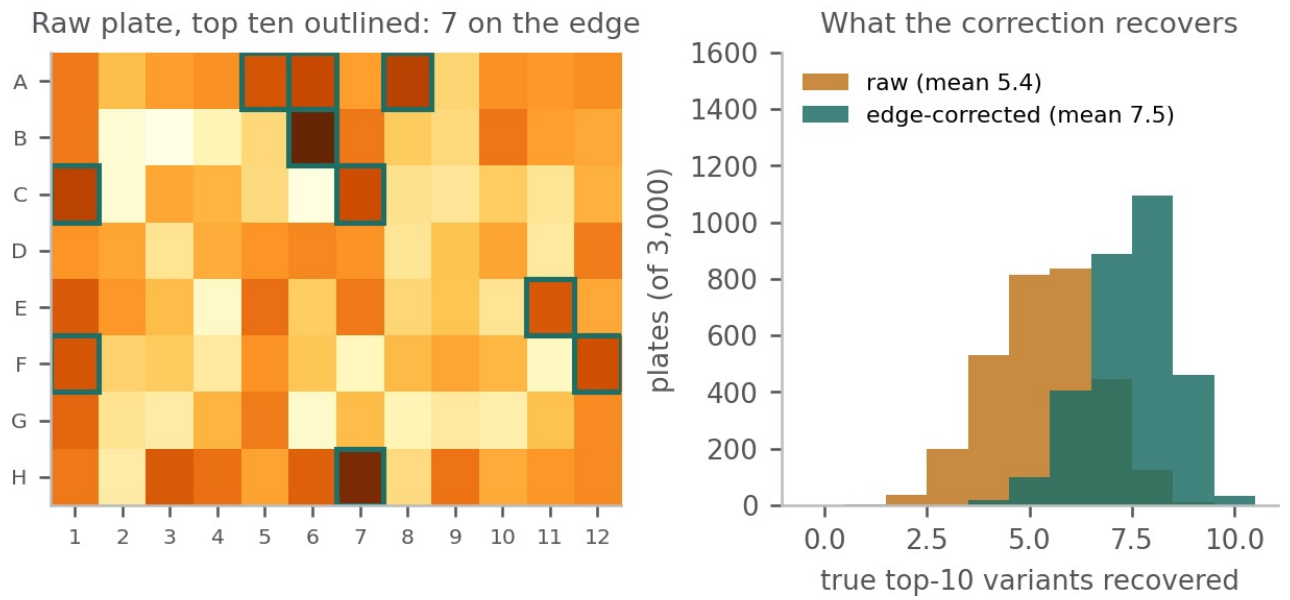


Figure 34.2 Left: one simulated raw plate with the ten highest readings outlined — seven of them are edge wells. Right: across 3,000 simulated plates, how many of the true top ten variants appear in the measured top ten, raw and after estimating the edge multiplier from the plate itself and correcting for it. Computed for this module; the 20% multiplier, 15% true spread and 5% noise are assumptions.

Averaged over 4,000 simulated plates, **8.2 of the ten highest raw readings are edge wells**, against 3.75 expected if position did not matter — the top of the ranking is dominated by evaporation. The raw ranking recovers 5.4 of the true top ten and has a Spearman correlation of 0.80 with the truth. Now estimate the multiplier from the plate itself, as the ratio of the median edge reading to the median inner reading, which works *because the variants were placed randomly* so that edge and inner wells hold statistically identical populations; the estimate comes out at 1.200, the true value. Dividing the edge readings by it recovers 7.5 of the true top ten and a correlation of 0.93.

The correction only works if the layout is randomised. In a blocked layout — one strain family filling the edge and another the inner wells — the same edge effect appears as a difference *between the families*. Simulating two identical families, one placed predominantly on the edge, gives an apparent family ratio of 1.15 where the truth is 1.00: a 15% "effect" produced by the plate, with nothing in the data to say so.

So the controls to add to the plate design are these. **Randomise the layout**, so position is not confounded with identity. **Put the same control strain in many positions** — in every row and column, and in edge and inner wells — which measures the position effect directly rather than inferring it, and also shows whether it is a simple edge multiplier or a gradient. **Leave the outer ring empty or filled with water or medium** if you can afford the wells, which turns the evaporation problem into a lost 37% of capacity. And **replicate the best candidates in a second plate** with a different layout, since a hit that survives a change of position is a hit that is not about position.

Identifier	`sample ID = plate barcode + well; joins must be one-to-one and fail loudly`
Master mix	`common µL per reaction × reactions × overage + dead volume`
Source well	`(uses of the part) × µL per use + dead volume`
Evaporation	`fractional loss ∝ 1/V at fixed open area; 1% → 5% → 20% per hour here`
Amdahl	`speed-up = 1/[(1 − f) + f/s]; limit 1/(1 − f)`
Edge multiplier	`median(edge)/median(inner), valid only for a randomised layout`

A foundry run of 960 variants in ten plates shows that the top hits cluster in columns 1 and 12. The group wants to take the top twenty into the fermenter. What do you advise?

That the clustering at the plate edges is the signature of an evaporation effect, so before anything goes into a fermenter the ranking needs to be separated from position. Check the layout first: if variants were placed in a design file order rather than randomly, position and identity are confounded and the correction in the chapter does not apply — the edge wells hold a particular subset. If the layout was randomised, estimate the edge multiplier per plate from the data and correct, then re-rank; the simulation says the correction recovers roughly two more of the true top ten, from 5.4 to 7.5 of 10.

In either case do a validation round before the fermenter: re-pick the top fifty by the *corrected* ranking, re-test them in a new plate layout with a control strain in every row and column, and in duplicate, and take forward only those that hold their position. That costs one more plate cycle and removes the class of hit that was a property of the plate. The fermenter, by contrast, costs weeks, and the lesson of part 2 chapter 17 is that the right time to find a false positive is before it is expensive.

Three things people get wrong

- ‘The robot is the risk.’ The risk is an identifier, a join, a dead volume or a position effect — the pipeline around the robot.
- ‘The simulator passed, so the protocol works.’ It checks geometry and tip counts; calibration, viscosity, tip behaviour and evaporation are only found on the machine.
- ‘Automating assembly makes the cycle ten times faster.’ If it is 30% of the cycle, it makes it 1.37 times faster; the unautomated step sets the ceiling.

Self-check — answer on paper first

1. Compute the source volume of each part and the master-mix volume for a 4 × 4 × 3 = 48-variant assembly in 10 µL reactions with 10% overage and 10 µL dead volume. What is the first thing that limits you?

Each of the four promoters appears in 4 × 3 = 12 reactions, so 12 + 10 = 22 µL; each of the four coding sequences in 4 × 3 = 12 reactions, 22 µL; each of the three terminators in 4 × 4 = 16 reactions, so 16 + 10 = 26 µL. The master mix is 7 µL × 48 × 1.10 = 369.6 µL plus dead volume, which is far more than a PCR-plate well holds, so it must go into a 1.5 mL tube — which the tube rack in the skeleton provides. The first limit is the tips: 48 × 3 = 144 fresh-tip transfers for the parts is more than one 96-tip rack, so the deck needs two, not the part wells or the master mix.

2. A fluorescence plate shows its highest values in the corner wells. Give two explanations that are not biology, and one control that separates them from a real effect.

Evaporation raising the concentration of the fluorophore and the cells in edge wells, and a temperature or aeration gradient from the plate's position in the incubator or reader. A control strain placed in a grid of positions across the plate gives a position map: if the control reads high in the corners, the effect is position, and it can be measured and divided out; if it reads uniform, the corner values are properties of the strains placed there.

3. Why does miniaturising from 10 µL to 500 nL save twenty-fold in reagent but not twenty-fold in time?

Reagent scales with volume, but time is set by the slowest step in the cycle, and the steps that cannot be miniaturised — transformation, plating, colony picking, growth, and the assay read — keep their duration. By Amdahl's law, if the assembly set-up is 30% of the cycle time, even an infinite speed-up of it gives a 1.43× overall gain. And the smaller volume brings evaporation and dispensing-accuracy problems that add repeats and controls, which can erode the gain further.

What to open a book for. K-Biofoundry and the Global Biofoundry Alliance sites for what a foundry offers; the Opentrons Python API documentation and the PyLabRobot paper for protocols as code; Seong et al., Trends Biotechnol. (2026), the biofoundry workflow section; and pandas' "10 minutes to pandas" for the joins. Part 2 chapter 11 is the chapter whose bottleneck argument this one puts numbers to.

Closing notes to part 9

The three weeks in one page

Week 32. A plasmid lost at 10^{-3} per division with a 5% burden is below half carrying by generation 79, and at 10^{-2} by generation 33 — inside a seed train. Serine integrases make an irreversible insertion into a landing pad, and the pad's position sets dose: 4× origin-to-terminus at a 20-minute doubling time. Q30 is a 0.1% error; at 100× and independent 10^{-3} errors, three-read agreement still yields about 83 false calls per genome; and a 10%-frequency variant needs about 60× to be called 95% of the time. The colony counts of the pUC19 and vector-only controls decide whether a failed assembly is the cells, the vector or the DNA.

Week 33. One gene at 20% of the mRNA makes every unchanged gene look 0.32 $\log_2$ down until median-of-ratios corrects it, and if most genes change the same method inverts the result. At n = 3, power for a two-fold change at a mean of 100 is 0.23, a fold-change cutoff is redundant with significance, and significant genes' fold changes are inflated by about 20%. Unadjusted $p < 0.05$ returned 234 false calls of 484. A transcript is a weak proxy for flux.

Week 34. A 24-variant plan needs 184.8 µL of master mix, 16 to 22 µL per part well because of dead volume, and 96 transfers. Miniaturising to 500 nL saves 20× in reagent and costs 20% an hour in evaporation. Automating 30% of a cycle by ten times speeds it 1.37×. Edge wells filled 8.2 of the top ten raw readings, and a correction estimated from the plate itself recovered 7.5 of the true top ten against 5.4 — but only for a randomised layout.

On the numbers in this part

The sequencing-depth, false-call, plasmid-retention, dosage and colony-count figures are analytic or iterated recursions with assumptions stated beside them — the loss rates, the burden, the 40-minute replication time, the transformation fractions. The RNA-seq results and the edge-effect results are simulations with known truth: 5,000 genes, a lognormal mean-expression distribution, a dispersion of 0.1, a 10% truly changed set for the threshold study; and a 96-well plate with a 20% edge multiplier. The power numbers use a delta-method standard error rather than a full negative-binomial fit, so they are good to perhaps a few percentage points and should be rerun with real data. Nothing here was produced by breseq, DESeq2, salmon or the Opentrons simulator, which I could not run; the commands and code in those sections are marked not run, and should be checked against current documentation before use.

What remains

Written so far: weeks 1 to 34 and 45 to 61 — 51 of the tracker's 66. Remaining: weeks 35 to 44 and 62 to 66. The next block is the modelling software: constraint-based modelling and genome-scale models in COBRApy, reconstructing a model, and the MATLAB toolbox (weeks 35 to 38), where the part 2 chapter 13 material becomes code you run. After that, the design-algorithm, machine-learning and fluxomics weeks with the process weeks (39 to 44), and then the five professional weeks. I will take weeks 35 to 38 next.

Part 10 — Constraint-based modelling in practice: solving it yourself, COBRApy, reconstruction and MATLAB (weeks 35 to 38)

Part 2 chapter 13 set up S·v = 0 and said that stoichiometry constrains while measurements or an objective decide. These four weeks turn that into software you run. The tracker's route is COBRApy, and that is the right tool; this environment has no network access, so I could not install it. Rather than write four chapters of untested commands, I built a 40-reaction core model of *E. coli* central metabolism by hand and solved every analysis in the weeks' lists — FBA, flux variability, deletions, pFBA, shadow prices, quality control and gap filling — with `scipy.optimize.linprog`, and ran them.

That turns out to be the better teaching order. Writing the linear programme yourself makes it impossible to treat FBA as a black box: you see the equality constraints, the bounds, the objective vector and the duals, and when something is wrong you have to find it. While building the model I made three real errors, each of which a tool would have hidden. The chapters below keep them, because diagnosing them is the skill: a phosphate bookkeeping inconsistency that let an ATP-wasting cycle run free, a missing soluble transhydrogenase that silently blocked the whole TCA cycle and capped growth at less than half, and a vestigial exchange reaction that the blocked-reaction check in chapter 37 caught.

Where the chapters give COBRApy or MATLAB code, it is marked not run and should be checked against the current documentation. Where a number appears, it came out of the scripts in this part. The model is mine and not the published core model, so its numbers are close to but not identical with the tracker's reference values, and I say which is which.

Ch.	Week	Chapter
35	35	Flux balance analysis, written as the linear programme it is
36	36	Working with models: media, loops, pFBA, thermodynamics and reproducibility
37	37	Reconstruction and quality control: drafts, gap filling and what a check detects
38	38	MATLAB and the COBRA Toolbox: reading other people's code

Flux balance analysis, written as the linear programme it is

FBA is three lines of mathematics and a solver. Writing those three lines yourself once is worth a month of using a package, because everything that goes wrong later is a bound, a medium or an objective — never the mathematics.

The optimisation problem, in full

Part 2 chapter 13's balance plus bounds plus an objective is a linear programme:

```
maximise    cᵀv                  c selects the objective — usually the biomass reaction
subject to  S·v = 0              one equality per internal metabolite: production = consumption
            lb ≤ v ≤ ub          irreversibility, measured uptake rates, capacity limits
```

Three features of that statement carry the whole method. It is **linear**, so it solves in milliseconds even at genome scale, and the solution sits at a vertex of the feasible polytope. It needs **no kinetic parameters**: no K_M, no k_{cat}, no concentrations appear anywhere. And the price of that is what it cannot tell you — **no concentrations, no time, no regulation, and no statement that the cell will actually do this**. The objective is a hypothesis about what the cell optimises, as part 2 chapter 13 insisted; FBA returns a flux distribution that is *consistent* with the network, not one that is observed.

The model used throughout this part

Forty reactions and twenty-nine metabolites covering glycolysis, the pentose phosphate pathway, the TCA cycle, anaplerosis, respiration, the fermentative branches, nitrogen assimilation through glutamate dehydrogenase, maintenance, and a biomass reaction. The biomass coefficients are rounded from the published *E. coli* core model's, the P/O assumptions are 2.5 ATP per NADH and 1.5 per quinol, and maintenance is fixed at 8.39 mmol ATP·gDW^{-1}·h^{-1}. Glucose uptake is bounded at 10 mmol·gDW^{-1}·h^{-1}. Building it is the whole of the code; solving it is four lines:

```
import numpy as np
from scipy.optimize import linprog

S  = np.zeros((len(mets), len(rxns)))            # filled from a dict of {reaction: {metabolite: coeff}}
c  = np.zeros(len(rxns)); c[idx['BIOMASS']] = -1 # linprog minimises, so negate to maximise
res = linprog(c, A_eq=S, b_eq=np.zeros(len(mets)),
              bounds=list(zip(lb, ub)), method='highs')
growth = -res.fun
```

Rank and degrees of freedom first, because they say what the network leaves undetermined: 29 metabolites, 40 reactions, rank 29, so **11 degrees of freedom**. Eleven numbers must come from outside — the uptake bounds, the maintenance demand and the objective supply them, which is exactly part 2 chapter 13's three ways to close a system, now made concrete.

Growth, aerobic and anaerobic

With glucose at 10 and oxygen unconstrained, the model grows at **0.754 h^{-1}**; with oxygen set to zero, at **0.217 h^{-1}**, a ratio of 3.5. The tracker's reference values from the published core model are about 0.87 and 0.21, a ratio of about 4.2. Mine is lower aerobically because my lumped respiratory chain and my rounded biomass coefficients are not theirs; the anaerobic figure agrees closely. If you reproduce this with COBRApy on the real core model, expect the tracker's numbers, and treat a mismatch as a difference in medium, bounds or objective before suspecting the solver — which is chapter 38's rule as well.

The anaerobic solution closes its redox balance exactly as part 1 chapter 6 said it must: ethanol at 8.9, acetate at 5.4 and formate at 16.5 mmol·gDW^{-1}·h^{-1}, with pyruvate formate-lyase carrying 16.5. The fourfold drop in growth is the ATP yield difference, and the model reproduces it without being told anything about fermentation.

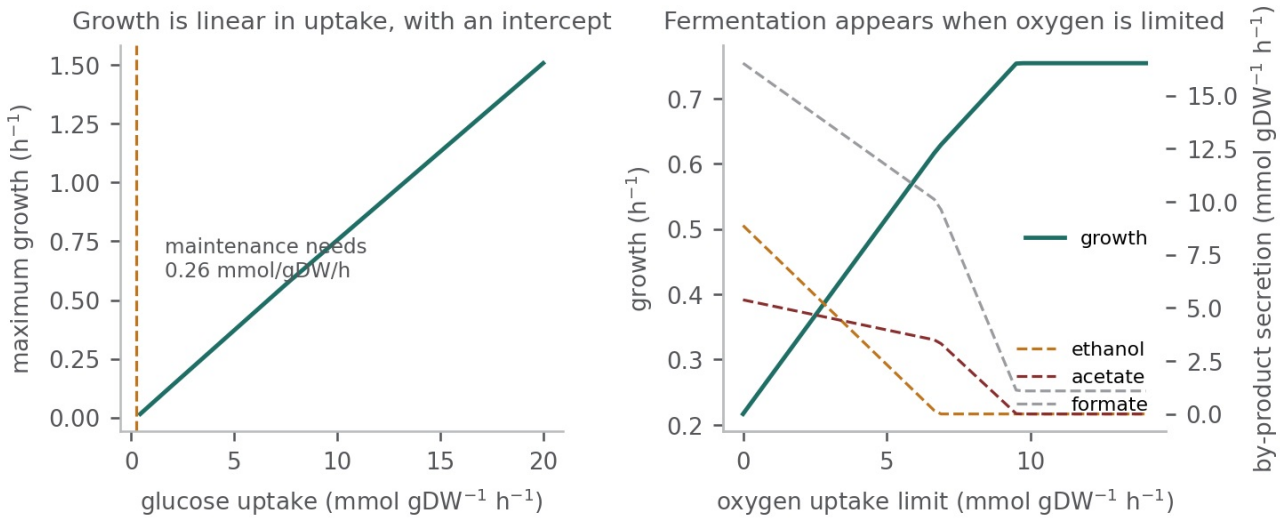


Figure 35.1 Left: maximum growth against the glucose uptake bound; the line is straight and meets zero at 0.26 mmol·gDW^{-1}·h^{-1}, the uptake that pays maintenance alone. Right: growth and by-product secretion as the oxygen bound is raised, showing the fermentative to respiratory transition. Computed with the hand-built model described in the text.

What the uptake sweep shows

Growth against glucose uptake is a straight line of slope 0.0754 gDW per mmol — the yield — and it crosses zero at 0.26 mmol·gDW^{-1}·h^{-1}. That intercept is the maintenance demand: the first fraction of a millimole of glucose buys no growth at all because 8.39 mmol of ATP must be paid first. Convert the slope and you get a biomass yield of **0.42 g per g of glucose**, which is in the region of measured aerobic yields for *E. coli* — a useful check that the biomass coefficients and the P/O assumptions are not wildly wrong.

The linearity is worth pausing on, because it is a property of linear programming and not of biology: with every bound except glucose non-binding, doubling the substrate doubles everything. Real cells deviate — overflow metabolism at high uptake (part 1 chapter 6), which plain FBA cannot predict because it has no proteome constraint, as part 4 chapter 51 explained.

The oxygen sweep, and a transition the model gets right

Raising the oxygen bound from zero to saturation walks the model through the fermentation-to-respiration transition:

O₂ uptake limit	Growth (h⁻¹)	Acetate	Ethanol	Formate
0	0.217	5.4	8.9	16.5
2	0.337	4.8	6.3	14.6
5	0.517	4.0	2.3	11.7
8	0.683	1.9	0	6.0
10 and above	0.754	0	0	1.1

Ethanol disappears first, then acetate, and growth saturates once oxygen stops limiting. Two honest observations. The model secretes a little formate even aerobically, because FBA knows nothing about pyruvate formate-lyase being oxygen-sensitive — a curation point, and the standard fix is to bound that reaction to zero in aerobic simulations. And the transition is a consequence of the redox balance alone: with limited oxygen the cell cannot reoxidise all its NADH through respiration, so it must dispose of electrons in a reduced product. No regulation is involved.

Deletions, and what a lethal prediction means

Knocking out each reaction in turn — setting its bounds to zero — gives ten lethal deletions out of thirty-nine:

Lethal: glucose and ammonium exchange, the glucose transporter, phosphofructokinase–aldolase, glyceraldehyde-3-phosphate dehydrogenase, enolase, transketolase, citrate synthase–isocitrate dehydrogenase, PEP carboxylase, glutamate dehydrogenase. **Growth-reducing:** oxygen exchange and NADH dehydrogenase to 0.22, phosphoglucose isomerase to 0.51, the transhydrogenase to 0.67, pyruvate dehydrogenase to 0.71.

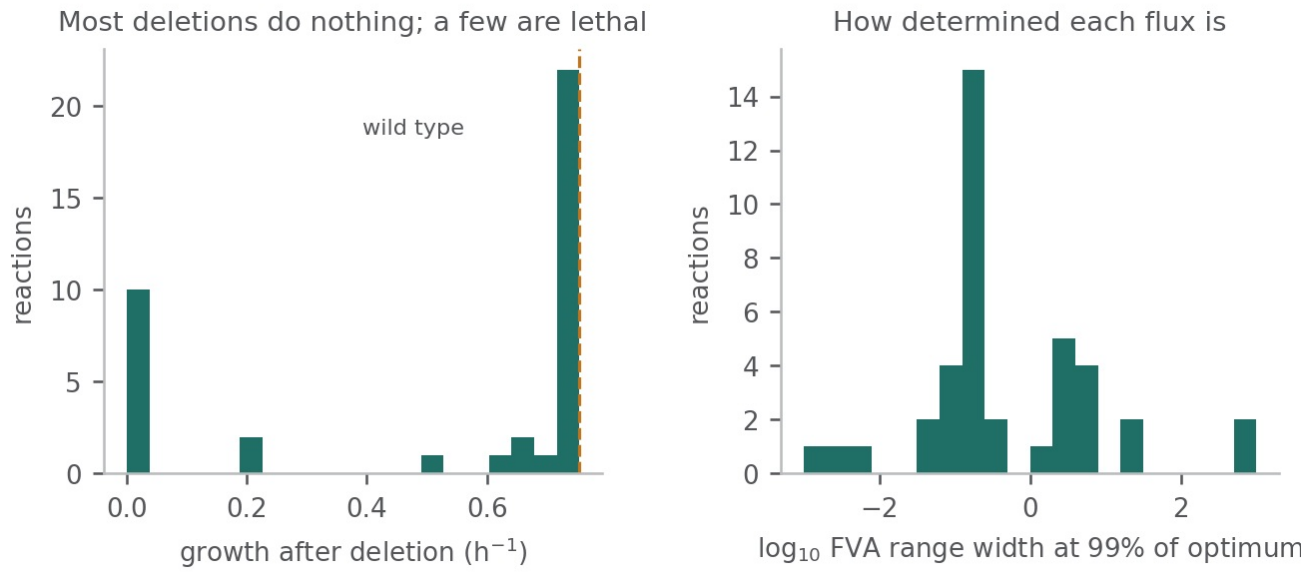


Figure 35.2 Left: the distribution of growth rates after single reaction deletions — most are silent, a few are lethal. Right: the distribution of flux-variability range widths at 99% of optimal growth, on a log scale; the two reactions at 1,000 are the loop of chapter 36. Computed for this module.

Check two against the biology, as the week asks. **Enolase** is lethal because there is no bypass from 3-phosphoglycerate to phosphoenolpyruvate in this network, and the tracker names exactly this class — the glycolytic trunk without a parallel route. **Phosphoglucose isomerase** is not lethal but costs a third of the growth rate, because carbon is forced through the oxidative pentose phosphate pathway and back, losing a carbon as CO_2 per turn — which is part 1 chapter 5's NADPH-versus-carbon trade appearing as a number. **Transketolase** is lethal in my model for a reason worth noticing: it is the only source of erythrose-4-phosphate, which biomass requires. In a real cell there are other routes, so this is a model artefact of the kind the tracker warns about — if a deletion you expected to be survivable is lethal, look for the route the organism has and the model does not, and vice versa.

Flux variability, and reading a wide range

Flux variability analysis fixes the objective at a fraction of its optimum and then minimises and maximises every flux in turn — two extra linear programmes per reaction. At 99% of optimal growth, the widest ranges in this model are:

Reaction	Minimum	Maximum	Reading
Succinate dehydrogenase	0	1,000	A loop — chapter 36
Fumarate reductase	0	1,000	The other half of the same loop
Transhydrogenase (NADH→NADPH)	7.5	26.0	Must carry flux, amount undetermined
Soluble transhydrogenase	0	18.1	Optional, and compensates the first
Maintenance ATP	8.4	14.4	At 99% of optimum, spare ATP can be burned
PEP carboxylase	6.4	12.3	Partly set, partly free

What a wide range means is precisely this: **the data and the objective do not determine that flux.** A reaction whose range spans zero may or may not carry flux in a solution that is just as optimal, so quoting its value from a single FBA solution is quoting an arbitrary choice. The honest way to report an FBA flux map is with its variability, which is why part 2 chapter 13 called flux variability analysis the honest way to report an FBA result.

Note the two transhydrogenases, which have wide ranges that compensate each other: the model is free to run NADH→NADPH and NADPH→NADH at various levels because only the net matters. That is the same structure as the loop below, in a milder form, and it is why *coupled* variability — how fluxes co-vary — is sometimes more informative than individual ranges.

Even at exactly 100% of the optimum, fourteen reactions still have a non-zero range, which means this model has alternative optima: several distinct flux distributions give identical maximum growth. Picking one and interpreting it biologically is the commonest misuse of FBA.

Shadow prices: what another millimole is worth

The dual variables of the equality constraints are the shadow prices — the increase in the objective per unit increase in

the availability of each metabolite. For this model at the aerobic optimum, glucose's shadow price is **0.0754 gDW per mmol**, which is exactly the slope of the uptake sweep. Oxygen's shadow price is **zero**, because oxygen is not binding: there is as much as the model can use, so another millimole buys nothing.

That contrast is the lesson. A shadow price of zero identifies a constraint that is not limiting, so effort spent relaxing it is wasted; a large shadow price identifies the resource to buy more of. Part 5 chapter 13 made this argument in the abstract; here it is one line of code — `res.eqlin.marginals` — and it tells you, for instance, that in an oxygen-limited process the shadow price of oxygen is what another unit of transfer capacity is worth in growth or product, directly comparable with the cost of supplying it.

The programme	max $c^T v$ s.t. $S \cdot v = 0$, $lb \le v \le ub$
Degrees of freedom	40 reactions − rank 29 = 11, closed by bounds, maintenance and the objective
Growth here	0.754 h^{-1} aerobic, 0.217 anaerobic (core model: ~0.87 and ~0.21)
Yield	slope 0.0754 gDW per mmol glucose = 0.42 g/g; intercept 0.26 mmol is maintenance
FVA	two LPs per reaction at a fixed objective; a wide range means undetermined
Shadow price	dual of the metabolite balance; zero means the constraint is not binding

Your FBA says a deletion is lethal and the experiment says the mutant grows. Give the four explanations in the order you would check them.

First, the medium: the model's exchange bounds may not match what the mutant was actually given, and a component present in the real medium — an amino acid, a vitamin — can bypass the blocked step entirely. This is the commonest cause and it costs nothing to check. Second, a missing route in the model: the organism has an isozyme, a bypass or a promiscuous enzyme that the reconstruction lacks, which is what my transketolase result above would be in a real cell; the test is to look for the compound's alternative route in a database and add it. Third, the gene–protein–reaction rule: a deletion of one gene may not remove the reaction if an isozyme shares it, so the lethality may be of the reaction and not the gene. Fourth, the biomass composition: if biomass demands a component the mutant can do without under those conditions — a specific lipid or cofactor whose coefficient is carried over from another strain — the lethality is an artefact of the objective.

The ordering matters because the first two are cheap and the fourth is laborious. And the result is worth having either way: a model that predicts a lethality the organism survives has told you there is a route you do not know about, which is a more useful outcome than agreement.

Three things people get wrong

- ‘FBA predicts the flux distribution.’ It returns one optimal vertex; fourteen reactions here have alternative optima at exactly the same growth rate.
- ‘No kinetic parameters means no assumptions.’ The objective, the bounds and the biomass composition are the assumptions, and they decide the answer.
- ‘The model says this deletion is lethal.’ It says so given this medium and this biomass equation. Check both before believing it.

Self-check — answer on paper first

1. Why does FBA need no kinetic parameters, and what does it therefore lose?

Because it imposes only the steady-state mass balances and capacity bounds, which are properties of the network's stoichiometry, and chooses among the resulting solution space by optimising a linear objective — none of which refers to rates as functions of concentration. What it loses is every question whose answer depends on concentration or time: it cannot predict a metabolite pool, a transient, a regulatory response, an intermediate's accumulation or the effect of an inhibitor, and it cannot tell you that the cell is actually at the optimum it computes.

2. The glucose shadow price is 0.0754 and the oxygen shadow price is zero. What would you spend money on?

Not oxygen transfer: at this operating point oxygen is not limiting, so another unit of k_La buys nothing in the model. Glucose is the binding resource, so the levers are the uptake rate itself and the efficiency with which the carbon is used. The caution is that the shadow prices are local to this solution — raise glucose far enough and oxygen will become binding, at which point its shadow price jumps from zero, which is exactly what the oxygen sweep shows from the other direction.

3. A flux you care about has an FVA range of −12 to +18 at 99% of optimal growth. What can you report?

That the network and objective do not determine either its magnitude or its direction, so no value from a single FBA

solution should be quoted for it. What you can report is the range itself, and what would narrow it: a measurement of that flux or of a flux coupled to it, a tighter uptake constraint, or a thermodynamic constraint fixing the direction if ΔG′ at physiological concentrations settles it — which is chapter 36's subject.

What to open a book for. Orth, Thiele & Palsson, "What is flux balance analysis?", Nat. Biotechnol. 28:245 (2010) — it is six pages and it is the paper to read before anything else. Then the COBRApy documentation and BiGG Models for the core model and iJO1366. Part 2 chapter 13 of this module is the derivation this chapter runs.

Working with models: media, loops, pFBA, thermodynamics and reproducibility

A thermodynamically impossible cycle can satisfy S·v = 0 perfectly. Finding one in your own model, and then removing it two different ways, is the exercise that teaches what a constraint-based model is and is not.

The object model, and building a network by hand

COBRApy represents a model as objects: a `Model` holding `Reaction`, `Metabolite` and `Gene` objects, with gene–protein–reaction rules linking genes to reactions. Building a reaction means creating it, setting bounds, and adding metabolites with signed coefficients — negative for substrates, positive for products. The code, not run here:

```
from cobra import Model, Reaction, Metabolite
m   = Model('toy')
A   = Metabolite('A', compartment='c'); B = Metabolite('B', compartment='c')
r   = Reaction('v2'); r.bounds = (0, 1000)
r.add_metabolites({A: -1, B: 1})
m.add_reactions([r])
m.objective = 'v4'
sol = m.optimize();  print(sol.objective_value)
```

The exercise is to build part 2 chapter 13's five-reaction toy network this way and check it against the hand solution — and the tracker's note about the usual culprit is right: if the answers differ, a sign convention on one reaction is reversed. My own experience building the 40-reaction model was the same lesson at larger scale, and the fastest way to find such an error is the carbon balance check of chapter 37, which catches a flipped sign immediately.

Media and bounds, and changes that undo themselves

The right way to set a carbon source is through the exchange reactions, which COBRApy exposes as `model.medium`: a dictionary of exchange reaction identifiers to uptake limits. Setting the medium rather than editing bounds by hand matters because it is the medium that defines what the organism is being offered, and because an exchange left open by accident is the commonest source of a model that grows on nothing. Switching between aerobic and anaerobic should be a change to the medium alone — removing oxygen — which in my model reproduces 0.754 and 0.217.

The context manager is the second habit:

```
with model:                      # everything inside is reverted on exit
    model.reactions.PGI.knock_out()
    print(model.slim_optimize())
print(model.slim_optimize())     # back to the wild type
```

Why this matters in a loop over hundreds of deletions is simple and brutal: without it, every knockout accumulates, so the second deletion is tested in the background of the first, the hundredth in the background of ninety-nine others, and the result is a plausible table of numbers that means nothing. When I ran the deletion scan in my own code I copied the bound arrays for each deletion for exactly this reason — `lb.copy()` inside the loop, never a mutation of the originals. The same discipline, by hand.

Loops: a cycle that satisfies the balances and cannot happen

My model contains succinate dehydrogenase (succinate → fumarate + quinol) and fumarate reductase (fumarate + quinol → succinate). Run them together at equal flux and the net conversion is nothing: every metabolite balances, S·v = 0 is satisfied exactly, and the pair can carry any flux at all up to whatever bound you set. The flux variability analysis of chapter 35 found exactly this — **both reactions range from 0 to 1,000**, the arbitrary upper bound, at 99% of optimal growth.

Why it cannot happen is thermodynamic, not stoichiometric. Going round the cycle returns every metabolite to its starting concentration, so the total ΔG around the loop is zero by construction; a net flux around a closed loop with zero driving force would be a perpetual motion machine. This is the class of error that part 2 chapter 14 called an internal cycle, and the point to absorb is that **S·v = 0 does not forbid it**: mass balance is indifferent to direction, and direction is set by free energy.

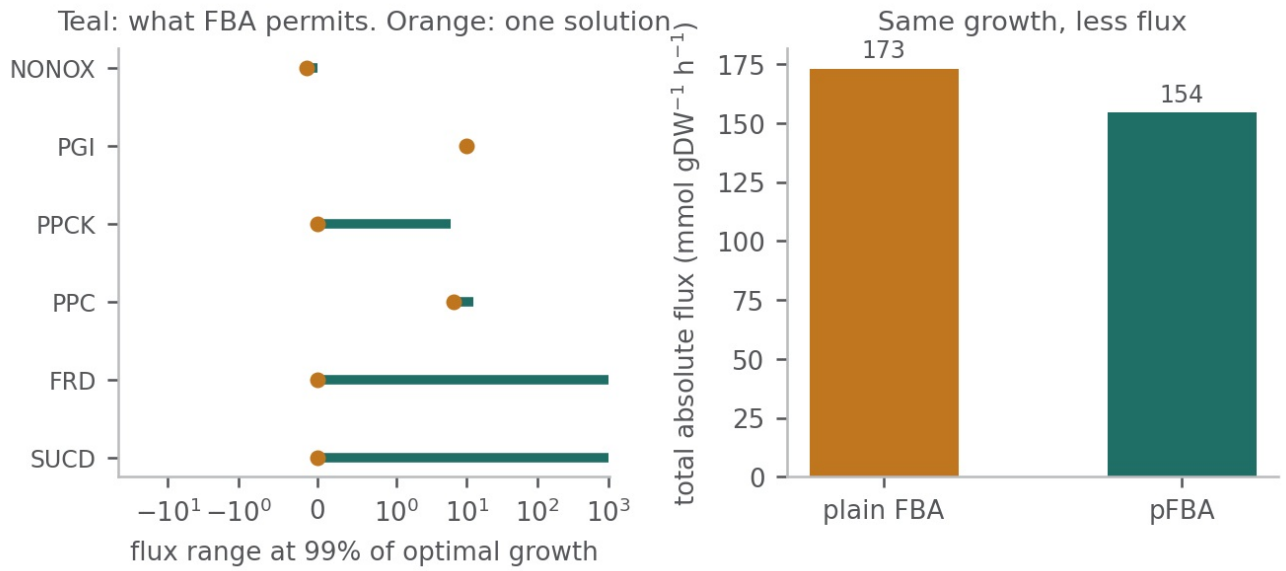


Figure 36.1 Left: for six reactions, the range flux variability analysis permits at 99% of optimal growth (bars, symmetric log scale) against the value one FBA solution happened to return (points). Succinate dehydrogenase and fumarate reductase run to the arbitrary 1,000 bound — the loop. Right: total absolute flux of a plain FBA solution and of a parsimonious one at the same growth rate. Computed for this module.

Two ways to remove it, and what each one is

Loopless FBA adds constraints that forbid net flux around any internal cycle, by requiring the existence of a thermodynamically consistent set of potentials. It is the cheap special case of thermodynamics-based flux analysis: it uses only the fact that a loop's ΔG is zero, with no actual thermodynamic data. The cost is that the problem becomes a mixed-integer programme, so it is much slower than FBA.

pFBA, parsimonious FBA, instead fixes the objective at its optimum and minimises the sum of absolute fluxes — the hypothesis being that the cell does not waste enzyme. It is a second linear programme, with each reaction split into forward and reverse parts so that the absolute value is linear. In my model it brings the total absolute flux from **172.9 to 154.4** at identical growth, and sets both halves of the loop to zero. It also changed the split between pyruvate dehydrogenase and formate-lyase, which is the alternative-optima point again: pFBA chooses among equally optimal solutions by a secondary criterion, which is a modelling assumption and not a fact.

pFBA removes a loop as a side effect of being parsimonious, and it does not *prove* the loop is absent; a loop that happened to lie on a minimal-flux solution would survive. Loopless FBA is the principled removal and pFBA is the cheap one, which is exactly how the tracker frames them.

From ΔG as a diagnostic to ΔG as a constraint

Part 2 chapter 14 used ΔG′ to decide whether a step can run. The step up is to put it into the optimisation as a bound. The procedure, which the week asks you to do with eQuilibrator values for a handful of core reactions:

- Get ΔG°′ for each reaction at the right pH and ionic strength, and compute ΔG′ at the concentration range the cell can hold — conventionally 1 µM to 10 mM, with cofactor ratios fixed, as in part 2 chapter 14.
- Where the sign of ΔG′ is the same across the whole feasible concentration range, the direction is determined: set the reaction's lower bound to zero (or its upper bound, for the reverse case). Where the sign can change, leave it reversible — that is an honest statement that thermodynamics does not decide it.
- Rerun and report what changed. In a well-constrained model the answer is often "nothing to the objective, something to the variability", which is the useful result: eliminating directions narrows FVA ranges and removes spurious solutions without changing the optimum.

Thermodynamics-based flux analysis does this systematically, carrying metabolite concentrations as variables and the ΔG of every reaction as a constraint linked to them. It is more expensive and it needs free energies for everything, which is why the loopless formulation — which needs none — is where most people stop.

The analysis functions, and exporting results

The functions the week lists all return DataFrames, which is the point: they are tables you join to other tables rather than printouts.

Function	What it returns	Watch for
`model.optimize()`	A solution object with fluxes, objective value, shadow prices and reduced costs	Use `slim_optimize()` in loops — it returns only the objective value and is much faster
`model.summary()`	Uptakes, secretions and the objective in a readable block	The fastest sanity check that the medium is what you think
`pfba(model)`	A parsimonious solution	It is a second optimisation with its own assumption
`loopless_solution(model)`	A solution with internal loops removed	Mixed-integer; slow at genome scale
`flux_variability_analysis(model, fraction_of_optimum=0.99)`	Minimum and maximum per reaction	Two LPs per reaction; the fraction is a choice you must report
`single_gene_deletion`, `double_gene_deletion`	Growth per deletion or pair	Gene, not reaction: isozymes mean the two differ

Export both the FVA and the deletion DataFrames to CSV and plot the deletion results as a histogram, as in figure 35.2 — the shape, with a spike at the wild-type rate and a few at zero, is what a deletion scan looks like in any model, and recognising a departure from it is how you spot a broken medium.

SBML, and what survives a round trip

SBML is the interchange format: an XML standard with an FBC extension that carries bounds, objectives and gene–protein–reaction rules. Write with `cobra.io.write_sbml_model`, read back, and compare reaction and metabolite counts, bounds and the objective. I could not test a round trip here — libsbml is not installed in this environment — so what follows is what to check rather than what I observed: counts and identifiers survive; bounds and the objective survive through the FBC extension; **annotations, notes and anything the writer put in free-text fields are the usual casualties**, along with any custom attribute your own script added to a reaction object. JSON round-trips more faithfully between COBRApy versions and is not portable to other tools, which is the trade. SBML is still what everyone uses because it is the only format every platform reads, which is chapter 38's reason for caring.

Escher draws a flux vector on a pathway map: load a core-model map, overlay your solution's fluxes as a dictionary of reaction identifier to value, and export. It is worth doing once for a solution you understand, because seeing where the flux goes catches errors that a table hides — a cycle running at high flux is obvious on a map and invisible in a sorted list.

Reproducibility, and a data management plan

The week's last session is the one that pays later. One notebook that runs top to bottom from a fresh kernel; pinned requirements; any randomness seeded; a README saying what the notebook shows and what the headline numbers are. Then the test that actually tests it: clone the repository into a clean directory and run it there. Everything else is a rehearsal.

A data management plan, which funders now require, fits on a page and answers five questions. **What data** — models, flux solutions, sequencing reads, fermentation time courses, each with its format and rough volume. **Where it lives** — raw data immutable and backed up, processed data derived by scripts that can be rerun, with the rule from part 9 chapter 32 that analysis reads from raw and never writes to it. **How it is named and versioned** — the identifier scheme, and version control for code and models. **What is deposited where on publication** — sequencing to a sequence archive, models to BioModels or a repository with a DOI, code to a tagged release. **How it meets the FAIR principles**: findable, through a persistent identifier and metadata; accessible, through an open repository; interoperable, through standard formats, which for a metabolic model means SBML with proper annotations; and reusable, through a licence and enough documentation for someone else to run it. Writing it before you are asked is cheap practice, and it also forces the identifier decisions early, which is when they are free.

Medium	`set the carbon source through exchange bounds, not reaction bounds`
Temporary change	`with model: – otherwise deletions accumulate through a loop`
Internal loop	`satisfies S·v = 0, forbidden by ΔG = 0 around a cycle; FVA shows ±bound`
pFBA	`fix the objective, minimise Σ|v|; here 172.9 → 154.4 at the same growth`
Thermodynamic bound	`if sign(ΔG′) is fixed over the feasible concentration range, fix the direction`
Round trip	`counts, bounds and objective survive SBML; annotations and notes often do not`

Your genome-scale model predicts a respectable growth rate and, on inspection, carries 500 mmol·gDW^{-1}·h^{-1} of flux around a three-reaction cycle. Does the growth prediction need to be thrown away?

No — but the flux map does. The growth rate is the objective value, and a loop carrying flux does not change it: the loop has no net conversion, so it neither consumes nor produces anything the biomass reaction needs. What is wrong is the *solution*, one of many optimal vertices, and any interpretation of individual fluxes from it. So the growth number stands, and every statement of the form "flux goes this way through that branch" has to be redone.

What to do, in order: run pFBA, which will almost certainly set the loop to zero and gives a flux map that is at least parsimonious; run flux variability analysis to see which other fluxes are undetermined, since a model with one loop usually has several; and if the loop involves reactions whose direction is thermodynamically decidable, fix those directions as bounds, which removes the loop permanently rather than solution by solution. Then check whether the loop is telling you about a curation error — a reaction duplicated in both directions, or a reaction whose reversibility was set to the default rather than considered, which is where most loops come from.

Three things people get wrong

- ‘If it satisfies the mass balances it is possible.’ A closed cycle balances perfectly and is thermodynamically forbidden.
- ‘pFBA is the correct solution.’ It is one choice among equally optimal ones, made by a secondary assumption about minimal enzyme.
- ‘I knocked out the genes in a loop over the list.’ Without a context manager the knockouts accumulate, and every row after the first is wrong.

Self-check — answer on paper first

1. Explain in two sentences why a thermodynamically impossible cycle satisfies S·v = 0.

Because S·v = 0 states only that each metabolite's production equals its consumption, and in a closed cycle every intermediate is both produced and consumed in equal measure, so the balance holds for any flux magnitude. Direction and feasibility come from free energy, which the stoichiometric matrix does not contain: going once round a cycle returns every concentration to its starting value, so the total ΔG is zero and a net flux would require work from nothing.

2. You need a flux map for a paper. Which analyses do you run, and what do you report?

FBA for the objective value; then pFBA or loopless FBA for a flux map without internal cycles; then flux variability analysis at a stated fraction of the optimum so that each reported flux carries a range. Report the objective, the fluxes from the parsimonious solution, and the FVA ranges alongside them, with the medium, the uptake bounds and the objective stated explicitly — and say which reactions had ranges wide enough that their values should not be interpreted.

3. Why is setting `model.medium` safer than editing reaction bounds directly?

Because the medium is the complete statement of what the organism is offered, so setting it makes every uptake explicit and any exchange you did not mention is closed; editing individual bounds leaves the rest at whatever they were, which is how a model ends up quietly consuming a nutrient you never intended to supply. It is also self-documenting: the medium dictionary printed in a notebook is the experiment's definition, whereas a sequence of bound edits has to be reconstructed by reading the code.

What to open a book for. The COBRApy documentation's building-a-model, simulating, deletions and FVA pages, which are the curriculum for this week, and its flux-analysis guide for pFBA and loopless FBA. BiGG Models for the core model and iJO1366 in both SBML and JSON, and Escher for the map. Part 2 chapter 14 is the chapter to reread before the thermodynamics section.

Reconstruction and quality control: drafts, gap filling and what a check detects

A quality report is only useful if you can say which failures would change a flux prediction and which are metadata. The way to learn that is to run the checks on a model you built yourself and see what they catch.

Where models come from

Manual reconstruction, as the Thiele and Palsson protocol describes it, is roughly a year of work: draw the draft from the genome annotation, then go reaction by reaction through the literature, fixing stoichiometry, cofactor usage, directionality, compartment assignment and gene associations, building a biomass equation from measured composition, and testing against growth phenotypes. The steps that cannot be automated are the ones that require judgement about evidence: deciding whether a paper's enzyme assay in a related organism justifies adding a reaction, choosing a direction when the thermodynamics are marginal, and deciding what the biomass equation should contain for *this* organism in *these* conditions.

Which is why automation exists. CarveMe starts from a universal reaction database and *carves* a species model out of it using genome evidence as scores — a top-down approach that produces a model that grows, fast. gapseq works bottom-up from annotation and pathway evidence, assembling reactions whose presence is supported and then filling gaps. A draft from either takes minutes to hours instead of a year and is correspondingly less trustworthy. And the first thing to check for your own organism is whether a curated model already exists in BiGG; for most non-model organisms it does not, which is the situation these tools are for.

Scale sets expectations. My hand-built core model has 40 reactions; an automated draft of a bacterial genome typically has of order a thousand to fifteen hundred; a curated genome-scale model of *E. coli* has around 2,600. A core model answers questions about central carbon and nothing else; a draft answers questions about capability; a curated model is the only thing that should be trusted for a gene-level prediction.

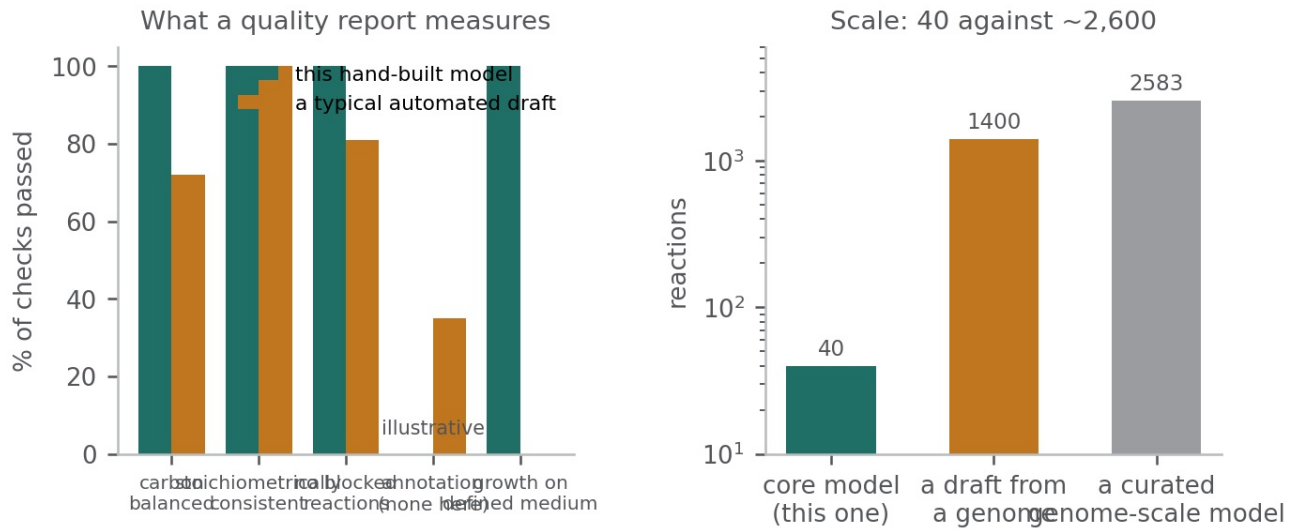


Figure 37.1 Left: the categories a quality report covers, with the results for the hand-built model of this part and illustrative values for a typical automated draft — the draft numbers are indicative, not measured, and are marked so. Right: reaction counts at three levels of reconstruction. Computed and drawn for this module.

Running the checks on my own model

MEMOTE scores a model on stoichiometric consistency, mass and charge balance, annotation coverage and SBML conformance, and produces an HTML report. I could not run it here, so I implemented the three checks that can change a prediction and ran them on the 40-reaction model. The results were not all what I expected, which is the point of the exercise.

Mass balance. Assign each metabolite its carbon count and check that every internal reaction's coefficients sum to zero in carbon. All **28 of 28** internal reactions balance, and the biomass reaction drains **70.87 carbons per unit of biomass** — which is not a failure but the definition of a sink, and is worth computing because it gives the carbon-limited ceiling directly: 10 mmol of glucose is 60 mmol of carbon, so no more than 0.85 h^{-1} of growth is possible however

efficient the energetics, against the 0.754 the model achieves. A real MEMOTE run checks full elemental formulas and charge, which catches protons and water that a carbon-only check misses.

Stoichiometric consistency asks whether there exists a strictly positive mass vector m with $S^{T}m = 0$ over the internal reactions — that is, whether a consistent set of molecular masses can be assigned so that every reaction conserves mass. **My model fails it.** The reason is instructive and not a mistake in the usual sense: ATP, NADH, NADPH and quinol are carried as bare currency metabolites with no ADP, phosphate or oxidised partner, so a reaction like NADH + ½O_2 → 2.5 ATP + H_2O cannot conserve mass under any assignment. Restricting the check to the carbon metabolites and excluding the currencies, the model **passes**, and the mass vector the solver returns is literally the carbon count of each metabolite — 6 for glucose and glucose-6-phosphate, 3 for pyruvate, 2 for acetyl-CoA, 5 for α-ketoglutarate, 1 for CO_2. That is a pleasing confirmation that the carbon skeleton is consistent, and a clear statement of what the lumping cost: a model like this is fit for carbon and energy accounting and would fail a real quality report on consistency grounds.

Blocked reactions — reactions that cannot carry flux anywhere in the feasible space, found by maximising and minimising each one without fixing the objective. The check found exactly one: a phosphate exchange reaction left behind when I removed phosphate from the model's bookkeeping. That is a genuine find in my own work, and it is the typical kind: a vestigial reaction from an edit, harmless to the predictions and a sign that the model and the edit history have diverged. In a draft model, blocked reactions are usually in the hundreds, and they mark pathways whose other half is missing — the gaps that gap filling is for.

Gap filling is a hypothesis, demonstrated

Automated gap filling adds reactions from a universal database until biomass can be produced, choosing the smallest set that does it. To see what that means, I deleted enolase from the model — which chapter 35 found lethal — and offered a candidate pool of six reactions: the true gap, a plausible shortcut (3-phosphoglycerate straight to pyruvate), PEP synthetase, malic enzyme, a decoy citrate lyase and acetate uptake.

Candidate added	Growth restored
Enolase — the true gap	**0.754 h^{-1}**, the original rate
Each of the other five, alone	0 — still infeasible
PEP synthetase *and* the 3-phosphoglycerate-to-pyruvate shortcut, together	**0.723 h^{-1}**

So a minimal-set gap filler, given this pool, would have two defensible answers: one reaction that is correct, and a pair that restores 96% of the growth by a route the organism may not have at all. If the true gap happened to be missing from the universal database — which for a non-model organism's unusual chemistry it often is — the filler would return the pair, the model would grow, and the model would be wrong in a way that no further simulation could reveal.

Hence the discipline the week asks for: list the reactions gap filling added, look each one up for evidence in *this* organism — a gene with a credible annotation, an enzyme assay, a growth phenotype — and treat every one without evidence as **a prediction to test, not a finding**. A gap-filled reaction is a claim that the organism has an enzyme; it is testable by assay, by growth on a diagnostic substrate, or by looking harder at the genome, and reporting the list is what lets a reader judge the model.

The biomass objective, and how wrong it can be

The biomass equation is a set of measured coefficients: macromolecular composition from a dry-weight analysis, amino acid and nucleotide fractions from the protein and nucleic acid content, lipid from the membrane composition, and the ATP requirement from polymerisation costs plus a fitted growth-associated maintenance term. For a non-model organism, almost none of that has been measured, and the universal practice is to copy a related organism's equation.

How much that matters depends on the question. For a **yield ceiling** it matters little: the carbon content of biomass is similar across bacteria, so the ceiling moves by a few per cent. For an **ATP-limited prediction** it matters a great deal, since the growth-associated maintenance coefficient is fitted and can differ twofold. And for a **gene-essentiality prediction** it is decisive, because essentiality is usually decided by whether a specific biomass component can be made — so a component that this organism does not actually require makes its synthesis pathway spuriously essential, and a component it does require but the equation omits makes a real essential gene look dispensable. My transketolase result in chapter 35 is the first of those failure modes in miniature.

Which failures change a prediction

Finding	Changes a flux prediction?	What to do
Mass or charge imbalance in a reaction carrying flux	Yes — it can create or destroy matter, and FBA will exploit it	Fix it; this is the first thing to repair

Stoichiometric inconsistency	Yes, if it permits a mass-generating cycle	Find the offending reactions; exempt deliberate currency lumping and say so
Reaction reversible when it should not be	Yes — loops and impossible routes	Fix directions from thermodynamics (chapter 36)
Blocked reactions	No, by definition — but they mark gaps	Investigate the pathway they belong to; delete vestigial ones
Missing or wrong annotations and identifiers	No	Fix before publication: it is what makes the model reusable, not what makes it right
SBML conformance warnings	No	Fix, so other tools can read it
No growth on a medium the organism grows on	Yes — the model is incomplete	Find whether the medium definition or the network is at fault, in that order

The three I would fix in my own model, with reasons: the stoichiometric inconsistency, by documenting the currency lumping explicitly in the model's notes so that nobody mistakes it for an error, since fixing it properly means adding ADP, phosphate and the oxidised cofactors; the blocked phosphate exchange, by deleting it, which takes a second; and the aerobic formate secretion from chapter 35, by bounding pyruvate formate-lyase to zero in aerobic simulations, which is a real curation decision and should be recorded with its justification. The ones I would leave: annotations, which this model does not have and does not need for a teaching purpose, and the absence of compartments, which is a deliberate scope limit.

When is a draft good enough?

For a **yield ceiling or a stoichiometric feasibility question**, often yes: those depend on the network's carbon and cofactor topology, which a draft usually gets right for central metabolism, and the answer is robust to peripheral errors. For a **gene-deletion prediction**, usually not: essentiality depends on isozymes, on gene–protein–reaction rules, and on the biomass equation, all of which are exactly what a draft gets wrong. In between — a flux map, a by-product prediction, a medium requirement — judge by whether the prediction depends on the parts of the model that were curated or the parts that were inferred, and say which in the paper.

Scale	`core ~40 reactions · draft ~1,000–1,500 · curated ~2,600`
Mass balance	`Σ coefficient × element count = 0 for every internal reaction`
Consistency	`∃ m > 0 with Sᵀm = 0; currency lumping breaks it by design`
Blocked	`max |v| = 0 over the whole feasible space; marks gaps, not errors`
Gap filling	`smallest set restoring biomass; here one right answer and one wrong pair at 96% of growth`
Good enough for	`yield ceilings often; gene essentiality rarely`

Your draft model of a non-model bacterium does not grow on the defined medium the organism grows on. How do you find out whether the medium or the model is at fault?

Separate the two by making the medium generous. Open every exchange reaction — let the model take up everything — and optimise. If it still does not grow, the network is at fault and the problem is a gap in biomass precursor synthesis; find which precursor by setting the biomass reaction aside and maximising the production of each component in turn, which localises the gap to one branch. If it does grow with everything open, the medium definition is at fault, and the way to find the missing component is to add the defined medium's exchanges back one at a time, or to remove them from the open medium one at a time until growth fails.

Then the two causes of a medium problem, which need different fixes: an exchange reaction that does not exist in the model for a nutrient that is in the medium, which is a naming or an annotation problem; and a transport reaction missing between the extracellular and cytosolic compartments, which is a real gap and a common one, since automated reconstruction handles transport worse than it handles metabolism. The second is worth checking first in any draft that will not grow.

Three things people get wrong

- ‘The model passed quality control, so it is right.’ The checks are necessary, not sufficient: consistency, balance and annotation say nothing about whether the organism has these reactions.
- ‘Gap filling completed the model.’ It proposed the smallest set that makes biomass producible; in my test a wrong pair restored 96% of growth.
- ‘A draft is fine for predicting deletions.’ Essentiality is the prediction most sensitive to isozymes, gene rules and biomass composition — the three things a draft gets wrong.

Self-check — answer on paper first

1. Why does a carbon-balance check pass on a model that fails stoichiometric consistency?

Because the carbon check only verifies that carbon is conserved reaction by reaction, while consistency asks for a single positive mass assignment that makes *every* element conserved simultaneously across the whole network. A model that tracks ATP or NADH as bare currency metabolites — with no ADP, phosphate or oxidised partner — conserves carbon perfectly and cannot conserve mass, because those currencies appear on one side of reactions with nothing to balance them. The practical reading is that a consistency failure is only a bug if it is unintentional; deliberate lumping should be documented so that a reader can tell the difference.

2. A gap filler adds three reactions to your draft. What do you do before using the model?

Look each one up for evidence in this organism: a gene with a credible annotation, a characterised enzyme in a close relative, or a growth phenotype that implies the activity. Report all three in the paper as gap-filled, with the evidence or its absence. Then test whether your conclusions depend on them — delete each and see whether the prediction you care about survives — because a conclusion that rests on an unsupported gap-filled reaction is a conclusion about the filler's database, not about the organism.

3. Which quality-report failures would you fix before computing a theoretical yield, and which would you leave?

Fix anything that lets the model create or destroy matter or energy: mass and charge imbalances in reactions that can carry flux, inconsistencies that permit a mass-generating cycle, and reversibility errors that open thermodynamically impossible routes — all of these can inflate a yield. Leave annotation coverage, identifier mapping and SBML conformance, which affect reusability and not the number; leave blocked reactions, which cannot carry flux by definition; and leave compartment metadata unless the yield depends on transport. Then state the biomass equation's provenance, because for a yield the carbon content of biomass is the assumption that matters most.

What to open a book for. Thiele & Palsson, Nat. Protoc. (2010) for what manual reconstruction involves — skim it to understand why automation exists. Machado et al., Nucleic Acids Res. (2018) for CarveMe, Zimmermann et al., Genome Biol. (2021) for gapseq, and Lieven et al., Nat. Biotechnol. (2020) for MEMOTE, read for what the checks are rather than how to run them.

MATLAB and the COBRA Toolbox: reading other people's code

Half the metabolic modelling literature ships MATLAB. You do not need to like the language; you need to be able to read a script, run it, change one thing, and reconcile its answer with yours.

Why this week exists

The COBRA Toolbox came first, and a great deal of published strain-design work — OptKnock and its descendants in week 39, ^{13}C flux analysis packages in week 43, many supplementary scripts — exists only as MATLAB. Being unable to read it means being unable to reproduce it. This chapter is written for someone who already writes Python, so it is organised as a translation rather than a tutorial. I could not install MATLAB or Octave in this environment, so every snippet here is **not run**: treat them as the shape of the code, and check the function signatures against the current Toolbox documentation.

MATLAB for a Python user

	Python / NumPy	MATLAB
Indexing	`v[0]`, zero-based, `v[-1]` for the last	`v(1)`, one-based, `v(end)` for the last
Default type	Arrays are objects; scalars are scalars	Everything is a matrix; a scalar is 1×1
Element-wise	`a * b` is element-wise; `a @ b` is matrix product	`a .* b` is element-wise; `a * b` is matrix product
Solve A·x = b	`np.linalg.solve(A, b)`	`A \ b`
Null space	`scipy.linalg.null_space(S)`	`null(S)`
Least squares	`np.linalg.lstsq(A, b, rcond=None)`	`A \ b` (same operator, overdetermined)
Structured data	dict, DataFrame	struct, cell array, table
Functions	Any file, any number per file	A script runs; a function needs its own file or a `function` block at the end
Stiff ODEs	`solve_ivp(..., method='LSODA')`	`ode15s` (`ode45` for non-stiff)

Two traps beyond syntax. **Row against column vectors**: MATLAB distinguishes 1×n from n×1 and many functions require one or the other, so a dimension mismatch is the first thing to check when a script fails. And **copy-on-write**: MATLAB passes by value, so modifying a model struct inside a function does not change the caller's copy — the opposite of Python's behaviour with mutable objects, and the reason Toolbox functions return a modified model rather than mutating it. That difference is also a small mercy: it makes MATLAB's equivalent of chapter 36's context-manager problem less dangerous, because `changeRxnBounds` returns a new struct instead of editing the original in place.

The translation exercise the week sets is the part 2 chapter 12 linear-algebra script. The same four calls, the same answers:

```
% MATLAB — not run here
S    = [ 1 -1 -1  0  0 ;  0  1  0 -1  0 ;  0  0  1  0 -1 ];
rankS = rank(S);              % 3
N     = null(S);              % 5x2 basis of the null space
v     = A \ b;                % in place of np.linalg.solve(A, b)
fprintf('rank %d, nullity %d\n', rankS, size(N,2));
```

The Toolbox, and the same analyses under other names

Installation is `initCobraToolbox`, which also chooses a solver; with no Gurobi academic licence, `changeCobraSolver('glpk','LP')` is the fallback and is adequate for everything in these weeks. Loading a model gives a struct whose fields are the linear programme of chapter 35, which is worth inspecting once because it makes the correspondence concrete: `S` the stoichiometric matrix, `lb` and `ub` the bounds, `c` the objective vector, `rxns`, `mets` and `genes` the names, `rules` or `grRules` the gene–protein–reaction logic.

COBRApy	COBRA Toolbox
`cobra.io.read_sbml_model(path)`	`readCbModel(path)`

`model.optimize()`	`optimizeCbModel(model)`
`model.objective = 'RXN'`	`changeObjective(model, 'RXN')`
`model.reactions.RXN.bounds = (lo, hi)`	`changeRxnBounds(model, 'RXN', lo, 'l')`
`single_gene_deletion(model)`	`singleGeneDeletion(model)`
`flux_variability_analysis(model, fraction_of_optimum=0.99)`	`fluxVariability(model, 99)`
`cobra.io.write_sbml_model(model, path)`	`writeCbModel(model, 'fileName', path)`

One difference catches everyone: `fluxVariability` takes the fraction as a **percentage** (99) where COBRApy takes a fraction (0.99). A result that looks absurdly tight or loose compared with your Python run is usually this.

The reconciliation exercise is the valuable one. Put the COBRApy and Toolbox results side by side and account for every difference. In order of likelihood: the **medium** — a different default for an exchange bound, especially oxygen; the **objective** — different biomass reaction variants exist in the same model file, one with and one without the growth-associated maintenance term; a **bound** — the glucose uptake default differs between distributions of the same model; and only then the **solver**, which matters for which alternative optimum is returned but not for the objective value. The rule to carry: a mismatch in the objective value is a model difference, and a mismatch in individual fluxes with the same objective value is alternative optima.

ODEs, and where Simulink fits

For the kinetic model of part 2 chapter 16, the translation is direct. MATLAB's stiff solver is `ode15s` and its non-stiff workhorse is `ode45`, corresponding to LSODA and RK45; the stiffness argument of part 2 chapter 12 applies identically, and a metabolic model should go to `ode15s` for the same reason it goes to LSODA.

```
% MATLAB — not run here
function dc = rhs(~, c, p)
    v  = rates(c, p);
    dc = [v(1)-v(2); v(2)-v(3); v(3)-v(4)];
end
opts = odeset('RelTol', 1e-10, 'AbsTol', 1e-12);
[t, y] = ode15s(@(t,c) rhs(t, c, p), [0 20000], [0.1; 0.1; 0.1], opts);
```

Note the column vectors — `[0.1; 0.1; 0.1]`, not `[0.1, 0.1, 0.1]` — which is the row-against-column trap in its most common form. The check the week asks for is that the steady state and the flux control coefficients agree to three decimal places with the Python version; if they do not, the suspects are the initial condition's shape, a transcription error in a rate law, and the tolerances, in that order.

Simulink is a block-diagram environment for dynamic systems: you assemble a model from blocks representing integrators, transfer functions, gains and feedback paths, and simulate it. Process and control engineers reach for it because a control loop is naturally a diagram — sensor, controller, actuator, plant — and because it connects to hardware-in-the-loop testing and code generation for real controllers. For metabolic modelling it is rarely used, and knowing what it is for is enough; the one place it genuinely fits is the dynamic-control analysis of part 7 chapter 24, where the question is a loop's stability and settling time.

Crossing platforms deliberately

Write a model to SBML from one platform, read it in the other, and compare reaction counts, metabolite counts, bounds, objective and gene rules. What to expect, with the caveat from chapter 36 that I could not test it here: counts, bounds and the objective survive, carried by the FBC extension; gene rules usually survive but their formatting differs, so a string comparison will show spurious differences; annotations, notes and anything stored in free-text fields are the casualties. The direction that is safer is generally the one with fewer custom attributes — a model freshly read from BiGG travels well in either direction, and a model that has been manipulated by a script with its own conventions does not.

Committing both versions of the same analysis with a README explaining why both exist is a real portfolio item: it says you can work in the toolchain the field actually uses, which includes code older than you are.

Indexing	`MATLAB is one-based; end is the last element`
Solve	`A \ b for both determined and least-squares systems`
FVA argument	`Toolbox takes 99 (per cent), COBRApy takes 0.99 (fraction)`
Stiff ODEs	`ode15s ≈ LSODA; column vectors for initial conditions`
	`objective differs → model difference; fluxes differ at equal objective → alternative`

Mismatch rule	`optima`

A published MATLAB script reports a growth rate you cannot reproduce in COBRApy with the same model file. Work through it.

First, print both media. `model.summary()` in COBRApy and the exchange bounds from the Toolbox struct — the most common cause is a different default on oxygen or on glucose uptake, and the second most common is an exchange the script opened with `changeRxnBounds` somewhere earlier in the file. Second, check which reaction is the objective: many model files contain two biomass reactions, with and without the growth-associated maintenance term, and they give noticeably different growth rates. Third, check the maintenance reaction's lower bound, which some distributions set to zero and others to the fitted value.

If the objective values then agree but individual fluxes do not, stop worrying: that is alternative optima, and the two solvers have picked different vertices of the same optimal face. Confirm it by running pFBA in both, which selects among them by a stated criterion and should bring them close. If the objective values still differ after the medium, the objective and the maintenance bound match, suspect the model file itself — compare reaction counts, and check whether one platform silently dropped reactions on import, which is the SBML round-trip problem from the other direction.

Three things people get wrong

- ‘The solvers disagree.’ They almost never do on the objective value. Check the medium, the objective reaction and the maintenance bound first.
- ‘`fluxVariability(model, 0.99)`.’ The Toolbox wants 99. A tiny argument difference, a wildly different answer.
- ‘I will avoid MATLAB.’ Then you cannot reproduce a large fraction of the strain-design and flux analysis literature.

Self-check — answer on paper first

1. Translate: `np.linalg.solve(A, b)`, `null_space(S)`, `v[-1]`, `a * b` element-wise.

`A \ b`; `null(S)`; `v(end)`; `a .* b`. And note that `A \ b` also covers the over-determined case that needs `lstsq` in NumPy, which is convenient and occasionally hides the fact that you are solving a least-squares problem rather than an exact one.

2. Your Toolbox and COBRApy runs give the same growth rate and different acetate secretion. What is happening and how do you confirm it?

Alternative optima: the optimal face of the linear programme contains more than one vertex, and the two solvers have returned different ones, which chapter 35 found for fourteen reactions in a 40-reaction model. Confirm it by running flux variability analysis on the acetate exchange at 100% of the optimum in either platform — if its range spans both values, the difference is a property of the model and not of the software. Then, if you need a single reportable number, use pFBA in both, and report the range alongside it.

3. Why does modifying a model inside a MATLAB function not affect the caller's model, and what is the Python equivalent danger?

Because MATLAB passes arguments by value with copy-on-write, so a function receives its own copy and must return the modified struct for the change to persist — which is why Toolbox functions are written as `model = changeRxnBounds(model, ...)`. The Python danger is the mirror image: COBRApy model objects are mutable and passed by reference, so a function that knocks out a reaction changes the caller's model permanently, which is exactly the accumulating-knockout bug of chapter 36 and the reason for the `with model:` context manager.

What to open a book for. The COBRA Toolbox documentation and installation guide; MATLAB's getting-started material, read with the translation table above in hand; GNU Octave if you have no licence, which runs most of what you need outside the Toolbox itself; and the MATLAB ODE solver documentation for `ode45` and `ode15s`.

Closing notes to part 10

The four weeks in one page

Week 35. FBA is max $c^{T}v$ subject to $S \cdot v = 0$ and bounds. A 40-reaction core model has 11 degrees of freedom, grows at 0.754 h^{-1} aerobically and 0.217 anaerobically, yields 0.42 g biomass per g glucose, and needs 0.26 $mmol \cdot gDW^{-1} \cdot h^{-1}$ of glucose before it grows at all. Ten of thirty-nine deletions are lethal; the pentose-phosphate detour forced by deleting phosphoglucose isomerase costs a third of the growth rate. Glucose's shadow price is 0.0754 and oxygen's is zero, because oxygen is not binding.

Week 36. Succinate dehydrogenase and fumarate reductase together form a cycle that satisfies the mass balances exactly and cannot happen; flux variability shows both running to the 1,000 bound. pFBA removes it and cuts total flux from 172.9 to 154.4 at identical growth; loopless FBA removes it on principle. Thermodynamics becomes a bound when the sign of $\Delta G'$ is fixed across the feasible concentration range. Use the medium, not bounds, and the context manager, not accumulated knockouts.

Week 37. The checks found three things in my own model: all 28 internal reactions carbon-balanced with biomass draining 70.87 C per unit; stoichiometric consistency failing on the whole model and passing on the carbon subnetwork, where the recovered mass vector is exactly the carbon counts; and one blocked reaction, a vestigial phosphate exchange. Deleting enolase and offering a candidate pool, the true gap restored growth alone and a wrong pair restored 96% of it — gap filling is a hypothesis.

Week 38. A translation table, the Toolbox's percentage-versus-fraction trap, and the reconciliation rule: a different objective value is a model difference, the same objective value with different fluxes is alternative optima.

On the numbers in this part

Every figure came from the scripts described: a 40-reaction model built by hand, solved with `scipy.optimize.linprog`, with flux variability, deletions, pFBA, shadow prices and the quality checks implemented directly. Four things to hold against them. The model is mine, not the published core model, so its aerobic growth of 0.754 differs from the tracker's reference 0.87 — the anaerobic figure agrees. The P/O assumptions, 2.5 ATP per NADH and 1.5 per quinol, set the ATP yield of 32 per glucose and therefore the growth rate. The biomass coefficients are rounded from the core model's and the carbon content, 70.87 per unit, sets the yield ceiling. And the draft-model percentages in figure 37.1's left panel are illustrative rather than measured, and are labelled so on the figure.

COBRApy, MEMOTE, CarveMe, gapseq, MATLAB and Octave could not be installed here, so every snippet involving them is marked not run. The three errors I made while building the model — phosphate bookkeeping, the missing soluble transhydrogenase, and the vestigial exchange — are described where they occurred rather than quietly fixed, because each is the kind of error a package hides and a hand-built model forces you to find.

What remains

Written so far: weeks 1 to 38 and 45 to 61 — 55 of the tracker's 66. Remaining: weeks 39 to 44 and 62 to 66. The next block is strain-design algorithms, retrosynthesis and machine learning, Bayesian optimisation, ^{13}C flux analysis and the fermentation and scale-up weeks (39 to 44) — which is where the modelling of this part becomes a design tool and where part 3's transport arithmetic returns. Then the five professional weeks. I will take weeks 39 to 44 next.

Part 11 — From model to design: strain-design algorithms, machine learning, labelling and the bioreactor (weeks 39 to 44)

These six weeks are where the modelling of part 10 becomes a design tool and where the transport arithmetic of part 3 comes back to constrain it. The core model built by hand in part 10 is reused throughout: it supplies the production envelopes and the knockout search in chapter 39, the proteome-constrained overflow in the same chapter, and — in chapter 44 — the oxygen exchange bound that a k_La calculation finally fixes, which is the point at which the transport calculation and the genome-scale model meet.

Almost everything quantitative here was run. The strain-design search, the Bayesian optimisation against a factorial grid, the overfitting and leakage simulations, the ^{13}C fitting and the oxygen-bound sweep are all executed code with the parameters stated. The exceptions, marked where they occur, are the retrosynthesis tools in chapter 40 and the flux-fitting packages in chapter 43, which need network access or software this environment does not have; for those the chapters give the arithmetic and the reading rather than output.

Two results worth flagging before you start. The knockout search in chapter 39 found no growth-coupled design at all under aerobic conditions and plenty under anaerobic — which is not a bug but the reason real succinate strains are anaerobic, and it only became obvious because the search failed first. And the Bayesian optimisation in chapter 41 beat a 16-run factorial grid decisively in the median and still got stuck on a local optimum in one seed out of forty, which is the honest version of a method that is usually sold without its failure mode.

Ch.	Week	Chapter
39	39	Strain-design algorithms: growth coupling, searched by hand
40	40	Retrosynthesis and pathway prediction: expanding a network backwards
41	41	Bayesian optimisation: choosing the next experiment from a dozen noisy points
42	42	Machine learning for biology: overfitting, leakage and the split that matches the question
43	43	^{13}C metabolic flux analysis: what labelling resolves that stoichiometry cannot
44	44	Bioreactor engineering and scale-up: mixing, transfer, feeding and the cost sketch

Strain-design algorithms: growth coupling, searched by hand

Growth coupling means arranging the network so that the cell cannot grow without making your product. Then evolution works for you instead of against you — which is the only way to win against a population that has 10^9 members and a generation time of twenty minutes.

OptKnock, and the bilevel structure

Burgard, Pharkya and Maranas posed strain design as a **bilevel optimisation**. The outer problem chooses which reactions to delete, subject to a limit on how many; the inner problem is the cell's own — given those deletions, it distributes flux to maximise growth. The outer objective is the product flux *at the inner problem's optimum*. In words: find the deletions such that, when the cell does the selfish thing, the product comes out anyway.

Growth coupling in one sentence for a non-specialist: *we break the metabolic routes the cell would otherwise use to balance its books, so that the only remaining way for it to grow is to make and excrete our product.* The reason it matters is the one from part 1 chapter 10: a strain that makes a product at a cost to growth is losing a race against its own revertants, whereas a growth-coupled strain is winning it, because a mutant that stops producing also stops growing.

Searching the design space by enumeration

OptKnock solves the bilevel problem as a mixed-integer programme. With a 40-reaction model, you can do something cruder and more instructive: enumerate. For every candidate deletion set, solve the inner problem for maximum growth, then minimise the product flux at that growth — the **guaranteed** product, the amount the cell must make even in its best case for itself. That minimisation is the step that distinguishes coupling from possibility, and it is what the production envelope shows graphically.

```
def coupled(ko, prod='EX_succ', frac=0.2):
    l, u = ANA.copy(), ub.copy()                # anaerobic base case
    for k in ko: l[idx[k]] = 0; u[idx[k]] = 0    # the deletions
    g = opt(idx['BIOMASS'], l, u)               # inner problem: the cell maximises growth
    if g is None or g < frac*gmax: return None   # reject designs that barely grow
    return opt(idx[prod], l, u, maximize=False, fix={'BIOMASS': g}), g   # guaranteed product
```

The first run found nothing. Searching single and double deletions aerobically returned no design with any guaranteed succinate: with oxygen available the cell can always oxidise its substrate completely and balance its redox books without the product. That failure is the result — it is why real succinate strains are anaerobic or micro-aerobic, and it is a cleaner demonstration of the principle than any successful design would have been. Rerunning anaerobically:

Deletions	Guaranteed succinate (mmol gDW^{-1} h^{-1})	Growth (h^{-1})
none (wild type)	0	0.217
ΔPYK (pyruvate kinase)	8.23	0.134
ΔPFL	2.94	0.111
ΔADH	1.98	0.123
ΔACK (acetate kinase)	1.75	0.155
ΔPYK ΔACK	9.05	0.072
ΔPYK ΔACK ΔPFL	9.36	0.048

Four of twenty-seven single deletions couple at all; fifty-two of the pairs do. The pattern is the trade every strain designer meets: **more coupling costs more growth**. ΔPYK gives 8.23 at 0.134 h^{-1}; adding ΔACK buys 10% more product and halves the growth rate; the triple buys 3% more and halves it again. Which point on that curve you want depends on the process — a slow strain needs a longer fermentation and a bigger seed train — and the model cannot tell you, because the choice is economic.

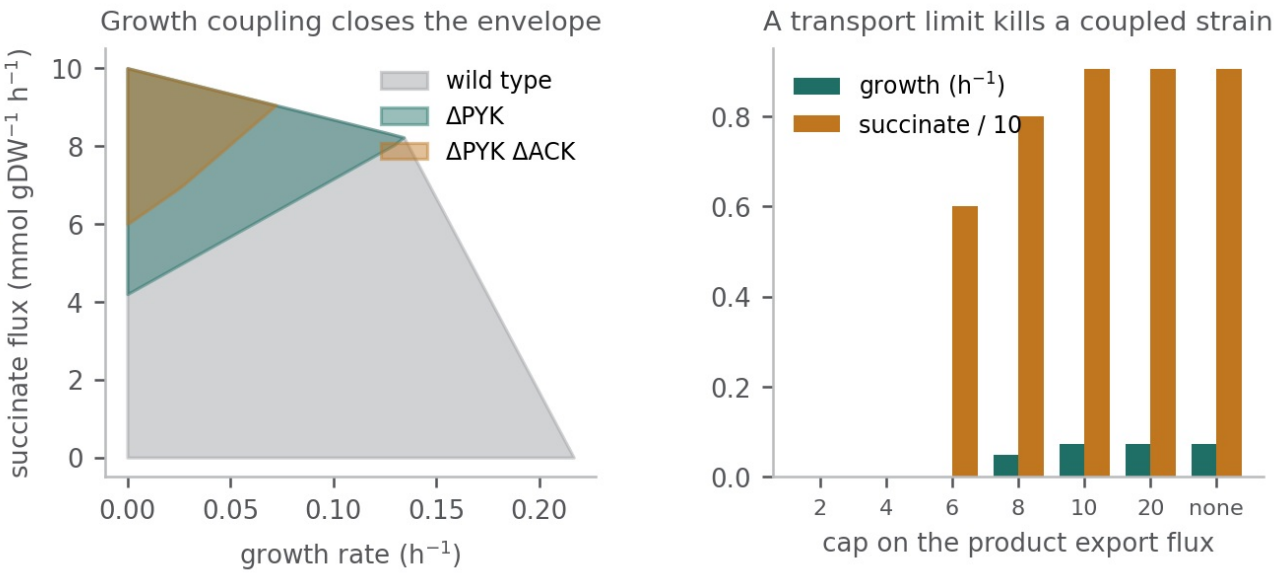


Figure 39.1 Left: production envelopes — the range of succinate flux possible at each growth rate — for the wild type and two knockout strains, anaerobically. Coupling is the closing of the envelope's lower edge: at every growth rate the strain must make product. Right: what happens when the product's export flux is capped. Computed with the hand-built core model of part 10.

Comparing with the literature, as the week asks: the published anaerobic succinate strains delete the competing fermentative routes — lactate dehydrogenase, pyruvate formate-lyase, acetate kinase and phosphotransacetylase — which my search also found, and they additionally engineer the carboxylation step and the glucose uptake system, which my model cannot suggest because its PEP-consuming phosphotransferase system is the only uptake route it has. That is the honest difference: the model agrees on the deletions it can see and is silent about the interventions outside its scope. What the experiment found that the model missed is usually of this kind — a transporter, a regulator, or an evolved tolerance.

The export bound: a transport limit made numerical

The right-hand panel is the week's most useful exercise. Take the ΔPYK ΔACK strain and cap the succinate export flux, as a real transporter's V_{max} would:

Export cap	Growth	Guaranteed product
none	0.072	9.05
10	0.072	9.05
6	0.000	6.00
4 and below	infeasible — no solution at all	

Below about 6 the strain cannot grow, and below 4 the model has no feasible steady state: the deletions have made product formation obligatory, so a cell that cannot export its product cannot balance its internal pools and dies. **Growth coupling and a transport limit interact catastrophically.** That is the quantitative version of part 3 chapter 45's argument, and it is the reason an export measurement belongs in a design campaign before the knockouts, not after.

What the proteome constraint adds: overflow, from first principles

Part 4 chapter 51 claimed that adding a proteome budget makes a constraint-based model predict overflow metabolism instead of needing to be told about it. I tested that on the core model by assigning each reaction a crude enzyme cost per unit flux — respiration and the TCA cycle expensive, fermentative routes cheap, exchanges free — and constraining the total. The costs are illustrative, not measured.

Glucose uptake	No proteome constraint		Budget fixed at the cost of the solution at 10			
	growth	acetate	growth	acetate	ethanol	O_2 uptake
10	0.754	0	0.754	0	0	9.5
12	0.906	0	0.827	0.18	0.82	9.6
15	1.131	0	0.777	0.75	7.20	7.4

20	1.508	0	0.673	4.34	16.6	3.1

Without the constraint, growth rises linearly and nothing is ever excreted. With it, by-products appear above a threshold uptake and **oxygen uptake falls as the uptake rate rises** — the cell switching from protein-expensive respiration to protein-cheap fermentation, which is exactly the proteome-allocation account of overflow from part 1 chapter 6. The growth decline beyond 12 is an artefact of my hard budget with no maintenance scaling, so take the onset as the result and not the shape at high uptake.

^{13}C-MFA, and the three kinds of flux

The distinction the week wants, which chapter 43 develops: an **FBA-feasible** flux is consistent with stoichiometry and an assumed objective — a hypothesis. A flux **constrained by measured exchange rates** is pinned at the boundary and still undetermined inside, as part 10's flux variability showed, where fourteen reactions had ranges at exactly optimal growth. A **^{13}C-measured** flux is inferred from labelling patterns and resolves the internal splits that neither of the others can: the division at glucose-6-phosphate, the reversibility of exchange fluxes, and anaplerotic against cyclic TCA flux.

What kinetic and enzyme-constrained models add, and cost

Model class	Adds	Costs
FBA	Yields, feasibility, deletion predictions	An objective assumption; no concentrations, no time
Enzyme-constrained (GECKO-style)	Overflow, growth-rate-dependent proteome allocation, a protein cost per flux	A k_{cat} and a molecular weight per reaction — and literature k_{cat} values are frequently wrong by an order of magnitude (part 4 chapter 51)
Kinetic	Transients, pool concentrations, control coefficients, regulatory behaviour	Three to five parameters per reaction, most unidentifiable from available data (part 2 chapter 16)

Bilevel	`outer: choose deletions · inner: cell maximises growth · objective: product at the inner optimum`
Guaranteed product	`min(product) subject to growth = max growth — the lower edge of the envelope`
Coupling trade	`ΔPYK 8.23 at μ 0.134; ΔPYK ΔACK 9.05 at 0.072; triple 9.36 at 0.048`
Export cap	`below 6 the coupled strain cannot grow; below 4 no feasible solution exists`
Proteome budget	`Σ (cost × flux) ≤ P reproduces overflow and falling O₂ uptake`

Your OptKnock design predicts a coupled yield of 70% of theoretical with four deletions. What do you check before building it?

Four things, in order of how cheaply they can kill the design. **The oxygen condition**: my search found nothing aerobically and plenty anaerobically, so check which the design assumes and whether the process can be run that way — an anaerobic design in an aerobic process is not a design. **The export flux**: cap the product exchange at a realistic transporter V_{max} and rerun, because the table above shows a coupled strain dying when export is limited; if the yield collapses, the transporter is the first thing to engineer. **The growth rate**: 70% of theoretical at what μ? A design at 0.05 h^{-1} is a fourteen-hour doubling time and a long, contamination-prone fermentation. **Alternative optima**: run flux variability at the predicted optimum to check the product flux is genuinely forced and not merely possible.

Then the things the model cannot see, which decide whether four deletions become a strain: whether each deleted gene has an isozyme that the gene–protein–reaction rules missed, whether the quadruple mutant is viable at all on the medium you will use, and whether the strain evolves its way out — which, for a truly coupled design, it cannot do without losing growth, and that is the whole point.

Three things people get wrong

- ‘The model says the strain can make the product.’ Can is the upper edge of the envelope. Coupling is about the lower edge — what it must make.
- ‘More knockouts, more product.’ More coupling, less growth: 8.23 at μ 0.134 against 9.36 at 0.048.
- ‘Transport is a detail for later.’ Cap the export at a realistic rate and a coupled strain can become infeasible outright.

Self-check — answer on paper first

1. Explain the bilevel structure and why the inner problem is the cell's.

The outer problem is the engineer's: choose a set of deletions, within a budget, to maximise product. The inner problem is the cell's: given those deletions, distribute flux to maximise growth. The nesting matters because the engineer does not control the flux distribution — the cell does, and it will do whatever is best for itself. A design evaluated by the product the cell *could* make is wishful; a design evaluated by the product it must make while maximising growth is a prediction about a strain that will stay that way under selection.

2. Why did the aerobic search find no coupled design?

Because with oxygen freely available the cell can oxidise substrate completely to CO_2, generating ATP through respiration and balancing redox without any fermentative product, so there is always a growth-maximising solution with zero product. Coupling requires that every route for balancing carbon, redox and energy pass through the product, which under aerobic conditions would need the respiratory chain itself restricted — an oxygen limit, or deletions in respiration, which carry their own costs. Hence anaerobic or micro-aerobic designs for fermentative products.

3. A coupled design is predicted at 9 mmol gDW^{-1} h^{-1} of product. Your transporter can manage 5. What happens?

By the table above, somewhere between 4 and 6 the strain stops growing: the deletions make product formation obligatory, so if the product cannot leave, its intracellular pool cannot balance and there is no feasible steady state. Practically the strain will either not grow, accumulate the product internally to a toxic concentration, or acquire suppressor mutations that break the coupling — all three of which look like "the design did not work". The fix is to raise export capacity first, by overexpressing or screening a transporter, and only then to install the knockouts.

What to open a book for. Burgard, Pharkya & Maranas, Biotechnol. Bioeng. 84:647 (2003) for OptKnock, read for the bilevel formulation rather than the solver. A ^{13}C-MFA methods review for chapter 43's material, and the GECKO methodology paper for enzyme constraints. Part 10's chapters 35 and 36 are the machinery this chapter uses.

Retrosynthesis and pathway prediction: expanding a network backwards

A retrosynthesis tool proposes routes. What it cannot propose is an enzyme that exists, is active on your substrate, and folds in your host — which is where every one of its routes actually fails.

Reaction rules, and expanding backwards

RetroPath2.0 and its relatives work from **reaction rules**: generalised patterns abstracted from known enzymatic reactions, encoding which bonds change and in what chemical context, at a chosen level of specificity. Applied in reverse to a target molecule, a rule proposes precursors; applied again to those, it proposes their precursors, and so on, expanding a network backwards until it reaches compounds the host already makes.

The arithmetic of that expansion is why ranking matters more than generation. If each compound matches on average r applicable rules and each application yields one precursor set, a breadth-first expansion grows as r per generation: at r = 5, four generations is 625 leaf compounds and several thousand candidate routes; at r = 20, 160,000. Tools control this with rule-specificity settings, with a maximum number of steps, and by pruning against a sink of compounds known to exist in the host. The output is therefore not "the route" but a ranked list whose length is set by the settings you chose.

Ranking criteria, and what each is worth:

Criterion	What it measures	Limit
Thermodynamic feasibility	$\Delta G'$ per step and the max-min driving force of the route (part 2 chapter 14)	Group-contribution estimates carry real uncertainty, especially for unusual chemistry
Enzyme availability	Whether a characterised enzyme catalyses each step, and on how similar a substrate	A rule that fired does not mean an enzyme exists; this is the binding constraint
Route length	Number of heterologous steps	A proxy for build effort and for the chance that one step fails
Host fit	Whether the precursor is abundant in the chosen host and the cofactors match	Needs the part 5 constraint sheets: which of the twelve precursors, and what cofactor demand
Toxicity of intermediates	Predicted or known toxicity	Usually unknown for novel intermediates

I could not run RetroPath2.0 or any retrosynthesis tool here — it needs network access and a rule database — so this chapter gives the reasoning and the arithmetic rather than output. The exercise the tracker sets is in any case a manual one, and it is the more valuable half.

Proposing a route by hand, and where the risk is

Take a non-natural target and propose two routes from a central metabolite. The method: identify the target's functional groups and carbon skeleton; find the nearest central-metabolism intermediate with the same skeleton or one bond away (the twelve precursors of part 1 chapter 7 are the candidate list); then work backwards one chemical transformation at a time — a reduction, a transamination, a decarboxylation, a condensation — writing down what enzyme class each would need.

Then source a candidate enzyme for every step from BRENDA or MetaCyc, and **mark every step for which you cannot find one**. Those marks are the project's real risk, and the arithmetic is unforgiving: if each step with a plausible enzyme has an independent probability p of actually working in your host, a route of n such steps works with probability p^n. At p = 0.8, a five-step route is 33%; at p = 0.6, 8%. A step with *no* candidate enzyme is not a low probability, it is a research project of its own — an enzyme discovery or design problem, chapter 29's territory — and a route containing two of them is a thesis rather than a pathway.

That is why route length and enzyme availability dominate the ranking in practice, and why a thermodynamically elegant six-step route with two unknown enzymes loses to a clumsy three-step route with three characterised ones.

Where machine learning actually enters the loop

The honest map, which chapters 41 and 42 then make concrete:

- **Design**: ranking retrosynthesis routes; predicting enzyme–substrate compatibility; predicting which variants of an enzyme are worth building (the variant-effect scores of part 8 chapter 29, which enrich a library roughly thirteen-fold

at a correlation of 0.5).

- **Build**: almost nothing. This is chemistry and robotics.
- **Test**: image and spectrum analysis, peak calling, outlier detection in plate data.
- **Learn**: the surrogate model that maps a design to a measured outcome and recommends the next batch — chapter 41 — which is where the phrase "machine-learning-guided DBTL" actually lives.

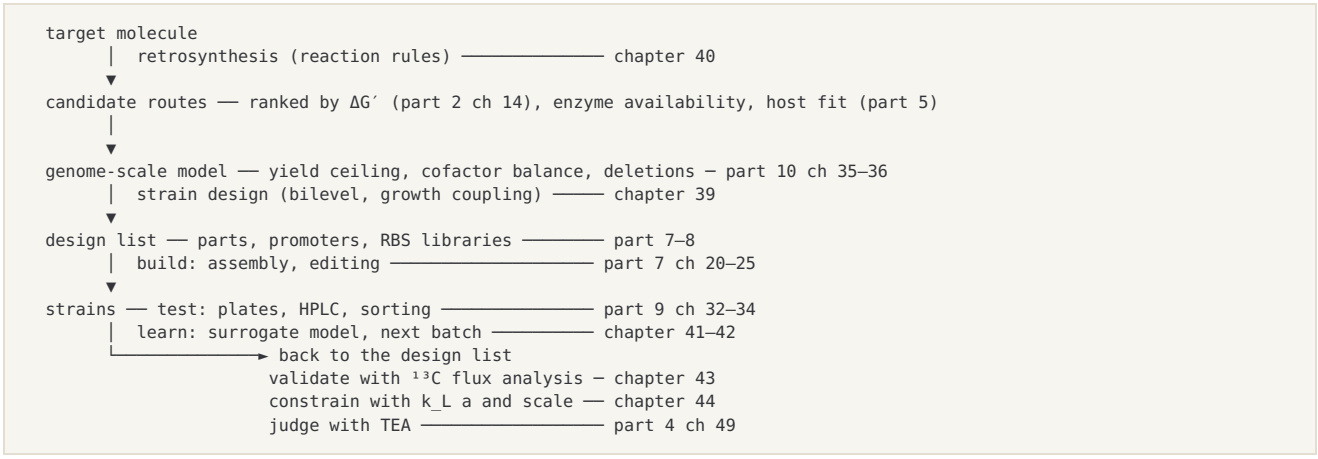
What it needs is the part most reviews skip: **data with identifiers that survive the pipeline** (part 9 chapter 34), enough of them, and a readout that tracks what you care about. Tens of experiments is the normal scale, which is why the methods that work here are the small-data ones of chapter 41 rather than the deep networks of the courses.

The tracker asks you to mark every forward-looking claim in a review as demonstrated or projected, as in part 6 chapter 61. In a 2025 review of AI-guided biofoundries expect the demonstrated set to be narrow — automated build and test at scale in specific well-instrumented facilities, surrogate models recommending batches in published campaigns — and the projected set to be most of the rest, especially anything described as autonomous, self-driving or closed-loop across the whole cycle.

What a biofoundry round actually looks like

Part 9 chapter 34 did the pipeline arithmetic; here the point is the division of labour. Automated: DNA assembly set-up, transformation plating in some facilities, colony picking, inoculation, plate-reader and sometimes chromatographic assay, data capture. Not automated: deciding what to build, designing the parts, troubleshooting a failed assembly, judging whether a hit is real, and the fermentation characterisation of a promising strain. For a PhD student's daily work this means the time saved is in repetitive execution, and the time spent moves to design and interpretation — which is exactly where the value of understanding parts 2 to 10 lies.

The computational stack, in one diagram



Expansion	r applicable rules per compound → r^n leaves after n generations
Route risk	P(route works) ≈ p^n; 0.8^5 = 33%, 0.6^5 = 8%
Unknown enzyme	not a low probability — a separate discovery project
Ranking that matters	enzyme availability and length first, thermodynamics second, elegance never
ML in the loop	design (ranking, variant priors) and learn (surrogate, next batch); not build

A tool returns 400 routes to your target. How do you get to three worth building?

Filter before you rank. First, **length**: discard anything longer than about five heterologous steps, which removes most of the list and costs little, since long routes fail on the p^n arithmetic anyway. Second, **precursor**: keep only routes starting from a precursor your host supplies well — part 5's constraint sheets say which, and a route from erythrose-4-phosphate in a host that has little of it is a project about precursor supply, not about the route. Third, **enzyme evidence**: for each remaining route, look up every step in BRENDA or MetaCyc and count the steps with a characterised enzyme on a substrate within one functional group of yours. Routes with any step at zero go to the bottom.

Then rank what survives, perhaps a dozen routes, by max-min driving force (part 2 chapter 14) and by cofactor demand against the host's supply (part 2 chapter 19) — and pick three that differ from each other, rather than three variations of the same chemistry, so that a shared failure does not kill all three. Finally, write down for each the single step you most expect to fail and how you would know: that sentence is worth more than the ranking.

Three things people get wrong

- ‘The tool found a route.’ It found a sequence of rule applications. Whether an enzyme exists for each is a separate question, and it is the one that decides the project.
- ‘Shorter is better because it is elegant.’ Shorter is better because the probability of the whole route working is roughly p to the power of its length.
- ‘Machine learning designs the pathway.’ It ranks candidates and recommends the next batch; the constraints that kill routes are thermodynamic, enzymatic and host-specific.

Self-check — answer on paper first

1. Why does ranking matter more than generation in retrosynthesis?

Because generation is combinatorial and cheap: a few rules applied over a few generations produce hundreds to hundreds of thousands of candidate routes, nearly all of which are unbuildable. The scarce resource is the attention to evaluate them and the years to build one, so the value of a tool lies in how well it orders its output by the things that actually kill routes — missing enzymes, thermodynamically stuck steps, precursors the host lacks and toxic intermediates — not in how many routes it can enumerate.

2. Two routes: four steps with all enzymes characterised, or two steps with one unknown enzyme. Which do you build?

The four-step route, in almost every case. At a generous $p = 0.8$ per characterised step it has a 41% chance of working as a whole, and every step has a literature starting point and a measurable activity. The two-step route's unknown enzyme is not a 50% risk, it is an open-ended discovery or design problem — screening homologues, directed evolution or computational design (part 8 chapters 29 and 30) — which can consume the whole project. The two-step route becomes attractive only if the unknown step is a well-precedented reaction class where a homologue is likely to be found quickly.

3. Where in the DBTL cycle does machine learning have the least to offer, and why?

Build. Assembly, editing and transformation are deterministic chemistry and robotics whose failure modes are physical — a bad overhang set, an inhibitor in the DNA, poor competence — and are diagnosed by the controls of part 7 chapter 20 and part 9 chapter 32, not by a model. The leverage is in design, where a prior narrows a library, and in learn, where a surrogate chooses the next batch from expensive measurements.

What to open a book for. Delépine et al., Metabolic Engineering 45:158 (2018) for RetroPath2.0 and how reaction rules expand a network; a 2025 review on machine learning or AI-powered biofoundries, read with a pen for demonstrated against projected; and BRENDA and MetaCyc, which are where the enzyme-availability question is actually answered.

Bayesian optimisation: choosing the next experiment from a dozen noisy points

The constraint in strain engineering is not data volume, it is experiment count. Twenty runs, each a week of work and a few per cent noisy, is the problem — and it is the opposite of the machine learning most courses teach.

The problem this is actually solving

The Automated Recommendation Tool paper's framing is the one to absorb: a handful of expensive experiments, each producing one noisy number, with a design space too large to grid and interactions between factors that make one-at-a-time optimisation wrong (part 2 chapter 17). That is not a big-data problem. It is a **sequential decision problem under uncertainty**, and the method that fits it is a surrogate model with calibrated uncertainty plus a rule for choosing where to look next.

Count the experiments in your own plan. If the answer is a few dozen, every method that needs thousands of examples is irrelevant, and the question becomes how to spend the few you have.

Gaussian process surrogates

A Gaussian process is a distribution over functions: given observations, it returns at every untried point a posterior **mean** — the best guess — and a posterior **standard deviation** — how unsure it is. The kernel encodes the assumption about smoothness: a Matérn kernel with a length scale of 0.25 on a unit-scaled factor says that the response changes appreciably over about a quarter of the factor's range and is continuous but not infinitely smooth. Far from any data the posterior mean reverts to the prior and the standard deviation grows to the prior's — which is the property that makes sequential design possible at all, because a model that reports "I do not know, here" can be asked to go and find out.

```
from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import Matern, ConstantKernel, WhiteKernel
k = ConstantKernel(1.0) * Matern(length_scale=0.25, nu=2.5) + WhiteKernel(noise**2)
gp = GaussianProcessRegressor(kernel=k, normalize_y=True, n_restarts_optimizer=3).fit(X, y)
mu, sd = gp.predict(Xgrid, return_std=True)
```

The WhiteKernel term matters: it tells the model the measurements are noisy, so it does not interpolate every point exactly. Without it, a noisy replicate becomes a spike the model believes.

Choosing the next experiment

An acquisition function turns the posterior into a decision. **Expected improvement** is the expected amount by which a point would beat the best result so far, integrating over the posterior:

```
EI(x) = (μ(x) − y_best − ξ)·Φ(z) + σ(x)·φ(z),      z = (μ(x) − y_best − ξ)/σ(x)
```

The first term rewards a high predicted mean — exploitation — and the second rewards high uncertainty — exploration; ξ is a small constant that biases toward exploration. **Upper confidence bound**, μ + κσ, does the same more bluntly with κ setting the balance. Either way, the next experiment is the arg max, and the loop is: fit, maximise the acquisition, run that experiment, repeat.

The comparison, run forty times

I built a two-factor medium optimisation: a response surface with a narrow global optimum (titer 3.03) and a broader local one (2.1), with measurement noise of 0.12 — about 4% at the optimum. Then I ran Bayesian optimisation from five random initial points against a 16-run factorial grid on the same experimental budget, forty seeds each.

Method and budget	Median best titer	10–90% range	Reaching ≥95% of the best
Bayesian, 8 runs	2.32	1.93–2.99	22%
Bayesian, 11 runs	2.98	2.08–3.02	62%
Bayesian, 16 runs	3.01	2.09–3.03	68%
16-run factorial grid	2.35	2.35–2.35	0%

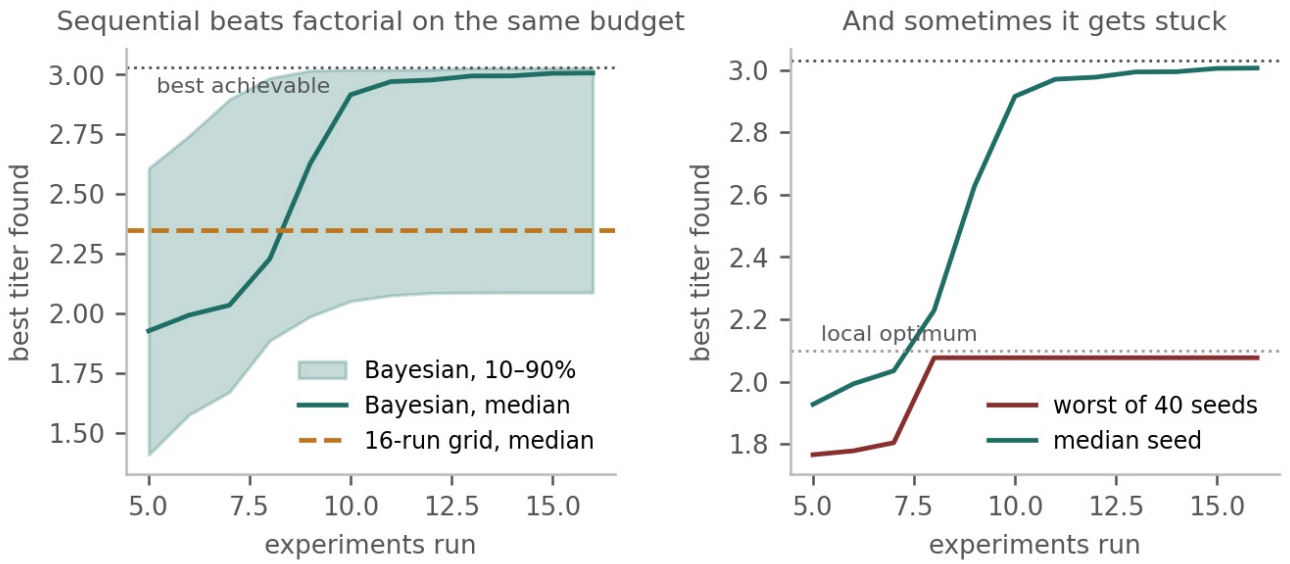


Figure 41.1 Left: best titer found against experiments run, median and 10–90% band over 40 seeds, against the 16-run grid. Right: the median seed against the worst of the 40, which converged on the local optimum and stayed there. Computed for this module; the response surface, the noise level and the kernel are stated in the text.

Three readings. **The sequential method wins decisively in the median** — by eleven runs it is at 98% of the achievable best, which the grid never reaches. **The grid's spread is zero** because it always visits the same points: it is reproducible and reproducibly wrong, missing the narrow optimum because no grid node lands near it. And **the spread of the sequential method is large**: the 10th percentile at sixteen runs is 2.09, barely above the local optimum. Reporting your best seed would be dishonest in exactly the way part 2 chapter 17 warns about.

Where it fails: the stuck seed

In the worst of forty seeds the run ends at 2.09 — 69% of the achievable best — having converged on the local optimum. What the acquisition function did is the general failure: once several points cluster around a good region, the posterior standard deviation there collapses, the mean is high, and expected improvement keeps selecting nearby points because each has a small but positive chance of beating the current best. Meanwhile the unexplored region retains high uncertainty but a *low mean*, and with ξ small, exploitation wins. The defences are a larger ξ, an upper-confidence-bound rule with a generous κ, batch selection that deliberately includes a far-away point, and more initial design points before the loop starts.

When not to use it

- **Fewer than about a dozen runs in total.** With five initial points and a handful of iterations the surrogate is barely constrained; a careful factorial or a one-factor sweep is more defensible and easier to explain.
- **A noisy or drifting assay.** The method assumes the noise model it was given. A drift — part 9 chapter 34's plate position effects, or chapter 50's instrument drift — is read as signal, and the optimiser will chase it.
- **Confounded factors.** If two factors always move together, the surrogate cannot separate them and will recommend along the confounded direction.
- **A readout that does not track what you care about.** The optimiser maximises what you measure, which is part 8 chapter 28's rule in a new form: a biosensor proxy optimised for twenty runs returns a strain that is excellent at the proxy.

From factors to sequences

Medium components are continuous and low-dimensional; a promoter or ribosome binding site choice is categorical and high-dimensional. To let a model learn across variants you must **encode the genotype**, and the options run from crude to subtle: one-hot indicators per part, which treats every part as unrelated; measured or predicted strengths as continuous features, which lets the model interpolate between a weak and a strong promoter and is usually the best value for the effort; physicochemical or sequence descriptors for an RBS; and embeddings from a pretrained sequence model for a protein variant (part 8 chapter 29). The tryptophan work the tracker names is the hybrid case worth reading: a mechanistic model of the pathway supplies structure, and the learned component covers what the mechanism does not.

Two sentences for a capstone build-and-test plan, which most proposals leave as the word "learn": *After each round of twelve strains we will fit a Gaussian process to titer as a function of the three promoter strengths, encoded as measured relative promoter units, and select the next twelve by expected improvement with a deliberately large exploration constant, keeping two of the twelve as replicates of the current best for drift control. We will report the median and spread across rounds, and stop when the predicted improvement falls below the assay's reproducibility.*

Surrogate	`Gaussian process: posterior mean and standard deviation at every untried point`
Noise	`a WhiteKernel term, or the model interpolates noise`
Expected improvement	`(μ − y_best − ξ)Φ(z) + σφ(z): exploitation plus exploration`
Result here	`11 runs reached 98% of the best; a 16-run grid reached 78% and never more`
Failure	`1 seed in 40 stuck at the local optimum, 69% of the best`
Report	`the spread over seeds, not your best run`

A colleague has run eleven medium experiments by Bayesian optimisation and wants to stop, claiming the optimum is found. What do you ask?

Three questions. **What does the surrogate predict for the best untried point, and with what uncertainty?** If the predicted improvement is smaller than the assay's reproducibility, stopping is defensible; if the model still sees a region with a high upper bound, it is not. **Where did the eleven points go?** If nine of them cluster in one region, the result may be a local optimum like my worst seed — the test is cheap: run two deliberately distant points and see whether the model was right that they are poor. **What is the replicate noise?** Without it there is no way to know whether the best point is genuinely best or a lucky draw, and part 2 chapter 17's winner's-curse argument says the best of eleven noisy measurements is biased upward.

Then the honest framing for a report: not "the optimum is 3.0" but "the best measured condition gave 3.0 ± noise, the surrogate predicts no untried point above that by more than the assay can resolve, and the search covered this region of the factor space" — with the coverage shown, because a reader cannot otherwise tell a thorough search from a lucky one.

Three things people get wrong

- ‘We used machine learning, so we need a lot of data.’ This method is for tens of runs. The kind that needs thousands is a different problem.
- ‘The grid is safer because it is systematic.’ It is reproducible and it missed the optimum in 40 of 40 seeds, because no node landed near it.
- ‘It converged.’ One seed in forty converged on the wrong peak. Report the spread, and test the model's claim that the unexplored region is poor.

Self-check — answer on paper first

1. Why does a surrogate need to report uncertainty for sequential design to work at all?

Because the decision is where to spend the next experiment, and that requires knowing not only where the response is predicted to be high but where the prediction could be wrong. A model that returns only a mean cannot distinguish a point it knows to be mediocre from a point it has no information about, so it has no basis for exploring. The posterior standard deviation is what turns "I predict 2.0 here" into "I predict 2.0 ± 0.1 here and 1.8 ± 0.9 there", and the second point may well be worth the run.

2. Your acquisition function keeps proposing points next to the current best. Diagnose and fix.

The exploitation term is dominating: the cluster of points has collapsed the local uncertainty, the mean there is the highest known, and expected improvement with a small ξ will keep picking neighbours because each has a positive chance of a marginal gain. Fix by raising the exploration constant ξ, switching to upper confidence bound with a generous κ, selecting a batch that is required to include a point far from the existing data, or simply adding a few space-filling points by hand. The underlying question to ask is whether the response surface plausibly has a second peak — if it does, as in my simulation, the cost of not exploring is 31% of the achievable titer.

3. When would you choose a factorial design over sequential optimisation?

When you need to *understand* rather than optimise: a factorial estimates main effects and interactions, which is what you report when the question is which factors matter and how they interact. Also when the total run count is very small, where a surrogate is barely constrained; when factors are categorical with few levels, where there is nothing to interpolate; and when the runs must be executed in one batch for logistical reasons, since sequential methods need rounds. The two combine well: a small factorial to find which factors matter, then sequential optimisation over those.

What to open a book for. Radivojević et al., Nat. Commun. (2020) for the Automated Recommendation Tool and its framing; Zhang et al., Nat. Commun. (2020) on tryptophan for the mechanistic-plus-learned hybrid; and the scikit-learn Gaussian process documentation, which is the practical reference for the kernels and the `return_std` argument that makes all of this work.

Chapter 42 · tracker week 42

Machine learning for biology: overfitting, leakage and the split that matches the question

Two simulations settle most of this week. One shows a model with a perfect training score and no predictive power. The other shows the same data giving R^2 of 0.67 or −0.44 depending only on how the test set was chosen.

A model, a feature, a baseline

A model is a function fitted to examples. What varies in biology is the **features** — the numbers that stand for the thing. For a **sequence**: one-hot encoding per position, physicochemical descriptors per residue, k-mer counts, or an embedding from a pretrained protein language model. For a **strain**: which promoter and ribosome binding site in which position, as measured strengths rather than labels (chapter 41), plus the deletions as indicators. For a **plate**: well position, row, column, edge indicator, plate identifier and run date — which are features precisely because they predict the outcome, and that is the problem this chapter is about.

The baseline comes first. Fit the mean, then a linear or ridge model, then nearest neighbours, and report the held-out error of each. A complex model has to beat them on held-out data to be worth anything, and in small biological datasets it frequently does not. This is not modesty; it is the only way to know whether the fancy model learned anything, because an R^2 of 0.6 is impressive against a mean-only baseline of 0 and unimpressive against a ridge baseline of 0.58.

How much data is enough: the overfitting simulation

Sixty features, of which five carry real signal, with noise, fitted by ridge regression and evaluated on 400 held-out samples, averaged over thirty repeats:

Training samples	Training R^2	Held-out R^2
20	1.00	0.19
40	1.00	0.40
80	0.97	0.64
160	0.93	0.82
320	0.91	0.87

At twenty samples and sixty features the model fits its training data perfectly and predicts almost nothing: **a training R^2 of 1.00 with a held-out R^2 of 0.19**. The gap closes as samples accumulate, and the learning curve flattens around 160 to 320 — which for this noise level and this number of features is where more data stops paying.

Then the control that makes the point unarguable: **shuffle the labels** and refit. With n = 40 the training R^2 is still 1.00 and the held-out R^2 is −1.67 — worse than predicting the mean. A model can fit pure noise perfectly whenever it has enough parameters, so a high training score is not weak evidence of learning, it is *no* evidence. Run the shuffle on your own data as a matter of routine: if the shuffled model's held-out performance is not near zero, your validation is broken.

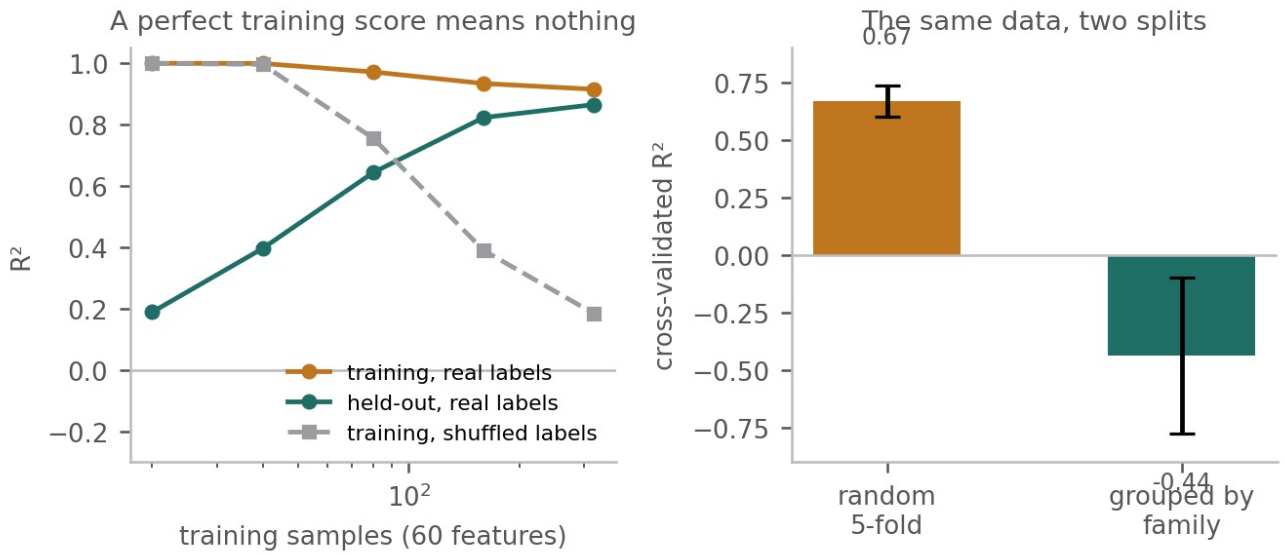


Figure 42.1 Left: training and held-out R² against training-set size for ridge regression with 60 features and 5 carrying signal, with the shuffled-label control. Right: the same dataset of 40 families × 6 variants evaluated by random five-fold cross-validation and by grouping folds by family. Computed for this module; parameters in the text.

Leakage: the same data, two answers

Now the simulation that matters most for sequence work. Forty families, six variants each, where the family identity accounts for most of the variance and the within-family differences are small — which is exactly the structure of a protein dataset built from homologues, or a strain dataset built from a few parental backgrounds. A random forest, evaluated two ways:

Cross-validation scheme	R²	What it means
Random 5-fold	**0.67 ± 0.07**	Relatives of every test point are in the training set; the model recognises the family and reports its mean
Grouped by family	**−0.44 ± 0.34**	Whole families held out; the model has learned nothing that generalises to a new family

The same data, the same model, and a swing from "works well" to "worse than the mean". The random split leaks because the information that predicts a test point is not the biology but the presence of its siblings in the training set. A negative R² under the grouped split is the honest answer to the question "will this predict a family I have not seen", and it is *no*.

Which split matches which question, which is the week's real content:

The question	The split
Rank variants within families I already have data for	Random split is legitimate — the use case matches the evaluation
Predict a family I have never measured	Group by family, cluster or sequence identity threshold
Predict next week's plate, or a new batch	Split by plate, by run date, or forward in time

One way my own data could leak: strains built from the same parental background and measured on the same plate share both a genetic and a positional effect, so a random split over wells reports a model that has learned the plate. What to report so a reader can tell: the split rule, in one sentence in the methods — *folds were grouped by parental strain so that no two variants of a background appear in both training and test, and plate identity was additionally held out* — plus both numbers when they differ, which is more informative than the better one alone.

Pretrained models and active learning

A **zero-shot score** is a prediction made with no training on your protein at all: a protein language model assigns each substitution a likelihood from patterns learned across evolution, and that correlates partially with measured fitness (part 8 chapter 29, where a correlation of 0.5 enriched the true top 1% thirteen-fold and cut a 30,000-clone screen to about 2,300). What it cannot tell you is anything about *your enzyme in your host*: expression, solubility, the non-natural substrate you care about, or the cofactor you swapped — because none of that is in the evolutionary record it learned from.

Active learning and Bayesian optimisation are the same idea with different emphasis. Chapter 41's acquisition function chooses the next point to **maximise an outcome**; active learning chooses the next batch to **reduce the model's**

uncertainty, which is what you want when the goal is a predictive model rather than one good strain. Both need the same thing: calibrated uncertainty.

When a model prior and a structure prior agree or conflict: they agree when a substitution is both evolutionarily unusual and buried in the core — strong evidence against it. They conflict when a substitution is evolutionarily rare but structurally innocuous, which is common for positions under selection for something other than stability, such as a regulatory interaction you do not care about. The resolution is that the structure prior answers "will it fold" and the sequence prior answers "has nature tolerated it", and for an engineering goal — a new substrate, a swapped cofactor — nature's answer may be irrelevant because nature was not trying to do what you are.

A notebook that stays honest

The one page the week asks for, with the seven headings filled: **the question** — stated as a prediction task, including what a new data point looks like; **the data** — how many, from where, with what replication; **the split** — the rule and why it matches the question; **the baseline** — mean, linear, nearest neighbours, with their held-out scores; **the metric** — and why, since R^2 on a skewed target flatters a model that predicts the bulk; **the uncertainty** — the spread across folds and seeds, not the best; and **what would change your mind** — stated in advance, which is the hardest and most valuable line.

Add seeds and pinned versions, and check the notebook runs from a fresh clone, as part 10 chapter 36 required. And the methods sentence about the split, written once and reused.

On AI assistants, which the tracker raises explicitly: they write plausible code and plausible prose, and the failure mode here is specific — code that touches the split or the metric can be subtly wrong in a way that inflates the score, and a wrong citation is invisible until a reviewer checks. The checks worth doing, which I applied to my own code in this part: run it on a case with a known answer (the shuffled-label control is exactly that), read every line that touches the split, the leakage structure or the metric, and confirm that any citation exists. Write down what the assistant did and what you checked, because that sentence is what makes the work yours.

Overfitting	`n = 20, p = 60: training R² 1.00, held-out 0.19`
Shuffle control	`shuffled labels still fit perfectly in training; held-out R² −1.67`
Leakage	`random 5-fold 0.67 against grouped-by-family −0.44, same data`
Split rule	`match the split to what you must predict: variant, family, or plate`
Zero-shot score	`a prior worth an order of magnitude in screening; silent about your host`
Baseline	`mean, linear, nearest neighbours — reported, before anything complex`

A collaborator reports $R^2 = 0.8$ predicting enzyme activity from sequence, on 300 variants from twelve homologue families, with random five-fold cross-validation. What do you conclude?

That the number is almost certainly a family-recognition score rather than a prediction of activity. With twelve families and 300 variants there are twenty-five variants per family, so a random split puts relatives of every test point in the training set, and my simulation shows exactly this structure giving 0.67 randomly and −0.44 when grouped. The first thing to ask for is the grouped-by-family number. If it is near zero, the model predicts the family mean and nothing else, which is still useful for ranking within a known family and useless for a new one.

Then two further checks: the shuffled-label control, which should give a held-out score near zero and will reveal any residual leakage in the pipeline; and a baseline, because if a ridge model on family identity alone reaches 0.78, the sequence features added two points. And the question behind all of it — what will this model be used for? If the plan is to rank variants of a measured enzyme, the random split is the right evaluation and the number stands. If the plan is to pick a new homologue to engineer, it does not.

Three things people get wrong

- ‘The training score is excellent.’ It is also excellent on shuffled labels. Only held-out performance is evidence.
- ‘We cross-validated.’ With what split? Random folds over related sequences measure family recognition, as 0.67 against −0.44 shows.
- ‘The language model scored this variant highly.’ It scored it against evolution, which was not trying to make your product in your host.

Self-check — answer on paper first

1. Why does shuffling the labels diagnose a broken validation?

Because with shuffled labels there is, by construction, no relationship between features and target, so any correctly validated model must score near zero on held-out data. If it does not — if the shuffled model still appears to predict — then information is reaching the test set that should not be: leakage through related samples, a preprocessing step fitted on the whole dataset before splitting, or a feature that encodes the target. The training score, meanwhile, can be perfect on shuffled labels whenever the model has enough capacity, which is why it is not evidence of anything.

2. You must predict titer for strains built from a new parental background. Design the evaluation.

Group the folds by parental background, so that every test fold contains backgrounds entirely absent from training — that matches the deployment. Report the per-fold scores and their spread, not the pooled number, because backgrounds differ and a single average hides a fold where the model failed completely. Include a baseline that predicts the global mean and one that predicts from the background's own mean where available, to show what the sequence or design features add. And if plates differ between backgrounds, hold plate out as well, or the background effect and the plate effect are confounded.

3. When is a random split the right choice?

When the intended use really is to interpolate within the population you sampled: ranking further variants of the same enzyme families you have measured, or predicting an untested combination of promoter strengths within a library whose parts are all represented in training. The split should mirror the deployment, and if the deployment involves known families then relatives in the training set are a feature rather than leakage. The error is not using a random split; it is using one and then claiming the model generalises to something it was never tested on.

What to open a book for. The literature searches the tracker names — machine-learning-guided directed evolution, protein language models and fitness prediction, and data leakage with homology-aware splitting — are the sources, and the third is the one to read first. Part 4 chapter 17 and part 8 chapter 29 of this module are the chapters this one builds on.

^{13}C metabolic flux analysis: what labelling resolves that stoichiometry cannot

Stoichiometry says what is possible, measured uptake rates pin the boundary, and labelling tells you what actually happened inside. The third is the only one that can falsify a flux map.

Three kinds of flux, and what only labelling settles

Part 10 chapter 35 found fourteen reactions in a 40-reaction model with non-zero flux variability at exactly optimal growth: the network and the objective did not determine them. Measured exchange rates — the uptake of glucose and oxygen, the secretion of by-products — pin the boundary fluxes and leave much of the interior free. Labelling resolves the interior, and three things in central metabolism depend on it entirely.

- **The split at glucose-6-phosphate** between glycolysis and the oxidative pentose phosphate pathway, which decides NADPH supply and carries the carbon cost of part 1 chapter 5.
- **Reversible exchange fluxes**: a near-equilibrium reaction's net flux is a small difference between two large opposing fluxes (part 2 chapter 14), and only labelling sees the exchange.
- **Anaplerotic against cyclic TCA flux**: carbon entering oxaloacetate by carboxylation against carbon arriving round the cycle, which stoichiometry cannot distinguish because both produce the same metabolite.

Tracer design

[1-^{13}C] glucose labels one specific carbon, and its value is positional: carbon 1 is released as CO_2 by 6-phosphogluconate dehydrogenase in the oxidative branch, and survives through glycolysis to become a methyl carbon of pyruvate. So the labelling of downstream metabolites reports the split directly, and it is the classic tracer for that question. **[U-^{13}C] glucose**, uniformly labelled, reports on how intact carbon skeletons are cleaved and recombined — bond breaking and forming — because a fragment's mass shift tells you how many of its carbons came from the same original molecule; it is what resolves anaplerosis and the cyclic flux. Mixtures — commonly a defined ratio of [1-^{13}C] and [U-^{13}C] with unlabelled glucose — are used because no single tracer resolves everything, and the mixture is chosen to maximise the information for the specific fluxes in question.

For the pentose phosphate split, the justification in two sentences: *[1-^{13}C] glucose places the label on the carbon that the oxidative branch decarboxylates, so any flux through that branch removes label from the pool while glycolysis retains it, and the fractional labelling of pyruvate therefore varies monotonically with the split. A mixture with [U-^{13}C] would be added if anaplerotic flux also had to be resolved in the same experiment.*

Atom transitions, and the carbon that is lost

The exercise the tracker sets is to write the transitions carbon by carbon. Glucose carbons 1 and 6 become the methyl carbons of the two pyruvates; carbons 2 and 5 the carbonyl carbons; carbons 3 and 4 the carboxyl carbons, which are the ones released as CO_2 by pyruvate dehydrogenase. In the oxidative pentose phosphate branch, glucose-6-phosphate is oxidised at carbon 1 and then **carbon 1 is released as CO_2** by 6-phosphogluconate dehydrogenase, leaving a five-carbon ribulose-5-phosphate whose carbons derive from glucose 2 to 6. That asymmetry is the entire basis of the measurement.

A **mass isotopomer distribution** is the set of fractions M+0, M+1, M+2 … of a measured fragment containing zero, one, two or more heavy atoms. It is what a mass spectrometer reports, and it must be **corrected for natural abundance** before it means anything: carbon is 1.07% ^{13}C naturally, so a three-carbon fragment has an M+1 of about 3.2% before any tracer is added, and larger fragments more. Silicon and sulfur in derivatisation reagents contribute further. Skipping the correction biases every flux.

Fitting a split, with its confidence interval

I built the simplest possible version to make the statistics concrete: a model in which a fraction f of glucose-6-phosphate goes through the oxidative branch, with [1-^{13}C] glucose, predicting the M+1 fraction of pyruvate including the natural-abundance contribution.

Oxidative PP fraction	Predicted pyruvate M+1
0.00	0.511

0.10	0.463
0.30	0.367
0.50	0.271
0.80	0.128

Fitting f from a measurement generated at a true value of 0.30 with 0.4% absolute noise on the MID gives **f = 0.298 ± 0.003**, with a bootstrap 95% confidence interval of **0.289 to 0.312**. The sensitivity is excellent because the response is steep: a 1% change in the split moves M+1 by about 0.005, which is larger than the measurement noise.

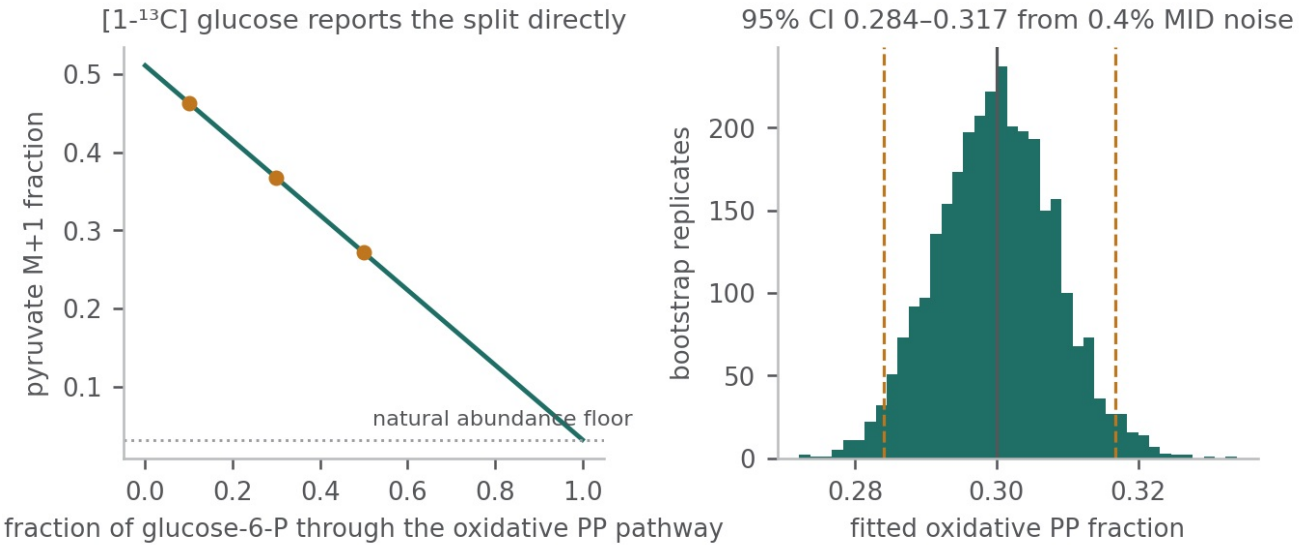


Figure 43.1 Left: predicted pyruvate M+1 fraction against the oxidative pentose phosphate fraction for [1-¹³C] glucose, with the natural-abundance floor marked. Right: the bootstrap distribution of the fitted split from 3,000 synthetic measurements at 0.4% MID noise. Computed for this module with a deliberately minimal atom-transition model — a real analysis uses the full network and software such as INCA, OpenFlux or mfapy, which I could not run here.

What a real fit does is the same thing on a network: simulate the labelling of every measured fragment from a candidate flux vector through the full atom-transition network, compare with the measured MIDs, and minimise the weighted sum of squared residuals over the free fluxes — **nonlinear least squares**, because labelling depends nonlinearly on fluxes. Confidence intervals come from the curvature at the optimum or, better, from a parameter scan: fix each flux at a grid of values, re-optimise the rest, and find where the residual rises past a chi-square threshold.

Reading the statistics honestly

A **poor goodness of fit** means one of two things and they need opposite responses. Either the data are bad — a mis-corrected natural abundance, a contaminated fragment, an unreached isotopic steady state — or **the network is missing a reaction**, in which case no flux vector can reproduce the labelling and the fit is telling you something valuable. The way to distinguish them is to look at which fragments are badly fitted: scattered residuals suggest data, while a consistent failure on fragments downstream of one node suggests a missing route there.

And the caution that matters for reading papers: **a narrow confidence interval on a badly specified network is worth nothing.** The interval reports how well the data constrain the fluxes *within the assumed network*, so a model lacking a real reaction can fit its own structure beautifully and precisely. What establishes the network is the goodness of fit together with the chemistry; what the intervals establish is precision given that.

Two assumptions underlie everything: **isotopic steady state**, that the labelling has stopped changing, which takes a few residence times and is slow for large pools; and **metabolic steady state**, that the fluxes themselves are constant during the labelling. Neither holds in a fed-batch, in a shifting culture, or in a photoautotroph with a slow-turning Calvin cycle — which is why non-stationary ¹³C-MFA exists, fitting the time course of labelling and thereby needing far more data and computation.

What labelling adds	`the G6P split, exchange fluxes, anaplerotic against cyclic TCA`
[1-¹³C]	`positional: C1 lost as CO₂ in the oxidative branch, retained through glycolysis`
[U-¹³C]	`bond breaking and forming; resolves anaplerosis`
MID	`M+0, M+1 … fractions; natural abundance is 1.07% ¹³C — correct it`

Fitting	`nonlinear least squares over the atom-transition network; CIs from a parameter scan`
Poor fit	`bad data or a missing reaction — look at which fragments fail`

A paper reports fluxes with tight confidence intervals and a goodness of fit that fails its chi-square test. How do you read it?

The failed fit is the headline and the intervals are decoration. A chi-square failure means no flux vector in the assumed network reproduces the measured labelling within the stated measurement errors, so either the errors are understated, the data are biased — most commonly by natural-abundance correction or by an unreached isotopic steady state — or the network omits a reaction that is carrying flux. The tight intervals are conditional on that network and simply say that, if the network were right, the data would pin the fluxes; they cannot detect the structural error.

What I would want: the residuals by fragment, to see whether the misfit is scattered or concentrated downstream of one node; confirmation that the culture was at isotopic and metabolic steady state, which for a batch culture sampled during growth is a real question; and the authors' own account of what they tried adding to the network. A paper that reports a failed fit, says so, and discusses the candidate missing reactions is more trustworthy than one that reports a perfect fit with no diagnostics — and the tracker's observation is usually right, that the measured fluxes contradict something in the authors' model, which is the most interesting sentence in the paper.

Three things people get wrong

- ‘The flux map is measured.’ Fluxes are *inferred* by fitting labelling to an assumed network; the network is an assumption the fit can test but not replace.
- ‘M+1 means one labelled carbon.’ Before natural-abundance correction, a three-carbon fragment has an M+1 of about 3% with no tracer at all.
- ‘Tight intervals mean reliable fluxes.’ They mean precision within the assumed network. Check the goodness of fit first.

Self-check — answer on paper first

1. Show why carbon 1 of glucose reports on the glucose-6-phosphate split.

In the oxidative branch, glucose-6-phosphate is oxidised to 6-phosphogluconate and then decarboxylated by 6-phosphogluconate dehydrogenase, and the carbon released as CO_2 is carbon 1 — so label at that position is lost from the carbon skeleton entirely. Through glycolysis, carbon 1 is retained: the hexose is cleaved so that carbons 1, 2 and 3 become one triose and 4, 5 and 6 the other, and carbon 1 ends as a methyl carbon of pyruvate. Therefore the fraction of pyruvate carrying the label falls monotonically with the fraction of glucose-6-phosphate routed through the oxidative branch, which is the measurement.

2. Why must MIDs be corrected for natural abundance, and what happens if you do not?

Because 1.07% of all carbon is naturally ^{13}C, along with heavy isotopes of hydrogen, oxygen, silicon and sulfur in the analyte and any derivatisation group, so every fragment shows M+1, M+2 and so on before any tracer is added — about 3.2% M+1 for three carbons, and more for larger fragments. Without correction the apparent labelling is systematically too high, so the inferred flux through the labelled route is overestimated, and the bias is larger for larger fragments, which distorts the *relative* information between measurements and therefore the whole flux map rather than just scaling it.

3. Your fit is poor. Give the two explanations and the diagnostic that separates them.

Bad data or a missing reaction. Look at the residuals fragment by fragment: if the misfit is scattered across unrelated fragments with no pattern, suspect the data — natural-abundance correction, instrument calibration, co-eluting contaminants, or a culture that had not reached isotopic steady state. If the misfit is concentrated in fragments derived from metabolites downstream of a particular node, suspect the network at that node — a missing bypass, an unaccounted exchange flux or a reaction with the wrong atom mapping. Adding the candidate reaction and seeing whether the fit becomes acceptable is the test, with the caveat from chapter 37 that a reaction added to make a model fit is a hypothesis to verify, not a finding.

What to open a book for. An Antoniewicz review of ^{13}C-MFA, which is the clearest account of the method and its statistics; a ^{13}C-MFA experimental protocol paper for the sampling, quenching and derivatisation that decide whether the data are any good; and a review of non-stationary ^{13}C-MFA if your organism is slow or your culture is not at steady state. The software — INCA, OpenFlux, mfapy — is where the real fitting happens, and none of it ran here.

Chapter 44 · tracker week 44

Bioreactor engineering and scale-up: mixing, transfer, feeding and the cost sketch

This week is mostly translation, not new theory — the transport phenomena you already have, applied to a vessel with cells in it. The single most important number is that mixing time grows with scale even at constant power per volume.

Power and mixing

For a stirred tank, the impeller Reynolds number is $Re = \rho n d^2/\mu$, with n the rotational speed and d the impeller diameter, and above about 10^4 the flow is fully turbulent and the power number N_p is constant — about 5 for a Rushton turbine. Then

```
P = N_p · ρ · n³ · d⁵
```

Compute two vessels at comparable power per volume. A 1 L vessel with a 4 cm impeller at 500 rpm (8.33 s^{-1}) draws 0.3 W, which is **296 W/m³**, with a tip speed of 1.05 m/s. A 100 m³ vessel with a 1.8 m impeller at 42 rpm (0.70 s^{-1}) draws 32 kW, which is **324 W/m³** — comparable power per volume — with a tip speed of 3.96 m/s.

Now the scaling that matters. Under geometric similarity the volume goes as d^3, so holding P/V constant requires $n^3d^5/d^3 = n^3d^2$ constant, giving $\mathbf{n \propto d^{-2/3}}$. Tip speed nd then rises as $d^{1/3}$ and the mixing time, which scales roughly as 1/n for a given geometry, **grows as $\mathbf{d^{2/3}}$**:

Impeller diameter	Speed at constant P/V	Tip speed	Mixing time, relative
0.04 m (1 L)	8.33 s^{-1}	1.05 m/s	1.0
0.40 m (~1 m³)	1.80 s^{-1}	2.26 m/s	4.6
1.80 m (~100 m³)	0.66 s^{-1}	3.72 m/s	12.7

So a vessel that mixes in a second or two at bench scale takes tens of seconds at cubic-metre scale at the same power per volume, and the only ways out are more power — which costs money and shear — or accepting the gradients. **That ratio is the whole scale-up problem in one number**, and the cell-level consequence is the next section but one.

Gas–liquid transfer, and the number that stops the process

The oxygen transfer rate is $OTR = k_La\cdot(C^* - C_L)$: the volumetric transfer coefficient times the driving force between saturation and the bulk dissolved concentration. k_La depends on power per volume, superficial gas velocity, and the medium's coalescence behaviour — correlations of the form $k_La = C(P/V)^{\alpha}v_s^{\beta}$ with exponents around 0.4 to 0.7 and 0.2 to 0.5 are standard; C* depends on temperature, pressure and the gas composition and is about 0.21 mM for air at 37 °C.

The demand side: a culture at 50 gDW/L with a specific oxygen uptake of 5 mmol·$gDW^{-1}\cdot h^{-1}$ consumes **250 mmol·$L^{-1}\cdot h^{-1}$**. Holding C_L at 0.05 mM leaves a driving force of 0.16 mM, so the required k_La is 250/0.16 = **1,560 h^{-1}**, against the roughly 100 to 500 h^{-1} a large stirred tank achieves. The culture cannot be supplied with air at atmospheric pressure.

Cell density	Oxygen uptake rate	Required k_La
10 gDW/L	50 mmol·$L^{-1}\cdot h^{-1}$	310 h^{-1} — feasible
20 gDW/L	100	625 h^{-1} — at or beyond the limit
30 gDW/L	150	940 h^{-1}
50 gDW/L	250	1,560 h^{-1}

The options, in the order a process engineer would consider them: **oxygen enrichment**, which raises C* in proportion to the oxygen mole fraction and can give a factor of four; **head pressure**, which raises C* proportionally — two bar absolute doubles it; **more power and gas flow**, which raises k_La sublinearly and costs utilities and shear; and **lowering the demand**, by feeding more slowly so the specific uptake rate falls. The last is usually the answer, and it is why nearly every industrial fermentation is a fed-batch — part 1 chapter 6's conclusion arriving as a reactor constraint.

Feeding the constraint into the model

This is the week's best exercise and the point where the transport calculation and the genome-scale model meet. Convert an achievable OTR into a per-biomass flux and impose it as the oxygen exchange bound in the core model of part 10, at 10 gDW/L:

k_La	OTR	Oxygen available	Growth	Acetate	Ethanol
100 h^{-1}	16 mmol·L^{-1}·h^{-1}	1.6 mmol·gDW^{-1}·h^{-1}	0.313 h^{-1}	4.92	6.78
200	32	3.2	0.409	4.47	4.68
300	48	4.8	0.505	4.01	2.58
500	80	8.0	0.683	1.90	0
1,500	240	24.0	0.754	0	0

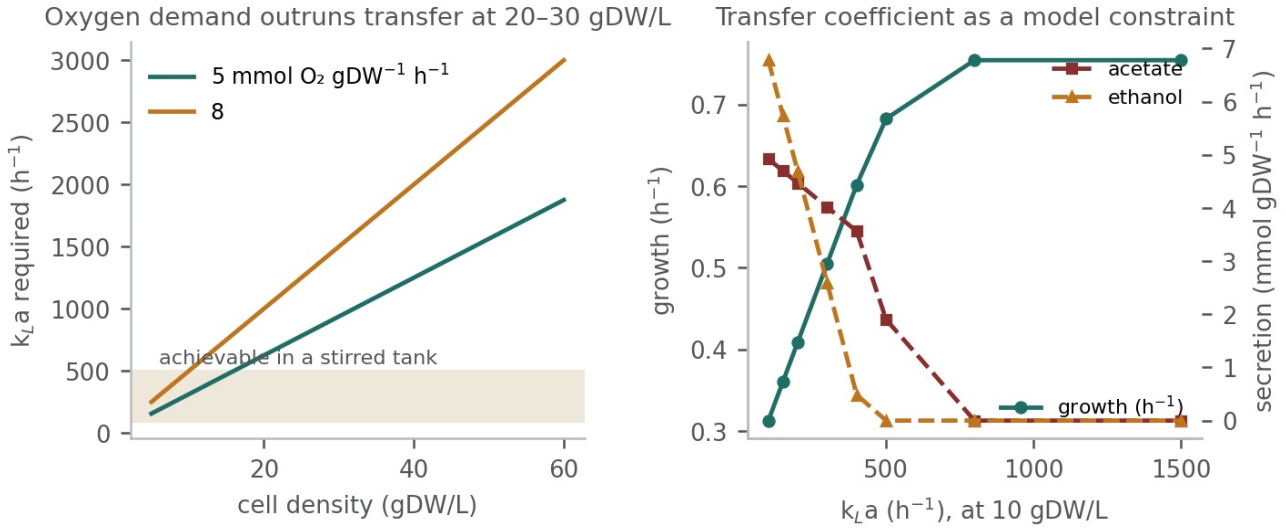


Figure 44.1 Left: the k_La required against cell density for two specific oxygen uptake rates, with the achievable range shaded — demand outruns transfer between 20 and 30 gDW/L. Right: growth and by-product secretion predicted by the core model when the oxygen exchange bound is set from the achievable transfer rate at 10 gDW/L. Computed for this module.

Read the right-hand panel as the answer to "what does my k_La cost me". At 100 h^{-1} the model loses 58% of its growth and excretes nearly 12 mmol of reduced by-products; at 500 it loses 9% and excretes only acetate; at 1,500 it reaches the unconstrained optimum. **The by-products are the diagnostic**: a fermentation secreting ethanol and formate aerobically is reporting an oxygen limitation, and the model quantifies how much.

What the cell actually experiences

Mixing time tens of seconds means a cell circulating through a large vessel passes through regions of different substrate and oxygen concentration on that timescale: high substrate near the feed point, oxygen-rich near the sparger, depleted elsewhere. Compare with the cell's own response times, which part 1 chapter 1 and part 2 chapter 16 put at seconds for metabolite pools, minutes for transcription and mRNA turnover, and a generation for protein dilution. So the cell is **fast enough to respond to a gradient it circulates through** — pools and fluxes adjust within a pass — but too slow to re-tune its proteome, and the result is repeated feast–famine cycling with a transcriptional stress response superimposed.

A **scale-down reactor** reproduces that in 1 L: two compartments, a stirred tank and a plug-flow section connected in a loop, with the residence time in the plug-flow leg set to the large vessel's circulation time, and the feed introduced at one point. The cells then experience the same feast–famine period in a vessel you can afford to run in replicate. What measurement would show it mattered: by-product formation and yield against the well-mixed control at identical average conditions, plus a stress reporter or transcript panel; if the two-compartment run secretes more acetate and yields less at the same average feed rate, the gradient is the cause of the scale-up loss.

Feeding strategy

The substrate balance for a fed-batch, with V the volume, F the feed rate and S_{feed} its concentration:

```
d(SV)/dt = F·S_feed − (μ/Y_XS + m)·X·V
```

To hold μ constant while biomass grows exponentially, the feed must grow at the same rate, which gives the **exponential feed** $F(t) = F_0 \cdot e^{\mu t}$. The consequence is the one worth stating in an interview: **the pump, not the organism, sets the growth rate**, and that is the point — it keeps the specific uptake rate below the overflow threshold of part 1 chapter 6 and the oxygen demand below the transfer limit above.

Strategy	What it does	Failure mode
Constant feed	Fixed F; μ falls as biomass accumulates	Early overflow, late starvation; simple and rarely optimal
Exponential	$F_0 e^{\mu t}$ holds μ at a set point	Open-loop: any deviation in yield or viability compounds, and the feed outruns the culture
DO-stat	Feed when dissolved oxygen rises above a threshold (substrate exhausted)	Oscillation; relies on the DO probe; confounded if oxygen transfer changes
pH-stat	Feed on the pH rise caused by substrate exhaustion	Only works when the organism's acid production tracks substrate; base addition confounds it

Choosing from physiology rather than convention: a Crabtree-positive yeast or an *E. coli* with a sharp overflow threshold needs the specific uptake rate held below that threshold, so exponential feeding at a μ below the critical value, with DO-stat as a safety net. An organism with no overflow metabolism — *C. glutamicum*, part 6 chapter 59 — tolerates a more generous feed and may be limited by oxygen instead. An organism making a growth-decoupled product (part 5 chapter 56's PHA) wants a two-phase feed: exponential during growth, then carbon-only with nitrogen withheld.

Downstream, and the cost sketch

The unit operations in order, with the property each exploits: **centrifugation or filtration** — density or size, to separate cells from broth; **homogenisation** — mechanical, if the product is intracellular; **clarification** — size again, to remove debris; **extraction or adsorption** — partitioning (part 3 chapters 45 and 46); **chromatography** — specific binding, for high-value products; **distillation** — volatility; **crystallisation** — solubility; **drying** — volatility again. Which of these you need is set by where the product is and what it is, which is part 6 chapter 59's localisation column.

Then the water, which is the number that decides the train:

Titer	Broth per kg of product	Water to remove
5 g/L	200 L	199 kg
20 g/L	50 L	49 kg
50 g/L	20 L	19 kg
100 g/L	10 L	9 kg

At 20 g/L you move fifty kilograms of water per kilogram of product, and evaporating it would take about 113 MJ — against the 20 to 30 MJ of chemical energy in a kilogram of most organic products. That is the part 6 chapter 60 dewatering argument in a fermenter, and it is why the cost sketch of part 4 chapter 49 is dominated by feedstock and separation, and why the titer below which a process cannot work is set by the separation rather than by the biology.

The back-of-envelope, with part 4 chapter 49's model: feedstock cost is substrate price divided by yield, so at $0.40/kg glucose and 0.40 g/g it is $1.00/kg of product at the theoretical yield and $1.60 at a realistic 0.25 g/g; capital and fixed costs scale roughly as 1/titer. Setting those against the product's market price gives the titer below which the process cannot work — which for a $1.80/kg commodity in that model was around 100 g/L.

Power	`P = N_p ρ n³ d⁵; N_p ≈ 5 for a Rushton turbine in turbulent flow`
Constant P/V	`n ∝ d^(−2/3); tip speed ∝ d^(1/3); mixing time ∝ d^(2/3)`
Transfer	`OTR = k_L a(C* − C_L); C* ≈ 0.21 mM for air at 37 °C`
The wall	`50 gDW/L at 5 mmol gDW⁻¹ h⁻¹ needs k_L a ≈ 1,560 h⁻¹ against 100–500 achievable`
As a model bound	`OTR/X in mmol gDW⁻¹ h⁻¹ becomes the oxygen exchange bound`
Exponential feed	`F(t) = F₀e^(μt): the pump sets the growth rate`
Water	`1,000/titer litres of broth per kg; 50 kg of water per kg at 20 g/L`

Your 1 L fermentation reaches 40 gDW/L and 25 g/L of product. At 50 m³ the same protocol gives 22 gDW/L, 11 g/L, and acetate in the broth. Diagnose it.

Oxygen first, because the numbers point there. At 40 gDW/L and 5 mmol·gDW^{-1}·h^{-1} the demand is 200 mmol·L^{-1}·h^{-1}, needing a k_La of 1,250 h^{-1}, which a 1 L vessel at high stirrer speed can approach and a 50 m³ vessel cannot — so the large culture is oxygen-limited, and the acetate is exactly what the core model predicts when the oxygen bound is tightened, as the table above shows. The biomass stopping at 22 gDW/L is consistent: growth is limited by the oxygen the vessel can deliver.

What to check to confirm it, in order: the dissolved-oxygen trace, which should be at or near zero during the productive phase; the off-gas, for an oxygen uptake rate that plateaus while the feed continues; and the acetate against the model's prediction at the measured OTR. Then the fixes, cheapest first — reduce the feed rate so the specific uptake rate falls below what the vessel can support, which costs time; enrich the inlet oxygen or raise the head pressure, which costs utilities; or accept a lower cell density and raise the specific productivity instead. And the thing not to do is to assume the strain degenerated: the gradient and transfer explanation is testable in a scale-down reactor before any strain work begins.

Three things people get wrong

- ‘Hold power per volume constant and the scale-up is matched.’ Mixing time still grows as $d^{2/3}$ — 13-fold from 1 L to 100 m³.
- ‘Add more air.’ k_La rises sublinearly with power and gas flow; at 50 gDW/L you need 1,560 h^{-1}, which air at atmospheric pressure cannot give.
- ‘The strain degenerated at scale.’ Check the oxygen and the gradients first — both are testable in a 1 L scale-down vessel.

Self-check — answer on paper first

1. Derive why mixing time grows with scale at constant power per volume.

Power is $P = N_p \rho n^3 d^5$ and, under geometric similarity, volume scales as d^3, so $P/V \propto n^3 d^2$. Holding that constant requires $n \propto d^{-2/3}$. Mixing time for a given geometry scales roughly as $1/n$, so it scales as $d^{2/3}$ — a 45-fold increase in impeller diameter, from 4 cm to 1.8 m, gives a mixing time about 12.7 times longer. Tip speed $nd \propto d^{1/3}$ rises at the same time, so the larger vessel is simultaneously worse mixed and more shear-intense at the impeller.

2. A culture needs 150 mmol·L^{-1}·h^{-1} of oxygen and your vessel manages k_La = 400 h^{-1}. What are the options, quantitatively?

With $C^* = 0.21$ and $C_L = 0.05$ mM the available OTR is $400 \times 0.16 = 64$ mmol·L^{-1}·h^{-1}, well short of 150. Enriching the inlet gas to 50% oxygen raises C^* to about 0.5 mM, giving a driving force of 0.45 and an OTR of 180 — sufficient. Raising the head pressure to 2 bar absolute roughly doubles C^*, giving about 134 — nearly sufficient, and combinable with modest enrichment. Raising power would need k_La up by a factor of 2.3, which at an exponent near 0.5 on P/V means about five times the power, with the shear and cost that implies. Or reduce the demand: feed more slowly, which lowers the specific uptake rate and is usually the cheapest answer.

3. Why does an exponential feed hold the growth rate constant, and what is its failure mode?

Because at quasi-steady state the substrate supplied per unit time must match consumption, which is $(\mu/Y_{XS} + m) \cdot X \cdot V$; with X growing as $e^{\mu t}$, matching requires the feed to grow at the same exponential rate, so $F(t) = F_0 e^{\mu t}$ fixes μ at the chosen value. The failure mode is that it is open-loop: it assumes the yield, the maintenance coefficient and the viability stay as designed, and if the culture underperforms — a lower yield, a stress response, partial loss of viability — the feed keeps rising on schedule while consumption does not, substrate accumulates, and the culture tips into overflow metabolism or substrate inhibition. Hence a DO or substrate measurement as a safety net.

What to open a book for. Doran's *Bioprocess Engineering Principles*, mass-transfer and reactor chapters, worked rather than read — this week is the one where working the problems is the content. Shuler, Kargi & DeLisa on reactor operation and scale-up, Blanch & Clark for depth, a scale-up and scale-down review for the gradient material, and your own transport phenomena notes, which cover most of the theory already.

Closing notes to part 11

The six weeks in one page

Week 39. Growth coupling is the lower edge of the production envelope. An enumerated search found no coupled design aerobically and 52 coupled pairs anaerobically; ΔPYK gives 8.23 mmol succinate at μ 0.134, the triple 9.36 at 0.048 — more coupling, less growth. Capping the product export at 6 kills growth and below 4 there is no feasible solution at all. A crude proteome budget reproduces overflow and falling oxygen uptake.

Week 40. Rule-based expansion is combinatorial, so ranking matters more than generation, and the ranking that matters is enzyme availability and length: a route of n steps each with probability p works with probability p^n — 33% at five steps and p = 0.8. A step with no candidate enzyme is a discovery project, not a low probability.

Week 41. Bayesian optimisation reached 98% of the achievable titer in eleven runs where a 16-run factorial grid reached 78% and never more, across forty seeds — and one seed in forty stuck at 69% on a local optimum. Report the spread.

Week 42. Twenty samples and sixty features gave a training R^2 of 1.00 and a held-out R^2 of 0.19; with shuffled labels, 1.00 and −1.67. The same family-structured dataset scored 0.67 under random cross-validation and −0.44 when grouped by family. The split must match what you need to predict.

Week 43. [1-^{13}C] glucose reports the glucose-6-phosphate split because carbon 1 is the carbon the oxidative branch decarboxylates; fitting a split from MIDs with 0.4% noise gave 0.298 with a bootstrap interval of 0.289–0.312. Natural abundance alone gives 3.2% M+1 for a three-carbon fragment. A narrow interval on a badly specified network is worth nothing.

Week 44. At constant power per volume, mixing time grows as $d^{2/3}$ — 12.7-fold from 1 L to 100 m³. A culture at 50 gDW/L needs a k_La of 1,560 h^{-1} against 100–500 achievable. Feeding that transfer limit into the core model as an oxygen bound: at k_La 100 the model loses 58% of its growth and excretes 12 mmol of reduced by-products; at 1,500 it reaches the unconstrained optimum.

On the numbers in this part

The strain-design search, the proteome-constrained sweep, the Bayesian optimisation comparison, the overfitting and leakage simulations, the ^{13}C fit and the oxygen-bound sweep were all executed. Their parameters are the assumptions: the core model of part 10 with its P/O ratios and rounded biomass coefficients; enzyme costs in the proteome constraint that are illustrative rather than measured, so the onset of overflow is the result and the behaviour at high uptake is an artefact of a hard budget; a response surface, noise level and kernel in chapter 41 that I chose; a feature count and signal structure in chapter 42 that set how fast the learning curve flattens; a deliberately minimal atom-transition model in chapter 43, where a real analysis uses the full network and software I could not run; and in chapter 44 a Rushton power number, a specific oxygen uptake of 5 $mmol \cdot gDW^{-1} \cdot h^{-1}$, and C* = 0.21 mM.

Not run, and marked where they appear: RetroPath2.0 and any retrosynthesis tool, the ^{13}C fitting packages, and COBRApy. The two results I would most want checked against real data are the proteome cost vector, because it decides where overflow begins, and the specific oxygen uptake rate, because every k_La number in chapter 44 scales with it.

What remains

Written so far: weeks 1 to 44 and 45 to 61 — 61 of the tracker's 66. What is left is weeks 62 to 66: how publishing works, writing a proposal, the application and interview phase, and the capstone. They are about how the profession works rather than about metabolic engineering, and as I said when part 6 went out, they are the ones nobody writes down for an engineer arriving from another discipline. That is the last part.

Part 3 — Getting the product out: toxicity, in situ removal, two-phase modelling and immobilisation (weeks 45 to 48)

These four weeks have no textbook. Not in the sense that the books are old or badly pitched, which is the complaint about phase 2, but in the sense that no book covers them: in situ product removal lives in a few hundred papers, two decades of the Daugulis group's polymer work, and scattered chapters of bioprocess texts that treat it as an aside. The modelling in week 47 exists mostly as supplementary material. So this part is the primary text rather than a spine, and it is written accordingly — the derivations are here in full, the models are complete enough to run, and where I give a number I also give the parameters that produced it.

The good news is that this is your own subject. Partition coefficients, isotherms, intraparticle diffusion, Thiele moduli, effectiveness factors, washout and dilution rate are chemical engineering, and the biologists in the field are mostly learning them from scratch. If you want one block of the curriculum where a chemical engineer arrives with an unfair advantage, it is this one. The things to be careful about are not the transport calculations but the biology wrapped around them: that the cells are in one phase and not the other, that a solvent can be eaten, and that a bead full of cells runs out of oxygen long before it runs out of sugar.

A warning about vocabulary, because the literature is loose with it and the confusion is the commonest error in this area. In situ product removal is the *goal*. A two-phase partitioning bioreactor is one *reactor* that achieves it. Liquid–liquid means a solvent; solid–liquid means polymer beads or resin. Immobilising the cells is a different thing entirely, aimed at a different problem, and it is week 48. Keep those separate and half the papers become easier to read.

Ch.	Week	Chapter
45	45	Product toxicity, export and loss: deciding what actually limits the titer
46	46	In situ product removal: choosing a strategy, and sizing a sorbent
47	47	Modelling a two-phase fermentation, and fitting it to data
48	48	Cell immobilisation: diffusion, dead cores and continuous operation

Product toxicity, export and loss: deciding what actually limits the titer

Three different things stop a titer, they look identical on an HPLC trace, and they call for three different interventions. Before choosing a lever, find out which one you have — it takes a week and saves a year.

The three limits, and the experiments that separate them

Toxicity means the product inhibits the cells: growth slows, specific productivity falls, and the culture plateaus with substrate left in the broth. **Export** means the product is made but cannot leave: intracellular concentration rises, flux falls as the pathway is inhibited or the exporter saturates, and the broth titer plateaus at a value that is low relative to what the cells contain. **Loss** means the product leaves the vessel or decomposes: production continues at full rate and the titer plateaus anyway, because what you are measuring is the balance between making and losing.

All three produce the same curve. Three experiments distinguish them, and none takes more than a few days.

- **Spike the product into the non-producing host.** Grow the parent strain with product added at the concentrations you are reaching, and measure growth rate and, if you can, the specific productivity of some other function. If growth falls at the titer you plateau at, toxicity is real; if it does not, toxicity is not your limit, whatever the review articles say about this compound class.
- **Measure intracellular and extracellular product separately.** Quench, separate cells from broth quickly, extract both. A high intracellular concentration with a modest broth titer is an export limit. The measurement is fiddly — the product partitions during the separation — but the answer is unambiguous enough to be worth it, and it is the only direct evidence for an export limitation.
- **Compare a closed vessel with a vented one.** For anything volatile, run the same fermentation with the off-gas condensed or trapped and measure what the trap caught. The difference between the broth titer and the total made is the loss, and the experiment also tells you how much of your "poor strain" was actually a collection problem.

Write down, before you run them, what each result would mean and what you would do next in each case. That discipline — the decision tree first, the experiment second — is the difference between diagnosis and data collection, and it is the habit this week exists to install.

Why hydrophobic products are toxic, and what log P is for

A hydrophobic product does not poison a specific target. It partitions into the lipid bilayer in proportion to its hydrophobicity, swells and fluidises the membrane, and the membrane stops holding a proton gradient. That mechanism explains the pattern you will see in the literature: toxicity tracks the compound's hydrophobicity rather than its chemistry, with the worst offenders being small amphiphilic molecules — short-chain alcohols, terpene alcohols, phenolics — that are soluble enough in water to reach the cells and hydrophobic enough to accumulate in the membrane once there. A very hydrophobic compound, such as a long-chain hydrocarbon or a carotenoid, is often less toxic simply because its aqueous concentration can never get high.

Hence the two numbers of this week, which are routinely conflated and are not the same thing. **The solvent's log P** — its own octanol–water partition coefficient — is a rule-of-thumb guide to whether the cells tolerate the solvent: below about 2 solvents tend to be toxic, above about 4 they tend to be tolerated, with a grey zone between and genuine exceptions. **The product's partition coefficient K** between the solvent and water is what decides how much product the second phase captures. You need a solvent that scores well on both, and the two are only loosely related, so a solvent chosen for one can fail on the other. Write both numbers down for every candidate solvent before you order any of them.

Efflux pumps, and the honest record

If the limit is export or toxicity, a pump is the obvious genetic lever. Three families matter.

Family	Architecture	Energy	Notes for a production strain
RND	Tripartite: inner-membrane pump, periplasmic adaptor, outer-membrane channel (AcrAB–TolC)	Proton motive force	Gram-negatives only; spans both membranes, so it exports to the medium in one step. All three components must be present and balanced
MFS	Single inner-membrane protein	Proton motive	Widely distributed, simple to express; in Gram-negatives it only reaches the periplasm unless coupled to an outer-membrane partner

		force	
ABC	Transmembrane domains plus nucleotide-binding domains	ATP	Works in eukaryotes; in yeast the pleiotropic drug resistance pumps are the usual candidates. Direct ATP cost, so it shows in the energy balance

The result to know is Dunlop and colleagues' 2011 screen: 43 pumps tested against seven biofuel compounds in *E. coli*. Pumps that improved tolerance were found for five of the seven, and none helped for n-butanol or isopentanol. Read that as the field's honest base rate. Pump engineering is a screening problem, not a design problem: you cannot predict from a compound's structure which pump will move it, so you screen a panel, and sometimes the answer is that no pump in the panel works.

Pumps also cost. Expression burden is the obvious one — these are large membrane proteins, and over-expressing a membrane protein competes for the insertion machinery as well as the proteome. Substrate range is the subtler one: a pump that exports your product may also export a nutrient or an intermediate, and a tripartite pump needs its adaptor and channel at matched levels or the assembled complex is a leak in the outer membrane rather than a pump. Measure whether it paid the way you would measure any intervention: growth rate with and without, product in and out of the cell, and the titer against the same strain without the pump.

Membrane composition is the other lever, and it is worth knowing which parts are established. Shifting the saturated-to-unsaturated fatty acid ratio and cis–trans isomerisation of unsaturated chains are real, documented adaptations to solvents in several bacteria and have been used deliberately. Hopanoid and sterol engineering is research-stage. Treat the first as an option and the second as a literature you can cite rather than a tool you can use.

The second phase, arithmetically

Now the calculation that makes this week quantitative. Put a volume fraction φ of solvent into the vessel, with the product partitioning at $K = C_{org}/C_{aq}$ at equilibrium. The cells live in the aqueous phase and see only C_{aq}, so if toxicity caps the aqueous concentration at P*, the total product per litre of aqueous phase is

```
P_total/L_aqueous = P* · R,        R = 1 + K·φ/(1 − φ)
```

and per litre of whole broth — the number a plant cares about — it is P*·[(1 − φ) + Kφ]. Those two differ by a factor of (1 − φ) and mixing them up is the commonest arithmetic error in this subject; say per litre of what, every time.

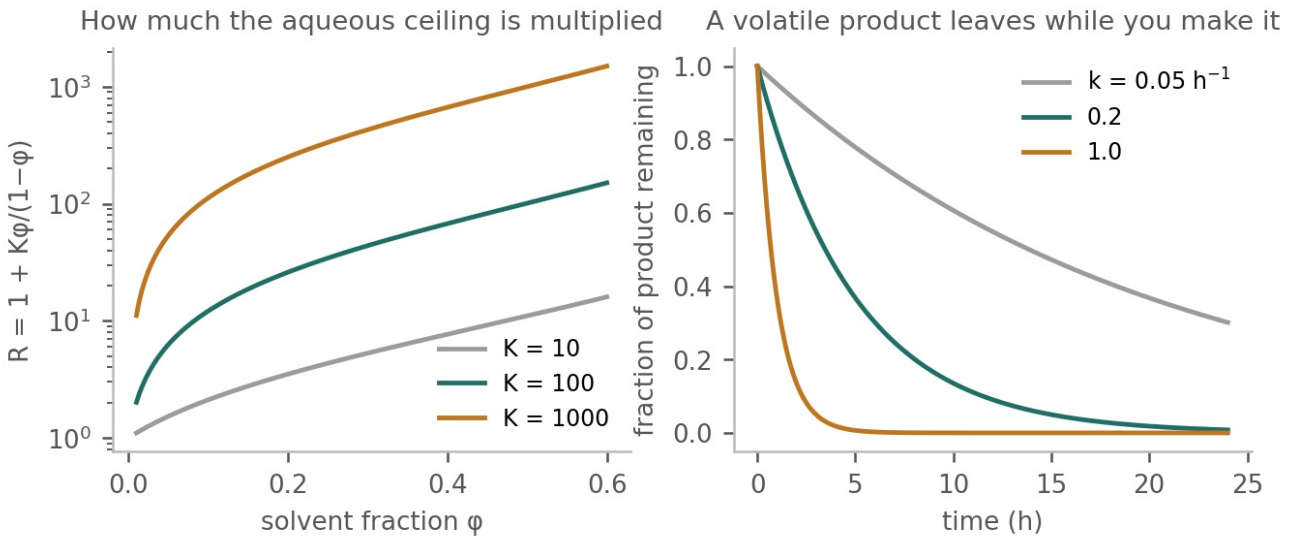


Figure 45.1 Left: the factor by which a second phase multiplies the aqueous ceiling, R = 1 + Kφ/(1 − φ), for three partition coefficients. Right: the fraction of a volatile product remaining in the broth under first-order loss at three rate constants. Both are arithmetic; the loss constants are illustrative, and measuring your own is a one-day experiment.

The numbers are striking and that is the trap. At K = 100 and φ = 0.2, R = 26: an aqueous ceiling of 2 g/L becomes 52 g/L per litre of aqueous phase, or 41.6 g/L of whole broth. At K = 1,000 and φ = 0.3, R = 430. Three things stop this being free money, and you should check all three before quoting such a number.

- **Solubility in the solvent.** R assumes the organic phase can hold $K \cdot C_{aq}$. At K = 1,000 and an aqueous concentration of 1 g/L, the organic phase must hold 1 kg/L, which no solvent can. The capture is capped by the product's solubility in the solvent, and above that the partition coefficient is irrelevant.

- **Displaced working volume.** The cells are in the aqueous phase, so a 30% solvent fraction removes 30% of your productive volume. Per litre of whole broth, output rises with φ at first and then falls; chapter 47 computes where the optimum sits for one specific model, and it was at a lower φ than the partition arithmetic alone suggests.
- **Everything else the solvent does.** Oxygen transfer through an organic layer, emulsions that will not separate, foaming, solvent loss to the off-gas, and the recovery step that has to get your product back out of the solvent. These are the subject of chapter 46 and week 49.

Loss, and the titer you measure from a culture that is producing well

For a volatile product, treat loss as first order in the broth concentration with a rate constant k that bundles stripping by the air flow, headspace loss and any degradation. Then a culture producing at a constant volumetric rate q reaches a steady titer of q/k, not a rising one. At q = 0.5 g/L/h, a loss constant of 0.05 h^{-1} gives 10 g/L, 0.2 h^{-1} gives 2.5 g/L, and 1 h^{-1} gives 0.5 g/L. The strain is identical in all three cases.

That is the single most useful equation in this chapter, because it says a plateau is not evidence that production stopped. The fraction remaining after a hold with no production is e^{-kt}: at k = 0.2 h^{-1}, 82% survives an hour, 30% survives six hours and essentially nothing survives overnight. So a sample held on the bench, or a fermentation sampled at the end of a shift, can misreport a good strain as a poor one — which is why the closed-vessel comparison is on the list of three experiments above, and why aeration rate belongs in the methods section of any volatile-product paper.

What the solvent costs

Solvent inventory is φ·V, so a 100 m³ vessel at φ = 0.2 holds 20 m³ of solvent. At a dodecane-like price of a few dollars a litre that is tens of thousands of dollars of inventory, which is acceptable as a one-off and ruinous if it is consumed. The cost per kilogram of product therefore depends almost entirely on how many cycles the solvent survives and how much is lost per cycle: inventory price divided by (kilograms of product per cycle × number of cycles), plus the make-up for losses. Work it both ways — perfect recycling and 5% loss per batch — and you will usually find that loss, not purchase, dominates. This is the same arithmetic as the resin cycles in chapter 46 and it is worth doing before the lab work rather than after.

Capture factor	`R = 1 + K·φ/(1 − φ), per litre of aqueous phase`
Per litre of broth	`P_total = P*·[(1 − φ) + K·φ]`
Volatile steady state	`P_ss = q/k; a plateau is loss, not a stopped pathway`
Hold-time loss	`P(t) = P₀·e^(−kt)`
Two different numbers	`solvent log P: will the cells live. product K: will it be captured`
Solvent cost	`inventory φ·V; cost/kg ≈ price/(kg per cycle × cycles) + make-up`

Classify four products by their main limit, and say what each calls for: a limonene-like monoterpene, a volatile sesquiterpene, a short-chain organic acid, and a flavonoid.

Monoterpene. Toxicity, and severe: it is hydrophobic enough to load the membrane and soluble enough to reach it, which is the worst combination, and aqueous ceilings of tens to a few hundred milligrams per litre are usual. Calls for a second phase from the start — an overlay is standard — plus tolerance work; a pump is worth screening but the base rate in the Dunlop data is sobering.

Volatile sesquiterpene. Loss first, toxicity second. It is less water-soluble than the monoterpene, so the cells see less of it, but it leaves with the off-gas. Calls for an overlay, which conveniently solves both — this is exactly the two-phase amorphadiene work — and for the closed-vessel experiment before concluding anything about the strain.

Short-chain organic acid. Export and toxicity, and a solvent is the wrong answer: it is charged at fermentation pH, so it does not partition into an organic phase at all. Calls for the week 14 analysis — an exporter is mandatory, diffusion will not do it — plus pH control, and if removal is wanted, ion exchange or electrodialysis rather than a solvent.

Flavonoid. Usually neither toxicity nor loss: the limits are precursor supply and, often, degradation or oxidation in the broth. Calls for checking stability in medium before anything else — spike it, incubate it without cells, and measure the decay — because if the half-life in the broth is hours, an adsorbent resin earns its place and no amount of pathway work will.

Three things people get wrong

- ‘The titer plateaued, so the product is toxic.’ Plateau is equally consistent with loss at q/k, or with an export limit. Three experiments distinguish them.

- ‘High log P means the solvent captures the product well.’ log P describes the solvent's tolerance by the cells; K describes capture. Different numbers.
- ‘Add a pump.’ The published screen found nothing useful for two of seven biofuels. A pump is a hypothesis to screen, not a fix to apply.

Self-check — answer on paper first

1. Your culture plateaus at 1.2 g/L with 40 g/L glucose left. Give the three candidate explanations and the cheapest experiment for each.

Toxicity — spike 1.2 g/L of product into the parent strain and measure growth rate. Export limitation — quench and measure intracellular against extracellular product. Loss — repeat with the off-gas trapped, or simply measure the product's half-life in medium at process temperature with no cells. The residual glucose is itself informative: it says the cells stopped consuming, which points toward toxicity or a stalled pathway rather than pure loss, since a culture losing product to the headspace would keep eating.

2. A solvent has log P = 1.8 and your product partitions into it at K = 400. Would you use it?

Not directly. A log P below about 2 puts it in the range where solvents are usually toxic, and the excellent partition coefficient does not help if the cells die — and worse, a solvent in that range is often water-miscible enough to reach the cells rather than staying as a separate phase. Options: find a more hydrophobic solvent with an acceptable K, use the toxic solvent in an external loop where the cells never contact it, or use a solid sorbent instead, which sidesteps solvent toxicity entirely and is chapter 46's main subject.

3. Derive, in one line, why the total product per litre of whole broth can fall as you add more solvent even though R rises.

Because the cells occupy only the aqueous fraction: output per litre of broth is proportional to $(1 - \varphi)$ times what the aqueous phase can hold, which is $P^{*} \cdot R$. R rises roughly as $K\varphi$ while the productive volume falls as $(1 - \varphi)$, so the product of the two has a maximum at an intermediate φ, and beyond it the displaced working volume costs more than the extra capture earns.

What to open a book for. There is no book. Read Dunlop et al., Mol. Syst. Biol. (2011) for the pump screen and its negative results, and the two amorphadiene papers — Newman et al., Biotechnol. Bioeng. (2006) and Tsuruta et al., PLoS ONE (2009) — for a two-phase process reported with real numbers. Doran's chapters on mass transfer are worth having open for the transport side, and week 14 of this module is the other half of the export argument.

In situ product removal: choosing a strategy, and sizing a sorbent

The product's own properties choose the removal method. Partition coefficient, volatility, charge, solubility and heat sensitivity decide it — not what the last paper you read happened to use.

Four reasons to remove a product as it is made

They are different reasons, they are often confused, and the one that applies determines what counts as success.

- **It inhibits the cells.** Removal raises the achievable total by keeping the aqueous concentration below the inhibition threshold. Success is measured as total product, and the control that matters is the same strain without removal.
- **It is volatile and lost.** Removal is capture: you were making it all along and now you keep it. Success is measured as recovery, and a large apparent "improvement" here is really a correction to your previous measurement.
- **It degrades in the broth.** Removal shortens the time the product spends in a hostile aqueous environment. Success is measured as product surviving to harvest, and the relevant pre-experiment is the half-life in medium without cells.
- **It inhibits its own pathway, or holds back an equilibrium.** Removal lowers Q in the week 14 sense and pulls the reaction forward. Success shows as a higher rate, not merely a higher total, and the diagnostic is whether the intracellular intermediate pools fall.

Note that only the first and fourth change what the cells do. The second and third change what you collect. A paper that conflates them will report a fivefold improvement that is partly bookkeeping, which is one of the questions to ask of the paper you read on day 5.

The menu, chosen by the product

Method	Suits a product that is	Main costs and failure modes
Organic overlay (liquid–liquid)	hydrophobic, uncharged, with a high K into a tolerable solvent	solvent toxicity and loss, emulsions, oxygen transfer through the layer, recovery by distillation
Polymer beads or resin (solid–liquid)	moderately hydrophobic; also where a solvent would be toxic or eaten	capacity ceiling, intraparticle diffusion, fouling, cost per cycle, abrasion
Gas stripping	volatile, with a reasonable Henry coefficient	huge gas flows, foaming, water and substrate co-stripping, condensation duty
Pervaporation	volatile, where gas flow is unacceptable	membrane area, fouling, modest fluxes, capital
External column in a loop	anything the cells must not contact the capture phase for	pumping, cells in the loop, shear, control of two coupled systems
Crystallisation / precipitation	sparingly soluble at some pH or temperature	nucleation control, cells in the solids, filtration
Ion exchange / electrodialysis	charged at fermentation pH — acids and amino acids	competing ions in the medium, regeneration chemicals, salt disposal

The single most useful line in that table is the last one. If your product is charged at the fermentation pH — an organic acid, an amino acid, a phosphorylated compound — a solvent is almost always the wrong answer, because the species in the broth does not partition into an organic phase at all. People try it anyway, because overlays are what the ISPR literature is mostly about, and then report a disappointing partition coefficient. Check the pK_a against the fermentation pH first; it takes one line, and it eliminates the largest category of wasted effort in this subject.

The second most useful line is the beads one. A solid phase cannot be metabolised, cannot be toxic in the way a solvent is, and can be used with mixed or undefined cultures where a solvent cannot. That is why the polymer-bead literature exists and why it is worth more of your attention than its citation count suggests.

Isotherms, and the sorbent mass you actually need

Two isotherms cover nearly everything. The linear one, $q = K_p \cdot C$, with loading q in grams per kilogram of sorbent and C the aqueous concentration, is the low-concentration limit and is adequate when you operate far below saturation. The Langmuir form,

```
q = q_max · C / (K_d + C)
```

adds the thing the linear form hides: a capacity ceiling. q_{max} is the loading at saturation, and K_d is the aqueous concentration at half of it — which, usefully, is the same algebra as Michaelis–Menten with the same two readings. Below K_d the sorbent behaves linearly with $K_p \approx q_{max}/K_d$; above it, adding sorbent is the only way to take up more.

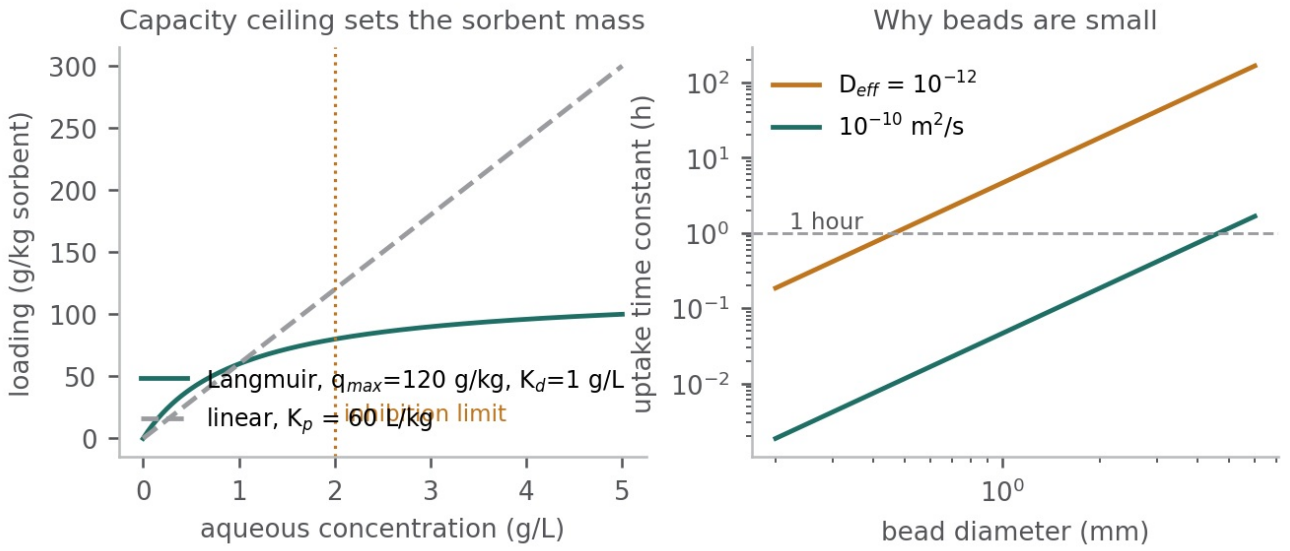


Figure 46.1 Left: a Langmuir isotherm with q_{max} = 120 g/kg and K_d = 1 g/L against its low-concentration linear approximation, with the aqueous inhibition limit marked; the sorbent must do its work to the left of that line. Right: the intraparticle uptake time constant $R^2/15D_{eff}$ against bead diameter for two effective diffusivities spanning what polymers give. Both computed for this module; the isotherm parameters are illustrative and the diffusivities are order-of-magnitude class values.

The sizing calculation is then three lines, and it is the one to do before ordering anything. You must keep the aqueous concentration at or below the inhibition limit — say 2 g/L — so the loading you can rely on is the isotherm evaluated there: with the figure's parameters, q = 120 × 2/(1 + 2) = 80 g/kg. To hold one kilogram of product you therefore need 1,000/80 = 12.5 kg of sorbent. For a 100 m³ fermentation making 20 g/L — two tonnes of product — that is 25 tonnes of resin, which is where the arithmetic usually stops a project unless the resin is cheap and reusable.

Hence the cycle question, which is the same as the solvent arithmetic in chapter 45. A resin at $50/kg used for one batch adds $625 per kilogram of product and is absurd for anything but a pharmaceutical; the same resin surviving 100 cycles adds $6.25, which is tolerable for a specialty. So the number to establish experimentally is not only the isotherm but the loading retained after tens of cycles of adsorption, elution and regeneration in real broth. Fouling by medium components, cell debris and antifoam reduces capacity cycle by cycle, and a resin that loses 2% of its capacity per cycle is finished in 35 cycles whatever its catalogue specification says.

Inside the bead: equilibrium limit against rate limit

Everything above is equilibrium. Whether equilibrium is reached is a transport question, and for a solid sorbent the resistance is almost always inside the polymer rather than in the liquid film outside it, because the solute's effective diffusivity in a dense polymer is orders of magnitude below its value in water. The characteristic time for uptake into a sphere is

```
τ ≈ R² / (15·D_eff)
```

and the figure plots it. For a 1 mm bead, τ is about 3 minutes at $D_{eff} = 10^{-10}$ m²/s and about 4.6 hours at 10^{-12}; for a 3 mm bead the same two diffusivities give 25 minutes and 42 hours. Since the production timescale is tens of hours, a 1 mm bead is at equilibrium with the broth throughout and the equilibrium arithmetic above holds, while a 3 mm bead in a slow polymer is not — its interior is still filling when you harvest, so you have paid for capacity you never used.

This is why beads are small, and it is also the stress test worth running on any ISPR design: double the production rate and ask whether the sorbent can still keep up. Capacity is a stock and uptake is a flow; a design can be sized correctly on the stock and fail on the flow. The rough criterion is to compare the volumetric production rate with the sorbent's uptake rate per unit volume of broth, which for a linear isotherm is roughly (phase fraction)·(K_p/τ)·(aqueous concentration). If production exceeds that, the aqueous concentration rises above equilibrium and the cells see more than you designed for — the same phenomenon that chapter 47 models properly for a liquid phase.

The polymer choice itself is a screening problem rather than a design one. Partition coefficients for one compound differ by orders of magnitude between commercial polymers, which is the central finding of the solid–liquid two-phase literature, and the differences are not reliably predictable from the product's structure. So the method is to screen a panel of a dozen polymers at a few grams each, measure the isotherm of the two or three best in real medium rather than buffer, and only then size the reactor.

Recovery, reuse, and the external loop

The product has to come back off. From a solvent, that is usually distillation or evaporation, and the governing question is whether the product is more or less volatile than the solvent and whether it survives the temperature. From a resin, it is elution with a solvent or a pH or temperature swing, and the governing question is the volume of eluate: a resin that needs ten bed volumes of methanol to strip has moved your separation problem rather than solved it. From a gas stream, it is condensation or an absorber, and the duty is set by the gas flow, which is why stripping is cheap in principle and expensive in practice.

Internal versus external is the last structural choice. An internal phase — solvent or beads in the vessel — is simple, has no pumping, and exposes the cells to the capture phase, whose abrasion can shear them and whose presence complicates oxygen transfer and foam. An external column in a recirculation loop keeps the capture phase away from the cells, can be regenerated without stopping the fermentation, and allows a toxic solvent to be used at all; the costs are the pump, the cells that spend time in the loop under different oxygen and shear conditions, the dead volume, and the fact that you are now controlling two coupled systems. For a toxic solvent or an expensive resin the loop usually wins; for a benign overlay it rarely does.

Linear isotherm	`q = K_p·C, with K_p ≈ q_max/K_d at low C`
Langmuir	`q = q_max·C/(K_d + C); the ceiling is what sizes the sorbent`
Sorbent mass	`kg sorbent per kg product = 1000 / q(C_inhibition)`
Cost per kg	`resin price / (kg product per cycle × cycles survived)`
Uptake time	`τ ≈ R²/(15·D_eff): 3 min to 4.6 h for a 1 mm bead`
Charged product	`pKa versus fermentation pH decides whether a solvent can work at all`

A paper reports that adding 10% w/v of a polymer raised the titer of a phenolic from 0.9 to 4.1 g/L, a 4.6-fold improvement. What do you check before believing it is a 4.6-fold improvement in production?

First, what the reported titer is per litre of. If the 4.1 g/L counts the product extracted from the polymer and is expressed per litre of aqueous phase while the 0.9 g/L is per litre of broth, part of the gain is bookkeeping. Second, the control: is it the same strain in the same vessel without polymer, run for the same time, or is it a historical number from a different run? Third, whether the product was previously being lost or degraded — if the phenolic has a few hours' half-life in medium, part of the gain is recovery rather than production, which is still useful but is a different claim. Fourth, substrate: if the no-polymer run left substrate unused and the polymer run consumed it all, production genuinely increased and the yield comparison will show it.

Then the scale-up questions the paper almost certainly does not answer: oxygen transfer with 10% solids in the vessel, bead abrasion and fines over a long run, fouling in real medium, how the product comes off the polymer and at what eluate volume, and the polymer cost per kilogram of product at a realistic number of cycles. None of these invalidates the result; all of them decide whether it matters.

Three things people get wrong

- ‘ISPR means a solvent.’ It means removing the product as it is made; beads, stripping, crystallisation and ion exchange are all ISPR, and for a charged product a solvent is the one that cannot work.
- ‘The isotherm is linear.’ It is, until you reach the capacity ceiling — which is exactly where a correctly sized system operates.
- ‘Capacity is what matters.’ Capacity is a stock and uptake is a flow. A big slow bead fails at the flow while looking fine on the stock.

Self-check — answer on paper first

1. Your product inhibits at 3 g/L aqueous and your sorbent gives 50 g/kg at that concentration. How much sorbent does a 10 m³ fermentation need to reach 30 g/L of whole broth, and what does that tell you?

Thirty grams per litre of 10 m³ is 300 kg of product; at 50 g/kg that is 6 tonnes of sorbent, which in a 10 m³ vessel is physically impossible — the solids would occupy more than the working volume. The calculation tells you the design is

wrong at the concept stage: either the sorbent must have a far higher loading at 3 g/L, or the removal must be external with continuous regeneration so that the same sorbent is reused many times within one batch, or the target titer must come down. This is the normal outcome of the sizing calculation and it is why it is worth doing first.

2. Distinguish an equilibrium limit from a rate limit in a solid-phase ISPR, and give the experiment that tells you which you have.

An equilibrium limit means the sorbent is saturated at the aqueous concentration you are holding — more sorbent would help, smaller beads would not. A rate limit means the sorbent is far from saturated but cannot take product up fast enough, so the aqueous concentration rises above the equilibrium value — smaller beads or more sorbent surface would help, more capacity would not. The experiment: harvest beads during the run and measure their loading against the isotherm prediction for the aqueous concentration at that moment. Loading at the predicted value means equilibrium; loading well below it means the rate limits.

3. Why can a solid phase be used with a mixed culture where a solvent usually cannot?

Because a solvent is itself an organic compound available to the organisms: in a mixed or undefined culture something will metabolise it, which consumes the capture phase, generates biomass and by-products, and changes the partitioning as the solvent composition drifts. A crosslinked polymer is not bioavailable, so it cannot be consumed, and it also cannot dissolve into the aqueous phase to reach the membrane. The trade is lower capacity per unit volume and intraparticle diffusion instead of the fast equilibration of a dispersed liquid.

What to open a book for. Again, no book. The Daugulis group's polymer papers — Prpich and Daugulis on phenol delivery (Biotechnol. Prog. 2004) and on 3-methylcatechol (Biotechnol. Bioeng. 2007), and Rehmann, Sun and Daugulis on polymer selection — are the primary literature and are short. For the transport side, any mass-transfer text's treatment of adsorption in porous media covers intraparticle diffusion better than any biology source will.

Modelling a two-phase fermentation, and fitting it to data

This is the week where your own discipline does the work. The model is four ordinary differential equations, the errors are all in the volume bookkeeping, and the honest conclusion is usually that the data identify fewer parameters than the paper claims.

The single-phase model, and the three rate laws in it

Start with one phase. Three constitutive relations and three balances:

```
μ   = μ_max · S/(K_S + S) · (1 − P_a/P*)        growth: Monod, with Levenspiel product inhibition
q_P = α·μ + β·(1 − P_a/P*)·S/(K_S + S)          production: Luedeking–Piret, growth- and non-growth-associated
dX/dt  =  μ·X
dS/dt  = −(μ/Y_XS + q_P/Y_PS)·X
dP_a/dt =  q_P·X
```

Each parameter does one thing, and you should be able to say what. μ_{max} and K_S are the growth kinetics; P* is the aqueous concentration at which growth stops, so it is the inhibition ceiling; α is the part of production that happens only while growing and β the part that continues in stationary phase; Y_{XS} and Y_{PS} partition substrate between biomass and product. The Levenspiel factor is the simplest inhibition form that goes to zero at a finite concentration rather than asymptotically, which is what toxicity actually looks like.

Run it with the parameters in the listing below — μ_{max} 0.40 h^{-1}, P* 2 g/L, 120 g/L substrate — and the answer is the characteristic shape of a toxicity-limited process: the aqueous product rises to 2.0 g/L, growth and production stop, and **91 of the 120 g/L of substrate is left in the broth**. The strain is not the problem; the ceiling is.

Before interpreting anything, check the substrate balance closes: substrate consumed should equal biomass formed over Y_{XS} plus product formed over Y_{PS}. In the run above it closes to better than 0.01%, which is what you want to see. That check catches sign errors and missing terms and proves nothing about whether the model is right — it is an arithmetic check, not a validation, and the distinction is worth stating in your own methods sections.

Adding the second phase, and the bookkeeping where the errors live

The second phase holds product and nothing else: no cells, no substrate. Transfer between the phases is driven by the distance from equilibrium, not by the concentration itself —

```
N = k_L a · (P_a − P_o/K)        mass transfer, per litre of aqueous phase per hour
```

— and the two new balances are

```
dP_a/dt = q_P·X − N
dP_o/dt = N · (1 − φ)/φ
```

That (1 − φ)/φ factor is the whole difficulty. N was defined per litre of aqueous phase; the organic phase is smaller or larger than the aqueous phase by exactly that ratio, so the same quantity of product arriving in the organic phase changes its concentration by a different amount. Get the factor upside down and the model runs, conserves nothing, and gives plausible-looking curves. Hence the rule: write on the page what each concentration is per litre of, and check total product — $P_a(1 − φ) + P_o φ$ per litre of whole broth — against cumulative production.

```
import numpy as np
from scipy.integrate import solve_ivp

P = dict(mumax=0.40, Ks=0.5, Yxs=0.40, Yps=0.30, alpha=0.10, beta=0.05,
         Pstar=2.0, S0=120.0, X0=0.10, K=100.0, phi=0.20, kla=50.0)

def rhs(t, y, p):
    X, S, Pa, Po = y
    S   = max(S, 0.0)
    inh = max(0.0, 1.0 - Pa/p['Pstar'])
    sat = S/(p['Ks'] + S)
    mu  = p['mumax']*sat*inh
    qp  = p['alpha']*mu + p['beta']*sat*inh
    N   = p['kla']*(Pa - Po/p['K']) if p['phi'] > 0 else 0.0
    return [mu*X,
            -(mu/p['Yxs'] + qp/p['Yps'])*X,
            qp*X - N,
            N*(1 - p['phi'])/p['phi'] if p['phi'] > 0 else 0.0]

s = solve_ivp(rhs, [0, 80], [P['X0'], P['S0'], 0.0, 0.0], args=(P,),
              method='LSODA', rtol=1e-9, atol=1e-11, dense_output=True)
X, S, Pa, Po = s.y[:, -1]
print(Pa, Po, Pa*(1-P['phi']) + Po*P['phi'])      # 0.32  32.0  6.65 g per L of broth
```

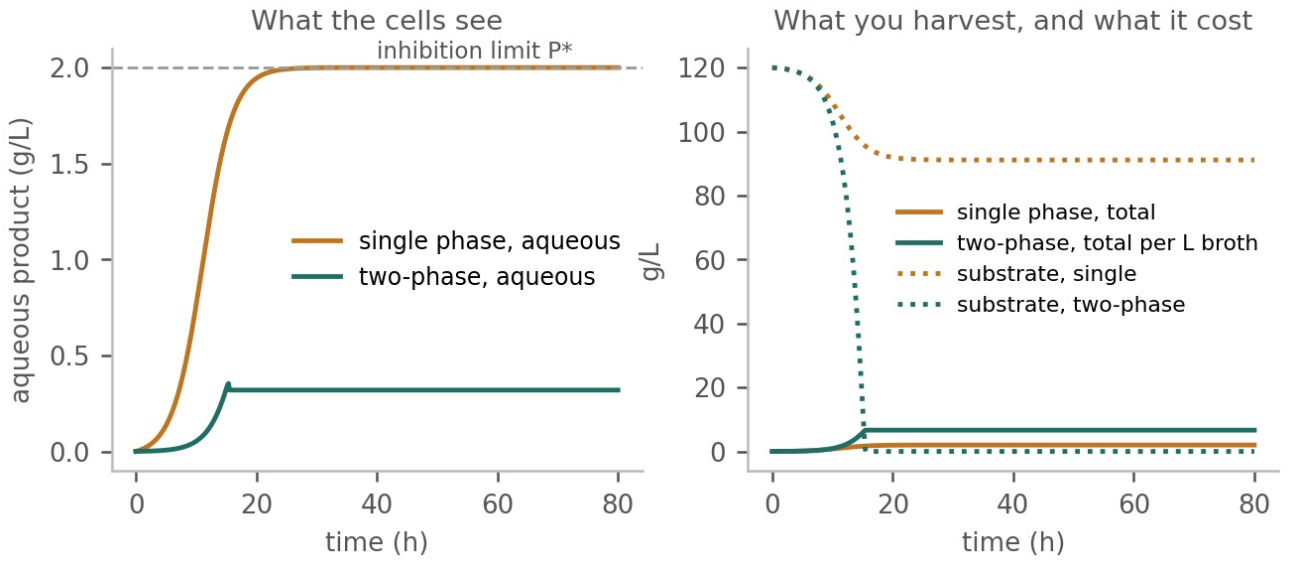


Figure 47.1 The same strain with and without a second phase (K = 100, φ = 0.2, k_La = 50 h^{-1}). Left: what the cells see — the single-phase culture sits against its inhibition ceiling, the two-phase culture never approaches it. Right: what you harvest, and the substrate it cost: the single-phase run leaves 91 g/L of substrate unused, the two-phase run consumes all of it. Computed from the model listed above; all parameters are assumed.

What the model says, and the row where more solvent is worse

With the second phase, the aqueous concentration peaks at 0.32 g/L instead of 2.0, the culture consumes all 120 g/L of substrate, and the harvest is 6.65 g per litre of whole broth against 2.0 — a 3.3-fold gain, obtained without touching the strain. That is the case for ISPR in one sentence, and it is also a warning: the gain came from removing a ceiling, so it is bounded by what the strain and the substrate can actually make, not by the partition coefficient.

Sweep the solvent fraction and the point becomes sharp:

φ	R = 1 + Kφ/(1−φ)	peak aqueous (g/L)	harvest per L of broth (g/L)
0.05	6.3	1.33	7.89
0.10	12.1	0.69	7.48
0.20	26.0	0.32	6.65
0.30	43.9	0.19	5.82
0.50	101.0	0.08	4.15

R rises twentyfold across that range and the harvest *falls* by half. Once the second phase is large enough to keep the aqueous concentration well below P*, extra solvent buys nothing — the ceiling is no longer binding — and every additional litre of it displaces a litre of aqueous phase containing the cells that make the product. The optimum here is

around 5%, far below what the capture arithmetic alone would suggest. This is the row the tracker asks you to explain, and the explanation is that two different quantities are being optimised: R maximises capture per litre of aqueous phase, while a plant maximises product per litre of vessel.

The equilibrium shortcut, and when it fails

If transfer is fast relative to production, the phases stay at equilibrium, $P_o = K \cdot P_a$, and the whole second phase collapses into a single algebraic factor: the inhibition ceiling behaves as though it were $P^* \cdot R$ with $R = 1 + K\varphi/(1 - \varphi)$. You do not need the fourth differential equation at all. Most of the time this is the right model, and deriving it is worth doing once so that you trust it.

When does it fail? Define the relaxation rate at which an offset from equilibrium decays:

```
λ = k_L a · (1 + (1 − φ)/(K·φ))
```

A production rate $q_P X$ must be carried across the interface, and carrying it requires an offset of roughly $q_P X/\lambda$. So the shortcut is good when that offset is small compared with P^*, and bad when it is not. The model confirms the estimate: the measured offset tracks $q_P X/\lambda$ across three orders of magnitude of $k_L a$.

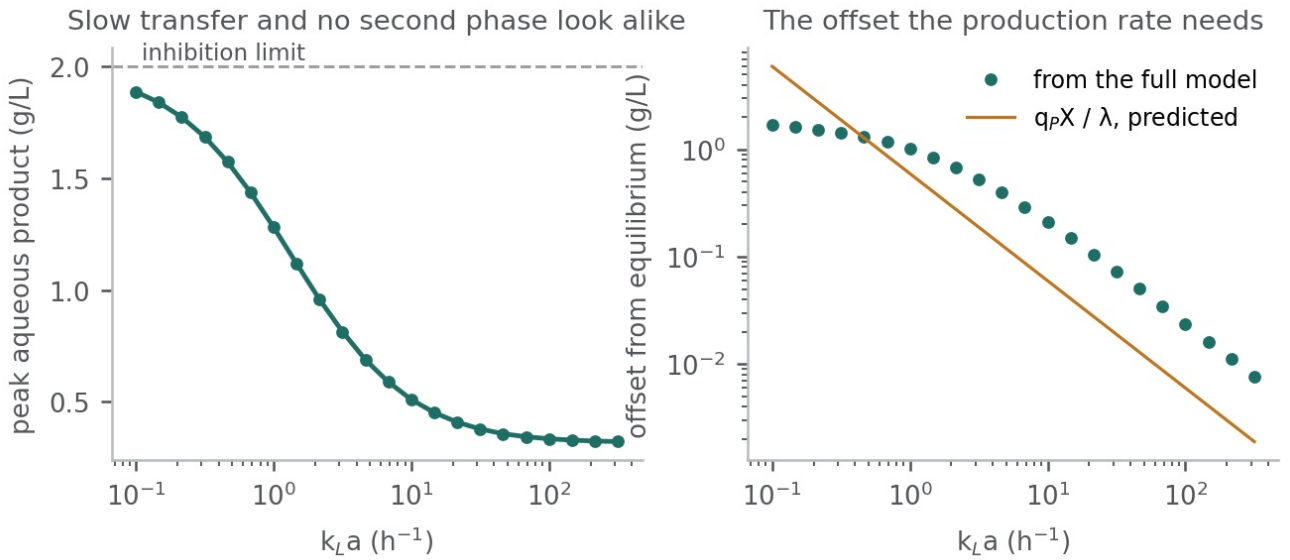


Figure 47.2 Left: the peak aqueous concentration the cells experience against the transfer coefficient. At $k_La = 0.1\ h^{-1}$ the culture reaches 1.89 g/L — almost the inhibition limit — despite a second phase with a capture factor of 26; at $10\ h^{-1}$ it reaches 0.51 and at $100\ h^{-1}$, 0.34. Right: the offset from equilibrium against the prediction q_PX/λ. Computed from the model; q_PX is taken at its mid-run value of about 0.62 g/L/h.

The practical reading is the left panel, and it is the result of the week. **Slow transfer and no second phase look the same to the cells.** A process with an excellent partition coefficient and a poorly dispersed solvent — a thick unmixed overlay, large droplets, or beads too big — delivers the toxicity of a single-phase process while incurring all the cost of the second one. Dispersion is therefore not a detail of implementation, it is part of the design, and the measurement that settles it is the aqueous concentration during the run, not the total at harvest.

A numerical note for week 12's benefit: a large k_La makes this system stiff, because transfer then equilibrates in minutes while the fermentation runs for days. That is exactly the structure chapter 12 described, and it is why every listing here uses LSODA. If you must explore k_La over orders of magnitude, use the equilibrium shortcut in the fast regime instead of integrating through it — which is model reduction doing its job.

Fitting, and what the data actually identify

Now the uncomfortable part. Generate synthetic aqueous-phase data from the two-phase model, add 2% noise, and fit K and k_La back. The fit is excellent to the eye and returns $K = 107 \pm 2$ — within 7% of the true 100 — and $k_La = 28 \pm 11$ against a true value of 50. Profile the residual against k_La and the reason is obvious: it falls steeply up to about $10\ h^{-1}$ and is then essentially flat, so any value above 15 fits the data equally well.

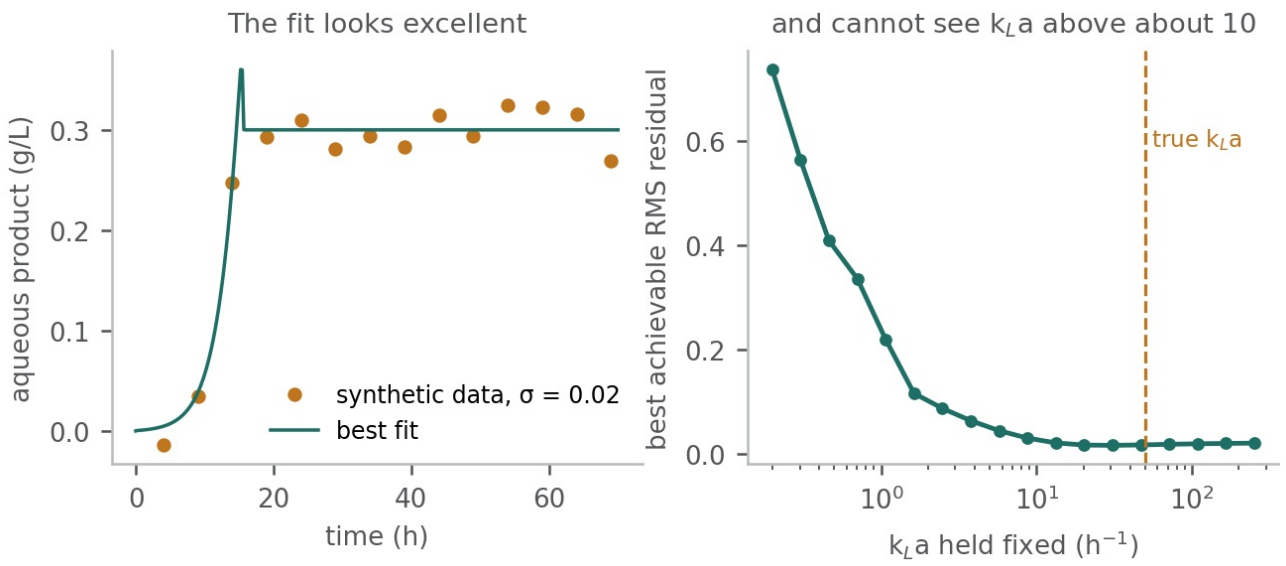


Figure 47.3 Left: the fit to synthetic aqueous-phase data, which looks excellent. Right: the profile of the best achievable residual against k_La held fixed, showing that the data constrain it only up to about 10 h^{-1}. The partition coefficient is determined to a few per cent; the transfer coefficient is not determined at all in the fast regime. Computed for this module with `least_squares`.

That is not a defect of the optimiser, it is the physics: once transfer is fast enough to hold equilibrium, the system's behaviour depends on K and φ and not on how fast equilibrium is reached, so the data contain no information about k_La. The methods-section sentence that follows is the deliverable of this week: *K was estimated from aqueous-phase concentrations by least squares, with the interval from the covariance of the fit; k_La was not identifiable above approximately 10 h^{-1} and was therefore fixed at a value in the fast-transfer regime, which the data cannot distinguish from any larger value.* That sentence is more credible than a table of four parameters with tight intervals, and reviewers who know the subject will read it as a sign you understood your own model.

Two further habits. Fit from two different starting points: agreement is weak evidence of a unique optimum, disagreement is strong evidence of a problem. And ask what a second experiment would add — here, a run at low k_La, where the offset is large and the parameter is visible, which is the experiment the sensitivity analysis of chapter 16 would have told you to do.

Growth	`μ = μ_max·S/(K_S+S)·(1 − P_a/P*)`
Production	`q_P = α·μ + β (Luedeking–Piret)`
Transfer	`N = k_L a·(P_a − P_o/K); the driving force is distance from equilibrium`
Bookkeeping	`dP_o/dt = N·(1 − φ)/φ; say per litre of what, every time`
Harvest	`per litre of broth = P_a(1 − φ) + P_o·φ`
Shortcut	`fast transfer → ceiling × R, R = 1 + Kφ/(1 − φ)`
When it fails	`offset ≈ q_P X/λ, λ = k_L a·(1 + (1 − φ)/(Kφ))`

A collaborator's two-phase model predicts a 26-fold titer gain from an overlay at φ = 0.2, K = 100. Your experiment gives 3.2-fold. Reconcile them.

They are both right, and the factor of eight between them is three separate things. First, the 26 is R — the multiplication of the *aqueous ceiling*, per litre of aqueous phase — and the experiment measured product per litre of broth; that accounts for a factor of (1 − φ) = 0.8 on its own. Second, R only pays off if the ceiling is what binds all the way to the end. In my model the second phase removes the ceiling after a few hours and the process becomes substrate-limited instead: the gain is then set by how much product the strain can make from the available substrate, which was 6.65 against 2.0 g/L — a factor of 3.3, almost exactly what your experiment found. Third, if the actual k_La or dispersion is poor, the cells still see a significant aqueous concentration and part of the ceiling remains in force.

How to tell which applies, in one measurement: the residual substrate at harvest. If the two-phase run consumed all the substrate, you are substrate-limited and the gain cannot be increased by more solvent — the next intervention is the strain or the feed. If substrate remains and the aqueous concentration was high during the run, transfer or dispersion is the limit. If substrate remains and the aqueous concentration stayed low, something else stopped the culture and neither model applies.

Three things people get wrong

- ‘More solvent is better.’ Past the point where the ceiling stops binding, extra solvent only displaces productive volume. In the model the optimum was 5%.
- ‘The transfer term is $k_La \cdot P_a$.’ It is $k_La \cdot (P_a - P_o/K)$. A driving force is a distance from equilibrium.
- ‘The model fits, so the parameters are measured.’ In the fast-transfer regime the data carry no information about k_La at all. Profile it and say so.

Self-check — answer on paper first

1. Write the two-phase product balances from memory and state what each concentration is per litre of.

$dP_a/dt = q_PX - k_La(P_a - P_o/K)$, with P_a and X per litre of aqueous phase; $dP_o/dt = k_La(P_a - P_o/K) \cdot (1 - \varphi)/\varphi$, with P_o per litre of organic phase. The factor converts a quantity expressed per litre of aqueous phase into a concentration change in a phase of different size. The harvest figure, per litre of whole broth, is $P_a(1 - \varphi) + P_o\varphi$.

2. Your aqueous concentration sits at 1.9 g/L during a run with a second phase whose capture factor is 26. What is happening and what do you change?

Transfer is too slow to carry the production rate, so the aqueous phase runs far from equilibrium with the organic phase and the cells experience almost the full inhibition ceiling. The offset needed is q_PX/λ with $\lambda = k_La(1 + (1 - \varphi)/(K\varphi))$, so the fix is anything that raises k_La: better dispersion through agitation, smaller droplets or beads, or more interfacial area at the same phase fraction. Adding solvent is the wrong move — it raises R, which is not what is binding.

3. Why does the substrate-balance closure check prove so little, and what would a real validation require?

Because closure is a property of the arithmetic, not of the biology: a model with the wrong rate laws, wrong parameters and wrong mechanism still conserves mass if the stoichiometric coefficients are consistent. It catches sign errors and missing terms, which is worth having. A real validation requires prediction of data the model was not fitted to — a different phase fraction, a different initial substrate, a different k_La — with the parameters fixed at the values obtained from the first dataset.

What to open a book for. Nothing covers this. Newman et al., Biotechnol. Bioeng. (2006) is the two-phase case study to read for real numbers, and the 2012 Bioprocess Biosyst. Eng. paper on substrate transport in two-phase partitioning bioreactors is the closest thing to a modelling reference. The SciPy documentation for `solve_ivp` and `least_squares` is, for this week, more useful than any textbook — which is itself the reason this chapter exists.

Chapter 48 · tracker week 48

Cell immobilisation: diffusion, dead cores and continuous operation

Immobilising cells retains biomass. It does not remove product — that was the last two chapters — and confusing the two is the commonest error in this area. The question this week answers is whether retention is worth what it costs you in oxygen.

What it is for, and what it is not for

Entrapping or attaching cells does four things, and you should be able to say which of them you are buying.

- **Retention.** Biomass is no longer washed out by the feed, so you can run continuously at a dilution rate above the organism's maximum growth rate. This is the main reason, and it is the only one that changes what is thermodynamically possible rather than merely convenient.
- **Reuse of the biocatalyst.** For a conversion rather than a growth process — a whole-cell biotransformation — the beads are a catalyst you recover and use again.
- **Easier separation.** Millimetre beads settle or are screened out; cells are centrifuged. For a clean broth this is worth real money downstream.
- **Some protection.** From shear, and partly from inhibitors, because the bead interior is a different microenvironment from the bulk — which is also the source of every problem below.

What it does not do is lower the product concentration in the broth. If your problem is product toxicity, immobilisation does not solve it and may make it worse, since the product must diffuse out through the same polymer. And it requires cells that work while largely non-growing, because growth inside a bead bursts it. So the first question about any candidate product is whether its production is growth-associated: in the Luedeking–Piret language of chapter 47, immobilisation needs a large β and tolerates a small α. A strictly growth-associated product is a poor candidate however attractive continuous operation looks.

Materials, and the failure each one has

Material	Gentleness	Strength	The specific weakness
Calcium alginate	Very gentle — beads form by dropping into $CaCl_2$ at room temperature	Weak	Dissolves when calcium is removed: phosphate buffers, citrate and other chelators, and high sodium all strip the crosslinks. Rules out most standard media without modification
κ-Carrageenan	Gentle, gels with potassium on cooling	Weak to moderate	Needs K^+ maintained; softens with temperature; similar ion sensitivity
Polyvinyl alcohol	Moderate — freeze–thaw or boric acid routes	Tough, reusable	The boric acid route is harsh for some strains; freeze–thaw costs a cycle; more opaque to oxygen transfer design
Polyacrylamide	Harsh	Tough	Acrylamide monomer is toxic to the cells being entrapped and to the operator; largely historical for living cells

The alginate–phosphate incompatibility deserves emphasis because it catches everyone once. A defined medium with phosphate as a buffer and a nitrogen or trace-element source will slowly dissolve calcium alginate beads, and the symptom is cells appearing in the broth over hours to days, which looks like leakage or bead breakage and is actually chemistry. Either reformulate the medium, add calcium continuously, or use a tougher polymer.

Diffusion inside the bead: the Thiele modulus

A bead is a catalyst pellet with live catalyst in it, so the analysis is the one you already know. For a first-order reaction in a sphere of radius R with effective diffusivity D_{eff}, a steady-state shell balance gives the concentration profile, and the dimensionless group that governs it is

```
φ = R·√(k/D_eff)            (Thiele modulus: reaction rate ÷ diffusion rate)
```

with the effectiveness factor — the rate the bead achieves divided by the rate the same cells would achieve fully exposed —

```
η = (3/φ²)·(φ·coth φ − 1),    η → 1 for φ ≪ 1,    η → 3/φ for φ ≫ 1
```

Read the two limits physically. Small φ means reaction is slow relative to diffusion: the substrate reaches the centre at nearly bulk concentration and the whole bead works, η ≈ 1. Large φ means reaction is fast relative to diffusion: the substrate is consumed in a thin outer shell, η ≈ 3/φ, and the interior is a passenger. In that regime halving the bead diameter doubles the rate, which is the design lever.

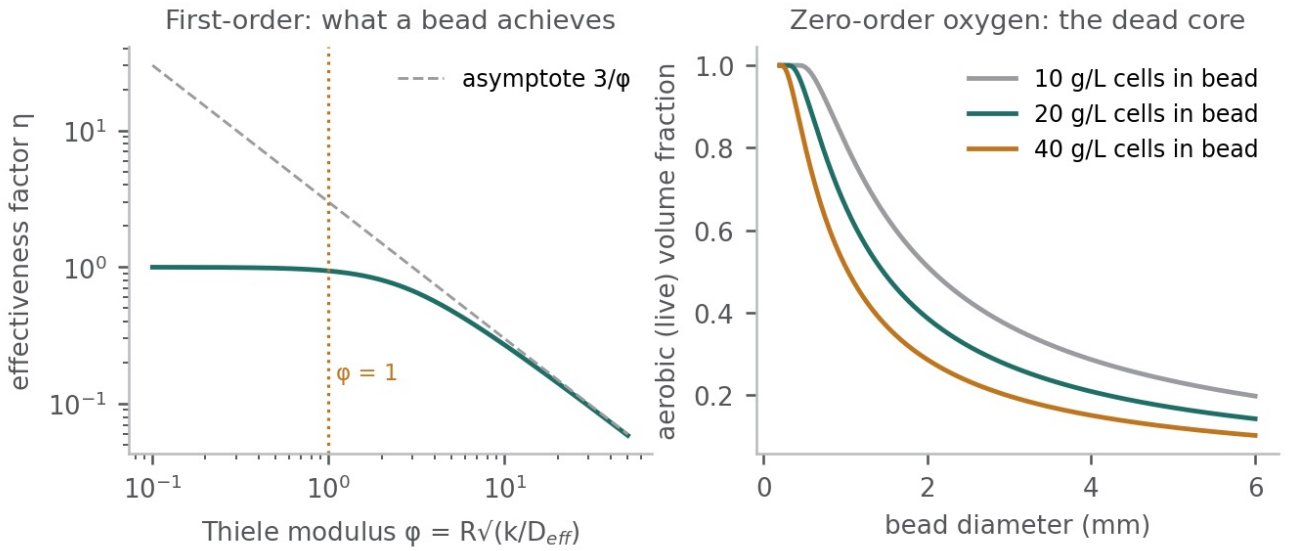


Figure 48.1 Left: the sphere effectiveness factor against the Thiele modulus with its 3/φ asymptote. Right: the aerobic fraction of a bead against diameter for three cell loadings, from the zero-order dead-core calculation with air-saturated bulk oxygen (0.21 mol/m³), $D_{eff} = 1.5 \times 10^{-9}$ m²/s and a specific uptake of 5 mmol O_2 per gram of cells per hour. Assumptions stated; arithmetic computed.

Dead cores, and why it is always oxygen

Oxygen consumption is better described as zero order than first order, because cells respire at their maximum rate down to very low oxygen concentrations and then stop. With zero-order consumption at rate k_0 per unit bead volume, the oxygen is exhausted at a finite depth rather than decaying asymptotically, and that penetration depth comes from the same shell balance:

```
L = √(2·D_eff·C_s / k₀)
```

Put numbers in. A bead loaded at 20 g/L of cells respiring at 5 mmol O_2 per gram per hour consumes 100 mol·m^{-3}·h^{-1}, or 0.028 mol·m^{-3}·s^{-1}. With air-saturated bulk oxygen at 0.21 mol/m³ and $D_{eff} = 1.5 \times 10^{-9}$ m²/s, L = **151 µm**. At 10 g/L it is 213 µm and at 40 g/L it is 106 µm. So in a 2 mm bead at 20 g/L, only the outer 0.15 mm is aerobic and the live fraction is $1 - ((R - L)/R)^3 \approx 0.39$ — most of your catalyst is sitting in an anaerobic core doing nothing, or doing something you did not design.

Now do glucose for comparison, because the contrast is the lesson. At 20 g/L in the broth, glucose is 111 mol/m³ — more than 500 times the oxygen concentration — and even with a lower diffusivity in the gel and a consumption rate of 10 mmol per gram per hour, the penetration depth comes out at about **1.1 mm**. Glucose reaches the centre of a 2 mm bead comfortably; oxygen does not come close. The reason is entirely the solubility: oxygen's driving force is small because its saturation concentration in water is small, and no amount of agitation changes that. Agitation raises the transfer rate to the bead surface, not the concentration at it.

Three levers shrink a dead core, in increasing order of cost. Smaller beads: L is fixed, so the aerobic shell is a larger fraction of a smaller sphere — going from 2 mm to 0.8 mm at 20 g/L takes the live fraction from about 0.39 to about 0.75. Lower cell loading: fewer cells consume oxygen more slowly and L rises as the inverse square root, so halving the loading raises L by 41% — but you have also halved the catalyst. More oxygen in the bulk: enriched air or elevated pressure raises C_s, and L rises as its square root, which works and costs utilities and capital.

And do not take the dead core from the model alone. The measurement is a microelectrode profile through a bead, which is standard in the biofilm literature, or a redox or oxygen-sensitive dye that reports the anaerobic zone visually. The model tells you what to look for; it does not establish that it is there, since D_{eff} in your particular gel at your particular cell loading is the parameter you are least sure of.

Continuous operation: what retention actually buys

Here is the comparison that decides the week, and it is less favourable to immobilisation than the literature's enthusiasm

suggests. Free cells in a chemostat wash out above a critical dilution rate, which for μ_{max} = 0.5 h^{-1}, K_S = 0.5 g/L and 20 g/L feed is 0.488 h^{-1}. Below it, biomass is $X = Y_{XS}(S_0 - S)$ with $S = K_S D/(\mu_{max} - D)$, and volumetric productivity is $q_P X$ with $q_P = \alpha D + \beta$. That curve peaks at **0.65 g/L/h at D ≈ 0.39 h^{-1}** and then collapses at washout.

Immobilised cells do not wash out, so productivity is set by capacity rather than by growth: beads occupying 30% of the volume at 40 g/L of cells inside, with an effectiveness factor of 0.45 from the dead-core calculation, give an effective 5.4 g/L of working biomass, and at β = 0.05 g per gram per hour that is **0.27 g/L/h — flat in D** once the feed supplies enough substrate.

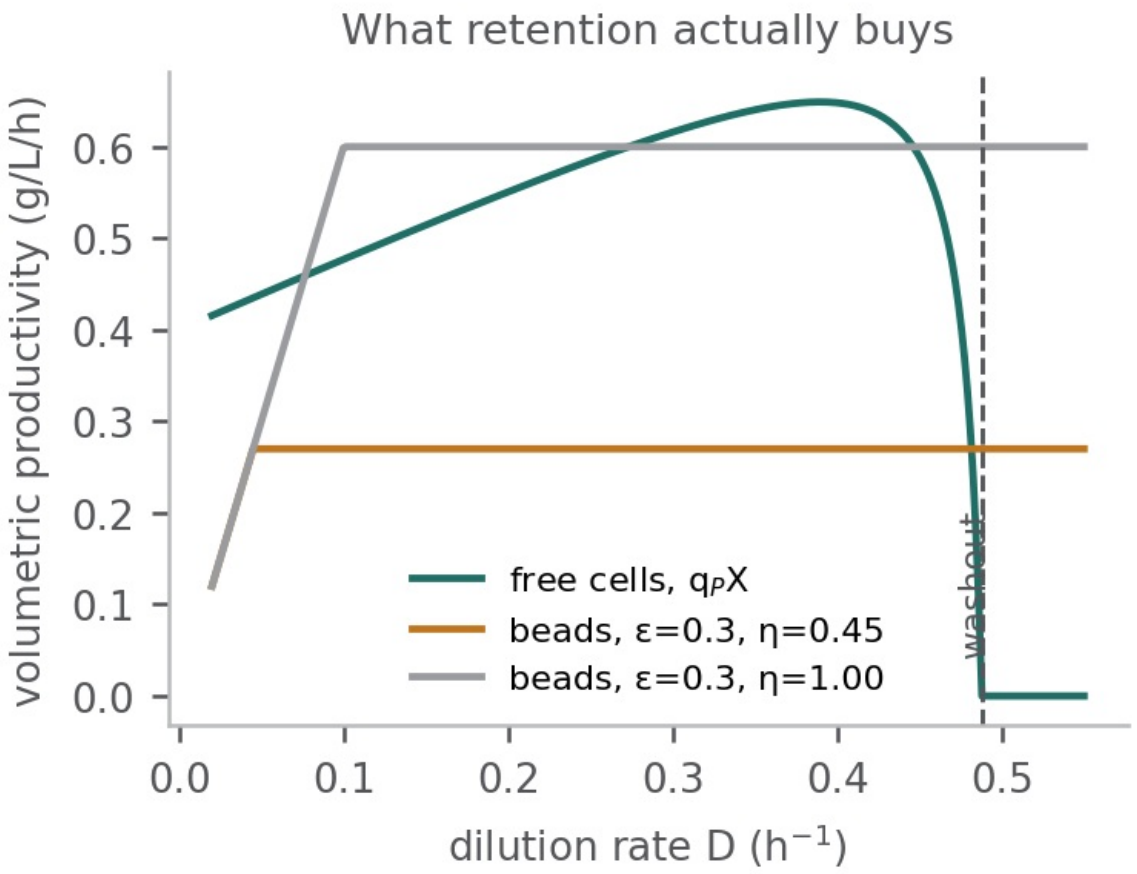


Figure 48.2 Volumetric productivity against dilution rate for free cells and for two immobilised cases, with the free-cell washout marked. The immobilised lines rise with the substrate the feed supplies and then flatten at the capacity of the retained cells. Parameters in the text; the comparison is model arithmetic and its conclusion depends entirely on β and η.

So free cells win, by more than twofold, and that is the normal result. The reason is visible in the arithmetic: retention gives you biomass that cannot wash out, but the retained cells are non-growing, so they produce only at β, and a fraction η of them are in an anaerobic core. A free-cell chemostat run just below washout has nearly all of its biomass producing at αD + β with full oxygen access.

What would have to change for immobilisation to win? Any of four things, and they are the questions to ask of any immobilisation proposal. The non-growth-associated rate β would have to be high — a product made efficiently by resting cells, which is exactly the whole-cell biotransformation case. The product would have to inhibit growth strongly, so that free cells cannot be maintained at a high dilution rate at all while resting cells keep producing. The organism would have to grow slowly, so that washout constrains free cells at an uneconomic dilution rate — immobilisation is far more attractive for a slow-growing organism than for *E. coli*. Or downstream separation of free cells would have to be unusually expensive relative to the productivity lost.

There is also a stability argument that the arithmetic above hides. The free-cell optimum sits just below washout, and that is an uncomfortable place to run a plant: a transient in the feed rate, a temperature excursion or a small drop in μ_{max} from a medium batch takes you over the cliff and the vessel empties itself over a few hours. A retained-biomass system has no cliff. Trading some productivity for an operating point that cannot catastrophically fail is a legitimate engineering decision, and it is the strongest honest argument for immobilisation in a growth-based process.

How these systems fail

- **Bead breakage.** Cells grow inside, gas bubbles form from respiration or fermentation, and the bead cracks. Monitor by sieving samples and counting intact beads, and by free cell density in the effluent.
- **Leakage.** Cells escape from the surface layer continuously, which is normal at a low rate and a failure at a high one. A rising effluent cell count that is not accompanied by broken beads is leakage rather than breakage, and the distinction matters because the fixes differ.
- **Fouling and channelling.** In a packed bed, biomass and precipitate block the interstices, flow finds a path, and part of the bed stops contributing. The signature is a rising pressure drop with a falling conversion — monitor both, and a tracer residence-time distribution will show channelling directly.
- **Lost working volume.** Beads at 30% of the vessel displace 30% of the liquid, exactly as a solvent does in chapter 45. It belongs in the productivity comparison, and in the figure above it already is.
- **Oxygen at scale.** A packed bed cannot be sparged effectively, a fluidised bed can but abrades the beads, and an airlift is gentle but has limited transfer. The reactor form is chosen by the oxygen requirement, and for a strictly aerobic process that choice is usually the binding constraint.

Thiele modulus	$\varphi = R\cdot\sqrt{(k/D_{eff})}$
Effectiveness	$\eta = (3/\varphi^2)(\varphi\cdot\coth\varphi - 1)$; $\eta \approx 3/\varphi$ for large φ
Dead core	$L = \sqrt{(2\cdot D_{eff}\cdot C_s/k_0)}$; live fraction $= 1 - ((R-L)/R)^3$
Oxygen vs glucose	151 µm against 1.1 mm, at the stated loadings — it is the solubility
Free-cell washout	$D_{crit} = \mu_{max}\cdot S_0/(K_S + S_0) = 0.488\ h^{-1}$ here
Retained productivity	$\beta\cdot X_{bead}\cdot\varepsilon\cdot\eta$, flat in D; 0.27 against 0.65 g/L/h free

You are asked to immobilise a strain making a growth-associated product in 3 mm alginate beads in a packed bed, fed a phosphate-buffered defined medium. Write the three objections, in the order you would raise them.

First, the product. Growth-associated production means α is large and β is small, and immobilised cells must be largely non-growing or they destroy the beads. The process is therefore asking resting cells to do something they do at a low rate, and the productivity comparison above will favour free cells by a wide margin. This objection is fatal to the concept rather than to the implementation, so it goes first.

Second, the beads. Three millimetres with any realistic cell loading gives an aerobic shell of about 0.15 mm, so roughly three quarters of the entrapped biomass sits in an anaerobic core. If the process is aerobic, you are paying for catalyst that is not working; if the organism is facultative, the core will ferment and excrete by-products into your product stream. Submillimetre beads, or a different reactor, would be needed.

Third, the chemistry. Calcium alginate in phosphate buffer dissolves: phosphate sequesters the calcium that crosslinks the gel. The beads will soften and shed cells over days, which in a packed bed also means fouling and channelling. Either the medium's buffer changes, calcium is fed continuously, or the material changes to PVA.

What I would propose instead, if continuous operation is the real objective: cell retention by membrane filtration or settling rather than entrapment, which keeps the biomass growing and fully oxygenated, has no diffusion limit and no dead core, and achieves the same decoupling of dilution rate from growth rate. Immobilisation is the right answer for a resting-cell biotransformation, not for a growth-associated fermentation.

Three things people get wrong

- ‘Immobilisation and ISPR are both ways of improving a toxic-product process.’ Immobilisation retains biomass; it does nothing to the product concentration the cells experience.
- ‘Glucose limitation inside the bead is the problem.’ It is oxygen, by an order of magnitude, and for the same reason in every system: solubility.
- ‘Retention beats washout, so it beats free cells.’ Retained cells are non-growing and partly anaerobic. Do the productivity comparison; it usually favours free cells.

Self-check — answer on paper first

1. Derive the oxygen penetration depth for zero-order consumption and compute it for a bead at 30 g/L of cells, using the chapter's other assumptions.

A shell balance with constant consumption gives $D_{eff}\cdot d^2C/dx^2 = k_0$ in the slab limit, and integrating with $C = C_s$ at the surface and both C and dC/dx vanishing at the front gives $L = \sqrt{(2\cdot D_{eff}\cdot C_s/k_0)}$. At 30 g/L and 5 mmol·g^{-1}·h^{-1}, $k_0 = 150$ mol·m^{-3}·h^{-1} $= 0.0417$ mol·m^{-3}·s^{-1}, so $L = \sqrt{(2 \times 1.5 \times 10^{-9} \times 0.21/0.0417)} \approx 123$ µm. In a 2 mm bead that is a live fraction of $1 - (0.877)^3 \approx 0.33$.

2. A colleague proposes doubling the cell loading in the beads to double the productivity. What happens?

Productivity rises by much less than twofold and may not rise at all. Doubling k_0 reduces the penetration depth by a factor of √2, so the aerobic shell thins; the live biomass per bead is roughly the loading times the live fraction, and in the large-bead limit — where the live volume is a thin shell of thickness L — the product of loading and L scales as √(loading), so the gain is about 41% at best. It also raises the oxygen demand per bead, so the bulk oxygen concentration falls and the whole calculation shifts further against you. Smaller beads are the better lever.

3. State the honest condition under which immobilisation beats a free-cell chemostat.

When the non-growth-associated productivity of the retained cells, multiplied by the retained biomass and by the effectiveness factor, exceeds the free-cell volumetric productivity at its best dilution rate — which requires a product made well by resting cells, or an organism whose growth rate is too low to permit an economic dilution rate, or a product that inhibits growth so strongly that free cells cannot be maintained at all. Add to that the stability argument: a retained system has no washout cliff, which can justify accepting lower productivity for an operating point that cannot fail catastrophically.

What to open a book for. Here a book does exist, and it is a chemical engineering one: any reaction-engineering text's treatment of the Thiele modulus and effectiveness factor for catalyst pellets covers the mathematics of this week exactly, and better than the biological literature does. For the biology, the reviews of alginate and PVA immobilisation and of oxygen limitation in immobilised-cell beads are the sources; read them for the materials and the failure modes rather than the theory.

Closing notes to part 3

The four weeks in one page

Week 45. Three different limits — toxicity, export, loss — look identical on a plot and need three different interventions. Three experiments separate them, and none takes a week. A plateau at q/k is a loss, not a stopped pathway.

Week 46. The product's properties choose the removal method, and the first filter is charge: if it is ionised at fermentation pH a solvent cannot work. For a sorbent, the Langmuir ceiling at the inhibition concentration sizes the mass, the cycle count sets the cost, and intraparticle diffusion decides whether the capacity you bought is reachable.

Week 47. Four ODEs, one transfer term driven by the distance from equilibrium, and a volume factor that is where the errors live. Past the point where the ceiling stops binding, more solvent lowers output per litre of vessel. Slow transfer and no second phase look the same to the cells. And in the fast-transfer regime the data do not identify k_La at all, which you should say rather than hide.

Week 48. Immobilisation retains biomass and does nothing about product concentration. Oxygen penetrates about 150 µm at realistic loadings while glucose penetrates a millimetre, so most of a large bead is anaerobic. The honest productivity comparison usually favours free cells, and the real argument for retention is the absence of a washout cliff.

On the numbers in this part

Every figure here was computed by a script and every parameter is stated. The two-phase model of chapter 47 is listed in full and generated figures 47.1, 47.2 and 47.3 as well as the phase-fraction table and the identifiability result; its parameters are assumed, not measured, and the captions say so. The transport numbers in chapters 46 and 48 use order-of-magnitude effective diffusivities — the quantity you are least sure of in any real system — and the conclusions that depend on them are stated as comparisons (oxygen against glucose, 1 mm against 3 mm beads) rather than as absolute predictions, because comparisons survive an error in D_{eff} that absolute numbers do not. The efflux-pump result is from the published screen and is quoted as a base rate rather than a prediction.

Where a number here would decide a purchase or an experiment, measure it: the partition coefficient and the isotherm in real medium rather than buffer, the loss constant by a hold experiment with no cells, and the effective diffusivity by the uptake curve of a single bead. Each is a day's work and each replaces an order-of-magnitude assumption with a number.

What is still to come

Working forward in tracker order from here, the remaining weeks with no textbook behind them are techno-economic analysis and life-cycle assessment (week 49), analytical method validation (50), the product-class weeks (53 to 57, including the PHA case study and the C1 and CO_2 feedstocks), and then the host weeks — yeasts in 58, the Gram-positives *C. glutamicum* and *B. subtilis* in 59, and the phototrophs in 60, with the panorama and cell-free systems in 61. Those host chapters are the ones that will need the most original work, because what you need from them is not a species description but a comparison: what each host costs you in tools, growth rate, cofactor supply, secretion and regulatory acceptance, against the arithmetic of parts 2 and 3. That is the next part, and the immobilisation and ISPR results here are inputs to it — week 58's compartment arithmetic and week 60's photobioreactor light penetration are the same transport calculations as chapter 48's dead core, with light in place of oxygen.

Part 4 — Trusting the numbers: economics, method validation, omics as constraints, and getting DNA into a wild strain (weeks 49 to 52)

These four weeks are about whether your numbers mean anything. Week 49 asks whether the process clears its market price, which is the calculation that kills more strains than any reviewer. Week 50 asks whether the titer you reported is real. Week 51 asks what an omics dataset is actually evidence for, and how to turn it into a constraint rather than a figure. Week 52, which the tracker marks optional, asks why a wild industrial strain destroys your plasmid on arrival and what a pooled screen can and cannot tell you.

Like part 3, none of this has a textbook behind it. Unlike part 3, the reason is not that the subject is young: analytical validation and discounted cash flow are old and well codified, but the codification lives in regulatory guidance documents and process-economics texts that no one hands a biologist, and the translation into what a strain engineer actually needs has not been written down. So the chapters do the translation, with the arithmetic worked.

Two of these weeks will make you unpopular if you use them carelessly. A techno-economic analysis done honestly on a colleague's favourite strain usually says no, and a method validation done honestly on your own early data usually says the first six months of titers were not measurable. Both are worth doing anyway, and both are much cheaper done early. The way to make them useful rather than destructive is in chapter 49's last section: report what would have to be true for the answer to change, rather than only the answer.

Ch.	Week	Chapter
49	49	Techno-economic analysis and life-cycle assessment: the calculation that kills strains
50	50	Analytical method validation: trusting your own numbers
51	51	Omics, screening, and turning data into model constraints
52	52	Host defences and pooled screens: getting DNA into a wild strain, and reading 50,000 of them at once

Techno-economic analysis and life-cycle assessment: the calculation that kills strains

A techno-economic analysis is not a business exercise appended to a science project. It is the only calculation that tells you which of titer, rate and yield to work on next, and it takes an afternoon.

What a flowsheet model contains that week 18's envelope did not

Week 18 gave you the back-of-envelope: feedstock cost divided by yield, broth volume per kilogram, annual output per vessel. That is enough to rank strains and not enough to decide a process, because it omits four things that frequently dominate.

- **Capital, properly sized.** Not just the fermenters, but the seed train, the separation sequence, the utilities, and the installation that turns a $200,000 piece of equipment into a $700,000 installed item.
- **Utilities with a physical basis.** Agitation power, air compression, cooling duty — all of which follow from the oxygen demand you computed in chapter 19 rather than from a guess.
- **Waste.** Spent broth, biomass, salt from pH control, solvent losses from chapter 45. In a commodity process this is a cost line and occasionally a permit problem.
- **The cost of money.** Capital spent now against product sold over twenty years, which is a discounted cash flow and not a division.

The tool for this is BioSTEAM, which is free, Python, and comes with complete biorefinery models you can run unmodified before building your own. Run one of those first. The value of running someone else's model is seeing the line items you would have forgotten.

The model this chapter uses, stated in full

Everything below comes from one small model, and it is stated so you can disagree with it. Target output 10,000 t/yr. Fermenters of 200 m³ working volume, 48 h batch plus 8 h turnaround, 8,000 operating hours a year. Yield 0.40 g/g on glucose at $0.40/kg. Installed capital of $3M per fermenter train slot, plus $120 per m³ of annual broth throughput for the downstream train. Utilities $0.15/kg of product, labour $3M/yr, maintenance at 4% of capital, a 10% discount rate over a 20-year life. The capital recovery factor is then

```
CRF = i(1 + i)^n / ((1 + i)^n − 1) = 0.1175 at i = 0.10, n = 20
```

so each dollar of capital costs about 11.75 cents a year, and the capital charge per kilogram is CRF × capital ÷ annual output. Minimum product selling price is the sum of feedstock, utilities, fixed costs and that capital charge — the price at which net present value is zero, which is the number to compare against the market.

What it says: three titers, one conclusion

Titer	Fermenters	Broth per year	Capital	MPSP	Breakdown ($/kg)
20 g/L	17.5	500,000 m³	$112.5M	**$3.22/kg**	feed 1.00, util 0.15, fixed 0.75, capital 1.32
50 g/L	7.0	200,000 m³	$45.0M	**$2.16/kg**	feed 1.00, util 0.15, fixed 0.48, capital 0.53
100 g/L	3.5	100,000 m³	$22.5M	**$1.80/kg**	feed 1.00, util 0.15, fixed 0.39, capital 0.26

Against an incumbent price of $1.80/kg — a typical commodity number — the 20 g/L process loses money by 79%, the 50 g/L process by 20%, and the 100 g/L process breaks even exactly. That is the whole argument for the industrial obsession with titer, and notice what produces it: the feedstock term does not move at all, because yield is unchanged. Titer moves capital and fixed costs, by reducing the number of vessels and the volume of broth to process. Yield moves feedstock. They are different levers acting on different lines, which is why the question "should we work on titer or yield" has an arithmetic answer rather than a cultural one.

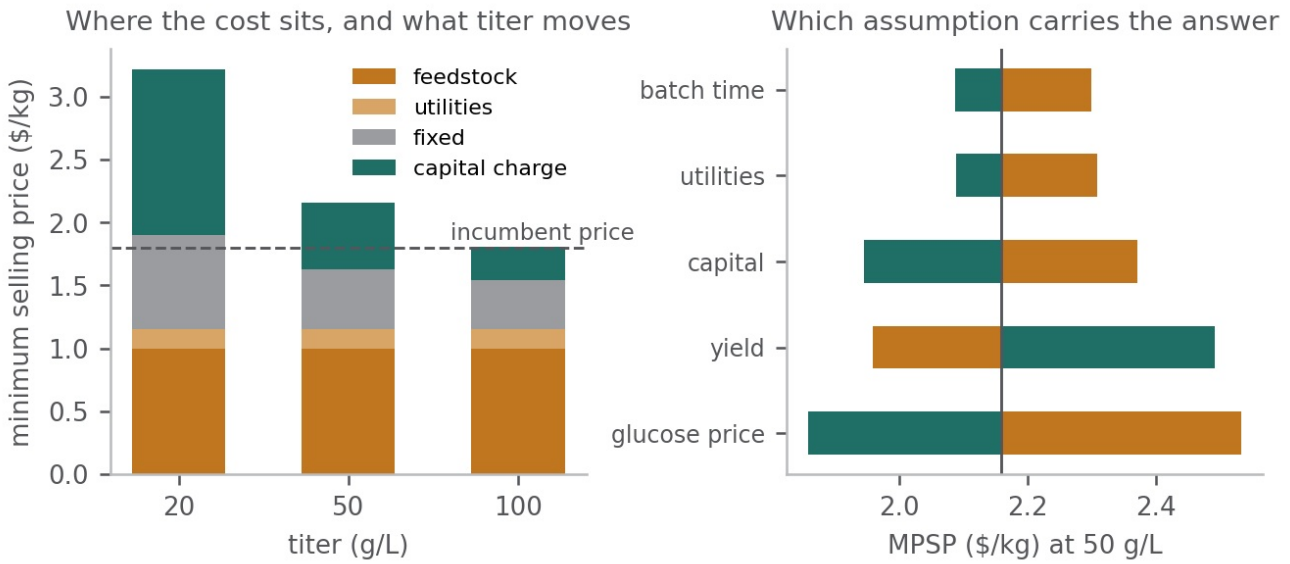


Figure 49.1 Left: the cost breakdown at three titers, against an assumed incumbent price. Right: the tornado plot at 50 g/L, each bar spanning the stated range of one input with everything else held at base. Computed from the model stated above; every parameter in it is an assumption, and the point of the right-hand panel is to show which assumptions matter.

Sensitivity: which assumption is carrying the conclusion

The tornado plot ranks the swings at 50 g/L: glucose price between $0.28 and $0.55/kg moves MPSP by $0.68/kg, yield between 0.30 and 0.50 g/g by $0.53, capital ±30% by $0.43, utilities by $0.22, and batch time between 36 and 72 h by $0.21. So the conclusion at 50 g/L is carried by a feedstock price you do not control and a yield you do. That is the sentence to put in the discussion, and it also tells you that arguing about batch time is arguing about the fifth-largest term.

Beyond ranking, there is the question of what to report. A single MPSP implies a precision you do not have. Monte Carlo over the inputs you genuinely cannot pin down — here yield, glucose price and capital, each as a triangular distribution over its plausible range — gives a distribution instead: P10 $1.96, median $2.21, P90 $2.49, and a 1% probability of coming in below the incumbent's $1.80. That last number is the honest answer to "is this process viable at 50 g/L", and it is more useful than either optimism or a flat no.

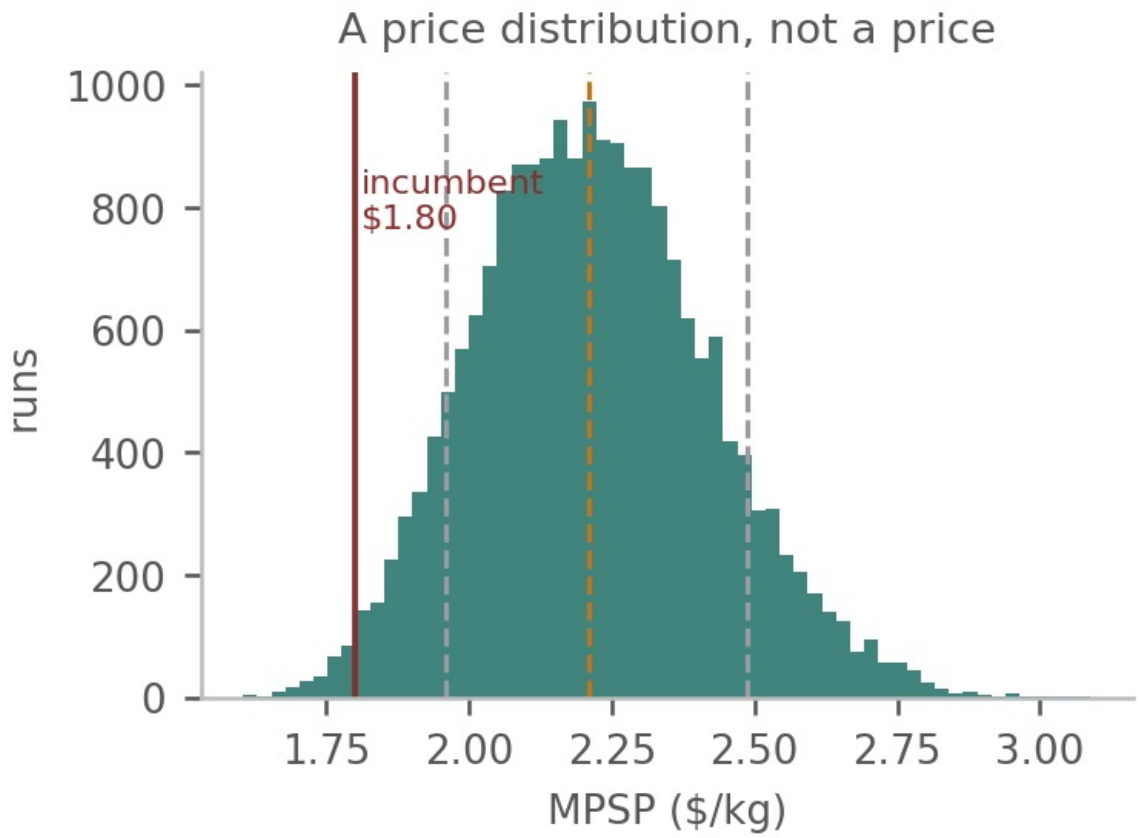


Figure 49.2 Twenty thousand Monte Carlo runs at 50 g/L over yield, glucose price and capital, with P10, median and P90 marked. The distribution sits almost entirely above the assumed incumbent price: the process does not fail because of an unlucky assumption, it fails structurally at this titer. Computed for this module.

Then the constructive step, which distinguishes a useful analysis from a discouraging one: **name the assumption carrying the conclusion and say what experiment retires it**. Here, yield — because unlike the glucose price it is ours to change, and because the model says each 0.05 g/g is worth about $0.13/kg. The experiment that retires it is a carbon balance on the current strain showing where the other 60% of the carbon goes, which is a week's work and changes what the next eighteen months are spent on.

Life-cycle assessment, and why the boundary decides the answer

Three definitions carry the whole subject. The **functional unit** is what you are comparing — one kilogram of product at a stated purity, not "a process". The **system boundary** says what is counted: cradle-to-gate means from the field or well to your factory gate, excluding use and disposal. **Allocation** decides how burdens are split when a process makes more than one saleable thing, and it is where honest people disagree: by mass, by energy content, by economic value, or by system expansion, and the choice can move the answer by a factor of two.

A first pass from your own mass and energy balances takes ten minutes. With glucose at an assumed 0.6 kg CO_2e/kg, a yield of 0.40 g/g contributes 1.50 kg CO_2e per kilogram of product; with 2.5 kWh/kg of electricity for agitation, compression and separation, at 0.40 kg CO_2e/kWh, energy contributes 1.00. Total 2.50 kg CO_2e/kg against a fossil incumbent assumed at 3.0 — a 17% advantage.

Now notice how fragile that is. A grid at 0.6 kg/kWh instead of 0.4 erases it entirely. Counting the nitrous oxide from fertiliser on the corn, or land-use change, raises the feedstock term. Giving the spent biomass credit as animal feed lowers it. The boundary choice, not the biology, decides whether the headline says 17% better or 20% worse — which is why reviewers go to the boundary first, and why your own LCA paragraph should state what it excludes before anyone asks.

Reading a published TEA critically

Four questions, in order. **What yield and titer does it assume, and are they achieved or projected?** Many published analyses model a target strain rather than an existing one, which is legitimate and must be stated. **What does the separation train cost, and was it designed or assumed?** Downstream is often half the capital and is frequently a single line item. **What feedstock price, and from what year?** Sugar prices swing by a factor of two, and

an analysis built on a 2015 price is a different analysis today. **What is the discount rate and plant life?** Moving from 10% over 20 years to 15% over 10 changes the capital charge per kilogram by about 70% and can be buried in a methods sentence.

Capital recovery	$CRF = i(1+i)^n/((1+i)^n - 1)$; 0.1175 at 10%, 20 y
MPSP	feedstock + utilities + fixed + CRF·capex/annual output
Feedstock term	substrate price ÷ yield; moved by yield only
Capital term	moved by titer and rate, through vessels and broth volume
Equipment scaling	$C = C_0(S/S_0)^{0.6}$, then an installation factor; Lang is a first pass
LCA first pass	feedstock CO_2e/yield + energy CO_2e per kWh × kWh/kg

Your strain is at 35 g/L and 0.28 g/g. Your supervisor wants a year of fermentation optimisation to reach 70 g/L. What does the model say, and what do you propose?

Run both. At 0.28 g/g the feedstock term alone is \$0.40/0.28 = \$1.43/kg, so even with capital and fixed costs driven to zero by an infinitely good titer the process cannot clear \$1.80 once utilities and labour are added — doubling the titer takes MPSP from roughly \$3.00 to \$2.30 and still loses money. Raising the yield to 0.40 g/g at the current titer of 35 g/L takes the feedstock term to \$1.00 and MPSP to roughly \$2.40; doing both gets to about \$1.75 and clears. So the answer is that neither alone is sufficient and yield is the one that must happen.

What I would propose: the carbon balance first, because at 0.28 g/g — probably half of theoretical — the carbon is going somewhere identifiable, and by-product measurement costs a week against a year of fermentation optimisation. Then the titer campaign, with the TEA rerun at each milestone so that the project has a stated pass mark rather than a direction. And the sentence that makes this a collaboration rather than an argument: the model says a year of titer work raises the price we can tolerate by \$0.70/kg, which is worth doing *after* the yield work and not instead of it.

Three things people get wrong

- ‘TEA is for the business school.’ It is the only calculation that ranks titer against yield against rate, which is the central strain-engineering decision.
- ‘The MPSP is \$2.16.’ It is a distribution; report P10, median and P90 and the probability of clearing the market price.
- ‘Our process is greener because it is biological.’ The boundary and the grid decide that, and a first-pass number from your own balances will tell you in ten minutes.

Self-check — answer on paper first

1. Explain why doubling the titer reduces cost but doubling the yield reduces it differently, in terms of the four cost lines.

Titer reduces the number of vessels needed for a given output and the volume of broth the separation train must handle, so it moves the capital charge and the fixed costs that scale with plant size; it leaves the feedstock term untouched, because the substrate consumed per kilogram of product is unchanged. Yield moves only the feedstock term, which in a commodity process is usually the largest single line. They are complementary rather than alternatives, and which to pursue first depends on which line currently dominates — which the breakdown tells you directly.

2. A published TEA reports \$1.20/kg for a product you know is made at 15 g/L. What do you look for?

Almost certainly a projected rather than achieved performance, so first the assumed titer and yield in the methods rather than the abstract. Then the separation train, since at 15 g/L the broth volume per kilogram is 67 litres and the downstream capital should be large; if it is a single assumed line item, that is where the \$1.20 comes from. Then the discount rate and plant life, and whether co-products have been credited by an allocation choice that moves most of the burden elsewhere. None of these makes the paper wrong; all of them are needed to know what it claims.

3. Your LCA gives 2.5 kg CO_2e/kg against a fossil incumbent at 3.0. Why is that not yet a claim you can publish?

Because the comparison rests on boundary and allocation choices that have not been justified: whether land-use change and fertiliser emissions are in the feedstock figure, what grid intensity the electricity is costed at, whether spent biomass receives a co-product credit, and whether the two routes deliver the same functional unit at the same purity. A 17% advantage is well inside the range those choices move the answer, so the publishable version states the boundary, shows the result under the two or three defensible allocation choices, and says which of them the conclusion survives.

What to open a book for. No book, but real software: install BioSTEAM, run a Bioindustrial-Park configuration unchanged,

then build your own. Cortes-Peña et al., ACS Sustain. Chem. Eng. (2020) is the paper to cite and to read for what the platform does with uncertainty. Peters, Timmerhaus and West, or any process-economics text, covers capital estimation properly if you want the engineering rather than the API.

Analytical method validation: trusting your own numbers

Every conclusion in your thesis is a number from an instrument, multiplied by a calibration, corrected by a recovery, in a matrix that is not water. This week is about the multiplication.

Calibration, and the fit that lies to you at the bottom of the range

Build the curve from a dilution series spanning the concentrations your samples actually occupy, plot the residuals, and ignore R². Here is why, computed rather than asserted. Take a genuinely linear detector — true slope 12,500 area units per g/L, true intercept 150 — with noise that is 4% proportional plus a small constant term, which is what a chromatographic detector really does. Standards from 0.05 to 50 g/L. Fit it two ways.

	Slope	Intercept	Back-calculated error at 0.05 / 0.1 / 0.25 g/L
True values	12,500	150	—
Unweighted least squares	11,767	3,431	−551% / −260% / −98%
Weighted (1/σ²)	12,382	229	−11% / +7% / +6%

Both fits have R² above 0.999. The unweighted fit is dominated by the 25 and 50 g/L standards, because their absolute residuals are hundreds of times larger than those at the bottom, and it pays for its fit up there with an intercept of 3,431 — which, when you back-calculate a sample at 0.05 g/L, returns a negative concentration. A curve that reports negative concentrations for real samples is not a rare pathology; it is the normal consequence of unweighted regression over two orders of magnitude.

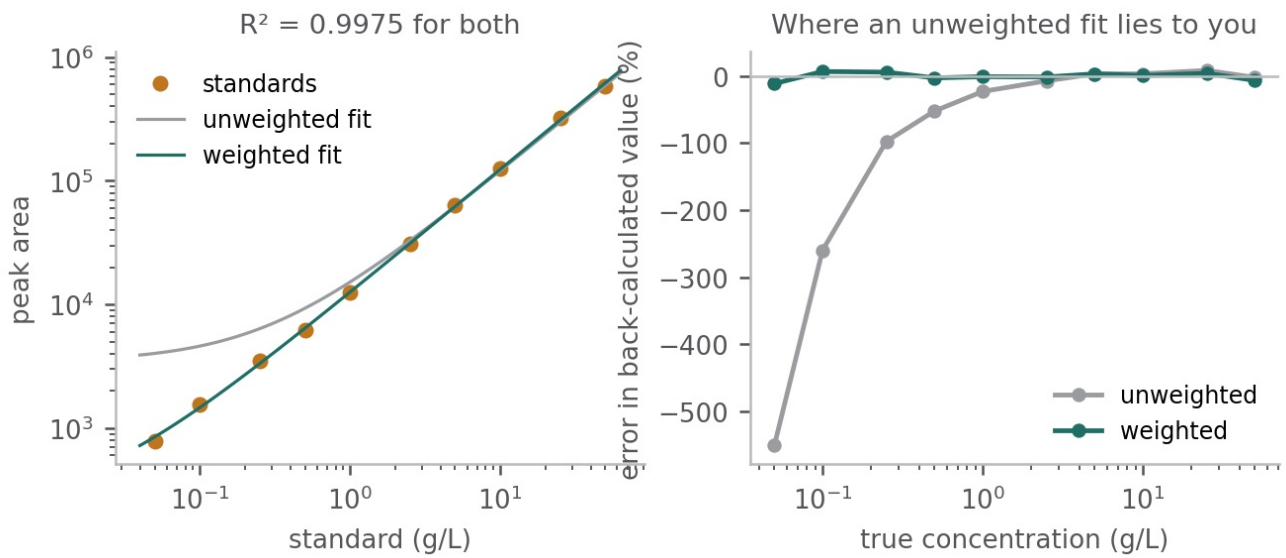


Figure 50.1 Left: the same ten standards fitted with and without weighting; both have R² above 0.999 and the lines are indistinguishable on a log plot. Right: the error in the back-calculated concentration of each standard. The unweighted fit is unusable below about 1 g/L. Synthetic data with 4% proportional noise, computed for this module.

Three rules follow. **Weight by 1/σ²** when the error is proportional — in practice 1/x or 1/x² weighting, which every chromatography data system offers and almost nobody switches on. **Do not force through zero** unless you have evidence the intercept is zero; forcing it hides exactly the baseline offset and carryover that a non-zero intercept is telling you about. And **state the working range**, with a rule for what happens outside it: samples above the top standard get diluted and re-injected, samples below the quantification limit get reported as below it, not as a number.

Limits of detection and quantification, and the standard abuse

With σ the standard deviation of the blank response, or of the residuals near zero, and m the calibration slope, the conventional definitions are

```
LOD = 3.3·σ/m        LOQ = 10·σ/m
```

In the example above, σ ≈ 60 area units and m = 12,382, so LOD = 0.016 g/L and LOQ = 0.048 g/L. What each permits you to claim is different and the difference is the point. Above the LOD you may say the analyte is present. Above the LOQ you may say how much. Between them you have evidence of presence and no measurement — and the abuse, which you will see in print constantly, is a table of values at a third of the LOQ carried through to a yield calculation with three significant figures and an error bar derived from triplicate injections of the same unmeasurable sample.

The honest handling of a trace result is to report it as < LOQ, to say what the LOQ is, and — if the trace matters to the conclusion — to concentrate the sample, change the method, or state that the experiment cannot answer the question. A negative result at a stated LOQ is a measurement; a small number below the LOQ is not.

Accuracy, precision, and four words that mean four experiments

Repeatability is the spread of replicate injections of the same sample, on the same instrument, in one session: it measures the instrument. **Intermediate precision** is the spread across days, operators, columns and calibration curves: it measures your laboratory, and it is typically two to five times worse than repeatability. Quoting repeatability where intermediate precision is meant is how a method comes to look better than the numbers it produces over a six-month project.

Accuracy is closeness to the true value and cannot be assessed by replication at all, only by recovery of a known addition. **Recovery** is the fraction of a spike you get back, and the single most important instruction of this week is that it must be measured in fermentation broth and not in water. Broth contains salts, protein, antifoam, residual sugar, pigments and cell debris, all of which suppress or enhance ionisation, bind analyte, co-elute, or foul the column. A method that recovers 99% in water and 62% in broth is a method that reports 62% of your titer, and the only way to find out is to spike the broth.

Where matrix effects are strong — which for mass-spectrometric quantification of intracellular metabolites is always — an internal standard becomes necessary rather than good practice, and the right one is an isotope-labelled version of the analyte: it experiences the same suppression, the same extraction losses and the same instrument drift, and the ratio cancels them. A structurally different internal standard cancels injection volume and little else.

Before the instrument: where the sample is actually lost

For an intracellular measurement, most of the error happens before anything is injected. Metabolite pools turn over in seconds — chapter 13's argument for the pseudo-steady state, read backwards — so the sample must be quenched faster than that, usually into cold methanol, and every known quenching method has an artefact: cold methanol leaks small metabolites out of the cells, fast filtration takes time, and whole-broth extraction measures intracellular and extracellular pools together. There is no method without a loss; there is only a method whose loss you have characterised.

The rule that makes this tractable is the one worth memorising: **a 30% extraction loss applied evenly is survivable; a loss that varies between strains invalidates the comparison**. A systematic 30% under-recovery shifts every pool by the same factor, so ratios, fold changes and flux interpretations survive. But if your producing strain has a different membrane composition, a different cell size or a different lysis behaviour from the control — which engineered strains often do — then the recovery differs between the two and the comparison is measuring extraction, not biology. Hence: measure recovery separately in each strain being compared, and say so in the methods.

Carryover, drift and the sample queue

Two instrument behaviours corrupt long runs, and both are revealed by queue design rather than by inspection. **Carryover** is analyte from the previous injection appearing in the current one; detect it by placing a blank immediately after your highest standard and quantifying what appears in it, expressed as a percentage of the high standard. **Drift** is a slow change in response across a run; detect it by placing a quality-control sample of known concentration every ten injections and plotting its measured value against position in the queue.

Then the design that protects the conclusion: randomise the order of samples across treatments, so that a drift cannot align with your experimental groups. This is week 17's randomisation argument applied to the instrument instead of the plate, and it is the reason a queue that runs all the controls first and all the engineered strains afterward can manufacture a difference out of a drifting detector. Blocking works too — alternate control and test — and is easier to defend.

The method section, as a checklist

A methods section a stranger can evaluate contains: instrument and column with dimensions and particle size; mobile phase and gradient, or oven programme; detector and wavelength or ion transitions; standards, their source and purity,

and how the stock was prepared; internal standard and its concentration; calibration range, number of levels, replicates, weighting, and whether the intercept was forced; LOD and LOQ with how they were determined; recovery in matrix, by strain if strains are compared; repeatability and intermediate precision with the experiments that produced them; the queue structure including blanks, QCs and randomisation; and sample handling — quenching, extraction, storage temperature and time to analysis. It is a long list and most of it is one line each.

Weighted calibration	`weight 1/σ² (in practice 1/x or 1/x²) when error is proportional`
Limits	`LOD = 3.3σ/m, LOQ = 10σ/m; 0.016 and 0.048 g/L in the example`
Precision	`repeatability = one session; intermediate = days, operators, columns`
Accuracy	`only from recovery of a known spike, in broth and not in water`
Extraction loss	`even loss survivable; strain-dependent loss invalidates the comparison`
Queue	`blank after the top standard (carryover), QC every 10 (drift), randomise order`

A collaborator reports that strain B makes 0.31 g/L against strain A's 0.09, a 3.4-fold improvement, from triplicate injections with an R^2 of 0.9997 calibration. What do you ask, and what is your guess?

Ask four things. Whether the calibration was weighted and what the LOQ is — if the LOQ is near 0.1 g/L, strain A's value is below it and the fold change is a ratio involving an unmeasurable number. Whether the triplicates are injections or independent cultures; three injections of one culture is n = 1 in the week 17 sense. Whether recovery was measured in broth for both strains, since a strain-dependent difference in extraction would produce exactly this result. And whether the two strains were run in the same queue and in what order.

My guess, in order of likelihood: the low value sits below the quantification limit, so the true statement is "strain A was below the LOQ of 0.1 g/L and strain B measured 0.31 g/L", which is still a real and publishable improvement but is a different claim with a different error structure. Second most likely is an unweighted calibration making the low end unreliable. The improvement is probably real; the number 3.4 probably is not, and the fix — rerun both with a weighted low-range calibration, independent cultures and a matrix spike — costs two days.

Three things people get wrong

- ‘R^2 = 0.999, so the calibration is good.’ R^2 is insensitive to exactly the low-end bias that ruins trace measurements. Plot residuals, and weight the fit.
- ‘Below the LOQ but we reported it anyway.’ Below the LOQ you have presence, not a value. Report it as such.
- ‘Recovery was 98%.’ In water. Broth is not water, and the number you need is per strain if strains are being compared.

Self-check — answer on paper first

1. Why does an unweighted calibration over two orders of magnitude systematically mis-measure the low end, and what does the symptom look like?

Because least squares minimises absolute residuals, and with proportional error the absolute residuals at the top of the range are orders of magnitude larger than at the bottom, so the top standards determine the line. The fit buys accuracy there at the cost of an inflated intercept, and the symptoms are a positive intercept far larger than the blank, back-calculated standards at the bottom that are badly wrong or negative, and real low-concentration samples reporting negative concentrations — all while R^2 remains excellent.

2. Design the experiment that establishes recovery for an intracellular metabolite in two strains being compared.

Spike a known amount of the analyte — ideally the isotope-labelled version, so it is distinguishable from what the cells contain — into the cell suspension of each strain immediately before quenching, then carry both through the full quench, extraction and analysis. Recovery is the measured spike over the added spike, computed separately for each strain, with replicates. Spiking after extraction instead measures only matrix suppression at the instrument, which is also worth knowing: the ratio of the two spikes separates extraction loss from ionisation suppression, and reporting both is what a careful methods section does.

3. Your QC samples drift downward by 12% across a 90-sample queue. What does that do to your results, and what should have been done?

If the samples were ordered by treatment, it manufactures a difference of up to 12% between groups that is purely instrumental, and if the real effect is of that size the conclusion is unsafe. The corrections available after the fact are to normalise against the QC trend or against an internal standard, both of which assume the drift affects samples and QCs

identically. What should have been done is randomisation or blocking of the run order, plus the internal standard — which would have cancelled most of the drift automatically and turned a salvage operation into a non-event.

What to open a book for. No textbook; the sources are a chromatographic method-validation tutorial and a review of quenching and extraction for microbial metabolomics. Read one published methods section in your own product area alongside them and list what it omitted — that exercise teaches more than either source, and the list becomes your own template.

Omics, screening, and turning data into model constraints

An omics dataset is not evidence about flux. It becomes useful at the moment you can say which parameter or bound in a model it sets — and for three of the four layers, the honest answer is none directly.

The four layers, and what each cannot tell you

- **Transcriptomics** measures mRNA abundance. It is cheap, genome-wide and fast, and it measures the layer furthest from flux: mRNA predicts protein only loosely, protein predicts activity only loosely, and activity predicts flux only in combination with substrate concentrations. What it is genuinely good for is detecting a regulatory response — a stress programme firing, a catabolite repression you did not expect, a promoter silenced in stationary phase.
- **Proteomics** measures protein abundance, and is the first layer that constrains anything quantitatively, because an enzyme's abundance times its turnover number is an upper bound on the flux it can carry. Absolute quantification is harder than relative, and relative is what most datasets are.
- **Metabolomics** measures pool concentrations. It cannot give flux — a pool can be large and static or small and fast — but it gives two things nothing else does: the mass-action ratios that decide thermodynamic feasibility and direction (chapter 14), and the accumulating intermediate that localises a bottleneck.
- **Fluxomics**, meaning ^{13}C metabolic flux analysis, measures flux itself, including the split ratios and reversibilities that every stoichiometric model has to assume. It is the only layer that is evidence about what happened rather than about what the cell was equipped to do, and it is the most laborious.

For a strain that grows well and makes nothing, measure metabolites first. Growth being normal rules out a gross burden or energy problem, so the question is where in the installed pathway the carbon stops — and an accumulating intermediate answers it directly and cheaply, while a transcript dataset would tell you that your genes are expressed, which you already believe.

Analytics, and the throughput that matters

HPLC with UV or refractive index detection is the workhorse for extracellular titers, sugars and organic acids: robust, quantitative, and slow at a few minutes to half an hour per sample. GC-MS needs volatility or derivatisation and is the standard for fatty acids, terpenes, alcohols and derivatised amino acids. LC-MS covers polar, charged and thermally fragile compounds and is what intracellular metabolomics is done on, at the cost of matrix effects severe enough to make chapter 50's isotope-labelled internal standards mandatory. The sample-preparation step that most often ruins metabolomics is the quench, for the reasons in the last chapter.

Then screening throughput, which is the week-11 test bottleneck quantified:

Format	Throughput	What it measures
Shake flask with HPLC	tens per week	the product, properly, with yields
Microtitre plate with a colorimetric or fluorescent proxy	10^3–10^4 per day	a correlate of the product
Biosensor-coupled FACS	10^7 per hour	a transcription-factor response to an intracellular metabolite
Growth coupling or selection	10^9 and above	survival, which you have arranged to require production

Five orders of magnitude separate the top and bottom rows, and each step down the table trades fidelity for throughput: the thing measured gets further from the thing you want. That trade is the reason the library-coverage arithmetic of chapter 17 matters — a 10^6 library can only be searched by the bottom two rows — and the reason every selection campaign needs the confirmation rate reported, because cells optimise what is selected and will find ways to light up a biosensor without making product.

From a dataset to a constraint: the arithmetic

Here is the calculation that turns proteomics into modelling, and it is one line. An enzyme's maximum flux is its abundance times its turnover number:

```
v ≤ k_cat · [E]
```

Put real units on it. Cell protein is about 0.5 g per gram of dry cell weight. An enzyme at 1% of that protein, with a molecular weight of 50 kDa, is present at 5 mg/gDCW = 1×10^{-7} mol/gDCW. With $k_{cat} = 10\ s^{-1}$, the ceiling is 10^{-6} mol·s^{-1} per gram of cells, which is **3.6 mmol·gDCW^{-1}·h^{-1}**. Invert it and you have the more useful form: carrying 10 mmol·$gDCW^{-1}$·h^{-1} through an enzyme with that k_{cat} requires 2.8% of the cell's entire protein, for one step.

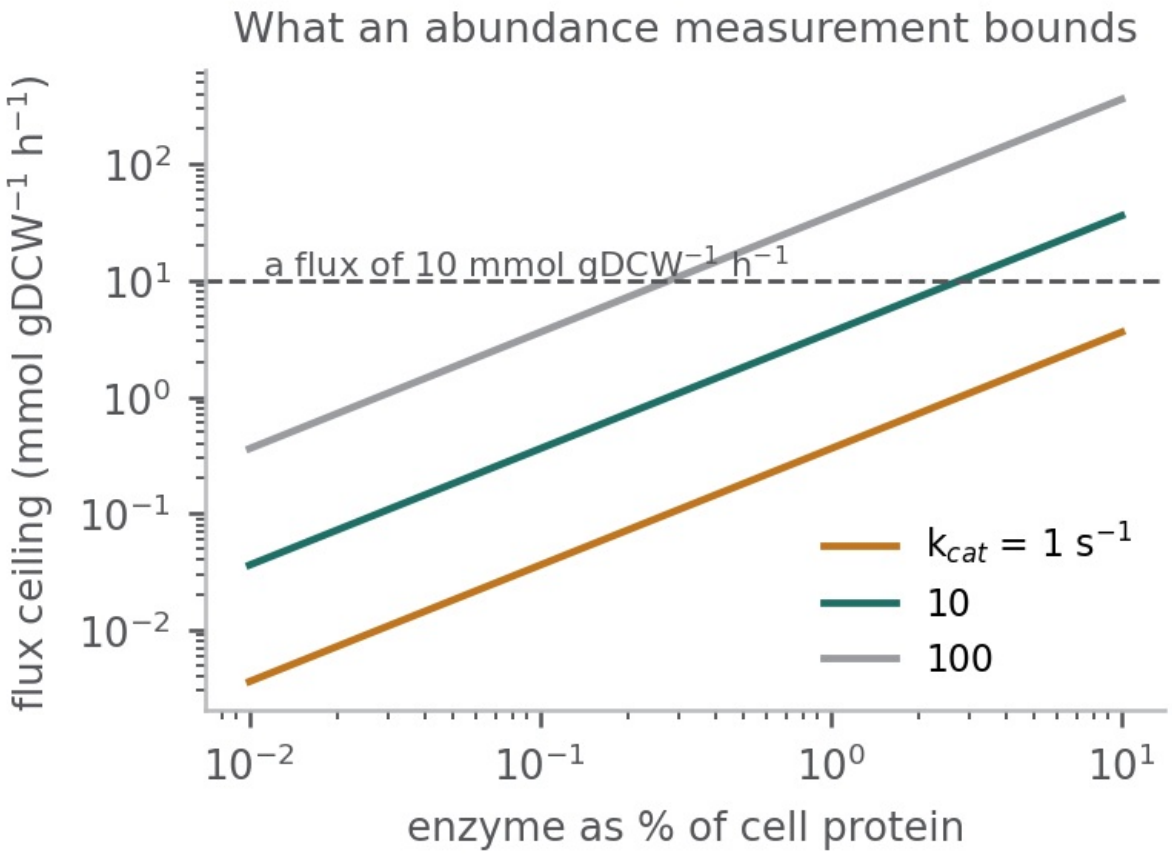


Figure 51.1 The flux a single enzyme can carry, against its share of cell protein, for three turnover numbers, assuming 0.5 g protein per gDCW and a 50 kDa enzyme. The dashed line is a flux of 10 mmol·$gDCW^{-1}$·h^{-1}, roughly a tenth of glycolytic flux in a fast-growing cell. Arithmetic, computed for this module; k_{cat} and molecular weight are the assumptions.

Two things follow immediately. First, a low-k_{cat} enzyme is not merely slow, it is expensive: at $k_{cat} = 1\ s^{-1}$ the same flux needs 28% of the proteome, which is impossible, so the pathway cannot run at that rate however hard you express it. That is a design rejection you can make from a single literature k_{cat}, before any cloning. Second, summing the requirement across all enzymes gives the proteome budget — $\Sigma\ v_i \cdot MW_i / k_{cat,i} \le$ total protein — which is what enzyme-constrained models such as GECKO add to flux balance analysis. That single extra constraint is what makes a constraint-based model predict overflow metabolism and the growth laws of week 9 instead of needing them imposed by hand, because fermentation is cheaper in protein per ATP than respiration.

Layer by layer: what each can and cannot set

Layer	Can set	Cannot set
Transcriptomics	Which reactions to switch off in a condition-specific model; a qualitative on/off prior	Any flux bound quantitatively — the mRNA-to-flux chain has two lossy steps in it
Proteomics	An upper bound $v \le k_{cat}[E]$ per reaction, and the total proteome budget	The actual flux, which may be far below the bound; and k_{cat} is not measured here
Metabolomics	Mass-action ratios, hence $\Delta G'$ and direction bounds; saturation state against K_M	Flux, and the distinction between a static pool and a fast one
Fluxomics	The fluxes themselves, with split ratios and reversibilities	Much beyond central metabolism, where labelling information runs out

The honest caveats on the proteomics route are worth stating because they decide how much to trust the resulting model. The k_{cat} values come from in vitro assays, often on the enzyme from a different organism, at a temperature and

pH unlike the cytoplasm, and they can be wrong by an order of magnitude. Absolute protein quantification requires labelled standards or careful intensity-based methods. And an upper bound is not a prediction: an enzyme present at 1% of protein and carrying 1% of its ceiling is perfectly normal, so a model constrained this way tells you what is impossible rather than what is happening.

Which brings the phase 3 checkpoint's best question: the difference between a flux that FBA calls feasible and one that ^{13}C-MFA calls actual. FBA returns a flux vector consistent with stoichiometry, the bounds you imposed and an assumed objective — a hypothesis about what could happen. ^{13}C-MFA infers fluxes from measured labelling patterns — evidence about what did happen, including the split ratios and reversibilities a stoichiometric model must assume. A design justified only by FBA should be validated by labelling before it is believed, and the sentence to be able to say is that the two methods answer different questions and the second is the one that can falsify a strain hypothesis.

Enzyme ceiling	`v ≤ k_cat·[E]; 1% of protein at k_cat 10 s⁻¹ → 3.6 mmol gDCW⁻¹ h⁻¹`
Protein cost of a flux	`fraction of proteome = v·MW/(k_cat · 0.5 g protein per gDCW)`
Proteome budget	`Σ v_i·MW_i/k_cat,i ≤ total protein — the GECKO constraint`
Screening ladder	`flask 10¹/wk · plate 10³–10⁴/d · FACS 10⁷/h · selection 10⁹`
FBA against ^{13}C-MFA	`feasible under an assumed objective, against inferred from labelling`

Your strain's titer is a third of what your FBA predicts. Proteomics shows the pathway enzymes are abundant and transcriptomics shows the genes are strongly expressed. What do you conclude, and what do you measure next?

That the limitation is not expression, which is what those two datasets establish and the only thing they establish. Four candidates remain, and they are distinguishable. The enzymes may be abundant and inactive — misfolded, missing a cofactor, or mis-assembled — which an in vitro activity assay on the lysate settles in a day. The pathway may be thermodynamically or kinetically stalled at a particular step, which metabolomics settles by showing an accumulating intermediate and a mass-action ratio near equilibrium. The carbon may not be arriving, which a ^{13}C labelling experiment or simply a carbon balance with by-products settles. Or the FBA prediction may be wrong, which is likely if it assumed growth maximisation for a burdened engineered strain.

Order: lysate activity first, because it is a day and it bisects the space — abundant and active sends you to supply and thermodynamics, abundant and inactive sends you to expression conditions and cofactors. Then metabolite pools across the pathway. Then labelling, which is weeks of work and should be spent only once the cheaper layers have narrowed the question. And note what the enzyme-constraint arithmetic adds: if the abundant enzymes are carrying far less than $k_{cat}\cdot[E]$ allows, the limitation is definitely not the enzymes, which is a useful thing to be able to state quantitatively.

Three things people get wrong

- ‘The transcriptomics shows the pathway is upregulated, so flux increased.’ Two lossy steps sit between mRNA and flux. It shows regulation responded.
- ‘Metabolomics gives us the flux.’ It gives pool sizes. A large pool can be static and a small one fast; flux needs labelling.
- ‘An enzyme-constrained model predicts the flux.’ It bounds it. A model built from upper bounds tells you what is impossible.

Self-check — answer on paper first

1. Compute the fraction of cell protein needed to carry 5 mmol·gDCW^{-1}·h^{-1} through a 60 kDa enzyme with k_{cat} = 2 s^{-1}, and say what you would do with the answer.

The flux is $5 \times 10^{-3}/3600 = 1.39 \times 10^{-6}$ mol·s^{-1}·gDCW^{-1}, so the enzyme needed is that divided by k_{cat}, 6.9×10^{-7} mol/gDCW, which at 60 kDa is 42 mg/gDCW — about 8% of the cell's 0.5 g of protein, for one step. That is a serious but not impossible burden, and what I would do with it is treat the enzyme as the first candidate for protein engineering or replacement by a faster homologue, because every doubling of k_{cat} halves a proteome cost that is already competing with growth.

2. Why does adding a proteome budget to flux balance analysis make it predict overflow metabolism without being told to?

Because respiration requires far more protein per ATP produced than fermentation does — the respiratory chain, the TCA cycle and their enzymes are a large investment — so once total protein is a constrained resource, the optimal solution above a threshold uptake rate is to make part of the ATP by the protein-cheap fermentative route and excrete the product. Plain FBA, which has no protein constraint, sees no reason to waste carbon and predicts full respiration. The

overflow is therefore a consequence of the constraint rather than an observation added to the model, which is exactly the proteome-allocation account of week 9.

3. You can run either one ^{13}C-MFA experiment or twenty plate-based screens this quarter. How do you decide?

By what the current bottleneck is. If you have many untested designs and no idea which works, the screens buy parallelism and the labelling buys a detailed picture of a strain you may abandon. If you have one strain whose behaviour you cannot explain — product low with no accumulating intermediate and no by-product — the labelling buys the only evidence that will settle where the carbon goes, and twenty more screens will produce twenty more unexplained strains. The general rule from chapter 11 is that parallelism beats depth when the prior is poor, and depth beats parallelism when a single well-characterised failure would change the design.

What to open a book for. Ko, Lee et al., Chem. Soc. Rev. 49:4615 (2020) is the review to read for the omics layers and how a systems-metabolic-engineering group actually uses them. For enzyme-constrained models, read the GECKO methodology paper rather than a textbook; for ^{13}C-MFA, week 43's sources. No book covers the translation from dataset to constraint, which is why the table in this chapter exists.

Host defences and pooled screens: getting DNA into a wild strain, and reading 50,000 of them at once

A wild industrial strain has spent its evolutionary life defending itself against incoming DNA, and it is good at it. The arithmetic of how good takes five minutes and tells you whether your construct can survive the journey.

Why this week is optional, and when it stops being optional

The tracker marks this optional because a project in *E. coli*, *S. cerevisiae* or another laboratory host never meets the problem: those strains have had their restriction systems removed or characterised decades ago, and the cloning kits assume it. The week becomes mandatory the moment you work with a wild isolate, an industrial production strain, a non-model host from week 25, or anything a collaborator describes as "the strain our group uses" without being able to name its genotype. In those cases it is the first week to do, not the last, because the alternative is three months of transformations that return nothing and are blamed on competence.

Restriction–modification, in four lines

- **Type I.** A single complex that both methylates and cleaves, recognising a bipartite asymmetric site and cutting at a distance from it, often thousands of base pairs away. Hard to predict, and the cleavage position means site removal from the construct is not a reliable defence.
- **Type II.** Separate enzymes for cleavage and methylation, recognising a short usually palindromic sequence and cutting within or beside it. These are the familiar cloning enzymes and the easiest case: known motif, predictable sites, removable by codon-level sequence changes.
- **Type III.** A combined complex that requires two inversely oriented recognition sites to cut, so a construct carrying only one site survives — which is why orientation matters when you count sites.
- **Type IV.** The inverse logic: it cleaves DNA that *is* methylated, in specific patterns. So DNA prepared in a standard *E. coli* cloning strain, carrying Dam and Dcm methylation, can be destroyed precisely because it was methylated — and the fix is a methylation-free cloning host, which is the opposite of the Type I and II fix.

REBASE lists the systems found in sequenced genomes along with their recognition motifs, so the first action for a new host is to look it up and write down the motifs. If the genome is sequenced but the systems are only predicted, treat the prediction as a hypothesis: methyltransferase genes are easy to recognise and their specificities are not.

How often a motif occurs, and why a 6-base cutter is fatal

This is the calculation of the week. For a random sequence of equal base composition, a specific k-base motif occurs with probability $(1/4)^k$ at each position, and a double-stranded plasmid of length L offers about $2(L - k + 1)$ positions once both strands are counted. For a 6 kb plasmid:

Motif length	Expected sites in 6 kb	P(no sites at all)
4 bp	46.9	4×10^{-21}
5 bp	11.7	8×10^{-6}
6 bp	2.93	0.054
7 bp	0.73	0.48
8 bp	0.18	0.83

Read the third column, which is the probability that an unmodified plasmid gets past that one system untouched, taking the counts as Poisson. A single 6-base system gives your construct about a 5% chance; a 4-base system gives it none at all. And strains carry several systems — three independent 6-base specificities take the survival probability to $0.054^3 \approx$ **0.0002**, which is why the number of distinct systems matters more than the identity of any one of them. A strain with four or five systems destroys essentially everything foreign, and a transformation efficiency of a few colonies per microgram against 10^8 in *E. coli* is not poor technique, it is the defence working.

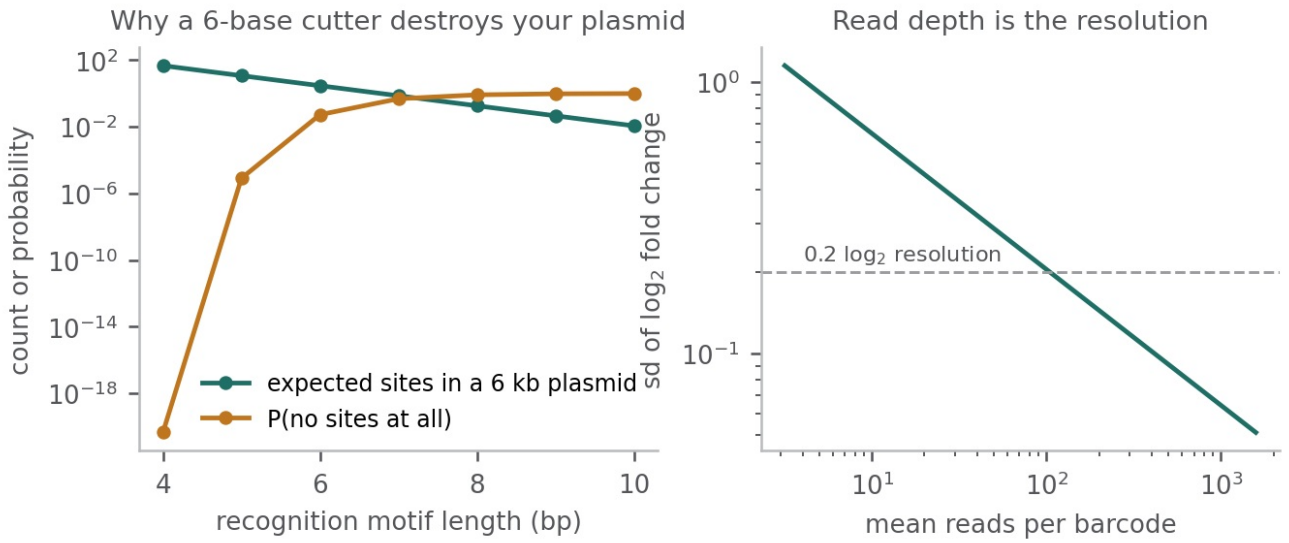


Figure 52.1 Left: expected sites and the probability of a site-free 6 kb plasmid against recognition-motif length, from the Poisson approximation to random sequence. Right: the standard deviation of a $\log_2$ fold change estimated from barcode counts at a given read depth, from counting statistics alone. Both computed for this module; the left panel assumes equal base composition, which real plasmids and real GC-skewed genomes violate, so treat it as the order of magnitude.

Getting past the barrier, and confirming that you did

Four routes, with different costs.

- **Methylate the plasmid in advance.** Express the host's own methyltransferases in your cloning strain, so the DNA arrives already marked as self. This is the general-purpose answer, it works against Types I to III, and it needs the methyltransferase genes cloned, which is a project of a few weeks and is reusable forever afterward.
- **Remove the sites.** Synonymous codon changes and non-coding edits to eliminate Type II motifs. Cheap when sites are few and the systems are Type II with known motifs; useless against Type I, whose cleavage is remote, and impossible when the motif count is in the dozens.
- **Change the cloning host.** Against Type IV, prepare the DNA in a *dam*$^{-}$ *dcm*$^{-}$ strain so there is no methylation to attack. One day of work, and the first thing to try if the host has a Type IV system.
- **Change the delivery.** Conjugation delivers single-stranded DNA, which some restriction systems handle poorly, and electroporation of large amounts of DNA can saturate the defences. Conjugation is the standard route into *Streptomyces* for exactly this reason, as week 10 noted.

The confirmation is a controlled comparison, and it is worth insisting on: transform the same plasmid, prepared methylated and unmethylated, side by side, and compare transformants per microgram. A tenfold to thousandfold difference is the defence being bypassed; no difference means the barrier was not your problem and you should look at competence, the origin of replication, the selection marker or the promoter instead. Without that control, a successful transformation after four changes tells you nothing about which change mattered.

Pooled screens: what a barcode buys

The second half of the week is a different subject with the same motivation — scale. A pooled screen constructs a library, attaches a sequencable barcode to each member, grows the whole pool under a condition, and reads the barcode abundances before and after. The fitness of each member is the log fold change in its count.

The structural insight, which is what the random barcoded transposon work made practical, is that barcoding **decouples library construction from fitness measurement**. You map barcodes to insertion sites once, expensively, by sequencing junctions; thereafter every experiment is a cheap amplicon sequencing of the barcodes alone. So the marginal cost of asking a new question of the same library is a PCR and a sequencing lane, and a library that cost months can answer dozens of questions.

What it cannot do is also structural. A pooled screen measures the average behaviour of each genotype in a mixed population, so it cannot see anything that depends on the clone being alone: product titer in its own vessel, morphology, anything cross-fed between neighbours — a mutant that cannot make a metabolite will be rescued by its neighbours and score as fit. It cannot report non-fitness phenotypes at all unless they are coupled to growth or to a sortable signal, and it cannot distinguish a genotype that grows well from one that grows slowly and lyses little. Those questions need isolated clones, which is why the last day of the week is about converting a pooled hit into a validated one.

Counting barcodes, and the read depth the question needs

Counts are Poisson, so precision follows from depth with no biology in it. For a barcode observed at mean depth d in each of two conditions, the standard deviation of the estimated $\log_2$ fold change is approximately $\sqrt{(2/d)}/\ln 2$:

Mean reads per barcode	sd of $\log_2$ fold change	Reads needed for a 50,000-member library
10	0.65	0.5 million
50	0.29	2.5 million
100	0.20	5 million
500	0.09	25 million

So detecting a twofold fitness effect — one $\log_2$ unit — reliably needs only tens of reads per barcode, while detecting a 15% effect, 0.2 $\log_2$ units, needs hundreds, and that is before biological variation between replicate cultures, which is usually larger than the counting noise and is the reason replicate pools matter more than extra depth. The practical sequence: decide the effect size that would change your decision, read off the depth, multiply by the library size, and compare with what a lane delivers. It is the same pre-experiment arithmetic as chapter 17's power calculation, wearing different clothes.

Analysis then follows week 33's RNA-seq machinery exactly, because the data have the same shape: normalise for sequencing depth and composition, model counts rather than ratios, shrink fold changes for low-count barcodes, and control the false discovery rate across thousands of tests rather than the per-test error. A volcano plot of fitness against significance is the standard output, and the barcodes with large effects and low counts — the top-left of the plot — are where the false positives live.

From a pooled hit to a phenotype you believe

A hit from a pooled screen is a hypothesis about one genotype, measured in competition, with a mutation whose identity comes from a barcode map. Validating it has three steps and they are not optional. **Reconstruct cleanly**: rebuild the mutation in the parent background, because the library member may carry passengers. **Measure it alone**: the phenotype in isolation is the one you will report, and it frequently differs from the competitive one — a mutant can be fit in a pool because of what its neighbours provide. **Measure it in competition with the parent**: this reproduces the screen's own measurement and tells you whether the pooled result was real.

How many replicates the follow-up needs comes straight from chapter 17: for an effect of about one pooled standard deviation, n = 3 has roughly a third of the power you want, so plan for six to eight independent cultures if the decision matters. And decide in advance what would make you drop the hit — for instance, no effect in isolation and no effect in a clean competition, which together say the barcode was mis-mapped or the member carried a passenger mutation. Writing that pass mark down before the follow-up is what stops a hit surviving on sunk cost.

Expected sites	`2(L − k + 1)·(1/4)^k; 2.93 for a 6-mer in 6 kb`
Site-free plasmid	`P ≈ e^(−expected); 0.054 for one 6-base system, 0.0002 for three`
Bypass	`pre-methylate (I–III), site removal (II only), dam⁻dcm⁻ host (IV), conjugation`
Confirmation	`transformants per µg, methylated against unmethylated, same prep`
Barcode precision	`sd(log₂FC) ≈ √(2/depth)/ln 2; 0.20 at 100 reads per barcode`
Sequencing budget	`library size × depth; 50,000 × 100 = 5 million reads per sample`

A collaborator's industrial *Bacillus* strain takes your plasmid at 30 colonies per microgram against 10^8 in *E. coli*. Plan the next two weeks.

Week one is arithmetic and lookup, not bench work. Pull the strain's genome from REBASE or, if it is not listed, annotate the restriction–modification genes and note their predicted motifs and types. Count the occurrences of each motif in the plasmid, on both strands and with orientation recorded for any Type III system. Compare the counts with the table above: if there are two Type II systems with a few sites each, the predicted survival probability is percent-level, which matches an efficiency of 30 per microgram and means the barrier is almost certainly the explanation rather than competence. If the strain has a Type IV system, note that your standard cloning prep is methylated and is therefore the problem.

Week two is the bypass and its control. The cheapest first test is a *dam*$^-$ *dcm*$^-$ prep, which costs a day and resolves the Type IV case outright. In parallel, order synthesis of the host's methyltransferase genes for a methylating helper plasmid in the cloning strain — the general solution, reusable for every subsequent construct in this host. Then run the comparison that proves it: the same plasmid, prepared both ways, transformed side by side, efficiency per microgram reported for each. If efficiency rises by orders of magnitude the barrier is beaten; if it does not, the problem was never restriction, and the next suspects are the replication origin, the marker and the promoter — all of which are cheaper to test once restriction has been excluded by a control rather than by hope.

Three things people get wrong

- ‘Low transformation efficiency means poor competent cells.’ In a wild strain it usually means the defences are working, and the arithmetic will tell you before the bench does.
- ‘We removed the restriction sites.’ Which works for Type II and not for Type I, whose cleavage happens far from the recognition site, nor for Type IV, which attacks methylation rather than sequence.
- ‘The pooled screen says this gene is important.’ In competition, in a pool, averaged over a genotype. Reconstruct it and measure it alone before believing it.

Self-check — answer on paper first

1. Compute the expected number of sites for a 6-base motif in a 10 kb plasmid and the probability it carries none, and say what you would conclude from a strain with two such systems.

About $2 \times 10{,}000/4{,}096 = 4.9$ expected sites, so the probability of none is $e^{-4.9} \approx 0.007$. With two independent systems of that kind the plasmid's chance of arriving intact is roughly 5×10^{-5}, which means essentially every molecule is cut and any transformants you obtain arrived in rare unmethylated-and-uncut molecules or by repair. The conclusion is that the construct cannot be delivered unmodified and that pre-methylation or a different delivery route is required, not more electroporation.

2. Why does barcoding decouple library construction from fitness measurement, and what does that change about experimental design?

Because the expensive step — determining which barcode corresponds to which mutation — is done once by junction sequencing, after which the identity of every member is known from a short barcode amplicon. Each subsequent experiment therefore costs a PCR and a sequencing lane rather than a library build. What it changes is that the marginal cost of a new condition collapses, so the design should plan many conditions against one library rather than one condition against many libraries, and the limiting resource becomes the conditions you can think of and the replicate cultures you can run, not the cloning.

3. You want to resolve a 15% fitness difference in a 20,000-member library. What depth and how many reads, and what will actually limit you?

Fifteen per cent is about 0.2 $\log_2$ units, so from counting statistics you need the standard deviation below roughly that, which takes about 100 reads per barcode — 2 million reads per sample for 20,000 members, which is trivial on any modern instrument. What will actually limit you is biological variation between replicate pools, which at this effect size exceeds the counting noise, so the budget should go into independent replicate cultures rather than deeper sequencing of the same ones. That is the same conclusion as chapter 17's biological-against-technical replicate argument.

What to open a book for. REBASE is the reference and it is a database, not a book. Wetmore et al., mBio (2015) is the RB-TnSeq paper and is worth reading in full for the design logic rather than the protocol. For the restriction barriers, a review of restriction–modification in non-model bacteria; Snyder & Champness, from week 10, remains the best background on how these systems were worked out.

Closing notes to part 4

The four weeks in one page

Week 49. Minimum product selling price is the number that ranks titer against yield against rate. In the model here, 20 / 50 / 100 g/L give $3.22 / $2.16 / $1.80 per kilogram against a $1.80 incumbent, with feedstock unmoved by titer and capital unmoved by yield. Report a distribution, name the assumption carrying the conclusion, and say what experiment retires it. For the life-cycle number, the boundary decides the answer, so state it before anyone asks.

Week 50. An unweighted calibration over two decades is unusable at the bottom — in the worked example, −551% error at 0.05 g/L with R^2 above 0.999. Above the detection limit you may say present; above the quantification limit you may say how much. Recovery is measured in broth, per strain when strains are compared, and an even extraction loss is survivable while a strain-dependent one invalidates everything.

Week 51. Omics becomes useful when it sets a bound. An enzyme at 1% of cell protein with $k_{cat} = 10\ s^{-1}$ can carry 3.6 $mmol \cdot gDCW^{-1} \cdot h^{-1}$, and carrying 10 would take 2.8% of the whole proteome — which is how a single k_{cat} rejects a design before cloning, and how enzyme-constrained models predict overflow instead of assuming it. Transcriptomics sets nothing quantitative; metabolomics sets thermodynamics; only labelling measures flux.

Week 52. A single 6-base restriction system gives an unmodified 6 kb plasmid a 5% chance of arriving intact, and three such systems reduce it to 0.02%. The bypass depends on the system type and the confirmation is a methylated-against-unmethylated efficiency comparison. In pooled screens, barcoding makes new conditions nearly free; counting statistics set the read depth, and biological replicates set everything else.

On the numbers in this part

The techno-economic model is listed in full in chapter 49 and every parameter in it is an assumption — vessel size, batch time, installed capital, utilities, discount rate — so the output is a structure rather than a forecast, and the tornado and Monte Carlo panels exist to show which parts of the structure carry weight. The calibration demonstration uses synthetic data generated with 4% proportional noise; the slopes and errors quoted are what the fits returned. The enzyme-constraint arithmetic assumes 0.5 g protein per gram of dry cell weight and a 50 kDa enzyme, and k_{cat} values in the real literature are frequently wrong by an order of magnitude, which is the dominant uncertainty in any enzyme-constrained model. The restriction-site counts assume random sequence with equal base composition, which real plasmids violate, so they are order-of-magnitude guides rather than site counts — count the actual motifs in your actual sequence, which takes one line of code.

What is still to come

Working forward in tracker order, the remaining weeks with nothing behind them are the product-class weeks 53 to 57 — pharmaceuticals and terpenoids, commodity chemicals and amino acids, biofuels and C1, the PHA case study, and CO_2, hydrogen and waste carbon as feedstocks — and then the host weeks: yeasts in 58, *C. glutamicum* and *B. subtilis* in 59, phototrophs in 60, and the panorama with cell-free systems in 61.

Those two blocks fit together, and that is how the next parts will be written. A product class is a set of constraints — the precursor it draws on, its cofactor demand, whether it is secreted, whether it is toxic, what price it must clear — and a host is a set of capabilities against exactly those constraints. So the product chapters will end in a constraint sheet for each class, and the host chapters will be a comparison against those sheets rather than four species descriptions: tools and editing efficiency, growth rate and medium cost, native precursor and cofactor supply, secretion and post-translational capability, tolerance, regulatory acceptance, and the specific arithmetic each host forces — compartment transport costs for the yeasts, which is chapter 48's dead-core calculation with membranes instead of gel, and light penetration for the phototrophs, which is the same calculation again with photons instead of oxygen.

Part 5 — What the field has actually made: five product classes and the constraints each one imposes (weeks 53 to 57)

Phase 4 asks you to build a mental library of what has been achieved. The risk in these weeks is that the library becomes a list of anecdotes — artemisinin, butanediol, LanzaTech — which is what most candidates arrive with and what no interviewer is impressed by. So these chapters are organised the other way round: each product class is treated as a set of constraints, and every case study is there because it illustrates one of them. Each chapter ends with a constraint sheet, which is the form the host chapters of part 6 will be written against.

The single most useful habit for this phase is the one from part 2 chapter 18: convert every reported achievement into titer, yield as a fraction of theoretical, and rate, and then ask what price the product must clear. Done consistently, that reduces fifty papers to a table you can reason about and makes the pattern visible — which is that the technical difficulty of a pathway and the commercial difficulty of a product are almost uncorrelated. Twenty-one heterologous genes in yeast is a spectacular achievement at micrograms per litre; a four-gene commodity route is a decade of process work at a hundred grams per litre, and only one of them was ever a business.

Everything quantitative here is computed from stoichiometry, with the balance written out so you can check it. The reported titers are from the papers the tracker names, and are marked as reported rather than recomputed.

Ch.	Week	Chapter
53	53	Pharmaceuticals, terpenoids and natural products: the expensive-molecule regime
54	54	Commodity chemicals and amino acids: the cheap-molecule regime
55	55	Biofuels, C1 and gas fermentation, and food ingredients
56	56	PHA, from pathway to polymer: the case study worked all the way through
57	57	CO_2, hydrogen and waste carbon: when electricity is the feedstock

Pharmaceuticals, terpenoids and natural products: the expensive-molecule regime

When the product sells for thousands a kilogram, yield stops mattering and everything else starts: whether the pathway works at all, whether the enzymes are specific, and whether the molecule is worth making biologically rather than chemically.

The regime, before the case studies

Part 2 chapter 18 established that for a commodity, yield binds. Invert the price and the constraints invert with it. At $1,000/kg, glucose at a yield of 0.05 g/g costs $8 per kilogram of product, which is a rounding error, so nobody optimises yield. What binds instead is whether the pathway can be assembled at all — twenty enzymes from four organisms, several of them membrane-bound plant cytochrome P450s — whether those enzymes are specific enough not to make three side products for every product, and whether the resulting titer clears the floor set by the cost of processing broth. That floor is real but low: at 1 g/L a pharmaceutical is a viable process, where a commodity at 1 g/L is a demonstration.

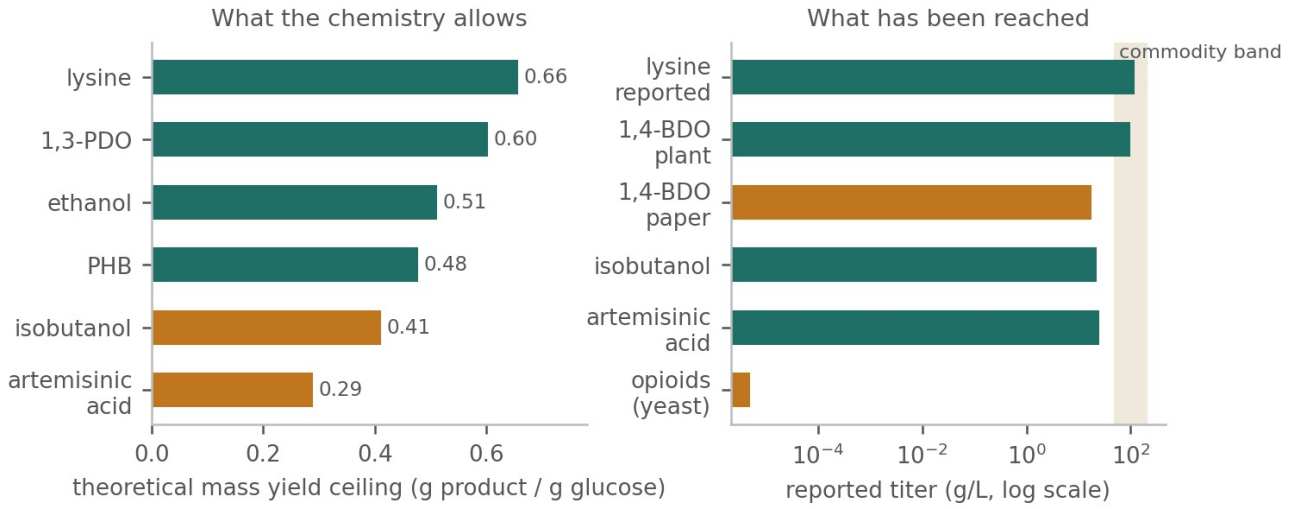


Figure 53.1 Left: theoretical mass-yield ceilings computed from the stoichiometry in this part, with the two natural-product-style targets in orange. Right: reported titers across the case studies of weeks 53 to 55, on a log scale, against the band a commodity process must reach. The spread is nine orders of magnitude, and it does not track pathway sophistication. Ceilings computed for this module; titers are as reported in the cited papers.

Artemisinic acid, and what 25 g/L actually required

The semi-synthetic artemisinin project is the field's set piece, and the reason to know it in detail is that it contains every lesson in the regime. The target: artemisinin, the frontline antimalarial, previously extracted from *Artemisia annua* with a supply chain that swung in price by factors of several. The route: engineer *S. cerevisiae* to overproduce farnesyl pyrophosphate through the mevalonate pathway, add amorphadiene synthase to cyclise it, add the plant cytochrome P450 CYP71AV1 with its reductase to oxidise the product three times to artemisinic acid, and then convert that chemically to artemisinin. Paddon and colleagues reported 25 g/L of artemisinic acid.

Work out what that number costs. Each five-carbon isoprene unit requires three acetyl-CoA, three ATP and two NADPH, and loses one carbon as CO_2 at the mevalonate decarboxylation step. A C15 sesquiterpene therefore takes nine acetyl-CoA, nine ATP and six NADPH before any oxidation. In yeast, cytosolic acetyl-CoA is made by the acetyl-CoA synthetase route, which cleaves ATP to AMP — two equivalents each, as part 1 chapter 4 insisted — so add eighteen more ATP equivalents. Then three P450 oxidations consume three more NADPH and three O_2. Totals per molecule of artemisinic acid:

```
9 acetyl-CoA  ·  27 ATP equivalents  ·  9 NADPH  ·  3 O₂  ·  3 CO₂ lost to the pathway itself
```

Nine acetyl-CoA means 4.5 glucose, so the theoretical mass yield is 234.3/(4.5 × 180.2) = **0.289 g/g** and the carbon yield is 56%. Reaching 25 g/L at the ceiling would consume 86 g/L of glucose, and the real process consumed considerably more, because 27 ATP equivalents per molecule is an enormous energy demand that has to be met by respiring carbon that then cannot become product. This is the quantitative content of the sentence everyone repeats about terpenoids being acetyl-CoA and NADPH sinks.

Two further lessons from the same paper. The P450 was the hard part — plant P450s in yeast need their reductase partner, haem, and a membrane environment, which is precisely the failure mode part 1 chapter 2 listed as the second hypothesis for an inactive band on a gel. And the process is *semi*-synthetic for a good reason: the final step from artemisinic acid to artemisinin is a photochemical oxidation that chemistry does cleanly and cheaply, and biology would do badly. The engineering judgement — make the complicated carbon skeleton biologically and the simple transformation chemically — is one of the most transferable ideas in the field, and it is the answer to give when asked where biology should stop.

Opioids in yeast, and what a complete pathway is worth

Galanie and colleagues reported complete biosynthesis of opioids in yeast from sugar, assembling 21 to 23 heterologous genes from plants, bacteria and rat. As an engineering feat it is extraordinary: the pathway spans two compartments, requires several P450s, an enzyme with a stereochemical inversion, and a reticuline epimerase that had only recently been identified. The reported titers were a few micrograms per litre.

Put that in the arithmetic of this module. At 5 µg/L you would need roughly 200,000 m³ of broth to make one kilogram of product — not a process, by five or six orders of magnitude. And yet the paper mattered, for two reasons worth separating. Scientifically, it established that the full pathway functions in a microbe, which converts the remaining work from discovery into optimisation. Politically, it triggered the field's standard dual-use discussion, because a yeast strain that makes a controlled substance from sugar is a different kind of object from a yeast strain that makes a terpene.

You should have a considered position on that, because it is an interview question and both available clichés are poor. Alarm ignores that the titers were a millionfold from usable, that poppy cultivation is not the limiting factor in opioid supply, and that the published strains are traceable, licensed and surveilled. Dismissal ignores that the gap between a microgram and a gram per litre has historically been closed in a few years, that strain distribution is harder to control than plant cultivation, and that governance regimes work better when designed before capability arrives than after. A defensible answer names both and lands somewhere specific: that the research should continue in licensed hands, that strain registries and sequence screening are proportionate, and that the field's own early engagement with the question is the reason the discussion has been calm.

The isoprenoid platform, and why the same pathway keeps appearing

Terpenoids are the largest class of natural products, and all of them are built from the same two five-carbon units — isopentenyl pyrophosphate and its isomer — by the same chemistry: condensation to C10, C15, C20 and higher prenyl chains, cyclisation by a terpene synthase, then decoration by P450s, dehydrogenases and transferases. This is why one engineered chassis serves many products: get a strain with a high flux to farnesyl pyrophosphate and you have a platform, and the product-specific work is one synthase plus a handful of tailoring enzymes.

Five commercial classes sit on that platform: flavours and fragrances such as nootkatone and valencene; pharmaceutical precursors like artemisinic acid and taxadiene; nutraceuticals and pigments including carotenoids such as astaxanthin and β-carotene; sterols and squalene for cosmetics and vaccine adjuvants; and cannabinoids, which are polyketide–terpenoid hybrids. The enzyme families recur across all of them — a prenyltransferase, a terpene synthase, a P450 with a reductase, a glycosyltransferase — which is why experience in one transfers.

Two routes exist to the five-carbon units and the choice has consequences from part 2 chapter 19: the mevalonate pathway, from acetyl-CoA, costs ATP and NADPH and loses a carbon; the methylerythritol phosphate pathway, from pyruvate and glyceraldehyde-3-phosphate, is more carbon-efficient but is tightly regulated in its native hosts and harder to de-repress. Yeast has the first, *E. coli* the second, and engineered strains frequently carry a heterologous copy of the other.

Technical success against market success

Semi-synthetic artemisinin worked and did not take the market. The reason is instructive and is not a failure of the engineering: plant-derived artemisinin remained competitive, partly because the agricultural supply responded to the same high prices that motivated the project, and partly because a fermentation route's cost floor includes capital that farmers do not pay. A programme that takes six years to deliver is competing against the price its own existence helped to stabilise.

The interview answer that uses this well avoids both cynicism and naivety. Something like: the project achieved what it set out to do technically and demonstrated that a complex plant natural product could be made at industrial scale, which changed what the field attempts; commercially it ran into the fact that an agricultural incumbent with a volatile price is a moving target, and the lesson the field took was to choose targets where the incumbent cannot respond — molecules that are hard to extract, hard to synthesise, or not available at all. That is also why the field's attention moved toward novel molecules and food ingredients rather than drop-in replacements for crops.

Constraint sheet — pharmaceuticals and natural products

Binding metric	Total output and purity. Yield on sugar is nearly irrelevant; titer matters only through processing cost.
Yield ceiling	Low by commodity standards — 0.29 g/g for a C15 terpenoid acid — and nobody optimises it.
Cofactor demand	Severe: 27 ATP equivalents and 9 NADPH per molecule for artemisinic acid. Acetyl-CoA and NADPH supply dominate strain design.
Pathway difficulty	High: 10–25 heterologous genes, plant P450s needing reductase, haem and membrane, compartment-spanning routes.
Host requirements	Eukaryote usually: endoplasmic reticulum for P450s, compartments, post-translational capability. Yeasts dominate.
Toxicity and removal	Hydrophobic and often membrane-toxic — part 3 chapter 45's regime; an overlay is standard, not optional.
Where it fails	Enzyme promiscuity producing side products, low-flux bottlenecks at tailoring steps, and an agricultural incumbent that can cut its price.

Three things people get wrong

- ‘Twenty-three genes in yeast is the hard part.’ Assembly is now routine; flux through the tailoring steps and enzyme specificity are the hard parts.
- ‘Artemisinin failed.’ It succeeded technically and lost commercially to an agricultural incumbent whose price moved — a different and more useful lesson.
- ‘Yield doesn't matter for pharmaceuticals, so stoichiometry doesn't either.’ The ATP and NADPH balance still decides whether the strain can run at all.

Self-check — answer on paper first

1. Compute the acetyl-CoA, ATP and NADPH demand of a C20 diterpene made through the mevalonate pathway in yeast, and say which supply you would engineer first.

Four C5 units: 12 acetyl-CoA, 12 ATP for the mevalonate steps, 8 NADPH, with 4 CO_2 lost at decarboxylation; plus 24 ATP equivalents if the acetyl-CoA comes through acetyl-CoA synthetase, giving 36 ATP equivalents in total. Acetyl-CoA first, because it is both the carbon and, through the synthetase route, most of the energy demand — which is why the standard interventions are a PDH-bypass replacement, routing through a cytosolic acetylating acetaldehyde dehydrogenase or an ATP-cheaper acetyl-CoA source, before touching the terpene synthase.

2. Why is a plant P450 the step most likely to fail, and what are the three fixes in order of cost?

Because it needs a membrane to insert into, haem, and a cytochrome P450 reductase partner at a matched ratio, and because its native environment is a plant endoplasmic reticulum at 25 °C. Cheapest first: co-express the reductase and supplement or upregulate haem synthesis. Then engineer the N-terminal membrane anchor, or swap to a homologue from a better-behaved organism. Most expensive: change host or compartment, or replace the P450 with a different enzyme chemistry altogether.

3. Give the argument for stopping the biology at artemisinic acid rather than going to artemisinin.

Because the remaining transformation is a photochemical oxidation and rearrangement that chemistry performs cleanly, cheaply and at scale, whereas biology would need several enzymes for a reaction with an unstable intermediate and would be slower and less selective. The general principle is to divide the synthesis where each discipline has the advantage: biology builds the chiral, polycyclic carbon skeleton from a cheap feedstock, chemistry performs the simple high-energy transformations. Insisting on a fully biological route is an aesthetic preference, not an engineering one.

What to open a book for. Three papers and no book: Paddon et al., Nature 496:528 (2013) for the build, Paddon & Keasling, Nat. Rev. Microbiol. 12:355 (2014) for the retrospective, and Galanie et al., Science 349:1095 (2015) for the opioid pathway and the governance discussion published around it. Log all three with the paper log's protocol; they recur in every interview in this field.

Commodity chemicals and amino acids: the cheap-molecule regime

Here yield is everything, the incumbent is a petrochemical plant with its capital already written down, and the strain is the easy part of a ten-year programme.

1,3-Propanediol: the archetype of a combined pathway

The DuPont bio-PDO process is the cleanest example of building a route that no organism has. The logic: *S. cerevisiae* has the genes to turn dihydroxyacetone phosphate into glycerol; *Klebsiella* has the genes to turn glycerol into 1,3-propanediol through a B_{12}-dependent dehydratase; neither organism does both. Put both halves into *E. coli* and you have glucose to PDO in one strain, which goes into Sorona polymer and won the 2003 Presidential Green Chemistry Challenge award for greener reaction conditions.

The interesting engineering is not the pathway, it is the energy. Compute the ceiling yourself, and the shape of the problem appears immediately. Glucose to two dihydroxyacetone phosphate costs 2 ATP; each DHAP to glycerol consumes one NADH; each glycerol to PDO consumes another. So two PDO per glucose requires **four NADH, and that route produces none**, because the NADH-generating step of glycolysis is downstream of the branch point. The redox balance must be closed by oxidising part of the feedstock: with x the fraction of glucose routed to product and full oxidation of the remainder yielding about 10 NADH per glucose, $4x = 10(1 - x)$ gives $x = 0.71$, so 1.43 mol of PDO per mole of glucose and a mass yield of **0.60 g/g**.

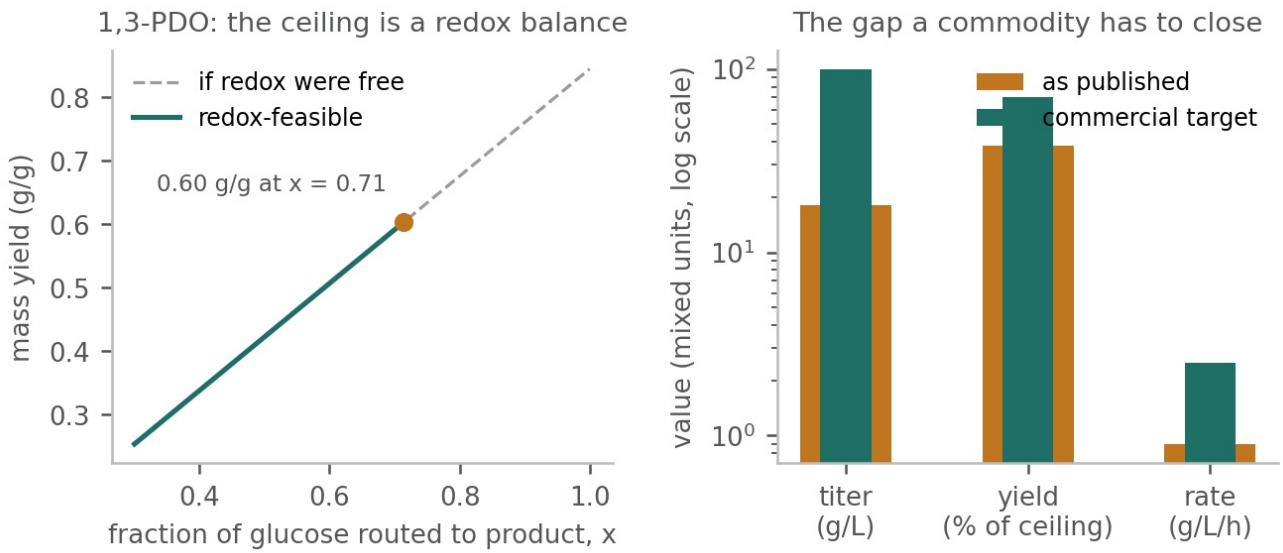


Figure 54.1 Left: the 1,3-propanediol ceiling as a redox balance — the dashed line is what carbon alone would allow, the solid line what redox permits, and the optimum sits where the NADH demand of the product exactly consumes the NADH from oxidising the rest. Right: the gap a commodity has to close, using the reported 1,4-butanediol numbers. Computed from the balances in the text; the right-hand panel's commercial figures are as reported rather than derived.

That is also why the published process re-engineered sugar uptake. The phosphotransferase system spends one phosphoenolpyruvate per glucose imported, and PEP is downstream of the branch point — so importing sugar by the native route burns carbon that the product cannot use, and replacing it with a permease plus a kinase changes the energy balance in the product's favour. Part 1 chapter 7 flagged this for aromatics; here it is the same lever for a different reason, and it is the kind of detail that distinguishes a process built by people who did the balance.

1,4-Butanediol: a molecule no organism makes

Yim and colleagues designed a route to BDO computationally — searching reaction databases for paths from central metabolism to a compound with no natural biosynthesis — then built it in *E. coli*, including an engineered oxidative TCA branch running under anaerobic conditions to supply the precursor and the reducing power. Reread it now with parts 2 and 4 behind you and three things stand out that are invisible on a first reading.

First, the knockouts were chosen by the strain-design algorithms of week 39, not by intuition: the deletions couple product formation to growth so that the strain cannot grow well without making BDO, which converts a screening

problem into a selection. Second, the reducing power comes from running part of the TCA cycle in the oxidative direction without oxygen, which is the kind of thing you only propose after writing the redox balance — it is week 6's anaplerosis argument used in reverse. Third, the paper reports 18 g/L and commercial operation required several times that, which the tracker quotes as above 100 g/L. That gap is the subject of part 2 chapter 18 and is the normal distance between a demonstration and a plant.

The case, written in the form the tracker asks for: host *E. coli*; target 1,4-butanediol, a non-natural product; pathway computationally designed from succinyl-CoA and α-ketoglutarate; tools genome-scale modelling, strain-design algorithms, growth coupling, multiplexed editing; TRY 18 g/L in the paper against above 100 g/L commercially; bottleneck precursor supply and redox under anaerobic conditions; commercial status produced at scale into polymer markets.

Why *C. glutamicum* owns the amino-acid market

Corynebacterium glutamicum was isolated as a glutamate excreter in the 1950s and has made amino acids at multi-million-tonne scale ever since. The properties that put it there are worth listing because they are what a host comparison in part 6 will be built from.

- **It excretes amino acids natively**, through dedicated exporters. Part 2 chapter 14's arithmetic says a charged product needs an exporter; this organism came with them, which is a larger advantage than any pathway engineering.
- **It tolerates very high product concentrations** — over 100 g/L of lysine — which is a tolerance phenotype nobody has had to engineer.
- **It has no significant overflow metabolism** of the acetate kind under normal industrial feeding, so the week 6 problem that dominates *E. coli* fed-batch is much reduced.
- **It is Generally Recognised As Safe in practice**, with decades of regulatory history for food and feed products, which matters enormously for amino acids destined for animal feed.
- **Its genetics are now routine**: genome sequence, strong promoter libraries, CRISPR tools and well-characterised integration sites, which was the missing piece thirty years ago.

The engineering story of lysine in this host is also the best single illustration of parts 1 and 2 working together. Part 1 chapter 3 said the first committed enzyme of an amino-acid pathway is feedback-inhibited by its end product and that a feedback-resistant allele is the single most effective change — in lysine that is aspartokinase, and the resistant allele is the foundation of every production strain. Part 1 chapter 5 computed the ceiling: 0.81 mol per mol, 0.66 g/g, limited not by carbon but by the NADPH the pathway demands and the carbon the pentose phosphate pathway burns to supply it. And the published systems-engineering strains report titers above 100 g/L at yields near half a gram per gram — roughly three quarters of the ceiling — assembled from a dozen defined genomic changes rather than from mutagenesis, which is the field's clearest demonstration that rational design works when the biology is known well enough.

Bioplastics, as the bridge to chapter 56

Cupriavidus necator accumulates polyhydroxybutyrate as a carbon and energy store when nitrogen or phosphorus runs out while carbon keeps arriving — up to the majority of its dry weight. That is a different kind of product from everything else in this chapter: it is intracellular, it is a polymer rather than a molecule, and it is made by deliberately stopping growth. Chapter 56 works it all the way through, because it is the case study with the most to teach about reading a fermentation, and because partitioning carbon between storage polymer and growth is a question the rest of this module has prepared you for exactly.

Constraint sheet — commodity chemicals and amino acids

Binding metric	Yield, then titer, then rate. Feedstock is usually the largest cost line; part 4 chapter 49's model says a strain below about 0.35 g/g cannot clear a $1.80/kg market whatever its titer.
Yield ceiling	High, and usually set by redox rather than by carbon: 0.60 g/g for PDO, 0.66 for lysine, 0.51 for ethanol. Write the NADH balance before the carbon balance.
Titer target	50–150 g/L, because broth volume per kilogram sets separation cost.
Pathway difficulty	Low — two to six genes. The difficulty is in balancing them and in the host, not in finding them.
Host requirements	Fast growth, cheap minimal medium, high tolerance, native export of a charged product, and a regulatory history if the product goes into food or feed.
Characteristic lever	Feedback-resistant first committed enzyme; sugar-uptake re-engineering when PEP is a precursor; growth coupling so that selection replaces screening.
Where it fails	Against an incumbent petrochemical plant with amortised capital and a volatile feedstock price; and in the decade between 18 g/L and 100 g/L.

Three things people get wrong

- ‘The ceiling is set by carbon.’ For most commodity routes it is set by redox — PDO's 0.60 g/g is a NADH balance, not a carbon count.
- ‘Non-natural products are the hard ones.’ BDO was designed and built in a few years; closing the titer gap took far longer.
- ‘*C. glutamicum* is a legacy host.’ Its advantages — native export, tolerance, little overflow, regulatory history — are exactly the ones that are hardest to engineer into something else.

Self-check — answer on paper first

1. Derive the 1,3-propanediol yield ceiling and explain in one sentence why the number is below the carbon limit.

Two PDO per glucose needs four NADH and the route to PDO generates none, since glycolysis makes NADH only below the branch point, so part of the feedstock must be fully oxidised to supply reductant: with roughly 10 NADH per glucose oxidised, $4x = 10(1 - x)$ gives $x = 0.71$ and 1.43 mol per mol, or 0.60 g/g. It is below the carbon limit of 2 mol per mol because the product is more reduced than the substrate per carbon, so electrons, not carbon atoms, are the scarce resource.

2. Why does replacing the phosphotransferase sugar-uptake system help a PDO strain, and what does it cost?

Because the phosphotransferase system consumes one phosphoenolpyruvate per glucose imported, and PEP lies below the branch point to glycerol, so the energy for uptake is taken from carbon the product cannot use — effectively a tax on every glucose. Replacing it with a permease plus glucokinase pays the tax in ATP instead, which the oxidative part of the metabolism can supply more cheaply. The costs are a slower uptake rate, which can limit productivity, and a loss of the regulatory coupling the native system provides.

3. Name the three properties of *C. glutamicum* that would be hardest to engineer into *E. coli*, and say why.

Native export of amino acids, because finding or engineering a specific exporter is the problem part 3 chapter 45 showed to be a screening exercise with a poor base rate; tolerance to above 100 g/L of product, because tolerance is polygenic and usually evolved rather than designed; and regulatory acceptance for food and feed, which is not a biological property at all and cannot be engineered on any timescale. The absence of acetate overflow is a fourth, and is the one that is most nearly addressable by engineering.

What to open a book for. Yim et al., Nat. Chem. Biol. 7:445 (2011), reread with parts 2 and 4 behind you; the EPA's 2003 Greener Reaction Conditions award summary for bio-PDO, which is short and unusually clear about why the process was built the way it was; and a *C. glutamicum* systems-metabolic-engineering review for the lysine strain lineage. Stephanopoulos chapters 2 and 3 remain the reference for the yield arithmetic.

Biofuels, C1 and gas fermentation, and food ingredients

Three regimes in one week, united by a single question: what is the feedstock, and what does getting it into the cell cost? For fuels the answer killed an industry; for gas it rebuilt one.

Higher alcohols from 2-ketoacids: one idea, a family of products

Atsumi, Hanai and Liao's non-fermentative route is the most elegant piece of pathway design in the fuels literature, and its logic is worth being able to draw from memory. Every amino acid is made from a 2-ketoacid, and every 2-ketoacid can be decarboxylated and then reduced — the Ehrlich degradation that yeast uses to make fusel alcohols — to give an alcohol one carbon shorter. So the amino-acid biosynthetic network, which cells already run at high flux, becomes a precursor network for alcohols: 2-ketoisovalerate from the valine pathway gives isobutanol, 2-keto-3-methylvalerate gives 2-methyl-1-butanol, phenylpyruvate gives 2-phenylethanol, and so on. Two enzymes — a broad-specificity ketoacid decarboxylase and an alcohol dehydrogenase — convert the whole family.

The reported isobutanol result was 22 g/L at 86% of theoretical. Check the theoretical: isobutanol is $C_4H_{10}O$, one mole per mole of glucose with two CO_2 lost, so 74.12/180.16 = **0.411 g/g**, as part 2 chapter 18 computed. Eighty-six per cent of that is a remarkable yield for a first demonstration and reflects the fact that the route is nearly redox-balanced on glucose, which is why it works anaerobically.

Two refinements the week asks you to understand. The route consumes NADPH in the ketol-acid reductoisomerase step while glycolysis makes NADH — part 2 chapter 19's cofactor mismatch, and the standard fix is the cofactor swap, which in this pathway has been done. And in yeast, moving the pathway into the mitochondrion raised titers: the precursor 2-ketoisovalerate is made there, so compartmentalising the downstream enzymes with it avoids exporting an intermediate and keeps the local concentration high — the same argument as part 4's compartmentalisation note, with a measured payoff.

Why advanced biofuels mostly did not happen

The honest account, which you should be able to give without sounding either bitter or naive. The technical programmes largely worked: isobutanol, farnesene, fatty alcohols and renewable diesel were all demonstrated at pilot scale by companies built around them. What killed or slowed them was arithmetic external to the biology. Fuels are the cheapest products by mass that anyone makes, so the margin per kilogram is cents; the incumbent is an oil price that fell by half in 2014 and has swung since; the capital for a fuel-scale plant is hundreds of millions, which must be raised against that price; and the yield ceiling of a reduced product on sugar is fixed by stoichiometry at a level that leaves little room.

Part 4 chapter 49's model makes the point precisely: at a product price near $1/kg, the feedstock term alone at a realistic yield consumes most of it, and there is no titer that fixes that. The companies that survived pivoted to higher-value products made on the same platforms — farnesene to cosmetics and flavours rather than diesel, for instance — which is not a failure of the science but a correct response to a cost structure. The lesson for your own target selection is the one in chapter 53: choose products whose incumbent cannot halve its price.

Gas fermentation: the acetogens and why ATP is so tight

Gas fermentation is the exception to the pessimism above, and it is instructive that the reason is feedstock rather than biology. *Clostridium autoethanogenum* and relatives grow on carbon monoxide, CO_2 and hydrogen through the Wood–Ljungdahl pathway, and LanzaTech's first commercial plant started in 2018 converting steel-mill off-gas to ethanol. The feedstock is a waste stream with a negative or near-zero price, which changes the whole cost structure.

The energetics are the thing to understand, because they explain the product range. Wood–Ljungdahl is the most ATP-efficient route to acetyl-CoA known, and that is faint praise: it is very nearly ATP-neutral, with acetogens living on fractions of an ATP per acetate formed, conserving energy through a membrane ion gradient generated by ferredoxin oxidation and flavin-based electron bifurcation rather than through substrate-level phosphorylation. A cell with almost no ATP margin cannot afford long biosynthetic routes, which is why the natural products are acetate and ethanol, why engineered products are mostly two- to four-carbon compounds close to acetyl-CoA, and why anything requiring significant ATP must have an external energy input.

Carbon efficiency follows from the chemistry of the feedstock, and the two cases differ dramatically:

```
6 CO + 3 H₂O  →  C₂H₅OH + 4 CO₂        only 33% of the carbon reaches the product
2 CO₂ + 6 H₂  →  C₂H₅OH + 3 H₂O        all of the carbon reaches the product
```

On CO alone, two thirds of the carbon is sacrificed to supply electrons, because CO is both the carbon source and the reductant. Add hydrogen as a separate electron donor and the carbon is fully conserved — which is why the field's trajectory is toward CO_2 plus H_2, and why chapter 57 is about hydrogen and electricity rather than about gas.

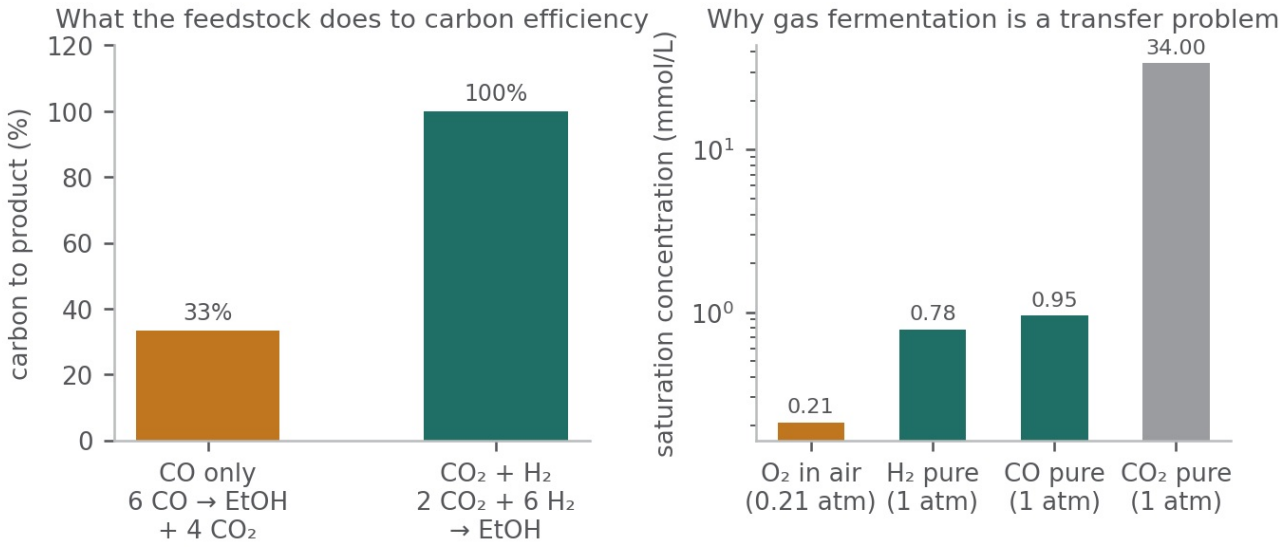


Figure 55.1 Left: carbon efficiency of the two gas-fermentation stoichiometries, from the balanced equations above. Right: saturation concentrations of the gases involved, on a log scale — the reason gas fermentation is a mass-transfer discipline. Values are standard Henry's-law figures at ambient conditions and are approximate; the comparison, not the third digit, is the point.

And the operational consequence, which is figure 55.1's right-hand panel: CO and H_2 are about as soluble as oxygen, which is to say barely, while CO_2 is two orders of magnitude more soluble. So a gas fermentation is limited by gas–liquid transfer of the sparingly soluble components, exactly like an aerobic fermentation but with two or three such components at once, and the reactor designs that dominate — loop reactors, bubble columns with long gas residence times — are chosen for transfer rather than for mixing.

C1 substrates, compared honestly with sugar

Methanol and formate are the other one-carbon feedstocks under development, and the comparison with sugar is worth having at hand.

Feedstock	Advantage	Cost in engineering
Glucose	Universal metabolism, high solubility, cheap at scale, no transfer problem	Competes with food and land; the price is volatile
Methanol	Liquid, fully miscible, made from natural gas or CO_2, more reduced than sugar so higher yields for reduced products	Toxic above a few per cent; native methylotrophs have modest tools; synthetic methylotrophy in *E. coli* or yeast is recent and slow-growing
Formate	Liquid, electrochemically made from CO_2, easy to transport, no transfer problem	Carries only one electron pair per carbon, so ATP and reductant are tight; toxic as formic acid; few hosts use it natively for growth
CO / CO_2 + H_2	Waste or atmospheric carbon; full carbon conservation with H_2	Gas transfer, very tight ATP, safety, and the hydrogen has to come from somewhere — chapter 57

The honest summary is that the one-carbon routes are not yet competitive with sugar for most products, and that their case rests on what happens to the price of renewable electricity and on whether carbon accounting is priced. Which hosts are being developed matters for part 6: methylotrophic yeasts and *Bacillus methanolicus* for methanol, engineered *E. coli* and *Cupriavidus* for formate and H_2/CO_2, acetogenic clostridia for CO.

Food ingredients, and why the host choice was a yeast

Soy leghemoglobin, the molecule that gives a plant-based burger its colour and flavour, is produced in *Komagataella phaffii* — the yeast formerly called *Pichia pastoris* — and was cleared through the FDA's GRAS notification route, notice GRN 737. The host choice is a teaching case in itself, and the week asks exactly why a yeast rather than a bacterium.

Four reasons, each from an earlier part of this module. It is a secreted protein, and *K. phaffii* secretes at grams per litre with a clean background, which is part 1 chapter 1's cell-volume argument: a secreted product has no intracellular ceiling. It is a haem protein, so the host must supply haem at a stoichiometry of one per molecule of a protein made at high titer, and the engineering of haem supply alongside the apoprotein is the core of the process. It is a eukaryotic protein, so folding and any modifications are more reliable in a eukaryote, and the reducing cytoplasm problem of part 1 chapter 2 does not arise for a secreted route. And regulatory acceptance: the host has a long history in food enzymes, which shortens the GRAS argument substantially, and as chapter 54 noted, regulatory history is the one host property that cannot be engineered.

Precision fermentation for food is now the field's fastest-growing commercial segment, for a reason that follows from chapter 53's lesson: the incumbent — an animal or a crop — cannot halve its price in response, and the product's value per kilogram sits comfortably above a commodity chemical's.

Constraint sheet — fuels, gas and food

Fuels: binding metric	Yield, and it is not enough. At cents-per-kilogram margins, the feedstock term alone consumes the price; no titer fixes it.
Fuels: ceiling	0.41 g/g isobutanol, 0.51 ethanol — fixed by the stoichiometry of a reduced product on sugar.
Gas: binding metric	Gas–liquid transfer and ATP. Carbon efficiency is 33% on CO alone and 100% with H_2 as the electron donor.
Gas: host requirement	An acetogen or a knallgas organism; products must sit close to acetyl-CoA because there is almost no ATP margin.
Food: binding metric	Regulatory clearance, purity and consistency; titer matters mainly through cost of goods.
Food: host requirement	Secretion at grams per litre, eukaryotic folding, cofactor supply (haem) at product stoichiometry, and an existing food-use history.
Where each fails	Fuels: the oil price. Gas: transfer rates, safety and the cost of hydrogen. Food: consumer acceptance and the length of a novel-food filing.

Three things people get wrong

- ‘Biofuels failed because the science didn't work.’ The science largely worked; the margin per kilogram and the oil price did not.
- ‘Gas fermentation is just anaerobic fermentation on a different substrate.’ It runs on a near-zero ATP margin and is limited by gas transfer, which constrains the product range.
- ‘CO_2 fixation means carbon-neutral.’ On CO alone, two thirds of the carbon leaves as CO_2; with H_2, the hydrogen's own footprint becomes the question.

Self-check — answer on paper first

1. Explain why one decarboxylase and one dehydrogenase give access to a whole family of alcohols.

Because every amino acid derives from a 2-ketoacid, and the Ehrlich route — decarboxylation then reduction — converts any 2-ketoacid into the corresponding alcohol with one carbon fewer. Both enzymes have broad specificity, so the product is set by which 2-ketoacid accumulates, which is set by which amino-acid branch is amplified. The engineering therefore moves from the two terminal enzymes to the precursor-supply branch, which is where the real work is.

2. Why does adding hydrogen to a CO-fed gas fermentation triple the carbon efficiency?

Because on CO alone the same molecule serves as carbon source and electron donor, so two of every three carbons must be oxidised to CO_2 to supply the electrons for reducing the third — the balanced equation is 6 CO → ethanol + 4 CO_2. Hydrogen supplies electrons without carbon, so the carbon can all be fixed: 2 CO_2 + 6 H_2 → ethanol + 3 H_2O. The cost moves from carbon to hydrogen, and therefore to electricity, which is chapter 57's subject.

3. A collaborator proposes making a 20-carbon product in an acetogen on CO_2 and H_2. What is your first objection?

ATP. Acetogens conserve barely a fraction of an ATP per acetate through a chemiosmotic mechanism, so there is almost no energy margin for a long biosynthetic route — a C20 product requires many ATP-consuming condensation and reduction steps and would not leave enough for maintenance, let alone growth. The options are an external energy input, a two-stage process in which a different organism takes the acetate the acetogen makes, or a different host with respiratory energy conservation such as a knallgas bacterium growing on H_2 and O_2, which is what chapter 57 uses.

What to open a book for. Atsumi, Hanai & Liao, Nature 451:86 (2008) for the 2-ketoacid route; a recent gas-fermentation or acetogen review for the energetics, since the quantitative account of electron bifurcation is newer than any textbook; and FDA GRAS notice GRN 737 for soy leghemoglobin, which is worth reading as an example of what a regulatory filing actually contains — it is not a biology document and that is the point.

PHA, from pathway to polymer: the case study worked all the way through

Three enzymes, one glucose, one monomer. This is the simplest pathway in the module and the richest case study, because everything that decides whether it works sits outside the pathway: the redox surplus, the nutrient feed, and the difference between content and yield.

The pathway, and its balance

Draw it once and you will not forget it. Glucose through glycolysis to two pyruvate, each oxidised by pyruvate dehydrogenase to acetyl-CoA. Then three enzymes: **PhaA**, a β-ketothiolase, condenses two acetyl-CoA to acetoacetyl-CoA; **PhaB**, an NADPH-dependent reductase, reduces it to (R)-3-hydroxybutyryl-CoA; **PhaC**, the synthase, polymerises it and releases coenzyme A. Mark every cofactor as you draw, and the net equation per monomer unit is

```
glucose + 1 NADPH  →  (3-hydroxybutyrate unit) + 2 CO₂ + 4 NADH + 2 ATP + 1 NADPH consumed

carbon: 4 of 6 reach the polymer (67%)        mass yield ceiling: 86.09/180.16 = 0.478 g/g
cofactors: +2 NADH (glycolysis) +2 NADH (PDH) −1 NADPH  →  a surplus of 3 reduced cofactors
```

That surplus is the thing to carry out of this week. The polymer is slightly *less* reduced per carbon than glucose, so making it leaves the cell with electrons it must dispose of, and the only disposal route is respiration: three reduced cofactors per glucose is 1.5 O_2 per monomer unit, which is **0.56 g of oxygen per gram of PHB**, or about 35 $mmol \cdot L^{-1} \cdot h^{-1}$ of oxygen demand at a productivity of 2 g/L/h, on top of whatever the residual biomass requires for maintenance. PHB production is therefore an aerobic process with a real oxygen duty, and oxygen transfer is one of the two things that limit it at scale.

Why can a strain not reach 0.478 g/g even with perfect enzymes? Two reasons, both structural. Carbon must also build the residual biomass — the catalytic cells doing the work — and that carbon never becomes polymer. And the cell must pay maintenance throughout the accumulation phase, which in a non-growing culture is met by oxidising substrate to CO_2 with no product at all. Both are substrate consumed for something other than polymer, so the achievable yield is the ceiling minus the biomass investment minus maintenance, and the longer the accumulation phase runs, the larger the second term grows.

The hosts, and what each one costs you

Host	What it gives	What is harder than in *E. coli*
Cupriavidus necator	The native short-chain-length pathway, very high accumulation (majority of dry weight), tolerance of nutrient limitation, and autotrophic growth on $H_2 + CO_2 + O_2$ — which is chapter 57	Slower growth, conjugation rather than simple transformation for DNA delivery, fewer characterised promoters and integration sites, and restriction barriers in wild isolates (part 4 chapter 52)
Pseudomonas putida	Medium-chain-length PHA from β-oxidation of fatty acids or de novo fatty-acid synthesis, giving elastomeric polymers; robust to solvents and aromatics	mcl-PHA is a mixture of monomers rather than a defined polymer; the carbon source is usually fatty acids, which are more expensive than sugar
Recombinant *E. coli*	Every genetic tool in the module, fast growth, no native depolymerase so the polymer is not re-consumed, and high content from a three-gene operon	No native accumulation physiology, so the nutrient-limitation trigger has to be imposed by the process; lower tolerance of the very high cell densities the process wants

Note the pattern, because it recurs in part 6: the host with the best native physiology has the worst genetic tools, and the host with the best tools has no native physiology. Which to choose depends on how much of the process you intend to engineer and how much you intend to inherit.

Reading a fed-batch run, which is the real skill

Here is a run. Six time points, and the only measurements are total dry cell weight, PHB content as a percentage of that weight, and cumulative glucose consumed — which is all most papers give you.

Time (h)	0	12	24	36	48	60
Dry cell weight (g/L)	1.0	15	45	70	95	110
PHB content (% of DCW)	0	5	20	45	62	72

Glucose consumed (g/L)	4	40	110	190	280	350
PHB (g/L)	0	0.8	9.0	31.5	58.9	**79.2**
Residual biomass (g/L)	1.0	14.2	36.0	38.5	36.1	**30.8**

Now the four numbers that matter, none of which the table gives directly.

- **Residual biomass** — dry weight minus polymer — is the catalyst, and it plateaus at 36–38 g/L around 24 to 36 h and then declines slightly. That plateau is the nutrient feed doing its job: nitrogen ran out, growth stopped, and carbon kept flowing into polymer. The small decline afterward is maintenance consuming cell material. If residual biomass had kept rising, the limitation had not been imposed and you would be making biomass instead of polymer; if it had collapsed, the cells were dying and the run should have been harvested earlier.
- **Volumetric productivity** is 79.2/60 = **1.32 g/L/h** over the whole run, and (79.2 − 9.0)/36 = **1.95 g/L/h** over the accumulation phase alone. Quote the first; the second is the one papers quote, and the difference is the growth phase, as part 2 chapter 18 warned.
- **Overall yield** is 79.2/350 = **0.226 g/g**, which is 47% of the ceiling — and that is the honest process number, because the growth-phase glucose was paid for too.
- **Accumulation-phase yield** is (79.2 − 9.0)/(350 − 110) = **0.293 g/g**, 61% of the ceiling. The remaining 39% is the gap worth decomposing: maintenance of 36 g/L of cells over 36 hours, the CO_2 from the two decarboxylations, and any residual growth.

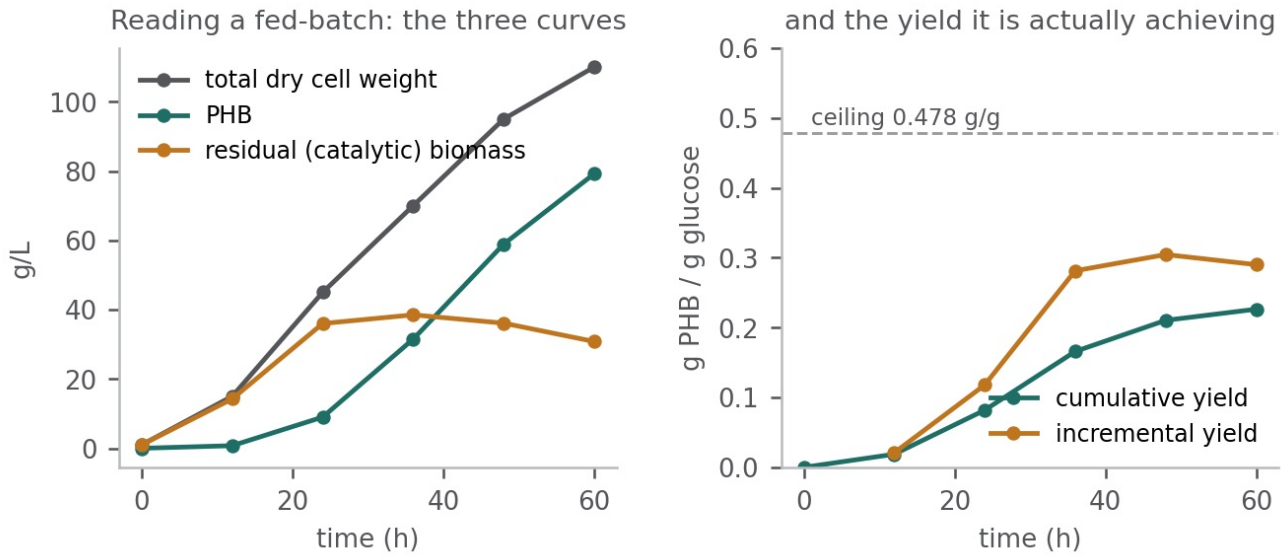


Figure 56.1 Left: the three curves to extract from any PHA fed-batch table — total dry weight, polymer, and residual catalytic biomass, whose plateau marks the nutrient limitation. Right: cumulative and incremental yield against the 0.478 g/g ceiling; the incremental yield peaks at 0.30 g/g around 48 h and then falls as maintenance consumes a larger share. Computed from the table above, which is constructed for this module rather than taken from a paper.

Look at the incremental yield in the right-hand panel: 0.28 between 24 and 36 h, 0.30 between 36 and 48 h, and 0.29 thereafter — it is already turning down by the last interval. That is the signal to harvest, and it is invisible in the cumulative number, which is still rising. **Incremental yield falling below the point where the marginal glucose pays for itself is the harvest criterion**, and computing it from the table is the single most useful thing you can do with someone else's fermentation data.

What to change in the next run, from these numbers alone: the growth phase consumed 110 g/L of glucose to build 36 g/L of residual biomass, which is nearly a third of the total substrate for catalyst rather than product, so a shorter growth phase to a lower residual biomass — or a richer inoculum — would raise the overall yield even if it lowered the peak productivity. And since the incremental yield turns down by 60 h, harvesting at 48–54 h loses a little content and gains yield and productivity. The measurement that tells you it worked is the incremental yield curve, not the final content.

Copolymers: why composition is the product specification

Poly(3-hydroxybutyrate) alone is a stiff, brittle polymer with a melting point uncomfortably close to its degradation temperature, which is the reason PHB has never displaced polypropylene on its own. Incorporating a second monomer fixes it: poly(3-hydroxybutyrate-*co*-3-hydroxyvalerate) with a few mole per cent of the C5 unit has a lower melting point, a wider processing window and much better toughness. So for this product class **the specification is a composition,**

not a titer — a strain making 80 g/L of the wrong copolymer fraction has made the wrong product.

The engineering handle is the supply of the C5 precursor, propionyl-CoA. Classically that meant feeding propionate or valerate, which is expensive and somewhat toxic. The more interesting route makes it from sugar: 2-ketobutyrate, from the threonine pathway or from citramalate, is the entry point, and Jo and colleagues showed that overexpressing acetolactate synthase in *C. necator* H16 modulates the 3-hydroxyvalerate fraction from unrelated carbon — which is precisely the kind of result that only makes sense if you can see the amino-acid network as a precursor network, exactly as chapter 55's 2-ketoacid route does.

Feedstocks, inhibitors, and what "robust" means

Lignocellulosic hydrolysate is the feedstock everyone wants and few strains tolerate, and the reason is a short list of inhibitors generated by the pretreatment itself. Four to measure, with the assay for each:

- **Furfural**, from pentose dehydration — HPLC with UV detection, or GC-MS.
- **5-hydroxymethylfurfural**, from hexose dehydration — the same HPLC run.
- **Acetic acid**, from deacetylation of hemicellulose, and the one that matters most at neutral-to-low pH for the weak-acid reasons of part 3 chapter 45 — HPLC with refractive index or UV, on the organic-acid column you already run.
- **Phenolics**, from lignin depolymerisation, which are the most variable and the least characterised — total phenolics by a colorimetric assay for a quick number, GC-MS or LC-MS for the individual compounds when it matters.

"Robust", applied to a strain, should mean something measurable: growth rate and yield maintained at a stated concentration of each inhibitor, or in a stated hydrolysate, against the parent. Sun and colleagues' report of efficient PHB production from lignocellulosic hydrolysate with a *C. necator* strain is the kind of claim to read with that definition in hand — what concentrations, which inhibitors, against what control, and whether the improvement was evolved or engineered. Those are the questions a member of that lab would expect you to ask, and asking them is the point of the week's reading.

Recovery and characterisation, and what to trust

PHA content and monomer composition are measured by gas chromatography after methanolysis — the polymer is digested into its methyl esters, which are quantified against standards, with benzoic acid or a similar compound as internal standard. Molecular weight comes from gel permeation chromatography, thermal properties from differential scanning calorimetry, and composition can be confirmed by NMR. Recovery from the cells is by solvent extraction, typically chlorinated, or by digesting the non-polymer biomass with hypochlorite or enzymes — and the choice matters for the molecular weight, because aggressive digestion cleaves chains.

Now connect this to part 4 chapter 50, because a PHA titer is a long chain of multiplications: a GC peak area, a calibration against a methyl ester standard, a methanolysis efficiency that is assumed rather than measured, a dry-weight determination, and a content percentage derived from both. Before trusting a reported titer I would want: the calibration range and whether it was weighted; the methanolysis recovery measured on a commercial polymer standard of known mass; the dry weight method; and the agreement between content × dry weight and an independent gravimetric extraction. A content figure above 90% should prompt the question of whether the residual biomass was measured or inferred, since at that level a small dry-weight error swings the content by several points.

Constraint sheet — PHA and intracellular polymers

Binding metrics	Three that trade against each other: polymer content (% of dry weight), volumetric productivity, and yield on substrate. A strain is only good if all three are.
Yield ceiling	0.478 g/g on glucose, 67% of carbon; unreachable because residual biomass and maintenance consume substrate without making polymer.
Cofactor balance	A surplus of three reduced cofactors per glucose — this is an oxygen duty of 0.56 g O_2 per g PHB, not a redox deficit.
Process requirement	Two phases: growth, then nutrient limitation with carbon still fed. Residual biomass must plateau; if it rises, the limitation was not imposed.
Product specification	Composition, not just quantity: the 3HV fraction sets the material properties, and the engineering handle is propionyl-CoA supply.
Host requirements	Native accumulation physiology and tolerance of nutrient limitation (*C. necator*), or full genetic tooling and an imposed process (*E. coli*). No host gives both.
Where it fails	Against polypropylene on price, unless the feedstock is waste or the polymer's biodegradability is valued; and on molecular weight after aggressive recovery.

Three things people get wrong

- ‘High PHB content means a good strain.’ Content is one of three metrics; a 90% content at 0.15 g/g and 0.3 g/L/h is a poor process.
- ‘PHB production needs reducing power.’ It produces a surplus of three reduced cofactors per glucose, so what it needs is oxygen.
- ‘The copolymer is a variant of the same product.’ The 3HV fraction is the specification; without it the polymer is too brittle to sell.

Self-check — answer on paper first

1. Write the glucose-to-PHB net equation from memory with its cofactors, and state the oxygen demand per gram of polymer.

Glucose → 2 pyruvate (2 ATP, 2 NADH) → 2 acetyl-CoA (2 NADH, 2 CO_2); PhaA condenses them, PhaB reduces with 1 NADPH, PhaC polymerises. Net per monomer: one glucose in, one C4 unit plus 2 CO_2 out, 4 NADH and 2 ATP generated, 1 NADPH consumed — a surplus of three reduced cofactors. Clearing that surplus by respiration takes 1.5 mol O_2 per mol of monomer, which is 48 g O_2 per 86 g of polymer, or 0.56 g of oxygen per gram of PHB.

2. In the table above, residual biomass declines from 38.5 to 30.8 g/L between 36 and 60 h. What does that mean and does it matter?

It means the catalytic cells are consuming their own protein and other components to meet maintenance, which is what happens when nitrogen has been withheld for a long accumulation phase. It matters in two ways: the specific productivity of the remaining cells sets the volumetric rate, so a shrinking catalyst means a falling rate even if content keeps rising, and the material consumed was built from substrate you paid for, so it depresses the overall yield. Combined with the incremental yield turning down, it is the argument for harvesting earlier.

3. A paper reports 92% PHB content and 0.40 g/g yield in *C. necator* on glucose. What is your reaction?

The content is plausible for this organism; the yield is 84% of the stoichiometric ceiling of 0.478, which is implausibly high once residual biomass and maintenance are accounted for, so I would check how the yield was computed. The likely explanations are that it is an accumulation-phase yield rather than an overall one, that the substrate consumed excludes the growth phase or a co-substrate, or that the dry-weight and content measurements are inconsistent in the direction that flatters both. The test is whether the carbon balance closes: polymer plus residual biomass plus CO_2 should account for the glucose, and at 0.40 g/g overall there is very little room for the CO_2 the pathway must release.

What to open a book for. No book. A *C. necator* PHA review for the hosts and physiology; Sun et al., Green Chemistry (2026) for the lignocellulosic strain and Jo et al., Int. J. Biol. Macromol. (2023) for the copolymer handle — both from the lab whose recruiting notice names them, so read them as the group would expect, with one question each that a lab member could answer in two sentences. Part 2 chapter 18 and part 4 chapter 50 are the two chapters of this module to have open alongside them.

CO_2, hydrogen and waste carbon: when electricity is the feedstock

A CO_2 route's feedstock is not carbon dioxide, which is nearly free. It is hydrogen, and therefore electricity, and therefore a number in kilowatt-hours per kilogram that decides everything.

The stoichiometric floor, and what it leaves out

Start with the balance the week asks you to write, for a 3-hydroxybutyrate unit made from CO_2 and hydrogen:

```
4 CO₂ + 9 H₂  →  C₄H₆O₂ + 6 H₂O
carbon 4 = 4 · hydrogen 18 = 6 + 12 · oxygen 8 = 2 + 6        the equation balances
```

Nine hydrogen per monomer unit, which is 105 mol of H_2 per kilogram of polymer. This is a floor and not an estimate, and it is important to understand why: it counts **reducing equivalents only**. It assumes the cell spends no energy fixing the carbon, which is false — the Calvin cycle costs ATP — so the real demand is higher by however much ATP the organism must make, and ATP in a knallgas organism is made by respiring hydrogen with oxygen.

Now tally the pathway, which is the week's second exercise and produces a satisfying check. Fixing six CO_2 to one hexose by the Calvin cycle costs 18 ATP and 12 NADPH. Glycolysis and pyruvate dehydrogenase then return 2 ATP and 4 NADH while releasing 2 CO_2, and PhaB consumes 1 NADPH. Net per monomer unit:

```
16 ATP   and   13 NADPH − 4 NADH = 9 net reducing equivalents
```

Nine — exactly the floor. The stoichiometric equation and the pathway tally agree on reductant, as they must, and the pathway reveals what the equation hid: **16 ATP that the equation does not count**. Supplying those by respiring hydrogen adds 16/(ATP per H_2) more hydrogen, and that assumption is the one that moves the answer most:

ATP per H_2 oxidised	Extra H_2 for ATP	Total H_2 per unit	mol H_2 per kg PHB	Electrolysis energy
1.0	16.0	25.0	290	31.9 kWh/kg
1.5	10.7	19.7	228	25.1 kWh/kg
2.0	8.0	17.0	197	21.7 kWh/kg
3.0	5.3	14.3	166	18.3 kWh/kg

Electrolysis taken at 0.11 kWh per mole of H_2, roughly 55 kWh/kg of hydrogen, which is a reasonable figure for alkaline or PEM electrolysis including balance of plant. The ATP yield per H_2 is the organism's P/O ratio question from part 1 chapter 4, and the table is the quantitative reason that chapter insisted you treat it as an assumption rather than a constant.

So the energy demand sits between roughly 18 and 32 kWh per kilogram of polymer depending on a parameter you cannot look up for your organism. Nearly a factor of two, from one assumption. This is the honest state of any CO_2-route calculation, and stating the range with its cause is a much stronger position than quoting a single number.

Electricity as the real feedstock, and the break-even price

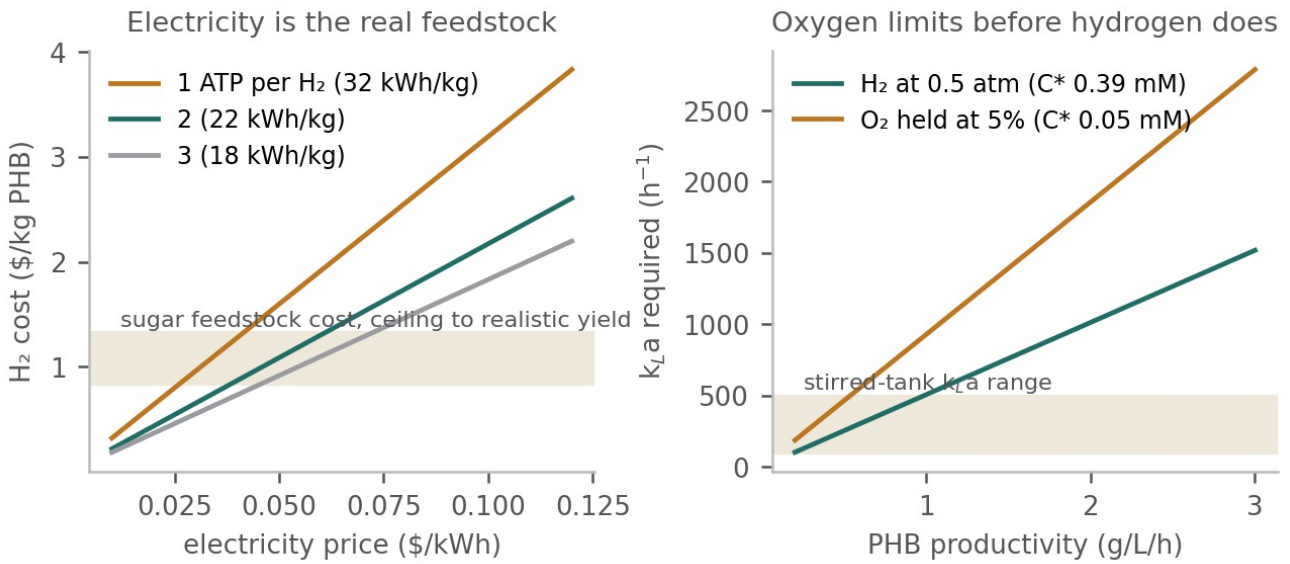


Figure 57.1 Left: hydrogen cost per kilogram of PHB against electricity price for three ATP-yield assumptions, with the sugar route's feedstock cost band — $0.84/kg at the stoichiometric ceiling to $1.33/kg at a realistic 0.30 g/g — shaded. Right: the gas-transfer coefficient required for a target productivity, showing that oxygen, held low for safety, limits before hydrogen does. Computed from the balances in this chapter; electrolysis energy, gas solubilities and the safety-limited oxygen fraction are the stated assumptions.

At 2 ATP per H_2 — 21.7 kWh/kg — electricity at $0.03/kWh puts the hydrogen cost at $0.65/kg of polymer, $0.05/kWh at $1.09, and $0.10/kWh at $2.17. Against the sugar route's feedstock cost of $0.84/kg at the ceiling and $1.33/kg at a realistic yield, the break-even electricity price is about **$0.04/kWh against a perfect sugar process and about $0.06/kWh against a real one**. Those are prices that exist today in a few places with curtailed renewable generation, and nowhere near the average industrial tariff.

What that number leaves out is as important as the number. Electrolyser capital, which at the scale required is a plant in its own right. CO_2 capture and supply, which is free only when co-located with a concentrated source. Gas handling, compression and recycle. Downstream processing, which is identical to the sugar route's and therefore does not favour either. And the safety engineering below. In part 4 chapter 49's framework, the hydrogen cost enters as the feedstock line and the electrolyser as capital, and the right way to present a CO_2 route is in that model rather than as a kilowatt-hour figure — because the kilowatt-hour figure alone makes the route look better than it is.

Moving the gas into the liquid, which is where it actually fails

Part 3 chapter 48's lesson applies again, with hydrogen added to oxygen. At a target of 2 g/L/h of PHB and 17 H_2 per monomer, hydrogen demand is 395 mmol·L^{-1}·h^{-1}, and the ATP-related hydrogen must be respired with oxygen at 93 mmol·L^{-1}·h^{-1}. Required transfer coefficients follow from $N = k_La \cdot C^*$:

```
H₂ at 0.5 atm partial pressure, C* ≈ 0.39 mM  →  k_L a ≈ 1,000 h⁻¹
O₂ held at 5% of the gas phase, C* ≈ 0.05 mM  →  k_L a ≈ 1,900 h⁻¹
```

A well-agitated stirred tank reaches a few hundred per hour. Both numbers are therefore out of reach at that productivity, and **oxygen is the binding one** — not because it is needed in larger quantity, it is not, but because its concentration in the gas phase must be kept low for safety, which cuts its driving force by a factor of four relative to air. That is the counter-intuitive result of the week and it is worth being able to state: in a hydrogen-oxidising fermentation, the gas that limits is the one you are deliberately starving the system of.

What raises k_La, and what each costs: more power input per volume, which costs electricity and shears the cells; smaller bubbles through sparger design or a loop reactor, which costs pressure drop; higher pressure, which raises C* proportionally and costs vessel capital and compression — and is the lever gas fermentation companies actually use; and gas recycle with a high recirculation rate, which costs compression. The design answer is usually a pressurised loop reactor rather than a stirred tank, which is why gas-fermentation plants do not look like fermenters.

Safety, which shapes the design rather than annexing it

Hydrogen and oxygen in the same vessel is a flammable mixture over a very wide composition range, and the standard mitigation is to hold the oxygen fraction below the limiting oxygen concentration — for hydrogen mixtures, around 5%, which is exactly the constraint that made oxygen the transfer bottleneck above. So safety is not a paragraph appended to the proposal; it is the reason the process operates where it does.

The statement to put at the top of any proposal involving these gases, and to know who reviews it: the process operates with hydrogen and oxygen in a single gas phase; the oxygen fraction is controlled below the limiting oxygen concentration with redundant analysers and automatic interlocks; the vessel and gas loop are rated and relieved for the credible overpressure case; and the design is reviewed by the institution's safety committee, with a HAZOP or equivalent, before any gas is connected. In most institutions that review involves the laboratory safety officer, the engineering or facilities group for the gas supply and ventilation, and a formal committee for the pressure system — and it happens before work starts, not alongside it.

Bioelectrochemical and hybrid routes

If electricity is the real feedstock, the obvious question is why make hydrogen at all. Microbial electrosynthesis supplies the electrons directly: cells grow at or near a cathode and take up reducing equivalents from it, either directly or through hydrogen generated in situ at the electrode surface. The architecture is a cathode chamber with the culture and the medium, an ion-exchange membrane, and an anode chamber where water is oxidised — a schematic worth drawing, because the engineering constraints are visible in it.

Two of those constraints are the research frontier. **Current density** limits volumetric productivity directly: electrons per second per square metre of cathode, against a vessel volume, which means electrode area per unit volume becomes the scaling parameter — the same surface-to-volume problem as the gas transfer above, in a different currency. And **biocompatibility** of the catalyst means something specific here: an electrocatalyst for hydrogen evolution must work at near-neutral pH in a salty, protein-rich medium, must not leach metals at concentrations that poison the cells, and must not foul as biofilm forms on it. Moon and colleagues' Cu/NiMo composite catalyst is an example of that specification being treated as a materials problem rather than a biological one, and Lee and colleagues' integrated system making microbial protein from CO_2 with recycled nitrogen is an example of what the architecture is for.

Waste carbon, and the table you would defend

The other route to cheap carbon is waste. Firmansyah and colleagues' approach is a hybrid: oxidise polyethylene waste chemically to short-chain dicarboxylic acids, then feed those to an organism that converts them to a biodegradable polymer. The division of labour is the one from chapter 53 in another form — chemistry breaks the inert polymer, biology does the selective synthesis — and the engineering questions are what the oxidation actually produces, in what distribution and with what contaminants, and whether the organism grows on that mixture rather than on the single acid the abstract names.

Feedstock	Pretreatment	Inhibitors	Achievable yield	Cost driver	Footprint
Glucose	None	None	0.478 g/g ceiling for PHB; 0.23–0.30 observed	Sugar price	Agriculture: fertiliser, land use
Lignocellulosic hydrolysate	Acid or steam explosion, enzymatic saccharification	Furfural, HMF, acetate, phenolics	Similar ceiling; lower in practice, mixed sugars consumed sequentially (part 1 chapter 9)	Enzymes and pretreatment capital	Lower than glucose if residues are genuinely waste
Plastic-waste-derived acids	Chemical oxidation	Oxidation by-products, metals from catalysts, unreacted polymer	Depends on the acid distribution; must be measured	Oxidation step and feed consistency	Credits for diverted waste; depends on allocation
CO_2 + H_2	Electrolysis and CO_2 supply	CO and sulfur in industrial CO_2 streams	18–32 kWh/kg; carbon fully conserved	Electricity price and electrolyser capital	Entirely determined by the grid

Of those cells, you can fill the yields and the CO_2 energy figure from your own calculations — they are in this chapter and chapter 56 — and must look up or measure the pretreatment costs, the inhibitor concentrations in a specific hydrolysate, and the acid distribution from a specific oxidation. Knowing which is which is the point of the exercise.

Constraint sheet — CO_2, H_2 and waste carbon

Binding metric	Electricity price, through hydrogen. 18–32 kWh/kg of PHB depending on the organism's ATP yield per H_2.
Break-even	About \$0.04–0.06/kWh against the sugar route's feedstock cost — before electrolyser and CO_2 capital.
Physical ceiling	Gas–liquid transfer: ~1,000 h^{-1} k_La for H_2 and ~1,900 h^{-1} for O_2 at 2 g/L/h, against a few hundred in a stirred tank. Pressurised loop reactors, not fermenters.
Safety	Oxygen held below the limiting oxygen concentration (~5% for H_2 mixtures), which is itself the transfer bottleneck. Formal review before work starts.
Host requirements	Knallgas metabolism with respiratory energy conservation (*C. necator*) rather than an acetogen, if the product needs ATP; tolerance of the gas regime.
Hybrid routes	Chemistry for the inert step, biology for the selective one. Judge them on feed consistency and contaminants, not on the headline conversion.
Where it fails	On the grid price and on gas transfer, not on the pathway — which is why the research frontier is electrodes and reactors rather than enzymes.

Three things people get wrong

- ‘CO_2 is the feedstock.’ Hydrogen is, and therefore electricity. CO_2 is nearly free and is not what the process is buying.
- ‘The stoichiometric floor is the energy demand.’ It counts reducing equivalents only; the ATP for fixation nearly doubles it, and the factor depends on an assumption.
- ‘Hydrogen transfer is the problem.’ Oxygen is, because safety forces its partial pressure down and with it the driving force.

Self-check — answer on paper first

1. Write the CO_2-to-3HB floor and explain why the pathway tally gives a larger hydrogen demand.

$4\ CO_2 + 9\ H_2 \rightarrow C_4H_6O_2 + 6\ H_2O$, which balances in C, H and O. It counts only reducing equivalents, and the pathway tally agrees on those — 13 NADPH consumed minus 4 NADH generated is 9 — but also shows 16 net ATP required, mostly for the Calvin cycle, which the balanced equation does not represent. Supplying that ATP means respiring additional hydrogen with oxygen, adding 5 to 16 more H_2 per unit depending on the ATP yield per hydrogen, so the real demand is 14 to 25 H_2 against the floor's 9.

2. At what electricity price does a CO_2 route match the feedstock cost of a sugar route running at 0.30 g/g, and what does the comparison omit?

The sugar route's feedstock cost is \$0.40/0.30 = \$1.33 per kilogram of product. At 21.7 kWh/kg — the 2 ATP per H_2 case — matching that requires electricity at \$1.33/21.7 ≈ \$0.06/kWh. The comparison omits electrolyser capital, CO_2 capture and compression, the gas handling and recycle system, the safety engineering, and the fact that the CO_2 route's volumetric productivity is likely lower because of gas transfer, which raises its capital charge per kilogram in exactly the way part 4 chapter 49 describes.

3. Why does holding the oxygen fraction at 5% make oxygen rather than hydrogen the transfer bottleneck?

Because the transfer rate is $k_La \cdot C^*$ and C^* is proportional to partial pressure: dropping oxygen from 21% to 5% of the gas phase cuts its saturation concentration from about 0.21 to 0.05 mM, a factor of four, while hydrogen can be supplied at a high partial pressure. So although the oxygen demand is roughly a quarter of the hydrogen demand in moles, the required k_La for oxygen is about twice as large. The safety constraint and the transfer constraint are the same constraint seen twice.

What to open a book for. For the engineering, Doran's mass-transfer chapters and part 3 chapter 48 of this module cover the transport. For the rest, four papers from the lab this week is built around: Firmansyah et al., Chem. Eng. J. (2025) on upgrading polyethylene waste; Lee et al., Water Res. (2025) on microbial protein from CO_2 in an integrated bioelectrochemical system; Moon et al., Adv. Sci. (2024) on a biocompatible cathode catalyst; and a *C. necator* autotrophic-growth review. Read them with the constraint sheet above beside you and one question each.

Closing notes to part 5

The five classes, side by side

Class	Binds on	Ceiling	Titer needed	Host property that decides it
Pharmaceuticals and natural products	Whether the pathway works at all; purity	0.29 g/g and nobody cares	~1 g/L is a process	Eukaryotic membranes and compartments for P450s
Commodity chemicals and amino acids	Yield, then titer	0.60–0.66 g/g, set by redox	50–150 g/L	Native export, tolerance, cheap medium, regulatory history
Fuels	Yield, and it is still not enough	0.41–0.51 g/g	Irrelevant — the margin is the problem	Anaerobic operation and redox balance
PHA and intracellular polymers	Content, rate and yield together	0.478 g/g, with an oxygen duty	Content >70% at >1 g/L/h	Accumulation physiology under nutrient limitation
CO_2 and waste carbon	Electricity price and gas transfer	18–32 kWh/kg	Set by k_La, not by biology	Knallgas metabolism with respiratory ATP

The pattern worth taking from the table: in four of the five classes, the property that decides the outcome is not in the pathway. It is export, tolerance, accumulation physiology, membrane biology or gas transfer — which is exactly why the host chapters come next and why they are worth writing as a comparison rather than as five species descriptions.

On the numbers in this part

Every yield ceiling here was computed from a written-out balance: the terpenoid cofactor tally per C15 unit, the propanediol redox balance with its oxidation fraction, the PHB net equation with its three-cofactor surplus, and the CO_2-to-3HB floor with the Calvin-cycle ATP tally that reproduces it. The fed-batch table in chapter 56 is constructed for this module rather than taken from a paper, so that every derived quantity could be checked; the method for extracting residual biomass, incremental yield and a harvest criterion from it is what transfers, not the numbers. Reported titers — 25 g/L artemisinic acid, micrograms per litre for opioids, 22 g/L isobutanol, 18 g/L and above 100 g/L for butanediol, above 100 g/L for lysine — are as reported in the sources the tracker names, and are marked as reported wherever they appear in a figure.

The electrolysis energy, gas solubilities, limiting oxygen concentration and ATP yield per hydrogen in chapter 57 are stated assumptions, and the chapter's main result is the size of the range they produce rather than any single value inside it.

What is still to come

Part 6 is the host block: weeks 58 to 61 — the yeasts *S. cerevisiae*, *Y. lipolytica* and *K. phaffii*; the Gram-positives *C. glutamicum* and *B. subtilis*; the phototrophs, cyanobacteria and microalgae; and then the panorama with cell-free systems and the 2025–2026 trends. It will be written against the five constraint sheets in this part, with a single comparison table running through it: genetic tools and editing efficiency, growth rate and medium cost, native precursor and cofactor supply, secretion and post-translational capability, compartments and the transport tax they levy, tolerance, native export, and regulatory history.

Two of those weeks also carry a piece of arithmetic that is this module's own transport calculation in a new currency. Week 58's compartmentalisation question — whether putting a pathway in the mitochondrion or peroxisome pays — is part 3 chapter 48's dead-core problem with membranes instead of gel: a local concentration advantage against a transport cost on every substrate and cofactor that crosses. And week 60's photobioreactor is the same calculation again with photons, where light penetrates a dense culture a few millimetres and the dark volume is a dead core by another name. Both will be computed rather than described.

Part 6 — Hosts: what each organism gives you, what it charges, and how to choose (weeks 58 to 61)

Part 5 ended with the observation that in four of five product classes the property deciding the outcome is not in the pathway: it is export, tolerance, accumulation physiology, membrane biology or gas transfer. All of those are host properties. So this part is written as a comparison against the five constraint sheets rather than as a set of species descriptions, and the question it keeps asking is the only one that matters at a host-selection meeting: which property decides this, and what do I give up by choosing on it?

Three pieces of arithmetic run through these chapters, and each is a calculation from earlier in the module appearing in a new currency. The compartment question in week 58 — whether putting a pathway in a peroxisome or mitochondrion pays — is part 3 chapter 48's dead-core problem with membranes instead of gel: a local concentration gain against a transport duty on every substrate and cofactor that crosses. The intracellular-product ceiling in week 59 is part 1 chapter 1's cell-volume argument, recomputed honestly at industrial cell density, where the answer turns out to be much less discouraging than the textbook version. And week 60's light penetration is the dead core a third time, with photons: light reaches a few millimetres into a dense culture and the rest of the vessel is dark, which is the photobioreactor's version of an oxygen gradient.

An instruction for how to use this part. The tracker asks you to build one host table and keep adding to it rather than duplicating it in each week. Chapter 61 contains that table, filled as far as this module can defend, with the cells marked where a number should be looked up rather than taken from me. Treat it as a template to fill with sources, not as a reference to cite.

Ch.	Week	Chapter
58	58	Yeast hosts: compartments, the lipid ceiling, and choosing between three yeasts
59	59	Gram-positive hosts: the envelope, the lysine arithmetic, and secretion as the deciding factor
60	60	Phototroph hosts: light as the limiting substrate, and the cost of taking the water out
61	61	The host table, cell-free systems, and reading the future tense

Yeast hosts: compartments, the lipid ceiling, and choosing between three yeasts

A eukaryote gives you folding, glycosylation, secretion, low-pH tolerance and organelles you can put a pathway into. It charges you a membrane between your enzymes and their substrates, and a toolkit that only exists in full for one of the three species.

Why a yeast at all

Four reasons, and it is worth separating them because different products need different ones. A eukaryote folds and glycosylates proteins a bacterium cannot — the reducing-cytoplasm problem of part 1 chapter 2 does not arise on a secretory route that passes through the endoplasmic reticulum. Yeasts tolerate low pH and high osmolarity, which matters for organic acids and for contamination control in a non-sterile process. Several have Generally Recognised As Safe status for food use, the host property part 5 chapter 54 noted cannot be engineered. And they have organelles, which is both the opportunity and the tax of this chapter.

The compartments to know: the nucleus; mitochondria, where pyruvate dehydrogenase sits and where the TCA cycle runs; the endoplasmic reticulum and Golgi, the secretory route where signal-peptide-bearing proteins fold, are glycosylated and are packaged into vesicles; peroxisomes, where β-oxidation of fatty acids happens in yeast exclusively — mammals run it in both mitochondria and peroxisomes, so a pathway ported from a mammalian source may expect the wrong compartment; the vacuole; and a cell wall of mannoproteins and glucan that is thick enough to matter for large secreted proteins.

Where acetyl-CoA lives, and why that is the first thing you engineer

This is the single most consequential fact about yeast as a chemical host. Pyruvate dehydrogenase is mitochondrial, so the acetyl-CoA it makes is mitochondrial. Fatty acid synthesis and the mevalonate pathway are cytosolic. **Acetyl-CoA does not cross the mitochondrial membrane**, so a cytosolic pathway cannot use the pool the cell makes most of, and the cytosolic pool has to be supplied by one of two routes:

- **The citrate shuttle with ATP-citrate lyase.** Citrate leaves the mitochondrion and is cleaved in the cytosol to acetyl-CoA and oxaloacetate, at a cost of one ATP per acetyl-CoA. This is the oleaginous yeasts' route and it is why they are oleaginous: *Y. lipolytica* has it, *S. cerevisiae* essentially does not.
- **The pyruvate dehydrogenase bypass.** Pyruvate is decarboxylated to acetaldehyde, oxidised to acetate, and activated to acetyl-CoA by acetyl-CoA synthetase — which cleaves ATP to AMP, so **two ATP equivalents** each, as part 1 chapter 4 insisted. This is *S. cerevisiae*'s route and it is expensive.

Hence the standard first intervention in any yeast strain for lipids, isoprenoids or polyketides: engineer cytosolic acetyl-CoA supply, by introducing ATP-citrate lyase, by replacing the bypass with a cheaper acetylating acetaldehyde dehydrogenase, or by rerouting through a phosphoketolase that avoids the decarboxylation entirely. Doing the terpene synthase first, before this, is the classic beginner's order and it is why so many first attempts give milligrams.

The lipid ceiling, computed

Palmitate, $C_{16}H_{32}O_2$, is eight acetyl-CoA condensed with seven rounds of malonyl-CoA chemistry. The bill:

```
8 acetyl-CoA = 4 glucose          carbon: 16 of 24 → 67%, so a carbon-only ceiling of 0.356 g/g
7 ATP  for acetyl-CoA carboxylase (malonyl-CoA)
8 ATP  for ATP-citrate lyase, one per acetyl unit shuttled    → 15 ATP per palmitate
14 NADPH for the seven reduction cycles (two each)
```

Now pay for the NADPH, which is the step most back-of-envelope calculations skip. Routed through a fully cycling oxidative pentose phosphate pathway, one glucose yields 12 NADPH while releasing all six carbons as CO_2, so 14 NADPH costs 1.17 glucose. Total feedstock is 4 + 1.17 = **5.17 glucose per palmitate**, and the ceiling falls from 0.356 to **0.275 g/g**. For triacylglycerol, with its glycerol backbone, the same assumptions give about 0.28 g/g. Published ceilings for TAG on glucose sit a little higher, around 0.30 to 0.33 g/g, because some NADPH comes from malic enzyme or from a partially cycling pentose phosphate pathway that is less wasteful of carbon than my assumption — the difference between my number and theirs is entirely in how the reductant is sourced, which is exactly the point.

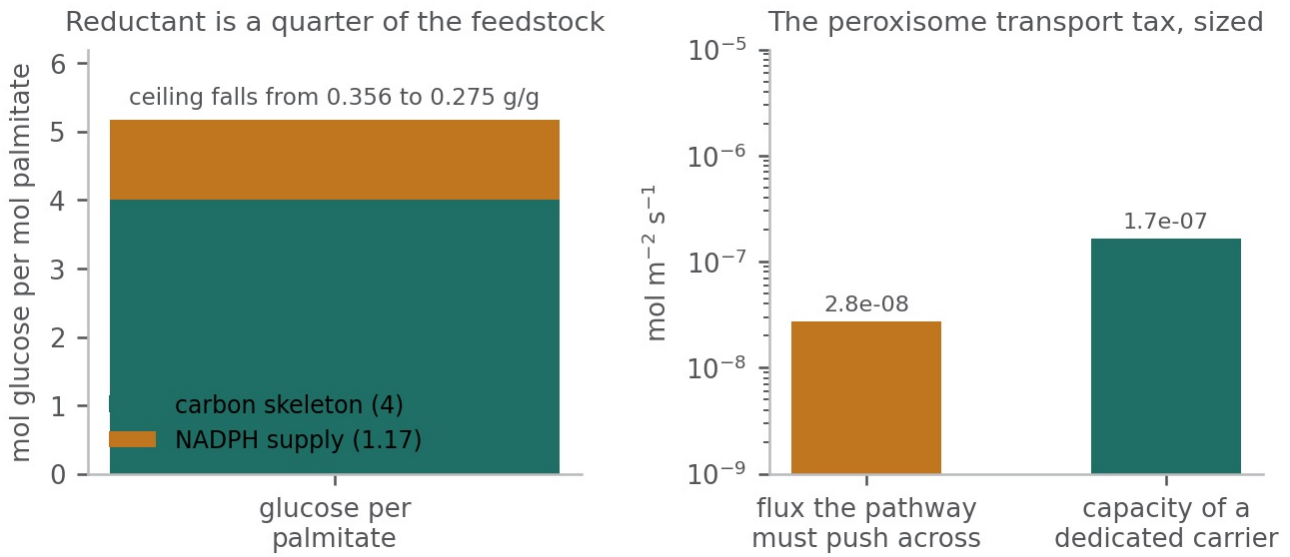


Figure 58.1 Left: the glucose bill for one palmitate, split into carbon skeleton and the glucose burned to supply NADPH — reductant is nearly a quarter of the feedstock, and it drops the ceiling from 0.356 to 0.275 g/g. Right: the peroxisome transport duty. A cell making 0.5 g/L/h at 20 gDCW/L must push about 2.8×10^{-8} mol·m^{-2}·s^{-1} across the organelle membrane, against roughly 1.7×10^{-7} for a carrier at 1,000 copies per µm² and k_{cat} 100 s^{-1} — feasible, with about sixfold margin, and only with a dedicated transporter. Computed for this module from the stated assumptions.

Three consequences follow, and they answer the week's questions directly. **Overexpressing fatty acid synthase alone rarely raises titer** for two reasons at once: part 2 chapter 15's control analysis says a single enzyme holds a fraction of the control, and the balance above says the binding constraints are cytosolic acetyl-CoA and NADPH, neither of which the synthase supplies. **The carbon-to-nitrogen ratio is the switch**: with nitrogen exhausted and carbon still fed, protein synthesis stops, AMP deaminase activity reduces the AMP pool, mitochondrial isocitrate dehydrogenase slows, citrate accumulates and is exported to the cytosol — which is the lipid-accumulation trigger, and it is set in a fed-batch by the ratio of carbon to nitrogen in the feed, not by a gene. And **the ATP bill is real**: 15 ATP per palmitate is paid by respiring carbon that cannot become product, which is part of why observed yields sit well below even 0.275.

Compartment engineering, with the transport tax sized

Putting a pathway into an organelle does two good things: it concentrates the enzymes and their intermediates in a small volume, and it keeps a toxic intermediate away from the cytosol. A peroxisome of 0.2 µm radius has a volume of about 0.034 fL against a yeast cell's 50 fL, so a cell with ten of them confines the pathway to under 1% of its volume — a hundredfold concentration gain for the same number of enzyme molecules, which for an enzyme operating below its K_M is a hundredfold rate gain, by part 1 chapter 3's arithmetic.

The tax is that every substrate, every cofactor and the product must cross a membrane. Size it as the figure does: a cell making 0.5 g/L/h of a 150 Da product at 20 gDCW/L carries about 1.4×10^{-20} mol·s^{-1} per cell, which across one peroxisome's 5×10^{-13} m² of membrane is a flux density of 2.8×10^{-8} mol·m^{-2}·s^{-1}. A dedicated carrier at 1,000 copies per µm² with a turnover of 100 s^{-1} delivers about 1.7×10^{-7} — six times the requirement, so it works, but only with a transporter expressed at high density and only for the species that has one. There is no carrier for your novel intermediate, and NADPH does not cross at all: the organelle must regenerate its own, which means installing a dehydrogenase and a source of its substrate inside the compartment.

So the honest rule, which is the same conclusion as the dead-core calculation: compartmentalisation pays when the intermediate is toxic or the precursor is already made inside that compartment — the isobutanol-in-mitochondria case of part 5 chapter 55 — and loses when it merely sounds elegant, because you have added three transport problems to solve a concentration problem that a stronger promoter might have solved.

Three yeasts, compared properly

	S. cerevisiae	*Y. lipolytica*	*K. phaffii*
Substrate range	Glucose, sucrose, maltose; not methanol; poor on alkanes	Glucose, glycerol, acetate, alkanes and fatty acids — hydrophobic substrates are a speciality	Glycerol and methanol; glucose less favoured industrially because the strong promoters are methanol-linked
Crabtree status	Positive: makes ethanol from sugar even with oxygen, so carbon leaks to ethanol at high flux	Negative: respires sugar fully, which is why high-density aerobic growth is straightforward	Negative

Lipid storage	Little — a few per cent of dry weight; engineering it for lipid means rewiring acetyl-CoA supply and storage	Oleaginous: tens of per cent of dry weight natively, with the citrate shuttle already present	Low
Secretion	Moderate; hyperglycosylates	Strongly secretory, with native lipases and proteases exported at high titer	Excellent — grams per litre with a clean extracellular background, which is why it dominates protein production
Tool maturity	The best toolkit in biology: markers, promoter libraries, efficient homologous recombination, CRISPR, synthetic chromosomes	Good and improving; non-homologous end joining dominates, so targeting is harder	Good for expression, thinner for pathway work
GRAS / food use	Yes, long history	Yes for several products and processes	Yes, with a history in food enzymes — the soy leghemoglobin precedent
Documented products	Ethanol, artemisinic acid, farnesene, resveratrol, insulin precursors	Lipids and oleochemicals, citric acid, erythritol, omega-3 oils, cosmetic ingredients	Secreted enzymes and proteins; soy leghemoglobin

One line each. *S. cerevisiae* is uniquely good at being engineered and uniquely bad at keeping carbon out of ethanol. *Y. lipolytica* is uniquely good at lipid and at hydrophobic substrates and uniquely awkward to target genetically and to keep morphologically consistent, since it is dimorphic and switches between yeast and hyphal forms with conditions — which changes viscosity, oxygen transfer and your optical density calibration at the same time. *K. phaffii* is uniquely good at secreting protein at high cell density and is not where you would build a twelve-gene small-molecule pathway.

Tools, and the three things that do not carry across

DNA delivery and maintenance differ in ways that matter for a project plan. In *S. cerevisiae*, episomal plasmids with centromeres or 2-micron origins are routine and homologous recombination is efficient enough that a 40 bp flank targets a locus — which is why assembly in yeast is itself a cloning technique. In *Y. lipolytica*, non-homologous end joining outcompetes homologous recombination, so targeting a locus needs long flanks, a *ku70*-deleted background, or a CRISPR cut to force repair at the right place. In *K. phaffii*, integration is the norm and expression is usually driven from the methanol-inducible promoter system, with methanol-free variants now available.

Three things that work in *S. cerevisiae* and do not simply carry across: the promoter library, because promoter strength does not rank the same way in another yeast and must be re-measured; the auxotrophic marker set, because the corresponding mutants may not exist in the new host and must be made; and recombination-based assembly in the host itself, which depends on the efficient homologous recombination the other two lack. The first three experiments to run in a yeast new to you follow from exactly that: measure growth rate and the cell-density-to-dry-weight calibration in your medium; test DNA delivery and quantify transformants per microgram with a known construct; and measure a small promoter and terminator set against a fluorescent reporter so that you have numbers rather than someone else's ranking.

Host sheet — the yeasts

Choose a yeast when	The product needs eukaryotic folding, glycosylation or a secretory route; when a P450 needs an endoplasmic reticulum; when low pH or high osmolarity is part of the process; when GRAS status is required; or when lipid or hydrophobic-substrate metabolism is the point.
The first thing to engineer	Cytosolic acetyl-CoA supply — via ATP-citrate lyase, a cheaper bypass, or phosphoketolase — before any pathway enzyme.
Ceiling to quote	Lipid on glucose: 0.356 g/g on carbon alone, 0.275 g/g once NADPH is paid for from the pentose phosphate pathway, with 15 ATP per palmitate on top.
The switch	Carbon-to-nitrogen ratio, set in the feed, not a gene.
Compartments	Pay when the intermediate is toxic or the precursor is already inside; the tax is a transporter for every species crossing, and NADPH must be regenerated locally.
Honest bottlenecks	Glycosylation that differs from the human pattern; proteases in the medium; plasmid instability; Crabtree overflow in *S. cerevisiae*; dimorphism in *Y. lipolytica*; and a toolkit that does not transfer between the three.

Three things people get wrong

- ‘Yeast has acetyl-CoA, so a cytosolic pathway can use it.’ Most of it is mitochondrial and does not cross; the cytosolic pool costs one or two ATP per unit to make.
- ‘Compartmentalisation concentrates the pathway, so it helps.’ It also adds a transport duty on every substrate and cofactor, and NADPH does not cross at all.

- ‘The *S. cerevisiae* toolkit carries over to other yeasts.’ Promoters, markers and recombination-based methods all have to be rebuilt.

Self-check — answer on paper first

1. Why does a cytosolic isoprenoid pathway in *S. cerevisiae* cost more ATP than the same pathway in *Y. lipolytica*?

Because the cytosolic acetyl-CoA comes from different routes. *S. cerevisiae* uses the pyruvate dehydrogenase bypass, whose acetyl-CoA synthetase step cleaves ATP to AMP and therefore costs two ATP equivalents per acetyl unit; *Y. lipolytica* has an active citrate shuttle with ATP-citrate lyase, which costs one. For a C15 terpene that is nine acetyl units, so eighteen ATP equivalents against nine — a difference that has to be met by respiring carbon, and therefore appears in the yield.

2. Compute the glucose required per mole of palmitate if half the NADPH comes from a source that costs no carbon, and say what that does to the ceiling.

Seven of the fourteen NADPH then cost 7/12 = 0.58 glucose, so the total is 4.58 glucose and the ceiling rises from 0.275 to 256.42/(4.58 × 180.16) = 0.311 g/g. That is roughly the published range for TAG, and it shows that the entire difference between my number and the literature's is the assumption about reductant sourcing — which is why the first question to ask of any lipid-yield claim is where the NADPH was assumed to come from.

3. A colleague proposes moving a six-enzyme pathway into the peroxisome to raise its local concentration. What do you ask?

What crosses the membrane, and how. Specifically: which substrate enters and is there a native transporter for it, or must one be found; whether the pathway consumes NADPH or ATP, since neither crosses and both would have to be regenerated inside; how the product leaves; and whether the peroxisomal targeting signal actually imports folded multi-subunit enzymes, which it does for some and not others. Then the comparison question: is the intermediate toxic, or is the precursor already made in there? If neither, the concentration gain is probably obtainable more cheaply with expression tuning in the cytosol.

What to open a book for. No single book. The literature searches the tracker names are the sources — *Y. lipolytica* as a lipid host, cytosolic acetyl-CoA supply, *K. phaffii* secretion, and subcellular compartmentalisation in yeast. Part 1 chapters 1 and 4 of this module cover the compartment and ATP-equivalent arithmetic, and part 3 chapter 48 is the transport analogy worth rereading before the compartment discussion.

Gram-positive hosts: the envelope, the lysine arithmetic, and secretion as the deciding factor

One membrane instead of two. That single structural fact decides what these hosts are for: a secreted protein crosses one barrier and arrives in the medium, and there is no endotoxin to remove afterward.

The envelope, and what it changes

Draw both. *E. coli*: inner membrane, then a periplasm containing the peptidoglycan layer, then an outer membrane whose outer leaflet is lipopolysaccharide. A Gram-positive: one cytoplasmic membrane, then a thick peptidoglycan wall, then the medium. A signal peptide in *E. coli* delivers a protein to the periplasm, and getting it to the medium needs a second, separate export step; in a Gram-positive the same signal peptide delivers it to the outside of the single membrane, from which it diffuses through the wall into the medium.

Three consequences. **Secretion is simpler and higher-yielding**, which is why *B. subtilis* and its relatives make most of the world's industrial enzymes. **There is no lipopolysaccharide**, so no endotoxin to remove — a significant advantage for any injectable product, and the reason endotoxin removal is a line item in *E. coli*-derived biologics and not in Gram-positive ones. And **the periplasm is gone**, which is occasionally a loss: it is a small oxidising compartment where disulfide bonds form and where a product can be concentrated and protected, and some *E. coli* processes depend on it.

What the wall still makes difficult: it is a thick crosslinked mesh, and very large or slowly-folding proteins can be retained or degraded in transit through it. Lysis for an intracellular product is also harder than in *E. coli*, since the wall resists mechanical and enzymatic disruption — a cost that only appears if you choose a Gram-positive and then keep the product inside.

Secretion as the deciding factor, with the cell-volume arithmetic redone

Part 1 chapter 1 computed that an intracellular product is capped by cell volume, using 10^9 cells per millilitre and arriving at 0.1% of the broth volume and therefore a titer of a fraction of a gram per litre. That calculation is correct and it is also, for an industrial process, pessimistic — and the week asks you to redo it at a realistic density. Take about 2 mL of cell volume per gram of dry weight:

Cell density	Cell volume fraction	Broth titer at 200 g/L inside the cells
1 gDCW/L (the textbook case)	0.2%	0.4 g/L
10 gDCW/L	2%	4 g/L
50 gDCW/L	10%	20 g/L
100 gDCW/L (high-cell-density fed-batch)	20%	40 g/L

So an intracellular product at very high cell density is not hopeless — 20 to 40 g/L is a real process, which is exactly why intracellular products like PHA work at all, as part 5 chapter 56 showed. The honest statement is the combined one: **the cell-volume ceiling scales with cell density, so an intracellular product forces you into high-cell-density operation**, with everything that implies for oxygen transfer, cooling and feed control, and then forces lysis, debris removal and a harder purification. A secreted product needs none of that.

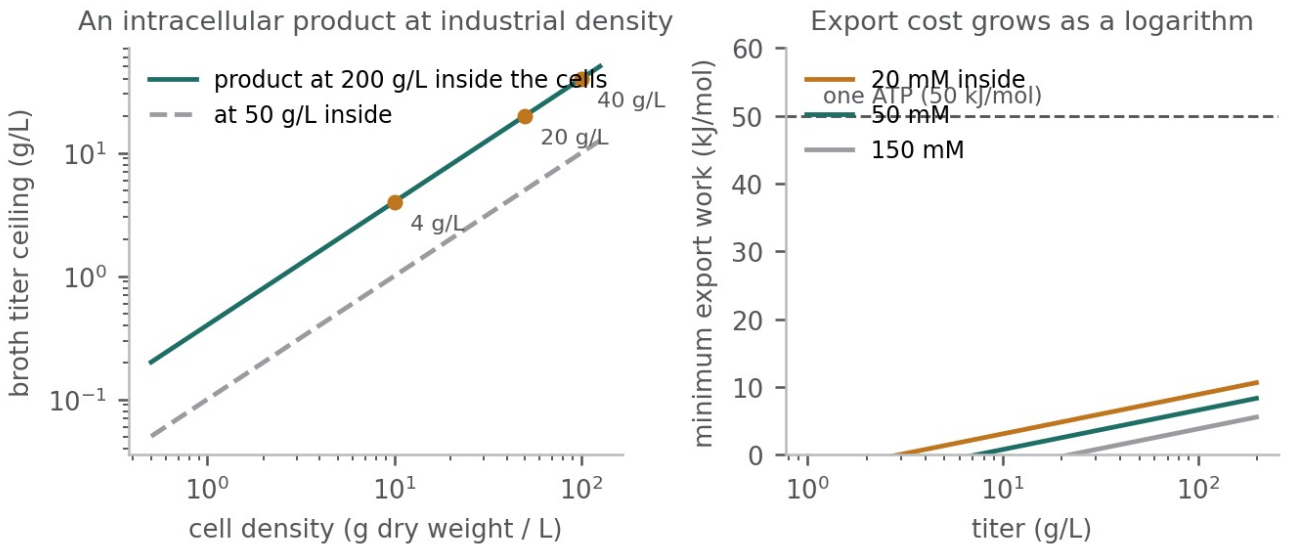


Figure 59.1 Left: the intracellular titer ceiling against cell density, at two intracellular concentrations — the textbook 0.1% case sits at the far left. Right: the thermodynamic minimum work of exporting a product against its own gradient, which grows only as the logarithm of titer: from 50 to 150 g/L of lysine it rises from 4.8 to 7.6 kJ/mol, a tenth to a seventh of an ATP. Computed for this module; 2 mL of cell volume per gram dry weight and a 50 mM intracellular concentration are the assumptions.

The lysine case, with the numbers that decide it

C. glutamicum makes glutamate and lysine at the million-tonne scale, and the lysine arithmetic is the best single illustration in this module of parts 1, 2 and 5 working together.

The ceiling. Reported theoretical maxima are 0.81 mol per mol of glucose for the native network and 0.86 with a transhydrogenase — which in mass terms is **0.657 and 0.698 g/g**. Note what moves it: not carbon, but a cofactor device. Lysine needs four NADPH per molecule, the oxidative pentose phosphate pathway supplies them at the cost of one carbon per two NADPH, and a transhydrogenase shifts part of that demand onto the NADH that glycolysis already makes — which is part 2 chapter 19's three-way choice, appearing as a 6% difference in the ceiling of a real industrial product.

The tension, stated properly. Carbon yield and reductant supply pull against each other: every glucose sent through the oxidative branch for NADPH loses a carbon as CO_2, and every glucose kept in glycolysis for carbon fails to supply NADPH. The literature addresses it three ways, and you should be able to name them: engineer a transhydrogenase, so NADH substitutes for NADPH at an energy cost; replace a glycolytic dehydrogenase with an NADP-dependent one so that glycolysis itself makes NADPH, which is the engineered NADPH-generating glycolysis the week's sources describe; or change the pathway enzymes' cofactor preference toward NADH. All three are cofactor-balance interventions rather than pathway interventions, which is the pattern.

Export, and whether it is a real term. The minimum work of moving lysine out against its own gradient is $RT \cdot \ln(c_{out}/c_{in})$: with 50 mM inside, 4.8 kJ/mol at 50 g/L, 6.6 at 100 g/L and 7.6 at 150 g/L — a tenth to a seventh of one ATP equivalent. So the thermodynamic term is small and, because it is logarithmic, **barely grows with titer**: tripling the titer adds 60% to a term that was already small. What is not negligible is the practical cost: real carriers are not reversible, so budget something closer to one ATP or one to two protons per molecule, which at 150 g/L is a few per cent of the cell's energy budget and belongs in the balance rather than in a footnote. The useful conclusion for a design discussion is that export energy is a real but second-order cost, and that the reason *C. glutamicum* wins is not the energy but that the carriers exist at all.

B. subtilis, and what its proteases do to your product

B. subtilis dominates industrial enzyme production because it secretes native amylases and proteases at commercial titers, has excellent classical and modern genetics, and is naturally competent — DNA uptake needs no electroporation, which makes library work unusually easy. It also secretes, natively, a set of extracellular proteases whose job is to degrade extracellular protein, and your heterologous product is extracellular protein.

That is why protease-deficient backgrounds exist, with several of the extracellular protease genes deleted. What they buy is a product that survives in the medium; what they cost is slower growth, some nutritional fastidiousness since the proteases also feed the cell, and sometimes a strain that sporulates or lyses differently. The failure mode to recognise is specific and the week asks for it directly: **a secreted protein that appears in the medium and then disappears is being degraded**, and the first checks are a protease-deficient background, a time course with protease inhibitors added

to samples, and a zymogram or activity assay on the spent medium to confirm proteolysis rather than aggregation or adsorption.

The other failure modes, with what to monitor: sporulation, which diverts part of the culture irreversibly late in a fed-batch — monitor by phase-contrast microscopy and by the sudden plateau in biomass; phage, which in a plant is a serious event rather than a nuisance; foaming from the secreted protein itself, which is worse the better your secretion works; and plasmid loss, which is why integration is standard for production strains.

Three experiments that tell you whether a new product can be secreted at all: express it with two or three different signal peptides, since signal-peptide performance is product-specific and is the single largest variable; run a time course measuring the product in the medium and in the cell fraction, to distinguish a secretion failure from a degradation one; and test it in a protease-deficient background, which separates the two remaining explanations at a stroke.

Host sheet — the Gram-positives

Choose one when	The product is a secreted protein, or an amino acid or other charged metabolite needing a native exporter; when endotoxin must be avoided; when GRAS status and a food or feed market matter.
Decisive property	One membrane: secretion crosses a single barrier and the product is harvested from the medium, with no lysis, no debris and no endotoxin removal.
Ceiling to quote	Lysine 0.81 mol/mol native, 0.86 with a transhydrogenase — 0.657 and 0.698 g/g. The difference is a cofactor device, not carbon.
Export cost	Thermodynamic minimum 5–8 kJ/mol across 50–150 g/L; practically about one ATP per molecule. Real, second-order, and logarithmic in titer.
Characteristic bottlenecks	Extracellular proteases degrading the product; sporulation; the wall limiting large proteins; phage; foaming from your own secreted protein.
Wrong choice when	The pathway needs eukaryotic folding or glycosylation, or the process depends on the oxidising periplasm that *E. coli* provides.

Three things people get wrong

- ‘An intracellular product is capped at a fraction of a gram per litre.’ At 1 gDCW/L, yes. At 100 gDCW/L the cells are a fifth of the broth and the cap is 40 g/L.
- ‘Export energy is negligible.’ The thermodynamic term is, and the carrier's practical cost of about one ATP per molecule is a few per cent of the budget at high titer.
- ‘*B. subtilis* secretes well, so my protein will accumulate.’ It also secretes proteases whose function is to degrade extracellular protein.

Self-check — answer on paper first

1. Explain why secretion changes the downstream process, in the order the unit operations occur.

For a secreted product: separate cells from broth by centrifugation or filtration, discard the cells, and purify from a relatively clean liquid. For an intracellular one: concentrate the cells, lyse them mechanically or enzymatically, remove the debris — which is fine, compressible and the single hardest solid–liquid separation in the sequence — and then purify from a solution containing the entire soluble contents of the cell, including nucleic acids that raise the viscosity and host proteins that must be resolved from the product. The secreted route removes two unit operations and starts the purification from a far simpler feed, which is usually worth more than any titer difference.

2. Why does a transhydrogenase raise the lysine ceiling by only about 6%, and why is that worth having?

Because it does not create reductant, it converts NADH to NADPH at an energy cost, so it substitutes a cheaper source for part of a demand that would otherwise be met by the oxidative pentose phosphate pathway at the cost of one carbon per two NADPH. Only the carbon saved shows in the yield, and only part of the demand can be shifted before energy becomes limiting — hence 0.81 to 0.86 mol/mol. It is worth having because in a million-tonne commodity made at a few hundred dollars a tonne, six per cent of the yield is an enormous sum, and because the intervention is a single gene.

3. A secreted enzyme reaches 1.2 g/L at 20 h and 0.3 g/L at 36 h. Give your diagnosis and the three experiments.

Proteolysis in the medium, most likely, since the loss is of product already successfully secreted. The experiments: repeat in a protease-deficient background, which is the definitive test; add protease inhibitors to samples at harvest and compare, which distinguishes degradation in the vessel from degradation during handling; and assay the spent medium for protease activity directly, ideally with your own product as substrate. If none of those explains it, the remaining candidates are aggregation or adsorption to cells and vessel surfaces, which a mass balance on the cell pellet and a

wash of the vessel would catch.

What to open a book for. Brock's cell-envelope chapter, which is genuinely the right source for the structural comparison and is worth reading with both envelopes drawn side by side. Then the reported theoretical maxima for lysine in *C. glutamicum* and the engineered NADPH-generating glycolysis paper in Appl. Environ. Microbiol. for the cofactor arithmetic, plus the literature on amino-acid exporters and on protease-deficient *B. subtilis* strains. Part 2 chapter 19 is the chapter of this module to have open beside them.

Phototroph hosts: light as the limiting substrate, and the cost of taking the water out

Light is a substrate you cannot stir. It arrives at a surface, is absorbed within millimetres, and leaves most of your vessel dark — which is why the unit is grams per square metre per day and why the water is the problem.

Light as the substrate, and the depth it reaches

Attenuation through a culture follows Beer–Lambert, with the biomass itself as the absorber:

$$I(z) = I_0 \cdot \exp(-k \cdot X \cdot z) \qquad k \approx 0.15\ \text{m}^2 \text{ per gram of biomass, } X \text{ in g/L, } z \text{ in metres}$$

Put numbers in and the problem is immediate. At a specific attenuation coefficient of 0.15 m²/g — a middle value for microalgae, and the assumption that matters most here — light falls to 10% of its surface value at the following depths:

Biomass density	Depth at 10% of surface light	Depth at 1%	Lit fraction of a 30 cm pond
0.25 g/L	61 mm	123 mm	20%
0.5 g/L	31 mm	61 mm	10%
1 g/L	15 mm	31 mm	5%
5 g/L	3 mm	6 mm	1%

At a density that would be considered dilute in any heterotrophic fermentation — one gram per litre, against a hundred in a fed-batch — light penetrates fifteen millimetres and 95% of a pond is dark. This is the structural fact of the field, and the comparison to make is with part 3 chapter 48's dead core: in both cases a substrate is consumed faster than it penetrates, leaving most of the catalyst idle. The difference is that oxygen can be raised by pressure and glucose can be fed, while photons arrive at a fixed areal rate set by the sun, and mixing cannot deliver more of them — it can only move cells between the lit layer and the dark one.

Which is why the unit is areal. You are not buying litres; you are buying illuminated area, and therefore land or glazing. **Grams per square metre per day is the only honest productivity unit for a phototroph**, and a paper reporting grams per litre per day for a thin photobioreactor is reporting a number that does not scale, because the thinness is what produced it.

What a pond can actually produce

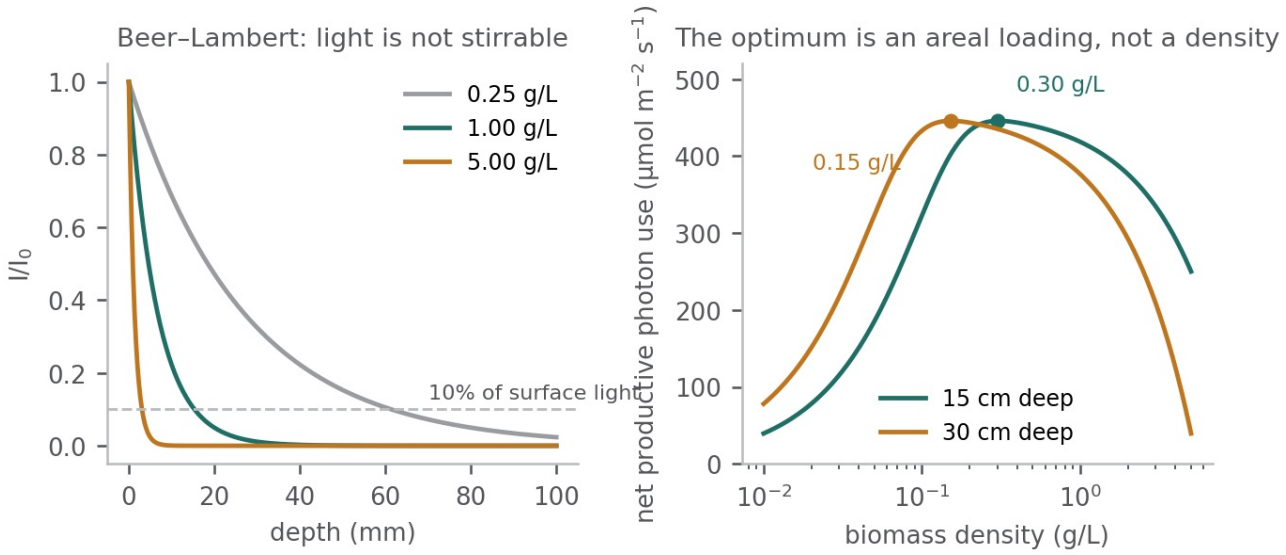


Figure 60.1 Left: Beer–Lambert attenuation at three densities, showing where the 10% depth falls. Right: net productive photon use against biomass density for two pond depths, with saturation wasting photons at high local irradiance and maintenance respiration consuming the dark biomass; the optima lie at 0.30 g/L in 15 cm and 0.15 g/L in 30 cm — the same areal loading of about 45 g/m². Computed for this module with k = 0.15 m²/g, surface irradiance 1,800 µmol·m⁻²·s⁻¹, saturation parameter 200 µmol·m⁻²·s⁻¹ and a maintenance coefficient of 0.28 µmol photon-equivalents per gram per second; those four assumptions set the result.

The right-hand panel contains the week's most useful result and it is not obvious. Once the culture is optically thick enough to absorb all the incident light — which happens at a few tenths of a gram per litre — adding more cells cannot raise areal productivity, because the photons were already all absorbed; it only redistributes them into a thinner surface layer, where high local irradiance wastes them to saturation, while the growing dark biomass consumes more in maintenance. The optimum therefore sits where those two losses balance, and it falls at a **fixed areal biomass loading of around 45 g/m² rather than a fixed concentration**: 0.30 g/L in a 15 cm pond, 0.15 g/L in a 30 cm one. That is why ponds are shallow and dilute, and it is the sentence to have ready when someone proposes to improve productivity by growing denser.

Now convert photons to biomass. A high-sun site delivers roughly 45 mol of photosynthetically active photons per square metre per day. At the theoretical minimum of about 10 photons per CO_2 fixed and about 24.6 g of biomass per C-mol, the photon-limited ceiling is around **110 g·m⁻²·day⁻¹**, and with the saturation and maintenance losses above, the model's productive photon use corresponds to roughly 50 g·m⁻²·day⁻¹. Reported outdoor productivities are 10 to 25 g·m⁻²·day⁻¹ — ten to twenty per cent of the photon ceiling. The gap is made of photoinhibition, which the saturating model does not contain at all: strong light damages photosystem II and the cell pays a continuous repair cost, so the response at midday is worse than saturation alone predicts. Then suboptimal temperature at the start and end of the day, night respiration, incomplete mixing between the lit and dark zones, and the fraction of the year the system runs at all.

In agricultural units, 20 g·m⁻²·day⁻¹ is 73 tonnes of dry biomass per hectare per year, against roughly 25 to 40 for a good land crop's dry matter. So phototrophs are genuinely two to three times more areally productive than crops — a real advantage, and nothing like the hundredfold sometimes implied — and the assumption in the model I would most want to measure is the specific attenuation coefficient k, because it sets the lit depth, the optimum loading and therefore everything downstream.

The two kinds of host, and what their genetics cost you

Cyanobacteria are prokaryotes: no organelles, photosynthesis on internal thylakoid membranes, a single circular chromosome, and in the model species natural competence and efficient homologous recombination, which makes targeted knockouts straightforward in principle. Eukaryotic microalgae have a nucleus, a chloroplast with its own genome and mitochondria, so a transgene can go into the nuclear genome or the chloroplast genome, and the two behave completely differently: chloroplast transformation gives high expression and homologous integration but no secretion and no eukaryotic processing, while nuclear transgenes integrate at random positions and are subject to silencing.

Two specific obstacles the week asks about. **Polyploidy**: *Synechocystis* carries multiple copies of its chromosome, so a knockout is not a knockout until every copy has segregated — which takes repeated restreaking under selection and must be verified by PCR showing complete loss of the wild-type allele. A partially segregated mutant behaves like a knockdown and has produced a quantity of irreproducible literature. **Silencing**: a nuclear transgene in *Chlamydomonas* is frequently silenced over generations, and the defences — codon optimisation, intron inclusion, screening many transformants, and using strains selected for expression — are all ways of managing a problem that has no clean solution.

Then the cycle time, which is what it actually costs a project. These organisms double in hours to a day rather than twenty minutes, transformation and segregation take weeks rather than days, and a growth experiment under light takes a week or two. A design–build–test cycle that is three days in *E. coli* is six to ten weeks in a phototroph, which by part 2 chapter 11's arithmetic is a factor of fifteen fewer hypotheses tested per year of funding. That is the single largest hidden cost of choosing a phototroph, and it should be stated in a project plan rather than discovered.

Three tool gaps to close before engineering a pathway in a host new to you: a characterised set of promoters and terminators with measured strengths in your conditions, since borrowed rankings do not transfer; a reliable selection and counter-selection system with markers that work and can be removed; and a validated transformation and segregation protocol with transformants per microgram quantified — which is the same first-three-experiments list as the yeasts, for the same reason.

Harvesting, which is where bulk products die

Cells are small, dilute, and close to neutrally buoyant, so getting them out of the water is a major cost term. Do the arithmetic once and you will never again read an algal-fuel projection the same way.

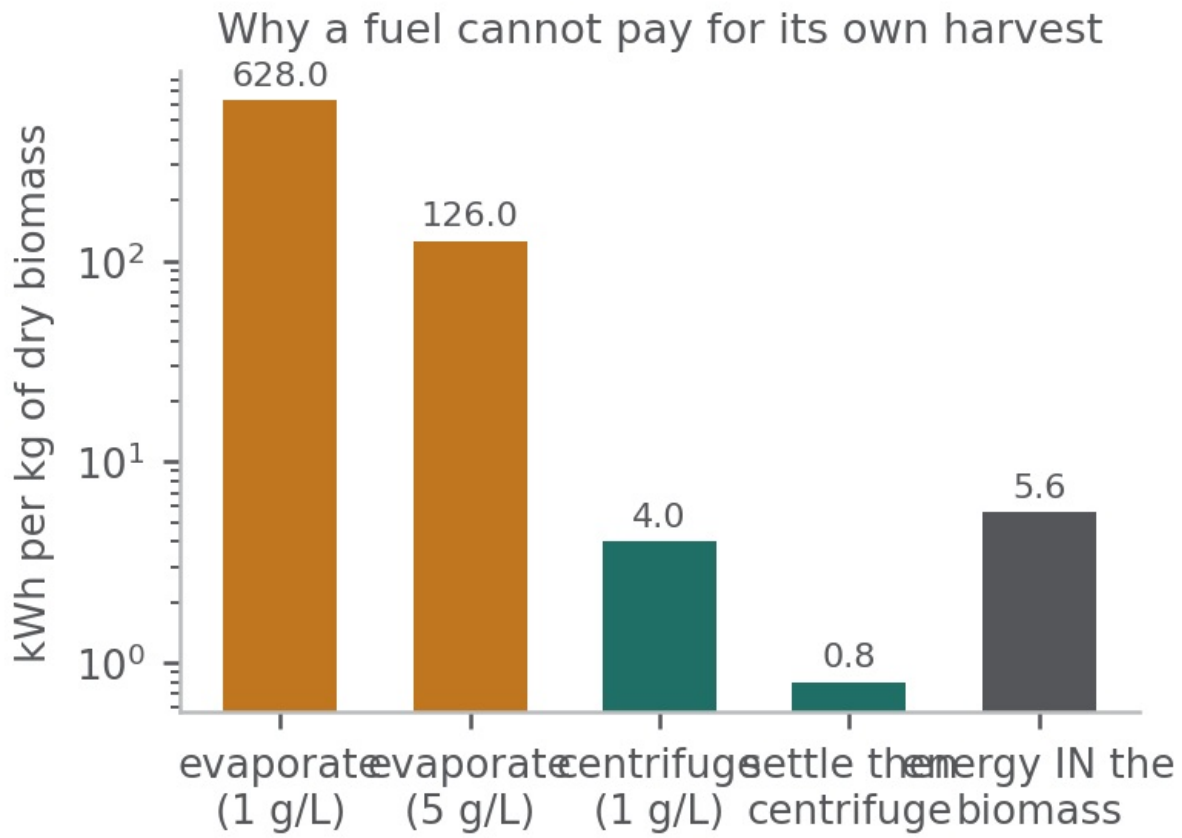


Figure 60.2 Energy per kilogram of dry biomass for different harvest routes, on a log scale, against the energy contained in the biomass itself. Evaporation is absurd by two orders of magnitude; centrifugation alone is of the same order as the biomass energy content; a staged route — flocculation or settling to concentrate, then centrifugation of a much smaller volume — is the only one that works. Computed from 2.26 MJ/kg for water evaporation and 1–8 kWh/m³ for centrifugation, with biomass taken at 20 MJ/kg.

At 1 g/L you must remove **1,000 litres of water per kilogram of dry biomass**. Evaporating it would take 2,260 MJ, or 628 kWh, against the roughly 20 MJ — 5.6 kWh — of chemical energy in the biomass: a factor of 113 against you. Centrifuging the whole volume at 1 to 8 kWh per cubic metre costs 1 to 8 kWh per kilogram, which is still of the same order as the energy in the product. Only a staged route works: flocculate or settle to concentrate the culture perhaps thirtyfold with very little energy, then centrifuge the small remaining volume, which brings the total under a kilowatt-hour per kilogram.

Connect that to part 4 chapter 49 and the conclusion is immediate. For a fuel — a product whose entire value is its energy content — a harvest step costing a comparable amount of energy cannot work, and no improvement in the strain changes it. For a product worth thousands per kilogram, the same harvest cost is irrelevant. **The dewatering arithmetic, not the biology, is what sorts phototroph products into viable and not.**

What a phototroph is actually for

Justified	Why	Not justified	The number that kills it
Astaxanthin, β-carotene and other pigments	Thousands of dollars per kilogram; the natural isomer commands a premium over the synthetic one; harvest cost is negligible against value	Transport fuels	Harvest energy of the same order as the fuel's energy content, before any conversion
Food and feed protein, and omega-3 oils	Value per kilogram in the tens of dollars; areal productivity two to three times a crop's; no agricultural land needed	Bulk commodity chemicals	A few dollars per kilogram against a dewatering and capital burden that a sugar fermentation does not carry
Carbon capture where capture itself is the point	The product can be low-value if the service is paid for	Bulk biomass for energy	The same as fuels, plus the land and water the system still needs

The week ends by asking you to write five honest sentences about your own microalgal work — glutathione and lipids — and the arithmetic above is how to do it without either dismissing it or overselling it. Something of this shape: the work was aimed at a product class where phototrophs can compete on value rather than on cost; the binding constraints were areal productivity and harvest, neither of which a strain improvement addresses; what a strain improvement can address

is content per unit biomass, which is exactly where glutathione and lipid engineering act; the honest reason to return to it would be a high-value product plus a co-product or service that pays for the harvest; and the honest reason not to is the design–build–test cycle time, which makes it a poor place to do iterative strain engineering. That is an answer with numbers in it, and it is also a much better interview answer than enthusiasm.

Host sheet — the phototrophs

Choose one when	The product is high-value and preferably accumulated in the biomass; when the CO_2 fixation itself is the point and is paid for; when a natural-source claim carries a premium.
Limiting substrate	Light, which cannot be stirred or fed. 15 mm of lit depth at 1 g/L; 95% of a 30 cm pond is dark.
Right unit	$g \cdot m^{-2} \cdot day^{-1}$. The optimum is an areal loading of about 45 g/m^2, so optimum density scales inversely with depth.
Realistic productivity	10–25 $g \cdot m^{-2} \cdot day^{-1}$ outdoors, 10–20% of the photon-limited ceiling, which is 36–90 $t \cdot ha^{-1} \cdot yr^{-1}$ against a crop's 25–40.
Harvest	1,000 L of water per kg of biomass at 1 g/L. Staged flocculation then centrifugation; evaporation is 113× the biomass energy content.
Tool cost	Polyploidy requiring full segregation in cyanobacteria; silencing and position effects for nuclear transgenes in algae; a DBTL cycle of six to ten weeks against three days in *E. coli*.
Wrong choice when	The product is a fuel or a bulk chemical, or the project needs many iterations of strain design.

Three things people get wrong

- ‘Grow the culture denser to raise productivity.’ Above full light absorption, more cells only waste photons to saturation and add maintenance; the optimum is an areal loading.
- ‘Phototrophs are vastly more productive than crops.’ Two to three times, areally — a real advantage and not an order of magnitude.
- ‘Harvesting is an engineering detail.’ At 1 g/L it is 1,000 litres of water per kilogram, and for a fuel it is the whole argument.

Self-check — answer on paper first

1. Compute the depth at which light falls to 1% of its surface value at 2 g/L, and say what it implies for mixing.

With $k = 0.15$ m^2/g, $kX = 300$ per metre, so the 1% depth is $\ln(100)/300 = 15$ mm. In a 30 cm pond that is 5% of the depth, so a cell spends about one twentieth of its time in usable light and the rest effectively dark. The implication is that mixing determines the *frequency* with which cells visit the lit layer, which matters because photosynthesis has a characteristic timescale: rapid light–dark cycling can use flashing light more efficiently than continuous dim light, which is the basis of the claimed flashing-light effect. Mixing cannot, however, increase the total photons available, so it redistributes rather than adds.

2. Why is the optimum biomass density halved when a pond is made twice as deep?

Because what the optimum fixes is the areal biomass loading — the grams of cells per square metre of illuminated surface — not the concentration. Once the culture absorbs all the incident light, areal productivity depends on how the fixed photon supply is distributed among a fixed areal quantity of cells, and the losses that create the optimum, photon waste at high local irradiance and maintenance of the dark biomass, both scale with that areal quantity. So $X \cdot L$ is the invariant, about 45 g/m^2 in the model here, and doubling L halves the optimal X.

3. A proposal claims 2 g/L/day in a 2 cm flat-panel reactor and projects it to a 100-hectare facility. What is wrong?

The volumetric figure was produced by the thinness of the panel and does not scale. Convert it: 2 g/L/day in a 2 cm path is 2 g per litre per day × 20 litres per square metre = 40 $g \cdot m^{-2} \cdot day^{-1}$, which is at the very top of what has been reported outdoors and is plausible only under optimal light and temperature — so the projection is assuming near-best-case areal performance across a hundred hectares and a full year. The questions to ask are what the areal figure is, over what period and with what light, and what the glazing and land capital cost per square metre is, since a flat-panel system at 100 hectares is an enormous capital number that a pond does not carry.

What to open a book for. Brock's phototrophy and microbial diversity chapters for the biology. Everything quantitative is in the literature the tracker points at: reviews of cyanobacteria as cell factories, of harvesting and dewatering energy, and of

areal productivity and light attenuation in photobioreactors, plus the genetic-tool literature for *Synechocystis* and *Chlamydomonas*. Part 3 chapter 48 of this module is the transport analogy and part 4 chapter 49 is where the harvest number belongs.

Chapter 61 · tracker week 61

The host table, cell-free systems, and reading the future tense

The deliverable of this week is one page: a table you could defend line by line in an interview. The skill it teaches is separating what has been demonstrated from what has been projected — which, in this field's reviews, is most of the text.

The host table

Here it is, filled as far as this module can defend. Cells marked with a dagger are where you should find a source rather than take my word: they are judgements rather than measurements, and the point of the exercise is to replace each with a citation of your own.

Host	Tool maturity	GRAS	Substrate range	Product localisation	Uniquely good at
E. coli	Highest	No	Sugars, glycerol, acetate; engineered C1	Cytoplasm or periplasm; poor secretion to medium	Being engineered fast — the shortest DBTL cycle in biology
S. cerevisiae	Highest among eukaryotes	Yes	Glucose, sucrose, maltose	Cytoplasm, organelles, or secreted	Eukaryotic pathways with P450s, plus the best eukaryotic toolkit
C. glutamicum	Good	Yes	Sugars; engineered on pentoses†	Secreted via native amino-acid exporters	Amino acids at million-tonne scale: native export plus tolerance
B. subtilis	Good	Yes	Sugars, starch	Secreted to medium across one membrane	Secreted enzymes at commercial titer; natural competence
Y. lipolytica	Moderate	Yes for several products†	Glucose, glycerol, acetate, alkanes, fatty acids	Stored as lipid bodies, or secreted	Lipids, oleochemicals and hydrophobic substrates
K. phaffii	Good for expression	Yes, food-enzyme history	Glycerol, methanol	Secreted, very clean background	High-density secretion of proteins — grams per litre
C. necator	Moderate	†	Sugars, fructose, plant oils, formate, H_2 + CO_2 + O_2	Stored as intracellular polymer	PHA, and autotrophic growth on gas
Cyanobacteria	Moderate; polyploidy complicates knockouts	Varies†	CO_2 and light	Cytoplasm or secreted†	Photoautotrophy with relatively tractable prokaryotic genetics
CHO cells	Industrialised but slow to engineer	n/a — clinical	Glucose and glutamine in defined medium	Secreted	Human-compatible glycosylation for therapeutic proteins

Two columns deserve comment because they decide more host arguments than the rest. **Product localisation** is the one most people leave out and the one that most often settles the question: secreted, retained or stored determines the downstream train, the titer ceiling by chapter 59's volume arithmetic, and whether toxicity is relieved. And **GRAS or regulatory history** is the only column that cannot be engineered on any timescale, which makes it decisive whenever the product goes into food, feed or a human.

The format for a host argument, which the phase 4 checkpoint asks for: name the property that decides it, name the alternative you rejected, and say what you gave up. "*K. phaffii*, because the product is a secreted haem protein and this host secretes at grams per litre with a clean background and a food-enzyme regulatory history; I rejected *E. coli*, which would have given a faster cycle and better tools but requires either periplasmic targeting or refolding and carries endotoxin that a food product cannot have; what I gave up is cycle time and the ability to use the promoter and assembly libraries I already know." An answer that says a host is standard or widely used is a default, not a choice, and interviewers are listening for exactly that distinction.

Constraining a phototroph model with photons

If a phototroph is on your list, the modelling step the week asks for is small and worth doing, because it makes chapter 60's arithmetic enter a flux model properly. A cyanobacterial genome-scale model has a photon exchange reaction — photons are a substrate like glucose — and its bound is normally set arbitrarily. Set it from your own attenuation calculation instead: compute the photons actually absorbed per gram of biomass per hour at your density and depth, which is the incident areal flux times the absorbed fraction divided by the areal biomass loading, and use that as the uptake bound. The model then predicts growth and product flux under a light limitation you can defend rather than one you assumed.

Two costs to note while you are there, because both are real and both are invisible in a stoichiometric model that has not

been told about them. RuBisCO is slow — a turnover number of a few per second, among the slowest central enzymes known — so a photoautotroph must invest an enormous fraction of its proteome in it, which by part 4 chapter 51's enzyme-constraint arithmetic is itself a growth limitation. And RuBisCO also fixes oxygen, so the carbon-concentrating mechanism exists to raise local CO_2 and suppress that, at a cost of roughly one ATP per CO_2 concentrated. Both belong in the model as constraints: a proteome cost and an ATP cost, respectively.

Cell-free systems: what removing viability buys

A cell-free system is a reaction containing the enzymes — as a lysate, as purified components, or as a transcription-translation system that makes them in situ — and not the cell. What that buys:

- **No competition for carbon.** There is no growth, so every substrate molecule is available to the pathway and yields can approach the stoichiometric ceiling rather than the ceiling minus biomass and maintenance, which parts 5's PHA chapter showed is a third of the gap.
- **No membrane.** Substrates, intermediates and cofactors can be added and measured directly, so the transport problems of part 2 chapter 14 and the compartment tax of chapter 58 disappear, and the concentrations are what you set rather than what the cell permits.
- **Tolerance of the intolerable.** A toxic intermediate or product cannot kill a system that is not alive, which is the single strongest reason to choose cell-free.
- **A short cycle.** Hours from DNA to measured function rather than days, which is why cell-free is used as a prototyping layer for parts and pathways before they are moved into cells.

What it costs: cofactors are no longer regenerated by a living metabolism, so ATP and NAD(P)H regeneration must be engineered into the reaction and is usually the binding constraint; the catalyst does not self-replicate, so enzyme preparation is a real cost per batch rather than a one-off; the system is unstable over long reactions as enzymes denature and phosphatases degrade cofactors; and scale is limited by the cost of lysate and of the energy source.

One product class where cell-free is genuinely the right answer: on-demand protein synthesis — vaccines, therapeutics and diagnostic reagents made at the point of need, including freeze-dried reactions rehydrated in the field — where the value per gram is high, the quantity is small, and the alternative is a cold chain. The second-best case is a short cascade with a toxic intermediate, where the tolerance argument dominates.

Reading the future tense

The last skill of this phase is reading a review sceptically, and it reduces to grammar. Mark every claim as **demonstrated** — it reports a result with a number, a host and a condition — or **projected**, which in practice means it contains *may*, *could*, *promises to*, *has the potential to*, *is expected to*, or *opens the way for*. Do this with a pen on a 2025 or 2026 review of self-driving biofoundries or AI-driven design and the proportion will surprise you; it is normal for the majority of the forward-looking sentences in a review of a fast field to be projections, and the failure mode is not the review's, it is quoting one of them in an interview as though it were a result.

Three current trends worth being able to discuss, with the demonstrated and projected parts separated. **Self-driving biofoundries**: demonstrated is high-throughput automated build and test with machine-learning-guided selection of the next round, in specific well-instrumented cases; projected is autonomy across the whole loop including hypothesis generation. **Enzyme-constrained and omics-integrated models**: demonstrated is that a proteome constraint reproduces overflow metabolism and improves growth prediction, as part 4 chapter 51 showed; projected is reliable prediction of engineered-strain phenotypes, which remains poor. **C1 and CO_2 valorisation**: demonstrated is gas fermentation at commercial scale for ethanol and growth on methanol and formate in engineered strains; projected is cost parity with sugar, which chapter 57 of part 5 showed to depend on an electricity price that mostly does not exist yet.

And the honest version of the question "which trend would you want to work inside": the one where the bottleneck matches what you are good at. For a chemical engineer who has done the arithmetic in this module, that is the modelling-and-constraints trend and the C1 and gas trend, because both are limited by transport, thermodynamics and cost accounting rather than by molecular biology, and both have more people who can clone than people who can close a balance.

The phase 4 checkpoint

1. Reconstruct artemisinin, 1,4-BDO and LanzaTech from memory in the seven-field format.

Artemisinin. Host *S. cerevisiae*; target artemisinic acid as a precursor to an antimalarial; pathway mevalonate plus amorphadiene synthase plus a plant P450 with its reductase; tools yeast genetics, promoter engineering, P450 expression; TRY 25 g/L with yield far below the 0.29 g/g ceiling and nobody minding; bottleneck P450 activity and acetyl-

CoA plus ATP supply; commercial status produced, then undercut by agricultural supply.
1,4-BDO. Host *E. coli*; target a non-natural four-carbon diol; pathway found computationally from succinyl-CoA and α-ketoglutarate; tools genome-scale modelling, growth-coupled strain design, multiplexed editing; TRY 18 g/L in the paper and above 100 g/L commercially; bottleneck precursor supply and redox under anaerobic conditions; commercial status produced at scale into polymers.
LanzaTech. Host *Clostridium autoethanogenum*; target ethanol and C2–C4 products; pathway native Wood–Ljungdahl; feedstock steel-mill off-gas containing CO, CO_2 and H_2; TRY driven by gas transfer rather than by pathway; bottleneck ATP — acetogen energetics leave almost no margin, which limits products to those close to acetyl-CoA; commercial status operating since 2018.

2. Defend your capstone host choice against one named alternative.

No fixed answer, but the structure is fixed: the decisive property, the alternative rejected, and what was given up. Name a measurable property — P450 support, native lipid flux, secretion titer, gas uptake rate, GRAS status, transformants per microgram — not a reputation.

3. Mark every forward-looking claim in a 2025 review as demonstrated or projected.

Check the marking against the tense. Demonstrated claims report a result with a number; projected claims use may, could, promises to, or has the potential to. Reviews in this field carry a great deal of the second kind, and quoting one as though it were the first is the fastest way to lose credibility in an interview.

What to open a book for. No book at all this week. A 2025 or 2026 review on AI-driven biofoundries or systems metabolic engineering, read with a pen for the demonstrated-versus-projected exercise, and a cell-free metabolic engineering review. The host table is your own document and should end the week with a citation in every cell that currently carries a dagger.

Closing notes to part 6

The four weeks in one page

Week 58. Yeast gives you folding, secretion, low pH, GRAS status and organelles. Its central fact is that acetyl-CoA is mitochondrial and your pathway is cytosolic, so cytosolic supply is the first thing to engineer — at one ATP per unit through ATP-citrate lyase or two through the pyruvate dehydrogenase bypass. The lipid ceiling is 0.356 g/g on carbon alone and 0.275 once NADPH is paid for, with 15 ATP per palmitate on top. Compartmentalisation buys a hundredfold concentration gain and charges a transporter for everything that crosses — about 2.8×10^{-8} mol·m^{-2}·s^{-1} against a carrier's 1.7×10^{-7}, so it works with roughly sixfold margin and only with a dedicated carrier.

Week 59. One membrane instead of two is the whole argument: secretion crosses one barrier, there is no endotoxin, and the downstream train loses two unit operations. The intracellular ceiling is not a fraction of a gram per litre but scales with cell density — 20 g/L at 50 gDCW/L, 40 at 100 — which is why intracellular products force high-cell-density operation. Lysine's ceiling moves from 0.81 to 0.86 mol/mol with a transhydrogenase, a cofactor device rather than a carbon one, and export work grows only logarithmically with titer: 4.8 to 7.6 kJ/mol across 50 to 150 g/L.

Week 60. Light reaches 15 mm at 1 g/L, so 95% of a pond is dark, and the unit is g·m^{-2}·day^{-1}. Above full absorption more cells cannot help, so the optimum is an areal loading of about 45 g/m² — 0.30 g/L in 15 cm, 0.15 in 30 cm. Realistic output is 10–25 g·m^{-2}·day^{-1}, ten to twenty per cent of the photon ceiling and two to three times a crop areally. And 1,000 litres of water per kilogram of biomass is what sorts the viable products from the rest.

Week 61. One table, with product localisation and regulatory history as the two columns that decide most arguments. Cell-free buys freedom from growth competition, membranes and toxicity at the cost of cofactor regeneration and a catalyst that does not replicate. And in a review, mark every claim demonstrated or projected before you quote it.

On the numbers in this part

The lipid ceiling, the peroxisome flux density and carrier capacity, the intracellular-titer table, the export work, the light attenuation depths, the areal optimum and the dewatering energies were all computed from the assumptions printed beside them. Four of those assumptions carry most of the weight and should be the first things you replace with measurements: the NADPH source in the lipid calculation, which is the entire difference between my 0.275 g/g and the literature's 0.30 to 0.33; the cell volume per gram of dry weight, taken here as 2 mL; the specific light attenuation coefficient k, taken as 0.15 m²/g, which sets the lit depth and the areal optimum together; and the maintenance coefficient in the pond model, which is what creates the optimum at all. Host properties in the tables that are judgements rather than measurements are marked with a dagger, and the week's exercise is to replace each with a source.

Where this leaves the module

Six parts now exist. Part 1 covered the biology foundations of weeks 1 to 10 as a spine beside the textbooks. Part 2 did the modelling and control framework of weeks 11 to 19 at twice that density. Part 3 covered in situ removal and immobilisation, parts 4 the economics, validation, omics and host defences, part 5 the five product classes with a constraint sheet each, and this part the hosts against those sheets. Together they cover weeks 1 to 19 and 45 to 61 — 36 of the tracker's 66 weeks, and most of the weeks with nothing behind them.

What remains unwritten, in tracker order: weeks 20 to 44, the technique and process phases, where the sources are protocols, software documentation and reviews that are genuinely good and genuinely specific — assembly and editing, screening, analytics, modelling software, fermentation and scale-up; and weeks 62 to 66, the publishing, proposal and application weeks, which are about how the profession works rather than about metabolic engineering. The technique weeks are the ones where a written module adds least, because a protocol you can run beats a chapter about it; the professional weeks are the ones where it might add a surprising amount, since nobody writes that down for engineers coming in from another discipline. If you want more, that is where I would go next, and I would ask which of the two you would actually use.

Part 12 — The profession: publishing, writing, planning, the proposal and the application (weeks 62 to 66)

These five weeks are not about metabolic engineering. They are about how the work becomes a career: how a manuscript becomes a paper and who decides what at each stage, how to write so an argument survives a reader, how to plan a multi-year project when half the duration is waiting, how to assemble the capstone proposal, and how to apply. When part 6 went out I said these were the weeks a written module might add the most to, because nobody writes them down for an engineer arriving from another discipline — and that is still my view. This is the last part.

Two things carry over from the rest of the module and make these weeks different from generic career advice. The first is that **you already have the planning skill**: critical paths, buffers, lead times and risk registers are engineering practice, and a research proposal with a realistic timeline reads differently from one with only science. Week 64 makes that quantitative, and it is the one week here with real arithmetic in it — a schedule simulation I ran, whose central result is that a plan built from most-likely durations had a **1% chance** of being met.

The second is that the whole curriculum has been producing **artifacts**: notebooks, a reproduced design, flux maps, a host table, a paper log, a proposal. Weeks 65 and 66 are about pointing at them. The test the tracker keeps returning to is whether a sentence would survive unchanged in someone else's application; if it would, it is about ambition rather than evidence, and the artifacts are the cure.

Ch.	Week	Chapter
62	62	How publishing works: review, authorship and the record
63	63	Writing and talking about your work
64	64	Planning a multi-year project: the critical path and what you cannot control
65	65	The capstone proposal, section by section
66	66	Applications: labs, statement of purpose, interviews

How publishing works: review, authorship and the record

A PhD is judged largely by papers, so knowing how they are made is not administrative trivia — it is what lets you ask a prospective supervisor the questions that matter.

From manuscript to paper, and who decides what

Stage	Who decides	What is decided	Typical time
Submission	You	Venue, framing, which claims are made	—
Editorial (desk) check	Handling editor	Whether it is sent out for review at all	Days to two weeks
Reviewer selection and reports	Editor chooses; reviewers write	Whether the claims are supported	Four to ten weeks, often longer
First decision	Editor, weighing the reports	Reject, major revision, minor revision	With the reports
Revision and response	You	What changes, what you defend	Weeks to months — usually the longest stage you control
Re-review and acceptance	Editor, sometimes the same reviewers	Accept, further revision, reject	Weeks

The time from submission to first decision is reported by many journals on their own pages and is worth looking up for the venue you have in mind: the answer is typically one to three months, and knowing it is how you plan around a thesis deadline.

Desk rejection is the editor declining without review, and it usually happens for one of three reasons: the work is out of the journal's scope, the advance is too incremental for that venue's threshold, or the manuscript is presented so poorly that the editor cannot see the claim. The first two are editorial judgements you can anticipate by reading the journal; the third is avoidable and is what chapter 63 is about.

A preprint changes three things: it timestamps the work publicly, it lets others read and cite it before review, and it exposes it to comment. It does not change whether the work is peer reviewed, how a hiring or funding panel weighs it in most systems, or your obligation to disclose it on submission. Most journals in this field accept preprinted manuscripts; a few do not, and that is worth checking before posting rather than after.

Reviewing a paper properly

The structure of a useful referee report, which is also the structure of how you should read any paper in your log:

- **Summary in your own words** — two or three sentences on what was done and claimed. If you cannot write it, you have not understood the paper, and saying so is itself a finding.
- **The claim**, stated as the authors would state it, so that everything after it is judged against that and not against the paper you would have written.
- **Do the data support it?** — the central question, and the one that makes the rest subordinate.
- **Controls** — part 7 chapter 22's list: the no-template, the empty vector, the loading control, the non-targeting guide, the parent strain assayed alongside.
- **Reproducibility** — could you repeat the method from the methods section, and are the data and code available (part 10 chapter 36)?
- **Limitations** — stated by the authors, or not.
- **Recommendation**, with major and minor points separated.

The discipline that makes a report useful rather than merely critical: **every major point must name what would fix it**. "The claim is unsupported" helps nobody; "the claim that flux was redirected needs either a carbon balance including by-products or a labelling experiment, since product up with uptake also up is consistent with higher throughput" tells the authors exactly what to do, and it is the standard this module has used since part 2 chapter 11.

Then the two questions you could not answer from the paper as written. Those are the most valuable sentences in a

report, because they are what the next reader will also not be able to answer.

Responding to reviewers

The response is a document with its own conventions: every point quoted, every reply saying **what changed and where** — page and line — or why you disagree, with evidence. Three habits make the difference. **Never ignore a comment**, including one you think is foolish; an unanswered point reads as an evasion. **Disagree in evidence, not in tone**: "we respectfully disagree; the control the reviewer asks for is shown in Figure S3, which we have now referenced in the main text" is a disagreement that wins. And **make the revision findable**: a marked-up manuscript plus a response that points to it costs you an hour and saves the editor an argument.

If a reviewer asks for an experiment you cannot afford, you have four honest moves and should choose explicitly rather than hoping: do a cheaper experiment that addresses the same doubt and say so; show that existing data already answer it; narrow the claim so the experiment is no longer needed — often the best option, and the one that improves the paper; or argue that it is beyond the scope, which works only if you also narrow the claim. What does not work is promising it as future work while keeping the strong claim.

Authorship and integrity

The widely used criteria — the ICMJE conditions — require **all** of: substantial contribution to conception, design, acquisition, analysis or interpretation; drafting or critically revising the intellectual content; approval of the final version; and agreement to be accountable for the work. Contribution alone is not enough, and neither is seniority. **First author** signals the person who did most of the work and usually wrote the draft; **corresponding author** signals who is answerable for the work and handles the submission, normally the supervisor. Order between the two is negotiated, and the time to negotiate it is at the start of a project, not at submission.

The three cases the week sets, with my answers and the reasoning:

Case	Decision	Why
A technician who ran every assay	Usually authorship; at minimum a substantial acknowledgement	Data acquisition is a listed contribution. The question is whether they also contributed intellectually and can approve and defend the work — if they designed or interpreted the assays, authorship is clear; if they executed a protocol exactly as specified and will not review the manuscript, a detailed acknowledgement is the honest answer. The failure mode to avoid is excluding them by default because of their job title.
A collaborator who supplied a strain	Normally acknowledgement, not authorship	Providing a reagent is not conception, analysis or interpretation. It becomes authorship if they also contributed to design or interpretation — for instance by advising on how the strain behaves in a way that shaped the experiments.
A supervisor who did not read the draft	Does not qualify as written, and the right response is to get them to read it	Approval of the final version and accountability are explicit criteria. In practice this is the most common gap between the standard and the norm, and the useful framing is not to police it but to insist on the reading — which is also what makes the paper better.

Image manipulation: adjustments applied uniformly to a whole image — brightness, contrast, colour balance — are legitimate if disclosed and if they do not obscure or create features. What is not legitimate: adjusting part of an image, splicing lanes without a visible divider and a note, removing a blemish, duplicating a region, or enhancing a band. Journals increasingly screen for this automatically, and the expectation is that you keep the unprocessed originals. The rule that covers every case: a reader must be able to reach your conclusion from the raw data, and any processing must be reversible in description.

The other integrity failures are statistical and were covered quantitatively earlier: **selective reporting** and **p-hacking** (part 2 chapter 17's multiplicity arithmetic, where 96 tests at $\alpha = 0.05$ give five false positives by construction), and **salami slicing**, dividing one result into several papers, which inflates a publication list and wastes readers' time.

Negative results matter for a reason this module has demonstrated repeatedly: part 8 chapter 29's honest base rate for pump engineering came from a screen that found nothing useful for two of seven compounds, and part 11 chapter 39's aerobic search found no coupled design at all — which was the result. A field that publishes only successes has a literature in which every intervention appears to work, so the next student repeats the failures privately. Reporting a negative result properly means stating what you would have been able to detect — part 2 chapter 17's power argument — which converts "it did not work" into a measurement.

Reading a lab through its papers

For each lab on your shortlist, count: papers in the last three years, where, and how often a student is first author. The pattern tells you things no website does.

Pattern	What it suggests
Many papers, students rarely first author	A large group where students contribute to others' projects; you may publish often and own little
Few papers, students always first author	Projects are owned and long; higher risk per student, more credit when it works
Consistent venue and topic	A defined programme; you will join a line of work rather than start one
Scattered topics and venues	Either a broad group or one following funding; ask which
Long gaps then a cluster	Projects that publish at the end — worth knowing before you are three years in

The question to ask a current student: *how long was it from starting your project to your first submission, and what delayed it?* The question to ask the supervisor: *what would a successful first year look like, and what would be publishable from it?* Both answers are concrete, and a supervisor who cannot answer the second has not thought about your project as a sequence of outputs.

Route	`submit → desk check → review → decision → revise → accept; one to three months to first decision`
Desk rejection	`scope, insufficient advance, or an unreadable claim`
Review report	`summary · claim · support · controls · reproducibility · limitations · recommendation`
Major point rule	`every major point names what would fix it`
Authorship	`ICMJE: contribution AND drafting or revising AND approval AND accountability — all four`
Image rule	`uniform, disclosed, reversible in description; keep the originals`

A reviewer asks for a ^{13}C labelling experiment to support your claim that a deletion redirected flux. You have no access to the instrumentation. What do you write?

First decide whether the claim needs it, because that decides everything else. The claim "flux was redirected" does need evidence beyond a titer, as part 11 chapter 43 argued — but a closed carbon balance including by-products and biomass, with substrate uptake measured, distinguishes redirection from higher throughput, and that you can do. So the response runs: we agree that a titer alone does not establish redirection; we have added a carbon balance (new Table 2) showing that total uptake was unchanged while the by-product fell by the amount the product rose, which is consistent with redirection and inconsistent with increased throughput; a labelling experiment would additionally resolve the split at the branch point, and we state that limitation explicitly in the discussion.

If even the carbon balance is unavailable, the honest move is to narrow the claim to what the data support — "titer increased by X with no change in substrate uptake" — and say in the discussion what would establish redirection. That is a better paper than one whose central claim rests on an experiment the reviewer correctly says is missing, and reviewers generally accept a narrowed claim where they will not accept a defended overreach.

Three things people get wrong

- ‘The reviewer is wrong, so I will ignore that point.’ An unanswered point reads as an evasion. Disagree with evidence, in writing.
- ‘They contributed, so they are an author.’ The standard requires contribution *and* drafting or revising *and* approval *and* accountability.
- ‘Negative results are unpublishable.’ They are publishable when you state what you could have detected, which turns a non-result into a measurement.

Self-check — answer on paper first

1. Your paper is desk-rejected in four days. What are the likely causes and what do you do?

Scope, insufficient advance for that venue, or a claim the editor could not find. Read the rejection letter for which — editors usually say, briefly — and then check the journal's recent contents for whether work like yours appears there at all. If it is scope, resubmit elsewhere unchanged. If it is the advance, either add the experiment that raises it or choose a venue whose threshold matches. If it is presentation, the abstract and the first paragraph are where to work, since those are what a desk decision is made on, and chapter 63's four-move abstract is the fix.

2. A collaborator supplied a plasmid and asks for authorship. How do you handle it?

Apply the criteria rather than the relationship: supplying a reagent is not by itself conception, design, analysis or interpretation, so the default is a specific acknowledgement. Then ask whether there is more — did they advise on the construct's behaviour, interpret a result, or will they review and stand behind the manuscript? If so, authorship is

justified. Handle it by raising the criteria early and in writing, with the supervisor involved, because the conversation is much easier before the data exist than at submission.

3. What does a lab's publication pattern tell you that its website does not?

Whether students own projects — visible in how often a student is first author rather than third — how long projects take to reach an output, from the gaps between a student joining and their first paper, and whether the group has a programme or follows funding, from the consistency of topic and venue. It also shows where the group's work is read, which affects who sees yours. None of this is on a website, and all of it predicts your own years there more reliably than a research-interests paragraph.

What to open a book for. The ICMJE recommendations on authorship, which are short and are the standard most journals point at; COPE's guidance and flowcharts, which show how real cases are handled; and the literature on peer review and on research integrity that the tracker names. Then read one journal's own guide for authors and reviewers in full — it is the most concrete source there is.

Chapter 63 · tracker week 63

Writing and talking about your work

A reader is looking for the argument and will stop looking if it is hard to find. The craft is not elegance; it is making the structure so obvious that the reader cannot lose it.

The shape of an argument

The pattern that organises almost every good paper is **context, content, conclusion**, applied recursively: to the paper, to each section, and to each paragraph. Context says why this matters and what is known; content says what was done and found; conclusion says what it means. And **one idea per paragraph**, with that idea in the first sentence — because a reader skimming first sentences should get the argument, and a reader who does not skim loses nothing.

The exercise that matters is this: write your capstone's **single central claim in one sentence**, then list the evidence under it. If the sentence needs an "and" joining two claims, you have two papers or one unfocused one. Then **mark every place where you asserted something the evidence does not reach** — and there will be several, because enthusiasm writes faster than evidence. In my own writing across this module the recurring form of that error was a confident number from a model whose assumptions I had not stated beside it, which is why every chapter here carries its parameters.

Two specific moves that make a metabolic-engineering argument legible, both of which the module has used throughout: state the **number that decides the question** early — the ceiling, the control coefficient, the k_La — because a reader who knows what the binding constraint is can follow everything after it; and **name the alternative you rejected**, which is what distinguishes a choice from a default, exactly as part 6 chapter 61 required for a host.

Figures that argue

A figure should carry **one message**, which should be the title or the first clause of the caption. The audit to run on your own figures:

- **Axes** with units and a readable range; log scales labelled as such, since a log axis flatters a small difference and a reader must be told.
- **Error bars defined** — standard deviation or standard error, and n, in the caption. Part 4 chapter 17's argument: they differ by √n and a caption that omits which is unreadable.
- **No decoration**: no three-dimensional bars, no gradients, no gridlines competing with the data, no colour carrying meaning that is not also carried by shape or position.
- **Every element earns its place**: if removing something changes no message, remove it.
- **A caption that stands alone**, including what was measured, n, what the error bars are, and the assumptions behind any computed line.

Rebuild the one figure that carries the capstone's argument to that standard. For a metabolic-engineering proposal that figure is usually the yield or flux analysis: the theoretical ceiling, the model prediction under a realistic constraint, and where the published state of the art sits — three numbers on one axis, which is an argument rather than a decoration.

Abstract and title

The four moves of an abstract, in order, with roughly a quarter of the words each: **problem** — what is wrong or unknown and why it matters; **approach** — what you did, specifically enough to be recognisable; **result** — the number, not the adjective; **consequence** — what follows.

Then cut it by a third. The words that go are the hedges, the throat-clearing first sentence, the restatement of the approach in the result, and every "novel" and "promising". The test the week sets is the right one: give it to someone unfamiliar and ask them what you are claiming. If their answer is not your central claim, the abstract has failed regardless of how well written it is.

Three candidate titles for a proposal on, say, a C4 diacid:

Type	Example	Judgement
Descriptive	"Growth-coupled production of a C4 diacid in *E. coli*: design, modelling and scale-up analysis"	Safe, searchable, says what is inside. Submit this.
Claim-making	"Oxygen transfer, not pathway flux, sets the ceiling for C4 diacid production"	Stronger if the evidence supports it — and in a proposal it usually does not yet, since the claim is what you propose to test

Too clever	"Breathing room: the hidden cost of aerobic diacid production"	Unsearchable and will read badly in a citation list

For a proposal, descriptive; for a paper whose central result is genuinely a reversal of the expected constraint, claim-making, because that is the finding. Never the third.

The ten-minute talk

Ten minutes is **one idea**. The structure that survives: thirty to sixty seconds of context — the minimum needed to make the question sensible, not a review; the question in one sentence on its own slide; the approach; the result, with the figure you rebuilt; and the limitation, **raised by you before anyone asks**. That last move is the one that changes how a room treats you: naming your own weakest point converts a cross-examination into a discussion, and it is the same argument as part 11 chapter 39's reporting of a search that found nothing.

Record it and watch it back. The two things to cut are everything you **skipped past** — a slide you waved at is a slide you do not need — and everything you **apologised for**: "sorry, this is a busy figure" means rebuild the figure, not warn the audience.

On questions you cannot answer, which is the week's last skill. The honest sequence: say you do not know; say what you would do to find out; and, if you have a partial answer, mark clearly where the evidence stops and the speculation begins — "what I know is X; what I would expect is Y, and the experiment that would settle it is Z." That answer is better received than a confident guess, and in a PhD interview it is close to the only answer that is.

Structure	`context · content · conclusion, recursively; one idea per paragraph, in the first sentence`
The test of a claim	`one sentence, no "and"; evidence listed under it`
Figure	`one message · units · defined error bars with n · nothing decorative · standalone caption`
Abstract	`problem · approach · result · consequence, then cut by a third`
Talk	`one idea; raise your own limitation; cut what you skipped or apologised for`
Unanswerable question	`I don't know · here is how I'd find out · here is where evidence stops`

You have ten minutes to present the capstone to a group that works on a different host. What goes in, and what gets cut?

In: the target and why it needs engineering rather than chemistry, in two sentences; the host choice with the deciding property named and the alternative rejected; the yield analysis figure — ceiling, constrained prediction, state of the art; the one experiment that would falsify the design; and the limitation you would raise yourself, which for most proposals is either a step with no good candidate enzyme or an oxygen-transfer ceiling.

Out: the pathway diagram with every intermediate labelled, which costs two minutes and conveys less than "four steps from acetyl-CoA, all with characterised enzymes except step three"; the literature survey; the methods you have not yet used; and every slide that exists because it was work. The hard cut is the last one — the modelling notebook took a week and gets one figure, because the audience is deciding whether the design is sound, not whether you can use COBRApy. Point at the repository for the rest, which is what artifacts are for.

Three things people get wrong

- ‘The reader will follow the logic.’ They will follow the first sentences of paragraphs. Put the idea there.
- ‘The figure shows the data.’ A figure should show one message; data with no message is a table.
- ‘I'll handle the weakness if someone asks.’ Raise it first. It converts an interrogation into a conversation.

Self-check — answer on paper first

1. Write the four moves of an abstract for a result you already have — the part 10 core model will do.

Problem: constraint-based models are used routinely to predict yields and deletions, but their assumptions are hidden inside software. Approach: a 40-reaction core model of *E. coli* central metabolism was built by hand and every standard analysis — flux balance, variability, deletions, parsimonious flux, shadow prices and quality control — implemented directly as linear programmes. Result: the model grows at 0.754 h^{-1} aerobically and 0.217 anaerobically, ten of 39 deletions are lethal, fourteen reactions have alternative optima at exactly maximal growth, and the implementation exposed three modelling errors that a package would have hidden. Consequence: writing the programme yourself is a better first exercise than using a package, because every later failure is a bound, a medium or an objective.

2. Your figure has six panels and reviewers keep asking what it shows. Diagnose it.

Six panels is at least two figures and possibly six: a figure carries one message, and a reader asking what it shows is telling you the message is not identifiable. Split it by message, promote the panel that carries the central claim to its own figure, and move the rest to supplementary material unless each is load-bearing. Then write each caption as a standalone sentence beginning with the message — if you cannot, that panel has no message and should go.

3. In a talk, someone asks a question that reveals a flaw you had not considered. What do you say?

Acknowledge it directly — "that is a real problem and I had not considered it" — then think aloud about what it would take to find out: which measurement would distinguish the possibilities, and what you would expect under each. Do not defend, do not improvise a mechanism, and do not promise it is fine. The reason this is the right answer is that the room is evaluating how you handle being wrong, which in research is the more frequent condition, and a candidate who can say "I don't know, here is how I'd find out" is demonstrating the thing the job actually requires.

What to open a book for. The ten simple rules for structuring papers, which is short and is the standard reference for the context-content-conclusion pattern; a figure-design guide; and then your own figures from across this curriculum, audited against the list above — that audit is the content of the week, not the reading.

Chapter 64 · tracker week 64

Planning a multi-year project: the critical path and what you cannot control

A plan built from most-likely durations is usually missed, and by a predictable amount. One simulation shows why, and it is the week's whole argument.

From an aim to a task list

The work breakdown: every task gets a **duration**, its **predecessors**, and a **deliverable**. The deliverable matters because it is what makes a task finishable — "characterise the strain" is not a task, "a table of titer, yield and growth rate for three strains in triplicate" is.

Here is the first year of a capstone-style project as ten tasks, with optimistic, most-likely and pessimistic durations in weeks. Two are **waiting rather than working**, which is the distinction the week insists on:

Task	o / m / p (weeks)	Predecessors	Kind
A — literature review and design	2 / 3 / 5	—	work
B — order gene synthesis	3 / 4 / 8	A	**waiting**
C — build vectors	1 / 2 / 4	B	work
D — host toolkit (parallel)	3 / 5 / 10	A	work
E — transform and verify	1 / 2 / 5	C, D	work
F — shake-flask screen	2 / 3 / 6	E	work
G — analytical method	2 / 3 / 7	A	work
H — quantify and iterate	3 / 4 / 8	F, G	work
I — fermenter slot	2 / 4 / 12	H	**waiting**
J — fed-batch and write-up	3 / 4 / 7	I	work

The critical path, and the float nobody uses

With most-likely durations the project takes **26 weeks** and the critical path is A → B → C → E → F → H → I → J. The two off-path tasks have float: the host toolkit has **1 week** and the analytical method has **8 weeks**. Shortening the analytical method by a month changes the end date by nothing at all, which is the first lesson and the one most people violate by optimising whatever is in front of them.

But float computed from single durations is misleading, because durations are uncertain. Sampling each task from a triangular distribution over its three estimates and recomputing the critical path 20,000 times gives the **criticality index** — the fraction of runs in which a task lies on the critical path:

Task	Criticality	Reading
A, E, F, H, I, J	100%	Always critical — attention here always matters
B order synthesis, C build vectors	76%	Usually critical
D host toolkit	**24%**	The near-critical branch that quietly becomes the problem
G analytical method	0%	Never critical, despite 8 weeks of float being only nominal

The host toolkit has one week of nominal float and is on the critical path in a quarter of simulated runs, because its pessimistic duration is ten weeks. That is the "second near-critical branch" the week warns about: a task you have classified as non-critical and therefore stopped watching.

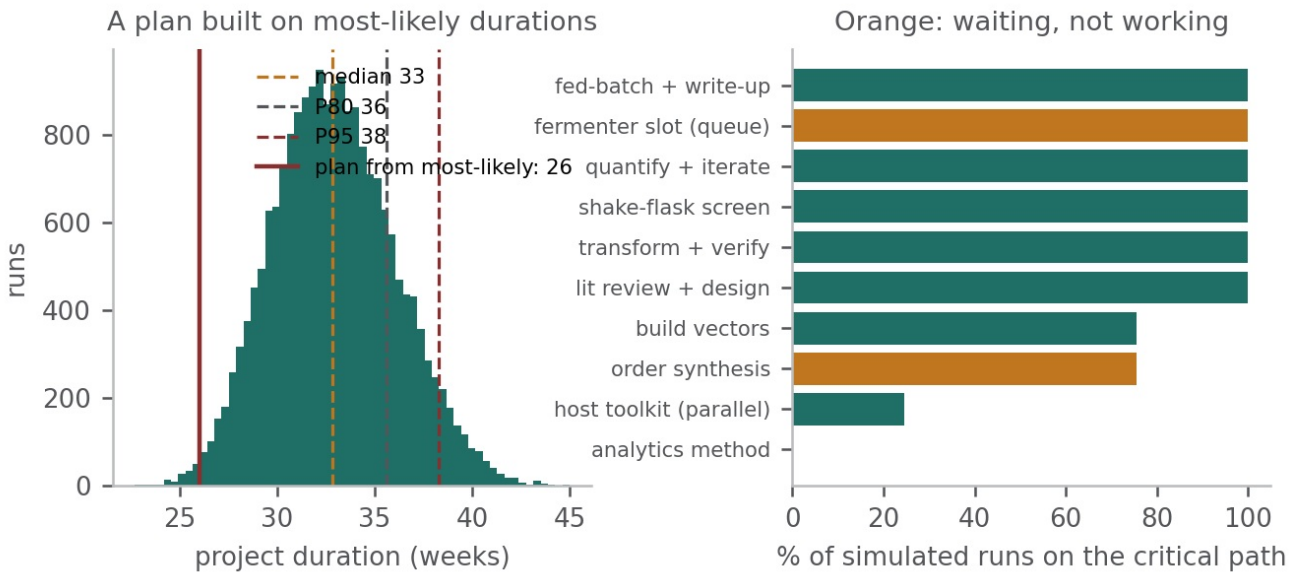


Figure 64.1 Left: simulated project duration over 20,000 runs with triangular task durations, against the 26 weeks a most-likely plan predicts. Right: criticality index per task, with the two waiting tasks in orange. Computed for this module from the task list above; the durations are illustrative and the triangular distribution is the assumption.

Why plans slip

The distribution is the week's central result:

Plan from most-likely durations	26.0 weeks
Simulated median	32.8 weeks
80th percentile	35.6 weeks
95th percentile	38.3 weeks
Probability of finishing within the planned 26 weeks	**1%**

A plan built by adding most-likely durations along the critical path has a one-in-a-hundred chance of being met, and the median overrun is 26%. Two mechanisms produce this and both are structural rather than behavioural. **Task durations are right-skewed**: a task can finish a little early and can be delayed a great deal, so its mean exceeds its mode. And **paths merge**: task E waits for both C and D, so it starts when the *later* of the two arrives, and the maximum of two uncertain quantities is larger than either's typical value. Every merge point in a network adds a little delay that no individual estimate contains.

So: **the date you put in a proposal and the date you privately expect should differ**, and you should know which is which. Put a date near the 80th percentile in the proposal — 36 weeks here — and plan your own work against the median. A proposal promising the 26-week date is not optimistic, it is innumerate, and a supervisor who has run projects will read it as such.

Where to spend attention: waiting against working

The week asks what halving a waiting time is worth against working faster. Simulated on the same task list:

Intervention	Median	Saving	80th percentile
Baseline	32.9 weeks	—	35.6
Halve the synthesis wait	31.5	1.4 weeks	34.2
Halve the fermenter queue	29.9	**3.0 weeks**	32.1
Halve both waits	28.5	4.3 weeks	30.7
Work 30% faster on every bench task	26.0	**6.8 weeks**	28.5

The honest result, which is not quite the one the week implies: in *this* task list, working 30% faster on everything saves more than halving both waits, because the waits are only 8 of the 26 nominal weeks. But the two interventions are not comparable in difficulty. Halving a fermenter queue means booking earlier or finding a second vessel — a phone call and

some planning. Working 30% faster on every bench task for nine months is not a decision, it is a different person. **Judge interventions by effort as well as by weeks**, and on that basis the queue is the best value on the table: three weeks for an email.

The general rule that survives: compute the criticality index, then look first at the critical tasks you do not control, because those are the ones where a small amount of foresight — ordering earlier, booking the slot before you need it, starting the approval paperwork in week one — buys weeks for nothing.

Buffers

Two ways to protect the same 5.2 weeks of padding on a 26-week plan: add 20% to every task, or hold one buffer of 5.2 weeks at the end. Simulated with the padded tasks subject to Parkinson's law — a task with a padded estimate takes at least that long, because work expands to fill the time available:

Approach	Median duration	Probability of meeting the 31.2-week committed date
Pad every task by 20%, pads consumed	35.2 weeks	**0%**
One buffer of 5.2 weeks at the end	32.8 weeks	**30%**

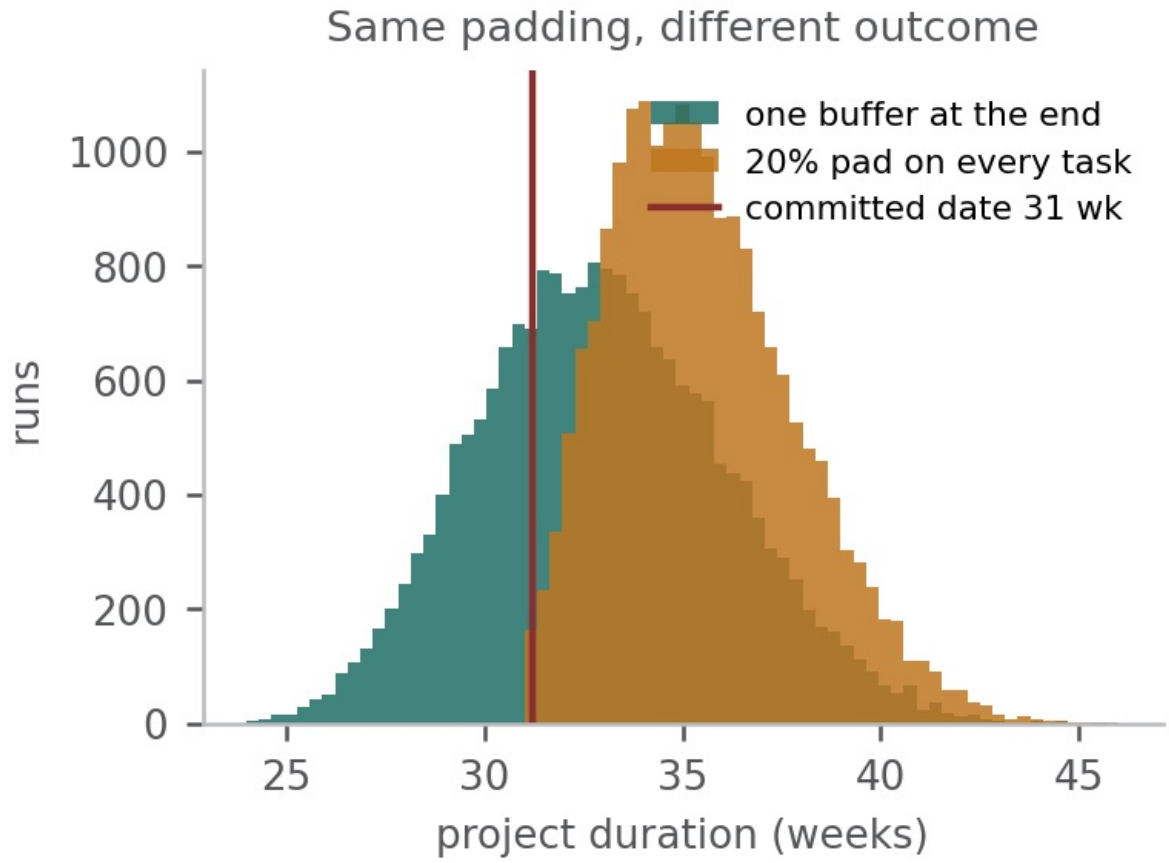


Figure 64.2 The same total padding, applied per task and consumed, against held as a single buffer at the end. Computed for this module; Parkinson's law is modelled as each padded task taking at least its padded estimate, which is an assumption about behaviour rather than a measurement.

Distributed padding is strictly worse, for two reasons. The pads are **consumed** rather than saved, because an early finish on a padded task is not passed on; and they are **invisible**, so nobody can see how much protection remains. A single buffer is visible, is consumed only by real overruns, and can be reported: "we are twelve weeks in, two weeks of the five-week buffer are gone" is a sentence a supervisor can act on.

And what to tell a supervisor three months into a six-month slip: the slip, the cause, the revised date with its uncertainty, and the decision you are asking for. Not an apology and not a promise to catch up — the arithmetic above says catching up does not happen. The useful version is: *the strain work is eight weeks behind because the toolkit took ten weeks rather than five; the revised 80th-percentile date is X; the options are to drop the second host, to run the fermentation with the existing strain, or to extend — I recommend the second, and here is why.*

Money, and what a student controls

Price one month of the planned work. The categories, with the pattern rather than numbers I would have to invent: **large and obvious** — gene synthesis, which scales with length and is the single biggest line for a pathway project; sequencing; enzymes and kits. **Hidden** — plasticware, media components, antibiotics, and the shipping and import charges on everything, which in a non-US lab can be a quarter of a reagent's cost and weeks of its lead time. **Shared facility charges** — analytical time per sample, fermenter day rates, sequencing per lane — which are billed to the project but queued by the facility, and therefore appear in both the budget and the schedule.

Look up current prices for your own institution rather than taking any figure from me: synthesis and sequencing prices in particular have moved by orders of magnitude over a decade, and a stale number in a proposal is a visible error. What a student actually controls is a small discretionary fraction; the rest is committed by the grant, which is why the next paragraph matters.

To see what funds a lab, read the **acknowledgements** of its recent papers: they name the programme and often the grant number. Then read what that programme is for — its call text states objectives in plain language — and write one sentence connecting your capstone to them. That sentence is worth disproportionate attention in an application, because it tells a supervisor that your project can be funded from money they already have.

Risk, and deciding in advance when to stop

Risk	Likelihood	Cost if it happens	Alternative
A pathway step has no active enzyme in the host	Moderate to high — part 11 chapter 40's p_n	Months; the central claim fails	Screen homologues; switch to the second route identified in the proposal
Product is toxic below the needed titer	Moderate	Caps the titer; TEA may fail	Two-phase removal (part 3 chapter 45); tolerance evolution
Oxygen transfer limits at the target density	Known in advance from chapter 44's arithmetic	Lower productivity, by-products	Lower feed rate, enrichment, or a lower-density higher-q_P design
Host toolkit takes longer than planned	24% critical in the simulation	Weeks on the critical path	Prototype in *E. coli* and transfer (part 7 chapter 25)
Strain instability at scale	Moderate	Invalidates the fermentation result	Integrate rather than carry a plasmid; sequence after the run (part 9 chapter 32)

Three go/no-go points with the triggering result, which is the discipline that stops two years disappearing into one idea: **month 4** — if no construct expresses a detectable product, stop and test whether the enzymes are active in lysate before building more strains; **month 8** — if the best strain is below a stated fraction of the modelled yield with no identified bottleneck, switch to the alternative route; **month 12** — if the techno-economic sketch at the achieved yield and titer cannot clear the market price under favourable assumptions, change the target rather than the strain.

And the sentence about risk for the proposal, which is the week's last point: *the highest risk is that step three has no characterised enzyme active on our substrate; we will test this in month two by expressing three homologues and assaying the lysate, and if none is active we will pursue the alternative route via [X], which is two steps longer but fully characterised.* Naming a risk strengthens a proposal because it demonstrates that you know where it might fail and have thought about what to do — which is indistinguishable, to a reader, from having run projects before.

Critical path	`the chain that sets the end date; float elsewhere is free time you cannot spend on it`
Criticality index	`fraction of simulated runs on the critical path; 24% for a task with 1 week of nominal float`
Why plans slip	`right-skewed durations and merge points: median 32.8 against a 26-week plan, P(met) = 1%`
Proposal date	`the 80th percentile; plan privately against the median`
Buffers	`one at the end, visible and managed, not 20% on every task`
Risk table	`what could stop it · likelihood · cost · alternative · and the result that triggers a stop`

Your supervisor asks for a one-year plan with a date. What do you give them?

Two dates and the reason they differ. The committed date at roughly the 80th percentile of a simulated schedule — in my example 36 weeks against a 26-week most-likely plan — with a single visible buffer that accounts for the difference; and the median as the internal working target. Alongside them: the critical path, the criticality index so they can see which tasks matter, the two or three tasks that are waiting rather than working with their lead times, and the go/no-go points with the results that trigger them.

What I would not give them is a Gantt chart of most-likely durations, because its date is met 1% of the time and everyone involved will spend the year explaining why it is late. The conversation to have instead is which of the three or four interventions is worth buying: in the simulation, halving the fermenter queue was worth three weeks for an email, while working 30% faster everywhere was worth seven weeks and is not available. Framing the plan as a set of purchasable improvements rather than a promise is the thing your engineering background lets you do that most applicants cannot.

Three things people get wrong

- ‘The plan says 26 weeks.’ It says the most-likely durations sum to 26; the simulation says the median is 33 and 26 is met 1% of the time.
- ‘That task has float, so it does not matter.’ One week of nominal float and a long tail made it critical in 24% of runs.
- ‘I padded every task to be safe.’ Padding is consumed and invisible; one buffer at the end gave 30% on-time against 0%.

Self-check — answer on paper first

1. Why does adding most-likely durations along the critical path underestimate the duration?

Two reasons. Task durations are right-skewed — a task can run a little early and a long way late — so each task's mean is greater than its most likely value, and the sum of means exceeds the sum of modes. And wherever paths merge, the successor waits for the *latest* predecessor, so the expected start is the expectation of a maximum, which is greater than the maximum of the expectations. Every merge point adds delay that no individual estimate contains, which is why the simulated median exceeded the deterministic plan by 26% here.

2. A task has four weeks of float. Under what circumstances should you still watch it?

When its distribution has a long right tail, which is exactly the host-toolkit case: one week of nominal float, a pessimistic duration twice the most likely, and a 24% criticality index. Float computed from single-point estimates tells you about the mode, not the tail, and a task that is non-critical in the plan and critical in a quarter of futures deserves attention. The quantitative test is the criticality index, and the practical sign is a task whose pessimistic estimate is more than about twice its most-likely one.

3. You are eight weeks behind at month six. What do you tell your supervisor?

The slip, its cause, a revised date with its uncertainty, and a decision to be made — with your recommendation. Not an apology, and not a plan to catch up, because the arithmetic says recovery does not happen: the remaining tasks have the same distributions they always had. The useful content is the options and their consequences — drop a scope item, run with what exists, or extend — each with what it costs, so that the person who has to decide can decide. And if a buffer was declared at the start, say how much of it is left, which turns the conversation from blame into inventory.

What to open a book for. No book: the tracker's literature searches on research project management and on Monte Carlo schedule risk analysis, plus the NumPy random-number documentation, are what this week needs. The simulation above is thirty lines, and building your own on your own task list is the content of the week.

The capstone proposal, section by section

Three to five pages that a principal investigator reads in ten minutes and either believes or does not. What makes it believable is not the science, which anyone can read up — it is that the numbers are yours and the risks are named.

What the proposal is for

It is an argument that a specific molecule should be made in a specific host by a specific route, that the numbers support it, and that you know where it will fail. Seven sections, roughly: motivation, host, pathway, the numbers, build-and-test, scale-up and economics, risks. The order matters because each section's conclusion is the next one's premise — the molecule sets the price it must clear, the price sets the titer and yield required, those set the host and the route, the route sets what must be built, and the economics decides whether any of it was worth it.

Section 1 — target and motivation, one page

Why this molecule: what it is used for, the market size with a source, and **the incumbent route**, because that is what you are competing against. Part 5 chapter 53's lesson is the governing one — a target whose incumbent can halve its price in response is a bad target however good the biology — so name the incumbent's cost structure, not just its existence: a petrochemical plant with amortised capital, an agricultural supply that responds to price, or an extraction from a scarce natural source that cannot scale.

Then the sustainability argument, **honestly, including where it is weak**. The honest version almost always includes: the feedstock is agricultural and therefore carries land and fertiliser burdens; the electricity mix decides the result (part 5 chapter 57's 17% advantage that a grid at 0.6 kg CO_2e/kWh erases); and the boundary you chose is the one that makes your number look as it does. A proposal that states this is more credible than one claiming a clean win, and a reviewer goes to the boundary first anyway.

Section 2 — host, with the deciding property named

Use your own host table from part 6 chapter 61, and write the choice in the format that chapter required: **the property that decides it, the alternative you rejected, and what you gave up**. "We will use *K. phaffii* because the product is a secreted haem protein and this host secretes at grams per litre with a clean extracellular background and a food-enzyme regulatory history; we rejected *E. coli*, which would have given a faster design–build–test cycle and a far better toolkit but requires periplasmic targeting or refolding and carries endotoxin a food product cannot have; what we gave up is cycle time."

Then the **localisation subsection**, which part 6 chapter 59 argued decides more than it is given credit for: secreted, retained or stored. Secreted means no intracellular ceiling and a simpler downstream train. Retained means the cell-volume ceiling of part 1 chapter 1, recomputed at industrial density — 20 g/L at 50 gDCW/L, 40 at 100 — and therefore a high-cell-density process plus lysis and debris removal. Stored means an accumulation physiology and a nutrient-limitation trigger (part 5 chapter 56). Say which, and say what it forces on the host.

Section 3 — the pathway, every enzyme sourced

From a named central metabolite — one of the twelve of part 1 chapter 7 — to the product, step by step, with an enzyme named and cited for each from BRENDA, MetaCyc or KEGG, including the organism it came from and, where known, its activity on a substrate like yours. Then the sentence that makes this section honest: **mark the steps with no good candidate enzyme**. Part 11 chapter 40's arithmetic says a route of n steps each with probability p works with probability p^n, so five steps at p = 0.8 is 33%, and a step with no candidate at all is a discovery project rather than a low probability. Marking them is not a weakness in the proposal; it is the difference between a route and a wish.

Section 4 — the numbers, worked by hand and then in a model

This is the section that distinguishes your proposal from a biologist's, and it has three parts.

The hand calculation. Theoretical maximum mass and molar yield on glucose, with the carbon, NADH, NADPH and ATP balances closed *explicitly and separately* — part 2 chapter 19's order, and the point at which most back-of-envelope balances fail by lumping the two hydride carriers. Show the working: part 5's ethanol at 0.511 g/g with its redox closing exactly, isobutanol at 0.411 with its cofactor mismatch, PHB at 0.478 with a surplus of three reduced cofactors and therefore an oxygen duty, are the models for how this looks.

The model. Add the heterologous reactions to a genome-scale model — metabolites, stoichiometry, bounds, objective — and **confirm it reproduces the hand yield**. That confirmation is the point: if the model and the hand calculation disagree, one of them is wrong, and finding out which is the exercise. Part 10 chapter 36's warnings apply: set the medium through the exchange reactions, use a context manager for temporary changes, and check for the loops and alternative optima that make a single flux vector meaningless.

The constrained prediction. Then the step that almost no student proposal contains: constrain oxygen to a value your own k_La calculation justifies (part 11 chapter 44) and rerun. In my core model, going from unconstrained oxygen to the transfer a 100 h^{-1} vessel delivers at 10 gDW/L cost 58% of the growth rate and produced 12 mmol of reduced by-products. **Report the yield you would actually get in a vessel**, not the one the unconstrained model gives, and say which is which.

Push the notebook to the repository, commented so a PI can read it without running it, and put its headline figure — ceiling, constrained prediction, state of the art — in the proposal. That figure is the one to rebuild to chapter 63's standard, because it carries the argument.

Section 5 — build and test, with a reason per tool

Assembly method, editing tool, analytics, screening readout, and the number of design–build–test iterations you expect. The rule the week sets is the right one: **no tool without a reason, in one clause**. "MoClo-standard Golden Gate, because six parts will be swapped combinatorially (part 7 chapter 20); CRISPR–recombineering with Cas9 counterselection, because the host has no efficient end joining and the edits are deletions (chapter 23); CRISPRi for the two essential competing genes, because a knockout is not recoverable (chapter 24); HPLC with a weighted low-range calibration for titer, because the first strains will be near the quantification limit (part 4 chapter 50)."

Then the **transport strategy**, which part 3 chapter 45 argued belongs on the first page rather than the last: if the product is secreted, which efflux pump you would mine, express or evolve, and the base rate you expect from the published screens; if it is retained, the lysis and extraction train and what it costs. And state **what result would make you abandon the design** — a sentence most proposals omit, and the one that makes a reader trust the rest.

Section 6 — scale-up, downstream and the economics

Fermentation mode with a physiological justification rather than a conventional one (part 11 chapter 44): exponential fed-batch below the overflow threshold if the host has one, two-phase with nitrogen limitation if the product is stored, gas-phase considerations if it is a knallgas organism. The scale-up risks, which you can now name quantitatively: mixing time growing as $d^{2/3}$, the k_La wall at 20 to 30 gDW/L, and the extra generations in the seed train with their instability risk (part 9 chapter 32's plasmid-retention recursion).

The downstream train as a list of unit operations with the property each exploits, and the water: 1,000 divided by the titer in litres of broth per kilogram of product, which at 20 g/L is fifty kilograms of water to move per kilogram made. Then the back-of-envelope from part 4 chapter 49: feedstock cost as substrate price over yield, capital and fixed cost scaling as 1/titer, and **the titer below which the process cannot work** — which in that model, for a $1.80/kg commodity, was around 100 g/L. State it as a target, because it converts the whole proposal into a number the strain work has to hit.

Section 7 — risks and alternatives, named specifically

Part 64's table, filled for this project: what could stop it, how likely, what it would cost, and what you would do instead, plus the go/no-go points with their triggering results. Specific means naming the step, the organism and the measurement — "step three may have no active enzyme" rather than "enzyme activity may be insufficient".

Reading it once, aloud

The last session's instruction is to read the whole draft aloud and mark the weakest section. Reading aloud finds three things that silent reading does not: sentences you cannot finish in one breath, which are too long; claims you hesitate before, which are the ones the evidence does not reach; and transitions that do not exist, where you have put two paragraphs next to each other and hoped. The weakest section in most of these proposals is section 5, because it is a list of tools and the reasons are the thing that is missing — which is why the one-clause rule exists.

Section 1	`molecule · market · incumbent's cost structure · honest sustainability, with its weak point`
Section 2	`deciding property · rejected alternative · what you gave up · localisation`
Section 3	`central metabolite → product, every enzyme cited, unknown steps marked`

Section 4	`hand yield with C, NADH, NADPH, ATP closed separately → model reproduces it → oxygen-constrained rerun`
Section 5	`one clause of justification per tool; transport strategy; the abandonment criterion`
Section 6	`feeding from physiology · mixing and k_L a limits · water per kg · the titer below which it fails`
Section 7	`risks with likelihood, cost and alternative; go/no-go triggers`

A PI reads your proposal and says "this is a nice review of the field, but where is your contribution?" What went wrong?

Almost certainly sections 1 and 3 swallowed the document and section 4 is thin. The failure mode is writing what is known — the pathway, the host's properties, the published titers — at the length it deserves in a thesis introduction rather than in a four-page proposal, so that the reader finds a competent summary and no argument. The contribution in a proposal of this kind is not the pathway, which is assembled from the literature; it is **the analysis**: the closed balance done by hand, the model that reproduces it, the oxygen-constrained yield, the titer the economics demands, and the named risks with their triggers. Those are the parts nobody else's proposal has.

The fix is mechanical. Cut sections 1 and 3 to one page combined, with the enzyme table as an appendix. Expand section 4 until the three numbers — ceiling, constrained prediction, required titer — are each derived in the text and visible in one figure. And add the two sentences that are almost always missing: what result would make you abandon the design, and which assumption, if wrong, changes the conclusion. A reader who can see those two sentences knows they are reading an engineer's proposal rather than a summary.

Three things people get wrong

- ‘The pathway is the proposal.’ The pathway is the easy part and is mostly literature. The analysis is the contribution.
- ‘Quote the model's yield.’ Quote the oxygen-constrained yield and say what constrained it; the unconstrained number is the ceiling, not a prediction.
- ‘Naming a risk weakens it.’ A named risk with an alternative reads as experience; an unnamed one reads as inexperience the moment a reviewer thinks of it.

Self-check — answer on paper first

1. Your model gives a higher yield than your hand calculation. What do you do?

Find the discrepancy before writing anything, because one of the two is wrong and it matters which. The usual causes, in order: the model has a route your hand balance did not consider — often a transhydrogenase or an alternative cofactor specificity that relieves the redox constraint you imposed; the model's biomass reaction is not being used, so carbon that should go to cells is going to product; a loop or an alternative optimum is inflating the exchange flux (part 10 chapter 36); or the hand calculation lumped NADH and NADPH, which is the commonest arithmetic error and the one part 2 chapter 19 warns about. Reconciling them is the exercise, and the reconciliation is worth a paragraph in the proposal because it demonstrates you did both.

2. What makes a host justification defensible rather than a default?

Naming the property that decides it, the alternative you rejected, and what you gave up by rejecting it. "*E. coli* because it is standard" is a default; "*E. coli* because the design needs twenty DBTL iterations and it has the shortest cycle in biology, accepting that the product is retained and will need lysis and that endotoxin removal will be required if this ever goes to a clinical product" is a choice. The test is whether the sentence would still be true if you had chosen differently — if it would, it is not a justification.

3. Which section do you expect to be weakest, and how do you tell?

Section 5, the build-and-test plan, because it is where a list of tools is easiest to write and hardest to justify. Tell by applying the one-clause rule mechanically: go through every tool named and check that a reason follows it that refers to *this* project — the number of variants, the essentiality of the gene, the expected concentration range. Any tool whose clause would be identical in someone else's proposal is a default. Reading aloud finds them, because you will hesitate before the ones you cannot justify.

What to open a book for. Your own paper log, and two recent papers from each target lab — the proposal has to touch what they are actually doing. Everything else you need is in parts 1 to 11 of this module, and the cross-references in the section list above are the route back to each calculation.

Chapter 66 · tracker week 66

Applications: labs, statement of purpose, interviews

Evidence over enthusiasm. Every sentence that would survive unchanged in someone else's application is a sentence doing no work.

Finding the labs

Three sources, in this order. The **editorial board** of *Metabolic Engineering* and of one or two adjacent journals, which is a curated list of the field's active groups. The **last year of titles** in two journals, read as titles only, which takes an hour and shows you what the field is currently doing rather than what it did when the textbooks were written. And the **acknowledgements and affiliations** of the papers already in your log, which is how you find the groups whose work you have been reading without noticing whose it was.

Shortlist eight to ten groups with, for each, one recent paper and one sentence on why it interests you. Then cut to three or four you would actually join — and the cut is the work, because a list of ten is a list of names and a list of three is a decision. What should decide the cut: whether the group's current direction touches your proposal, whether students lead papers there (chapter 62's pattern analysis), whether the techniques you have been learning are the ones they use, and whether you would be glad to spend four years on their questions rather than your own, because that is what happens.

Reading their work properly

For each shortlisted lab, read one recent paper closely and log it. Then two specific outputs. Write **how your proposal touches that lab's current direction** — not the field's, the lab's, in one sentence naming the paper. And note **the one key figure** of each paper, because that is what you will be asked about and because identifying it forces you to decide what the paper's claim was.

The test, from part 6 chapter 61's checkpoint: if you cannot name each lab's recent paper and its key figure, that lab is not on your shortlist yet — it is on a list of names.

The statement of purpose

Five parts, and the discipline is that each must point at something that exists.

Part	What it does	The failure mode
The problem	A concrete problem you want to work on, specific enough to be arguable	"Sustainability" — a topic, not a problem
Why this field now	Why metabolic engineering is the way to solve it, and why the moment is right	A paragraph about the field's importance that any applicant could write
Evidence of preparation	What you have actually done, **named**: the notebooks, the reproduced design, the flux analysis, the proposal	Listing coursework, or claiming interest
Lab fit	Which group, which paper, which direction yours touches	Praise for the group's reputation
Goals	What you would want to be able to do in four years	A career ambition rather than a capability

Then the cut that matters: **remove every sentence that would survive in someone else's application**. "I am fascinated by the potential of synthetic biology to address climate change" survives in ten thousand applications and therefore conveys nothing. "I rebuilt a 40-reaction core model of central metabolism as a linear programme to understand where flux balance analysis's assumptions sit, and found three of my own modelling errors doing it" survives in one.

The bridge from chemical engineering, which is this curriculum's whole premise, should be made in the language of the thing you bring rather than the thing you lack: stoichiometric and energy balances closed explicitly, kinetics and rate laws, reactor design and residence time, transport and the k_La arithmetic, separations, and techno-economic analysis. Part 11 chapter 44's oxygen-bound calculation and part 4 chapter 49's minimum selling price are the two artifacts that demonstrate it best, because almost nobody entering from biology has done either.

Polish and send

Revise the proposal's weakest section — chapter 65 says which it usually is — and tighten the whole to three to five pages. Send it to one person for comment, which is worth more than three rounds of self-editing. Then two short emails to principal investigators, and the instruction is exact: **a real question, not an introduction**.

What distinguishes them: a real question refers to a specific result in a specific paper and asks something that could not be answered by reading it again. "I was struck that your 2026 strain reached X with no change in uptake rate — did the carbon balance close, or is some of the gain recovery of a by-product you had previously lost?" is a question a PI can answer in two lines and will remember. "I am very interested in your work on Y and would like to discuss opportunities" is not, and answering it costs them more effort than it conveys.

The interview drill

Three questions you should be able to answer cold, and what a good answer contains.

"Why metabolic engineering, and why you?" — sixty seconds, containing a concrete problem, the reason metabolic engineering is the way to solve it, and at least one artifact by name. The test, again, is whether it would survive unchanged in someone else's mouth.

"Walk me through a paper you have read." — pick one from the log and give it in the structure of chapter 62's referee report: what was claimed, whether the data support it, what the key figure shows, and the two questions you could not answer from the paper. The last part is what distinguishes a reader from a summariser.

"What would you do differently?" — about your own work, and the honest answer is the one to give. Across this curriculum mine would be: build the model before reading more about the pathway, because the balance tells you what matters; and measure the transport capacity before designing the knockouts, because part 11 chapter 39's export cap made a coupled strain infeasible outright. An answer that names a specific technical decision and what it cost is worth more than a reflection on time management.

And the question you cannot answer, which is chapter 63's skill and will come up: say you do not know, say how you would find out, and mark where the evidence stops.

The final checkpoint

Can you say in sixty seconds why metabolic engineering, and why you?

Your own answer, but it must contain a concrete problem, the reason this field is the way to solve it, and an artifact by name. If it would survive unchanged in someone else's application, it is not yet specific enough.

Can you name three PIs, their recent papers and how your proposal touches each?

If you cannot name each lab's recent paper and its key figure, that lab is on a list of names rather than a shortlist.

Is your repository readable by a stranger in five minutes?

Open it as a stranger would — README first. If it does not say what the project is, what to run and what the result was, fix that before anything else. Then check that it runs from a fresh clone, which is the only real test (part 10 chapter 36).

Does your statement of purpose cite an artifact rather than an ambition?

The test is whether the document points at something that exists — a notebook, a reproduced design, a proposal, a flux analysis — rather than at an intention.

Then the last instruction in the tracker, which is the one I would keep: reread your week 1 notebook page, and write what you will do in the next six months. The point of rereading it is to see how much of what looked difficult in week 1 is now routine, which is the only reliable evidence that the sixty-six weeks worked — and the point of writing the next six months down is that the curriculum ends and the work does not.

Finding labs	`editorial boards · a year of titles · the acknowledgements in your own log`
Shortlist test	`name the paper and its key figure, or it is a list of names`
Statement	`problem · why now · evidence (named artifacts) · lab fit · goals`
The cut	`delete every sentence that would survive in someone else's application`
Emails	`a real question about a specific result, answerable in two lines`
Interview	`why you (60 s, with an artifact) · a paper as a referee report · what you would do differently`

A PI replies to your email with "interesting — what would you actually do in the first six months?" What do you send?

The chapter 64 answer, compressed to a paragraph and a table. A task list for the first six months with durations, the two or three tasks that are waiting rather than working with their lead times, the critical path, and a committed date at the 80th percentile with the median as the internal target. Then the first go/no-go point: the result at month four that would make you stop and test whether the enzymes are active rather than build more strains.

What makes this the right reply is that it answers a question most applicants cannot. A biologist's first-six-months plan is usually a list of experiments in sequence; yours is a schedule with uncertainty, float and a decision point, which is the thing your previous career actually taught you and the thing the proposal's reader cannot get from another candidate. Keep it to half a page, attach the proposal, and offer the repository — and then the conversation is about the project rather than about you, which is where you want it.

Three things people get wrong

- ‘I will explain my enthusiasm for the field.’ Every applicant does. Point at something that exists.
- ‘I will email to introduce myself.’ Introductions cost the reader effort and convey nothing. Ask a real question about a specific result.
- ‘I will avoid mentioning weaknesses in interview.’ The strongest answer to "what would you do differently" is a specific technical decision and what it cost.

What to open a book for. The *Metabolic Engineering* editorial board and recent issues of *ACS Synthetic Biology*, read as lists of names and titles rather than as papers. Everything else this week is your own output: the log, the repository, the proposal and the statement.

Closing notes to part 12 — and to the module

The five weeks in one page

Week 62. The route from submission to acceptance and who decides at each stage; desk rejection's three causes; a referee report whose every major point names what would fix it; the four authorship conditions, all of which must hold; and reading a lab through its publication pattern, which tells you whether students own projects.

Week 63. Context, content, conclusion, recursively, with one idea per paragraph in the first sentence. One message per figure, error bars defined with n. The four-move abstract, then cut by a third. A ten-minute talk is one idea with your own limitation raised first, and the honest answer to an unanswerable question has three parts.

Week 64. The week with the arithmetic. A ten-task first year with a most-likely duration of 26 weeks had a simulated median of 32.8, an 80th percentile of 35.6, and a **1% chance** of meeting the planned date — because durations are right-skewed and paths merge. A task with one week of nominal float was critical in 24% of runs. Halving the fermenter queue was worth three weeks for an email. And the same padding held as one buffer gave 30% on-time delivery against 0% when spread across tasks.

Week 65. Seven sections, each one's conclusion the next one's premise. The contribution is section 4: the hand balance with the cofactors separated, the model that reproduces it, and the oxygen-constrained yield you would actually get in a vessel.

Week 66. Evidence over enthusiasm, artifacts by name, and a real question rather than an introduction.

On the numbers in this part

The schedule simulation is the only computed content here and it was run: ten tasks with triangular durations over 20,000 iterations, with the critical path and criticality index recomputed in each. The task list and its three-point estimates are illustrative — they are a plausible first year, not a measurement — and the Parkinson's-law model in the buffer comparison is an assumption about behaviour rather than something I measured. The qualitative results are robust to the estimates: a plan summed from modal durations is met rarely, merge points add delay no single estimate contains, and distributed padding is consumed. The specific weeks are not.

Costs in week 64 are deliberately described as patterns rather than figures, because synthesis and sequencing prices move fast enough that a number from me would be wrong by the time you read it.

The module, finished

Twelve parts, 66 weeks. Part 1 covered the biology foundations as a spine beside the textbooks; part 2 the modelling and control framework at twice that density; parts 3 and 4 in situ removal, immobilisation, economics, validation, omics and host defences; part 5 the five product classes with a constraint sheet each; part 6 the hosts assessed against those sheets; part 7 assembly, assays and genome engineering; part 8 parts, circuits, libraries and protein design; part 9 integration, sequencing, RNA-seq and the biofoundry; part 10 constraint-based modelling built from first principles; part 11 strain design, machine learning, labelling and the bioreactor; and this part, the profession.

What held it together, if anything did, was a single habit: **compute the number that decides the question, and state the assumption it rests on beside it**. The capture factor and the displaced working volume; the summation theorem and the ceiling it puts on a single enzyme; the 24 usable four-base overhangs; the proteome cost of a slow enzyme; the oxidative branch's half-carbon per NADPH; 0.67 against −0.44 for the same dataset under two splits; a k_La of 1,560 against 500 achievable; and a 1% chance of meeting a plan built from most-likely durations. None of those is hard to compute and each one changes a decision.

Three cautions to carry out of it. The arithmetic here is mine and was run in a container without network access, so where a number came from memory rather than a calculation I marked it, and those are the ones to verify — eQuilibrator for free energies, BioNumbers for physiology, the primary paper for anything attributed. The models are small and their parameters are stated; they are teaching instruments, not descriptions of *E. coli*. And the errors are mine: three in the core model alone, each described where it happened because finding them is the skill the module was trying to teach.

What is left is the part no document can do. Go and run the sessions.

Part 13 — The gaps: separations, deconstruction, enzymology, consortia, measurement and control

Six supplementary chapters covering what an audit of the tracker found missing. They are numbered S1 to S6 rather than by week, because each extends a week that already exists rather than replacing it, and the heading of each says which. The order is mine and differs from the audit's: I put downstream separations first because it is the largest and most defensible gap, and formal control theory last because it serves argument more than it serves the work.

The audit was broadly right and partly wrong. Right about separations, deconstruction, two-substrate kinetics, consortia, untargeted and absolute-quantitative omics, optimal control and genetic code expansion. Wrong that biosafety and dual use are absent — part 9 chapter 32 covers containment, the institutional committee and the Korean LMO Act, and part 5 chapter 53 works the opioid case — and imprecise about bistability, which part 8 chapter 26 and part 7 chapter 24 treat informally; what was actually missing there is the eigenvalue analysis, which chapter S6 supplies. Two items I have left out deliberately: genetic code expansion and quantitative genetics, which are real gaps but are reading rather than calculation, and the Tier 3 list, which is a literature-awareness exercise.

Everything quantitative here was computed, with the parameters printed beside it, and the assumptions that carry the most weight are named at the end of each chapter. Two numbers in chapter S1 and one in S2 are deliberately presented as sensitivity ranges rather than values, because the inputs — resin price and cellulase price per kilogram of protein — move by an order of magnitude between sources and a single figure from me would be worse than a range.

Ch.	Extends	Chapter
S1	wk 44, 49	Downstream processing: choosing and sizing a train
S2	wk 55–57	Biomass deconstruction: pretreatment, saccharification and the severity trade
S3	wk 3	Two-substrate kinetics and allostery: mechanisms, MWC and KNF
S4	wk 9, 26–27	Synthetic consortia: division of labour, and why it does not stay divided
S5	wk 43, 51	Measuring the pools and the proteins: quantitative metabolomics and proteomics
S6	wk 16, 27, 44	Control theory for pathways: stability, bistability, integral feedback, optimal feeding

Chapter S1 · extends weeks 44 and 49

Downstream processing: choosing and sizing a train

The module has told you repeatedly that separation decides whether a titer is worth having. It has never told you how to choose the train or size a column. This chapter does, and the first result is that recovery is a product.

Recovery is multiplicative, and that is the whole design constraint

Every unit operation loses product. Overall recovery is the product of step recoveries, so the number of steps matters more than the quality of any one:

Recovery per step	3 steps	5 steps	8 steps	12 steps
95%	86%	77%	66%	54%
90%	73%	59%	43%	28%
80%	51%	33%	17%	7%

Read it the other way and it becomes a specification: to deliver 90% overall you need 96.5% per step in a three-step train, 97.9% in five and 98.7% in eight. **Shortening the train is almost always worth more than improving a step**, which is the opposite of where effort usually goes, and it is the reason an upstream decision that removes a step — secretion instead of retention (part 6 chapter 59), a product that crystallises, in situ removal that delivers a partially purified stream (part 3 chapter 46) — is worth more than any chromatography optimisation.

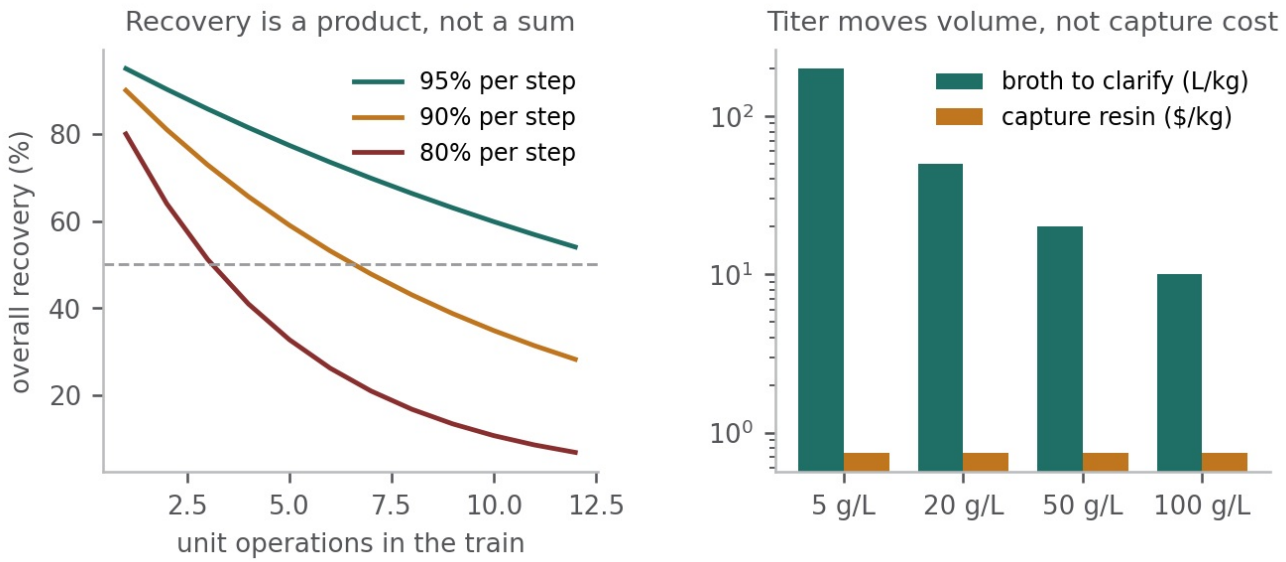


Figure S1.1 Left: overall recovery against train length for three per-step recoveries. Right: what titer does and does not move — the broth volume to clarify scales as 1/titer while the capture resin cost per kilogram does not, for the reason derived below. Computed for this module.

Choosing the train from the product's properties

The sequence is not a matter of taste. Each operation exploits one physical property, so the product's properties select the operations and their order: remove the largest contaminant first, work from high-volume cheap steps to low-volume expensive ones, and never put a capacity-limited step before a volume-reduction step.

Property of the product	What it selects	What it rules out
Secreted, soluble	Clarify, then capture directly from broth	No lysis, no debris removal — two steps saved
Intracellular, soluble	Harvest, lyse, clarify, then capture	Adds the hardest solid–liquid separation in the sequence
Intracellular, insoluble (inclusion body, polymer granule)	Harvest, lyse, wash the solids; for PHA, digest the non-polymer biomass	Chromatography is irrelevant — this is a solids process
Charged at process pH (organic acid, amino acid)	Ion exchange, electrodialysis, reactive extraction, crystallisation of a salt	Solvent extraction of the free species (part 3 chapter 46)

Volatile	Distillation, stripping, pervaporation	Anything requiring the product to stay in solution
Hydrophobic, uncharged	Liquid–liquid extraction, adsorption onto polymer	Ion exchange
Sparingly soluble at some pH or temperature	Crystallisation — the cheapest purification there is	—
Heat-labile	Membranes, chromatography, lyophilisation	Distillation, spray drying at high inlet temperature
A protein needing >99% purity	Capture, intermediate purification, polish — three chromatographic steps	Single-step purification; the purity is not reachable

Sizing the first solid–liquid separation

Cells are small and nearly neutrally buoyant, which is the whole difficulty. Stokes' law for a 1 µm cell with a 100 kg/m³ density difference in water gives a settling velocity of **5.5×10^{-8} m/s — 0.2 mm per hour**. A centrifuge is rated by its sigma factor, the equivalent gravitational settling area that would do the same duty: $\Sigma = Q/v_g$. For 1 m³/h you need **$\Sigma \approx$ 5,100 m²**, and for 10 m³/h, 51,000 m².

A disc-stack centrifuge reaches sigma factors of this order by stacking many short settling paths at high g, which is why it exists. The arithmetic also explains two things the literature mentions and rarely derives: velocity goes as the *square* of diameter, so a 0.5 µm cell needs four times the sigma, which is why some hosts are far harder to harvest than others; and flocculation works by raising the effective diameter, so it buys a quadratic improvement for a small reagent cost. Microfiltration is the alternative, where the duty is set by permeate flux against transmembrane pressure and the enemy is the gel layer, so the design variable is crossflow velocity rather than area.

Sizing a chromatography step, and the cost result that surprised me

A bind-elute step is sized by **dynamic binding capacity** — grams of product bound per litre of resin at the chosen residence time and at a stated breakthrough, conventionally 10%. It is lower than the static capacity, often by half, because mass transfer into the bead limits it; the tool is a breakthrough curve, and the usable figure comes from it rather than from the brochure.

For a 10 m³ batch:

Titer	Product in the batch	Resin at DBC 40 g/L	At DBC 60 g/L	Cycles on a 100 L column at DBC 50
5 g/L	50 kg	1,250 L	833 L	10
20 g/L	200 kg	5,000 L	3,333 L	40
50 g/L	500 kg	12,500 L	8,333 L	100

Either you buy the resin or you cycle a smaller column, and cycling is what everyone does — which turns column volume into campaign time and buffer volume. At ten column volumes of buffer per cycle, the 50 g/L case consumes 100 m³ of buffer, which is itself a utility, a cold room and a waste stream.

Now the result I did not expect until I wrote it down. The resin needed scales with the product mass, and the cost is amortised over the resin's cycle life, so

```
capture cost per kg of product = resin price per litre / (DBC in g/L × cycles the resin survives)
```

and **the titer cancels**. Four cases, as a sensitivity because resin prices span an order of magnitude:

Resin class	Price	DBC	Cycle life	Capture cost per kg
Ion exchange, cheap	$300/L	50 g/L	200	$0.03
Ion exchange, dearer	$1,000/L	50 g/L	200	$0.10
Affinity-like	$3,000/L	40 g/L	100	$0.75
Protein-A class	$8,000/L	40 g/L	200	$1.00

So raising titer does not reduce the capture cost per kilogram at all; what it reduces is the broth volume to clarify, the buffer, the tank farm and the campaign time. The levers on the capture step itself are **cycle life** and **DBC** — each doubling halves the cost — and both are measured, not quoted. Part 4 chapter 49 modelled separation as a cost scaling with 1/titer, which is right for the volume-driven steps and wrong for capture; treat that chapter's separation line as the volumetric part only.

Crystallisation, drying, and the cheap end

Crystallisation is the cheapest purification available and the most underused in fermentation papers: it combines purification and concentration in one step, and its yield follows directly from the solubility curve — cool or acidify from a concentration C_1 to a solubility C_2 and the yield is $(C_1 - C_2)/C_1$. Supersaturation has to be controlled, because the metastable zone decides whether you get crystals you can filter or fines you cannot; seeding at a controlled supersaturation is how that is done. For an organic acid, acidifying below the pK_a to crystallise the free acid is the standard route and is the reason part 3 chapter 45's pH arithmetic matters downstream as well as upstream.

Drying closes the train, and the comparison is thermal: spray drying is cheap and fast and exposes the product to a hot inlet; lyophilisation is gentle and costs roughly an order of magnitude more energy per kilogram of water removed. Which you need is set by heat lability, and it should be decided before the formulation, not after.

Recovery	`overall = Π(step recoveries); 90% over 8 steps needs 98.7% each`
Centrifuge duty	$\Sigma = Q/v_g$; $v_g = \Delta\rho \cdot g \cdot d^2/18\mu = 5.5 \times 10^{-8}$ m/s for a 1 µm cell
Chromatography sizing	`resin volume = mass/DBC; cycles = mass/(DBC × column volume)`
Capture cost	`resin price ÷ (DBC × cycle life) — independent of titer`
Crystallisation yield	$(C_1 - C_2)/C_1$ from the solubility curve
What titer moves	`volume-driven steps only: clarification, buffer, tankage, campaign time`

Your product is a secreted 25 kDa protein at 8 g/L, needed at 98% purity, and heat-labile. Design the train and name its largest cost.

Four steps, and the aim is to keep it to four. **Clarify** by disc-stack centrifugation followed by depth filtration — 125 litres of broth per kilogram of product at this titer, so the duty is set by volume and the sigma arithmetic above. **Capture** by ion exchange, chosen over affinity because the cost table says affinity costs ten to thirty times more per kilogram and 98% does not need it; size it from a measured dynamic binding capacity at your residence time, not the brochure's static figure. **Polish** by a second chromatographic step on an orthogonal property — hydrophobic interaction after ion exchange — since 98% from a single capture step is not reachable. **Formulate** by ultrafiltration/diafiltration into the final buffer and lyophilise, because spray drying is ruled out by heat lability.

Overall recovery at a realistic 90% per step is 66%, which is the number to carry into the techno-economic model — and the reason to ask whether the polish step can be avoided by a better capture. The largest cost is not the resin: at $1,000/L, DBC 50 and 200 cycles it is $0.10/kg. It is the **buffer and the water** — tens of cubic metres per batch, with the cold storage, the WFI-grade preparation if this is a biologic, and the waste — plus the lyophiliser, which is capital-intensive and slow and therefore sets the plant's throughput. That inversion, where the exotic-sounding step is cheap and the utilities dominate, is the usual finding and the reason to cost a train rather than intuit it.

Three things people get wrong

- ‘We will improve the chromatography yield.’ Removing a step is worth more: 90% per step over eight steps is 43% overall.
- ‘Higher titer makes purification cheaper per kilogram.’ It makes the volume-driven steps cheaper. Capture cost per kilogram is resin price ÷ (DBC × cycle life) and does not contain titer.
- ‘Chromatography is how you purify.’ For an intracellular polymer or a crystallisable acid it is irrelevant, and crystallisation purifies and concentrates in one step.

Self-check — answer on paper first

1. A train of six steps recovers 55% overall. Where do you look first?

At the number of steps, then at the worst step. Six steps at 55% implies about 90.5% each if they are equal, which is unremarkable — so the question is whether any step can be deleted, combined with its neighbour, or moved upstream where the volume is larger and the loss cheaper. Only then look for the one step well below the others, which a step-by-step mass balance identifies; and note that a step losing product to a side stream may be recoverable by recycle, which is a different fix from improving the step itself.

2. Derive why the capture cost per kilogram does not depend on titer, and say what that implies for a project plan.

The resin volume needed is the product mass divided by the dynamic binding capacity, so it scales with mass; the cost is the resin price times that volume, amortised over the number of cycles the resin survives; dividing by the mass processed cancels the mass entirely, leaving price ÷ (DBC × cycle life). It implies that a titer campaign should be

justified by volume-driven costs — clarification, buffer, tankage and campaign time — and that the two things worth measuring early for the capture step are the dynamic binding capacity at your real residence time and the resin's lifetime in your real feed, since fouling is what actually ends a resin's life.

3. Why can a 0.5 µm host be much more expensive to harvest than a 1 µm one?

Because settling velocity goes as the square of the diameter, so halving the diameter quarters the velocity and quadruples the required sigma factor — a bigger or slower centrifuge, or a switch to microfiltration where the flux is set by fouling. It is also why flocculation is attractive: raising the effective particle diameter by aggregation buys a quadratic improvement for the cost of a polymer, at the price of a contaminant in the product stream that must then be removed.

What to open a book for. Harrison, Todd, Rudge & Petrides, *Bioseparations Science and Engineering*, which is the standard text and is organised exactly as this chapter is — by property and by step. Doran's chapters on unit operations for the transport side. For chromatography scale-up specifically, a vendor's process-development handbook is more useful than any textbook, and the breakthrough-curve protocol in it is the one measurement this chapter rests on.

Chapter S2 · extends weeks 55 to 57

Biomass deconstruction: pretreatment, saccharification and the severity trade

Part 5 told you which inhibitors to measure in a hydrolysate. It never said where they come from: they are made by the same step that releases the sugar, which makes pretreatment a trade rather than a maximisation.

What is in the biomass, and the sugar ceiling

Lignocellulose is three polymers and some ash. For a representative composition — 40% cellulose, 25% hemicellulose, 22% lignin, 13% ash and extractives — the sugar ceiling follows from hydrolysis adding a water molecule per bond: glucan to glucose multiplies by 180.16/162.14 = 1.111, and xylan to xylose by 150.13/132.12 = 1.136.

```
per kg of dry biomass:  0.444 kg glucose + 0.284 kg xylose = 0.729 kg of fermentable sugar
```

That ceiling matters because it is the number every downstream yield multiplies. At 0.729 kg of sugar and ethanol's 0.511 g/g from part 2 chapter 18, the biomass-to-ethanol ceiling is 0.373 kg/kg — and only if the pentoses are fermented, which is the catabolite-repression problem of part 1 chapter 9. Realistic saccharification and recovery of 80% brings it to 0.583 kg of sugar and 0.298 kg of ethanol; at 65%, to 0.474 and 0.242.

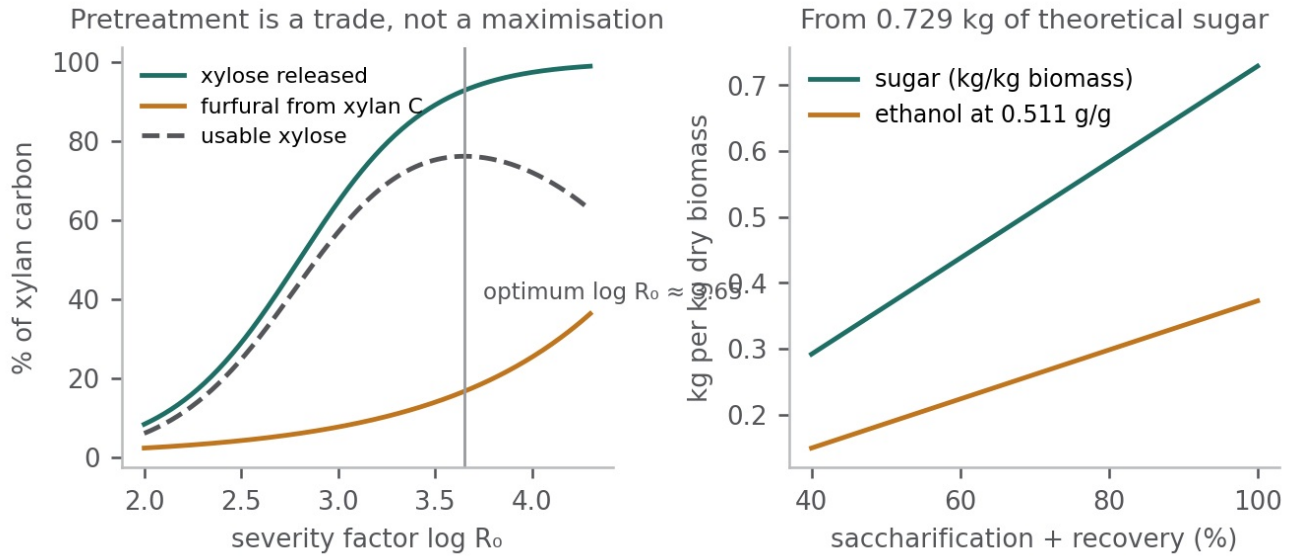


Figure S2.1 Left: the severity trade — xylose released and furfural formed against the severity factor, with the usable difference peaking at log R₀ ≈ 3.5. The two curves are illustrative shapes fitted to the qualitative behaviour, not measured data, and the optimum's position depends entirely on them. Right: sugar and ethanol per kg of dry biomass against the fraction of the 0.729 kg ceiling actually recovered. Computed for this module.

Pretreatment, and the severity factor

Native cellulose is crystalline and sheathed in hemicellulose and lignin, so enzymes cannot reach it: pretreatment exists to raise accessibility. The main routes and what each does:

Pretreatment	What it removes or changes	Cost
Steam explosion	Solubilises hemicellulose, disrupts structure by explosive decompression	Furfural and HMF formation; acetic acid released from hemicellulose
Dilute acid	Hydrolyses hemicellulose to monomeric pentoses	Most degradation products of any route; neutralisation produces gypsum
Alkaline (lime, NaOH)	Removes lignin, swells cellulose	Alkali recovery; lower sugar degradation, so fewer furans
Organosolv	Dissolves lignin into an organic phase, giving a recoverable lignin stream	Solvent recovery and the capital it needs
AFEX (ammonia fibre expansion)	Decrystallises cellulose, leaves the solids nearly dry, few inhibitors	Ammonia recycle

The common currency is the **severity factor**, which collapses temperature and time into one number: log R₀ = log₁₀[t ·

exp((T − 100)/14.75)], with t in minutes and T in °C. It is an Arrhenius-type combination, so a 14.75 °C rise is worth a factor of e in time:

Condition	log R_0	Condition	log R_0
160 °C, 10 min	2.77	190 °C, 2 min	2.95
180 °C, 10 min	3.36	200 °C, 5 min	3.64
210 °C, 3 min	3.72	220 °C, 1 min	3.53

The trade is the chapter's point. Raising severity releases more hemicellulosic sugar and makes cellulose more accessible, and it also degrades the sugar it released: pentoses dehydrate to furfural, hexoses to 5-hydroxymethylfurfural, and the acetyl groups of hemicellulose leave as acetic acid — the four inhibitors part 5 chapter 56 told you to measure. So **the optimum is interior**, and it is set by the ratio of release kinetics to degradation kinetics for your feedstock, not by a number from a paper on a different one. Detoxification — overliming, activated carbon, resins, laccase treatment, or simply evolving a tolerant strain — is the alternative to backing off the severity, and it costs a step in the sense of chapter S1.

Enzymatic saccharification

The cocktail is three activities working in series on an insoluble substrate: **endoglucanases** cut internally and create new chain ends, **cellobiohydrolases** processively release cellobiose from those ends, and **β-glucosidase** hydrolyses cellobiose to glucose — which matters disproportionately because cellobiose inhibits the cellobiohydrolases, so the whole cascade stalls behind a missing β-glucosidase. **Lytic polysaccharide monooxygenases** oxidatively cleave crystalline regions and need a reductant and oxygen, which is why they were missed for decades and why they raised cocktail performance when added. Hemicellulases and accessory esterases remove the sheath.

Three features make the kinetics unlike anything in part 2 chapter 3. The substrate is **insoluble**, so the relevant concentration is accessible surface rather than molarity, and conversion slows as the easy fraction is consumed — the characteristic hyperbolic-with-a-long-tail curve. Enzymes **adsorb** to the substrate, including unproductively to lignin, so part of your dose is lost to a polymer that is not the target. And product inhibition is strong, which is the argument for simultaneous saccharification and fermentation: the yeast consumes glucose as it appears and relieves the inhibition, at the cost of a temperature compromise between the enzymes' optimum near 50 °C and the organism's near 35.

Enzyme cost is the line that decides a cellulosic process, and it is also the number I cannot give you, because the price per kilogram of enzyme protein spans an order of magnitude between sources. As a sensitivity, with 80% saccharification of the composition above:

Loading	$10/kg protein	$50/kg	$200/kg	$500/kg
10 mg/g glucan	$0.13/kg ethanol	$0.67	$2.68	$6.71
20 mg/g glucan	$0.27	$1.34	$5.37	$13.42

Against an ethanol price near $0.60/kg, the table says the process is viable only in its top-left corner — cheap enzyme at low loading — which is exactly why enzyme cost reduction and on-site enzyme production have been the central engineering effort in cellulosic ethanol, and why **consolidated bioprocessing** — one organism that secretes the cocktail and ferments the sugar — is the standing ambition. Its difficulty is the combination: a good cellulase secretor is usually a poor fermenter and vice versa, so CBP is either a cellulolytic organism engineered to ferment or a fermenter engineered to secrete a cocktail, and the second runs into the secretion-capacity limits of part 6 chapter 59.

Sugar ceiling	`1.111 × glucan + 1.136 × xylan; 0.729 kg/kg for 40/25 composition`
Ethanol ceiling	`0.729 × 0.511 = 0.373 kg/kg of dry biomass, pentoses included`
Severity	`log R₀ = log₁₀[t·exp((T−100)/14.75)]; 14.75 °C ≈ a factor of e in time`
The trade	`release and degradation rise together — the optimum is interior`
Cocktail	`endoglucanase + cellobiohydrolase + β-glucosidase + LPMO; cellobiose inhibits`
Enzyme cost	`loading × glucan fraction × price; viable only at low loading and cheap enzyme`

A collaborator proposes raising pretreatment severity from log R_0 3.0 to 3.7 to get more sugar. What do you say, and what would you measure?

That it will indeed release more sugar and that the question is what the extra sugar costs. On the illustrative curves in figure S2.1 the released xylose rises from about 69% to 94% of the xylan while furfural rises from 8% to 18% of the xylan carbon, so the *usable* xylose barely improves and the fermentation inherits more than double the inhibitor load. That load does not appear in the saccharification yield — it appears later as a longer lag, a lower growth rate and a lower final titer, which is where it will be misdiagnosed as a strain problem.

What to measure, in this order: the four inhibitors by HPLC and total phenolics, at both severities, reported as concentrations in the hydrolysate as fed; growth rate and final titer of your actual strain on both hydrolysates, against a synthetic-medium control, which is the only test that matters; and the sugar yield itself, so the trade is quantified rather than assumed. Then the decision is between backing off the severity, detoxifying — which adds a step, with chapter S1's multiplicative penalty — or evolving tolerance, which part 8 chapter 28 says is the method with the better record for a polygenic phenotype like this.

Three things people get wrong

- ‘Maximise sugar release.’ The same step that releases sugar degrades it; the optimum is interior and feedstock-specific.
- ‘The hydrolysate is a sugar solution.’ It is a sugar solution plus furfural, HMF, acetate and phenolics, and the last of those is the least characterised and the most variable.
- ‘Enzyme cost is a detail.’ At a plausible price and loading it exceeds the value of the ethanol it makes.

Self-check — answer on paper first

1. Compute the sugar and ethanol ceiling for a feedstock that is 45% cellulose and 20% hemicellulose.

Glucose 0.45 × 1.111 = 0.500 kg/kg; xylose 0.20 × 1.136 = 0.227; total 0.727 kg of sugar per kg of dry biomass — almost identical to the 40/25 case, because the two polymers convert at nearly the same mass ratio. Ethanol at 0.511 g/g is 0.372 kg/kg if the pentose is fermented, and 0.256 kg/kg if only the glucose is, which is the single largest reason pentose utilisation is a research priority.

2. Why does a missing β-glucosidase stall the whole cocktail rather than just slowing the last step?

Because cellobiose, the product of the cellobiohydrolases, inhibits them. Without β-glucosidase to hydrolyse it to glucose, cellobiose accumulates and shuts down the processive enzymes that are doing most of the work on crystalline cellulose, so the cascade stops well short of conversion even though endoglucanase and cellobiohydrolase are both present and active. It is a product-inhibition cascade of exactly the kind part 2 chapter 3 describes, and it is the reason β-glucosidase is supplemented into commercial cocktails and the reason simultaneous saccharification and fermentation works.

3. What makes consolidated bioprocessing hard, in one sentence, with the module reference?

It requires one organism to be both a high-capacity protein secretor and an efficient fermenter of mixed sugars, and those are different physiologies — a cellulolytic organism engineered to ferment runs into its own tolerance and pathway limits, while a good fermenter engineered to secrete a multi-enzyme cocktail runs into the secretion-capacity and proteome-cost limits of part 6 chapter 59 and part 1 chapter 9.

What to open a book for. A current review of lignocellulose pretreatment for the severity comparisons, and one of cellulase cocktails and LPMOs for the enzymology — this is a field where the reviews are better than the textbooks because the LPMO work is recent. The NREL biomass compositional analysis procedures are the reference for the composition numbers, and your own feedstock's composition is the first thing to measure rather than assume.

Chapter S3 · extends week 3

Two-substrate kinetics and allostery: mechanisms, MWC and KNF

Almost every enzyme you will engineer has two substrates, and single-substrate Michaelis–Menten cannot describe it. The mechanisms are distinguishable by a pattern you can see in two minutes of plotting.

Why one substrate is not enough

Part 2 chapter 3 derived $v = V_{max}[S]/(K_M + [S])$ for $E + S \rightleftharpoons ES \rightarrow E + P$. Now count the enzymes in a pathway you would actually build: every NAD(P)H-dependent oxidoreductase has two substrates, as does every transaminase, every kinase, every ligase and every acyltransferase. For those, a single-substrate fit gives you an *apparent* K_M that depends on the concentration of the other substrate — which is why two groups measuring the same enzyme report different numbers, and why a cofactor-swap campaign (part 2 chapter 19) needs the two-substrate form to be meaningful at all.

Two mechanisms, and the plot that tells them apart

Ternary-complex mechanisms — ordered or random Bi-Bi — require both substrates to be bound at once before chemistry happens. The rate law is

```
v = V[A][B] / (K_ia·K_b + K_b[A] + K_a[B] + [A][B])
```

Ping-pong (substituted-enzyme) mechanisms take the first substrate, release the first product carrying a group on the enzyme, then bind the second — so the two substrates are never on the enzyme together, and the $K_{ia}K_b$ term disappears:

```
v = V[A][B] / (K_b[A] + K_a[B] + [A][B])
```

That one missing term is visible. Plot 1/v against 1/[A] at several fixed [B] and read the slopes. Computed for V = 1, K_a = 0.5, K_b = 0.4, K_{ia} = 0.6:

Fixed [B]	Ternary complex		Ping-pong	
	slope	intercept	slope	intercept
0.2	1.700	3.000	0.500	3.000
0.5	0.980	1.800	0.500	1.800
1.0	0.740	1.400	0.500	1.400
4.0	0.560	1.100	0.500	1.100

Ternary complex gives intersecting lines; ping-pong gives parallel lines. The reason is in the algebra: the slope with respect to 1/[A] is $(K_a + K_{ia}K_b/[B])/V$ for the ternary mechanism and K_a/V for ping-pong, so only the first depends on [B]. Transaminases and many flavin enzymes are ping-pong, because the group is parked on the cofactor between half-reactions; most NAD(P)-dependent dehydrogenases are ordered ternary, with the cofactor binding first.

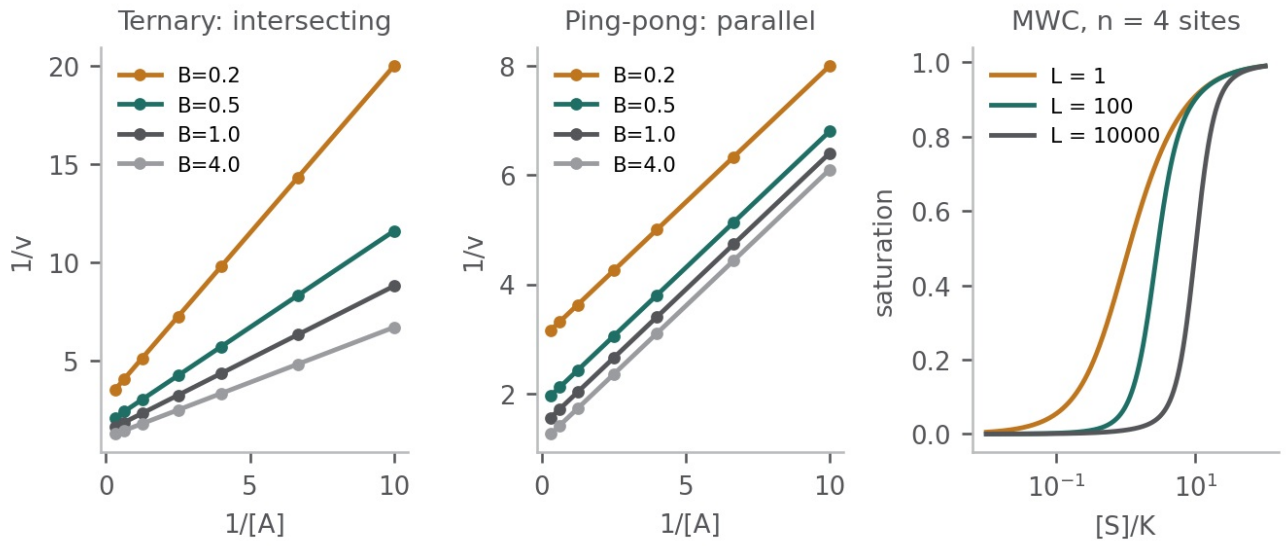


Figure S3.1 Left and centre: double-reciprocal plots for the two mechanisms at the stated parameters — intersecting against parallel. Right: the MWC saturation curve for an enzyme with four sites at three values of the allosteric constant L. Computed for this module; these are exact rate laws rather than fitted data.

Two practical consequences. The **order of a fit matters**: fitting a ping-pong enzyme with a ternary rate law will return a K_{ia} near zero and a large standard error, which is the model telling you which mechanism it is. And the **design question differs**: in an ordered mechanism where the cofactor binds first, a cofactor swap changes the first binding event and therefore the whole catalytic cycle, while in a ping-pong mechanism the two half-reactions can be engineered more nearly independently.

Where the Hill coefficient comes from

Part 2 chapter 3 used the Hill equation as an empirical description of a sigmoid and said n is a measure of cooperativity rather than a number of sites. The two models behind it say why.

MWC (Monod–Wyman–Changeux), the concerted model. The enzyme exists in two conformations, T (tense, low affinity) and R (relaxed, high affinity), and *all subunits switch together* — no hybrid states. Three parameters: n, the number of sites; L, the equilibrium constant [T]/[R] with no ligand bound; and c = K_R/K_T, the affinity ratio. Substrate binds preferentially to R, so each binding event shifts the T⇌R equilibrium toward R, which raises the affinity of the remaining sites. That indirect coupling is the whole mechanism — the sites never touch.

Computed for n = 4 sites and c = 0.01:

L (T/R with no ligand)	Half-saturation at [S]/K	Apparent Hill coefficient
1	1.11	1.12
10	1.59	1.47
100	2.85	2.04
1,000	5.34	2.58
10,000	10.00	2.78

Three things to take from that table. The apparent Hill coefficient **never reaches 4**, the number of sites — it approaches it only in a limit that real enzymes do not occupy, which is the quantitative justification for part 2 chapter 3's warning. Cooperativity requires the protein to *start* mostly in T: at L = 1 the enzyme is effectively non-cooperative. And an allosteric activator or inhibitor works by changing L — an inhibitor raises it, pushing the enzyme into T and both lowering the rate and sharpening the switch, which is why feedback inhibition and cooperativity so often appear on the same enzyme.

KNF (Koshland–Némethy–Filmer), the sequential model. Subunits change conformation one at a time, on binding, and the change alters the interaction energy with neighbours. Because the subunits are allowed to differ, KNF can produce **negative** cooperativity — a Hill coefficient below 1, where each binding event makes the next harder — which MWC structurally cannot. So the experimental signature that distinguishes them is not a better fit to a sigmoid; it is the sign of the cooperativity, with negative cooperativity ruling MWC out.

For engineering, the useful consequence is the one part 2 chapter 3 already used without derivation: a **feedback-**

resistant variant is a mutation at the allosteric site that stops the effector binding, so L no longer responds to the end product. The enzyme keeps its catalytic site and loses its regulation, which is why the usual outcome is full activity in the presence of the product and why those variants are the single most effective change in amino-acid strains.

Ternary complex	$v = V[A][B]/(K_{ia}K_b + K_b[A] + K_a[B] + [A][B])$
Ping-pong	$v = V[A][B]/(K_b[A] + K_a[B] + [A][B])$
Diagnosis	1/v against 1/[A] at several [B]: intersecting = ternary, parallel = ping-pong
MWC	all subunits switch together; parameters n, L, c; apparent Hill < n always
KNF	sequential; permits negative cooperativity, which MWC cannot
Feedback resistance	a mutation that stops the effector changing L

You measure an NADPH-dependent reductase and want to swap it to NADH. What do you measure, and in what form?

Not a single apparent K_M. Measure the full two-substrate surface — initial rates over a grid of NADPH and ketone concentrations — and fit the ternary rate law, since a dehydrogenase of this class is almost certainly ordered with the cofactor binding first. That gives V, K_a, K_b and K_{ia}, and the double-reciprocal pattern confirms the mechanism before you trust the parameters. Then repeat with NADH, and report the **specificity constants k_{cat}/K_M for each cofactor**, because the swap is a change in their ratio and nothing else is comparable between variants.

Two things the two-substrate form tells you that the single-substrate one hides. If the cofactor binds first in an ordered mechanism, its K_{ia} — the dissociation constant from the binary complex — is the quantity your pocket redesign is actually changing, and it is invisible in an apparent K_M. And because the apparent K_M for the ketone depends on the cofactor concentration, a variant that looks improved at a saturating cofactor concentration may be worse at the intracellular one — which, with part 1 chapter 4's point that the two cofactor pools sit at very different ratios, is the difference between a variant that works in a cuvette and one that works in a cell.

Three things people get wrong

- 'The K_M of the dehydrogenase is 40 µM.' For which cofactor concentration? A two-substrate enzyme has no single K_M.
- 'The Hill coefficient is 2, so there are two sites.' MWC with four sites gives apparent Hill coefficients from 1.1 to 2.8 depending on L alone.
- 'MWC and KNF are interchangeable descriptions.' Negative cooperativity is possible in KNF and impossible in MWC, which is the experiment that separates them.

Self-check — answer on paper first

1. Your double-reciprocal plots at four fixed [B] are parallel. What mechanism, and what does it predict about inhibition?

Ping-pong: the slope with respect to 1/[A] is K_a/V and contains no [B], which is what parallel lines mean. It predicts that the first product is released before the second substrate binds, so that product should be a *competitive* inhibitor with respect to the first substrate — and testing that is the confirmation, because the parallel-line pattern alone can also arise from a ternary mechanism with a very small K_{ia}. It also means the enzyme passes through a covalently modified intermediate, which is often assayable in its own right.

2. Why can an MWC enzyme with six sites never show a Hill coefficient of six?

Because the apparent Hill coefficient measures the steepness of the observed curve, and the observed curve is a weighted mixture of the T and R binding curves rather than a pure all-or-nothing transition. Only in the limit of infinite L and infinitesimal c — the enzyme starting entirely in T and R binding infinitely more tightly — does the mixture approach a step with slope equal to the number of sites; at any finite L a fraction of the enzyme is already in R before ligand arrives, and that fraction binds non-cooperatively and flattens the curve. The computed table shows the approach: n = 4 gives 2.78 at L = 10,000.

3. A feedback-resistant variant has the same k_{cat} and no inhibition by the end product. In MWC terms, what changed?

The effector can no longer bind, or can no longer couple its binding to the conformational equilibrium, so the end product stops raising L. The enzyme's catalytic site and therefore its k_{cat} are untouched, which is exactly what you want and exactly what makes these variants effective; what is lost is the regulation that protected the cell from accumulating its own end product, which is the cost part 2 chapter 3 names.

What to open a book for. Cornish-Bowden, *Fundamentals of Enzyme Kinetics*, chapters on multi-substrate mechanisms and on cooperativity — this is the book the module has cited four times and this is the chapter where you actually need it. Segel's *Enzyme Kinetics* if you want every mechanism derived. For the allosteric models, the original MWC paper (J. Mol. Biol., 1965) is short, readable and still the clearest statement of the concerted idea.

Chapter S4 · extends weeks 9, 26 and 27

Synthetic consortia: division of labour, and why it does not stay divided

Splitting a burdensome pathway across two strains is a direct answer to the proteome problem of week 9. It is also an ecology problem, and the ecology has a cheater in it.

The argument, quantified

Part 1 chapter 9 established that expressing a pathway costs proteome and that growth falls roughly in proportion to the fraction diverted. If a pathway costs a fraction φ of the proteome and you split it across two strains, each carries φ/2 and each grows faster:

Total pathway proteome cost	One strain, μ	Split across two, μ each	Gain
10%	0.540 h^{-1}	0.570	+6%
20%	0.480	0.540	+13%
30%	0.420	0.510	+21%

The gain is real and modest, and it is largest exactly where the single-strain route hurts most. Three other arguments usually matter more than the growth number. **Incompatible conditions**: a pathway needing an anaerobic step and an aerobic one, or two enzymes with pH optima three units apart, can be split so each half runs where it works. **Compartment-free compartmentalisation**: a toxic intermediate can be made in one strain and consumed in another, which is part 2 chapter 19's problem solved by ecology rather than by a membrane. And **modularity**: each strain is a smaller construct, so the design–build–test cycle per module is shorter, which by part 2 chapter 11's arithmetic is the highest-value property a project can have.

The cost is the interface. The intermediate must leave one cell and enter the other, which is part 2 chapter 14's transport problem twice over — and if it is charged, it needs an exporter and an importer, not diffusion.

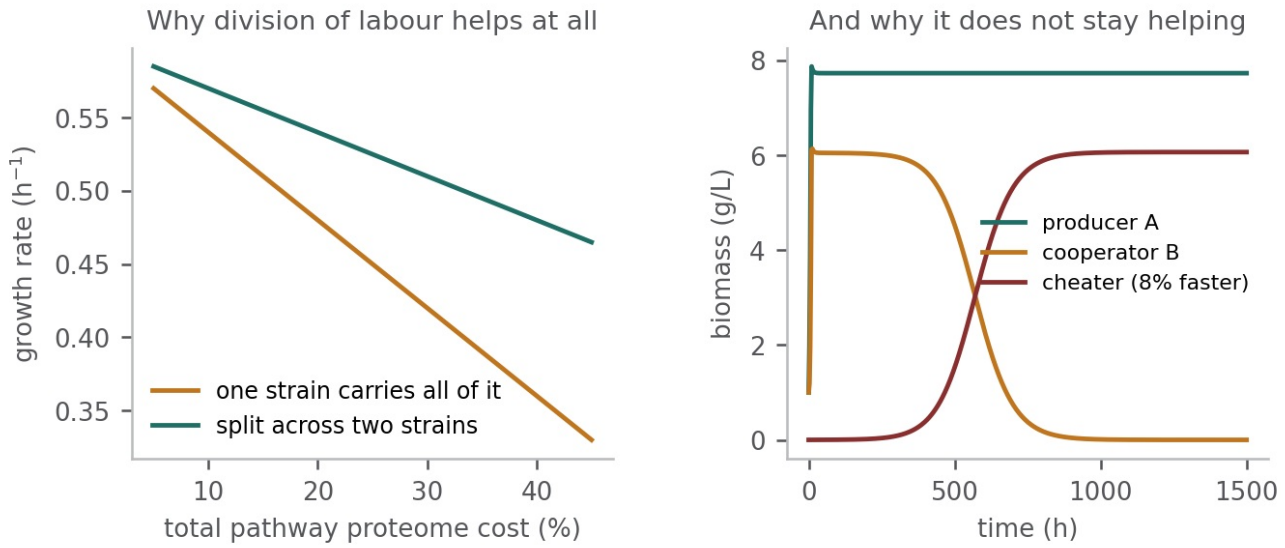


Figure S4.1 Left: growth rate against total pathway proteome cost, carried by one strain or split across two. Right: a chemostat simulation of obligate cross-feeding invaded by a cheater that stops making its own costly enzyme and grows 8% faster. Computed for this module from the model in the text; the 8% advantage and the burden slope are assumptions.

Keeping the ratio where you want it

The population ratio is not a design parameter you set — it is a steady state the ecology chooses. For obligate cross-feeding, where strain A makes an intermediate that strain B needs, a chemostat model makes this concrete:

```
dS/dt = D(S₀ − S) − r_A·A/Y_A
dI/dt = −D·I + α·r_A·A − r_B·B/Y_B          I is the shared intermediate
dA/dt = (r_A − D)·A        dB/dt = (r_B − D)·B
```

At D = 0.2 h^{-1} with the parameters in the code, the steady state is A = 7.73 and B = 6.05 g/L, a ratio of **1.28** — set by the yields and the intermediate's stoichiometric coefficient α, not by the inoculum. Changing the dilution rate barely moves it (1.25 at D = 0.05, 1.28 at 0.2) until washout approaches, where it climbs to 2.56 at D = 0.45 because the slower partner is washing out first.

Two readings. The ratio is **self-correcting** under obligate mutualism, which is the attractive property: if B falls behind, the intermediate accumulates, B grows faster and the ratio restores. And the **washout condition is set by the slower partner**, so the consortium's operating window is narrower than either strain's alone — a point to check before proposing continuous operation.

The cheater, which is the real objection

Mutualism is only stable if cooperating is not optional. Add to the same chemostat a variant of B that has lost its costly contribution — a leaky auxotroph complementation, a deleted export enzyme, a silenced pathway module — and grows 8% faster as a result:

Time	Producer A	Cooperator B	Cheater	Cheater share of the B-type population
0 h	1.00	1.000	0.0001	0%
200 h	7.73	6.036	0.017	<1%
500 h	7.73	4.533	1.525	25%
1,000 h	7.73	0.010	6.059	100%

From one part in ten thousand to complete takeover in about a thousand hours — forty days, which is inside a long continuous campaign and well inside a scale-up seed train plus production run. An 8% advantage is not large; it is the same order as the burden numbers above, which is the uncomfortable symmetry: **the thing that made division of labour worth doing is exactly what makes cheating profitable**.

What stops it, and what each costs:

- **Obligate mutual dependency.** Make each strain auxotrophic for something only the other supplies, so a cheater cannot grow at all rather than merely paying less. This is the only defence that is structural rather than quantitative, and it is why engineered consortia are usually built from reciprocal auxotrophs. It costs a growth penalty on both strains and makes the medium a design parameter.
- **Growth coupling within each strain**, as in part 11 chapter 39: if the contribution is on the path to the strain's own growth, losing it is not an advantage.
- **Spatial structure.** Biofilms and droplets keep a cooperator's product near its relatives, so the benefit is not freely shared; microfluidic droplet compartments do this deliberately. It costs the homogeneity a stirred tank provides.
- **Short campaigns and fresh inocula**, which is the honest industrial answer: run batches short enough that the cheater cannot sweep, and sequence the end of the run — part 7 chapter 32's instability argument, now with a second mechanism.

Designing one, and the vocabulary reviewers expect

The terms to use correctly: **division of labour** for splitting a function; **cross-feeding** for one strain's metabolite supporting another, **obligate** if the recipient cannot grow without it and **facultative** otherwise; **mutualism**, **commensalism** and **parasitism** for who benefits; **auxotroph complementation** for the reciprocal-dependency design; **cheater** or **defector** for the variant that takes the benefit without paying; and **coexistence** as the property to demonstrate rather than assume. The ecological point reviewers will test is whether you know that a consortium is a population with its own dynamics, not a pair of strains in the same vessel.

The minimum experiment that demonstrates one works: the population ratio over time by a strain-specific marker — differential plating or qPCR on a unique locus, not optical density, which cannot distinguish them; the titer against the best single-strain control, since a consortium that only matches it has bought nothing; and the ratio and titer after a deliberately long run or a serial-transfer experiment, which is the only test for a cheater sweep. Reporting the first two and not the third is the commonest gap in consortium papers.

Burden split	`μ(φ) → μ(φ/2) per strain; +13% growth at φ = 20%`
Cross-feeding steady state	`ratio set by yields and the stoichiometric coefficient, not the inoculum — 1.28 here`
Washout	`set by the slower partner; the window is narrower than either strain's`
Cheater sweep	`an 8% advantage goes from 10⁻⁴ to fixation in about 1,000 h`

The only structural defence	`reciprocal auxotrophy — a cheater cannot grow, rather than merely paying less`

A proposal splits a twelve-enzyme pathway across three strains to reduce burden. What are your three questions?

What crosses between them, and how? Two interfaces for three strains, each needing an intermediate to leave one cell and enter another — and if either intermediate is charged, that is two exporters and two importers to find or engineer, which is part 3 chapter 45's screening problem four times over. A proposal that does not name the transporters has not costed the interface.

What stops the cheater? With three strains there are three ways to defect and the simulation above says an 8% advantage fixes in about a thousand hours. If the answer is not reciprocal auxotrophy or growth coupling, the answer is a short batch and sequencing at harvest, which should be stated.

What is the control? The comparison is against the best single-strain strain, tuned, not against an untuned one — and against a two-strain split, because the burden gain from splitting is sublinear while the number of interfaces grows linearly, so three strains may well be worse than two. I would also ask for the ratio measurement method up front, since optical density cannot see it.

Three things people get wrong

- ‘We set the population ratio at inoculation.’ The ratio is a steady state the ecology chooses; the inoculum sets only how long it takes to get there.
- ‘The consortium is stable because it is mutualistic.’ Mutualism is stable against extinction and not against cheating, and cheating is profitable in proportion to the burden you were trying to avoid.
- ‘Splitting reduces burden, so it increases titer.’ It increases growth rate per strain; whether titer follows depends on the interface, and the control is a tuned single strain.

Self-check — answer on paper first

1. Why is reciprocal auxotrophy qualitatively better than growth coupling for consortium stability?

Because it changes the cheater's payoff from a quantitative penalty to zero growth. Under growth coupling the defector still grows, just less well, so the defence depends on the coupling being tight enough that the advantage is negative — a quantitative claim that a suppressor mutation can undo. Under reciprocal auxotrophy the cheater lacks something it cannot make and no partner will give it, so it cannot grow at all in that medium. The cost is that both strains grow more slowly than prototrophs and the medium must not contain the complemented nutrients, which constrains the feed.

2. Your two-strain consortium drifts from 1:1 to 4:1 over a fermentation. Is that a failure?

Not necessarily — the ratio is a steady state set by the yields and the intermediate stoichiometry, so a drift away from the inoculated ratio toward the model's prediction is the system working as designed. It is a failure if the drift continues past the predicted steady state, which points to a cheater sweep or to one strain losing its construct, and the test distinguishes them: plate and screen for the phenotype, or sequence the locus. The thing not to do is re-inoculate to 1:1 and call it fixed.

3. What is the minimum evidence that a consortium beats a single strain?

Titer and yield against a *tuned* single-strain control carrying the whole pathway, run in the same vessel and medium; the population ratio over time by a strain-specific marker, showing it reaches and holds its steady state; and the same two measurements after a long run or serial transfer, which is the only test of whether the advantage survives the ecology. Without the third, a consortium paper reports a result that may not exist at the end of a campaign.

What to open a book for. A recent review of synthetic microbial consortia for the design patterns and the vocabulary; the reciprocal-auxotrophy literature for the stability argument; and, if you want the dynamics properly, any mathematical-ecology text's treatment of chemostat competition — the equations are the ones in this chapter and the analysis is a century old.

Chapter S5 · extends weeks 43 and 51

Measuring the pools and the proteins: quantitative metabolomics and proteomics

Part 2's thermodynamics and part 4's enzyme constraints are only as good as the concentrations and abundances fed into them. This chapter is about where those numbers come from and how wrong they are.

Quenching: the measurement that is over before you start

Part 2 chapter 13 argued that metabolite pools are small and turn over fast, which is what justifies the pseudo-steady state. Read it backwards and it is a measurement problem: a pool's turnover time is its size divided by the flux through it, and anything slower than that changes the pool while you are sampling it.

Pool	Flux	Turnover time	Fraction turned over in a 1 s quench
0.1 mM	1.0 mM/s	0.10 s	essentially all of it
0.5 mM	2.0 mM/s	0.25 s	essentially all of it
2.0 mM	1.0 mM/s	2.0 s	about 40%
10 mM	0.5 mM/s	20 s	about 5%

So for central-metabolism intermediates, **sampling must be faster than a second**, which is why the standard method is a quench directly into cold methanol at −40 °C and why rapid-sampling devices exist on fermenters. The two artefacts to know, because they pull in opposite directions: cold methanol **leaks** small polar metabolites out of the cells, so a separated cell pellet under-reports them; and whole-broth extraction avoids the leakage and then measures intracellular and extracellular pools together, which for a secreted product is meaningless. There is no method without an artefact — only a method whose artefact you have quantified, which is the same conclusion as part 4 chapter 50's extraction discussion and is worth stating in a methods section rather than hiding.

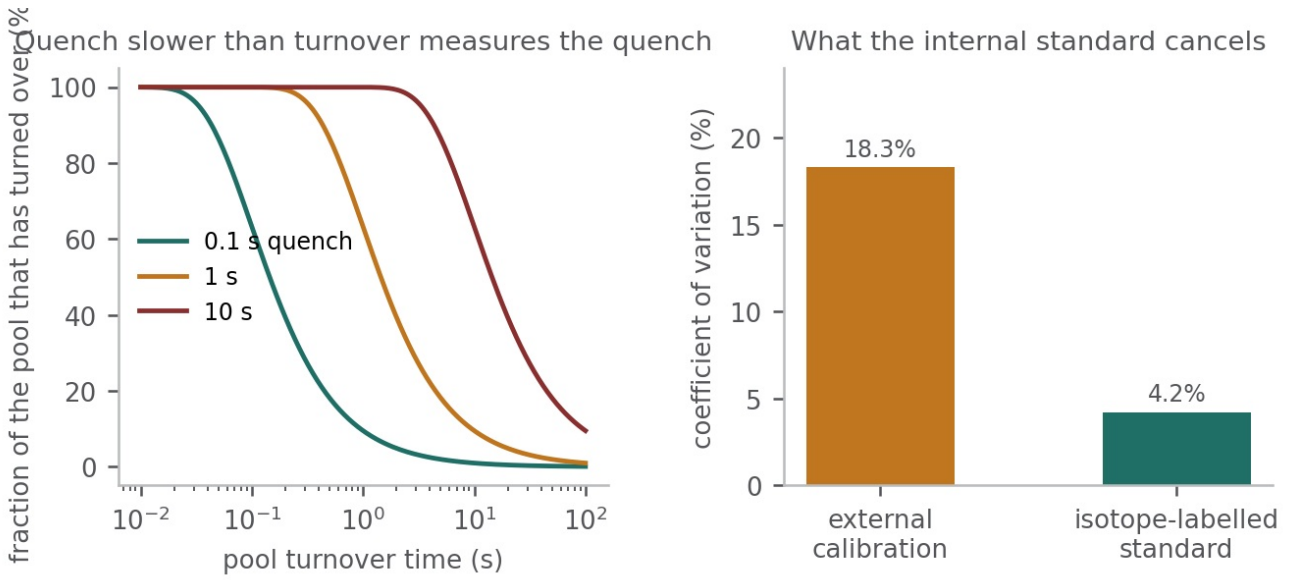


Figure S5.1 Left: the fraction of a pool that turns over during a quench of 0.1, 1 and 10 seconds, against the pool's turnover time. Right: the coefficient of variation of a quantification with external calibration alone and with an isotope-labelled internal standard, from the error model in the text. Computed for this module.

What the internal standard actually buys

Part 4 chapter 50 said an isotope-labelled internal standard is mandatory for mass-spectrometric quantification. Here is the arithmetic. Model four independent error sources — extraction recovery at 8%, ionisation suppression at 15%, injection volume at 5% and detector response at 3% — and compare a quantification against an external calibration curve with one against a co-eluting labelled standard that experiences extraction, suppression and injection identically:

```
external calibration only  : CV 18.3%
isotope-labelled standard  : CV  4.2%
```

A factor of four, and it comes from cancellation rather than from precision: the labelled standard shares every error that acts on both species, so the ratio retains only what differs between them. That is also the limit of the method — it cannot cancel an error the standard does not share, such as incomplete quenching, degradation before the standard is added, or a matrix effect that differs across the chromatographic peak. **Add the standard as early in the workflow as possible**, ideally into the quenching solvent, because every step before it is unprotected.

Targeted against untargeted, and identification confidence

	Targeted (MRM on a triple quadrupole)	Untargeted (high-resolution MS)
What it measures	A defined list, typically tens to a few hundred compounds	Thousands of features, most unidentified
Quantification	Absolute, with labelled standards per analyte	Relative, and semi-quantitative at best
Use	Feeding a model: thermodynamics, kinetics, flux fitting	Hypothesis generation; finding the unexpected accumulation
Main risk	You only see what you looked for	Identification, and the statistics below

Identification is the part that engineers under-rate. A feature is a mass and a retention time, not a compound, and the community's confidence levels run from a mass match alone — the weakest, and the most commonly over-claimed — through a spectral-library match, to confirmation against an authentic standard run on the same instrument, which is the only level at which "we detected X" is a measurement. **A pathway intermediate reported from an untargeted run without a standard should be treated as a candidate**, and confirming it is a day's work with a purchased standard.

The statistics of many metabolites and few samples

Simulate 200 metabolites with four replicates, where 20 have truly changed by 0.8 in $\log_2$ and the measurement CV is 35%:

Standard error of $\log_2$ fold change	Raw $p < 0.05$	False among them	BH $q < 0.05$	False	True changes found
0.36	18 calls	9	3 calls	0	3 of 20

Half the raw calls are false, and correcting properly leaves three of the twenty real changes — this is part 2 chapter 17 and part 9 chapter 33 in a new dataset, and the conclusion is the same: **four replicates of 200 metabolites is a survey, not a test**. Two further habits specific to metabolomics: normalise to a biomass measure rather than to a volume, since the pool concentration is per cell and dry weight is the denominator your balances need; and treat the data as compositional if the normalisation is to a total, because a large change in one abundant metabolite moves every other one, which is part 9 chapter 33's composition argument applied to pools rather than transcripts.

Proteomics, and the uncertainty it feeds into GECKO

Part 4 chapter 51 used $v \le k_{cat} \cdot [E]$ as a constraint and part 11 chapter 39 built a proteome-budget model on it. Both need absolute protein abundances, so here is how they are obtained.

Bottom-up is the default: digest the proteome with trypsin, separate the peptides, and infer proteins from the peptides identified — which carries the inference problem that a peptide shared between homologues cannot be assigned. The quantification options, in increasing order of effort: **label-free**, comparing intensities between runs, cheapest and least precise; **isobaric labelling** (TMT, iTRAQ), multiplexing samples into one run with good precision and a known ratio-compression bias; **data-independent acquisition** (DIA/SWATH), fragmenting everything in windows and quantifying from a library, which gives better reproducibility across many samples; and **absolute quantification** — AQUA peptides or QconCAT constructs, which spike known amounts of labelled peptide and are the only way to get molecules per cell rather than a ratio.

That last distinction is the one that matters for a model. An enzyme-constrained model needs **absolute** abundance, and most published proteomics is relative, so the usual practice is to anchor a relative dataset to a few absolutely quantified proteins or to a total protein mass — and the resulting uncertainty propagates straight into every flux bound. Taken together with part 4 chapter 51's warning about k_{cat} values:

```
v ≤ k_cat·[E]:   a k_cat wrong by 10× makes the bound wrong by 10×
                 an abundance with a 25% CV gives a ±25% band on every bound
```

Since the two multiply and the k_{cat} error dominates, an enzyme-constrained model's bounds are good to a factor of a few at best. That is not a reason to avoid them — a bound good to a factor of three still rejects a design needing 28% of the proteome — but it is a reason to report the model's conclusions as what they are: statements about what is impossible, with an uncertainty set by the worst k_{cat} in the pathway.

Quench speed	must beat the pool's turnover time, pool/flux — under a second for central metabolism
Internal standard	CV 18.3% → 4.2%; cancels only errors it shares, so add it into the quench
Identification	mass match < library match < authentic standard; only the last is a measurement
200 metabolites, n = 4	9 of 18 raw calls false; BH leaves 3 of 20 real changes found
Proteomics for models	absolute quantification (AQUA/QconCAT), not relative, or the bound has no units
Bound uncertainty	k_cat error × abundance error; the first dominates and is often 10×

You need intracellular concentrations for the max-min driving force calculation of part 2 chapter 14. What do you actually do?

Targeted, with labelled standards, on the dozen or so intermediates the pathway contains — not an untargeted run, because what you need is absolute concentrations for a specific list and untargeted gives relative intensities for thousands of features. The sequence: rapid sampling into cold methanol at −40 °C with the labelled standards already in the quenching solvent; a decision about whole-broth against separated cells, made and stated, with the extracellular concentration measured separately so the intracellular can be inferred if you chose whole broth; extraction, with recovery measured per compound by a second spike; and LC-MS/MS in MRM mode against a calibration in matrix.

Then the two conversions people forget. Concentrations come out per gram of dry weight or per millilitre of extract, and $\Delta G'$ needs **molar concentrations in the cytoplasm**, so you need the cell volume per gram of dry weight from sheet 2 of the lookup appendix — an assumption that carries directly into every $\Delta G'$. And cofactor ratios, not absolute cofactor concentrations, are what enter Q for a redox reaction, so NAD^+/NADH and $NADP^+$/NADPH must be measured in the same sample, which is harder than it sounds because the reduced forms degrade quickly at low pH and the oxidised forms at high pH — the usual answer being two extractions in parallel.

Three things people get wrong

- ‘We quenched quickly.’ A 0.1 mM pool at 1 mM/s turns over in 0.1 s; anything slower measured the quench, not the cell.
- ‘The feature at m/z 259.02 is the intermediate.’ A mass is a candidate. Only an authentic standard on the same instrument makes it a measurement.
- ‘Our proteomics gives the enzyme abundances for the model.’ Relative abundance has no units; a flux bound needs absolute, and the k_{cat} is still the dominant error.

Self-check — answer on paper first

1. Why does an isotope-labelled internal standard reduce the CV fourfold, and what does it fail to cancel?

Because it is chemically identical to the analyte and therefore experiences the same extraction recovery, the same ionisation suppression and the same injection variability, so taking the ratio removes every error the two share and leaves only what differs — essentially the detector's own noise on two nearby masses. It fails to cancel anything that happens before it is added, which is why it goes into the quenching solvent, and anything that affects the two species differently: incomplete quenching, degradation of the analyte by a route the standard does not take, and isotope effects on chromatography, which can separate the labelled and unlabelled peaks enough that they see different matrix.

2. A collaborator reports 40 significantly changed metabolites from 200 measured, n = 3, raw p < 0.05. What do you say?

That about ten of them are expected by chance from the nulls alone — 5% of roughly 180 unchanged metabolites — and that the list has had no multiple-testing correction. Ask for Benjamini–Hochberg-adjusted values and expect the list to shrink to a handful, as the simulation in this chapter did from 18 to 3. Then ask two metabolomics-specific questions: what the normalisation was, since normalising to a total makes the data compositional and a large shift in one abundant metabolite moves all the others; and whether the identifications are standard-confirmed, since an unidentified feature cannot be interpreted however significant it is.

3. What is the uncertainty on a flux bound in an enzyme-constrained model, and how should it be reported?

The product of the k_{cat} uncertainty and the abundance uncertainty, dominated by the first: literature k_{cat} values are frequently wrong by an order of magnitude for the organism and conditions you care about, while absolute abundance might carry 25%. So a bound is good to a factor of a few. Report it as a bound with its assumed k_{cat} and source named, state the conclusion as what the model excludes rather than predicts, and show the sensitivity — if the conclusion survives a tenfold change in the worst k_{cat}, say so, because that is the statement a reader can rely on.

What to open a book for. A review of microbial metabolomics sample preparation — the quenching and extraction comparison is the content, and the reviews are better than any textbook because the artefact comparisons are experimental. For proteomics, a bottom-up methods review covering label-free, TMT and DIA, and the AQUA/QconCAT papers for absolute quantification. The metabolomics community's identification-confidence levels are worth reading in the original short note that defined them.

Chapter S6 · extends weeks 16, 27 and 44

Control theory for pathways: stability, bistability, integral feedback, optimal feeding

Part 2 integrated models and part 7 raised loop stability without analysing it. The analysis is two lines of linear algebra, and it answers questions the simulation cannot: whether a design oscillates, whether it switches, and whether it can reject a disturbance perfectly.

Stability from the Jacobian

Integrating a model tells you what it does from one initial condition. Linearising it tells you what it does near a steady state, for every initial condition. Take the three-enzyme pathway of part 2 chapter 16, compute the Jacobian J = ∂f/∂c numerically at the steady state, and read its eigenvalues: **negative real parts mean stable, a positive real part means unstable, and a non-zero imaginary part means the approach oscillates**. The real part's magnitude is the decay rate, so the slowest eigenvalue sets the settling time — which is the formal version of part 8 chapter 26's observation that a loop is as fast as its actuator.

Feedback cooperativity n	Eigenvalue real parts (h⁻¹)	Oscillatory?	Reading
1	−0.144, −0.369, −0.920	no	Overdamped — monotone return
2	−0.268, −0.268, −1.009	yes	Damped oscillation; fastest settling of the set
4	−0.257, −0.257, −1.176	yes	Still stable
8	−0.210, −0.210, −1.407	yes	Less damped
16	−0.123, −0.123, −1.677	yes	Approaching the imaginary axis

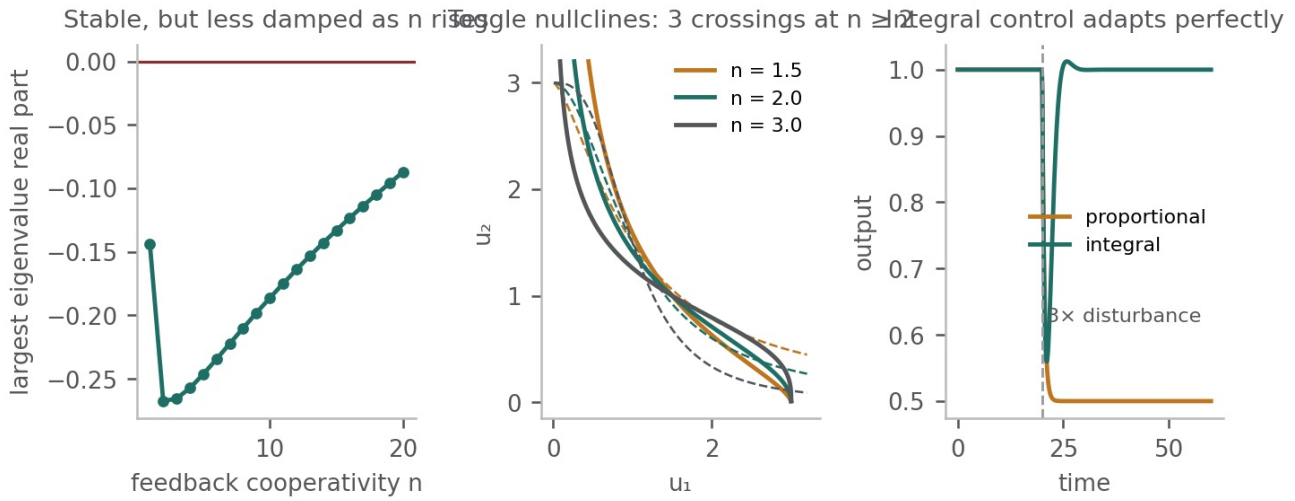


Figure S6.1 Left: the largest eigenvalue real part against feedback cooperativity for the part 2 chapter 16 pathway — stable throughout, but increasingly lightly damped. Centre: toggle-switch nullclines, which cross once at n = 1.5 and three times at n ≥ 2. Right: proportional against integral feedback under a threefold disturbance. Computed for this module.

The pattern is the one control engineers expect and biologists usually discover empirically: **sharper feedback makes a loop faster to respond and less damped**, and past some gain it would oscillate outright. This model stays stable because its feedback acts instantaneously; real circuits add the delay of transcription, translation and dilution, and delay is what converts light damping into sustained oscillation. That is the formal reason part 7 chapter 24 insisted a dynamic controller's gain and delay be considered before building it — and the measurement that tests it is the disturbance response, not the titer.

Bistability, and the condition for a switch

The toggle switch — two repressors, each repressing the other — is the canonical bistable circuit:

```
du₁/dt = α₁/(1 + u₂ⁿ) − u₁        du₂/dt = α₂/(1 + u₁ᵐ) − u₂
```

Its steady states are where the nullclines cross, and the number of crossings is the question. With α = 3 for both:

Cooperativity n = m	Steady states found	Values
1.0	1	(1.303, 1.303) — a single symmetric state
1.5	1	(1.251, 1.251)
2.0	3	(0.382, 2.618), (1.213, 1.213) unstable, (2.618, 0.382)
3.0	3	(0.108, 2.996), (1.164, 1.164) unstable, (2.996, 0.108)

Bistability appears between n = 1.5 and n = 2, and that threshold is the design rule: a switch requires cooperativity, because without it the nullclines are too shallow to cross three times however strong the repression. The middle state is a saddle — unstable, and the boundary between the two basins of attraction — so a perturbation that pushes the system past it flips the switch permanently, which is what memory means in a circuit. The transition as n crosses the threshold is a **pitchfork bifurcation**, and the vocabulary matters because "the circuit becomes bistable above a cooperativity of about 2" is a statement a reviewer can check, while "the circuit shows switching behaviour" is not.

Oscillators follow the same analysis with a different answer: a repressilator or a delayed negative-feedback loop loses stability through a **Hopf bifurcation**, where a complex eigenvalue pair crosses into the right half plane and a limit cycle appears. Glycolytic oscillations in yeast are the natural example, and the engineering relevance is that a dynamic metabolic controller with too much gain and a generation-time delay sits uncomfortably close to that boundary.

Integral feedback and perfect adaptation

Here is the result worth taking from control theory into circuit design. A proportional controller — one whose action is proportional to the error — leaves a steady-state offset: the error is what drives the action, so it cannot be zero. An **integral** controller acts on the accumulated error, so the only steady state it admits is zero error, whatever the disturbance. Simulating both under a threefold step disturbance:

Controller	Output before	Output after a 3× disturbance	Steady-state error
Proportional	1.000	0.500	**50%**
Integral	1.000	1.000	**0%**

This is **robust perfect adaptation**, and the robustness is structural: it does not depend on the parameter values, only on the integral being there. Biology implements it with an **antithetic** motif — two species that annihilate each other, so that their difference integrates the error — and bacterial chemotaxis is the textbook natural example, where the methylation state integrates the receptor occupancy and the cell returns exactly to its baseline tumbling frequency at any background attractant concentration.

For a pathway, the consequence is the design target part 7 chapter 24 named without the theory: if you want an intermediate held at a set point regardless of how the feed or the growth rate drifts, a proportional sensor-actuator loop will not do it — it will hold the intermediate near a value that moves with the disturbance — and an integral implementation will. It costs two more components and a slower response, since integral action trades speed for accuracy, and it can destabilise a loop that already has delay, which is the standard trade in any controller.

Optimal feeding, briefly

Part 11 chapter 44 gave feeding strategies — constant, exponential, DO-stat — as a menu. Optimal control asks the harder question: given the dynamics and a constraint on the feed, what is the feed profile that maximises product at a fixed time? Pontryagin's maximum principle answers it by introducing adjoint variables and showing that the optimum is **bang–singular–bang**: feed at the maximum rate, then on a singular arc where an interior condition holds, then at the maximum or zero again. The singular arc is usually the interesting part, and for a fed-batch it typically corresponds to holding the specific growth rate — or the specific uptake rate — at a constant value, which is precisely the exponential feed of part 11 chapter 44. **So the conventional strategy is the optimal one under those assumptions**, which is worth knowing both because it justifies the convention and because it tells you when the convention fails: when the objective is not simply final product, when the constraint is oxygen rather than feed rate, or when the product inhibits, the singular arc is no longer a constant specific rate.

Model predictive control is the implementable version: solve the optimisation over a horizon, apply the first move, measure, and re-solve. It needs a model good enough to predict a few hours ahead, which for a fermentation usually means a simple kinetic model updated from online measurements rather than a genome-scale one — and it is where a chemical engineer's training transfers most directly into a bioprocess group.

Stability	`eigenvalues of J = ∂f/∂c: negative real parts stable, imaginary parts oscillatory`

Settling time	`set by the slowest eigenvalue; sharper feedback is faster and less damped`
Bistability	`three nullcline crossings; appears between n = 1.5 and 2 for the toggle switch`
Bifurcations	`pitchfork into bistability; Hopf into a limit cycle`
Integral feedback	`zero steady-state error, structurally; antithetic motifs implement it`
Optimal feed	`bang–singular–bang; the singular arc is usually a constant specific rate`

Your dynamic controller holds the intermediate at 0.8 mM in a batch and at 1.4 mM in a fed-batch. Diagnose it in control terms.

It is a proportional controller meeting a different disturbance. The loop's action is proportional to the deviation from its implicit set point, so the steady-state error depends on the load — the rate at which the intermediate is being produced — and a fed-batch loads it differently from a batch. That is not a tuning failure; it is the structural property the simulation above shows as a 50% offset under a threefold disturbance, and no amount of gain removes it, although more gain shrinks it and brings the loop closer to oscillation.

The fix, if the set point genuinely matters, is integral action: an antithetic pair whose difference integrates the error, which gives zero steady-state error regardless of the load, at the cost of two components and a slower response. Before building it, two checks. Measure the **disturbance response** — step the feed and record the intermediate's trajectory and settling time — because that is what identifies the loop's dynamics and it is the measurement part 7 chapter 24 says almost no paper reports. And estimate the **delay**: if the actuator acts by dilution at a generation time, adding integral action to a delayed loop is the standard route to a sustained oscillation, and the right answer may be a degradation tag first and integral control second.

Three things people get wrong

- ‘The simulation is stable, so the design is stable.’ One trajectory from one initial condition; the eigenvalues tell you about all of them, and the imaginary parts tell you whether it rings.
- ‘Stronger feedback makes the loop more stable.’ It makes it faster and less damped; with delay, past some gain it oscillates.
- ‘We tuned the controller and the offset remains.’ A proportional loop has a load-dependent offset by construction. Only integral action removes it.

Self-check — answer on paper first

1. Your Jacobian has eigenvalues −0.05 ± 0.4i and −2.0. Describe the behaviour and the risk.

Stable but very lightly damped: the complex pair has a small negative real part, so a perturbation decays with a time constant of 1/0.05 = 20 time units while oscillating with angular frequency 0.4, meaning about a dozen visible cycles before it settles. The third eigenvalue is fast and irrelevant. The risk is that any additional delay, or any increase in feedback gain, pushes that pair across the imaginary axis into a Hopf bifurcation and a sustained oscillation — so the design has very little margin, and in a real circuit, where transcription, translation and dilution all add delay the model may not contain, it may already be oscillating.

2. Why does a switch need cooperativity, in terms of nullclines?

Because bistability requires the two nullclines to cross three times, and the number of crossings depends on how steeply each variable's nullcline falls. With n = 1 the repression curve is a shallow hyperbola and the nullclines can only cross once, whatever the promoter strength α. Cooperativity steepens them into a sigmoid, so each nullcline has a flat–steep–flat shape and the two can intersect at a low-u_1 state, an unstable middle state and a high-u_1 state. The computed threshold here is between n = 1.5 and n = 2, which is why toggle switches are built from dimerising repressors.

3. If the exponential feed is already optimal, why learn optimal control?

Because it is optimal only under the assumptions that make the singular arc a constant specific rate: a single feed constraint, a product objective at a fixed time, no product inhibition and no binding oxygen limit. Change any of those — add product inhibition, make oxygen transfer the constraint rather than the feed, or value the productivity rather than the final titer — and the optimal profile is no longer exponential, and only the optimisation tells you what it is. Knowing the framework also tells you what to measure to implement it: model predictive control needs a model that predicts a few hours ahead, which is a different and much more achievable thing than a model that predicts a flux map.

What to open a book for. Strogatz, *Nonlinear Dynamics and Chaos*, for stability, nullclines and bifurcations — it is the clearest treatment there is and chapters 2 to 8 cover everything in this chapter. Del Vecchio & Murray, *Biomolecular Feedback Systems*, which is free online and written for exactly this audience. Briat, Gupta & Khammash on antithetic integral feedback for the perfect-adaptation result. And your own process-control notes for Pontryagin and MPC, which is the one

section here you may already have.

Closing notes to part 13

The six chapters in one page

S1 Downstream. Recovery is multiplicative: 90% per step over eight steps is 43% overall, so deleting a step beats improving one. A 1 µm cell settles at 0.2 mm/h, which is why a centrifuge needs a sigma factor of 5,100 m² for 1 m³/h. And the capture cost per kilogram is resin price ÷ (DBC × cycle life), in which titer does not appear — titer moves the volume-driven steps only.

S2 Deconstruction. A 40/25 composition gives 0.729 kg of sugar per kg of dry biomass and a 0.373 kg ethanol ceiling. Severity collapses temperature and time into log R_0, and because release and degradation rise together the optimum is interior. Enzyme cost spans $0.13 to $13 per kg of ethanol across a plausible price range, which is why cellulosic processes live or die on it.

S3 Enzymology. Ternary-complex mechanisms give intersecting double-reciprocal lines and ping-pong gives parallel ones — the slopes computed exactly. MWC with four sites gives apparent Hill coefficients from 1.12 to 2.78 depending on L alone, which is why a Hill coefficient is not a site count, and a feedback-resistant variant is a mutation that stops the effector changing L.

S4 Consortia. Splitting a 20% pathway burden across two strains buys 13% growth. The population ratio is a steady state set by yields and stoichiometry, not by the inoculum. And a cheater with an 8% advantage goes from one part in ten thousand to fixation in about a thousand hours, which is why reciprocal auxotrophy is the only structural defence.

S5 Measurement. A 0.1 mM pool at 1 mM/s turns over in 0.1 s, so sampling must beat a second. An isotope-labelled standard takes the CV from 18.3% to 4.2% by cancellation, and cancels nothing that happens before it is added. Two hundred metabolites at n = 4 gives nine false calls out of eighteen raw ones and finds three of twenty real changes. And an enzyme-constrained bound is good to a factor of a few, dominated by k_{cat}.

S6 Control. Eigenvalues of the Jacobian give stability, oscillation and settling time; sharper feedback is faster and less damped. The toggle switch becomes bistable between n = 1.5 and 2. Integral feedback gives zero steady-state error where a proportional loop gives 50% under the same disturbance. And the exponential feed is the optimal singular arc — under assumptions worth knowing, because they fail often.

On the numbers in this part

Everything was computed with the parameters printed beside it. Four results rest on assumptions I chose and would want replaced: the illustrative release and degradation curves in figure S2.1, whose shapes set where the severity optimum falls; the 8% cheater advantage and the burden slope in S4; the four-component error model in S5's internal-standard comparison; and the biomass composition in S2, which is feedstock-specific and is the first thing to measure. Three numbers are deliberately ranges rather than values — resin price, cellulase price per kilogram of protein, and enzyme loading — because the sources disagree by an order of magnitude and a single figure from me would mislead.

Nothing here was verified against the literature, for the same reason as the rest of the module: no network access. The mechanisms, models and formulas are standard and derivable, and the figures I quote from them are my own arithmetic; the *parameter values* are the part to check, and sheet 10 of the lookup appendix is where they belong.

What is still not here

Four items from the audit I left out, with the reason. **Genetic code expansion** and **quantitative genetics** are real gaps and are reading rather than calculation — a chapter of mine would be a literature summary, which is the thing this module is least good at. **Mass-spectrometry instrumentation** is a half-day of reading that would make chapter S5 and weeks 43 and 51 read differently, and I would get it from an analytical-chemistry text rather than from me. And the **Tier 3 list** — synthetic CO_2-fixation cycles, droplet microfluidics, immobilised enzymes, single-cell omics, biophysical characterisation, intellectual property — is an awareness exercise: know the terms, know one example each, and know which would matter for your capstone. That last one, IP and freedom to operate, is the one I would actually read about before an interview with an industrial group, because it is the question a company asks that a university does not.

Part 19 — Protein products, and the theory the problems had outrun

Two things, and the second is the more important. First, protein products — secretion, inclusion bodies, refolding, glycosylation — which the whole module had treated as an aside because it was built around small molecules and polymers. Second, an audit I should have run earlier: whether the module actually teaches the theory my own problems assume. It did not, in five places.

The audit, and what it found

I searched all fifteen module parts for the theory behind every problem I wrote. Most was there. These were not:

Theory a problem assumes	Problem	In the module before
Analysis of variance and interaction testing	X13	**Absent**
Variance components and nested replication	X15	**Absent**
Equivalence testing	X16	**Absent**
Gas holdup and impeller flooding	X11	**Absent**
Dissolved CO_2 and its pH consequences	X9	**Absent**
Response-surface designs	X14	Thin — named, never derived
Diafiltration and sieving coefficients	Z4	Thin
Protein secretion, refolding	PR1–PR8	Thin — scattered mentions

That is a fair criticism of how this was built: I wrote problems faster than the chapters that justify them, so for eight problems the worked solution was the only explanation available. Chapters T1 to T5 fix it, and PR1 to PR8 are the new protein problems — taking the set to **125**.

Chapter T1 · extends weeks 8, 9 and 30

Protein products I: expression, folding and inclusion bodies

A protein product is the one case where the thing you are making is also the thing doing the making. Everything that follows comes from that: the proteome budget, the folding competition, and the fact that a failed product is not lost carbon but a precipitate.

What makes a protein product different

For a small molecule, the pathway is the catalyst and the product is separate. For a protein, the ribosome makes the product directly, so **yield is set by the proteome budget, not by a carbon ceiling**. Part 14 chapter B9 gives the arithmetic: a cell holds about 0.5 g of protein per gram dry weight, and a heterologous product competes with that directly. At 30 gDW/L a product at 5 g/L is a third of one proteome-equivalent, and at 20 g/L it exceeds the cell's own protein content — possible only because the product leaves the cell while the cells keep growing, which is the first argument for secretion.

The second difference is that a protein can be made and still be worthless. A small molecule is right or absent; a protein can be **expressed, insoluble, misfolded, truncated, proteolysed, or correctly folded and wrongly modified**. So the quantity to measure is never "how much protein" but "how much active, correctly folded protein", and part 7 chapter 22's warning that a band is not an activity is the governing rule.

The folding competition, quantitatively

A newly synthesised chain either folds or aggregates, and the two differ in order. Folding is first order in the unfolded chain; aggregation requires two chains to meet, so it is second order. With rate constants k_f and k_a and an unfolded pool [U], the soluble fraction is approximately

```
soluble fraction ≈ k_f /(k_f + k_a[U]) = 1/(1 + (k_a/k_f)[U])
```

k_a/k_f	[U] = 0.1 µM	1 µM	5 µM
0.5	95%	67%	29%
2.0	83%	33%	9%
10	50%	9%	2%

The asymmetry in order is the whole of inclusion-body engineering. Because aggregation depends on [U] and folding does not, **anything that lowers the unfolded pool raises the soluble fraction** — and the unfolded pool is set by the synthesis rate. That is why every one of the standard remedies works:

Intervention	What it does to the arithmetic	Cost
Lower the induction level	Halves [U], roughly doubles the soluble fraction at $k_a/k_f = 2$	Less total protein — the trade is soluble yield against total
Drop 37 °C to 25 or 16 °C	Slows synthesis several-fold and slows aggregation more than folding	Slower growth, longer run
Co-express chaperones	Raises k_f and holds chains in a foldable state	Proteome, computed below
Fusion partners (MBP, SUMO, thioredoxin)	Raise intrinsic solubility; SUMO is cleavable without a scar	Tag removal is a step, with its own yield
Periplasmic targeting	Oxidising environment lets disulfides form; different chaperones	Translocation capacity, and a volume ceiling (T2)

Chaperones illustrate how to decide. Co-expression costs proteome and buys solubility, so the net gain is the solubility factor times the burden penalty of part 14 chapter B9, $\mu/\mu_0 = 1 - \varphi/0.55$. At a 10% chaperone load the penalty is 0.82, so **the break-even solubility gain is 1.22-fold** — below that you are worse off. A threefold gain at a 10% load nets 2.45-fold; the same gain at a 2% load nets 2.89.

When inclusion bodies are the plan

They are not always a failure. Inclusion bodies are dense, nearly pure, protease-resistant, and easy to separate by

centrifugation — so for a robust protein with no disulfides and a known refolding route, the deliberate route is **express into inclusion bodies at maximum rate, wash, solubilise, refold**, and the overall yield can beat a soluble route that expresses weakly. The decision turns on whether the protein refolds, which is T1's companion question and chapter T2's arithmetic.

What makes them unattractive: a protein with several disulfides, a protein that needs a cofactor inserted co-translationally, anything where refolding yield is below about 30%, and any case where the regulatory burden of demonstrating correct refolding is heavier than the fermentation saving.

Proteome limit	`product per gDW against 0.5 g/gDW; 20 g/L at 30 gDW/L exceeds one proteome-equivalent`
Folding competition	`soluble ≈ 1/(1 + (`k_a/k_f`)[U]); aggregation is second order, folding first`
The lever	`lower [U] — induction level, temperature, promoter strength`
Chaperone break-even	`gain > 1/(1 − φ/0.55); 1.22-fold at a 10% load`
Inclusion bodies as a plan	`viable when refolding yield is high and disulfides are few`

What to open a book for. A review of recombinant protein production in *E. coli* for the solubility interventions and their published success rates; Baneyx & Mujacic, Nat. Biotechnol. 22:1399 (2004) for folding and inclusion bodies, read for the mechanisms rather than the protocols.

Chapter T2 · extends weeks 58, 59 and 49

Protein products II: secretion, refolding, glycosylation and formulation

Secretion removes the intracellular ceiling and replaces it with a translocation ceiling. Refolding trades concentration against yield. Glycosylation decides whether the molecule is a drug or a liability.

Secretion, and the three ceilings it moves

A secreted product escapes the volume ceiling of part 6 chapter 59 entirely — there is no intracellular pool to saturate. What replaces it is **translocation capacity**. In the Sec pathway an unfolded chain is threaded through a translocon, so the flux per cell is the number of translocons times the threading rate divided by the chain length. At roughly 100 translocons per cell† and 20 residues per second, a 300-residue protein can leave at about 24,000 molecules per cell per hour, which at 10^9 cells per millilitre is of order **1 g·L⁻¹·h⁻¹ at full occupancy** — enough for grams per litre over a long run, and only just, which is why secretion titers plateau where they do.

Three routes, with different constraints. **Sec** takes unfolded chains and is the high-capacity default. **Tat** takes *folded* chains, which is the only option for a protein with a co-translationally inserted cofactor, and has far lower capacity. In yeasts the route is the **ER and Golgi**, which adds glycosylation and quality control: misfolded chains are retro-translocated and degraded, so secretion yield depends on folding in a compartment you cannot easily see.

For an *E. coli* product, the usual compromise is the **periplasm**: oxidising, so disulfides form, with DsbA and DsbC to catalyse and reshuffle them. The ceiling is volumetric — the periplasm is roughly 15%† of cell volume, so at 100 gDCW/L and 2 mL/gDCW it is 30 mL per litre of broth, and a protein at 5 mM there is **5 g/L of broth**. That is a real ceiling and it is why true secretion into the medium, where *E. coli* is poor and *Bacillus* and *K. phaffii* are good, is worth a host change.

Refolding: the dilution trade

Solubilised inclusion bodies are refolded by removing the denaturant, and the same competition as T1 applies with the same orders: folding first, aggregation second. So the yield falls with concentration:

Refolding concentration	Yield	Protein recovered per litre
0.01 g/L	93%	0.009 g
0.1 g/L	56%	0.056 g
1.0 g/L	11%	0.111 g

Note what the third column does: **protein per litre of vessel rises monotonically while yield collapses**, so there is no interior optimum and the choice is purely vessel volume against material loss. The engineering answer is to decouple the two by **pulse or fed-batch refolding**: add the denatured protein in portions so the local concentration stays low while the total is high. Adding 1 g in twenty pulses at 0.05 g/L each recovers **0.714 g (71%)** against 0.111 g (11%) in one shot — a six-fold improvement from the addition schedule alone. The same logic underlies continuous dilution refolding and size-exclusion refolding.

Glycosylation

N-linked glycans are attached at Asn-X-Ser/Thr sequons and matter for three reasons: **serum half-life**, since poorly sialylated glycoproteins are cleared quickly; **immunogenicity**, since non-human glycans — yeast high mannose, plant xylose and core fucose, the mammalian Neu5Gc and α-gal epitopes — provoke responses; and **activity**, since glycans at the Fc region modulate effector function.

Two numbers worth holding. **Occupancy is never complete**: at 85% per site, a protein with two sites is 72% fully occupied, with three sites 61%, with five sites 44% — so a multi-sequon protein is intrinsically a mixture, and "the product" is a distribution you must specify. And yeast **hypermannosylation** is a mass problem, not just an immunogenicity one: a human core glycan of ten mannose adds about 1.6 kDa per site, while a yeast outer chain of 50 to 150 adds 8 to 24 kDa — enough to change the molecule's chromatographic behaviour and its apparent mass.

Which host follows from that. **Bacteria do not glycosylate**, so for a glycoprotein they are out unless the glycan is dispensable. **Yeasts** glycosylate wrongly unless humanised — deleting OCH1 and installing a human pathway, each step costing growth. **CHO cells** glycosylate acceptably and are slow and expensive. That is the table in part 6 chapter 61, and glycosylation is the row that usually decides it.

Formulation, briefly, because it decides the specification

Aggregation continues after purification, second order in concentration, so $1/C - 1/C_0 = kt$. At $k = 10^{-3}$ L·g^{-1}·day^{-1} a 1 g/L formulation loses 10% in 111 days and a 10 g/L one loses it in **11 days** — tenfold concentration costs tenfold stability. This is why formulation screens are run at the final concentration and why a high-concentration product needs excipients. For an engineer it matters because **the fermentation titer and the formulation concentration are different numbers**, and a process optimised for the first can be inconvenient for the second.

Secretion capacity	translocons × rate ÷ length; ~1 g·L^{-1}·h^{-1} at full occupancy†
Periplasmic ceiling	~15% of cell volume → about 5 g/L of broth at 100 gDCW/L and 5 mM
Refolding	yield falls with concentration; pulsed addition gives 71% against 11%
Glycan occupancy	0.85^n fully occupied; 44% at five sites
Storage	second order: tenfold concentration, tenfold faster loss

† Translocon count, threading rate and periplasmic volume fraction are recalled from memory, not computed — verify them and record them on sheet 10 of the lookup workbook. Everything computed from them is arithmetic and is reliable given the inputs.

What to open a book for. A review of secretory protein production in *K. phaffii* and in *Bacillus*; the glycoengineering literature for humanised yeast; and, for refolding, a methods review on dilution and on-column refolding — the pulse-addition result above is the single most useful idea in it.

Chapter T3 · extends week 17

The statistics the problems assumed: ANOVA, response surfaces, variance components, equivalence

Part 2 chapter 17 taught replication, power and multiplicity. These four tools are what problems X13 to X16 actually use, and none of them was in the module.

Analysis of variance, and why it exists

With two factors you could run a t-test per comparison, and the multiplicity arithmetic of part 2 chapter 17 then eats you. ANOVA instead **partitions the total variation** into named parts and tests each against the residual. For two factors A and B with n replicates per cell:

```
SS_total = SS_A + SS_B + SS_AB + SS_error,     F = MS_effect / MS_error
```

Each mean square is its sum of squares divided by its degrees of freedom, and the F statistic is compared with the F distribution on those degrees of freedom. What matters for design is the **interaction term, SS_{AB}**: it is the variation that neither factor explains alone, and it exists only if you ran the combinations. One-factor-at-a-time generates no cell in which both factors are changed together, so it cannot estimate the interaction at any sample size — that is a structural blindness, not an efficiency problem, and it is what problem X13 demonstrates with a p = 0.0061 interaction.

Two cautions. A significant F says the factor explains variation, not that the effect is large: report the effect size and the fraction of variance explained alongside. And ANOVA assumes roughly equal variances and independent residuals, which a nested design (chapter T3's third section) violates.

Response-surface designs

A factorial finds which factors matter; a response surface finds *where the optimum is*, by fitting a quadratic:

```
y = β₀ + Σβᵢxᵢ + Σβᵢᵢxᵢ² + Σβᵢⱼxᵢxⱼ
```

A **central composite design** supplies the runs: the factorial corners give the interaction terms, axial points at ±α give the quadratic terms, and replicated centre points give an estimate of pure error. Three things then come out that a grid cannot give. The **stationary point**, from solving ∇y = 0. Its **character**, from the eigenvalues of the matrix of second derivatives — all negative is a maximum, mixed signs a saddle, and a saddle mistaken for an optimum sends the next round in a useless direction. And a **lack-of-fit test**, comparing the residual variation with the pure error from the centre replicates: if lack of fit is significant, the quadratic is the wrong model and the optimum it reports is not trustworthy.

Variance components and nested replication

When measurements are nested — injections within flasks within strains — the variance decomposes:

```
Var(total) = σ²_strain + σ²_flask + σ²_analytical
```

and each level is estimated from the mean squares of a nested ANOVA. The standard error of a strain mean with f flasks and a injections each is $\sqrt{(\sigma^2_{flask}/f + \sigma^2_{analytical}/(f \cdot a))}$, which is why **the level you replicate must be the level that varies**. Injections are *pseudo-replicates*: they estimate analytical variance, and if that is 2% of the total — as in problem X15 — they buy almost nothing. The general rule is to compute the components before designing the next experiment, because the cheapest replication is almost never the useful one.

Equivalence testing

A non-significant difference is not evidence of equivalence; it is usually evidence of a small experiment. To claim equivalence you invert the hypotheses. Choose an **equivalence margin δ** — the largest difference that would not change a decision — and run **two one-sided tests**: that the difference exceeds −δ, and that it is below +δ. Both must reject for equivalence to be claimed, which is identical to requiring the **confidence interval to lie entirely inside ±δ**.

The margin is a process decision made *before* the data, and the defensible choice is usually the run-to-run variability of the process, since a difference below that is unobservable in manufacturing. Problem X16 shows the consequence: the

same five-replicate comparison is "not equivalent" at ±0.5 g/L and "equivalent" at ±1.0, and the conventional t-test says nothing either way.

ANOVA	partition SS; $F = MS_{effect}/MS_{error}$; interaction needs the combinations
Response surface	quadratic fit; stationary point; eigenvalues classify it; centre points give lack of fit
Variance components	$SE = \sqrt{(\sigma^2_{flask}/f + \sigma^2_{analytical}/(f \cdot a))}$; replicate what varies
Equivalence	TOST at a pre-registered margin; CI inside ±δ

What to open a book for. Box, Hunter & Hunter, *Statistics for Experimenters*, for factorials and response surfaces — it is the standard and it is readable. For equivalence testing, a short methods paper on TOST is enough.

Chapter T4 · extends week 44

The reactor physics the problems assumed: heat, shear, holdup and dissolved CO_2

Part 11 chapter 44 did mixing, oxygen transfer and feeding. These four decide whether a vessel that can supply oxygen can also be cooled, aerated and kept at pH.

Heat: the same calculation as oxygen

Aerobic metabolism releases about **460 kJ per mole of oxygen consumed**†, nearly independent of substrate — which is the useful fact, because it means the oxygen calculation you already did is also the heat calculation. Heat duty is OUR × volume × 460 kJ/mol, and the area to remove it is Q/(U·ΔT).

Two consequences the problems computed. A glucose fed-batch at 40 gDW/L and 4 mmol·gDW^{-1}·h^{-1} releases 2.04 MW in 100 m³ and needs 408 m² against roughly 110 m² of jacket, so **internal cooling is needed above about 11 gDW/L**. And a methanol process at 30 gDW/L releases 4.28 MW — twice as much for less biomass, because a reduced substrate demands more oxygen per gram. Heat, not oxygen, is often what sizes the vessel, and it is a capital decision taken at design time.

Shear: a question of length scales

Turbulent energy cascades to ever smaller eddies until viscosity dissipates it, at the Kolmogorov scale $\lambda = (\nu^3/\varepsilon)^{1/4}$, with $\varepsilon = (P/V)/\rho$. Because the dependence is a fourth root, **λ falls very slowly with power** — 56 µm at 100 W/m³, 32 µm at 1,000, 15 µm at 20,000.

Compare the scale with the particle. A cell much smaller than λ is carried by the eddy rather than deformed by it, so **a 1 µm bacterium is not damaged by shear in a stirred tank at any realistic power**, and reports to the contrary usually concern the air–liquid interface or foam. A 15 µm animal cell is comparable to λ at high power, which is why those cultures run at tens of W/m³. A 100 µm floc, pellet or immobilisation bead is larger than λ everywhere and will erode — so pellet processes are limited by impeller design and tip speed, not by mean power. Local dissipation near the impeller is 20 to 50 times the mean, dividing λ by about 2.3, and cells circulate through that zone repeatedly.

Gas holdup and flooding

Holdup is the volume fraction of gas in the aerated broth, rising roughly as the superficial gas velocity to the power 0.7. It matters twice: it is working volume you no longer have — 12% at 0.1 m/s — and it is where the interfacial area for transfer comes from, so the same gas that costs volume buys k_La.

Flooding is the limit. Above a flow number Q/(ND³) of about 0.25 the impeller can no longer disperse the gas; bubbles coalesce and rise as a column past the blades, and k_La *stops rising with gas flow*. CO_2 stripping stops improving at the same point, because it depends on the same interfacial area. Flooding arrives sooner at low stirrer speed — exactly when a control system is trying to save power — so the practical rule is to check the flow number before increasing aeration, and to accept that beyond it the only levers are oxygen enrichment, head pressure and lower demand.

Dissolved CO_2, the variable nobody controls

At RQ = 1 the vessel must strip CO_2 as fast as it supplies oxygen, and the same k_La governs both, so the excess dissolved CO_2 at steady state is **CER/k_La**. At 250 mmol·L^{-1}·h^{-1} and k_La 100 h^{-1} that is 2.5 mM — about 0.1 atm equivalent — and it does three things. It **acidifies**, through the CO_2/bicarbonate equilibrium at $pK_1 \approx 6.1$, so at pH 7 each 1.5 mM of dissolved CO_2 carries about 12 mM of bicarbonate and consumes base. It **inhibits** some organisms directly. And it **rises with everything you do to help oxygen**: head pressure multiplies dissolved CO_2 and O_2 in the same ratio, so a vessel pressurised for oxygen transfer gives its cells proportionally more CO_2.

This is the classic scale-up variable that bench vessels hide, because a small vessel with a short liquid path strips CO_2 easily. Watch base consumption and the pH control output, which move before growth does.

Heat	`Q = OUR·V·460 kJ/mol; area = Q/(U·ΔT); jacket runs out near 11 gDW/L`
Shear	`λ = (ν³/ε)1/4; bacteria safe, 100 µm pellets not`
Holdup, flooding	`holdup ∝ v_s0.7; flooding above Q/(ND³) ≈ 0.25`
Dissolved CO_2	`excess = CER/k_La; pK₁ 6.1; rises with head pressure`

What to open a book for. Doran, the mass-transfer and heat-transfer chapters, worked rather than read; a mixing handbook for the power and flooding correlations, where the exponents are tabulated by impeller type.

Chapter T5 · extends chapter S1

The downstream theory the problems assumed: breakthrough, diafiltration, disruption

Chapter S1 gave the train and the costs. These are the three unit operations whose governing equations the problems used without deriving them.

Breakthrough and dynamic capacity

Loading a column is a moving concentration front. Ahead of it the resin is empty; behind it the resin is saturated; between them is a **mass-transfer zone** whose length is set by how fast the solute can diffuse into the bead relative to how fast the liquid moves past it. Breakthrough begins when the leading edge of that zone reaches the column outlet, which is before the bed is full — so **dynamic binding capacity is always below static capacity**, and the deficit is the unsaturated part of the zone.

Two design consequences follow. The unused length is roughly independent of bed height, so a **taller bed wastes a smaller fraction** — hence tall, narrow capture columns. And since the zone narrows as residence time rises, dynamic capacity increases with contact time, which is why DBC must always be quoted with its flow rate. Breakthrough also defines the yield: a capacity measured at 10% breakthrough means 10% of the feed leaves unbound at the end of loading, so capacity and loss are two readings of one curve.

Diafiltration: two sieving coefficients

Washing a small impurity out of a retained product by adding buffer at the permeate rate gives a first-order washout in **diavolumes**, N, the buffer volume divided by the retentate volume. For a species with sieving coefficient S — the ratio of its permeate concentration to its retentate concentration — the concentration falls as

$$C/C_0 = \exp(-S \cdot N)$$

so the impurity needs $N = \ln(F)/S_i$ diavolumes for an F-fold clearance while the product is retained as $\exp(-S_p N)$. **A membrane is therefore specified by two coefficients, not one**, and they compound badly: a membrane that partly retains the impurity needs more diavolumes, and every extra diavolume loses more product. The worst case in problem Z4 keeps under half the product. The remedies are a better-matched membrane, a cascade with permeate recycle, or changing the buffer so the impurity's sieving rises.

The operational cost is buffer and time: seven diavolumes on 20 m³ is 140 m³ of buffer, and at a modest permeate flux that is days, which makes diafiltration the schedule bottleneck in many trains.

Cell disruption

High-pressure homogenisation releases intracellular product approximately first order in the number of passes, $1 - e^{-kn}$, with k set by pressure and organism — Gram-positives and yeasts are far tougher than *E. coli*. Two things make the optimum interior rather than "as many passes as needed": **product damage accumulates per pass**, and **fine debris is generated**, which makes the subsequent clarification worse. With 8% quality loss per pass and k = 0.9, the net optimum is three passes at 73%, and a fifth pass destroys product (problem PR-adjacent, computed in PP10).

The alternatives trade differently. **Bead milling** suits tough cell walls and generates fine debris. **Chemical or enzymatic lysis** is gentle, selective and expensive, and leaves the lysing agent as an impurity. For a polymer such as PHA the choice changes the product itself — hypochlorite digestion is cheap and lowers molecular weight, which for a material is the specification rather than a detail.

Breakthrough	DBC < SBC by the unsaturated mass-transfer zone; quote DBC with its residence time
Diafiltration	$C/C_0 = \exp(-S \cdot N)$; impurity needs ln F/S_i; product retained $\exp(-S_p N)$
Disruption	release $1 - e^{-kn}$; damage per pass makes the optimum interior

What to open a book for. Harrison, Todd, Rudge & Petrides, *Bioseparations Science and Engineering*, the chromatography, membrane and disruption chapters; a vendor's process-development handbook for breakthrough-curve protocols.

Problems PR1 to PR8 — protein products

PR1 · extends chapter T2 · 1.5 h

How much can a cell secrete?

Take 100 Sec translocons per cell at 20 residues per second, a 300-residue 33 kDa product, and 10^9 cells per millilitre.

1. Compute the export rate per cell per hour and per litre of broth at full occupancy.
2. Repeat at 10% and 30% occupancy, and give the titer over a 48-hour run.
3. Published secretion titers reach grams per litre. Is that consistent?
4. What would you measure to find whether translocation is your limit?

Worked solution. 100 × 20 ÷ 300 × 3,600 = **24,000 molecules per cell per hour**, which at 10^9 cells/mL is **1.32 g·L⁻¹·h⁻¹** at full occupancy — **0.13 at 10% and 0.40 at 30%**, giving **6.3 and 19 g/L over 48 hours**. So grams per litre is consistent, but only at high cell density and long runs, and only if a substantial fraction of translocons is working on your product rather than the cell's own envelope proteins.

To test whether translocation limits you: compare the **intracellular precursor** (with signal peptide uncleaved) against the mature secreted form. Accumulating precursor means translocation is saturated, and the fixes are a different signal peptide, lower synthesis rate, or co-expression of translocation components. If precursor does not accumulate and product is low, the limit is upstream — transcription, translation or folding — and secretion engineering will not help.

What it teaches: secretion replaces a volume ceiling with a flux ceiling, and the uncleaved precursor is the measurement that tells you which one you are against.

PR2 · extends chapter T1 · 1.5 h

Inclusion bodies: compute the soluble fraction

Soluble fraction ≈ $1/(1 + (k_a/k_f)[U])$.

1. Tabulate the soluble fraction for k_a/k_f of 0.5, 2 and 10 at [U] of 0.1, 1 and 5 µM.
2. What does halving the synthesis rate do at $k_a/k_f = 2$ and [U] = 5 µM?
3. Rank the five standard interventions by what they change in the equation.
4. When is the right answer to stop fighting and go to inclusion bodies deliberately?

Worked solution. At $k_a/k_f = 2$ the soluble fraction is **83%, 33% and 9%** at the three unfolded pools; at 10 it is 50%, 9% and 2%. Halving the synthesis rate halves [U], taking 5 µM to 2.5 and the soluble fraction from **9% to 17%** — a near-doubling of soluble protein from a weaker promoter, at some cost in total.

By mechanism: **induction level and temperature lower [U]**; **chaperones raise k_f**; **fusion partners lower k_a**; **periplasmic targeting changes the compartment** and so both constants. Go to inclusion bodies deliberately when refolding yield is high, disulfides are few, proteolysis in the cytoplasm is a problem, or when the purity of the washed pellet saves a chromatography step — noting that the regulatory burden of demonstrating correct refolding is real for a therapeutic.

What it teaches: solubility is a kinetic competition with a different reaction order on each side, so the dominant lever is always the unfolded pool.

PR3 · extends chapter T2 · 2 h

Refolding: the trade with no interior optimum

1. With yield = 1/(1 + 8C), tabulate yield and protein recovered per litre at 0.01 to 1.0 g/L.
2. Is there an optimum concentration? Explain the shape.
3. Compute the total recovered when 1 g is added in 1, 5 and 20 pulses.
4. Name three process formats that exploit this, and the cost of each.

Worked solution. Yield falls from **93% at 0.01 g/L to 11% at 1.0 g/L**, while protein per litre rises from 0.009 to 0.111 g — so **there is no interior optimum**: yield wants dilution and vessel economics want concentration, and the two never cross. The resolution is to separate local from global concentration. Adding 1 g in **20 pulses at 0.05 g/L each recovers 0.714 g (71%)** against 0.111 g (11%) in a single shot.

Three formats exploit it: **pulse or fed-batch refolding**, cheap and slow; **continuous dilution** into a stirred vessel, which needs good mixing or the local concentration is not what you think; and **on-column refolding**, where the protein is immobilised so chains cannot meet, which gives the highest yield and the lowest throughput. All three are the same idea: keep the chains apart while they fold.

What it teaches: when yield and throughput pull in opposite directions with no optimum, the answer is usually a change of format rather than a better set-point.

PR4 · extends chapter T2 · 1.5 h

Glycosylation: occupancy, mass and host choice

1. A 50 kDa protein has two N-glycan sites, each adding about 2 kDa. What fraction of the molecule is carbohydrate?
2. At 85% occupancy per site, what fraction of molecules is fully glycosylated at 2, 3 and 5 sites?
3. Compute the glycan mass for 10, 50 and 150 mannose residues per site.
4. Choose a host for a therapeutic glycoprotein and justify it from these numbers.

Worked solution. Two sites at 2 kDa on a 50 kDa protein is **7% carbohydrate**. At 85% occupancy the fully glycosylated fraction is **72% at two sites, 61% at three and 44% at five** — so a multi-sequon protein is intrinsically a mixture, and the product specification must describe a distribution rather than a molecule.

Glycan mass: 10 mannose is **1.6 kDa** per site, 50 is **8.1**, and 150 is **24.3** — so yeast hypermannosylation can add half the protein's own mass again, changing both its chromatography and its clearance. For a therapeutic glycoprotein: **bacteria are excluded** (no glycosylation at all), wild-type **yeast is excluded** on immunogenicity and mass, and the choice is between a **humanised yeast** — faster and cheaper, with each pathway edit costing growth — and **CHO**, slower and dearer with acceptable glycans and a regulatory precedent. If the glycan is dispensable for function, the whole argument collapses and *E. coli* returns, which is why that question is asked first.

What it teaches: glycosylation turns "the product" into a distribution, and it is usually the row that decides the host table.

PR5 · extends chapter T1 · 1 h

What a protein product costs the proteome

1. At 30 gDW/L, express titers of 1, 5 and 20 g/L as product per gDW and as a fraction of the 0.5 g/gDW proteome.
2. How is more than one proteome-equivalent possible?
3. Compute the product synthesis rate at 5 g/L over 48 h and compare with total protein synthesis at $\mu = 0.1\ h^{-1}$.
4. What does that ratio imply about where to look for the limit?

Worked solution. 1, 5 and 20 g/L at 30 gDW/L are **0.03, 0.17 and 0.67 g per gDW**, which is **7%, 33% and 133%** of one proteome-equivalent. More than 100% is possible only because the product *leaves* and the cells keep growing — the cell never holds that much at once, it accumulates in the broth. For an intracellular product the same titer would be impossible, which is the whole secretion argument in one number.

The rate tells a different story: 5 g/L over 48 h at 30 gDW/L is $0.003\ g \cdot gDW^{-1} \cdot h^{-1}$ against total protein synthesis of 0.050 at $\mu = 0.1\ h^{-1}$ — the product is **7% of protein synthesis**. So at this titer the ribosome is not the limit, and looking for a limit in transcription, folding or secretion is better founded than raising promoter strength. At 20 g/L the ratio reaches 27% and the proteome argument becomes real.

What it teaches: the same titer is a trivial or an impossible proteome demand depending on whether the product is secreted, and the synthesis-rate ratio says whether expression is even the right thing to push.

PR6 · extends chapter T1 · 1 h

Is chaperone co-expression worth its proteome?

1. Using $\mu/\mu_0 = 1 - \varphi/0.55$, compute the net benefit of chaperones at 2%, 5% and 10% of the proteome for solubility gains of 1.5×, 2× and 3×.
2. What is the break-even gain at a 10% load?
3. Why is the net gain not simply the solubility gain?
4. What experiment establishes the actual gain and the actual load?

Worked solution. Net benefit is the gain times $(1 - \varphi/0.55)$. At a 2% load a 1.5× gain nets **1.45×** and a 3× gain nets

2.89×; at 10% those become **1.23× and 2.45×**. The break-even gain at a 10% load is **1.22×** — so a chaperone set that improves solubility by only 20% at a 10% proteome cost is not worth expressing.

The net is not the raw gain because the chaperones displace the growth-supporting proteome, so you get more soluble protein per cell and fewer cells. The experiment that settles it is two measurements, not one: **soluble active product per litre with and without the chaperones** — which captures both effects at once — and the chaperone fraction of total protein by gel or proteomics, which tells you whether the load is 2% or 10%. Reporting a solubility fold-change from a gel of the soluble fraction alone, as is common, measures the numerator and ignores the denominator.

What it teaches: any co-expressed helper has to clear a break-even set by its own proteome cost, and the right readout is product per litre rather than solubility.

PR7 · extends chapter T2 · 1.5 h

Cytoplasm, periplasm or medium: three ceilings

1. At 2 mL cell volume per gDCW and a periplasm that is 15% of it, compute the periplasmic volume per litre of broth at 30 and 100 gDCW/L.
2. At 5 mM protein in the periplasm (33 kDa), what broth titer is that?
3. Compute the cytoplasmic ceiling at 20% of a 0.5 g/gDW proteome and 100 gDCW/L.
4. Rank the three compartments and say what each costs.

Worked solution. Periplasm is **9.0 mL per litre of broth at 30 gDCW/L and 30 mL at 100**; at 5 mM and 33 kDa that is **1.5 g/L and 5.0 g/L** of broth. The cytoplasmic route at 20% of the proteome and 100 gDCW/L gives **10 g/L**. Secretion into the medium has no intracellular ceiling at all.

So the ranking by ceiling is **medium > cytoplasm > periplasm**, and by ease of purification it is the reverse — the periplasm is a small, relatively clean compartment released by osmotic shock, the cytoplasm needs full lysis and gives you everything, and the medium is the cleanest of all but dilute. The periplasm's compensation is chemistry rather than capacity: it is oxidising, so disulfide-bonded proteins fold there and not in the cytoplasm. That is why a small disulfide-bonded therapeutic goes periplasmic despite the worst ceiling of the three.

What it teaches: compartment choice trades a volumetric ceiling against purification cleanliness and redox chemistry, and the smallest compartment is sometimes the right one.

PR8 · extends chapter T2 · 1 h

Formulation: why titer and concentration are different targets

1. With second-order aggregation, $1/C - 1/C_0 = kt$, compute the time to lose 10% at k of 10^{-4} and 10^{-3} $L \cdot g^{-1} \cdot day^{-1}$ for C_0 of 1 and 10 g/L.
2. State the scaling in one sentence.
3. Why are formulation screens run at the final concentration?
4. What does this mean for a process optimised for titer?

Worked solution. At $k = 10^{-4}$, 1 g/L loses 10% in **1,111 days** and 10 g/L in **111**; at $k = 10^{-3}$ those become 111 and **11 days**. The scaling is simple and brutal: **tenfold the concentration, tenfold faster loss**, because the rate is second order while the loss is measured as a fraction.

Hence screening at the final concentration: a buffer and excipient set that looks stable at 1 g/L can fail at 50 g/L, and the aggregation pathway itself can change — a protein that loses a few per cent as dimer at low concentration may form visible particulates at high. For the process, this decouples two targets that are easy to confuse. **Fermentation titer buys separation cost** (problem PP12); **formulation concentration is set by the dose and the delivery route**, and a high-titer process still has to be concentrated or diluted to that number. Optimising fermentation titer does not optimise the product, and the formulation concentration should be known before the downstream train is designed, because it decides the final ultrafiltration step.

What it teaches: stability scales against concentration, so the final concentration is a specification set outside the process and imposed on it.

Closing notes to part 19

Answering the second question directly

No, the module did not cover the theory behind all the problems — I wrote problems faster than chapters. The audit found **five topics absent** (ANOVA, variance components, equivalence testing, gas holdup and flooding, dissolved CO_2) and **three thin** (response surfaces, diafiltration, protein production). Chapters T1 to T5 close all eight. If you find another problem whose theory is not in a chapter, that is a reasonable thing to send back — the audit was keyword-based and will have missed cases.

What the five chapters add

Chapter	Covers	Serves problems
T1 Protein products I	Proteome limit, folding competition, inclusion bodies, chaperone economics	PR2, PR5, PR6
T2 Protein products II	Sec and Tat capacity, periplasm, refolding, glycosylation, formulation	PR1, PR3, PR4, PR7, PR8
T3 Statistics	ANOVA and interaction, response surfaces, variance components, equivalence	X13, X14, X15, X16
T4 Reactor physics	Heat duty, Kolmogorov shear, holdup and flooding, dissolved CO_2	X4, X9, X10, X11, X12
T5 Downstream	Breakthrough and the mass-transfer zone, diafiltration sieving, disruption	Z1, Z4, PP10

The protein results worth remembering

- **PR1** — translocation capacity is about 1 $g \cdot L^{-1} \cdot h^{-1}$ at full occupancy, so grams per litre needs high density and a long run; accumulating precursor is the diagnostic.
- **PR2** — halving the synthesis rate takes the soluble fraction from 9% to 17%, because aggregation is second order and folding first.
- **PR3** — refolding 1 g in twenty pulses recovers 71% against 11% in one shot.
- **PR4** — at 85% occupancy only 44% of a five-site protein is fully glycosylated; yeast hypermannosylation adds up to 24 kDa per site.
- **PR5** — 20 g/L secreted is 133% of one proteome-equivalent, possible only because it leaves.
- **PR6** — chaperones at a 10% proteome load must deliver 1.22-fold just to break even.
- **PR7** — the periplasm holds only 1.5 g/L of broth at 30 gDCW/L, and is still the right compartment for a disulfide-bonded protein.
- **PR8** — tenfold the formulation concentration, tenfold faster aggregation.

On the numbers

Everything computed here is arithmetic from the stated inputs. Four inputs are recalled rather than computed and are marked in the text: the translocon count and threading rate, the periplasmic volume fraction, the 460 kJ per mole of oxygen, and the glycan masses. They belong on sheet 10 of the lookup workbook. The folding, refolding and aggregation rate constants are illustrative parameters chosen to show the shape of each relation — the *shapes* are the content, the *values* are yours to measure.

Where the set stands

Source	Problems
Your curriculum appendix A1–W3	54
Part 15 X1–X24, Y1–Y6	30
Part 16 Z1–Z9	9
Part 17 MC1–MC5, R1–R5	10
Part 18 PP1–PP12	12
Part 19 PR1–PR8	8
Total	**123**

With G2 from the addendum and the module's own two construction exercises, the tracker counts 125.

Part 20 — Multiphase and solid-state fermentation, gel immobilisation, the pathway-modification playbook, and design of experiments properly

I audited the module for the four things you named before answering. Two were well covered, one was thin, and two were absent — and the absent ones are whole reactor classes.

What you asked about	In the module	Verdict
Metabolic pathway modification	Knockouts and overexpression (158 mentions), dynamic control (82), compartmentalisation (75), cofactor swaps (57), feedback resistance (17)	**Well covered**, but scattered across twelve parts with no single decision guide — chapter V3 consolidates it
In situ gel fermentation	Part 3 chapter 48: alginate, Thiele modulus, dead core, effectiveness factor (60 mentions)	**Covered as transport**, not as a gel process — bead making, leakage and gel stability were missing (V2)
Two-liquid-phase and gas fermentation	Part 3 chapters 45–47, part 5 chapter 55 (54 and 16 mentions)	Covered
Three-phase (gas–liquid–solid) reactors	Nothing	**Absent** — V1
Solid-state fermentation	Nothing	**Absent** — V1
Biofilm and attached growth	7 mentions in passing	Thin — V1
Degrees of freedom, factorials, blocking	Part 2 chapter 17, part 19 chapter T3	Covered
Response-surface methodology as a method	Named and fitted once	**Thin** — screening, steepest ascent, design choice and canonical analysis were missing (V4)

Four chapters, V1 to V4, and eight problems, MP1 to MP4 and DE1 to DE4, taking the set to **133**.

Chapter V1 · extends weeks 44 and 48

Multiphase and solid-state fermentation

A stirred tank is two phases with a third dissolved in one of them. Add a solid and three things change at once: the transfer coefficient, the working volume, and what the reactor can be.

What "multiphase" actually means here

Phases	Example	What governs it
Liquid + gas	Ordinary submerged fermentation	k_La, mixing (part 11 ch 44)
Liquid + liquid + gas	Two-phase in situ removal	Partition coefficient, phase volume (part 3 ch 45–47)
Liquid + solid + gas	Immobilised cells, biofilm, slurry reactor	Solids holdup, intraparticle diffusion, bed hydrodynamics
Solid + gas, no free liquid	Solid-state fermentation on bran, husk, bagasse	Water activity, evaporative cooling, conduction

Three-phase reactors: the three penalties

The transfer penalty. Solids reduce the liquid fraction and damp bubble breakup, so k_La falls roughly as $(1 - \varepsilon_s)^{2.5}$ with solids holdup ε_s. At 10% solids that is a 26% loss, at **20% a 43% loss**, at 30% 59%. Since you added the solids to retain biomass, you are trading oxygen transfer for cell retention, and whether that is favourable depends on whether the process was oxygen-limited to begin with.

The volume penalty. A 20% bead loading is 20% of the vessel that is not broth — on top of the gas holdup of chapter T4. Together, 20% solids and 10% gas leave 70% of the vessel working.

The hydrodynamic constraint. Whether a bed is packed, fluidised or slurry depends on the liquid velocity against the particle's terminal velocity. For a 2 mm alginate bead with a 100 kg/m³ density difference, the terminal velocity is about **0.077 m/s**, and a 4 mm bead about 0.109 m/s. A column run at a typical 0.01 m/s superficial velocity is therefore **a packed bed, not a fluidised one** — and the difference matters, because a packed bed channels, compacts under its own weight and cannot be aerated through the bed without flooding it.

Which configuration to choose follows from those three. **Stirred tank with beads** for a small bead loading and an oxygen-demanding organism; **packed bed** for an anaerobic or microaerobic process with a clean feed; **fluidised or expanded bed** when you need both retention and mass transfer and can supply the velocity; **airlift with beads** when shear must be low, which chapter T4 says matters for the beads rather than the cells.

Biofilm and attached growth

A biofilm is the same problem with the solid self-assembled. The governing number is the **oxygen penetration depth**, $\sqrt{(2DC^*/R_V)}$, with R_V the volumetric uptake rate. For a film at 100 kg/m³ of wet biomass consuming 4 mmol $O_2 \cdot gDW^{-1} \cdot h^{-1}$:

Film density	Oxygen penetration
50 kg/m³	106 µm
100 kg/m³	**75 µm**
200 kg/m³	53 µm

So at 100 kg/m³ the active layer is 75 µm, and a film thicker than that is partly dead weight: a 300 µm film is **75% inactive** and a 1 mm film **92% inactive**. That is why biofilm reactors are designed around **thickness control** — shear to slough the excess, or a support geometry that keeps films thin — rather than around maximising attached biomass, which is the intuitive but wrong objective. The dead interior is not merely idle: it ferments, acidifies and eventually detaches as a plug.

Solid-state fermentation

With no free water the familiar levers disappear. There is no k_La, because oxygen moves through the gas phase in the inter-particle voids and then a short way into a wet particle; there is no jacket cooling worth the name, because the bed conducts badly; and pH cannot be controlled by addition.

Heat is the whole problem. At 460 kJ per mole of oxygen and 2 kg O_2 per kg of biomass, a bed growing at 3 $kg \cdot m^{-3} \cdot h^{-1}$ releases **86 $MJ \cdot m^{-3} \cdot h^{-1}$, which is 24 kW/m³** — against the 1 to 5 kW/m³ of a stirred tank. It leaves by

evaporation, which costs water: **38 kg of water per m³ per hour**, so a bed at 50% moisture holding 250 kg/m³ of water **dries out in 7 hours** at that growth rate, and in 20 hours at 1 kg·m^{-3}·h^{-1}. Hence the two controls that define SSF practice — **humidified air at high flow**, and **water replenishment** — and the reason productivity is capped by aeration and humidity rather than by oxygen transfer in the usual sense.

Conduction sets the depth. With a bed conductivity near 0.2 W·m^{-1}·K^{-1}, a heat generation of 1 kW/m³ and conduction only, the centre of a bed rises by $qL^2/2k$: **6 K at a 5 cm half-depth, 56 K at 15 cm, and 225 K at 30 cm** — the last two being thermally impossible, which is why static trays are shallow and anything deeper is forced-aerated, rotated or agitated. The reactor classes follow: trays (shallow, labour-intensive, easy), packed beds with forced aeration (deeper, with an axial temperature and moisture gradient), and rotating drums or intermittently mixed beds (uniform, with shear on the mycelium).

Solids penalty	$k_La \propto (1 - \varepsilon_s)^{2.5}$; 43% loss at 20% solids
Bed regime	compare superficial velocity with terminal velocity (0.077 m/s for a 2 mm bead)
Biofilm	active layer $\sqrt{(2DC^*/R_V)}$ = 75 µm at 100 kg/m³; a 1 mm film is 92% inactive
SSF heat	24 kW/m³ at 3 kg·m^{-3}·h^{-1}; 38 kg water/m³/h; bed dries in 7 h
SSF depth	$\Delta T = qL^2/2k$: 56 K at a 15 cm half-depth, 1 kW/m³

What to open a book for. Doran on heterogeneous reactions and particle mass transfer; a review of solid-state fermentation bioreactor design (Mitchell and colleagues) for the heat and water balances, which is the whole subject; and a biofilm-reactor review for thickness control.

Chapter V2 · extends week 48

In situ gel immobilisation as a process, not just a transport problem

Part 3 chapter 48 asked how deep oxygen gets into a bead. This chapter asks the questions that come before and after: how the bead is made, how cells get out of it, and what dissolves it.

Making the beads

The standard route is **ionotropic gelation**: a cell suspension in 1.5–3% sodium alginate is dripped or sprayed into a calcium chloride bath, where Ca^{2+} cross-links the guluronic acid blocks into an "egg-box" structure. Bead size comes from the droplet, so it is set by nozzle diameter, flow rate and whether you use vibration, coaxial air flow or an electrostatic field to break the stream. What matters downstream:

Bead radius	Volume	Surface/volume	Outer 50 µm as a share of volume
1.0 mm	4.19 µL	3,000 m^{-1}	**14%**
1.5 mm	14.1 µL	2,000 m^{-1}	10%
2.0 mm	33.5 µL	1,500 m^{-1}	7%

Smaller beads transfer better — chapter 48's Thiele modulus falls with radius — and have more surface per volume, which is the problem in the next section. The usual compromise is 1 to 2 mm, made by a vibrating nozzle at hundreds of beads a second.

Leakage: the cells get out

The failure mode nobody plans for is that **cells near the surface escape as they divide**. Treating the outer 50 µm as the escape zone, a 1 mm bead releases cells at roughly its outer-layer fraction times the growth rate: at $\mu = 0.3\ h^{-1}$ that is **0.042 g of free cells per gram of immobilised biomass per hour**, and 0.022 for a 2 mm bead. Over a day that is a substantial free-cell population, which matters three ways: the "immobilised" process is partly a free-cell process; the free cells are the ones your assays see; and in a continuous reactor they wash out, so you are continuously losing biomass.

The defences, in order of practicality: **a coating layer** (chitosan or poly-L-lysine, then a second alginate layer) which adds a diffusion barrier of its own; **larger beads**, trading transfer for retention; **growth limitation** inside the bead, which is the elegant answer — a nitrogen-limited feed lets the cells produce without dividing, so there is nothing to escape; and **accepting it** and designing the downstream for a mixed population.

What dissolves the gel

Calcium alginate is held together by an ion, so anything that takes the ion away destroys the bead. **Phosphate and citrate chelate calcium**, and both are ordinary medium components — a standard M9 or LB formulation will soften beads over days. Sodium and potassium at high concentration exchange with calcium and do the same more slowly. The fixes: formulate with the **lowest phosphate that supports growth** and feed phosphate separately if needed; include **a few millimolar of calcium in the medium** to keep the equilibrium on the right side; or cross-link with **barium or strontium**, which bind alginate more tightly, at the cost of toxicity considerations and regulatory awkwardness. Alternatives avoid the problem entirely — κ-carrageenan gelled with potassium, chitosan, or covalently cross-linked polyacrylamide and polyvinyl alcohol, each with its own toxicity and diffusivity.

When gel immobilisation actually pays

Part 3 chapter 48 computed the central trade: a retained-biomass reactor reached 0.27 $g \cdot L^{-1} \cdot h^{-1}$ against a free-cell chemostat's 0.65 at the same dilution rate, because the beads' diffusion limitation outweighed the higher biomass. So the honest default is that **immobilisation costs productivity**, and it has to be bought back by something else:

- **Operating above the washout dilution rate** — the one unambiguous win, since retained cells cannot wash out and a free-cell chemostat is hard-limited at μ_{max}.
- **Protecting cells from shear, solvent or inhibitor**, which is the argument in a two-phase process where the organic phase would otherwise contact the cells.
- **Reusing an expensive catalyst** across batches, which is the argument for immobilised *enzymes* more than cells.

- **Avoiding a downstream separation**, since the beads are removed by a screen rather than a centrifuge.

If none of those four applies, the calculation in chapter 48 says do not immobilise.

Gelation	1.5–3% alginate into $CaCl_2$; bead size from the droplet
Leakage	outer-layer fraction × μ: 0.042 g free cells per g per h for a 1 mm bead at μ 0.3
Gel killers	phosphate and citrate chelate Ca^{2+}; fix with low phosphate, added Ca, or Ba/Sr
When it pays	above washout, under shear or solvent, to reuse catalyst, or to avoid a centrifuge

What to open a book for. A review of cell immobilisation in alginate for the gelation chemistry and the stability data; part 3 chapter 48 of this module for the transport arithmetic, which this chapter assumes rather than repeats.

Chapter V3 · consolidates parts 2, 7, 8 and 11

The pathway-modification playbook: nine interventions and how to choose

The module teaches every one of these somewhere. What it never did was put them on one page with the number that tells you which to reach for.

Intervention	Use when	The number that decides it	Where
1. Overexpress a step	The step carries flux control	Flux control coefficient; gain = 1/(1 − C(1 − 1/m)), ceiling 1/(1 − C). At C = 0.1 a tenfold overexpression buys 9.9%	Part 2 ch 15; part 8 ch 30
2. Delete a competing branch	Carbon leaves through a known side reaction	Branch partition 1/(1 + r/F); at r = 1, raising F to 10 moves 67% → 91%	Part 8 ch 27; part 1 ch 6
3. Relieve feedback inhibition	The product inhibits its own first committed step	Removing feedback in the part 2 model gave +43%, against +8.3% from a tenfold overexpression	Part 2 ch 15; part 19 ch T3 of the enzymology
4. Swap the cofactor	The pathway demands the scarcer carrier	NADPH costs 0.5 C per molecule from the oxidative branch; a transhydrogenase costs ~0.33 ATP	Part 2 ch 19; part 14 ch B5
5. Push precursor supply	The pathway's precursor is a minor node	Biomass demand per gDW against pathway demand; acetyl-CoA is 3.75 mmol/gDW for biomass alone	Part 1 ch 7; part 14 ch B7
6. Engineer export	Product accumulates inside or inhibits	Intracellular ceiling 20 g/L at 50 gDCW/L; an export cap below 6 killed a coupled strain outright	Part 3 ch 45; part 11 ch 39
7. Dynamic control	The best expression differs between growth and production	Two-stage optimum: f = 0.078 of substrate to growth, 23% better than one stage	Part 8 ch 27; part 18 PP8
8. Growth coupling	Revertants will take over a long run	Guaranteed product at maximum growth; ΔPYK ΔACK gives 0.90 mol/mol at 0.072 h^{-1}	Part 11 ch 39; part 18 PP7
9. Compartmentalise or scaffold	An intermediate is toxic, volatile or shared	Peroxisomal flux 2.8 × 10^{-8} against carrier-mediated 1.7 × 10^{-7} mol·m^{-2}·s^{-1}	Part 6 ch 58; part 5 ch 53

The order to try them in

The order is not arbitrary; it follows from cost and from what the measurement tells you.

First, find out what limits it. Part 2 chapter 11's diagnosis: measure the intermediate pools. An intermediate that accumulates says the step after it is limiting; one that is absent says the step before it is. A product that rises with uptake rate says throughput, not pathway. Doing this first saves most of the work, because the common finding is that the limitation is not where the proposal assumed.

Then the cheap, high-leverage interventions: feedback relief (3) and branch deletion (2) are single point mutations or single knockouts with large effects. Feedback relief in particular gave five times the benefit of a tenfold overexpression in the part 2 model, and it costs one allele.

Then the supply-side interventions: cofactor (4) and precursor (5), which need a balance calculation first, because they trade carbon for reductant and can lower yield while raising rate.

Then export (6), which is cheap to test and catastrophic to ignore — part 11 chapter 39 showed a coupled strain made infeasible by a transport cap.

Only then expression tuning (1) and libraries, because without a control coefficient you are guessing, and the ceiling of 1/(1 − C) caps what tuning can ever deliver.

Dynamic control (7) and growth coupling (8) last, because they are architectural: they change how the process is run, not just what the strain contains, and they are worth the effort only once the pathway itself works.

Compartmentalisation (9) when a specific intermediate problem demands it — toxicity, volatility or competition for a shared pool — and not as a general improvement.

The honest caveat from part 11 chapter 39: a model proposes these, measurement decides them, and the published strains almost always contain one intervention the model could not suggest — a transporter, a regulator, or an evolved tolerance.

Chapter V4 · extends week 17 and chapter T3

Design of experiments, properly: from screening to a response surface you can defend

Chapter T3 gave the four tools. This is the method: how many runs, which design, how to walk to the optimum, and how to read the output without fooling yourself.

Degrees of freedom: the bookkeeping that decides whether a design works

Every run gives one degree of freedom. Every term you estimate spends one. What is left is the error degrees of freedom, and without it you have no test at all.

```
df_error = N − 1(mean) − k(main effects) − k(k−1)/2(two-factor interactions) − …
```

Design	Runs	df spent on the model	df left for error
2^2 with 4 replicates	16	4	12
2^3 with 2 replicates + 3 centre	19	7	12
2^4 unreplicated + 5 centre	21	11	10
2^4 unreplicated, no centre points	16	11	**5**
2^5 unreplicated	32	16	16

Two lessons. An unreplicated factorial with all interactions fitted leaves very little error — the 2^4 case leaves five — which is why unreplicated designs are analysed by pooling the high-order interactions into error or by a normal plot of effects instead of by F tests. And **centre points are the cheapest degrees of freedom you can buy**: three to five of them give a pure-error estimate independent of the model and let you test curvature before you commit to a quadratic.

Screening: fractional factorials and what gets confounded

With k factors a full factorial is 2^k runs, which is 128 at k = 7. A fraction assigns extra factors to interaction columns: set E = ABCD, and you get a half fraction in 16 runs. The price is **aliasing** — each effect is confounded with others, read off the defining relation.

Design	Runs	Generators	Resolution	What is confounded
2^7 full	128	—	—	Nothing
2^{7-1}	64	G = ABCDEF	VII	Nothing that matters
2^{5-1}	16	E = ABCD	**V**	Main with four-factor, two-factor with three-factor — all main effects and two-factor interactions estimable
2^{7-4}	8	E = ABC, F = BCD, G = ACD	**III**	A = BE = CF = DG — main effects confounded with two-factor interactions; screening only

The rule of thumb: **resolution III for screening many factors** when you only want to know which few matter and are willing to be fooled by an interaction; **resolution IV** when main effects must be clean of two-factor interactions; **resolution V** when you want the two-factor interactions too. Moving from 8 runs to 16 buys you from III to V at five factors, which is almost always worth it.

A worked screening, computed: five factors in a 2^{5-1}, with true effects A = 2.0, B = 1.2, C = 0.2, D = −1.5, E = 0.1 and an AB interaction of 0.7, measured with 0.4 of noise. The estimated effects (conventionally twice the coefficients) came out **A +4.11, D −2.67, B +2.32, AB +1.64**, with everything else under 0.3 — so the design correctly identifies A, B, D and the AB interaction and correctly rejects C and E. That is what screening is for: **reduce five factors to three** before spending runs on a surface.

Steepest ascent: getting to the right neighbourhood

A first-order model tells you the direction of improvement, not the optimum. The path of steepest ascent moves in proportion to the coefficients: with coded coefficients $b_A = 1.0$, $b_B = 0.6$ and $b_D = -0.75$, you move **one coded unit in A for every 0.60 in B and −0.75 in D**. Converting to real units with step sizes of 5 g/L, 0.5 °C and 100 rpm per coded unit gives a concrete path:

Step	A	B	D
1	+5 g/L	+0.30 °C	−75 rpm
3	+15 g/L	+0.90 °C	−225 rpm
5	+25 g/L	+1.50 °C	−375 rpm

You run single experiments along the path until the response stops improving, and **that is where the response-surface design goes**. Skipping this step is the commonest waste in an optimisation campaign: a central composite design centred in the wrong place fits a plane and tells you only to move, which is what the screening design already said.

Choosing the response-surface design

Factors	Quadratic coefficients	Central composite	Half-fraction CCD	Box–Behnken
2	6	13	—	—
3	10	19	—	15
4	15	29	21	27
5	21	47	31	46

The axial distance for a rotatable design is $\alpha = (2^k)^{1/4}$: **1.414 at two factors, 1.682 at three, 2.000 at four**. Rotatability means the prediction variance depends only on distance from the centre, which is what you want when you do not know the direction of the optimum. **Box–Behnken has no corner points**, so no run combines all factors at their extremes — use it when the corners are infeasible or dangerous, and accept slightly poorer estimation at the edges.

Reading the output: ANOVA, lack of fit, canonical analysis

A worked three-factor central composite design, 19 runs with five centre points, fitted to the full quadratic:

$$y = 42.01 + 3.05x_1 + 1.87x_2 - 0.43x_3 - 2.53x_1^2 - 1.85x_2^2 - 0.66x_3^2 - 1.32x_1x_2 + 0.38x_1x_3 + 0.04x_2x_3$$

Model test: $R^2 = 0.9941$, adjusted $R^2 = 0.9882$, $F = 168.0$ on 9 and 9 degrees of freedom, $p = 6 \times 10^{-9}$. Report the *adjusted* R^2, because the plain R^2 rises whenever you add a term; a large gap between the two means you have fitted noise.

Lack of fit: $F = 2.27$ on 5 and 4 degrees of freedom, $p = 0.223$ — **not significant, so the quadratic is adequate**. This is the test that most published response surfaces omit, and it is the only one that asks whether the *model form* is right rather than whether the coefficients are non-zero. A significant lack of fit with a high R^2 means the surface is real and your equation is the wrong shape for it.

Canonical analysis: write the quadratic terms as a matrix **B** and solve $x^* = -\tfrac{1}{2}B^{-1}b$ for the stationary point. Here that is **x* = (0.505, 0.323, −0.171)** with a predicted response of **43.12**, against a true optimum at (0.510, 0.313, −0.247) — recovered to within the noise. The eigenvalues of **B** are **−2.945, −1.458, −0.642**, all negative, so the stationary point is a **maximum** rather than a saddle or a minimum.

The eigenvalues say more than that. The smallest in magnitude, −0.642, with eigenvector **(0.12, −0.05, 0.99)**, identifies the **flattest direction** — a ridge along x_3 in which the response changes very little. That is operationally valuable: along a ridge you can move a factor freely to satisfy some other constraint — cost, a regulatory limit, an equipment limit — at almost no cost in response. A response surface's eigenvectors are where the process flexibility is, and almost nobody reports them.

Degrees of freedom	`df_error = N − 1 − k − k(k−1)/2 − …; centre points are the cheapest df`
Resolution	`III screening · IV clean main effects · V clean two-factor interactions`
Steepest ascent	`move in proportion to coded coefficients until the response stops improving`
Rotatable α	`(2k)1/4 = 1.414, 1.682, 2.000 at k = 2, 3, 4`
Adequacy	`adjusted R², then lack of fit against pure error from the centre points`
Canonical analysis	`x* = −½B⁻¹b; eigenvalue signs classify it; the smallest gives the ridge`

What to open a book for. Box, Hunter & Hunter for screening, aliasing and steepest ascent — chapters 5 to 12; Myers,

Montgomery & Anderson-Cook, *Response Surface Methodology*, for canonical analysis and design comparison, which is where the eigenvalue treatment above comes from.

Problems MP1–MP4 and DE1–DE4

MP1 · extends chapter V1 · 1.5 h

What do the beads cost you?

1. For solids holdup of 0, 10, 20 and 30%, compute the relative k_La from $(1 - \varepsilon_s)^{2.5}$ and the working volume lost.
2. With 10% gas holdup as well, what fraction of the vessel is broth at 20% solids?
3. Compute the terminal velocity of a 2 mm and a 4 mm bead at a 100 kg/m³ density difference.
4. At a superficial liquid velocity of 0.01 m/s, what reactor do you have, and what follows?

Worked solution. Relative k_La is 1.00, 0.74, **0.57** and 0.41 at the four loadings, so 20% solids costs **43% of oxygen transfer** and 20% of the volume. With 10% gas as well, only **70%** of the vessel is broth. Terminal velocities are **0.077 and 0.109 m/s**, both far above 0.01 m/s — so at that velocity the bed is **packed, not fluidised**. That matters because a packed bed channels, compacts under its own weight, and cannot be aerated through the bed without disrupting the flow: you must either raise the velocity by an order of magnitude to fluidise, aerate externally in a loop, or accept an anaerobic or microaerobic process.

What it teaches: adding a solid phase costs transfer and volume simultaneously, and the liquid velocity decides which reactor you actually have rather than which one you intended.

MP2 · extends chapter V1 · 1.5 h

How thick should a biofilm be?

1. Compute the oxygen penetration depth at biofilm densities of 50, 100 and 200 kg/m³, taking 4 mmol $O_2 \cdot gDW^{-1} \cdot h^{-1}$, $D = 1.5 \times 10^{-9}$ m²/s and $C^* = 0.21$ mM.
2. At 100 kg/m³, what fraction of a 50, 100, 300 and 1,000 µm film is active?
3. What is the design objective, and what are the two ways to achieve it?
4. What does the dead interior do, and why is that worse than being idle?

Worked solution. Penetration is **106, 75 and 53 µm** at the three densities. At 100 kg/m³ a 50 µm film is fully active, a 100 µm film 75%, a 300 µm film **25%** and a 1 mm film **8%** — so thicker is emphatically not better. The design objective is therefore **thickness control, not maximum attached biomass**, achieved either by shear that continuously sloughs the excess or by a support geometry that limits it.

The dead interior is worse than idle because it is anaerobic and metabolically active: it ferments, produces acids and gas, weakens the film's attachment, and eventually detaches as a plug that blocks the bed or appears in the product. A biofilm reactor that is not controlling thickness is accumulating a failure.

What it teaches: in any attached-growth system the active layer is tens of microns, so the engineering variable is thickness rather than quantity.

MP3 · extends chapter V1 · 2 h

Solid-state fermentation: the heat and water balance

1. At 2 kg O_2 per kg biomass and 460 kJ/mol O_2, compute the heat release for growth of 1, 3 and 10 $kg \cdot m^{-3} \cdot h^{-1}$, in $MJ \cdot m^{-3} \cdot h^{-1}$ and kW/m³.
2. If all of it leaves by evaporation, how much water per m³ per hour?
3. A bed at 50% moisture holds 250 kg of water per m³. How long until it dries at 1 and 3 $kg \cdot m^{-3} \cdot h^{-1}$?
4. With $k = 0.2$ $W \cdot m^{-1} \cdot K^{-1}$ and conduction only, compute the centre temperature rise at half-depths of 5, 15 and 30 cm for 1 kW/m³. What reactor designs does this force?

Worked solution. Heat is **28.8, 86.2 and 287.5 $MJ \cdot m^{-3} \cdot h^{-1}$ — that is 8.0, 24.0 and 79.9 kW/m³**, against 1 to 5 kW/m³ in a stirred tank. Evaporating all of it needs **12.7, 38.2 and 127.2 kg of water per m³ per hour**, so a bed holding 250 kg/m³ **dries out in 20 hours at the slow rate and 7 hours at the middle one**. That is the central fact of SSF: the process is limited by humidity and aeration, and water replenishment is a continuous operation rather than a formulation decision.

Conduction gives centre rises of **6 K at 5 cm, 56 K at 15 cm and 225 K at 30 cm** for only 1 kW/m³. The last two are impossible, which forces the reactor classes: **shallow trays** (passive, labour-heavy), **forced-aerated packed beds**

(deeper, with axial temperature and moisture gradients that must be managed), and **rotating drums or intermittently mixed beds** (uniform, at the cost of shear on mycelium and of mechanical complexity).

What it teaches: in solid-state fermentation the oxygen problem is replaced by a heat and water problem that is an order of magnitude harder, and bed depth is the variable it constrains.

`MP4` · extends chapter V2 · 1.5 h

Gel beads: leakage, stability and whether to bother

1. For beads of radius 1.0, 1.5 and 2.0 mm, compute the volume, the surface-to-volume ratio, and the outer 50 µm as a fraction of volume.
2. Taking that outer layer as the escape zone and $\mu = 0.3\ h^{-1}$, compute free-cell generation per gram of immobilised biomass per hour.
3. Name the two medium components that dissolve calcium alginate, and three fixes.
4. Given part 3 chapter 48's result that immobilisation cost productivity, state the four conditions under which it still pays.

Worked solution. The outer 50 µm is **14% of a 1 mm bead's volume, 10% of a 1.5 mm and 7% of a 2 mm**, and surface-to-volume falls from 3,000 to 1,500 m^{-1}. At $\mu = 0.3\ h^{-1}$ the free-cell generation is **0.042 g per g of immobilised biomass per hour for a 1 mm bead and 0.022 for a 2 mm** — a day of that is a substantial free population, so an "immobilised" process is partly a free-cell process and your assays will mostly see the escapees.

Phosphate and citrate chelate the calcium and soften the beads; ordinary M9 or LB will do it over days. Three fixes: minimise phosphate and feed it separately; include a few millimolar calcium in the medium; or cross-link with barium or strontium, accepting the toxicity and regulatory cost. The four conditions that justify immobilisation at all: **operating above the washout dilution rate**; **protecting cells from shear, solvent or inhibitor**; **reusing an expensive catalyst**; and **avoiding a centrifugation step**. Absent all four, chapter 48's productivity comparison says do not.

What it teaches: immobilisation is a retention technique with three running costs — diffusion, leakage and gel stability — and it needs a specific reason, not a general one.

`DE1` · extends chapter V4 · 1 h

Degrees of freedom: will this design even work?

1. For a 2^2, 2^3, 2^4 and 2^5 design with the replication and centre points stated in chapter V4, compute the degrees of freedom spent and left.
2. Which design leaves too few, and what are the two standard remedies?
3. How many centre points would you add to a 2^4, and what do they buy?
4. Why can an unreplicated factorial still be analysed?

Worked solution. Spent and left: 2^2 with four replicates, 4 and **12**; 2^3 with two replicates and three centre points, 7 and **12**; 2^4 unreplicated with five centre points, 11 and **10**; 2^4 unreplicated with none, 11 and **5**; 2^5 unreplicated, 16 and **16**.

The bare 2^4 is the weak one at five error degrees of freedom. Two remedies: **pool the high-order interactions into error**, on the argument that three- and four-factor interactions are rarely real, which recovers degrees of freedom at the cost of an assumption; or **add centre points**, which give pure error independent of the model. I would add **four to five centre points** to a 2^4 — they cost 25% more runs and buy an independent error estimate plus a curvature test that tells you whether to go quadratic at all. An unreplicated factorial is still analysable without any error estimate by a **normal plot of the effects**: real effects fall off the line that the inactive ones define.

What it teaches: count the degrees of freedom before running anything, because a design with none left cannot be tested, only described.

`DE2` · extends chapter V4 · 1.5 h

Screening five factors, and knowing what you confounded

1. Build a 2^{5-1} with E = ABCD and write the defining relation and the alias of A and of AB.
2. From the simulated data in chapter V4, rank the estimated effects and decide which are real.
3. Compare with a 2^{7-4} in eight runs: what is confounded, and when is that acceptable?
4. How many runs for a full 2^7, a half fraction, and the eighth fraction?

Worked solution. With E = ABCD the defining relation is I = ABCDE, so A = BCDE and AB = CDE: every main effect is

aliased with a four-factor interaction and every two-factor interaction with a three-factor one — **resolution V**, and for practical purposes everything of interest is clean. The estimated effects rank **A +4.11, D −2.67, B +2.32, AB +1.64**, then nothing above 0.29 — so A, B, D and the AB interaction are real and C and E are noise, which matches the truth used to generate the data.

A 2^{7-4} in eight runs has A = BE = CF = DG: **main effects confounded with two-factor interactions**, resolution III. That is acceptable only for a first screen where you expect a few large main effects and will follow up — and it is dangerous when interactions are likely, because a real AB can appear as a main effect of a factor that does nothing. Run counts: **128 full, 64 half, 8 for the eighth fraction**, which is the trade in one line.

What it teaches: a fractional design is defined by what it confounds, and you must be able to write the aliases before you trust the effects.

`DE3` · extends chapter V4 · 1 h

Walk the path of steepest ascent

1. Given coded first-order coefficients of 1.0 for A, 0.6 for B and −0.75 for D, write the direction of steepest ascent.
2. Convert to real units with step sizes of 5 g/L, 0.5 °C and 100 rpm per coded unit, for steps 1 to 5.
3. How do you know when to stop walking?
4. What happens if you skip this step and place a central composite design at the screening centre?

Worked solution. The direction is proportional to the coefficients, so normalising on the largest: **one coded unit in A for every 0.60 in B and −0.75 in D**. In real units that is +5 g/L, +0.30 °C and −75 rpm per step, reaching +25 g/L, +1.50 °C and −375 rpm at step five. You run **single, unreplicated experiments along the path** and stop when the response stops improving — two consecutive non-improving points is the usual rule — and that point is where the response-surface design goes.

Skipping it is the commonest waste in an optimisation campaign. A central composite design centred far from the optimum fits an almost-planar surface: the quadratic terms come out small and insignificant, the stationary point lands outside the design region where the model has no support, and the whole 19-run design tells you only to move in a direction the 16-run screen had already given you.

What it teaches: steepest ascent is cheap, unreplicated and the step that makes a response-surface design worth running.

`DE4` · extends chapter V4 · 2 h

Read a response surface: adequacy, optimum, and the ridge

1. From the fitted equation in chapter V4, report R^2, adjusted R^2 and the model F test, and say which you would quote.
2. Carry out the lack-of-fit test and state what a significant result would have meant.
3. Solve for the stationary point and classify it from the eigenvalues.
4. Identify the flattest direction and say what you would do with it.

Worked solution. $R^2 = 0.9941$, adjusted R^2 = **0.9882**, F = 168.0 on 9 and 9 df, $p = 6 \times 10^{-9}$. Quote the **adjusted** R^2, because plain R^2 rises with every term added; a large gap between them is the signature of fitted noise. Lack of fit gives **F = 2.27, p = 0.223** — not significant, so the quadratic form is adequate. A *significant* lack of fit with a high R^2 would have meant the surface is real and the equation is the wrong shape for it, and the response would be to add higher-order terms, transform a variable, or shrink the design region.

The stationary point is **x* = (0.505, 0.323, −0.171)**, predicted response **43.12**, against a true optimum of (0.510, 0.313, −0.247) — recovered within the noise. Eigenvalues are **−2.945, −1.458, −0.642**, all negative, so it is a **maximum**.

The flattest direction is the eigenvector of the smallest-magnitude eigenvalue, **(0.12, −0.05, 0.99)** — essentially x_3. Along that ridge the response barely changes, so **x_3 can be set by something other than the response**: cost, an equipment limit, a regulatory constraint, or operator convenience. Reporting the ridge is how a response surface becomes operationally useful rather than a single set-point, and almost no published surface does it.

What it teaches: the optimum is the least interesting output of a response surface; the adequacy test and the flat directions are what you actually use.

Closing notes to part 20

Direct answers to what you asked

Multiphase fermentation — the two-liquid-phase and gas-fermentation cases were covered (part 3 chapters 45 to 47, part 5 chapter 55). Three-phase gas–liquid–solid and solid-state were **not**, and biofilms were a passing mention. Chapter V1 and problems MP1 to MP3 cover them now.

In situ gel fermentation — the transport side was covered properly in part 3 chapter 48, and the *process* side was not: bead production, cell leakage, gel dissolution by phosphate, and the four conditions under which immobilisation pays. Chapter V2 and problem MP4.

Metabolic pathway modification — covered thoroughly but scattered over twelve parts. Chapter V3 is the consolidation: nine interventions, the number that decides each, where it is taught, and the order to try them in.

Design of experiments and response-surface methodology — degrees of freedom, factorials and blocking were covered; the *method* was not. Chapter V4 now runs screening → aliasing → steepest ascent → design choice → ANOVA → lack of fit → canonical analysis, with every number computed, and problems DE1 to DE4 work each stage.

The numbers worth keeping

- 20% solids loading costs **43% of k_La** and 20% of the working volume.
- A 2 mm bead's terminal velocity is **0.077 m/s**, so a 0.01 m/s column is a packed bed.
- A biofilm's active layer is **75 µm** at 100 kg/m³; a 1 mm film is **92% inactive**.
- Solid-state fermentation at 3 kg·m^{-3}·h^{-1} releases **24 kW/m³** and dries its bed in **7 hours**.
- A 15 cm half-depth bed at 1 kW/m³ runs **56 K** hot at the centre.
- A 1 mm alginate bead leaks **0.042 g of free cells per gram per hour** at $\mu = 0.3$.
- Feedback relief gave **+43%** where a tenfold overexpression gave +8.3%.
- A rotatable CCD's axial distance is **1.414, 1.682, 2.000** at two, three and four factors.
- The fitted surface recovered the optimum to **(0.505, 0.323, −0.171)** against a true (0.510, 0.313, −0.247), with lack of fit $p = 0.223$.

On the numbers

All computed here. The assumptions to challenge: the $(1 - \varepsilon_s)^{2.5}$ solids correlation, which is a rough empirical form and varies with bead density and sparger; the 50 µm escape zone in the leakage model, which is a plausible figure rather than a measurement; the bed conductivity of 0.2 W·m^{-1}·K^{-1} and the 2 kg O_2 per kg of biomass in the SSF balance; and the biofilm's wet-biomass density. The *forms* of all these relations are standard; the coefficients are yours to measure, and they belong on sheet 10 of the lookup workbook.

Where the set stands

Source	Problems
Curriculum appendix A1–W3	54
Parts 15–19 (X, Y, Z, MC, R, PP, PR)	69
Part 20 (MP1–MP4, DE1–DE4)	8
Total	**131**, with G2 and the two construction exercises making 133 in the tracker

Part 21 — Against a reviewer's checklist: the five things that were thin

You asked me to adjust the module to a PI's stated tiers. I checked the module against every item on that list before writing anything. Most of it was already covered, in several cases heavily. Five things were not, and one of them sits in Tier 1.

The audit against the three tiers

Tier	Requirement	Coverage	Verdict
1	Central carbon, redox balancing, theoretical yields	307 mentions; parts 2, 14, 15, 18	Strong
1	CRISPR Cas9, multiplex, markerless, integration	47 and 58 mentions; part 7, PP5, PP6	Covered
1	**Cas12a, base editors, prime editing**	3 and 1 mentions	**Thin — chapter W1**
1	Enzyme kinetics, directed evolution, active-site engineering	Kinetics strong; evolution and hotspots 8 mentions	Adequate, strengthened in W1's closing
1	k_La, OUR, fed-batch, burden	174 mentions; parts 11, 19, 20	Strong
2	FBA, COBRApy, genome-scale models	42 mentions; part 10, MC1–MC5, Y1	Strong
2	Multi-omics literacy	154 mentions; part 4, chapter S5, Z5, Z6, Z9	Strong
2	AlphaFold, ESM, ML-guided evolution	14 and 38 mentions; part 8, P1, P2, Y5	Adequate
2	Biosensors	33 mentions; part 8 ch 26	Covered
2	**FACS and droplet microfluidics**	4 and 10 mentions	**Thin — chapter W3**
3	Python automation, NGS pipelines, statistical DoE	Heavy; part 20 chapters V4, DE1–DE4	Strong
3	Non-conventional chassis: autotrophs, methanotrophs, thermophiles	42 mentions; part 6, X4	Covered
3	Techno-economic assessment	Part 4 ch 49, G1, X22, PP12	Strong
3	**Life cycle analysis**	7 mentions	**Thin — chapter W4**
—	Named products: terpenoids, biopolymers, rare amino acids	39, 568, 45 mentions	Strong
—	**Named products: flavonoids and polyphenols**	5 mentions	**Thin — chapter W2**

Four chapters, W1 to W4, and eight problems, EV1 to EV8, taking the set to **141**. One observation worth making: the reviewer's list is close to what the module already contained, which is a reasonable sign that the curriculum was well aimed. The gaps are specific rather than structural.

Chapter W1 · extends weeks 23 and 24 · Tier 1

The editing toolkit in full: Cas9, Cas12a, base editors, prime editing

Part 7 taught Cas9 and PP5 taught the counterselection arithmetic. A reviewer expects you to know which editor for which job, and that is a question about PAM, editing window and what repair pathway the host actually has.

The four tools, compared on what decides the choice

	Cas9	Cas12a (Cpf1)	Base editors	Prime editing
PAM	5′-NGG-3′, 3′ of the protospacer	5′-TTTV-3′, 5′ of it	Inherited from the Cas it is built on	Same
Cut	Blunt, 3 bp from the PAM	**Staggered, 4–5 nt overhangs**, distal to the PAM	Nick or none	Nick on one strand
Guide	crRNA + tracrRNA, or a 100-nt sgRNA	**crRNA alone, ~43 nt**	As the parent Cas	**pegRNA**: guide + primer binding site + reverse-transcription template
Multiplexing	One guide per promoter, or an array with processing	**Self-processes its own array** — one transcript, many guides	As parent	Hard: pegRNAs are long
Makes	A double-strand break	A double-strand break	**C→T (CBE) or A→G (ABE)**, no break	**Any substitution, small insertion or deletion**
Needs a donor?	Yes, in bacteria	Yes	**No**	No — the template is in the pegRNA

The fact that governs everything in bacteria: most bacteria have no efficient non-homologous end joining, so a double-strand break is **lethal, not mutagenic**. That is why Cas9 and Cas12a in *E. coli* are used as a *counterselection* — recombineering supplies the edit and the nuclease kills the unedited, which is PP5's arithmetic. In organisms that do have end joining (many actinomycetes and mycobacteria) the same cut gives small indels directly, and the strategy is entirely different. Check whether your host has *ku* and *ligD* before choosing.

Where each one wins

Cas12a for multiplexing. Because it processes its own CRISPR array, a single transcript can carry six or ten guides, where Cas9 needs a promoter or a processing system per guide. Its AT-rich PAM also complements Cas9's GC-rich one, which matters in a high-AT or high-GC genome where NGG sites may be scarce near your target. Its staggered cut leaves overhangs that favour insertion of a matching donor.

Base editors for point mutations at scale and for essential genes. A cytosine base editor makes C→T within a window of roughly positions 4 to 8 of the protospacer; an adenine base editor makes A→G similarly. No break, no donor, no counterselection — which is why base editing is the tool for making a **premature stop codon** (CAA, CAG, CGA → TAA, TAG, TGA) to knock a gene out without deleting it, and for installing a point mutation in an essential gene where a deletion would be lethal. The constraints are real and arithmetic: the target base must fall in the window, which means a PAM at the right distance; and **bystander editing** of other C's or A's in the window is the failure mode — a window containing three cytosines gives a mixture of edited products, which EV2 computes.

Prime editing for anything the other three cannot do: a substitution outside a base editor's chemistry, a small insertion, a short deletion, or a precise change with no bystanders. The cost is efficiency and complexity — the pegRNA must be designed with a primer binding site and template whose lengths are tuned empirically, and reported efficiencies in bacteria are low enough that a selection or screen is usually still needed.

Choosing, as a decision list

What you need	Tool	Why
Clean deletion of a gene or operon	λ-Red or RecET + Cas9 counterselection	Recombineering makes it, the nuclease enriches it (PP5, PP6)
Six knockouts at once	Cas12a array, or MAGE	Self-processing array; or MAGE's per-cycle arithmetic (PP5)
Knock out an essential gene's function	CBE, making a stop codon	No break, no lethality; or CRISPRi for knockdown
One point mutation, no scar	Base editor if in-window, else prime editing	Window and PAM decide it (EV2)

Tune expression without editing	CRISPRi with dCas9	Titratable, reversible, multiplexable
Insert a 3 kb cassette	Recombineering with long homology, or integrase	Beyond prime editing's size range

Two honest limits. Reported efficiencies for base and prime editing in non-model bacteria are **recalled and variable** — verify against a recent review for your host before planning a campaign. And the choice between editors is often settled by what is already working in the laboratory you join, which is a legitimate reason and worth saying out loud in an interview.

Chapter W2 · extends weeks 53 to 55 · the named products

Flavonoids: a second aromatic pathway with a different bottleneck

The module did terpenoids, polymers and amino acids. Flavonoids are the fourth product class a reviewer named, and they fail for a reason the other three do not: malonyl-CoA.

The pathway, and the yield ceiling

Naringenin ($C_{15}H_{12}O_5$, 272.26) is the branch point to most flavonoids. The route is: tyrosine → *p*-coumarate (tyrosine ammonia lyase, TAL) → *p*-coumaroyl-CoA (4-coumarate-CoA ligase) → condensation with **three malonyl-CoA** by chalcone synthase → naringenin chalcone → naringenin (chalcone isomerase). Downstream, flavonols, anthocyanins and so on come from oxidations and glycosylations, most of them by cytochrome P450s.

The carbon accounting has two tributaries. The C_9 aromatic ring comes from tyrosine, which needs erythrose-4-phosphate (4 C) plus two PEP (6 C), losing one CO_2 in the shikimate route — so **1.67 glucose per ring**. The remaining six carbons come from three acetyl units via malonyl-CoA, which is three pyruvate losing three CO_2 at pyruvate dehydrogenase — **1.5 glucose**. Total 3.17 glucose per naringenin, giving a mass yield ceiling of

```
272.26 / (3.17 × 180.156) = 0.477 g/g
```

Reported titers are typically tens to hundreds of milligrams per litre†, so a 0.5 g/L result from 20 g/L of glucose is about **5% of the ceiling** — an order of magnitude below where terpenoids and amino acids sit, and that gap is the subject.

Why malonyl-CoA is the bottleneck

Malonyl-CoA is made by acetyl-CoA carboxylase and consumed almost entirely by fatty-acid synthesis, where it is kept at a very low pool and tightly regulated. A flavonoid pathway needs three of them per product, and it is competing with the cell's own membrane synthesis for a metabolite that the cell deliberately does not accumulate. The demand arithmetic: 5 g/L of naringenin over 48 hours needs **55 mmol/L of malonyl-CoA**, which at 10 gDW/L is **0.115 mmol·gDW^{-1}·h^{-1}** — comparable with the cell's own fatty-acid demand of roughly 0.2. You are asking the cell to roughly double its malonyl-CoA flux and divert the increase.

The interventions, in the playbook form of part 20 chapter V3: **overexpress acetyl-CoA carboxylase** (the direct supply route, with a biotin requirement and an ATP cost); **inhibit fatty-acid synthesis** with cerulenin, which works in a flask and is far too expensive at scale, or by titrating *fabB* or *fabF* with CRISPRi, which is the scalable version; **raise acetyl-CoA** by the routes of part 6 chapter 58, including phosphoketolase; and **stage the process**, growing first and diverting malonyl-CoA later, which is PP8's two-stage calculation applied to a precursor rather than a product.

The second problem: cytochrome P450s

Most flavonoid decorations are P450 oxidations, and P450s are hard in bacteria for three compounding reasons. They are **membrane-anchored** in plants, so expression in *E. coli* needs truncation or a fusion and typically loses an order of magnitude or more of activity†. They need a **redox partner** — a cytochrome P450 reductase delivering two electrons from NADPH per turnover — which must be co-expressed at a matched ratio. And they **uncouple**: when the electron flow is not coupled to substrate oxidation, the electrons leak to peroxide or water, so the NADPH cost per product rises. At 100% coupling a turnover costs 2 NADPH; at 50% it costs 4; at **20% coupling it costs 10**, and a three-step P450 cascade at that coupling costs **30 NADPH per molecule of product** — which is why a decorated flavonoid is far harder than naringenin itself, and why yeast, with native ER membranes and a native reductase, is often the better host for the decorated compounds even though *E. coli* makes the core faster.

Naringenin ceiling	3.17 glucose per molecule → 0.477 g/g; reported titers ≈5% of it
Malonyl-CoA demand	3 per product; 5 g/L needs 0.115 mmol·gDW^{-1}·h^{-1} against ~0.2 for fatty acids
P450 cost	2 NADPH per turnover at full coupling; 10 at 20%; 30 for a three-step cascade
Host logic	*E. coli* for the core skeleton, yeast for the P450 decorations

What to open a book for. A review of microbial flavonoid production for the titers and the malonyl-CoA strategies — and read it specifically for what each group did about malonyl-CoA, which is the only question that separates the papers. For P450s, a review of their functional expression in microbial hosts.

Chapter W3 · extends weeks 29 and 30 · Tier 2

Sorting and droplets: screening at 10^7 instead of 10^3

A plate reader screens thousands. A sorter screens hundreds of millions. The arithmetic of what that buys — and of what it costs in false positives — is Tier 2 material and the module had four mentions of it.

FACS: throughput and enrichment

A cell sorter processes events at kilohertz to tens of kilohertz. At 5,000 events per second a four-hour run sees **72 million cells**; at 20,000 it sees **288 million**; at 50,000, **720 million**. Against problem X23's campaign of 387 strains and part 8 chapter 29's library of 30,000, that is a different regime entirely — the constraint stops being throughput and becomes **whether the thing you want is visible as fluorescence**.

Enrichment per round is the Bayes calculation. With a true-positive frequency f and a sort that is correct a fraction p of the time, the post-sort frequency is fp/(fp + (1−f)(1−p)):

Starting frequency	Purity 0.90	Purity 0.99
10^{-5}	9.0×10^{-5} (9×)	**9.9×10^{-4} (99×)**
10^{-4}	9.0×10^{-4} (9×)	9.8×10^{-3} (98×)
10^{-3}	8.9×10^{-3} (9×)	9.0×10^{-2} (90×)

Two readings. Enrichment per round is roughly p/(1−p) and **almost independent of the starting frequency** — which is why rounds compound: two rounds at 99% purity take 10^{-5} to **8.9%**, and three take it to dominance. And **purity matters far more than throughput**: going from 90% to 99% is a tenfold gain per round, where doubling the event rate is a twofold gain once.

The hard part is not the machine. It is **linking the product to fluorescence**: a transcription-factor biosensor (part 8 chapter 26), an enzyme-coupled reaction producing a fluorophore, or a FRET sensor. Each brings its own failure modes — the sensor can be gamed by a mutation that activates it directly rather than by making product, which is the single most common way a sorted library yields nothing. A **negative control strain carrying the sensor and no pathway**, sorted alongside, is what detects that.

Droplets: Poisson loading and what it buys

Water-in-oil droplets give each cell a private compartment, so a **secreted** product stays with its producer — which FACS cannot do, since a secreted product diffuses away. Loading follows Poisson statistics at mean occupancy λ:

λ	Empty	Exactly one cell	More than one	Singles as a share of occupied
0.1	90.5%	9.0%	0.5%	**95.1%**
0.3	74.1%	22.2%	3.7%	85.7%
1.0	36.8%	36.8%	26.4%	58.2%

The trade is between wasting droplets and corrupting them. At λ = 0.1 you discard 90% of your droplets as empty but **95% of occupied droplets hold exactly one cell**, so a positive droplet identifies a single genotype. At λ = 1.0 you use the droplets efficiently and 26% hold more than one cell, so a positive result is ambiguous and a strong producer can carry a passenger. For a *screening* experiment where you will re-isolate and re-test, λ near 0.3 is a reasonable compromise; for a *selection* where the droplet content is pooled and sequenced, λ must be low.

Volumes make it practical: 10^7 droplets of 50 pL is only **0.5 mL of emulsion**, which is why droplet screening at 10^7 is a bench-scale experiment while the equivalent in microtitre plates is 100,000 plates. What it costs is a bespoke device, a fluorogenic assay that works inside a droplet, and considerable trouble-shooting — which is why it belongs in Tier 2 rather than Tier 1.

FACS throughput	`72–720 million cells in a four-hour run at 5–50 kHz`
Enrichment	`fp/(fp + (1−f)(1−p)) ≈ p/(1−p) per round; 99× at 99% purity`
Droplet loading	`Poisson: λ = 0.1 gives 95% single-occupancy among occupied drops`
Scale	`$10^7$ droplets at 50 pL = 0.5 mL of emulsion`

What to open a book for. A review of biosensor-driven high-throughput screening in metabolic engineering; and, for droplets, a methods review on microfluidic droplet screening for enzymes — read both for the failure modes rather than the throughput claims.

Chapter W4 · extends week 49 · Tier 3

Life cycle analysis beside the techno-economic model

The module costed processes in dollars. A reviewer in industrial biotechnology expects you to cost them in carbon too — and the two calculations use the same inputs and reach different conclusions.

The four things that define an LCA

The functional unit — what you are comparing. "1 kg of product at the factory gate" is the usual cradle-to-gate choice; cradle-to-grave adds use and disposal, which for a biopolymer is where the interesting part is. **The system boundary** — whether you include agriculture, land-use change, transport, utilities, waste treatment. **The impact categories** — global warming potential is the one everyone reports, but eutrophication and water use frequently decide against a bio-route, because agriculture drives both. And **allocation** — how burden is split between a product and its co-products, by mass, energy content or economic value, which is the single most contestable choice in any LCA.

The calculation, with the module's own numbers

Taking illustrative factors of 0.6 kg CO_2e per kg of glucose, 0.4 per kWh of electricity and 0.07 per MJ of steam†, and PP12's propanediol process:

Case	Sugar	Steam for water removal	Power	Total
Y 0.40, titer 50 g/L	2.50 kg (1.50)	42.9 MJ (3.01)	2.0 kWh (0.80)	**5.31 kg CO_2e/kg**
Y 0.40, titer 135 g/L	2.50 kg (1.50)	14.5 MJ (1.01)	2.0 kWh (0.80)	**3.31**
Y 0.51, titer 135 g/L	1.96 kg (1.18)	14.5 MJ (1.01)	2.0 kWh (0.80)	**2.99**

Read the first two rows together. **Raising titer from 50 to 135 g/L cuts the carbon footprint by 38%** while changing nothing about the strain's yield — because at low titer the steam to evaporate water is the largest single term, larger than the sugar. That is the same conclusion PP12 reached in dollars, reached independently in carbon, and it is the most useful thing an engineer can take from an LCA: **titer is a sustainability variable, not just an economic one**.

Against a fossil benchmark of roughly 2 to 4 kg CO_2e per kg for a commodity chemical†, the low-titer case *loses* and the high-titer case is competitive. A bio-process is not automatically lower-carbon, and saying so in an interview is more convincing than assuming the opposite.

The two arguments that change the answer

Biogenic carbon. The product contains carbon that was fixed from the atmosphere. For a $C_3H_8O_2$ product that is **1.74 kg of CO_2 per kg embodied** in the molecule, and whether you may claim it depends on the end of life: if the product is burned or composted it returns, and a cradle-to-grave analysis cancels it; if it is a durable polymer in long-term use it is genuine sequestration for that period. This is where biopolymer LCAs are won and lost, and where they are most often overstated.

Allocation. A process yielding 1 kg of product and 0.5 kg of animal feed can assign the upstream burden either entirely to the product or partly to the feed. Allocating a third to the co-product takes the 135 g/L case from 3.31 to **2.82 kg CO_2e/kg**, and allocating half takes it to **2.56** — a 23% improvement from a bookkeeping decision with no change to the process at all. Hence the ISO guidance to *avoid* allocation where possible by expanding the system boundary, and the practical requirement to report results under more than one allocation rule so a reader can see the sensitivity.

Definition	`functional unit, system boundary, impact categories, allocation`
Titer effect	`50 → 135 g/L cuts GWP 38%, through steam rather than sugar`
Benchmark	`fossil commodity 2–4 kg CO₂e/kg: bio wins only at high titer`
Biogenic carbon	`1.74 kg CO₂ per kg of a C₃H₈O₂ product; claimable only with a durable end of life`
Allocation	`33% to a co-product moves 3.31 → 2.82 kg CO₂e/kg with no process change`

What to open a book for. ISO 14040 and 14044 for the framework, skimmed for the definitions rather than read; a published LCA of a bio-based chemical for how a real boundary is drawn; and the ecoinvent or GREET documentation for where emission factors come from — the factors above are illustrative and should be replaced by sourced ones.

Problems EV1 to EV8

EV1 · chapter W1 · 1.5 h

Choose the editor for six jobs

1. For each of six tasks — a clean operon deletion, six simultaneous knockouts, inactivating an essential gene, one scarless point mutation, tunable knockdown, and a 3 kb insertion — name the tool and justify it.
2. Which two tasks change answer completely if the host has *ku* and *ligD*?
3. Why is a double-strand break lethal rather than mutagenic in *E. coli*?
4. Your target has no NGG PAM within 10 bp. What are your three options?

Worked solution. Operon deletion: recombineering with Cas9 counterselection. Six knockouts: a Cas12a array (self-processing) or MAGE. Essential gene: a cytosine base editor making a stop codon, or CRISPRi for knockdown rather than knockout. Scarless point mutation: base editor if the target falls in the window, otherwise prime editing. Tunable knockdown: CRISPRi with dCas9. 3 kb insertion: recombineering with long homology arms or a site-specific integrase — beyond prime editing's size range.

The first two change completely with *ku* and *ligD*: in an end-joining host a cut alone produces indels, so you can knock out a gene with a guide and no donor at all. A break is lethal in *E. coli* because there is no end-joining pathway and homologous recombination needs a template — without one the chromosome is broken irreparably. With no NGG nearby: use **Cas12a** and its TTTV PAM, use a **PAM-relaxed Cas9 variant**, or **move the edit** — a synonymous change a few bases away is often as good, and is how domestication is done anyway (part 7 chapter 21).

What it teaches: editor choice is decided by PAM availability, repair pathways present in the host, and whether a break can be tolerated — not by which enzyme is newest.

EV2 · chapter W1 · 1.5 h

Base editing: windows, stop codons and bystanders

1. A cytosine base editor edits C→T at protospacer positions 4–8. Your target C is at position 6, and there are C's at positions 4 and 9. What products do you expect?
2. Which codons can a CBE turn into a stop, and which into a non-stop missense?
3. If each in-window C is edited independently with probability 0.6, what fraction of molecules carry the intended edit alone?
4. How would you remove the bystander problem without changing the editor?

Worked solution. The C at position 9 is outside the window and should be untouched; the C at position 4 is inside it and is a bystander. With independent editing at p = 0.6, the fraction carrying the target edit *and not* the bystander is 0.6 × 0.4 = **24%**, against 36% carrying both, 24% carrying only the bystander and 16% unedited. So **the single most likely outcome is the double edit**, and screening must sequence rather than assume.

A CBE makes stops from **CAA → TAA, CAG → TAG and CGA → TGA** — all on the sense strand, plus the reverse-complement equivalents when the guide targets the other strand, which doubles your options. Without changing the editor, the bystander problem is solved by **choosing a different guide** whose window places the target C alone, which is almost always possible given the PAM density, or by exploiting the genetic code: a bystander that is **synonymous** does not matter, so check the reading frame before rejecting the guide.

What it teaches: base editing is a window problem, and the guide is chosen by what else falls in the window, not just by what you want to change.

EV3 · chapter W2 · 2 h

The naringenin ceiling and where the gap is

1. Account for the carbon in naringenin from both tributaries and compute the glucose requirement.
2. Give the mass yield ceiling and express a 0.5 g/L titer from 20 g/L glucose as a fraction of it.
3. Compute the malonyl-CoA demand for 1 and 5 g/L over 48 h at 10 gDW/L, against a fatty-acid demand of 0.2 mmol·gDW^{-1}·h^{-1}.
4. Rank four interventions and say which you would test first.

Worked solution. The C_9 ring costs 1.67 glucose (erythrose-4-phosphate plus two PEP, one CO_2 lost in shikimate) and

the three acetyl units cost 1.5 (three pyruvate, three CO_2 lost at pyruvate dehydrogenase): **3.17 glucose per naringenin**, a ceiling of **0.477 g/g**. A 0.5 g/L titer from 20 g/L is 0.025 g/g, which is **5.2% of the ceiling** — the widest gap of any product in this module.

Malonyl-CoA demand is **0.023 mmol·gDW^{-1}·h^{-1} at 1 g/L and 0.115 at 5 g/L**, against roughly 0.2 for the cell's own fatty acids: at 5 g/L you are asking for a 57% increase in total malonyl-CoA flux, all of it diverted. Ranked: **titrate fabB/fabF with CRISPRi first** — it is reversible, tunable and tests the hypothesis that malonyl-CoA is limiting in one experiment; then **overexpress acetyl-CoA carboxylase**, which adds supply and costs ATP and biotin; then **raise acetyl-CoA** by phosphoketolase or the routes of part 6 chapter 58; and **stage the process** last, since that is architectural. Testing CRISPRi first is the cheap diagnostic: if knocking down fatty-acid synthesis does not move the titer, malonyl-CoA was not the limit and the other three are wasted work.

What it teaches: a product can be carbon-cheap and precursor-poor, and the first experiment should test which of the two the limitation actually is.

EV4 · chapter W2 · 1.5 h

What a P450 cascade costs

1. Compute NADPH per turnover at coupling efficiencies of 100%, 50% and 20%.
2. A decorated flavonoid needs three P450 steps at 20% coupling. What is the NADPH bill per product?
3. Compare that with the biomass NADPH demand of 13.03 mmol/gDW from part 14 chapter B5.
4. Give three reasons P450s underperform in *E. coli*, and say when you would switch host.

Worked solution. Two electrons per turnover means **2 NADPH at full coupling, 4 at 50% and 10 at 20%**. A three-step cascade at 20% coupling is **30 NADPH per product molecule**. For perspective, making a gram of biomass costs 13.03 mmol of NADPH; at 1 mmol/L of product (0.27 g/L of a 272 Da compound) the cascade would consume 30 mmol/L — more than twice the NADPH in a gram of cells per litre, which makes the cofactor supply the binding constraint rather than carbon.

Three reasons for underperformance: **membrane anchoring**, since a plant P450 is ER-bound and must be truncated or fused for *E. coli*, usually losing activity; **redox partner mismatch**, since the native reductase is absent and a co-expressed one must be ratio-matched; and **uncoupling**, which both wastes NADPH and generates peroxide that damages the enzyme. Switch to yeast when **two or more P450 steps are required** — it has ER membranes, a native reductase and the trafficking machinery, and the loss in growth rate and genetic tractability is worth it at that point.

What it teaches: a P450's cost is set by its coupling efficiency, and an uncoupled cascade turns a carbon problem into a cofactor problem.

EV5 · chapter W3 · 1.5 h

Sorting: how many rounds, and what can fool you

1. How many cells does a 4-hour sort see at 5,000, 20,000 and 50,000 events per second?
2. Compute the post-sort frequency and enrichment from 10^{-5}, 10^{-4} and 10^{-3} at purities of 0.90 and 0.99.
3. Where does two rounds at 99% purity take a 10^{-5} variant?
4. Name the commonest way a sorted library yields nothing, and the control that detects it.

Worked solution. 72, 288 and 720 million cells in four hours. Enrichment is about 9× at 90% purity and about **99× at 99%**, almost independent of the starting frequency — so purity, not throughput, is the lever: a tenfold gain per round from improving the gate, against a twofold gain from doubling the event rate. Two rounds at 99% take 10^{-5} to **8.9%**, and a third makes it the majority.

The commonest failure is **sensor gaming**: a mutation that activates the reporter directly — in the promoter, the transcription factor, or the fluorescent protein — is selected far more easily than one that raises product, because it needs no metabolic change. The control that detects it is a **sensor-only strain** with no pathway, sorted in the same run: anything it yields is a false positive by construction. The follow-up is to re-test sorted clones for *product* by chromatography, not for fluorescence, which is the X17 and R1 measurement.

What it teaches: sorting enrichment compounds with purity, and the biology that defeats it is a sensor that can be satisfied without making product.

EV6 · chapter W3 · 1 h

Droplets: choose the loading

1. At mean occupancies of 0.1, 0.3 and 1.0, compute the empty, single and multiple fractions and the single share of occupied droplets.
2. Which λ for a screen you will re-isolate from, and which for a pooled selection you will sequence?
3. Compute the emulsion volume for 10^7 droplets at 50 and 20 pL.
4. What can a droplet screen do that FACS cannot, and at what cost?

Worked solution. At λ = 0.1: 90.5% empty, 9.0% single, 0.5% multiple, and **95.1% of occupied droplets hold exactly one cell**. At 0.3: 74.1 / 22.2 / 3.7, with 85.7% singles. At 1.0: 36.8 / 36.8 / 26.4, with only 58.2% singles.

For a screen you will re-isolate and re-test, **λ ≈ 0.3** is a fair compromise — you use the droplets and can tolerate some ambiguity because the hit is verified afterwards. For a **pooled selection you will sequence**, λ must be low (0.1 or below), because a multiply-occupied droplet assigns a strong phenotype to a passenger genotype and that error is unrecoverable. Volumes: 10^7 droplets is **0.5 mL at 50 pL and 0.2 mL at 20 pL** — bench-scale, against 100,000 microtitre plates for the same count.

What droplets add over FACS is compartmentalisation: a **secreted** product stays with its producer, so you can screen for secreted enzymes and exported metabolites that FACS cannot see. The cost is a bespoke device, a fluorogenic assay that works in a 50 pL drop, and the trouble-shooting that goes with both.

What it teaches: droplet loading is a deliberate trade between wasted droplets and ambiguous genotypes, and the right λ depends on whether you will verify the hit afterwards.

EV7 · chapter W4 · 1.5 h

The carbon footprint of your own process

1. Using 0.6 kg CO_2e per kg glucose, 0.4 per kWh and 0.07 per MJ of steam, compute the footprint per kg of product at (0.40 g/g, 50 g/L), (0.40, 135) and (0.51, 135).
2. Which term dominates at low titer, and which at high?
3. Compare with a fossil benchmark of 2–4 kg CO_2e/kg.
4. What does this say about where to spend engineering effort?

Worked solution. 5.31, 3.31 and 2.99 kg CO_2e per kg. At 50 g/L the steam to evaporate water is 3.01 of the 5.31 — **the largest single term, bigger than the sugar**; at 135 g/L it falls to 1.01 and sugar becomes dominant. So raising titer from 50 to 135 g/L cuts the footprint **38%** with no change in yield.

Against 2–4 kg CO_2e/kg for the fossil route, the low-titer case **loses outright** and the high-titer case is competitive. A bio-process is not automatically lower-carbon, and the honest version of the claim is conditional on titer. On effort: titer improvements pay twice, in separation cost and in carbon, which is the same conclusion PP12 reached in dollars; yield improvements pay once, in sugar; and decarbonising the *utilities* — the 0.4 kg CO_2e per kWh — is often larger than anything the strain can deliver, and is outside the laboratory's control, which is worth knowing before claiming otherwise.

What it teaches: the carbon calculation uses the same inputs as the economic one and weights them differently, with water removal dominating at low titer.

EV8 · chapter W4 · 1 h

Allocation and biogenic carbon: the two arguments that move the number

1. A process gives 1 kg of product and 0.5 kg of animal feed. Compute the product footprint when 0%, 33% and 50% of upstream burden is allocated to the co-product.
2. Compute the biogenic carbon embodied in 1 kg of a $C_3H_8O_2$ product.
3. When may that carbon be claimed as a credit, and when not?
4. What does ISO guidance say to do about allocation, and what would you report?

Worked solution. From the 3.31 kg CO_2e baseline: **3.31, 2.82 and 2.56 kg CO_2e/kg** at the three allocations — a **23% improvement from a bookkeeping decision** with no change to the process. That is why allocation is the most contested choice in any LCA and why two studies of the same process can differ by more than any engineering intervention would deliver.

Biogenic carbon: a $C_3H_8O_2$ product of molar mass 76.09 containing three carbons embodies 3 × 44.01/76.09 = **1.74 kg of CO_2 per kg**. It may be claimed as sequestration **only if the end of life keeps it out of the atmosphere** — a durable polymer in long service, or something landfilled without degradation. If the product is burned, composted or

consumed, a cradle-to-grave analysis returns it and the credit vanishes, which is the commonest overstatement in biopolymer marketing.

ISO 14044 says to **avoid allocation where possible**, by subdividing the process or expanding the system boundary to include the function the co-product displaces, and only then to allocate — by physical relationship first, economic value last. What I would report: the result under **system expansion as the base case and under mass and economic allocation as sensitivities**, with the three numbers side by side, so a reader sees that the conclusion survives the choice or does not.

What it teaches: two defensible bookkeeping choices move an LCA result by more than most process improvements, so a result reported under one rule is not a result.

Closing notes to part 21

What the reviewer's list changed, and what it did not

Most of the list was already covered, several items heavily. That is worth stating plainly rather than pretending to a bigger overhaul: the curriculum you brought was well aimed at this kind of laboratory. What the audit found was five specific thin spots, now filled:

Gap	Tier	Now in
Cas12a, base editors, prime editing	**1**	Chapter W1, problems EV1, EV2
FACS and droplet microfluidics	2	Chapter W3, problems EV5, EV6
Life cycle analysis	3	Chapter W4, problems EV7, EV8
Flavonoids and P450s	product class	Chapter W2, problems EV3, EV4
Reading list organised by tier	all	`me-reading-list-by-tier.pdf`

One disagreement worth recording

The reviewer puts techno-economic assessment and non-conventional chassis in Tier 3, "easily learned in the first year". The module's own arithmetic suggests TEA belongs higher: problem G1 showed that the titer–separation relationship determines whether a project is viable at all, and X22 showed that the input dominating the uncertainty is the one a laboratory does not control. A candidate who can say "this strain cannot reach a viable selling price at any plausible yield, because the water removal costs more than the product sells for" is making an argument no amount of first-year catch-up substitutes for. I would treat Tier 3's TEA as Tier 2, and the module already does.

On the numbers in this part

All computed here. Recalled and flagged: the reported flavonoid titers, the magnitude of P450 activity loss in bacteria, the emission factors for glucose, electricity and steam, the fossil benchmark, and the reported efficiencies of base and prime editing in bacteria. The emission factors in particular are illustrative placeholders — a real LCA sources them from ecoinvent or GREET, and the conclusion about titer survives any plausible change to them, which is the useful kind of robustness.

Where the set stands

Source	Problems
Curriculum appendix A1–W3	54
Parts 15–20	77
Part 21 EV1–EV8	8
Total in the tracker	**141**

Part 22 — Strain stability as one chain, ALE as a designed campaign, the screening funnel, and the tank that crashes

A reviewer made two criticisms of this module: that strain stability is present but not integrated with scale-up, and that the analytical bottleneck behind high-throughput screening is never costed. Both check out against an audit. I added a third that neither of us had covered and that ends more campaigns than either.

What the audit found

Topic	Mentions	Verdict
Plasmid loss and retention arithmetic	26	Present (PP3, part 9 ch 32)
Cheater and revertant takeover	32	Present (Z8)
Mutation supply at scale	14	Present (part 14 ch B10)
IS elements and transposon activity	6	**Thin**
Hypermutators and mismatch repair	4	**Thin**
ALE as a technique	100	Well referenced
ALE as a designed campaign (population genetics)	—	**Absent as arithmetic**
Assay throughput in samples per day	**0**	**Absent**
The screening funnel / tiered assays	**0**	**Absent**
Contamination and phage, quantified	—	**Absent — my addition**

So the reviewer was right twice. The components of stability were all there and never multiplied together; and FACS throughput and chromatography throughput both appear in the module without their ratio ever being computed, which is precisely where the industrial friction sits. Four chapters, Y1 to Y4, and eight problems, SB1 to SB4 and SC1 to SC4, taking the set to **149**.

Chapter Y1 · integrates PP3, Z8 and part 14 ch B10

The stability chain: one calculation from vial to harvest

Each mechanism was already in the module. What follows is what happens when you multiply them along a real seed train — and the answer is that batch processes are safe and repeated-batch ones are not.

Stage by stage, with the generations counted

A seed train from a cryovial to a 100 m³ vessel is not one 27-generation step; it is five expansions plus a production phase, and plasmid loss compounds through all of them. With a loss rate of 5×10^{-4} per division and an 8% growth advantage for the cell that lost it:

Stage	Cumulative generations	Plasmid-positive fraction
Vial to 1 mL	0	1.000
1 mL to 50 mL	5.6	0.996
50 mL to 5 L	12.2	0.988
5 L to 500 L	18.8	0.976
500 L to 100 m³	26.4	**0.951**
Through the production phase	34.4	**0.907**

Two things follow that a single-step calculation hides. The strain is **91% plasmid-positive at harvest, not the 95% you would measure at the end of the seed train** — and the seed-train figure is the one most groups measure, because it is the last point before the big tank. And because biomass doubles every generation, **the last three generations contain 88% of all the cells**, so the loss is concentrated exactly where the product is made. The cell-weighted average plasmid content across the whole train is **0.875**, which is the number that actually multiplies your yield.

The mechanism the plasmid assay cannot see

Plating with and without antibiotic measures *plasmid presence*. It does not measure whether the pathway on the plasmid still works. The dominant route to a silently broken pathway in *E. coli* is **insertion-sequence transposition**: an IS element hops into a burdensome gene, the cell keeps the plasmid and its resistance marker, and it grows faster than its producing siblings. Such a cell is antibiotic-resistant, plasmid-positive, and useless.

The arithmetic decides when to worry. Without selection, an IS insertion arising at 10^{-6} per cell per generation reaches only **4×10^{-5} after 40 generations** and 4×10^{-4} after 400 — negligible. With an 8% growth advantage it still reaches only 10^{-11} by generation 40 from a single founder. But the time to sweep from 10^{-6} to half the population is

Growth advantage	Generations to 50%	Days at 10 generations/day
5%	283	28
8%	180	18
15%	99	10

So the conclusion is sharp and operationally useful: **a 40-generation batch train is safe, and anything running for hundreds of generations is not**. Repeated-batch operation makes the point concretely — re-inoculating from the previous batch, the plasmid-positive fraction falls from 0.876 after one batch to **0.233 after two and 0.001 after four**. The practice of carrying a culture forward to save a seed train destroys the strain in three passages, and it does so invisibly if you only assay the first batch.

Hypermutators: an asset in evolution, a liability in production

A *mutS* or *mutL* defect raises the point-mutation rate 30- to 100-fold. In a 100 m³ fermenter at 10^{16} cells, the number of new inactivating mutations arising in a single 1 kb pathway gene per generation goes from **10^{10} to 3×10^{11} at 30× and 10^{12} at 100×**. Those numbers are already enormous at the wild-type rate, which is part 14 chapter B10's point; what the hypermutator changes is the *supply of double mutants and of rarer events*, including the IS and deletion events that break pathways.

Hence the rule: a hypermutator is **deliberately useful in an ALE campaign**, where you want mutation supply, and must be **repaired before the strain goes to production**, where you want genetic stasis. Many published evolved strains carry mismatch-repair defects acquired during evolution, and checking for them in the sequencing output is part of characterising an evolved isolate.

Seed train	five expansions + production = 34 generations; 0.951 → 0.907 plasmid-positive
Weighting	last 3 generations hold 88% of cells; cell-weighted plasmid content 0.875
IS sweep	180 generations from 10^{-6} to 50% at an 8% advantage; batch safe, repeated batch not
Repeated batch	0.876 → 0.233 → 0.001 over four re-inoculations
Hypermutator	30–100× rate; asset in ALE, liability in production

Chapter Y2 · extends part 11 and problem Y6

Adaptive laboratory evolution as a designed campaign

The module referenced ALE a hundred times and never computed a campaign. These are the four numbers that design one: mutation supply, bottleneck size, fixation time, and how many lineages you need to tell cause from passenger.

Mutation supply: why ALE is a selection experiment

The number of new genome mutations entering a population each generation is $N_e \times \mu \times$ genome size. For *E. coli* at 2×10^{-9} per base per generation and 4.64 Mb:

Effective population	New genome mutations per generation	Times every base is hit
10^7	9.3×10^4	0.02
10^9	9.3×10^6	**2.0**
10^{11}	9.3×10^8	200

At $N_e \geq 10^9$ — a litre of culture at modest density — **every possible point mutation in the genome arises every generation**. That reframes the whole method: ALE is not a mutation experiment, it is a **selection experiment**, and the design variables are the selection pressure and the population structure, not the mutagenesis. Chemical mutagenesis or a hypermutator adds value only for multi-mutation events or at small population size.

Bottlenecks, time, and what a transfer regime costs

Serial transfer at 1:100 gives $\log_2(100) =$ **6.6 generations per cycle**, so a 500-generation campaign is **75 daily cycles — about eleven weeks**; at 1:1000 it is 10 generations per cycle and 50 days. The deeper dilution is faster in calendar time and *harsher* in population genetics, because each transfer is a bottleneck that discards rare beneficial lineages.

Fixation time for a beneficial mutation of advantage s, from a single cell in $N_e = 10^9$:

Advantage s	Generations to fixation	Probability of escaping drift (≈2s)
2%	1,036	4%
5%	414	10%
10%	207	20%
30%	69	60%

Two design consequences. A 2% improvement needs **more than a thousand generations** to fix and has a 96% chance of being lost to drift before it starts — so an ALE campaign of a few hundred generations can only find mutations worth about 5% or more, and if your objective offers only small gains, ALE is the wrong tool. And because mutation supply at 10^9 is enormous, **several beneficial lineages compete simultaneously** — clonal interference — so the mutation that fixes is the best of many contenders rather than the first to arise, which is why parallel replicate lineages often fix *different* mutations in the same gene.

Telling the causal mutation from the passengers

An evolved isolate typically carries several to tens of mutations, most of them hitchhikers that rose with a beneficial one. The statistical answer is **parallel independent lineages**. If a given passenger appears in any one lineage with probability 0.3, the chance it appears in all of n lineages is 0.3^n:

Independent lineages	Chance a passenger appears in all
3	2.7×10^{-2}
6	7.3×10^{-4}
12	**5.3×10^{-7}**

So a gene hit in all of six to twelve independent lineages is causal with high confidence, and that — not the number of generations — is what the experimental design should buy. **Run 6 to 12 parallel lineages rather than one deep one.** The confirmation remains reconstruction: put the candidate mutation into the clean parent and measure, as part 8 chapter 28 required, because parallelism establishes association and reconstruction establishes cause.

Mutation supply	$N_e\mu L$: every base hit twice per generation at $N_e = 10^9$
Transfer regime	1:100 = 6.6 generations/cycle; 500 generations ≈ 75 days
Fixation	$\ln(N_e)/s$ generations; 2s chance of escaping drift
Resolution limit	a few hundred generations finds s ≥ 5% only
Causality	6–12 parallel lineages, then reconstruct in the parent

Chapter Y3 · extends weeks 29, 30 and 34

The screening funnel: the analytical bottleneck is the design constraint

The module taught sorting at 10^7 and chromatography at 110 samples a day in different chapters and never divided one by the other. The ratio is the single most useful number in campaign design.

What each assay actually delivers in a working day

Assay	Per working day	Unit	What it measures
Sorting or selection	5×10^7	cells	A fluorescence proxy
Microtitre optical screen	3,000	clones	A colour, fluorescence or growth proxy
LC-MS with sample prep	**150**	samples	**The actual product**
HPLC, 12-minute method	110	samples	The actual product
Shake flask plus HPLC	24	strains	Product under growth conditions
Bench bioreactor	4	strains	Product under process conditions

The mismatch is seven orders of magnitude from end to end, and the one that bites is in the middle: a sorter hands you **10^5 candidates** and mass spectrometry verifies **150 a day**, a ratio of about **670 to 1**. You cannot verify what you sorted. That is not a resourcing problem to be fixed with another instrument; it is a structural feature, and the response is to design the campaign as a funnel whose tiers are sized by the capacity of the assay below them.

A funnel designed backwards from capacity

Tier	Input	Kept	Output	Instrument-days
Sort or select	10^7	0.1%	10,000	0.2
Optical plate screen	10^4	10%	1,000	3.3
HPLC confirmation	300	30%	90	2.7
Flask and bioreactor	20	—	20	0.8

Seven instrument-days in total, and note the structure: the tier that *sounds* most impressive, the sort, costs 0.2 days, while the optical screen costs 3.3 and the chromatography 2.7. **The cheap tier is the big one, and the expensive tiers are small by necessity.** Note also the discard between tiers two and three: 1,000 clones pass the optical screen and only 300 go to HPLC, which means the ranking from the proxy assay is being trusted to throw away 700 — and that is where a campaign silently loses its best strain.

Where the mass-spec time actually goes

Step, per 96-well plate	Hours
Quench and extract	2.0
Dry down	1.5
Derivatise (for GC-MS)	2.0
Resuspend and plate	0.5
Instrument run at 12 min per sample	**19.2**
Total	**25.2 h**

One plate is **25 hours end to end**, so a single operator with one instrument manages about **1.6 plates a week** — roughly 150 samples. That is the number the reviewer was pointing at, and it explains three industrial habits that look like laziness from outside and are arithmetic from inside: **shortening the chromatographic method** to two or three minutes, accepting worse separation for six times the throughput; **pooling** four wells into one injection, which multiplies throughput fourfold and destroys the identity of three in four wells unless you deconvolve; and **building an optical proxy for everything**, with all the sensor-gaming risk that EV5 quantified.

The design rule that falls out: **decide the bottom tier first**. Count how many true measurements you can make, multiply upward by each tier's kept fraction, and that tells you how large a library is worth building. Building 10^7 variants when the bottom tier is 20 bioreactor runs is not ambition; it is a funnel whose top is detached from its bottom.

The ratio	sort 10^5 candidates/day against MS 150/day = 670:1
MS reality	25.2 h per 96-well plate; ~1.6 plates per operator-week
Funnel	10^7 → 10^4 → 300 → 20 over about seven instrument-days
Design rule	size the library from the bottom tier upward, not the top down

Chapter Y4 · extends part 14 ch B10 · my addition

Contamination and phage: the failure that ends campaigns

Part 1 chapter 10 computed sterilisation and then said that filtration does not stop phage, without computing what that means. This is the chapter that asks what happens when tank three crashes on day four.

Contamination as a per-batch probability

Sterilisation arithmetic (part 14 chapter B10) answers "did we kill what was in the medium". Operational contamination is a different question — a seal, a sample port, a drain, an air filter — and it is best treated as a per-batch probability:

Per-batch probability	Expected losses in 50 batches	P(no loss at all)
2%	1.0	0.364
5%	2.5	0.077
10%	5.0	**0.005**

Read the third column. At a 5% contamination rate — not unusual for a poorly characterised process — the probability of completing a 50-batch campaign without a loss is **8%**, so you should plan for losses rather than hope. The planning consequence is V1's and X24's: build the expected losses into the schedule as extra batches, not as an optimistic assumption, and the economic consequence is that each loss carries the full cost of the medium, the vessel time and the seed train.

Phage, computed

Phage is worse than bacterial contamination for a reason that is arithmetic rather than mysterious: it replicates on your product organism, with a burst size near 100 and a latent period near 30 minutes. A single infective particle multiplies by 100 every half hour, so the time to sweep a population is $\ln(1/f_0)/(\ln(\text{burst})/\text{latent})$:

Starting free-phage fraction	Time to collapse
10^{-9}	**2.2 h**
10^{-7}	1.8 h

Against a 48-hour fermentation, **a phage entering at inoculation ends the run before the first shift change**, and the operational signature is unmistakable once you know it: optical density falls, the dissolved oxygen trace rises sharply as respiration stops, and the broth clears. That rising-DO signature is the same one problem X6 attributed to substrate inhibition, which is why the *first* diagnostic is a microscope or a plaque assay rather than a feed adjustment.

What actually works, in order

Defence	What it does	Limit
Do not filter and hope	A 0.2 µm filter removes bacteria; phage are 30–200 nm and pass	Needs 0.02 µm filtration or heat for any phage-risk stream
Segregate	Separate air, drains and personnel flow between seed train and production	Most plant outbreaks trace to a drain or a leaking seal, not the medium
Rotate strains	Different receptor profiles, so one phage cannot take the whole plant	Needs two qualified strains, which is a real development cost
CRISPR immunity	Spacers against the local phage population in the production strain	Phage escape by protospacer mutation; needs re-spacering
Receptor deletion	Delete the surface protein the phage uses	Often costs fitness, and selects for phage using another receptor
Lysogen screening	Check the production strain for inducible prophage before scale-up	A resident prophage can be induced by stress and is self-inflicted

The one most worth knowing for an interview: **sequence your production strain for prophages before you scale it**. A strain that carries an inducible prophage will crash under the stress of a large fermenter and the cause will look like contamination from outside.

Contamination	$(1-p)^n$: at 5% per batch, only 8% chance of 50 clean batches
Phage sweep	$\ln(1/f_0)/(\ln \text{burst/latent}) \approx 2$ h from 10^{-9}
Signature	falling OD, rising dissolved oxygen, clearing broth
Filtration	0.2 µm stops bacteria, not 30–200 nm phage

Problems SB1–SB4 and SC1–SC4

SB1 · chapter Y1 · 2 h

Carry plasmid loss along the whole seed train

1. Count the generations for each expansion from a vial to 100 m³, and compute retention stage by stage at a loss rate of 5 × 10^{-4} and an 8% burden.
2. What fraction is plasmid-positive at harvest, and how does that differ from the end-of-seed-train figure most groups measure?
3. What fraction of all cells exists in the last three generations, and what is the cell-weighted plasmid content?
4. Repeat for four rounds of repeated-batch re-inoculation.

Worked solution. Retention falls 1.000 → 0.996 → 0.988 → 0.976 → **0.951** at the end of the seed train, and **0.907** after the production phase. So the harvest figure is four points below the one usually measured, and in the wrong direction. Because biomass doubles each generation, **the last three generations hold 88% of the cells**, so the loss is concentrated where production happens; the cell-weighted plasmid content across the train is **0.875**, which is the figure that multiplies your titer.

Repeated batch is the striking case: re-inoculating from the previous culture gives **0.876, 0.233, then 0.001 by the fourth round**. Carrying a culture forward to save a seed train destroys the strain in three passages — and if you assay only the first batch, the collapse is invisible until the yield goes.

What it teaches: plasmid loss must be integrated along the whole train and weighted by where the cells are, and the standard end-of-seed-train assay measures the wrong point.

SB2 · chapter Y1 · 1.5 h

The reversion the antibiotic plate cannot see

1. Compute the fraction carrying an IS insertion in the pathway after 40, 200 and 400 generations with no selection, at rates of 10^{-7}, 10^{-6} and 10^{-5}.
2. Compute the generations for such a mutant to go from 10^{-6} to 50% at advantages of 5%, 8% and 15%.
3. Which operating modes are safe and which are not?
4. Why does plating with and without antibiotic miss this entirely, and what assay would catch it?

Worked solution. Without selection the frequencies stay small — **4 × 10^{-5} at 10^{-6} and 40 generations**, and only 4 × 10^{-3} at the highest rate and 400 generations. With selection the sweep times are **283, 180 and 99 generations** at the three advantages, which is **28, 18 and 10 days** at ten generations a day.

So: a 40-generation batch train is **safe**; repeated batch over hundreds of generations is **not**; continuous culture certainly is not, which is PP9's and Z8's conclusion arrived at by a third route. The antibiotic plate misses it because the cell *still has the plasmid and the marker* — only the pathway is broken. What catches it: **product per cell**, measured on individual colonies rather than the population; a **functional reporter** in the pathway; or **sequencing the construct** from end-of-run isolates, where an IS insertion appears as a new junction in exactly the evidence table problem Y6 told you to read.

What it teaches: genetic stability has two independent failure modes, and the routine assay detects only one of them.

SB3 · chapter Y2 · 2 h

Design an ALE campaign

1. Compute the new genome mutations per generation at effective populations of 10^7, 10^9 and 10^{11}, and say how many times each base is hit.
2. What does that imply about whether to add a mutagen or a hypermutator?
3. Compute generations per cycle at 1:100 and 1:1000 transfer, and the calendar time for 500 generations.
4. Compute fixation times and drift-escape probabilities for advantages of 2%, 5%, 10% and 30%. What is the smallest improvement your campaign can find?
5. How many parallel lineages, and why?

Worked solution. Mutations per generation are **9.3 × 10^4, 9.3 × 10^6 and 9.3 × 10^8**, so each base is hit 0.02, **2.0** and

200 times. At 10^9 and above **every possible point mutation arises every generation** — ALE is a selection experiment, and a mutagen adds nothing except for multi-mutation events or at small population size.

Transfers: 1:100 gives 6.6 generations per cycle so 500 generations is **75 days**; 1:1000 gives 10 and 50 days, faster but with harsher bottlenecks that discard rare lineages. Fixation needs **1,036 generations at s = 2% and 207 at 10%**, with only **4%** of 2% mutations escaping drift against 20% at 10% — so a few-hundred-generation campaign can realistically only find improvements of **about 5% or more**, and if your objective offers less, ALE is the wrong tool.

Run **6 to 12 parallel lineages**, not one deep one: a passenger appearing in any lineage with probability 0.3 appears in all six with probability 7×10^{-4} and in all twelve with 5×10^{-7}, so parallelism is what separates causal genes from hitchhikers — followed by reconstruction in the clean parent, because parallelism shows association and only reconstruction shows cause.

What it teaches: ALE's design variables are population size, bottleneck and number of lineages, and its resolution limit is a selection coefficient of a few per cent.

`SB4` · chapter Y1 · 1 h

Hypermutators: when to want one

1. At 30× and 100× the wild-type rate, compute the new inactivating mutations per generation in one 1 kb gene across a 10^{16}-cell fermenter.
2. Why is the wild-type number already large, and what does the hypermutator actually change?
3. State the rule for ALE against production.
4. You receive an evolved strain from a collaborator. What do you check first?

Worked solution. 3×10^{11} at 30× and 10^{12} at 100×, against 10^{10} at the wild-type rate. The wild-type figure is already enormous because the population is — part 14 chapter B10's point — so what the hypermutator changes is not whether single mutants exist (they always do) but the supply of **double mutants and of rarer events** such as IS transposition and deletions, which is exactly the class that breaks pathways.

The rule: **a hypermutator is an asset in an ALE campaign and a liability in production**, so repair the mismatch-repair defect before the strain is qualified. On receiving an evolved strain, check first for **mutations in *mutS*, *mutL*, *mutH* or *uvrD*** — many evolved isolates acquire them, and a strain that evolved fast may have done so because it was mutating fast, in which case it will keep mutating in your hands.

What it teaches: mutation rate is itself an evolvable trait, and an evolved strain's speed may be a warning rather than a feature.

`SC1` · chapter Y3 · 1.5 h

Compute your own screening funnel

1. Tabulate the daily throughput of sorting, optical plate screening, HPLC, flask culture and bench bioreactors.
2. Compute the ratio between what a sort delivers in a day and what mass spectrometry verifies.
3. Design a four-tier funnel from 10^7 to 20 strains and compute the instrument-days at each tier.
4. Which tier costs the most time, and which discard worries you most?

Worked solution. Per working day: sorting **5×10^7 cells**, optical screening **3,000 clones**, LC-MS **150 samples**, HPLC **110**, flask plus HPLC **24**, bench bioreactor **4**. A sort delivering 10^5 candidates against 150 verifications a day is a ratio of about **670 to 1**: you cannot verify what you sorted, so the funnel is forced rather than chosen.

The funnel: 10^7 sorted keeping 0.1% → 10,000; optical screen keeping 10% → 1,000; HPLC on 300 keeping 30% → 90; flask and bioreactor on 20. Instrument-days are **0.2, 3.3, 2.7 and 0.8** — so the glamorous tier is the cheapest and **the optical screen is the most expensive in time**. The discard that should worry you is between tiers two and three: 1,000 clones pass the optical screen and 300 go forward, so the proxy assay's ranking is trusted to discard 700, and if the proxy correlates imperfectly with product — which L1 and EV5 both quantify — your best strain is probably among them.

What it teaches: campaign design is arithmetic on assay capacity, and the tier that discards on a proxy is where the best strain is lost.

`SC2` · chapter Y3 · 1.5 h

Where the mass-spec week actually goes

1. Sum the hours per 96-well plate for quench and extraction, dry down, derivatisation, resuspension and a 12-minute instrument method.
2. How many plates can one operator with one instrument manage per working week?
3. What does shortening the method to 3 minutes do, and what does it cost?
4. What does pooling four wells per injection buy and destroy?

Worked solution. 2.0 + 1.5 + 2.0 + 0.5 + 19.2 = **25.2 hours per plate**, so one operator with one instrument manages about **1.6 plates — roughly 150 samples — per week**. Note the structure: the instrument run is 76% of it, so automating the prep helps less than it appears.

Shortening the method from 12 to 3 minutes cuts the run to 4.8 hours and takes the plate to **10.8 hours**, roughly **2.3× the throughput**. It costs chromatographic resolution, so co-eluting peaks — exactly R1's problem — become more likely, and the method must be re-validated for the specific matrix. Pooling four wells per injection multiplies throughput **fourfold** and destroys the identity of three in four wells: it works as a *coarse* tier where you only need to know which pool contains a hit and will re-run its members, and it fails when you need a value per clone. Both are legitimate; both must be stated in a methods section.

What it teaches: the instrument run dominates sample-prep time, so throughput is bought with method length or pooling, each with a specific cost.

`SC3` · chapter Y4 · 1.5 h

Plan a campaign that will be contaminated

1. At per-batch contamination probabilities of 2%, 5% and 10%, compute expected losses over 50 batches and the probability of none.
2. What does that do to a schedule built on 50 successful batches?
3. Compute the cost of a loss relative to a successful batch.
4. Where do plant contaminations usually originate, and what does that imply for where to spend?

Worked solution. Expected losses are **1.0, 2.5 and 5.0**, and the probability of a clean campaign is **0.364, 0.077 and 0.005**. At 5% the chance of 50 clean batches is **8%**, so a schedule built on 50 successes will be missed with 92% probability — the fix is to plan 53 batches, in the spirit of V1's 80th-percentile promise rather than a most-likely one.

A lost batch costs the full medium, the vessel time and the seed train, and it costs them *without* the product that would have paid for them — so a 5% contamination rate is roughly a 5% tax on cost of goods plus the schedule risk. On origin: medium sterilisation is usually the best-controlled step, because it is computed (part 14 chapter B10), while **seals, sample ports, drains and air filters** are the usual culprits. That is where to spend: on seal replacement schedules, sample-port design and air-filter integrity testing, rather than on longer sterilisation cycles that damage the medium for no gain.

What it teaches: contamination is a per-batch probability to be planned for, and the step that is computed is rarely the step that fails.

`SC4` · chapter Y4 · 1.5 h

Tank three is crashing. What is it, and what do you do?

1. Compute the time for a lytic phage to collapse a culture from a free-phage fraction of 10^{-9}, with a burst of 100 and a latent period of 30 minutes.
2. List the three instrument signatures of a phage infection and name the other problem that produces one of them.
3. Why does 0.2 µm filtration not protect you?
4. Rank six defences and name the one you would do before any scale-up.

Worked solution. The collapse time is **about 2.2 hours** from 10^{-9} and 1.8 from 10^{-7} — against a 48-hour fermentation, **a phage entering at inoculation ends the run before the first shift change**.

The signatures are **falling optical density, sharply rising dissolved oxygen as respiration stops, and a clearing broth**. Rising dissolved oxygen during active feeding is also the signature of substrate inhibition (problem X6), which is why the first diagnostic is a **microscope or a plaque assay**, not a feed adjustment — treating a phage infection as an inhibition problem loses the tank and contaminates the plant.

Filtration fails because phage are **30 to 200 nm** and a 0.2 µm membrane passes them; any phage-risk stream needs 0.02 µm filtration or heat. Ranked defences: **screen the production strain for inducible prophage before scale-up**

(first, because it is self-inflicted and cheap to check by sequencing); segregate air, drains and personnel between seed train and production; maintain seals and sample ports; qualify a second strain so rotation is possible; add CRISPR spacers against the local phage, accepting that escape mutants arise; and delete the receptor as a last resort, since it usually costs fitness and selects for phage using another receptor.

What it teaches: a phage infection is faster than any corrective action, so the whole defence is preventive — and the first prevention is sequencing your own strain.

Closing notes to part 22

On the feedback itself

Both criticisms were correct, and unequally so. **The analytics bottleneck was the sharper one**: "samples per day" and "screening funnel" each appeared zero times across twenty-one parts, while FACS throughput and chromatography throughput both appeared and were never divided by one another. That is a clean omission and chapter Y3 is the fix.

The stability criticism was about integration rather than coverage. Plasmid retention, cheater fixation and mutation supply were all present and all separate; what was missing was the product of them along a real seed train, and the two specific mechanisms the audit found thin — IS transposition at six mentions and hypermutators at four. Chapter Y1 integrates them, and the result is a cleaner statement than any of the three gave alone: **batch is safe, repeated batch collapses in three passages, continuous needs growth coupling or integration**.

The addition neither of us had made is contamination and phage. It is unglamorous, it ends real campaigns, and the module had computed sterilisation while never computing what happens when sterility fails.

The numbers worth keeping

- Plasmid-positive fraction falls to **0.951** by the end of the seed train and **0.907** at harvest; cell-weighted, **0.875**.
- Repeated batch: **0.876 → 0.233 → 0.001** over four re-inoculations.
- An IS mutant at an 8% advantage needs **180 generations** to reach half the population.
- At $N_e = 10^9$, **every base in the genome is mutated twice per generation**.
- A 2% beneficial mutation needs **1,036 generations** to fix and has a **4%** chance of escaping drift.
- A passenger appears in all of twelve parallel lineages with probability **5×10^{-7}**.
- Sorting delivers 10^5 candidates a day against **150** mass-spec verifications — **670 to 1**.
- One 96-well plate of GC-MS is **25.2 hours**; about 1.6 plates per operator-week.
- At 5% contamination per batch, the chance of 50 clean batches is **8%**.
- A phage collapses a tank in **2.2 hours** from a 10^{-9} starting fraction.

On the assumptions

All arithmetic computed here. The inputs to challenge: the plasmid loss rate of 5×10^{-4} and the 8% burden (both illustrative, and PP3 says to measure them with a plate count); the IS transposition rate, which varies by element, by host and with stress; the assay throughputs in chapter Y3, which depend heavily on your method and automation and should be replaced with your own laboratory's numbers; and the phage burst size and latent period, which are organism- and phage-specific. The *shapes* of these relations hold; the coefficients are local.

Where the set stands

Source	Problems
Curriculum appendix A1–W3	54
Parts 15–21	85
Part 22 SB1–SB4, SC1–SC4	8
Total in the tracker	**149**

Part 23 — Single cells are not the average, and downstream is not 50 to 70 per cent

Two more criticisms. The first is right, and the connection the reviewer drew is sharper than the gap itself. The second I partly disagree with, and the disagreement is worth more to you in a defence than agreement would be — so chapter X2 interrogates the number rather than repeating it.

The audit

Topic	Mentions	Verdict
Stochastic expression and noise	64	Present, but almost all in chapter S6 as *bistability* — a circuit property, not a population one
Bimodal or all-or-none induction	14	**Thin**, and never linked to sorting
Single-cell heterogeneity	10	**Thin**
Persisters and non-growing subpopulations	**0**	**Absent**
Recovery train and unit operations	199	The module's most developed theme
Titer linked to separation cost	**332**	Strong — PP12 and EV7 reach the reviewer's own conclusion
Downstream share of installed capital	—	**Never computed** — the actual gap
Foaming and antifoam	11	**Thin** — mentioned, never quantified
Debris and filter blinding	21	**Thin** — PP10 computes release, not the filtration consequence

So on downstream the criticism is not that the module neglects it — it is the most heavily covered theme in the whole set — but that three specific things inside it were never computed. Two chapters, X1 and X2, and six problems, HN1 to HN3 and DS1 to DS3, taking the set to **155**.

Chapter X1 · extends chapters S6 and W3

Noise, bimodality and persisters: why a sorted hit may not reproduce

Every expression number in this module so far has been a population average. A sorter does not see averages — it sees single cells — and the gap between those two facts is a specific, computable way for a screening campaign to fail.

Where the noise comes from

Protein copy number in a single cell varies for two separable reasons. **Intrinsic noise** is the randomness of transcription and translation themselves; for a two-stage model with transcriptional bursts of size b,

$$CV^2 \approx 1/\langle P\rangle + [b/(1+b)]\cdot(1/\langle M\rangle)$$

Protein copies	mRNA copies	Burst size	CV	Share from bursting
30,000	26	12	**19%**	>99%
1,000	5	12	44%	99%
100	2	12	70%	98%

Two readings. At the 30,000 copies of part 14 chapter B8's strong gene, the Poisson term 1/⟨P⟩ is negligible and essentially **all the noise is transcriptional bursting** — so reducing noise means changing transcription dynamics, not raising the copy number. And a lowly expressed regulator at 100 copies carries a 70% coefficient of variation, which is why the repressor calculation in B1 came with a warning about stochastic effects.

Extrinsic noise — differences in cell size, ribosome content, plasmid copy number and cell-cycle position — is correlated across all genes in a cell and typically adds a CV of 0.2 to 0.4. Combining in quadrature, a 19% intrinsic noise plus 30% extrinsic gives a **total CV of 36%**, and the extrinsic part dominates. That matters for screening because extrinsic noise affects your reporter and your product *together*, so it is partly correctable by ratiometric measurement against a constitutive second reporter — the single most useful noise-reduction trick in a sorting experiment.

Bimodality: when the population splits

Noise makes a distribution wide. Positive feedback makes it **two distributions**. An inducible system whose inducer enters through a transporter that is itself induced has exactly that structure, and solving for steady states of $dx/dt = \alpha x^n/(K^n + x^n) + \text{leak} - x$ shows where it appears:

Hill coefficient	Steady states	Behaviour
1.0	one, at 0.539	Graded — every cell responds proportionally
1.5	three: 0.058, 0.077, 0.567	Marginally bistable
2.0	three: 0.022, 0.354, 0.644	**Bistable — the population splits**
3.0	three: 0.020, 0.464, 0.852	Strongly bistable

With two stable states and an unstable one between them, individual cells sit at one or the other and the population is a mixture. **The population mean is then a number no cell occupies.** With 20% of cells induced at a twentyfold higher signal, the mean reads 4.8 arbitrary units while every cell is at either 1 or 20.

The failure this causes in a sort

Here is the mechanism problem EV5 missed. EV5 attributed failed sorts to **sensor gaming** — a mutation that activates the reporter. But if the inducible system is bimodal, a top-1% gate selects **cells that happened to be in the induced state**, which is a phenotypic condition and not a genotype. Re-grow them and they re-split into the parent distribution: 20% induced before, 20% induced after. **The hit does not reproduce, and nothing is wrong with the strain.**

Even without bimodality, noise alone corrupts a gate. Simulating a library with a lognormal genotype effect and independent expression noise:

Genotype spread	Noise spread	Of a top-1% gate, truly top-1% genotype
0.6	0.2	66%

0.6	0.6	**27%**
0.3	0.6	**11%**
0.6	1.0	13%

When noise equals the genotype signal, **73% of a top-1% gate are lucky cells**, and when noise exceeds it, 89% are. The remedy is not a tighter gate — it is **sort, regrow, re-sort**, because regrowth re-randomises the noise while the genotype persists. Simulated, with noise equal to signal, the mean genotype of survivors rises from **1.87 standard deviations after one round to 3.03 after two and 3.79 after three**; with noise above signal it starts at 1.36 and reaches 4.09 by the third. Two rounds is usually the right number, and it costs one culture per round.

Persisters

A small fraction of any culture is in a non-growing, stress-tolerant state — typically 10^{-6} to 10^{-2} of cells depending on conditions. In a 100 m^3 fermenter at 10^{10} cells/mL that is **10^9 to 10^{13} cells**, and they have three properties that matter here: they **do not make product**, so they are a yield drag proportional to their fraction; they **survive stress**, including antibiotics and starvation; and they **regrow** when the stress lifts.

The consequence for selection is the important one. **Any selection that kills growing cells enriches persisters**, so a colony surviving an antibiotic challenge may be a persister rather than a mutant. The distinguishing test is heritability: re-grow the survivor without selection and re-challenge it. A mutant re-survives; a persister gives a culture as sensitive as the parent. That test belongs in every selection protocol and is routinely omitted, and it is the same logic as the sort-and-regrow control above — **a phenotype that does not breed true is not a genotype**.

Intrinsic noise	`CV² ≈ 1/⟨P⟩ + [b/(1+b)]/⟨M⟩; 19% at 30,000 copies, nearly all bursting`
Extrinsic	`adds 0.2–0.4 CV, correlated — correctable by a ratiometric second reporter`
Bimodality	`Hill ≥ 2 with leak gives two stable states; the mean is a value no cell holds`
Gate purity	`27% true genotype at equal noise and signal; 11% when noise dominates`
Remedy	`sort–regrow–re-sort: 1.87 → 3.03 → 3.79 standard deviations of genotype`
Persisters	`10⁻⁶–10⁻² of cells; any kill-based selection enriches them; test heritability`

What to open a book for. Elowitz et al. on distinguishing intrinsic from extrinsic noise (Science, 2002) — the two-colour experiment, read for the design; Ozbudak et al. on bimodality in the lac system; and a review of bacterial persistence. All three are short and the first two are the clearest experiments in the field.

Chapter X2 · extends chapter S1 and problem PP12

What downstream actually costs, and where "50 to 70 per cent" is true

The claim is widely repeated and product-class-specific. Computing it for two plants shows where it holds, and a candidate who can do that is in a better position than one who recites it.

The claim, tested on a commodity plant

Take PP12's propanediol: 30,000 t/yr at 135 g/L and 0.51 g/g. That is **222,000 m³ of broth a year**, which at 80% fill and a 60-hour cycle needs about **21 vessels of 100 m³**. Sizing the evaporation from duty rather than guessing: 6.4 kg of water per kg of product is 192,000 t/yr, which at 2.26 MJ/kg is 434,000 GJ/yr — an average duty of **15 MW**, and a multi-effect train for that is of order $40M.

Item	$M	Section
Fermenters, 21 × 100 m³	126.0	Upstream
Seed train	31.5	Upstream
Media preparation and sterilisation	4.0	Upstream
Utilities: air, steam, chilling	12.0	Upstream
Centrifuge and clarification	6.0	Downstream
Evaporation train, 15 MW	41.0	Downstream
Distillation and ion exchange	14.0	Downstream
Crystallisation and drying	8.0	Downstream
Tankage and offsites	10.0	Other
Total	**252.5**	**69% upstream, 27% downstream**

The same claim on a biologic plant

Now 100 kg/yr of a protein at 5 g/L — **20 m³ of broth a year**, which is two single-use 2 m³ vessels and nothing else upstream.

Item	$M	Section
Fermenters, 2 × 2 m³ single-use	3.0	Upstream
Seed train and media	2.0	Upstream
Utilities	2.0	Upstream
Harvest: centrifuge and depth filtration	2.5	Downstream
Capture chromatography	6.0	Downstream
Viral inactivation and two polish columns	9.0	Downstream
Ultrafiltration, diafiltration, fill-finish	7.0	Downstream
Facility: cleanrooms and quality control	14.0	Other
Total	**45.5**	**15% upstream, 54% downstream, 31% facility**

The verdict, and what to say in a defence

"50 to 70 per cent of capital is downstream" is right for biologics and wrong for commodities. The computed shares are **54% for the protein plant and 27% for the chemical plant**, and the reason is structural rather than accidental: a commodity needs enormous fermentation volume because the product is cheap and the titer is in grams per litre, while a biologic needs almost no volume and a long chromatographic train because purity is the specification. Repeating the number without the product class invites the question "for which product?" — and the answer is the interesting part.

Two refinements worth having ready. For a commodity, **separation dominates operating cost rather than capital**:

at 135 g/L the steam for water removal is $0.12/kg against $0.78/kg of sugar, and at 50 g/L it is $0.34 — so the titer argument from PP12 and EV7 is an *operating* cost argument, which is where it belongs. And the biologic's capital is dominated as much by the **facility** — cleanrooms, air handling, quality control — at 31% as by any single unit operation, which is why single-use equipment and facility flexibility matter more than column efficiency in that business.

Two unit-operation failures the module had only mentioned

Foaming. Protein-rich broth foams, and the standard response, chemical antifoam, promotes bubble coalescence — which is exactly what reduces interfacial area. Taking an illustrative relation of k_La falling as $1/(1 + 0.004 \cdot ppm)$:

Antifoam	Relative k_La
10 ppm	0.96
50 ppm	0.83
200 ppm	**0.56**

So heavy antifoam dosing costs **44% of oxygen transfer** — it solves a foam problem by creating the oxygen problem of chapter T4, and the interaction is why antifoam is dosed on demand from a foam probe rather than added to the batch. The alternative, a mechanical foam breaker, costs power and nothing else. The failure mode if neither works is a **foam-out**: foam carries cells into the exhaust filter, blinds it, and raises headspace pressure — in a 100 m³ vessel, 15% foam is **15 m³ to disengage**, which the headspace does not have.

Debris. PP10 computed product release against homogeniser passes and stopped there. Each pass also **fragments the debris further**, and sub-micron debris blinds a depth filter by cake resistance. With filtrate flux falling as $1/(1 + 0.8 \cdot n)$:

Passes	Relative flux	Filter area needed
1	0.56	1.8×
3	0.29	**3.4×**
5	0.20	5.0×

So PP10's optimum of three passes, chosen on product quality alone, carries a **3.4-fold filter area** penalty that the product-release calculation never saw. The full optimum is lower than either calculation alone suggests, and the clean way to state it is that **disruption and clarification must be optimised together** — which is the general form of this chapter's argument and of chapter S1's.

Commodity capital	`69% upstream, 27% downstream at 30,000 t/yr and 135 g/L`
Biologic capital	`15% upstream, 54% downstream, 31% facility at 100 kg/yr and 5 g/L`
The claim	`"50–70% downstream" is a biologics number; for a commodity it is operating cost, not capital`
Antifoam	`200 ppm costs 44% of kLa; dose on demand or use a mechanical breaker`
Debris	`3 homogeniser passes need 3.4× the filter area of one`

The capital figures above are order-of-magnitude estimates built from scaling rules and my own judgement, not from vendor quotations — they are good enough to establish which section dominates and not good enough to put in a proposal. A real estimate comes from the NREL-style design reports referenced in the reading list. The antifoam and debris relations are illustrative forms chosen to show the shape; the coefficients are yours to measure.

Problems HN1–HN3 and DS1–DS3

HN1 · chapter X1 · 1.5 h

Where the noise in your reporter comes from

1. Using $CV^2 = 1/\langle P\rangle + [b/(1+b)]/\langle M\rangle$, compute the CV at 30,000, 1,000 and 100 protein copies with the mRNA and burst values given.
2. What share of the variance is Poisson and what share is bursting in each case?
3. Add extrinsic noise of 0.2 and 0.3 in quadrature to the 30,000-copy case.
4. Which component can a ratiometric measurement remove, and why?

Worked solution. CVs are **19%, 44% and 70%**, and bursting accounts for over 98% of the variance in all three — the Poisson term $1/\langle P\rangle$ matters only below a few hundred copies. Adding extrinsic noise gives **28% at 0.2 and 36% at 0.3**, so above about 10,000 copies the extrinsic component dominates the total.

A ratiometric measurement — your reporter divided by a constitutive second fluorophore in the same cell — removes the **extrinsic** component, because cell size, ribosome content and plasmid copy number affect both reporters together and cancel in the ratio. It cannot remove intrinsic noise, which is independent for each gene by construction. That is exactly what the two-colour experiment of Elowitz and colleagues was designed to separate, and it is why a sorting experiment with a single colour is measuring cell size as much as pathway activity.

What it teaches: noise has two components with different remedies, and the one a sort can correct for is the one that dominates at useful expression levels.

HN2 · chapter X1 · 2 h

The hit that will not reproduce

1. Solve for the steady states of $dx/dt = \alpha x^n/(K^n + x^n) + \text{leak} - x$ at Hill coefficients of 1, 1.5, 2 and 3, with $\alpha = 1$, $K = 0.5$, leak = 0.02.
2. With 20% of cells induced at a twentyfold signal, compute the population mean and say where the cells actually are.
3. What is a top-1% gate selecting, and what happens when you re-grow it?
4. Design the control that distinguishes this from a real genotype, and distinguish it from EV5's sensor gaming.

Worked solution. One steady state at n = 1 (0.539), three from n = 1.5 upward — at n = 2 they are 0.022, 0.354 and 0.644, with the middle one unstable, so cells occupy the low or high state and the population is a mixture. At 20% induced and twentyfold signal the mean is **4.8**, a value **no cell holds**.

A top-1% gate is therefore selecting **cells in the induced state**, not cells with a better genotype. Re-grown, they re-split to 20% induced and the apparent improvement vanishes. The control is **sort, regrow a clonal culture, and re-measure the distribution**: if it reproduces the parent's bimodal shape, you selected a state; if the whole distribution has shifted, you selected a genotype. That differs from EV5's sensor gaming in both mechanism and test — gaming is a heritable mutation in the reporter, so it *does* breed true and is caught by assaying **product** rather than fluorescence, whereas a phenotypic state does not breed true at all and is caught by re-measuring the distribution.

What it teaches: a sort can select a phenotypic state rather than a genotype, and the diagnostic is whether the distribution reproduces, not whether the clone is bright.

HN3 · chapter X1 · 1.5 h

How much of your gate is luck, and how many rounds fix it

1. For genotype and noise spreads of (0.6, 0.2), (0.6, 0.6), (0.3, 0.6) and (0.6, 1.0), compute what fraction of a top-1% gate truly has a top-1% genotype.
2. Why does a tighter gate not help?
3. Simulate sort, regrow and re-sort for three rounds at equal noise and signal, and report the mean genotype of survivors.
4. How many rounds would you run, and what does each cost?

Worked solution. True-genotype fractions are **66%, 27%, 11% and 13%** — so when noise equals signal, nearly three-quarters of your gate is lucky cells, and when the genotype spread is small relative to noise it is nine-tenths. A tighter gate does not help because it selects more extreme *observed* values, and the most extreme observations are the ones

with the largest noise contribution: narrowing from the top 1% to the top 0.1% makes the precision worse, not better.

Sort–regrow–re-sort works because regrowth **re-randomises the noise while the genotype persists**. Simulated at equal noise and signal, the mean genotype of survivors is **1.87 standard deviations after one round, 3.03 after two and 3.79 after three**; with noise above signal, 1.36, 2.48 and 4.09. I would run **two rounds** — the second adds more than the third and each costs one overnight culture plus one sorter session, which against the alternative of characterising hundreds of false positives is trivially worthwhile.

What it teaches: gate precision is set by the ratio of genotype spread to noise, and the fix is more rounds rather than a narrower gate.

`DS1` · chapter X2 · 2 h

Compute the capital split for your own process

1. For 30,000 t/yr at 135 g/L, compute annual broth volume, vessels required at 80% fill and a 60-hour cycle, and the evaporation duty from the water load.
2. Build an item list and compute the upstream and downstream shares of capital.
3. Repeat for 100 kg/yr of a protein at 5 g/L.
4. Is "50 to 70 per cent of capital is downstream" true? State the conditions under which it is.

Worked solution. The commodity needs **222,000 m³ of broth a year**, about **21 vessels of 100 m³**, and an evaporation duty of **15 MW** from 192,000 t of water. Its capital comes out **69% upstream and 27% downstream** on a $252M total. The biologic needs **20 m³ a year** — two small single-use vessels — and comes out **15% upstream, 54% downstream and 31% facility** on $45M.

So the claim is **true for biologics and false for commodities**, and the reason is structural: a cheap product at grams per litre needs enormous fermentation volume, while an expensive product at grams per litre needs almost none and a long purification train instead. The conditions under which it holds: **high product value, low volume, purity-driven specification, and a chromatographic train**. For a commodity the honest statement is different — separation dominates *operating* cost, at $0.12/kg of steam against $0.78/kg of sugar at 135 g/L, rising to $0.34 at 50 g/L — which is where PP12's and EV7's titer argument actually lands.

What it teaches: a widely quoted cost share is product-class-specific, and computing it for two classes is a better answer than quoting it for one.

`DS2` · chapter X2 · 1 h

Foam, and the oxygen it costs you

1. With k_La falling as $1/(1 + 0.004 \cdot ppm)$, compute the relative transfer at 10, 50 and 200 ppm of antifoam.
2. Why does antifoam reduce k_La at all?
3. What volume of foam is 15% of a 100 m³ vessel, and what does a foam-out do?
4. Compare chemical and mechanical foam control, and say how you would dose.

Worked solution. Relative k_La is **0.96, 0.83 and 0.56** — so heavy dosing costs **44% of oxygen transfer**, which for an oxygen-limited process is worse than the foam. Antifoam works by **promoting bubble coalescence**, and coalescence is precisely what destroys the interfacial area that k_La depends on: the mechanism that breaks the foam is the mechanism that cuts transfer. They are not separable.

Fifteen per cent of a 100 m³ vessel is **15 m³ of foam**, which the headspace does not have — the consequence is a **foam-out**: foam carries cells into the exhaust filter, blinds it, raises headspace pressure, and in the worst case forces sterile boundary failure. Chemical control is cheap, immediate and costs transfer; mechanical control — a foam-breaking impeller or a centrifugal breaker — costs power and capital and no transfer. The practice that follows: **dose on demand from a conductivity foam probe rather than batching antifoam in**, since batched antifoam pays the k_La penalty for the whole run to solve a problem that occurs for part of it.

What it teaches: foam control and oxygen transfer are the same physics with opposite signs, so the control strategy matters more than the agent.

`DS3` · chapter X2 · 1.5 h

Debris: the cost PP10 did not count

1. With filtrate flux falling as $1/(1 + 0.8 \cdot n)$ with homogeniser passes n, compute relative flux and filter area at 1, 3 and 5

passes.

2. PP10 chose three passes on product quality. What does that choice cost in clarification?
3. Combine the two and say qualitatively where the joint optimum lies.
4. What alternatives to homogenisation change this trade, and what do they cost?

Worked solution. Relative flux is **0.56, 0.29 and 0.20**, so area required is **1.8×, 3.4× and 5.0×** a baseline. PP10's three-pass optimum therefore carries a **3.4-fold filter area** penalty, or equivalently 3.4 times the filtration time, which its product-release calculation never saw.

Combining them: product recovery rises steeply from one to two passes and flattens, while filtration cost rises roughly linearly, so the **joint optimum is lower than PP10's three passes** — probably two — and the right way to find it is to cost both curves in the same units rather than optimise either alone. That is the general lesson: adjacent unit operations are coupled, and an optimum computed for one is not an optimum. Alternatives that change the trade: **bead milling**, which is worse for debris; **enzymatic or chemical lysis**, which is gentler and leaves larger fragments that filter better, at the cost of the lysing agent as an impurity; and **flocculation before filtration**, which aggregates the fines and recovers much of the flux for the cost of a polymer that must then be cleared.

What it teaches: optimising a unit operation in isolation exports its cost to the next one, and disruption exports to clarification.

Closing notes to part 23

On the two criticisms

The first was right and I had missed the connection. Noise was in the module as bistability — a circuit property — and not as population heterogeneity, and the link to sorting was absent. That link is the valuable part: a bimodal inducible system makes a top-1% gate select a *state* rather than a genotype, the hit does not reproduce, and the control is different from the one EV5 prescribes for sensor gaming. Add that noise alone leaves only **27% of a top-1% gate** truly top-1% when it equals the signal, and the practical conclusion is that **two rounds of sort-and-regrow should be the default**, not an afterthought.

The second was partly right about a theme that was already the most developed in the module. Downstream appears 199 times and the titer-to-separation link 332. What was genuinely absent was the capital split itself, and computing it gives a result that contradicts the suggestion as stated: **27% downstream for a commodity against 54% for a biologic**. I would rather you went into a defence able to compute that than able to quote "50 to 70 per cent", because the follow-up question — "for which product class?" — is the one a committee asks.

The two unit-operation gaps inside that theme were real and now computed: antifoam at 200 ppm costs **44% of oxygen transfer**, and PP10's three-pass optimum costs **3.4 times the filter area** it never counted.

The numbers worth keeping

- At 30,000 protein copies the CV is **19%**, and over 98% of the variance is transcriptional bursting.
- Extrinsic noise of 0.3 takes the total CV to **36%** and is removable by a ratiometric reporter.
- A Hill coefficient of **2** with a small leak splits the population into two states.
- At equal noise and signal, **27%** of a top-1% gate truly has a top-1% genotype; at noise above signal, **13%**.
- Sort–regrow–re-sort: **1.87 → 3.03 → 3.79** standard deviations of genotype over three rounds.
- Persisters are 10^{-6} to 10^{-2} of cells; any kill-based selection enriches them, so test heritability.
- Downstream capital share: **27% commodity, 54% biologic**; the biologic's facility alone is 31%.
- Antifoam at 200 ppm: **0.56** relative k_La.
- Three homogeniser passes: **3.4×** the filter area of one.

On the assumptions

The simulations and balances are computed. The inputs to challenge: the burst size of 12 and the extrinsic CV range, both recalled; the capital figures, which are order-of-magnitude estimates from scaling rules and judgement rather than quotations, and which are adequate to establish which section dominates and not to put in a proposal; and the antifoam and debris relations, whose *forms* are standard and whose coefficients are illustrative. For real capital numbers, the NREL-style design reports in the reading list are the source.

Where the set stands

Source	Problems
Curriculum appendix A1–W3	54
Parts 15–22	93
Part 23 HN1–HN3, DS1–DS3	6
Total in the tracker	**155**

Part 15 — Thirty more problems, worked: the second problem set

Three answers first, because two of your questions rest on a misunderstanding I caused by writing "6 of 54" without saying what the 54 were — and the count itself was wrong: the appendix holds 56.

Where the count comes from, and what it is not

56 is the number of problems in your own curriculum's appendix, labelled A1 to W3 (including B4 and G2, which an earlier extraction of mine missed) and starting at page 140 of `me-curriculum-wk01-66-and-56-problems.pdf`. It is not a number of datasets. I checked the whole document: **it supplies no data files at all** — no CSV, no accession to download, no spreadsheet. Every problem that needs numbers states them inline in its own text.

Of those 56, seven name a real paper and quote one measured parameter from it — the Fredens recoding count, the *Vibrio natriegens* generation time, the Newman amorphadiene partition coefficient, and so on. The other 47 use constructed numbers. That is what I meant, badly, by "6 of 54 use a real published dataset". See the corrigendum: the appendix holds 56.

Why not more? Two reasons, one of which is mine and one of which is not. I did not write those 56; they came with your curriculum, and I only verified their arithmetic — so their choice of constructed data was not a decision I made. And I cannot fix it by fetching real data now: this environment has no network access, so I can neither download a dataset nor check one. **If I wrote a problem claiming to use real measurements, I would be inventing them**, which is worse than a clearly labelled constructed example.

So the problems here come in two kinds, and both are honest

	Self-contained (X1–X24)	Download-required (Y1–Y6)
Data	Stated in the problem, or simulated with a known true answer	A real public dataset you download yourself
Advantage	You can check whether your method recovered the truth, which real data never lets you do	Real noise, real missing values, real file formats
Status here	**Every number was computed and is printed with its code**	Statement, source and code given; **I could not run them** — no network

The first kind is not a weaker substitute. A simulation with known truth is the only way to measure what a method *costs* — which is why X18 can tell you that a log-linear fit over the apparent exponential phase overstates μ by 97%, and no real growth curve could.

What the 24 new problems cover

Group	Problems	Skill
Mass, energy and redox balance	X1–X4	Yield ceilings with cofactors closed; route comparison; oxygen and heat duty
Kinetics and reaction engineering	X5–X8	Two-substrate fitting; substrate and product inhibition; thermal deactivation
Transport and reactor design	X9–X12	Dissolved CO_2 and pH; shear; gas holdup and flooding; heat removal
Experimental design and statistics	X13–X16	Two-way ANOVA with interaction; response surface; variance components; equivalence testing
Data analysis and instrument reading	X17–X19	Weighted calibration and carryover; growth-curve fitting; off-gas mass balance
Probability and library design	X20–X21	Coverage with uneven representation; barcode and UMI collisions
Economics and planning	X22–X24	Monte Carlo on selling price; campaign sizing under budget; vessel scheduling

They deliberately fill the gaps in the first 56: there was no ANOVA, no response-surface design, no variance-component analysis, no equivalence test, no heat balance, no shear or CO_2 calculation, no scheduling, and no Monte Carlo. Those are the eight things an examiner or a process group will ask about and the appendix did not cover.

Group 1 — Mass, energy and redox balance

X1 · extends weeks 18–19 · 1.5 h

Itaconic acid: a ceiling that carbon alone does not explain

Itaconic acid ($C_5H_6O_4$, 130.10 g/mol) is made from glucose through citrate: glucose → one pyruvate and one phosphoenolpyruvate; the PEP is carboxylated to oxaloacetate, the pyruvate is oxidised to acetyl-CoA losing a CO_2, the two condense to citrate, and cis-aconitate decarboxylase removes the second CO_2 to give itaconate.

1. Write the overall stoichiometry per mole of glucose and give the carbon and mass yield ceilings.
2. Count NADH, NADPH and ATP across the route. Is it redox-balanced?
3. Convert the surplus to an oxygen demand per gram of product.
4. Reported titers reach about 0.58, 0.72 and 0.86 of the ceiling in different hosts. What mass yields are those, and what would you measure to tell which constraint is binding?

Worked solution. One glucose gives one itaconate and loses one carbon as CO_2 at pyruvate dehydrogenase, while the other decarboxylation returns the carbon the PEP carboxylase fixed — so the net is **1 glucose → 1 itaconate + 1 CO_2**. Carbon yield 5/6 = **83.3%**; mass yield 130.10/180.156 = **0.722 g/g**. Glycolysis contributes 2 NADH and 2 net ATP, pyruvate dehydrogenase a third NADH, and neither the condensation nor the decarboxylation consumes reducing power, so the route leaves a **surplus of 3 NADH per product** and is not redox-balanced. Those electrons have to go to oxygen: 1.5 mol O_2 per mole of product, or **0.369 g O_2 per g of itaconate**. At 0.58, 0.72 and 0.86 of the ceiling the mass yields are **0.419, 0.520 and 0.621 g/g**.

Which constraint binds is answerable by measurement rather than argument: close a carbon balance including CO_2 and by-products — if carbon is unaccounted, it is leaving somewhere you have not looked; and measure the oxygen uptake rate against the 0.369 g/g the redox surplus demands — if the vessel cannot supply it, the ceiling is a transport ceiling and no pathway work will reach it.

What it teaches: a pathway can be carbon-efficient and redox-unbalanced, and the redox surplus converts directly into an oxygen duty that the reactor, not the strain, has to meet.

X2 · extends weeks 18–19 · 1.5 h

3-hydroxypropionate by two routes: which is actually better?

3-HP ($C_3H_6O_3$, 90.08) can be made from glycerol — dehydratase to 3-hydroxypropionaldehyde, then an aldehyde dehydrogenase — or from glucose through malonyl-CoA, where acetyl-CoA carboxylase fixes a CO_2 and two reductions give the product.

1. Give the carbon and mass yield ceiling of each route on its own substrate.
2. The malonyl-CoA route loses two CO_2 at pyruvate dehydrogenase. Why is its carbon ceiling still 100%?
3. Count ATP and NADPH per product for the malonyl-CoA route, and correct its ceiling for the carbon that supplying the NADPH costs.
4. At what glycerol-to-glucose price ratio do the two break even on feedstock cost alone?

Worked solution. The glycerol route is carbon-conservative — three carbons in, three out — giving 90.08/92.09 = **0.978 g/g**, and it generates one NADH rather than consuming reductant. The malonyl-CoA route loses two CO_2 making two acetyl-CoA, and then **acetyl-CoA carboxylase re-fixes one CO_2 per product**, so the six carbons of glucose end up in two three-carbon products: carbon yield **100%**, mass yield 2 × 90.08/180.156 = **1.000 g/g**. That is the headline number quoted for 3-HP, and it is real but conditional, because the route needs **1 ATP and 2 NADPH per product**. If the NADPH comes from a fully cycling oxidative pentose phosphate pathway at 12 NADPH per glucose, four NADPH costs a third of a glucose, so eight carbons are consumed for six in product and the **effective ceiling falls to 75%, about 0.75 g/g**.

Break-even on feedstock: cost per kg of product is price divided by yield, so the routes are equal when $P_{glycerol}/0.978 = P_{glucose}/0.75$, that is when glycerol costs **1.30 times** glucose per kilogram. Below that ratio the glycerol route wins on feedstock, which is why crude glycerol from biodiesel made this route attractive in the first place.

What it teaches: a 100% carbon yield can still be an overstatement once you charge the reductant to carbon, and the comparison between two routes is a price ratio, not a biology argument.

X3 · extends weeks 7 and 53 · 1.5 h

Isoprenoids: the mevalonate and MEP routes compared

Both routes make the five-carbon isopentenyl pyrophosphate. Mevalonate condenses three acetyl-CoA and loses one CO_2, spending 3 ATP and 2 NADPH. MEP condenses pyruvate and glyceraldehyde-3-phosphate, loses one CO_2, and spends a CTP, 3 NADPH and reduced ferredoxin.

1. Give the carbon ceiling to isoprene units for each route.
2. Amorphadiene is $C_{15}H_{24}$ (204.35). Compute its mass yield ceiling on glucose.
3. Tabulate ATP and reductant per amorphadiene for both routes.
4. Why do engineers import the mevalonate route into *E. coli*, which already has MEP?

Worked solution. Both routes convert six carbons into a five-carbon unit, so both have a carbon ceiling of **83.3%** — the common claim that MEP is intrinsically more carbon-efficient does not survive this count, because both lose exactly one CO_2 per C_5. Amorphadiene needs three C_5 units, so 15 carbons from 18, which is three glucose: **204.35/(3 × 180.156) = 0.378 g/g**. Per amorphadiene the mevalonate route spends **9 ATP and 6 NADPH**; MEP spends **3 CTP, 9 NADPH and 3 reduced ferredoxin** — cheaper in nucleoside triphosphate, dearer in reductant, and dependent on a ferredoxin pool that is tied to central metabolism in ways that are hard to push.

The reason for importing mevalonate is not thermodynamic but regulatory and practical: MEP in *E. coli* is feedback-controlled at its first step and embedded in native regulation, while a heterologous mevalonate operon is unregulated by the host and can be tuned by expression alone. That is the part 1 chapter 9 argument — **you import a pathway to escape regulation, not to improve stoichiometry**.

What it teaches: two routes with identical carbon ceilings can differ entirely in what they spend and in how controllable they are, and the engineering choice usually turns on the second.

X4 · extends weeks 18 and 44 · 2 h

Single-cell protein from methanol: the heat nobody budgets for

A 100 m³ fermenter grows a methylotroph on methanol to 30 gDW/L at $\mu = 0.15\ h^{-1}$, with a biomass yield of 0.40 g/g. Take biomass as $CH_{1.8}O_{0.5}N_{0.2}$ (24.6 g per C-mol, degree of reduction 4.2) and methanol as CH_4O (32.04 g/mol, degree of reduction 6).

1. From an electron balance, compute the oxygen demand per gram of biomass at yields of 0.35, 0.40 and 0.45 g/g.
2. Compute the biomass production rate and the oxygen uptake rate of the vessel.
3. Using 460 kJ of heat released per mole of oxygen consumed, compute the cooling duty.
4. With $U = 500\ W \cdot m^{-2} \cdot K^{-1}$ and a 10 K approach, what heat-transfer area is needed? A 100 m³ tank has roughly 110 m² of wall. What do you conclude?

Worked solution. The electron balance is $\gamma_S \cdot C_S = \gamma_X \cdot C_X + 4 \cdot O_2$. Per gram of methanol that gives $(6/32.04 - 4.2 \cdot Y/24.6)/4$ moles of O_2, so the oxygen demand per gram of biomass is **2.91, 2.38 and 1.96 g O_2 per g** at the three yields — methanol is a reduced substrate and costs far more oxygen than sugar does. At the design point the vessel makes $30 \times 0.15 \times 100$ = **450 kg of biomass per hour** and consumes 450×2.38 = **1,071 kg O_2 per hour**, which is 33.5 kmol/h. At 460 kJ/mol that is **4.28 MW of heat**, needing 368 m³/h of cooling water at a 10 K rise.

The area follows: 4.28 MW ÷ (500 × 10) = **855 m²**, against about 110 m² of jacket. The conclusion is structural, not a tuning problem — **a methanol fermentation at this density cannot be cooled through its wall** and needs internal coils or an external heat exchanger, which is why industrial methylotroph plants look the way they do. It is also why the same vessel on glucose, which demands roughly a third of the oxygen per gram, raises no such problem.

What it teaches: oxygen demand and heat release are the same calculation, and on a reduced substrate the heat, not the oxygen, is what sets the vessel design.

Group 2 — Kinetics and reaction engineering

X5 · extends week 3 and chapter S3 · 2 h

Fit a two-substrate enzyme, and see what a single-substrate fit hides

An NADPH-dependent reductase is assayed over a grid: six substrate concentrations from 0.05 to 1.6 mM at each of four cofactor concentrations from 0.05 to 2.0 mM, with 5% noise. The true parameters are V = 2.4, K_a = 0.30, K_b = 0.18 and K_{ia} = 0.45.

1. Fit the ternary-complex rate law and report parameters with standard errors.
2. Fit the ping-pong law to the same data and compare residual sums of squares.
3. Compute the apparent K_M for substrate A at the lowest and highest cofactor concentration.
4. What would a single-substrate fit at a saturating cofactor have reported, and why does that matter for a cofactor-swap campaign?

Worked solution. The ternary fit recovers **V = 2.43 ± 0.07, K_a = 0.347 ± 0.025, K_b = 0.193 ± 0.018, K_{ia} = 0.375 ± 0.073** — all within about one standard error of truth except K_{ia}, which is always the least identifiable parameter because it only shows up in the constant term. The ping-pong fit gives a residual sum of squares **5.9 times larger**, which with 24 points and one fewer parameter is a clear rejection; the diagnostic is the double-reciprocal pattern from chapter S3, intersecting against parallel.

The apparent K_M for A is $K_a + K_{ia}K_b/[B]$, which is **1.92 mM at [B] = 0.05** and **0.34 mM at [B] = 2.0** — a **5.6-fold difference in the same enzyme**. A single-substrate fit at saturating cofactor would report 0.34 and call it the K_M. For a cofactor swap that is the wrong number twice over: the quantity you are engineering is K_{ia}, the dissociation constant of the cofactor from the binary complex, which never appears in an apparent K_M; and a variant that looks improved at saturating cofactor can be worse at the intracellular concentration, which for NADPH sits far below saturation.

What it teaches: the K_M of a two-substrate enzyme is not a property of the enzyme, and the parameter a cofactor-swap campaign actually changes is invisible to the standard assay.

X6 · extends weeks 3 and 44 · 1 h

Substrate inhibition sets the feed set-point

An enzyme follows $v = V_{max}S/(K_M + S + S^2/K_{si})$ with K_M = 0.2 g/L and K_{si} = 8 g/L.

1. Derive the substrate concentration that maximises the rate and evaluate it.
2. What fraction of V_{max} is reachable at that optimum?
3. Tabulate the rate at 0.5, 1.3, 5, 20 and 50 g/L.
4. A fed-batch controller overshoots to 20 g/L. What has it cost, and how would you detect the overshoot without a substrate assay?

Worked solution. Differentiating and setting dv/dS = 0 gives **$S^* = \sqrt{(K_M K_{si})}$ = 1.26 g/L**, at which the rate is $V_{max}/(1 + 2\sqrt{(K_M/K_{si})})$ = **0.760 V_{max}** — note that the enzyme can never reach its own V_{max}, which is a feature of the rate law and not of the assay. The rate profile is **0.684, 0.760, 0.601, 0.285 and 0.138** of V_{max} at the five concentrations: the curve is asymmetric, falling slowly below the optimum and steeply above it.

An overshoot to 20 g/L runs at **37% of the achievable rate** — a 2.7-fold loss of productivity that looks exactly like a strain problem. Detecting it without a substrate assay is the useful part: the oxygen uptake rate falls as the metabolic rate falls, so the dissolved-oxygen trace *rises* while the feed is still running, and the respiratory quotient drifts. A rising DO during active feeding is the signature of inhibition or of a stalled culture, never of a healthy one.

What it teaches: with substrate inhibition the control objective is a set-point, not a maximum, and the cheapest instrument for finding the set-point is the dissolved-oxygen probe.

X7 · extends weeks 16 and 45 · 2 h

Product inhibition decides when to stop the batch

A fermentation follows $\mu = \mu_{max}(1 - P/K_P)\cdot S/(1 + S)$ with μ_{max} = 0.35 h^{-1} and K_P = 45 g/L, starting from 2 gDW/L and 120 g/L of substrate, with $Y_{P/S}$ = 0.45 g/g.

1. Integrate the system and tabulate biomass, product and substrate at 10, 20, 40, 60 and 80 h.
2. When does the culture reach 90% of the inhibition ceiling?
3. How much substrate is left at the end, and what does that mean for the yield on substrate fed?
4. What are the three engineering responses, and which does the model support?

Worked solution. The integration gives product at **34.9 g/L by 10 h, 44.7 by 20 h, and 45.0 thereafter** — the run is effectively over at 20 hours and the remaining sixty hours add nothing. It reaches 90% of the ceiling at **12.5 h**. Substrate stops at **70 g/L unconsumed** of the 120 fed, so the yield on substrate *fed* is 45/120 = **0.375 g/g** against a metabolic yield of 0.45 — the process throws away 58% of its feedstock, and the loss is invisible in a yield quoted on substrate consumed.

Three responses: harvest at 20 h and recycle the substrate, which fixes the feedstock loss and not the titer; remove the product in situ, which raises the ceiling (part 3 chapters 45 to 47); or engineer tolerance, which raises K_P. The model supports the first two immediately and says nothing about the third, because K_P is an input to it. The decisive measurement is whether the ceiling is tolerance or thermodynamic end-product inhibition: add product externally at the start and see whether growth stops at the same total.

What it teaches: a yield quoted on substrate consumed can hide a process that leaves most of its feed in the tank, and the harvest time is a design variable with a right answer.

X8 · extends weeks 3 and 22 · 1.5 h

Biocatalysis: the optimum temperature depends on the batch length

An enzyme's turnover has an activation energy of 50 kJ/mol and its thermal deactivation 250 kJ/mol, both referenced to 30 °C where k = 1.0 (relative) and k_d = 0.001 h^{-1}.

1. Tabulate the relative rate, the deactivation constant and the enzyme half-life from 25 to 60 °C.
2. Derive the total product formed in a batch of length t for a first-order decaying catalyst.
3. Find the temperature that maximises an 8-hour batch.
4. Repeat for 2, 24 and 72 hours. What is the general rule?

Worked solution. With activity decaying as $e^{-k_d t}$, the integrated product is proportional to $(k/k_d)(1 - e^{-k_d t})$. The table shows the trade sharply: at 40 °C the rate is 1.88 times the 30 °C value and the half-life is 29 h; at 50 °C the rate is 3.41 times but the half-life is **1.5 hours**; at 60 °C the enzyme is gone in minutes.

Batch length	Optimum temperature
2 h	47.4 °C
8 h	42.7 °C
24 h	39.1 °C
72 h	35.6 °C

The rule follows from the two activation energies: because deactivation is far more temperature-sensitive than catalysis, **the optimum falls as the batch lengthens**, by roughly 4 °C per tripling of duration here. A "optimum temperature" quoted from a ten-minute assay is therefore the wrong set-point for a day-long reaction — the assay optimum is near 50 °C and the 24-hour optimum is 39 °C. The same arithmetic is the argument for enzyme immobilisation, which usually raises k_d's activation energy and moves the whole curve.

What it teaches: an enzyme's optimum temperature is not a property of the enzyme but of the enzyme and the reaction time together.

Group 3 — Transport and reactor design

X9 · extends week 44 · 1.5 h

Dissolved CO_2: the variable nobody controls

CO_2 leaves the cells as fast as oxygen enters at RQ = 1, and it must be stripped by the same gas flow. Take Henry's constant for CO_2 as 0.025 M/atm at 37 °C and $pK_1 = 6.1$ for the CO_2/bicarbonate couple.

1. At CER values of 50, 150 and 250 mmol·L^{-1}·h^{-1} and k_La of 100 and 300 h^{-1}, compute the excess dissolved CO_2 at steady state.
2. Convert the worst case to an equivalent partial pressure.
3. Compute the bicarbonate in equilibrium with 1.5 mM dissolved CO_2 at pH 7.0.
4. Head pressure is often raised to 1.5 bar to help oxygen transfer. What does it do to CO_2, and what would you watch?

Worked solution. At steady state CER = $k_La(C - C^*)$, so the excess is CER/k_La. At 250 mmol·L^{-1}·h^{-1} and k_La = 100 h^{-1} that is **2.50 mM**, equivalent to **0.100 atm** of CO_2 — and at k_La 300 it falls to 0.83 mM. The bicarbonate in equilibrium with 1.5 mM dissolved CO_2 at pH 7.0 is $1.5 \times 10^{(7.0 - 6.1)}$ = **11.9 mM**, which is a substantial ionic load and consumes base in a pH-controlled vessel.

Raising the head pressure to 1.5 bar multiplies the dissolved CO_2 by 1.5 along with the oxygen, so the strain now sees **2.2 mM** where it saw 1.5. That is the trade the oxygen calculation usually ignores: pressure buys oxygen transfer and buys dissolved CO_2 at the same ratio. What to watch is base consumption and the pH control output, which rise before any growth effect appears, and — in a strain with a CO_2-sensitive carboxylation step — the by-product spectrum.

What it teaches: every measure that improves oxygen transfer also concentrates CO_2, and dissolved CO_2 is a scale-up variable that bench vessels never reveal.

X10 · extends week 44 · 1 h

Shear: what the impeller actually does to a cell

The smallest turbulent eddies have the Kolmogorov scale $\lambda = (\nu^3/\varepsilon)^{1/4}$, with $\nu = 10^{-6}$ m²/s and ε the dissipation rate per unit mass.

1. Compute λ at P/V of 100, 500, 1,000, 5,000 and 20,000 W/m³.
2. Compare with a 1 µm bacterium, a 15 µm animal cell and a 100 µm floc or pellet.
3. Local dissipation near the impeller is 20–50 times the mean. How does that change the picture?
4. What does this say about bacteria, animal cells and immobilised pellets?

Worked solution. The eddy scale is **56 µm at 100 W/m³, 31.6 µm at 1,000 and 15.0 µm at 20,000** — it falls only as the fourth root of power, so a 200-fold power increase shrinks the eddies by less than four times. Against that: a bacterium at 1 µm is smaller than the smallest eddy at any realistic power, so **bacteria are not damaged by shear in a stirred tank** and reports to the contrary are usually about the air–liquid interface or foam. An animal cell at 15 µm is comparable to the eddy scale at high power, which is why those cultures are run at a few tens of W/m³. A 100 µm floc or immobilisation pellet is larger than the eddies at every power in the table and **will be eroded**.

Near the impeller, dissipation is 20–50 times higher, dividing the scale by about 2.3 — so the 1,000 W/m³ case has 14 µm eddies locally, and the cells circulate through that zone repeatedly. The practical consequence is that pellet and floc processes are limited by impeller design and tip speed rather than by mean power, and that the standard advice to "reduce shear" for a bacterial culture is usually solving the wrong problem.

What it teaches: shear damage is a question of length scales, and for bacteria the answer is almost always that shear is not the problem.

X11 · extends week 44 · 1 h

Gas holdup and flooding: the limits of "add more air"

Take gas holdup as roughly $0.6 \cdot v_s^{0.7}$ with superficial velocity in m/s, and the flooding criterion for a Rushton turbine as a flow number $Q/(ND^3)$ above about 0.25, with N = 0.7 s^{-1} and D = 1.8 m.

1. Compute holdup at superficial velocities of 0.005, 0.02, 0.05 and 0.1 m/s, and the working volume lost.
2. Compute the flow number at gas rates of 0.05, 0.2, 0.5 and 1.0 m³/s and identify flooding.

3. Combine with X9: what happens to CO_2 stripping as you approach flooding?
4. State the three ways to raise oxygen transfer once flooding is reached.

Worked solution. Holdup rises from **1% at 0.005 m/s to 12% at 0.1 m/s** — the gas displaces broth, so a vessel pushed to high aeration loses more than a tenth of its working volume and its nominal titer per vessel falls with it. The flow numbers are **0.012, 0.049, 0.122 and 0.245**, so even 1 m^3/s is only just at the flooding threshold for this impeller at this speed; flooding is reached sooner at lower stirrer speed, which is exactly when a control system is trying to save power.

Beyond flooding the impeller no longer disperses the gas, bubbles coalesce and rise as a column, and k_La *stops rising with gas flow* — while CO_2 stripping, which depends on the same interfacial area, stops improving too. The three remaining levers are the ones from part 11 chapter 44: oxygen enrichment, which raises C* in proportion; head pressure, which does the same and worsens dissolved CO_2; and lowering demand by feeding more slowly.

What it teaches: aeration has a ceiling set by hydrodynamics, not by the compressor, and the holdup penalty means the last increment of oxygen transfer is paid for in working volume.

X12 · extends weeks 44 and 48 · 1.5 h

The sterilisation and cooling of a fed-batch, end to end

Using X4's methanol case as a contrast, take a 100 m^3 glucose fed-batch reaching 40 gDW/L with a specific oxygen uptake of 4 $mmol \cdot gDW^{-1} \cdot h^{-1}$, and 460 kJ released per mole of oxygen.

1. Compute the oxygen uptake rate, the required k_La at a 0.16 mM driving force, and the heat duty.
2. Compare the heat duty with X4's methanol case per unit volume.
3. How much cooling water at a 10 K rise, and what jacket area at U = 500 $W \cdot m^{-2} \cdot K^{-1}$?
4. At what cell density does the jacket alone stop being sufficient?

Worked solution. OUR = 40 × 4 = **160 $mmol \cdot L^{-1} \cdot h^{-1}$**, needing k_La = 160/0.16 = **1,000 h^{-1}** — already above what a large vessel delivers, which is the part 11 chapter 44 wall. The heat is 160 mmol/L/h × 100,000 L × 460 J/mmol = **2.04 MW**, against methanol's 4.28 MW in the same vessel: **methanol releases about twice the heat per unit volume** for a third less biomass, because it is the more reduced substrate.

Cooling water at a 10 K rise is 175 m^3/h, and the area required is 2.04 MW/(500 × 10) = **408 m^2** against roughly 110 m^2 of jacket. Setting the area equal to the jacket and solving backwards, the jacket alone suffices up to **about 11 gDW/L** at this specific uptake rate. That is the number worth remembering: **a high-cell-density process needs internal cooling from about 10 gDW/L upwards**, which is far below the densities everyone quotes as a target, and it is a capital decision taken at design time rather than a process adjustment.

What it teaches: the same oxygen number that sizes the sparger also sizes the cooling, and cooling runs out before oxygen does on a reduced substrate.

Group 4 — Experimental design and statistics

X13 · extends week 17 · 2 h

Two-way ANOVA: the interaction one-factor-at-a-time cannot see

Three media and two strains, four flasks each — 24 runs. Titers are generated with a true medium effect of 0 / 1.6 / 2.4 g/L, a strain effect of 1.1 g/L, an interaction of +0.9 g/L for strain 2 on medium 2, and a flask-to-flask standard deviation of 0.55 g/L.

1. Partition the total sum of squares into medium, strain, interaction and error, and build the ANOVA table with F and p.
2. What fraction of the variance does the model explain?
3. What would a one-factor-at-a-time campaign have concluded?
4. How many runs would the one-at-a-time route have used for the same information?

Source	SS	df	MS	F	p
Medium	22.09	2	11.04	50.25	<0.00001
Strain	6.73	1	6.73	30.64	0.00003
Interaction	3.01	2	1.51	**6.85**	**0.0061**
Error	3.96	18	0.220		

Worked solution. The model explains **89%** of the variance, and the interaction is significant at p = 0.0061 — the best medium is *not the same* for both strains. A one-factor-at-a-time campaign would have fixed one strain, found the best medium for it, then fixed that medium and compared strains, and would have reported both main effects correctly and **missed the 0.9 g/L interaction entirely**, because no run would have combined the second strain with the second medium at replication. It would then have recommended a strain–medium pairing that is not the best one available.

On run count, the one-at-a-time route needs 3 media × 4 + 2 strains × 4 = 20 runs to estimate the two main effects and still cannot estimate the interaction at all; the factorial spends 24 and estimates all three. That is the core design-of-experiments argument, and the honest caveat is that a factorial grows as the product of levels, so beyond three or four factors you move to a fractional design and accept confounding you choose deliberately.

What it teaches: interactions are the reason factorial designs exist, and one-at-a-time is not merely less efficient — it is blind to a whole class of effect.

X14 · extends weeks 17 and 41 · 2.5 h

Response surface: find an optimum in twelve runs and test the model

A central composite design in two factors — four factorial points, four axial points at ±1.414, and four centre replicates — on a true surface with an interior maximum and measurement noise of 0.25 g/L.

1. Fit the full quadratic model by least squares and report the coefficients.
2. Solve for the stationary point and classify it from the eigenvalues of the quadratic form.
3. Carry out a lack-of-fit test using the centre replicates.
4. Compare the information per run with a 5 × 5 grid.

Worked solution. The fit returns coefficients **9.15, +1.64 x_1, +1.29 x_2, −1.68 x_1^2, −1.28 x_2^2, −0.57 x_1x_2**. Solving $\nabla y = 0$ gives a stationary point at **x_1 = 0.417, x_2 = 0.411** with a predicted response of **9.75**, against the true optimum of 9.65 — recovered to within the noise from twelve runs. The eigenvalues of the quadratic form are **−3.66 and −2.26**, both negative, so the stationary point is a **maximum** and not a saddle; that check matters because a saddle point looks like an optimum in the coefficients alone and will send the next round in a useless direction.

The lack-of-fit test uses the four centre points to estimate pure error: **F = 0.53, p = 0.69**, so the quadratic model is adequate and there is no evidence of curvature it cannot capture. A 5 × 5 grid would need 25 runs to cover the same space, so the design delivers roughly **2.1 times the information per run** — and, unlike a grid, it gives a fitted surface, a stationary point and an adequacy test rather than a table of values.

What it teaches: the eigenvalue check and the lack-of-fit test are what separate a response-surface design from curve-fitting, and both come free with the centre replicates.

X15 · extends week 17 · 1.5 h

Variance components: replicate the right thing

Six strains, two flasks each, three HPLC injections per flask — 36 measurements. The true standard deviations are 0.80 g/L between strains, 0.35 between flasks and 0.12 between injections.

1. Estimate the three variance components from the nested mean squares.
2. Compute the standard error of a strain mean for the design as run.
3. Compare with three flasks × one injection and one flask × six injections, for the same 6 measurements per strain... and then for 3.
4. Where should the next unit of effort go?

Worked solution. The components come out at **analytical 0.014 (2%), flask 0.061 (8%), strain 0.730 (91%)** of the total variance. The standard error of a strain mean with two flasks and three injections is **0.181 g/L**. With three flasks and one injection each it is **0.158** — better, for half the chromatography; with one flask and six injections it is **0.252**, substantially worse despite the same number of numbers in the spreadsheet.

The rule the arithmetic enforces is that **injections are pseudo-replicates**: they estimate the analytical variance, which is 2% of the problem, and they do nothing about the flask variance that dominates what you can resolve between strains. The next unit of effort goes into more flasks, and the only reason to inject twice is to catch an instrument failure. The same logic recurs whenever a measurement is nested — wells within a plate, plates within a run, runs within a week — and the standard mistake is to treat the cheapest level as if it were replication.

What it teaches: replication at the wrong level buys precision you do not need; the variance components tell you which level to spend on.

X16 · extends week 17 · 1 h

Equivalence: proving two strains are the same

Strain A gives 12.1, 11.8, 12.4, 12.0, 11.9 g/L; strain B gives 12.3, 12.6, 11.9, 12.2, 12.4. You want to claim that a marker-free derivative is *equivalent* to the parent.

1. Run the conventional two-sample t-test and state what it does and does not establish.
2. Carry out two one-sided tests (TOST) against an equivalence margin of ±0.5 g/L, then ±1.0 g/L.
3. Report the 95% confidence interval for the difference and interpret it against both margins.
4. What margin would you pre-register, and on what basis?

Worked solution. The difference is **0.240 g/L** with **p = 0.16**, so the conventional test fails to reject — which establishes nothing, since absence of evidence at n = 5 is expected whatever the truth. TOST against ±0.5 g/L gives **p = 0.066, not equivalent**; against ±1.0 g/L it gives **p = 0.0006, equivalent**. The 95% interval is **−0.12 to +0.60 g/L**, which sits inside ±1.0 and not inside ±0.5 — exactly the same conclusion, read off the interval.

The margin has to be chosen before the data and on process grounds, not statistical ones: it is the largest difference that would not change a decision. For a marker-free derivative intended to replace the parent in production, a defensible margin is the run-to-run variability of the process itself, because a difference smaller than that is unobservable in manufacturing. Quoting "no significant difference" as equivalence is the error this problem exists to kill — and with these data it would have claimed equivalence at a margin the data cannot support.

What it teaches: failing to find a difference is not finding equivalence, and the equivalence margin is a process decision made before the experiment.

Group 5 — Data analysis and instrument reading

X17 · extends weeks 22 and 50 · 2 h

Calibration, weighting and the carryover that fakes a low-level peak

An HPLC calibration runs seven standards from 0.05 to 5.0 g/L with proportional noise — the standard deviation grows with concentration. The true slope is 57 area units per g/L with an intercept of 1.2.

1. Fit the calibration unweighted and with $1/\sigma^2$ weights; compare the slope, intercept and the back-calculated error on the three lowest standards.
2. Compute LOD and LOQ from a blank standard deviation of 0.05 area units.
3. At carryover of 0.1%, 0.5% and 2%, what does a 5 g/L sample leave in the following blank, as a fraction of the LOQ?
4. What injection order and what controls would you use?

Worked solution. Both fits recover the slope well — **57.26 unweighted, 57.46 weighted** — but the back-calculated error on the three lowest standards is **2.3% unweighted against 0.7% weighted**, because the unweighted fit lets the 5 g/L point dominate the line. LOD = 3.3σ/slope = **0.003 g/L** and LOQ = 10σ/slope = **0.009 g/L**.

The carryover result is the one worth internalising. A 5 g/L sample at 0.1% carryover leaves 0.005 g/L in the next injection, which is **57% of the LOQ**; at 0.5% it is **287%** and at 2% it is **1,149%** of the LOQ. In other words a quantifiable peak can be created entirely by the previous sample, which is how a trace product gets "detected" in a negative control. The defences are order and controls: inject from low to high concentration, put a blank after every high sample and *integrate it*, include a true negative control strain in the same sequence, and report carryover as a validated number rather than assuming it is negligible.

What it teaches: a detection limit is meaningless without a carryover figure, because carryover can exceed the LOQ by an order of magnitude.

X18 · extends weeks 10 and 17 · 1.5 h

Growth curves: fit the model, or misread the slope by a factor of two

Twenty-five hourly optical-density readings follow a Gompertz curve with a plateau of 2.2, μ_{max} of 0.42 h^{-1} and a lag of 3.0 h, with 0.03 of measurement noise.

1. Fit the Gompertz model and report all three parameters with standard errors.
2. Fit a straight line to ln(OD) over the window that looks exponential, 0.05 to 0.6, and compare the growth rate.
3. Explain the discrepancy.
4. Above what OD does Beer–Lambert fail, and what do you do about it?

Worked solution. The Gompertz fit recovers **A = 2.19 ± 0.01, μ_{max} = 0.420 ± 0.009 h^{-1}, lag = 2.98 ± 0.06 h** — all three parameters, to about 2%. The log-linear fit over the apparently exponential window returns **0.827 h^{-1}, which is 197% of the true value**.

The discrepancy is not noise and not a coding error: the window from OD 0.05 to 0.6 includes the *acceleration* out of lag, where the instantaneous slope of ln(OD) rises from zero toward μ_{max} and then falls again. Fitting a straight line through a sigmoid's inflection region gives the chord, not the tangent, and the chord here is nearly double. The fix is either to fit the full model, as in part 1, or to restrict the linear fit to a window verified to have constant slope by a rolling regression. On the instrument side, Beer–Lambert fails above about **OD 0.4** on a standard 1 cm path because of multiple scattering, so readings up to 2.2 must come from dilution or a short-path reader; using them raw flattens the curve and depresses the fitted plateau and rate together.

What it teaches: the "exponential phase" identified by eye is not exponential, and the most common growth-rate error in the literature is a factor of two from a linear fit over an inflection.

X19 · extends week 44 · 2 h

Off-gas analysis: OUR, CER and RQ from first principles

A 100 m^3 vessel is aerated at 0.5 vvm. Inlet air is 20.95% O_2, 0.04% CO_2 and the balance inert. Three outlet compositions are measured over a run.

1. Compute the inlet molar flow, then the outlet flow by an inert balance.

2. Compute OUR, CER and RQ for each case.
3. Compare with the naive calculation that assumes the outlet flow equals the inlet flow.
4. Convert the middle case to a specific uptake rate at 30 gDW/L and to the k_La it implies.

Outlet O_2 / CO_2	OUR (mmol·L⁻¹·h⁻¹)	CER	RQ	Naive OUR error
19.5% / 1.2%	20.4	15.5	0.76	−5%
18.2% / 2.8%	36.8	36.9	1.00	0%
16.8% / 4.5%	54.7	59.9	1.10	+2%

Worked solution. The inlet flow is 0.5 × 100,000 L/min × 60 ÷ 22.414 = **133,845 mol/h**. The outlet flow must be obtained from the inert balance, $F_{out} = F_{in} \cdot y_{inert,in} / (1 - y_{O_2} - y_{CO_2})$, because consuming oxygen and producing CO_2 at RQ ≠ 1 changes the total molar flow. The naive assumption is wrong by **−5% to +2%** here, which sounds small until you notice it is a systematic bias that varies through the run and so distorts the *shape* of the OUR trace, which is what you use to detect a feed problem.

The three RQ values read as a physiological narrative: **0.76** is growth on a substrate more reduced than the biomass with lipid synthesis under way, **1.00** is clean sugar oxidation, and **1.10** is the onset of overflow, with CO_2 appearing without a matching oxygen demand. The middle case at 30 gDW/L is **1.2 mmol·gDW⁻¹·h⁻¹** and implies a k_La of **230 h⁻¹** at a 0.16 mM driving force — comfortably achievable, which tells you this vessel is not oxygen-limited at this point and that the RQ excursion is physiological rather than a transfer artefact.

What it teaches: the off-gas analyser is the only instrument that reports metabolism continuously, and the inert balance is what makes its numbers trustworthy.

Group 6 — Probability and library design

X20 · extends weeks 28 and 30 · 1.5 h

Library coverage when the library is not uniform

A 10,000-member library is sampled at 30,000 and 50,000 clones. Representation is lognormal with coefficient of variation 0, 0.5, 1.0 and 2.0 — a realistic range for an oligo pool or a transformed library.

1. Derive the expected number of missing members for an arbitrary representation distribution.
2. Evaluate it at each CV and sampling depth.
3. By how much does the usual uniform-coverage rule understate the requirement?
4. How would you measure the CV of your own library, and what would you do about it?

CV of representation	Missing at 30,000 clones	at 50,000
0 (uniform)	498 (5.0%)	67 (0.7%)
0.5	979 (9.8%)	309 (3.1%)
1.0	1,843 (18.4%)	931 (9.3%)
2.0	3,389 (33.9%)	2,300 (23.0%)

Worked solution. For member *i* with probability p_i, the chance of never being sampled in k draws is e^{-kp_i}, so the expected number missing is Σe^{-kp_i} — which for a uniform library collapses to the familiar $Ne^{-k/N}$ and otherwise does not. At CV 1.0, **18.4% of the library is absent after 30,000 clones** where the uniform rule promises 5%; recovering the uniform figure needs about **2.6 times more clones**.

Measuring your own CV is a sequencing question: amplify the variable region from the pooled library and count reads per member, then take the coefficient of variation of the counts after correcting for the Poisson sampling of the sequencing itself. The remedies are upstream — normalise the oligo pool, use equimolar fragment assembly rather than a single ligation, and avoid growth steps between transformation and harvest, since a few hours of outgrowth imposes its own selection and widens the distribution further.

What it teaches: coverage rules assume uniformity, libraries are not uniform, and the correction is a factor of two to three in clones screened.

X21 · extends weeks 28 and 52 · 1 h

Barcodes and UMIs: when counting molecules stops working

A unique molecular identifier of length L has 4^L possible sequences. You tag n molecules.

1. Derive the expected number of collisions and the expected number of distinct identifiers recovered.
2. Evaluate for a 10-nt UMI at 10^5 and 10^6 molecules, a 12-nt at 10^6 and a 16-nt at 10^7.
3. By how much does a 10-nt UMI undercount 10^6 molecules?
4. What length would you specify for a barcoded strain library of 50,000 members, and why is the answer different from the UMI case?

Worked solution. Collisions follow the birthday calculation, $n(n-1)/2S$, while the distinct count recovered is $S(1 - e^{-n/S})$ — the saturation form. A 10-nt UMI (10^6 sequences) recovers **95.4% of 10^5 molecules but only 64.5% of 10^6**, so it **undercounts by 36%** at the higher load. A 12-nt UMI recovers 97.1% of 10^6, and a 16-nt recovers 99.9% of 10^7.

For a 50,000-member strain library the requirement is different and stricter: a UMI undercount is a quantitative bias you can correct for statistically, whereas two *strains* sharing a barcode is an identity error that silently merges two genotypes and cannot be corrected afterwards. You therefore design for essentially zero collisions rather than for acceptable saturation — with 50,000 members, a 12-nt barcode gives an expected 0.07 collisions and a 10-nt gives 1,250, so **12 nt is the minimum and 14 to 16 is the sensible specification**, with the additional requirement that barcodes differ at several positions so that a sequencing error cannot convert one into another.

What it teaches: the same birthday arithmetic has two different acceptance criteria depending on whether a collision biases a count or corrupts an identity.

Group 7 — Economics and planning

X22 · extends week 49 · 2 h

Monte Carlo on the selling price: which uncertainty actually matters

Take the part 4 chapter 49 model at a 50 g/L titer, 0.40 g/g yield, $0.40/kg sugar, $120M installed capital and 30,000 t/yr. Now treat five inputs as distributions: sugar price N(0.40, 0.06), yield N(0.40, 0.04), titer N(50, 8), capital lognormal with 25% spread, discount rate N(0.10, 0.02).

1. Compute the deterministic minimum selling price.
2. Run 20,000 Monte Carlo draws and report the median, P10 and P90.
3. Compute the probability of clearing $1.80, $2.00 and $2.50 per kg.
4. Rank the inputs by their correlation with the output and state where to spend the next engineering effort.

Worked solution. Deterministic MPSP is **$2.05/kg**. The Monte Carlo gives a median of **$2.08** with P10 **$1.76** and P90 **$2.47** — note that the median exceeds the deterministic value, because the lognormal capital distribution and the 1/titer term are both right-skewed, so a point estimate computed at the mean inputs is optimistic. Probabilities of clearing the three targets are **14%, 38% and 92%**.

Rank correlations with the output are **capital 0.66, sugar price 0.56, yield −0.38, titer −0.24, discount rate 0.23**, so the single biggest lever on the *uncertainty* is the capital estimate — not the yield, which is where a strain-engineering group would instinctively spend. That is the real use of the analysis: it separates what you can change from what dominates the risk, and it says that a firmer capital quote is worth more than a yield improvement of the size you could plausibly deliver. The complementary reading is that improving the yield shifts the whole distribution left, so it reduces the *level* while the capital estimate reduces the *spread*.

What it teaches: a point estimate at mean inputs is biased when the model is non-linear, and the input that dominates the risk is rarely the one the laboratory controls.

X23 · extends weeks 34 and 41 · 1 h

Sizing a campaign against a budget

A strain costs $120 to build and $35 to characterise. The budget for the round is $60,000.

1. How many strains does the budget buy?
2. For hit rates of 0.2%, 1% and 5%, how many strains give a 95% chance of at least one hit?
3. Which of those campaigns is adequately funded?
4. Halving the characterisation cost buys how many more strains, and is that the best use of the saving?

Worked solution. The budget buys **387 strains** at $155 each. A 95% chance of at least one hit needs ln(0.05)/ln(1 − p) strains: **1,496 at a 0.2% hit rate, 298 at 1%, and 58 at 5%**. So the campaign is adequately funded at 1% and 5% and **underfunded by a factor of four at 0.2%** — and 0.2% is a realistic hit rate for an unbiased library, which is precisely why the enrichment methods of part 8 chapter 29 matter: a thirteen-fold enrichment turns the hopeless case into the comfortable one.

Halving the characterisation cost raises the campaign from 387 to **436 strains**, a 13% gain, because build cost dominates. That is the answer to the fourth question: cheaper assays are the wrong place to optimise here, and the saving is better spent on anything that raises the hit rate, since the required number scales as 1/p while the affordable number scales only with the cost split. Before running the campaign at all, the number worth estimating is the hit rate itself — from a pilot of fifty clones, which costs $7,750 and bounds p well enough to decide.

What it teaches: screening campaigns fail on arithmetic more often than on biology, and the decision is between raising the hit rate and raising the throughput.

X24 · extends week 64 · 1.5 h

Scheduling two fermenters

Six runs are queued with durations of 7, 5, 7, 4, 6 and 5 days. Two identical vessels are available, and turnaround between runs is 1.5 days.

1. Compute the total vessel-days and the perfect-balance lower bound on the makespan.

2. Apply longest-processing-time first and give the resulting schedule.
3. Add turnaround and recompute.
4. What does this tell you about promising a date, and how does it connect to V1?

Worked solution. The runs total **34 vessel-days**, so the lower bound on two vessels is ⌈34/2⌉ = **17 days**. Longest-processing-time scheduling assigns A(7) + E(6) + D(4) = 17 days to one vessel and C(7) + B(5) + F(5) = 17 to the other — the heuristic hits the bound exactly, which it will not always do, and a quick check against the bound is how you know whether to look for a better schedule. Adding 1.5 days of turnaround between consecutive runs on the same vessel gives two changeovers per vessel and a makespan of **20 days**.

The connection to V1 is the one worth drawing: that 20 days is a *deterministic* figure built from most-likely run lengths, and V1 showed that such a plan has about a 1% chance of being met once durations are allowed to vary. A fermentation that overruns blocks its vessel, so delays propagate rather than absorbing — which means the right promise is the 80th percentile of a simulated schedule, and the right protection is a single visible buffer at the end rather than padding every run.

What it teaches: resource contention turns a list of durations into a scheduling problem with a computable lower bound, and the bound is what tells you whether to optimise the schedule or the plan.

Group 8 — Six problems on real public datasets

These are the "all from public datasets" problems you asked for, written as far as honesty allows. **I could not run them**: this container has no network access, so I cannot download the files, check their current format, or verify the answers. What follows is the source, the exact retrieval route, the questions, and code written against the documented file format — with my expectation of the answer stated as an expectation rather than a result. Treat the code as a starting point to debug, not as verified output.

	Dataset and source	What you do with it
Y1	**BiGG Models** — *E. coli* core model and iJO1366, as SBML and JSON	Reproduce the part 10 analyses on the real model: growth on glucose aerobically and anaerobically, the single-reaction deletion scan, flux variability at 99% of optimum, and pFBA. Compare with my hand-built model's 0.754 and 0.217 h^{-1} and account for every difference.
Y2	**Gene Expression Omnibus** — any bacterial RNA-seq count matrix with at least two conditions and three replicates	Normalise by median-of-ratios, test with a negative-binomial model, then repeat with the samples' labels shuffled. The shuffled run must return essentially nothing; if it does not, the pipeline leaks.
Y3	**BRENDA** — every k_{cat} and K_M reported for one enzyme of your capstone pathway	Tabulate by organism, temperature, pH and assay. Compute the spread of k_{cat} across reports, then propagate the worst case through the enzyme-constrained bound of chapter S5. Expect an order of magnitude.
Y4	**eQuilibrator** — $\Delta G'^{\circ}$ for every reaction of your pathway at pH 7.5, ionic strength 0.25 M	Compute the max-min driving force over the 1 µM–10 mM concentration window, identify the worst step, and test how much the answer moves when you take eQuilibrator's own uncertainty on each $\Delta G'^{\circ}$.
Y5	**ProteinGym** — a deep mutational scanning dataset for an enzyme	Score variants with a zero-shot predictor, compute the rank correlation with measured fitness, then compute the enrichment of true top-1% variants in your top 1,000 predictions. Compare with the 13-fold figure in part 8 chapter 29.
Y6	**Sequence Read Archive** — reads from an adaptive laboratory evolution experiment	Run breseq in polymorphism mode, read all three evidence tables including new junctions, and argue which one or two mutations are causal using the criteria of part 8 chapter 28.

Two notes on doing these well. **Record the accession, the version and the download date for every file** — sheet 9 of the lookup workbook has the fields, and a re-analysis six months later against a silently updated record is a real failure mode. And **run the self-contained problem first where one exists**: X22 before Y1's economics, X20 before Y5's enrichment, P1 and P2 before Y2. The simulated version tells you what the right answer looks like, so that when the real data give something different you can tell whether it is biology or a bug.

Closing notes to part 15

Answers to the four questions, in one place

Where does the count come from? It is the count of problems in your curriculum's appendix, A1 to W3, beginning at its page 140. Not datasets — the document supplies no data files at all.

What is the "datasheet" about? There isn't one. Each of the 56 states its own numbers inline, and seven of them quote one measured parameter from a named paper.

Why only those seven? I did not write the 56 — they came with your curriculum and I only verified their arithmetic, which all checked out. And I cannot retrofit real data now, because this environment has no network access; inventing data and calling it published would be worse than a labelled construction.

Can you make more? Yes — 24 new worked problems here, every number computed, plus 6 written against real public datasets with their retrieval routes, clearly marked as not run. That takes your set from 56 to **86**.

What the 24 add that the first 56 lacked

Gap in the original 56	Now covered by
No ANOVA, no interaction testing	X13
No response-surface or central composite design	X14
No variance components or nested replication	X15
No equivalence testing	X16
No heat balance or cooling duty	X4, X12
No dissolved CO_2, shear or gas holdup	X9, X10, X11
No off-gas mass balance	X19
No growth-curve model fitting	X18
No carryover or weighted calibration	X17
No Monte Carlo uncertainty propagation	X22
No scheduling or budget-constrained campaign sizing	X23, X24
No two-substrate kinetics or thermal deactivation	X5, X8

The six results most worth remembering

- **X4** — a methanol fermentation at 30 gDW/L needs 855 m² of heat-transfer area against about 110 m² of jacket. Cooling, not oxygen, sizes the vessel on a reduced substrate.
- **X8** — the optimum temperature falls from 47 °C for a 2-hour reaction to 36 °C for a 72-hour one. An assay optimum is the wrong set-point for a long batch.
- **X13** — a significant interaction ($p = 0.006$) that one-factor-at-a-time cannot detect at any sample size.
- **X15** — 91% of the variance is between strains and 2% is analytical, so extra injections buy almost nothing and extra flasks buy everything.
- **X18** — a log-linear fit over the apparent exponential phase overstates μ_{max} by 97%.
- **X22** — capital cost, not yield, dominates the uncertainty in the selling price.

On the numbers in this part

Every figure in X1 to X24 was computed in this session, and the parameters are printed in each problem. The ones that carry assumptions worth challenging: the 460 kJ per mole of oxygen in X4 and X12, which sets both heat duties; the holdup correlation in X11, which is a rough empirical form and varies with medium and sparger; the 20–50× local dissipation factor in X10; and the input distributions in X22, which I chose. Where a problem simulates data, the true parameters are stated so you can check whether your method recovered them — which is the whole point of the self-contained kind.

If you want more after these, the categories with the most room left are downstream separations (chapter S1 has the theory and only two problems), quantitative proteomics and metabolomics (chapter S5, no problems yet), and consortia

dynamics (chapter S4, no problems yet). Say which and I will work them the same way.

Part 16 — Nine more problems: separations, quantitative omics, consortia

The three categories that part 13 gave theory to and no problems. Same format as X1–X24: statement, stated assumptions, numbered questions, worked solution with every figure computed in this session. Together with the 56 in your appendix and the 30 in part 15, this takes the set to **95**.

Group	Problems	Extends
Downstream separations	Z1–Z4	Chapter S1, weeks 44 and 49
Quantitative proteomics and metabolomics	Z5–Z6, Z9	Chapter S5, weeks 43 and 51
Consortia dynamics	Z7–Z8	Chapter S4, weeks 9 and 26–27

Group A — Downstream separations

Z1 · extends chapter S1 · 2 h

Breakthrough, dynamic capacity, and the bed you are not using

A capture column holds 100 L of resin in a 200 mm bed. The resin's static binding capacity is 65 g/L; at 10% breakthrough and a 2-minute residence time its dynamic capacity is 45 g/L. Feed titer is 10 g/L and the campaign must process 500 kg of product.

1. Compute the fraction of the bed that is doing no work, and express it as a length of unused bed.
2. Dynamic capacity rises toward the static value as residence time increases, roughly as SBC$(1 - e^{-t/2.5})$ with t in minutes. Tabulate capacity and the number of cycles at residence times of 2, 4 and 6 minutes.
3. Doubling the residence time doubles the loading time per cycle. Does it reduce total campaign time?
4. What would you measure to get the real curve, and what does 10% breakthrough mean for product loss?

Worked solution. DBC/SBC = 45/65 = 0.69, so **31% of the bed is unused** — the mass-transfer zone that has not yet saturated when breakthrough begins — which is **62 mm of the 200 mm bed**. That length is the quantity a column designer actually works with: it is roughly independent of bed height, so a taller bed wastes a smaller *fraction*, which is why capture columns are tall and narrow rather than short and wide.

Residence time	Dynamic capacity	Load per cycle	Cycles for 500 kg
2 min	35.8 g/L	3.6 kg	140
4 min	51.9 g/L	5.2 kg	96
6 min	59.1 g/L	5.9 kg	85

Doubling residence from 2 to 4 minutes raises capacity by **45%** and cuts cycles from 140 to 96 — a 31% reduction — while doubling the load time per cycle. Since loading is only part of a cycle (equilibration, wash, elution, regeneration are the rest), the net effect is usually favourable, and the answer is obtained by writing the full cycle time rather than by intuition. The measurement to make is a **breakthrough curve at each candidate flow rate**: load at constant concentration and plot outlet against loaded mass; DBC is read at your chosen breakthrough, and the curve's steepness gives the mass-transfer zone directly. And 10% breakthrough means exactly that — **at the end of loading you are losing 10% of the feed to the flow-through**, so the capacity figure and the yield figure are two readings of the same curve.

What it teaches: dynamic capacity is a kinetic quantity, not a property of the resin, and the breakthrough criterion you choose sets both the capacity and the product you throw away.

Z2 · extends chapter S1 · 1.5 h

A three-step train: where purity comes from and what it costs

A secreted protein at 35% purity in clarified broth must reach 98%. The available steps are centrifugation plus depth filtration (93% yield, purifies 2-fold), ion-exchange capture (90% yield, 8-fold) and hydrophobic-interaction polish (88% yield, 22-fold), where "n-fold purification" means the impurity fraction falls n-fold.

1. Track purity and cumulative recovery through the train.
2. Is the 98% specification met, and with what margin?
3. What do one and two further polish steps buy, and cost?
4. How does recovery enter the cost per kilogram delivered?

Step	Yield	Purity after	Cumulative recovery
Clarification	0.93	67.5%	93.0%
Ion-exchange capture	0.90	95.9%	83.7%
HIC polish	0.88	**99.82%**	**73.7%**

Worked solution. The specification is met with room to spare — 99.82% against 98% — at an overall recovery of **73.7%**. Adding a fourth step takes purity to 99.97% and drops recovery to **66.3%**; a fifth gives 99.99% at **59.7%**. So each additional polish buys a decimal place of purity and costs roughly seven points of recovery, which is the quantitative form of chapter S1's argument that the train should be as short as the specification allows. Because the

product delivered is what you sell, the cost per kilogram scales as 1/recovery: at 73.7% it is **1.36 times** the single-pass cost of the materials and operations, and at 59.7% it is 1.68 times.

The design judgement is to *stop*: a group that habitually adds a final polish "for safety" is paying 7% of its product for purity nobody specified. The time to decide is when the specification is written, and the number to write beside it is the recovery the train can deliver.

What it teaches: purity and recovery trade against each other step by step, and over-purifying is a silent, recurring cost.

Z3 · extends chapter S1 · 1.5 h

Crystallisation: read the yield off the solubility curve

A product's solubility follows $c^* = 3.0 \cdot e^{0.035T}$ g/L with T in °C. The crystalliser is charged at 60 °C with a saturated solution.

1. Tabulate solubility at 60, 40, 25, 10 and 4 °C.
2. Compute the crystallisation yield for cooling from 60 °C to 25, 10 and 4 °C.
3. What does a 1.5-fold concentration step before cooling add?
4. With a metastable zone width of 6 °C, what constrains the cooling profile, and why is a linear ramp a poor choice?

Temperature	60 °C	40 °C	25 °C	10 °C	4 °C
Solubility	24.5 g/L	12.2	7.2	4.3	3.5

Worked solution. Yield is $(C_1 - C_2)/C_1$, so cooling to 25 °C gives **70.6%**, to 10 °C **82.6%** and to 4 °C **85.9%** — the last 6 °C of cooling buys only 3.3 points because the solubility curve is flattening, and it costs refrigeration. Concentrating 1.5-fold first raises the 10 °C yield to **88.4%**, which is the better lever: evaporation is cheaper than refrigeration per unit of yield here, and it also shrinks the vessel.

The metastable zone is what constrains the profile. Supersaturation must stay inside it — beyond the width, primary nucleation produces a shower of fines that will not filter. Since the yield is driven by the solubility *difference*, and that difference grows as the solution cools, a **linear temperature ramp generates supersaturation fastest at the end**, exactly when the crystal surface area is largest but the driving force is highest; it spends most of the batch near the metastable limit. The standard answer is to seed just below saturation and cool slowly at first, then faster — a cubic profile — so that supersaturation is held roughly constant as the growing crystal area consumes it.

What it teaches: crystallisation yield is pure thermodynamics off the solubility curve, while crystal quality is entirely a kinetic question about the cooling path.

Z4 · extends chapter S1 · 1.5 h

Diafiltration: diavolumes, sieving and what leaks through

An ultrafiltration step washes an impurity out of a 20 m³ retentate. The impurity passes the membrane with sieving coefficient S_i and the product with S_p; in N diavolumes the impurity falls as $e^{-S_i N}$ and the product is retained as $e^{-S_p N}$.

1. How many diavolumes for 10-, 100- and 1,000-fold impurity clearance with perfect sieving?
2. For four membrane cases, compute the diavolumes needed for 100-fold clearance and the product retained.
3. Compute the buffer volume and the time for 7 diavolumes at 2 m³/h of permeate.
4. A two-stage cascade recycles permeate. What does recovering 50% or 80% of the lost product do?

Worked solution. With perfect sieving the requirement is ln F: **2.3, 4.6 and 6.9 diavolumes**. The membrane cases show where the design actually lives:

S_i (impurity)	S_p (product)	Diavolumes for 100×	Product retained
1.0	0.00	4.6	100%
1.0	0.02	4.6	91.2%
0.8	0.05	5.8	75.0%
0.6	0.10	7.7	**46.4%**

The two coefficients compound viciously: a membrane that partly retains the impurity needs more diavolumes, and every extra diavolume loses more product, so the worst case keeps **under half** of what it started with. Buffer for 7

diavolumes on 20 m³ is **140 m³**, which at 2 m³/h of permeate takes **70 hours** — so the step is also the schedule bottleneck, and in a sterile process that is 70 hours of hold time to validate.

Recycling the permeate recovers part of the loss: at S_p = 0.10 and 7.7 diavolumes, retention rises from 46.4% to **73.2%** with 50% permeate recovery and **89.3%** with 80%. That is the argument for a cascade, and the cost is a second membrane stage and more complexity in a step that is already the longest in the train.

What it teaches: a membrane is specified by two sieving coefficients, not one, and the product you lose is set by the diavolumes the impurity's coefficient forces you to run.

Group B — Quantitative proteomics and metabolomics

Z5 · extends chapter S5 · 2 h

Absolute protein quantification, and the flux bound it feeds

You spike known amounts of isotope-labelled AQUA peptides and quantify three peptides per protein from their light-to-heavy ratios, with 5% or 15% CV on each ratio. The protein is a 50 kDa enzyme and the cell carries 0.5 g of protein per gram dry weight.

1. Recover the five spiked amounts at both CVs and report the mean absolute error.
2. Compute the protein-level CV from three peptides, and say why the median of three is used rather than the mean.
3. Convert abundance to a flux bound $v \le k_{cat}[E]$ for proteome fractions of 0.1, 1 and 5% and k_{cat} of 1, 10 and 100 s^{-1}.
4. State the uncertainty on the bound, and what the bound is still good for.

Worked solution. Recovery across 0.5 to 120 fmol has a mean absolute error of **7.3% at 5% ratio CV and 10.0% at 15%** — the method is accurate because the labelled standard is chemically identical, so the error is close to the ratio precision itself. Averaging three peptides gives a protein-level CV of $0.15/\sqrt{3}$ = **8.7%**, but the reason for taking the **median** rather than the mean is bias, not variance: a peptide with a missed cleavage or a partial modification reads systematically low — by 5%, 15% or 30% in the cases tabulated — and one bad peptide of three shifts a mean while leaving a median untouched.

Proteome fraction	k_{cat} = 1 s^{-1}	10 s^{-1}	100 s^{-1}
0.1%	0.04	0.4	3.6
1%	0.4	3.6	36.0
5%	1.8	18.0	180.0

Bounds are in mmol·gDW^{-1}·h^{-1}. The uncertainty is the product of the two errors and is dominated by k_{cat}: a literature value wrong by tenfold makes the bound wrong by **tenfold**, while a 25% abundance CV contributes 25%. So the bound is good to **about a factor of ten**. It is still worth computing, because the useful statement is an exclusion: a design needing 36 mmol·gDW^{-1}·h^{-1} from an enzyme with k_{cat} = 1 s^{-1} would need 100% of the proteome, and no plausible k_{cat} error rescues that.

What it teaches: absolute quantification is accurate and the flux bound built on it is not, because the k_{cat} dominates — so report bounds as exclusions with their assumed k_{cat} named.

Z6 · extends chapter S5 · 1.5 h

Quenching and extraction: choose your artefact deliberately

Four sampling methods are compared for intracellular metabolites, with their reported recoveries and whether they separate intracellular from extracellular pools.

1. Tabulate the four methods against recovery and separability, and say what each artefact does to a measurement.
2. Your product accumulates to twenty times its intracellular concentration in the broth. What fraction of an unwashed "intracellular" measurement is artefact?
3. Compute pool turnover times for four pool-and-flux combinations and the fraction lost in a 2-second quench.
4. Which method would you choose for a thermodynamic analysis, and what would you report?

Method	ATP recovery	Glucose-6-P	Separates pools?
Cold methanol, washed	0.45	0.60	yes
Cold methanol, unwashed	0.95	0.97	**no**
Fast filtration under nitrogen	0.80	0.85	yes
Whole broth, boiling ethanol	0.98	0.99	**no**

Worked solution. The trade is exact and unavoidable: washing separates the pools and leaks the metabolites, not washing retains them and measures both pools together. With the product twenty times more concentrated outside than inside, an unwashed measurement is **95% extracellular** — it is not a measurement of the cell at all. Turnover is the

other constraint: a 0.05 mM pool at 2 mM/s turns over in **0.03 s**, and a 2-second quench loses essentially all of it; only the 5 mM pool at 0.3 mM/s, turning over in 17 s, survives with 12% loss.

For a thermodynamic analysis the right choice is **fast filtration under nitrogen** — it separates the pools and recovers 80–85% — with two things reported alongside: the recovery measured per compound by a second spike, and the extracellular concentration measured separately so the intracellular figure can be corrected rather than assumed. And the honest statement in the methods is that no method is artefact-free; what makes the number usable is that the artefact was quantified.

What it teaches: every sampling method trades leakage against separability, and the pool you most want to measure is the one that turns over fastest.

Z9 · extends chapter S5 · 1.5 h

Instrument drift, and the pooled QC that catches it

A metabolomics batch of 120 injections loses 30% of its signal across the run as the source fouls. A pooled quality-control sample is injected every tenth position.

1. Compute the coefficient of variation of the QC injections before correction.
2. Correct the run by interpolating the QC signal and recompute the QC CV and the residual bias between the first and last ten samples.
3. What happens to a comparison between two conditions if the samples were injected in condition order?
4. State the three rules this implies for a batch design.

Worked solution. The uncorrected QC injections have a CV of **12.4%** — entirely drift, since the QC sample is the same material every time. Interpolating the QC trace and dividing it out brings the QC CV to **essentially zero** by construction and leaves a residual bias of **1.1%** between the first and last ten samples, which is the part the interpolation could not capture.

If the samples were injected in condition order — all controls then all treatments, as is natural when they come off the bench that way — the 30% drift maps exactly onto the comparison and appears as a **30% treatment effect**. Nothing in the data distinguishes it from biology, and no statistical test helps, because the confounding is in the design rather than the noise. Hence the three rules: **randomise the injection order** across conditions, **inject a pooled QC every ten samples** and correct against it, and **report the QC CV with the data** so a reader can judge the correction. A feature whose QC CV exceeds about 20–30% should not be interpreted at all.

What it teaches: instrument drift becomes a treatment effect whenever injection order follows condition, and the pooled QC is the only thing that separates them.

Group C — Consortia dynamics

Z7 · extends chapter S4 · 2 h

Designing the population ratio you actually want

Strain A consumes glucose and excretes α moles of an intermediate per mole consumed; strain B grows only on the intermediate. In a chemostat at D = 0.2 h^{-1} with S_0 = 20 g/L, both strains have μ_{max} = 0.5 h^{-1} and Y = 0.4 g/g, with K_S = 1.0 and K_I = 0.5 g/L.

1. Compute the steady state and the A:B ratio for the base case.
2. Vary α to 1.0 and 4.0, Y_B to 0.2, $\mu_{max,B}$ to 0.3 and D to 0.35. Tabulate the ratio.
3. Derive the steady-state intermediate concentration analytically and check it against the simulations.
4. You need a 1:1 ratio. Which parameter do you engineer, and how would you do it?

Case	A (g/L)	B (g/L)	A:B	Intermediate
Base	7.73	6.05	1.28	0.333 g/L
α = 1.0	7.73	2.96	2.61	0.333
α = 4.0	7.73	12.24	0.63	0.333
Y_B = 0.2	7.73	3.03	2.56	0.333
$\mu_{max,B}$ = 0.3	7.73	5.79	1.34	1.000
D = 0.35	7.07	5.19	1.36	1.167

Worked solution. The ratio responds to **α and Y_B and almost nothing else**: halving α doubles the ratio, and halving Y_B does the same, while changing B's growth rate barely moves it. The reason is in the third question's algebra. At steady state B grows at D, so $\mu_{max,B} I/(K_I + I) = D$, giving

```
I* = K_I·D/(μ_max,B − D)
```

which is the chemostat result applied to the intermediate — **0.5 × 0.2/0.3 = 0.333 g/L** for the base case, **0.5 × 0.2/0.1 = 1.000** when $\mu_{max,B}$ falls to 0.3, and **0.5 × 0.35/0.15 = 1.167** at D = 0.35, matching all three simulations exactly. B's kinetics set the intermediate *concentration*; the stoichiometry and yields set the *biomass ratio*. That separation is the design insight: the two are controlled by different parameters, so you can specify both.

To force 1:1 from the base case's 1.28 you raise B's biomass per unit of intermediate or lower A's excretion — practically, you tune α by changing the export or the branch flux in A (part 3 chapter 45), or you accept the ratio and tune the *inoculum of the subsequent batch*, which does nothing at steady state but matters in a batch that never reaches one.

What it teaches: in an obligate cross-feeding consortium the intermediate concentration and the population ratio are set by different parameters, and only the ratio is a design variable you can hit.

Z8 · extends chapter S4 · 1.5 h

How long before the cheater wins?

A defector that stops paying its share grows a fraction s faster. Starting from one part in a million in a chemostat at D = 0.2 h^{-1}, the defector's frequency follows the logistic replacement $p(t)/(1 - p(t)) = p_0/(1 - p_0)\cdot e^{sDt}$.

1. Compute the time to reach 50% and 99% of the population for advantages of 2, 5, 8 and 15%.
2. Express those as generations.
3. Compare with a 7-day batch and a 60-day continuous campaign.
4. Under reciprocal auxotrophy the defector cannot grow at all. How fast does it wash out at D = 0.05 and 0.2 h^{-1}?

Advantage s	To 50%	To 99%	Generations to 99%
2%	3,454 h	4,603 h	1,328
5%	1,382 h	1,841 h	531
8%	863 h	1,151 h	332

15%	461 h	614 h	177

Worked solution. Against a 7-day batch (168 h) every case is safe — a cheater starting at 10^{-6} cannot sweep in a week at any plausible advantage. Against a 60-day continuous run (1,440 h) the picture reverses: at an 8% advantage the defector is past 99% before the campaign ends, and at 15% it takes over in the first month. **The decisive variable is campaign length, not the advantage**, which is why batch processes tolerate designs that continuous processes cannot.

Reciprocal auxotrophy changes the kind of problem rather than its size. A strain that cannot grow has $\mu = 0$ and is simply diluted out at the dilution rate: **tenfold in 46 h and a thousandfold in 138 h at $D = 0.05$**, and **12 h and 35 h at $D = 0.2$**. So the defence is not that cheating is slow but that cheating is lethal — the frequency moves in the opposite direction, and faster at higher dilution. The caveat worth stating is that this holds only while the medium lacks the complemented nutrients; a rich feed or a lysing population restores cross-feeding and the protection with it.

What it teaches: a cheater's threat is set by the run length against its advantage, and reciprocal auxotrophy inverts the dynamics rather than slowing them.

Closing notes to part 16

The nine in one page

Z1 31% of a capture bed does no work at breakthrough, which is 62 mm of a 200 mm bed; doubling residence time raises capacity 45% and cuts cycles from 140 to 96. **Z2** a three-step train reaches 99.8% purity at 73.7% recovery, and each further polish buys a decimal place for seven points of recovery. **Z3** cooling 60 → 10 °C crystallises 82.6%, and a 1.5-fold concentration step beats the last 6 °C of refrigeration. **Z4** a membrane with $S_i = 0.6$ and $S_p = 0.10$ needs 7.7 diavolumes and keeps under half the product; permeate recycle restores it to 73–89%.

Z5 AQUA recovers spiked amounts to 7–10%, but the flux bound built on it is good only to a factor of ten because k_{cat} dominates. **Z6** an unwashed intracellular sample is 95% artefact when the product is 20-fold concentrated outside, and a 0.05 mM pool turns over in 0.03 s. **Z9** a 30% drift across 120 injections becomes a 30% treatment effect if injection order follows condition.

Z7 the intermediate concentration is $K_I D/(\mu_{max,B} - D)$, set by the consumer's kinetics, while the population ratio is set by α and the yields — two parameters, two controls. **Z8** a cheater at 8% advantage sweeps a 60-day campaign and cannot touch a 7-day batch; under reciprocal auxotrophy it washes out a thousandfold in 35 h.

On the numbers

All computed in this session. The assumptions worth challenging: the empirical capacity-against-residence form in Z1, which should be replaced by your own breakthrough curve; the fold-purification figures in Z2, which are illustrative and step-specific; the solubility expression in Z3; the recovery figures in Z6's method table, which are representative of the literature rather than measured here and are the kind of thing sheet 10 of the lookup workbook exists for; and the logistic replacement in Z8, which assumes a constant advantage and well-mixed conditions.

Where the problem set now stands

Source	Problems	Data
Your curriculum appendix, A1–W3	56	Stated inline; 7 quote a measured parameter from a paper
Part 15, X1–X24	24	Computed this session; simulated data with known truth where needed
Part 15, Y1–Y6	6	Real public datasets, retrieval routes given, **not run**
Part 16, Z1–Z9	9	Computed this session
Total	**95**	

Coverage against the nine skill categories is now even, with the thinnest remaining areas being genome-scale model curation, which needs software I cannot run, and anything depending on a real instrument's output. Both are better done at a bench or a terminal than on paper.

Part 17 — The last two gaps: model curation, and real instrument data

I said these two were better done at a terminal or a bench than on paper. That was half right: what I could not do was *fetch* real data. What I can do — and have done here — is generate instrument data with the answer built in, so you process genuine files and then check whether your method recovered the truth. Five datasets are attached as CSV. The curation problems are computed against the core model from part 10, with errors deliberately injected.

The data files

File	Size	Problem	What it is
`R1_chromatogram.csv`	15 kB	R1	A 12-minute HPLC trace at 10 ms resolution: four peaks, a drifting baseline, a solvent front and detector noise
`R2_qpcr.csv`	25 kB	R2	45-cycle amplification curves, two targets × two samples × three replicates
`R2_qpcr_standard_curve.csv`	0.5 kB	R2	A five-point dilution series for each target, in duplicate
`R3_plate_growth.csv`	139 kB	R3	96 wells × 73 time points of OD_{600}, with blanks, an edge effect and noise
`R4_ms_isotopologues.csv`	0.4 kB	R4	A pyruvate isotopologue cluster, M+0 to M+3, in triplicate
`R5_offgas.csv`	2.7 kB	R5	24 h of outlet gas composition, dissolved oxygen and feed rate at 15-minute intervals
`ANSWERS_do_not_open_first.json`	13 kB	all	The true parameters for every dataset. Open it after you have an answer, not before

The data are simulated, and the statements say so. The point is not to pretend they came off an instrument but to give you a file with noise, drift, overlap and a known answer — which is the one thing a real instrument file cannot give you.

Group A — Genome-scale model curation

MC1 · extends part 10 chapter 37 · 2 h

A reaction that balances on carbon and still breaks the model

Someone has added one reaction to the 40-reaction core model: **4 CO_2 → oxaloacetate**. It is written with a correct carbon count. Growth on 10 mmol of glucose rises from 0.754 to **10.00 h^{-1}**.

1. Check the reaction's carbon balance. Does a carbon-balance audit catch it?
2. Count the electrons on each side using degrees of reduction. What is created from nothing?
3. Design the one simulation that exposes it in a single line, and predict its result.
4. Why did the error inflate growth 13-fold rather than slightly?
5. List three further checks that would have caught this class of error before any simulation.

Worked solution. The carbon balance is **exactly zero** — four carbons in as CO_2, four out as oxaloacetate — so a carbon audit **passes it**, which is the whole lesson. The electron count does not: CO_2 is fully oxidised and contributes nothing, while oxaloacetate ($C_4H_4O_5$) carries $4\times4 + 4\times1 - 5\times2 =$ **10 electrons**, all of them created from nothing on every turn. Because the reaction also needs no ATP, it is simultaneously a free-reductant and a free-carbon generator.

The exposing simulation is one line: **set the glucose uptake bound to zero and optimise**. A correct model returns zero growth; this one still grows, because it can build biomass out of CO_2 and nothing else. That test — "can the model grow on no carbon source?" — belongs in every model's test suite, and it is the first thing to run after any edit.

The inflation is 13-fold rather than marginal because the injected reaction relieves *every* binding constraint at once: it supplies carbon, reductant and a four-carbon precursor, so the optimiser pushes it to 406 mmol·gDW^{-1}·h^{-1} and growth runs to whatever other bound it hits. An error that relaxes the binding constraint always shows up as a large number, which is why an implausibly good result is itself a diagnostic. The three further checks: a full **elemental balance** including H and O rather than carbon alone; a **charge balance**; and the **stoichiometric consistency** test of part 10 chapter 37, which asks for a positive mass vector satisfying $S^{T}m = 0$ and fails when any reaction creates mass.

What it teaches: carbon balance is necessary and nowhere near sufficient; the growth test on an empty medium is the cheapest and most powerful curation check there is.

MC2 · extends part 10 chapter 36 · 1.5 h

Separate the errors that change the answer from the errors that change the map

The same core model contains succinate dehydrogenase and fumarate reductase, which together form a closed loop. You want to know whether to care.

1. Compute the growth rate with the loop present and with fumarate reductase blocked. What changes?
2. At 99% of optimal growth, maximise the succinate dehydrogenase flux. What value do you get and why?
3. Classify: does this error affect the objective, the flux map, or both?
4. Give the order in which you would fix a model's errors, and justify it by this result.

Worked solution. Growth is **0.7541 h^{-1} either way** — identical to ten decimal places, because the loop has no net conversion and so neither consumes nor produces anything biomass needs. Maximising the flux through it at 99% of optimum returns **1000**, which is not a biological number at all: it is the arbitrary upper bound in the model file, and its appearance in an answer is the signature of a loop.

The classification matters: this error affects **the flux map and not the objective**. So every yield, every ceiling and every deletion prediction from this model is unaffected, while every statement of the form "flux goes this way through that branch" is unreliable. That gives the repair order: **first, errors that change the objective** — mass and charge imbalances, free-energy generators, wrong reversibility that opens an impossible route, because these corrupt the numbers you quote; **second, errors that change the map** — loops and alternative optima, because they corrupt the interpretation; **third, metadata** — annotations, identifiers, SBML conformance, which affect reusability and not correctness. A group that starts with annotations because they are easy has fixed the least important category.

What it teaches: not all model errors are equal, and knowing whether an error moves the objective or only the flux map tells you whether to stop the analysis or just stop interpreting the fluxes.

MC3 · extends part 10 chapter 37 · 2 h

A reversibility error, and the unbounded ATP it buys

Acetate kinase in the model is irreversible, as written. A curator "fixes" it to be reversible — the enzyme is reversible in vitro, after all — and the model is also given an acetate uptake exchange.

1. With acetate uptake allowed and the kinase irreversible, maximise ATP maintenance flux at zero growth.
2. Repeat with the kinase made reversible.
3. Now remove every carbon source. What do you get, and what does the answer tell you?
4. Explain why "the enzyme is reversible in vitro" is not sufficient grounds for the edit.

Worked solution. With acetate uptake and the irreversible kinase, maximum ATP production is **320 mmol·gDW^{-1}·h^{-1}** — a high but finite number set by the acetate uptake bound and the respiratory chain. With the kinase reversible it becomes **1000**, the bound again: the model has acquired a cycle that makes ATP without limit. Removing all carbon sources returns **zero**, so this is not a pure free-energy generator — it needs acetate to run — which is why it would survive the M1 test and has to be caught by the ATP test instead: **maximise ATP with the biomass reaction switched off and compare with the substrate's theoretical maximum**, which for acetate is about 10 ATP per molecule.

The in vitro argument fails because reversibility in a test tube is a statement about the enzyme at whatever concentrations the assay used, while reversibility in a model is a statement about the *direction the reaction can carry net flux at cellular concentrations*. Acetate kinase operates far from equilibrium in vivo — the cell holds a high ATP-to-ADP ratio and a low acetyl phosphate pool — so the reverse direction is thermodynamically shut even though the protein catalyses it. That is the part 2 chapter 14 distinction between $\Delta G^{\circ\prime}$ and $\Delta G'$, and it is why directionality should come from the thermodynamics at physiological concentrations rather than from an enzyme's catalytic competence.

What it teaches: reversibility in a model is a thermodynamic claim about cellular conditions, and the ATP-maximisation test is what catches a wrong one.

`MC4` · extends part 10 chapter 37 · 1.5 h

A biomass coefficient error, and why its direction decides whether you notice

The core biomass reaction charges 59.81 mmol of ATP per gram dry weight. Three variants are tested, with the coefficient multiplied by 0.5, 2 and 10.

1. Compute growth for each variant.
2. Which errors are visible in the growth rate and which are not? Explain.
3. What does this imply about a biomass equation copied from a related organism?
4. Which predictions are most sensitive to a biomass error, and which are robust?

ATP coefficient	Growth (h^{-1})	Fraction of correct
29.9 (half)	0.7541	**100%**
59.81 (correct)	0.7541	100%
119.6 (double)	0.6669	88%
598.1 (ten-fold)	0.3300	44%

Worked solution. Halving the coefficient changes the growth rate **not at all**, because at the correct value the model is carbon-limited rather than ATP-limited — the slack in the ATP budget absorbs the error completely. Doubling it costs 12% and a tenfold error costs 56%. So **an error that makes biomass cheaper is invisible** in the growth rate while an error that makes it dearer is visible, and only once it is large enough to make ATP the binding constraint.

For a copied biomass equation that is an uncomfortable asymmetry: the usual direction of error, borrowing coefficients from a better-studied relative, can sit undetected. The predictions sort accordingly. Robust: **carbon-limited yield ceilings**, which depend on the precursor demands rather than the ATP term. Sensitive: **anything ATP-limited** — maintenance behaviour, growth on poor substrates, the P/O comparisons of part 14 chapter B4 — and above all **gene essentiality**, since essentiality often turns on whether one specific precursor can be made at all, so a wrong coefficient for a minor component makes its pathway spuriously essential or spuriously dispensable.

What it teaches: a biomass error is only visible when it makes the constrained resource more expensive, so the direction of the error decides whether you will ever find it.

`MC5` · extends part 10 chapter 37 · 1 h

How much of this is there in a real model?

1. Your core model has 40 reactions and the injected error was found by two checks. Scale that to a 1,400-reaction automated draft with a reported 28% of reactions unbalanced, and to a 2,583-reaction curated model at 5%.
2. How many reactions need inspection in each case?
3. Which of MC1 to MC4's checks can be automated, and which cannot?
4. Write the test suite you would attach to your own model, in order.

Worked solution. At those rates a draft has **392 reactions** to inspect and a curated genome-scale model still has **129**. That is weeks of work in the first case, which is the honest reason automated drafts are used for capability questions and not for gene-level predictions (part 10 chapter 37). Automatable: elemental and charge balance, stoichiometric consistency, blocked-reaction detection, the no-carbon growth test, the ATP-maximisation test, and loop detection by flux variability — all of these are arithmetic on the stoichiometric matrix and need no judgement. **Not** automatable: deciding whether a reaction should exist in this organism, choosing a direction when the thermodynamics are marginal, and setting biomass coefficients — each needs evidence rather than computation.

The test suite, in the order to run it: (1) does the model grow on the defined medium, and *not* grow with no carbon source; (2) is maximum ATP at zero growth within the substrate's theoretical yield; (3) is every internal reaction elementally and charge balanced; (4) is the model stoichiometrically consistent, with any deliberate currency lumping documented; (5) which reactions are blocked, and does each belong to a pathway with a missing half; (6) which reactions appear at the arbitrary flux bound in variability analysis. Run as a script on every commit, this takes seconds and catches the entire class of error in MC1 to MC3.

What it teaches: curation splits cleanly into checks a script should run on every edit and judgements only evidence can settle, and conflating the two is why drafts stay uncurated.

Group B — Real instrument data, with the answer withheld

R1 · file `R1_chromatogram.csv` · 2.5 h

Integrate a chromatogram, including the peak that overlaps your product

A 12-minute trace sampled at 10 ms. Four peaks: an impurity at 2.41 min, the product at 3.86, a co-eluting compound at 4.12 and an internal standard at 6.73. The baseline drifts upward and there is a solvent front near 1.1 min. The response factor is 0.812 g·L^{-1} per unit of product-to-standard area ratio.

1. Fit and subtract a baseline using only regions you judge to be peak-free. Say how you chose them.
2. Integrate the product valley-to-valley, then over a window wide enough to include the co-eluting peak. Report both areas.
3. Compute the product-to-standard ratio and the concentration from each integration.
4. Deconvolve the 3.86/4.12 pair by fitting two exponentially modified Gaussians, and compare.
5. Open the answer file. By how much was each method wrong, and which error would you rather have?

Expected result. Integrating valley-to-valley **clips the product's tail and reads about 5% low**; a window wide enough to include the co-eluting peak reads about **23% high**. Both are wrong, and in opposite directions, from the same data. The deconvolution should land within a couple of per cent because the peak shapes are genuinely exponentially modified Gaussians — which is the case where fitting beats integration.

Two further points the data will make for you. The internal standard is recovered almost exactly by any sensible method, because it is baseline-resolved — so a **ratio is only as good as the worse of its two integrations**. And the answer you prefer depends on use: for a process decision a consistent 5% low bias is more useful than a variable high bias, because it is correctable by the calibration; for a purity claim neither is acceptable and the method must be changed to resolve the pair.

What it teaches: integration limits are a judgement that changes the answer by tens of per cent, and a co-eluting peak is a chromatography problem that no amount of arithmetic repairs.

R2 · files `R2_qpcr.csv`, `R2_qpcr_standard_curve.csv` · 2 h

Take a qPCR run from raw fluorescence to a fold change you can defend

Two targets, a pathway gene and a reference gene, in a control and an engineered strain, three replicates each over 45 cycles. A separate five-point dilution series in duplicate gives each target's efficiency.

1. Subtract the baseline and determine C_q for all twelve curves by a fixed threshold. Report means and spreads.
2. Fit each standard curve and convert the slope to an amplification efficiency.
3. Compute the fold change by the $2^{-\Delta\Delta Cq}$ method and by the Pfaffl method using the measured efficiencies.
4. Which is larger, by how much, and which would you report?
5. Open the answer file and check your C_q values and efficiencies.

Expected result. The threshold method recovers all four C_q values to within about 0.1 cycle, giving a $\Delta\Delta C_q$ near **3.07** against a true 3.00. The standard curves give slopes near **−3.44 and −3.34**, which convert to efficiencies of **1.95 and 1.99** — that is, 95% and 99%, not the 100% that $2^{-\Delta\Delta Cq}$ assumes for both.

The two methods then disagree: **$2^{-\Delta\Delta Cq}$ gives about 8.4-fold and Pfaffl about 7.8-fold**, so the simple method **overstates by roughly 8%** here. Report the Pfaffl figure with both efficiencies quoted, because the overstatement grows with ΔC_q and with the efficiency gap — at a ten-cycle difference and the same efficiencies it would be more than 20%. The deeper point is that the efficiencies must be measured on the same assay and matrix, not taken from the primer design software, and a target below about 90% efficiency should be redesigned rather than corrected for.

What it teaches: $2^{-\Delta\Delta Cq}$ assumes perfect, equal efficiency for both targets, and the standard curve is what tells you how much that assumption costs.

R3 · file `R3_plate_growth.csv` · 2.5 h

96 wells, a blank column, an edge effect and a growth rate

Seventy-three time points over 18 hours for every well of a 96-well plate. Column 1 contains medium blanks. There is an

edge effect, and the plate has not been randomised.

1. Build the blank correction from column 1, including its drift, and apply it.
2. Fit a Gompertz model to every well and extract μ_{max}, lag and plateau.
3. Compare the distribution of μ_{max} and of the plateau between edge and inner wells. Which parameter carries the edge effect?
4. Rank the wells by plateau, then by μ_{max}. How much does the ranking change, and which ranking would you trust?
5. Open the answer file. How close were your fitted rates, and did you correctly identify what the edge did?

Expected result. The blanks drift upward by about 0.0008 OD per hour — roughly 0.014 over the run, which is a third of the blank value itself — so a single initial blank subtraction leaves a systematic positive drift in every well. Gompertz fitting on properly corrected data should recover μ_{max} to **within about 1%** of truth, which you can verify well by well.

The diagnostic question is the third. The edge effect in this dataset is applied to the **plateau** (an 18% amplitude increase from evaporation concentrating the culture) and **not to the growth rate**. So ranking wells by final OD puts edge wells at the top and is confounded; ranking by μ_{max} is nearly unaffected. That is the generalisable lesson — **choose the parameter that the artefact does not touch** — and it is more powerful than correcting for the artefact, because it needs no model of the artefact at all. Where you must use the plateau, estimate the edge factor from the data as in part 9 chapter 34 and note that this plate was not randomised, so the correction rests on the assumption that edge and inner wells hold comparable strains.

What it teaches: an artefact usually affects one fitted parameter and not another, so the robust response is to choose your readout rather than to correct your data.

R4 · file `R4_ms_isotopologues.csv` · 1.5 h

From a mass spectrum to a flux split

The M+0 to M+3 intensities of a three-carbon pyruvate fragment, in triplicate, from a culture fed [1-^{13}C] glucose. Natural ^{13}C abundance is 1.07%.

1. Normalise the cluster and report the raw M+1 fraction.
2. Build the natural-abundance correction matrix for three carbons and invert it.
3. Convert the corrected M+1 to the oxidative pentose phosphate fraction using the relation from problem C1.
4. Compute how wrong you would have been without the correction.
5. Open the answer file and compare.

Expected result. The raw M+1 is about **0.351**; correction brings it to **0.337**, which converts to an oxidative fraction of **0.326** against a true **0.32** — recovered to within the 2% measurement noise. Skipping the correction would have given an oxidative fraction about **0.028 lower**, a relative error near 9% on the flux you are trying to measure, and the error is systematic rather than random, so replicates will not reveal it.

Two points worth drawing out. The correction's size scales with the number of carbons in the fragment, so a larger fragment needs a larger correction and the matrix is no longer nearly diagonal — a six-carbon fragment has a natural M+1 near 6.4%. And the correction must also account for any heavy isotopes in a derivatisation group, which is why silylated samples need a correction matrix built from the full elemental formula of the fragment rather than from its carbons alone.

What it teaches: natural-abundance correction is a small, systematic, mandatory step whose omission biases a flux in a direction replicates cannot expose.

R5 · file `R5_offgas.csv` · 2 h

Read a fermentation from its off-gas alone

Twenty-four hours of outlet oxygen and carbon dioxide, dissolved oxygen and feed rate, every 15 minutes, from a 100 m³ vessel at 0.5 vvm. Inlet air is 20.95% O_2 and 0.04% CO_2.

1. Compute OUR, CER and RQ at every time point using an inert balance for the outlet flow.
2. Identify the feed step from the data alone, and then the time at which the physiology changes.
3. Quantify the RQ before and after, and say what the change means.
4. Reconstruct the specific oxygen uptake rate before and after, given that biomass plateaus near 45 gDW/L.
5. What would you have done, and when, if you had been watching this run live? Open the answer file.

Expected result. OUR rises exponentially and plateaus near **180 mmol·L⁻¹·h⁻¹**, which is the vessel's transfer ceiling rather than the culture's demand. RQ sits at **1.00 before hour 14 and 1.18 after**, and the feed step is at hour 12 — so the physiological change **lags the feed step by about two hours**, which is the interval in which an operator can still act. The specific uptake rate rises from **4.0 to 5.6 mmol·gDW⁻¹·h⁻¹**, meaning the cells were asked for 40% more oxygen than before.

The live reading is the point of the problem. An RQ above 1 with a feed increase and a dissolved oxygen trace that has stopped falling means the culture is fermenting part of its carbon — overflow, as in part 1 chapter 6 — because the oxygen it would need is no longer available. The action is to **reverse the feed step**, not to raise the stirrer, since OUR has already plateaued at the transfer limit and more agitation buys little. Waiting for the acetate assay to come back from the laboratory costs hours during which carbon is being converted to by-product, which is why the off-gas analyser earns its cost.

What it teaches: off-gas gives you metabolism in real time, and the lag between an operating change and its physiological consequence is the window in which the run can still be saved.

Closing notes to part 17

How to work these

Do the R problems with the answer file closed. Write your analysis, record your number, *then* open `ANSWERS_do_not_open_first.json` — it holds the true growth rates, C_q values, efficiencies, per-well parameters, labelling fraction and RQ shift. Checking afterwards is how you learn what your method costs; checking first turns the exercise into a reading comprehension test.

The MC problems need COBRApy or any LP solver and the core model, either from BiGG or rebuilt from part 10's listing. They are stated so that the *diagnosis* is the work: each injected error has a signature, and the skill is knowing which test exposes which signature.

What the ten establish

- **MC1** — a carbon-balanced reaction inflated growth 13-fold, and the one-line test that exposes it is growth with no carbon source.
- **MC2** — a loop changes the flux map and leaves the objective identical to ten decimal places, which sets the repair order.
- **MC3** — one reversibility edit took maximum ATP from 320 to the arbitrary bound.
- **MC4** — halving the biomass ATP cost is invisible, doubling it costs 12%; the direction of the error decides whether you find it.
- **MC5** — 392 reactions to inspect in a draft, 129 in a curated model; which checks automate and which cannot.
- **R1** — the same chromatogram reads 5% low or 23% high depending on where you put the integration limits.
- **R2** — $2^{-\Delta\Delta Cq}$ overstates by 8% at measured efficiencies of 95% and 99%.
- **R3** — the edge effect lands on the plateau and not on μ_{max}, so the robust answer is to change readout, not to correct.
- **R4** — omitting natural-abundance correction biases the flux split by 9%, systematically.
- **R5** — physiology changes two hours after the feed step, and that lag is the operator's window.

Where the whole problem set now stands

Source	Problems	Data
Your curriculum appendix, A1–W3	54	Stated inline
Part 15, X1–X24	24	Computed here
Part 15, Y1–Y6	6	Real public datasets, not run
Part 16, Z1–Z9	9	Computed here
Part 17, MC1–MC5	5	Computed here against the part 10 model
Part 17, R1–R5	5	**Data files supplied**, truth withheld in a separate file
Total	**103**	

I verified that each of the five datasets returns its own truth when processed correctly, which is how I know the problems are solvable: the chromatogram's internal standard recovers to 0.1%, the qPCR C_q values to 0.1 cycle, the Gompertz fits to about 1%, the labelling fraction to 0.006, and the RQ shift exactly. Where your answer differs from the file by more than that, the difference is in your method, which is the whole point.

What genuinely remains out of reach on paper: operating an instrument, judging a chromatogram's suitability by eye, and anything needing a strain in a flask. Those are the bench, and no document substitutes.

Part 18 — The pipeline, end to end: twelve problems on the stages the set was missing

You asked whether the 105 problems cover the whole route from choosing a host to shipping a purified product. I audited them against your list instead of answering from memory. Most stages were well covered; three were not covered at all, and two were covered only sideways. This part fills them.

The audit, by pipeline stage

Stage	Problems before	Verdict
Pathway design and yield ceilings	15	Strong
Scale-up and transport	13	Strong
Statistics and design of experiments	13	Strong
Economics and planning	13	Strong
Analytics and data	12	Strong
Enzyme and protein engineering	10	Adequate
Model curation and software	10	Adequate
Extraction and recovery	9	Adequate for extracellular, **nothing on intracellular release**
Fermentation modes	7	Fed-batch and chemostat only; **no mode comparison, no two-stage**
Two-phase and multiphase	7	Adequate
Host and strain selection	6	**No decision problem** — all were single-host calculations
DNA assembly and plasmid construction	3	**Thin** — nothing on copy number, burden or stability
Expression tuning and regulation	2	**Thin** — nothing on combinatorial library sizing
Growth against product optimisation	2	**Thin** — consortia only, no growth coupling
Genome editing (CRISPR, recombineering)	**0**	**Absent entirely**

Zero editing problems in a set meant for someone whose stated interest includes gene editing is the worst of those, and PP5 and PP6 fix it. The twelve here are **PP1 to PP12**, taking the set to **117**.

On "real problems from published papers"

These are built on real cases and I have to be exact about what that means. **I have no network access**, so I cannot open a paper, pull its data or check a number. Where a problem names a published result, the figure is recalled from memory and marked `source: recalled, verify` — treat every one as a claim to check against the paper itself, which is listed. The *arithmetic* in each solution was computed here and is reliable; the *input taken from a paper* is not verified. That distinction is the honest version of what you asked for, and the lookup workbook's sheet 10 is where these belong until you have checked them.

Stage 1 — Choosing the host

PP1 · extends week 59 · 2 h

Lysine: why *Corynebacterium* and not *E. coli*

Case: industrial L-lysine by *C. glutamicum*. Reported around 0.55 g/g at over 120 g/L after systems metabolic engineering (Becker, Wittmann and colleagues). **Source: recalled, verify.**

You must choose a host for a lysine process. The carbon-limited ceiling is 0.75 mol/mol; allowing the pentose phosphate pathway to supply the four NADPH per lysine gives 0.81 mol/mol.

1. Convert both ceilings to g/g, and express the reported industrial yield as a fraction of each.
2. *E. coli* has no dedicated lysine exporter and *C. glutamicum* has LysE. Compute the broth titer a retained product could reach at 50 and 100 gDCW/L if the intracellular concentration caps at 200 mM.
3. List the three properties that actually decide this choice, and name what you give up.
4. Under what change of product would the answer flip to *E. coli*?

Worked solution. 0.75 mol/mol is **0.609 g/g** and 0.81 is **0.657 g/g**, so a reported 0.55 g/g is **84%** of the cofactor-corrected ceiling — close enough that further yield work has little room, which is itself the finding.

The export arithmetic is decisive. With 2 mL of cell volume per gDCW, a product held at 200 mM intracellular gives only **2.9 g/L at 50 gDCW/L and 5.8 g/L at 100** — two orders of magnitude below the 120 g/L the process needs. **Without an exporter the process is impossible at any yield**, which is why this is a host-selection problem and not a pathway problem.

The three deciding properties: a native exporter; tolerance to 120 g/L of a charged amino acid in the broth; and the absence of overflow metabolism, since *C. glutamicum* does not make acetate at high glucose the way *E. coli* does. What you give up is cycle time and toolkit depth — *E. coli*'s design–build–test loop is faster, which matters for the twenty iterations a new pathway needs.

The answer flips for a **secreted protein, a non-natural small molecule needing many heterologous enzymes, or anything where the campaign needs dozens of iterations** — there the toolkit outweighs the exporter, and if the product is uncharged and diffuses out, the export advantage disappears entirely.

What it teaches: host choice is usually settled by one property that the pathway cannot fix — here, export — and the yield comparison is a distraction until that is checked.

PP2 · extends weeks 18, 19 and 40 · 2.5 h

1,4-butanediol: designing a pathway that nature does not have

Case: 1,4-BDO in *E. coli* via succinyl-CoA and 4-hydroxybutyrate. Reported 18 g/L in the original paper and over 100 g/L commercially. **Source: recalled, verify.**

BDO is $C_4H_{10}O_2$ (90.12). The route runs glucose → oxaloacetate → succinate → succinyl-CoA → succinate semialdehyde → 4-hydroxybutyrate → 4-HB-CoA → BDO.

1. Compute the carbon and mass yield ceiling with and without anaplerotic CO_2 fixation.
2. Count the reductant the route consumes per BDO and compare with what glycolysis supplies.
3. The route needs an enzyme activity that did not exist. Which step, and how would you find one?
4. 18 g/L is what fraction of the ceiling, and what would you measure to decide whether to push titer or yield?

Worked solution. Four carbons out of six gives **0.500 g/g** with two CO_2 lost; with PEP carboxylase re-fixing CO_2 the balance improves to about **0.667 g/g** — and which applies depends on how much carbon enters through the carboxylation rather than pyruvate dehydrogenase, so the ceiling is a *design variable*, not a constant.

Reductant is the binding constraint. The reductive steps from succinyl-CoA to BDO consume roughly **4 NAD(P)H per molecule** while glycolysis to oxaloacetate supplies **2** — a deficit of two, which must come from further oxidation of carbon (lowering yield), from formate oxidation, or from a different split between the oxidative and reductive branches. That is the quantitative reason this product is hard, and it is visible before any strain exists.

The missing activity is the **CoA-dependent reduction of 4-HB to the aldehyde**, and the route to one is the chapter S3 and part 8 chapter 29 route: find enzymes acting on an analogous substrate (aldehyde dehydrogenases on acyl-CoAs), screen homologues, then evolve. 18 g/L against a 0.500–0.667 g/g ceiling and a typical 100 g/L target is **roughly**

a fifth of a commercially viable titer, so the next measurement is a carbon balance with by-products: if carbon is going to acetate and ethanol, the problem is the fermentative balance, not the pathway.

What it teaches: a novel pathway's hardest number is its redox deficit, and it is computable from the stoichiometry before a single gene is cloned.

Stage 2 — Building it: plasmid, expression, editing

PP3 · extends weeks 10 and 32 · 1.5 h

Plasmid or chromosome, decided by arithmetic

A seed train from a 1 mL vial to a 100 m³ vessel is about 27 generations. Three plasmid classes are available, with illustrative loss rates and burdens, against single-copy integration.

1. Compute the generations in the seed train, then plasmid retention at 27 and 40 generations for each class.
2. A 300-copy plasmid gives how much more gene dose than one integrated copy? What would it take to match it from the chromosome?
3. Which option would you choose for a 48-hour fed-batch, and which for a 30-day continuous run?
4. What single measurement decides it in practice?

Replicon	Loss/division	Burden	Retention at 27 gen	at 40 gen
pUC-type, high copy	10^{-3}	15%	**75%**	**32%**
pBR322-type, medium	5×10^{-4}	8%	95%	88%
pSC101-type, low	2×10^{-4}	3%	99%	98%
Chromosomal, single copy	—	—	100%	100%

Worked solution. $\log_2(10^8)$ = **26.6 generations** from vial to vessel. The high-copy plasmid is already a quarter lost before production begins and two-thirds lost by 40 generations — and the cells that lost it grow 15% faster, so they dominate the end of the run, which is where most of your product is made.

On dose: one integrated copy against 300 is a 300-fold deficit, and no promoter closes that — the strongest promoters span perhaps 10- to 100-fold over a medium one. Realistically you combine **a stronger promoter (10×) with three integrated copies (3×)** for 30 copy-equivalents, and accept that a plasmid remains ahead on dose while losing on stability.

So: for a **48-hour fed-batch** a medium-copy plasmid with selection is defensible — 95% retention over the train, and the run is short. For a **30-day continuous run** only integration survives the arithmetic of problem Z8. The deciding measurement is cheap and almost never done: **plate the end of a pilot run with and without antibiotic and count both**. The ratio is your retention, and if it is below about 90% the yield you measured is not the yield of your strain.

What it teaches: plasmid choice is a race between gene dose and retention over the generations the process actually runs, and both sides are measurable in an afternoon.

PP4 · extends weeks 8 and 28 · 2 h

Balancing three enzymes: how big must the library be?

Case: combinatorial promoter and ribosome-binding-site libraries for pathway balancing, as used for mevalonate and carotenoid pathways. **Source: recalled, verify.**

1. Compute the number of combinations for 3 enzymes × 5 levels, 3 × 8, 4 × 5 and 5 × 4.
2. For each, how many clones give 95% coverage?
3. What coverage do 96 and 384 wells give for the 125- and 625-member libraries?
4. If expression effects are lognormal with σ = 0.6, how much better than the median is the best of 24, 96 and 384 random clones?
5. Design the campaign you would actually run.

Design	Combinations	Clones for 95% coverage
3 enzymes × 5 levels	125	373
3 × 8	512	1,532
4 × 5	625	1,871
5 × 4	1,024	3,066

Worked solution. Coverage is the problem X20 calculation: 96 wells of a 125-member library covers **54%**, 384 covers **95%**, and 384 of a 625-member library covers only **46%**. So a single 96-well plate does not survey even the smallest

useful library — the common practice of picking one plate and reporting the best clone is sampling half the design space.

But the returns are blunt: the best of 24 random clones is **3.4-fold** above the median, the best of 96 is **4.6-fold**, and the best of 384 is **6.0-fold**. Sixteen times the screening effort buys 1.8 times the improvement, because you are sampling the tail of a lognormal.

The campaign that follows: **three enzymes at five levels, 384 clones, in two rounds** — one plate to estimate the spread and check the assay, then three more if the spread justifies it. Beyond three enzymes, move to a designed subset rather than full factorial coverage, and use the measured strengths as model inputs (chapter 41's surrogate) instead of screening the product space.

What it teaches: library size is set by coverage arithmetic, and the gain from screening more is logarithmic — so the design, not the plate count, is where the improvement comes from.

PP5 · extends weeks 23 and 24 · 2.5 h

CRISPR as a counterselection: what fraction of your colonies are edited?

Case: Cas9-assisted recombineering in bacteria, and multiplex automated genome engineering (MAGE), which reported 4.3 billion variants in three days. **Source: recalled, verify.**

In bacteria Cas9 is used to kill the unedited rather than to cut and repair, because there is no efficient end-joining. Recombination supplies the edit; Cas9 removes the background.

1. For recombination frequencies of 10^{-4}, 10^{-3} and 10^{-2}, and killing of 99%, 99.9% and 99.99%, compute the edited fraction among survivors and the clones to screen for a 95% chance of one.
2. If 1 in 10^4 cells escapes by losing the target site without being edited, what fraction of survivors are false positives at a recombination frequency of 10^{-3}?
3. For MAGE, compute the fraction of cells edited at all of 4, 10 and 24 loci after 20 cycles at per-cycle efficiencies of 10%, 20% and 30%.
4. When would you choose each method?

Recombination	Kill 99%	Kill 99.9%	Kill 99.99%
10^{-4}	1.0% (299 clones)	9.1% (32)	50% (5)
10^{-3}	9.1% (32)	50% (5)	90.9% (2)
10^{-2}	50% (5)	91% (2)	99% (1)

Worked solution. The edited fraction is f/(f + (1−f)(1−kill)), and the table shows why counterselection is transformative: at a recombination frequency of 10^{-4} — typical for a point mutation by oligo recombineering without selection — you would screen 30,000 colonies, and three nines of killing cuts that to **32**. The practical lesson is that **killing stringency matters as much as recombination efficiency**, and it is the easier of the two to improve.

Escapers set the floor. At 10^{-4} escape and 10^{-3} recombination, **9.1% of survivors are false positives** — colonies that lost the protospacer or the guide rather than taking the edit — so verification by sequencing is not optional, and a construct whose guide can be silently deleted will give you exactly this.

MAGE after 20 cycles reaches **95.5% of cells edited at all 4 loci at 20% per-cycle efficiency, 89.1% at 10 loci, and 75.7% at 24**; at 10% efficiency those fall to 27.4% at ten loci and 4.5% at twenty-four. So MAGE is for **many loci, tolerated imperfectly, with a selection or screen at the end**, while Cas9-assisted recombineering is for **one precise edit you must have in a clean background**.

What it teaches: editing is a probability calculation, and the number of colonies you must screen follows from recombination frequency, killing stringency and escape rate — all three measurable.

PP6 · extends weeks 23 and 32 · 2 h

Integrating a cassette, verifying it, and removing the marker

Case: λ-Red recombineering with FLP-removable markers, and *sacB* counterselection. **Source: recalled, verify.**

1. With 1 µg of a linear cassette, a transformation efficiency of 10^7 cfu/µg and recombination at 10^{-4} or 10^{-5}, how many integrant colonies do you expect?
2. You then plate 10^8 cells to counterselect marker loss on sucrose. At *sacB* escape rates of 10^{-3}, 10^{-4} and 10^{-6}, and 100 or 1,000 true excisants, what fraction of survivors are real?
3. Design the verification PCRs. Why is one PCR insufficient?

4. What does FLP removal leave behind, and what does that cost on the second round?

Worked solution. Expected integrants are **1,000 colonies at 10^{-4} and 100 at 10^{-5}** — ample, which is why λ-Red works at all.

The counterselection is where it goes wrong. Plating 10^8 cells with a *sacB* escape rate of 10^{-3} gives 10^5 escapers against 100 true excisants: **0.10% of survivors are real, so you screen about 1,000 colonies for one**. At 10^{-4} escape it is 101 colonies; only at 10^{-6} does it become **2 colonies per hit**. Two fixes follow directly: use a counterselection whose escape rate you have measured, and **plate a smaller fraction of the culture** — re-plating 1% cuts escapers a hundredfold while costing you proportionally fewer true events, which is favourable whenever escapers dominate.

Verification needs **two junction PCRs**, each with one primer in the chromosome outside the homology arm and one inside the cassette, plus one internal amplicon. One PCR cannot distinguish a correct single insertion from a tandem duplication or from a cassette that integrated elsewhere by a secondary homology. FLP removal leaves a **34 bp FRT scar**; a second round at a different locus leaves a second scar, and the two scars are substrates for recombination between them — which is the mechanism by which a strain with several rounds of marker removal can delete the intervening chromosome.

What it teaches: the integration step is easy and the marker-removal step is where the screening burden hides; the ratio of escapers to true events is the number to compute before plating.

Stage 3 — Making it produce

PP7 · extends week 39 · 2 h

Growth coupling for succinate: the deletions and what they cost

Case: anaerobic succinate strains of *E. coli*, built by deleting the competing fermentative routes and then evolving. **Source: recalled, verify.**

Using the 40-reaction core model anaerobically on 10 mmol of glucose, with growth-coupled production defined as the *minimum* product flux at maximum growth.

1. Compute the wild-type guaranteed succinate flux.
2. Evaluate ΔPYK, ΔPYK ΔACK, ΔPYK ΔACK ΔPFL, and ΔADH ΔLDH.
3. Express each as mol per mol of glucose, and state the growth cost.
4. Which design would you build, and what would the published strains have added?

Design	Growth (h^{-1})	Guaranteed succinate	mol/mol glucose
Wild type	0.217	0.00	0.00
ΔPYK	0.134	8.23	0.82
ΔPYK ΔACK	0.072	9.05	0.90
ΔPYK ΔACK ΔPFL	0.048	9.36	0.94
ΔADH ΔLDH	**lethal anaerobically**		

Worked solution. The wild type guarantees nothing: it can grow without making any succinate, so selection does not favour producers. Each deletion closes more routes and forces more product, at a growth cost that compounds — **0.82 mol/mol at 62% of wild-type growth, rising to 0.94 mol/mol at 22%**. ΔADH ΔLDH is lethal because it removes the cell's only remaining NADH sinks, which is the kind of answer a model gives cheaply and a strain-building campaign discovers expensively.

I would build **ΔPYK ΔACK**: 90% of the maximum coupled yield at nearly twice the growth rate of the triple, which matters because a slow strain means a longer fermentation and a bigger seed train. What the published strains added beyond the model: **evolution**, to recover growth rate after the deletions, and work on the **carboxylation step and glucose uptake** — neither of which this model can suggest, since its only uptake route is the PEP-consuming phosphotransferase system.

What it teaches: growth coupling is a trade curve, not a design, and the right point on it is chosen by the fermentation economics rather than by the maximum.

PP8 · extends weeks 44 and 56 · 2 h

Two-stage fermentation: where to put the substrate

A fixed budget of 200 g/L of glucose and 48 hours. Growth yield 0.45 g/g at $\mu = 0.45\ h^{-1}$; in stage two the cells do not grow and convert substrate to product at 0.40 g/g with a specific rate of 0.25 $g \cdot gDW^{-1} \cdot h^{-1}$. Start from 0.5 gDW/L.

1. Write the time needed for each stage as a function of the fraction f of substrate spent on growth.
2. Tabulate biomass, product and total time for f from 0.05 to 0.7.
3. Find the f that maximises product within 48 hours.
4. Compare with a one-stage growth-coupled process at 0.30 g/g.

f to growth	Biomass (gDW/L)	Growth time	Product (g/L)	Production time	Total
0.05	4.5	4.9 h	76.0	67.6 h	**72.4 h — too slow**
0.10	9.0	6.4	72.0	32.0	38.4 h
0.20	18.0	8.0	64.0	14.2	22.2 h
0.50	45.0	10.0	40.0	3.6	13.6 h

Worked solution. Substrate spent on cells is substrate not in the product, so product falls linearly with f — but the *time* to convert the rest falls as 1/f, because more catalyst works faster. The optimum sits where the 48-hour limit binds: **f = 0.078, giving 7.0 gDW/L, a switch at 5.9 h, and 73.8 g/L of product in exactly 48 hours** — an overall yield of

0.369 g/g and a productivity of **1.54 g/L/h**.

Against a one-stage growth-coupled process at 0.30 g/g, which gives 60 g/L and 1.25 g/L/h, the two-stage route is **23% better in titer and productivity** — because it spends almost no carbon on cells. The caveats are the ones the arithmetic does not contain: non-growing cells lose productivity over time, which this model ignores, and the switch has to be triggered by something real (nitrogen exhaustion, a temperature shift, an inducer), each of which has its own lag.

What it teaches: the two-stage switch time is a constrained optimum between carbon spent on catalyst and the time that catalyst needs, and it lands where the clock binds.

PP9 · extends week 44 · 2.5 h

Batch, fed-batch or continuous, for the same strain

One strain: μ_{max} 0.45 h^{-1}, K_s 0.5 g/L, $Y_{X/S}$ 0.45, specific productivity 0.30 $g \cdot gDW^{-1} \cdot h^{-1}$ falling linearly to zero at 60 g/L of product, maintenance 0.02. Turnaround between batches is 12 hours.

1. Simulate a batch from 200 g/L of glucose: titer, time, productivity, with and without turnaround.
2. Simulate an exponential fed-batch at $\mu = 0.12$ h^{-1} from a 500 g/L feed.
3. Compute the continuous steady state at $D = 0.05$ to 0.3 h^{-1} on a 50 g/L feed.
4. Tabulate all three on titer, yield, productivity and vessel-time, and choose.

Mode	Titer	Time	Productivity	With turnaround
Batch	46.4 g/L	14.3 h	3.26 g/L/h	1.77 g/L/h
Fed-batch, $\mu = 0.12$	53.7 g/L	34.7 h	1.55 g/L/h	1.15 g/L/h
Continuous, $D = 0.1$	60.0 g/L	—	**6.00 g/L/h**	6.00 g/L/h
Continuous, $D = 0.2$	33.5 g/L	—	6.70 g/L/h	6.70 g/L/h

Worked solution. Continuous wins on productivity by a factor of three to four, because it never stops and never spends time at low cell density — and that is exactly why it is rare. Its titer is capped at the steady-state product concentration, so the downstream train sees a dilute stream; it needs 30 to 60 days of genetic stability, which problem Z8 says a cheater at 8% advantage destroys; and contamination over that period is a serious risk.

Fed-batch reaches the **highest cell density (82 gDW/L)** and the highest titer of the two closed modes, at the cost of productivity, because most of the run is spent accumulating biomass. Batch is the most productive closed mode here and the simplest, but it leaves no room to control overflow.

The choice: **fed-batch** for a product that is inhibitory and needs titer for the separation — which is most of them — and **continuous** only when the organism is stable, the product is not inhibitory, and the downstream can take a dilute stream. The decisive comparison is not productivity but **cost per kilogram delivered**, which pulls toward titer.

What it teaches: mode selection is a four-way trade between titer, productivity, stability and downstream cost, and the mode with the best productivity is usually the wrong answer.

Stage 4 — Getting it out

PP10 · extends chapter S1 and week 56 · 2 h

An intracellular product: how many homogeniser passes?

Case: PHA recovery by solvent extraction, hypochlorite digestion, enzymatic digestion or surfactant. **Source: recalled, verify — the recovery and purity figures below are illustrative.**

Disruption releases product first-order in passes: released fraction = $1 - e^{-kn}$. The product also loses activity or quality at a fixed rate per pass.

1. Tabulate release after 1, 2, 3 and 5 passes for k = 0.6, 0.9 and 1.4.
2. With k = 0.9 and 3% or 8% quality loss per pass, compute the net recovered yield.
3. Find the optimum number of passes at 8% loss.
4. For a PHA at 60% of cell mass and 100 gDW/L, compare the four recovery routes on delivered product.

Passes	Release at k = 0.9	Net at 3% loss/pass	Net at 8% loss/pass
1	59%	58%	55%
2	83%	79%	71%
3	93%	85%	**73%**
5	99%	**85%**	65%

Worked solution. Release saturates while damage accumulates, so the net has an interior maximum: at 8% loss per pass the optimum is **3 passes for 73%**, and a fifth pass actively destroys product. At 3% loss the optimum is flat from three to five passes. Running "until lysis is complete" is therefore wrong whenever the product is fragile — and the measurement that settles it is a pass-by-pass assay of *active* product, not of released protein.

For the PHA, 60% content at 100 gDW/L is **60 g/L of polymer in the cells**; at 90% recovery you deliver 54 g/L. The four routes differ less in recovery (85–95%) than in what they cost elsewhere: solvent extraction gives the best purity and the worst solvent handling; hypochlorite is cheap and **degrades the polymer's molecular weight**, which for a material product is the specification, not a detail; enzymatic digestion preserves it and costs enzyme; surfactant is cheapest and leaves 8% impurity. The choice is made by the product specification, not the recovery column.

What it teaches: for an intracellular product the release step has an optimum, and for a polymer the recovery method changes the product itself rather than just how much you get.

PP11 · extends weeks 45 and 49 · 2 h

An extracellular product: two-phase capture through to crystals

Case: amorphadiene and artemisinic acid production with an organic overlay, then recovery. Partition coefficient around 107. **Source: recalled, verify.**

1. For solvent fractions of 0.02, 0.05, 0.10 and 0.20, compute the broth concentration relative to a single phase and the fraction of product held in the solvent.
2. Choose a solvent fraction and justify it against the displaced working volume.
3. With phase separation 99%, solvent stripping 97%, crystallisation 85% and drying 99%, compute the train recovery.
4. Give the overall recovery from fermenter to dry product, and name the step to improve first.

Solvent fraction	Broth concentration, relative	Product in the solvent
0.02	0.321	69%
0.05	0.159	85%
0.10	0.086	92%
0.20	0.045	96%

Worked solution. Even 2% solvent pulls **69%** of the product out of the aqueous phase and cuts what the cells see to a third, which is the whole point of in situ removal — relief of product inhibition comes cheaply. Going from 5% to 20% solvent moves the captured fraction from 85% to 96% while displacing four times the working volume, so **5% is the**

sensible choice: the last 11% of capture costs 15% of your vessel.

The downstream train multiplies to **80.8%**, and with 95% captured into the solvent the overall recovery from fermenter to dry product is **76.8%**. The step to improve first is **crystallisation at 85%**, which is both the worst step and the one with the most headroom — a second crop from the mother liquor typically recovers half of what is lost, taking the train to about 87% for the cost of one more vessel and a filtration.

What it teaches: in situ removal is bought cheaply at low solvent fraction, and the recovery train's worst step is where the next engineering hour belongs.

PP12 · extends weeks 44 and 49 · 2.5 h

A whole process at scale: 1,3-propanediol at 135 g/L

Case: bio-based 1,3-propanediol, reported around 135 g/L at about 0.51 g/g from glucose in an engineered *E. coli*. **Source: recalled, verify.**

1. Compute broth volume and water to remove per kg of product, and the feedstock cost at $0.40/kg glucose.
2. At 30 gDW/L and 4 mmol $O_2 \cdot gDW^{-1} \cdot h^{-1}$, compute OUR, the required k_La and the heat duty in a 100 m^3 vessel.
3. Propanediol boils at 214 °C and water at 100 °C. Compute the energy and cost of removing the water.
4. Compare separation with feedstock cost, and compute what the water cost would have been at 50 g/L.

Worked solution. At 135 g/L the broth is **7.4 L per kg** of product with **6.4 kg of water to remove** — against 20 kg at 50 g/L, which is the entire reason the titer target was set where it was. Feedstock is 1/0.51 = **1.96 kg of glucose per kg**, so **$0.78/kg** at $0.40/kg sugar.

Oxygen: OUR = **120 mmol·L^{-1}·h^{-1}**, needing a k_La of **750 h^{-1}** — at the top of what a large vessel achieves, so this process is close to its transfer limit and oxygen enrichment or a lower feed rate is likely in the real plant. Heat is **1.53 MW**, needing 132 m^3/h of cooling water and **307 m^2 of heat-transfer area** against roughly 110 m^2 of jacket: internal coils, as problem X12 predicted.

Water removal is **14.5 MJ per kg** of product, or **$0.12/kg** at $8/GJ — **13% of the variable cost** alongside $0.78 of feedstock. At 50 g/L it would have been **$0.34/kg**, nearly tripling. So the headline achievement of this process is not the yield, which is close to the theoretical maximum anyway, but the **titer**, and the number it bought is a separation cost.

What it teaches: a real process is judged on four numbers at once — yield for feedstock, titer for separation, rate for capital, and oxygen for the vessel — and titer is usually the one that pays for itself twice.

Closing notes to part 18

Where the set now stands against your pipeline

Stage	Before	Added	Now
Host and strain selection	6	PP1	7, with a decision problem
Pathway design	15	PP2	16, including a novel-enzyme route
Plasmid and construction	3	PP3	4, with copy number against stability
Expression tuning	2	PP4	3, with library sizing
Genome editing	**0**	PP5, PP6	**2**
Growth against product	2	PP7	3, with growth coupling
Fermentation modes	7	PP8, PP9	9, with two-stage and a mode comparison
Recovery, intracellular	**0**	PP10	**1**
Recovery, extracellular and purification	9	PP11	10, as a full train
Scale-up	13	PP12	14, with a whole process costed
Total problems	**105**	**12**	**117**

What is still not covered, honestly

- **Protein products** — secretion capacity, inclusion bodies, refolding, glycosylation. The set is built around small molecules and polymers. If your target labs do therapeutic proteins, this is the next gap to fill.
- **Regulatory and quality** — GMP, validation, batch records. Named in part 12 and never worked as a problem, because it is procedure rather than calculation.
- **Anything needing a real instrument or a real strain** — covered as far as synthetic data allows (R1 to R5), and no further.
- **Verified published numbers.** Every paper figure in this part is recalled, not checked. The arithmetic built on them is sound; the inputs need an afternoon with the papers and sheet 10 of the lookup workbook.

The eight results worth remembering

- **PP1** — without an exporter, a retained product caps at 2.9 g/L at 50 gDCW/L. Host choice decided by one property.
- **PP3** — a high-copy plasmid is 25% lost before production begins and 68% lost by 40 generations.
- **PP4** — 96 wells covers 54% of a 125-member library, and the best of 384 clones is only 1.8 times better than the best of 24.
- **PP5** — three nines of Cas9 killing cuts the screen from 30,000 colonies to 32.
- **PP6** — *sacB* at 10^{-3} escape means 1,000 colonies screened per true excisant.
- **PP7** — ΔPYK ΔACK gives 0.90 mol/mol at 0.072 h^{-1}; the triple buys 4% more product for half the growth.
- **PP9** — continuous is 3 to 4 times more productive and almost always the wrong choice.
- **PP12** — at 135 g/L water removal costs $0.12/kg; at 50 g/L it would cost $0.34/kg.

Reading it on an iPad and a Mac: what works, what cannot, and the half-hour that fixes it

Short version: the links do not work across devices because they are *file* links, and a file link points at a folder on one machine. On an iPad they do not work even on that machine, because Safari refuses to open a local file from inside another local file. The fix is to put the folder online once. After that the same tracker and the same links work on every device you own, including the iPad.

Why it fails now

Situation	Does a link work?	Why
Mac, every file in one folder, opened in Chrome or Firefox	**Yes**	A relative link resolves beside the tracker
Mac, files in different folders	No	Nothing is beside the tracker to find
Mac, Safari	Often no	Safari restricts what a local page may open
iPad, files in iCloud Drive, opened in Files or Safari	**No**	iOS gives the page no access to neighbouring files at all
Any device, folder hosted on the web	**Yes**	An `https://` address means the same thing everywhere

So hosting is not a nicety here; on an iPad it is the only route that works.

Route A — GitHub Pages, about half an hour, free and permanent

This is the one I would choose. It gives stable addresses, costs nothing, works on every device, and needs no app.

1. Create a free account at github.com, then click **New repository**. Name it `me-module`, make it **Public**, and create it.

2. On the repository page choose **Add file → Upload files**. Drag in all 24 files. For the data, make a folder first: in the upload box type `me-problem-data/` in the file-name field before dropping the seven CSV and JSON files, or simply drag the whole `me-problem-data` folder in — GitHub keeps the folder name. Commit.

3. Go to **Settings → Pages**. Under "Build and deployment" set **Source** to "Deploy from a branch", **Branch** to `main` and the folder to `/ (root)`. Save. Wait two or three minutes.

4. Your folder is now at `https://YOURNAME.github.io/me-module/`. Check it by opening `https://YOURNAME.github.io/me-module/00-INDEX-me-module.pdf` in any browser.

5. Open the tracker at `https://YOURNAME.github.io/me-module/me-tracker-wk01-66-enhanced.html` — bookmark that on both devices. Because the tracker is now served from the same folder as everything else, **the links already work and you do not need to set anything**. If you prefer to keep a local copy of the tracker, open its **Materials** tab, paste `https://YOURNAME.github.io/me-module/` into the folder box, press **Save folder**, then **Test a link**.

A public repository means anyone with the address can read the files. Everything here is your own study material, so that is usually fine. If you would rather it were not public, route B avoids it.

Route B — iCloud Drive, Dropbox or OneDrive, about ten minutes

Quicker to set up and less tidy, because these services do not serve a folder as plain web addresses.

- Put the whole folder in iCloud Drive, Dropbox or OneDrive and let it sync to both devices.
- **On the Mac** this works like any local folder: open the tracker in Chrome or Firefox (not Safari) and the relative links resolve.
- **On the iPad** the links will still not work, because iOS isolates each file. What you can do is open the tracker for ticking and planning, and open the PDFs directly from the Files app when you want to read one. You lose the one-click route from a week to its chapter, which is the part that made the integration worth doing.
- Dropbox and OneDrive can give a direct link per file, and you could paste a base address into the Materials box — but their link format differs per file, so a single folder address will not work. Treat this as the fallback, not the plan.

Route C — a local web server on the Mac, five minutes, for the Mac only

If you only want the Mac fixed today: open Terminal in the module folder and run

```
cd ~/Downloads/me-module
python3 -m http.server 8000
```

then open `http://localhost:8000/me-tracker-wk01-66-enhanced.html`. Every link works, because the page is now served over http. The iPad can reach it too — at `http://YOUR-MACS-IP:8000/…` — but only while the Mac is awake and on the same network, which is why route A is better.

The one thing hosting does not fix: progress

Your ticks, hours, quiz history and notes are stored by the browser, per device. The iPad and the Mac will each keep their own, and hosting does not change that — a web page cannot write to a shared file.

The tracker already has the fix at the foot of every page: **Export backup** writes a small file with your whole state, and **Import backup** reads it. The habit that works:

When	Do this
Finishing a session on one device	Export backup, save it to the shared folder
Starting on the other device	Import backup from that file
If you forget and tick on both	Keep the more complete export and re-import it; the other device's ticks for that period are lost

If that is too much friction, the honest alternative is to **tick on one device only** — the Mac, say — and treat the iPad as a reading device. You lose nothing except the tick, and the tick is the part you can redo in ten seconds.

Checklist

	Check
1	All 24 files in one folder, with the seven data files inside `me-problem-data/`
2	Folder uploaded and Pages enabled; `00-INDEX-me-module.pdf` opens at its web address
3	Tracker opens at its web address on both devices and is bookmarked
4	Materials tab → **Test a link** opens the index
5	A week's module pointer and a problem link both open their file
6	You know where the exported backup lives

If a link still fails after hosting, the cause is almost always a file name: GitHub is case-sensitive, so `ME-Module-wk01-10-part01.pdf` is a different file from `me-module-wk01-10-part01.pdf`. Compare the failing address with the file list on the Materials tab.